

ivdtools Handbook

R toolkit for performance evaluation of IVD reagents

hiox-tech

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Table of contents

Welcome	1
Reader' s Guide	1
Getting Help	1
Version and Citation	1
Project Information	1
1 ivdtools Package Overview	3
1.1 Functional Modules	3
1.2 Design Features	4
1.2.1 Standalone Workflows	4
1.2.2 Progressive Workflows	5
1.3 Quick Start	5
1.3.1 Method Comparison	5
1.3.2 ROC	5
1.3.3 Qualitative Agreement	6
1.3.4 Bottle-to-Bottle Difference	6
1.4 Dependencies	6
2 Installing and Using the Analysis Environment	7
2.1 Install R	7
2.1.1 Windows and macOS	7
2.1.2 Debian/Ubuntu	7
2.1.3 Fedora	7
2.2 Choose an Editor	7
2.3 Install R Packages	8
2.4 Create a Project	8
2.5 Import and Verify Data	9
2.5.1 CSV	9
2.5.2 Excel	10
2.6 Run Code	10
2.7 Export Results	11
2.7.1 Tables	11
2.7.2 Figures	11
2.8 Render and Save	11
3 Calibration Curve Fitting	13
3.1 Main Functions	13
3.1.1 View Built-in Equations with <code>list_equation()</code>	13
3.1.2 Summarize Replicates with <code>replicate_to_mean()</code>	14
3.1.3 Fit a Built-in or Custom Equation with <code>fit_equation()</code>	14
3.1.4 Compare Candidate Equations with <code>compare_equation()</code>	14
3.1.5 Get Fitted Parameters and Residuals with <code>coef()</code> and <code>residuals()</code>	15
3.1.6 Forward Response Prediction or Inverse Concentration Estimation with <code>predict()</code>	15
3.1.7 Plot the Fitted Curve and Interval with <code>plot()</code>	15
3.2 Analysis Examples	15

3.2.1	Fitting a Linear Calibration Curve	15
3.2.2	Fitting a 4PLC Calibration Curve for a Sandwich Assay	18
3.2.3	Constrained Fitting	20
4	Stability Studies	21
4.1	Main Functions	21
4.1.1	Plan a Study	22
4.1.2	Node Bias	22
4.1.3	Trend Regression	23
4.1.4	Regression Against the First Node <code>stability_regression()</code>	25
4.1.5	Time to an Allowable Limit	26
4.1.6	Arrhenius Accelerated Stability	27
4.1.7	Mean Kinetic Temperature	28
4.1.8	Split Multiple-Sample Long-Format Data	29
4.2	Analysis Cases	30
4.2.1	CLSI EP25-A2 Appendix B: Arrhenius Accelerated Stability	30
4.2.2	CLSI EP25-A2 Appendix C1: Reagent Stability	32
4.2.3	CLSI EP25-A2 Appendix C2.1: Calibrator Stability (Reference-Anchored Real-Time Study)	33
4.2.4	CLSI EP25-A2 Appendix C2.2: Calibrator Stability (Isochronous Study)	37
4.2.5	CLSI EP25-A2 Appendix C3.1: Control Stability (Reference-Anchored Real-Time Study)	40
4.2.6	CLSI EP25-A2 Appendix C3.2: Control Stability (Isochronous Study)	43
4.2.7	CLSI EP25-A2 Appendix C4: Transport Conditions	47
4.2.8	CLSI EP25-A2 Appendix C5: Outlier Evaluation	49
5	Measurement Uncertainty Analysis	53
5.1	Main Functions	53
5.1.1	Type A Components from Replicate Measurements: <code>uncertainty_type_a()</code>	54
5.1.2	Type B Components from Certificates or Specifications: <code>uncertainty_type_b()</code>	54
5.1.3	Long-Term IQC Components: <code>uncertainty_iqc()</code>	55
5.1.4	Between-Bottle Homogeneity: <code>uncertainty_homogeneity()</code>	56
5.1.5	Stability-Trend Components: <code>uncertainty_stability()</code>	57
5.1.6	Multi-Laboratory Characterization: <code>uncertainty_characterization()</code>	58
5.1.7	Single-Node Combination: <code>uncertainty_combine()</code>	58
5.1.8	Nonlinear Measurement Models: <code>uncertainty_propagate()</code>	60
5.1.9	Accumulating a Metrological Traceability Chain: <code>uncertainty_traceability()</code>	62
5.2	Analysis Cases	63
5.2.1	CLSI EP29-A Appendix A: Converting Type B Limits	63
5.2.2	CLSI EP29-A Appendix B: Cholesterol IQC	65
5.2.3	CLSI EP29-A Examples 1–3: Measurement-Model Propagation	66
5.2.4	CLSI EP29-A Example 4: One-Way ANOVA and IQC Uncertainty	67
5.2.5	CLSI EP29-A Examples 5–6: Bias Estimation and Bias-Correction Uncertainty	68
5.2.6	CLSI EP29-A Example 10: Creatinine-Clearance Uncertainty Propagation	69
5.2.7	CLSI EP29-A Example 11: Linear-Regression Prediction Equation	70
5.2.8	YY/T 1709: Calibrator Homogeneity, Stability, and Assigned-Value Uncertainty	71
6	Matrix Effects and Commutability	75
6.1	Main Function and Input Data	75
6.2	Statistical Paths	76
6.3	Analysis Examples	77
6.3.1	EP14-A3 Cholesterol on the Original Scale	77
6.3.2	EP30 Multi-Procedure Reference Materials	79
6.3.3	IFCC Part 2 Bias Difference and Uncertainty	212
6.3.4	IFCC Part 3 Calibration Validity	214
6.4	Interpretation	214

6.5	Detailed Standard Cases	215
6.5.1	CLSI EP14-A3 Appendix C Example C1: Cholesterol on the Original Scale	215
6.5.2	CLSI EP14-A3 Appendix C Example C2: Myoglobin on the log ₁₀ Scale	218
6.5.3	CLSI EP30-A Appendix B: Multi-Procedure Reference-Material Data	221
6.5.4	IFCC Part 2 Official Supplement: Bias Difference and Uncertainty	229
6.5.5	IFCC Part 3 Official Supplement: Calibration-Validity Assessment	233
6.5.6	Real-Application Comparison: POCT Glucose EQA Materials	239
7	Interference Studies	241
7.1	Main Functions	241
7.1.1	Replicate Planning	242
7.1.2	Paired-Preparation Studies	242
7.1.3	Dose-Response Studies	242
7.1.4	Patient-Sample Comparison	243
7.2	Analysis Examples	243
7.2.1	EP07 Replicate Planning and Paired Screening	243
7.2.2	EP07 Dose Response	244
7.2.3	Patient-Sample Comparison	245
7.3	Interpretation and Reporting	246
7.4	Full Standard Cases	246
7.4.1	CLSI EP07 Third Edition Section 5.1: Replicate-Count Design	246
7.4.2	CLSI EP07 Third Edition Table 3: Paired Screening	247
7.4.3	CLSI EP07 Third Edition Table 4: Curved Dose-Response and Point-to-Point Interpolation	249
7.4.4	CLSI EP07 Third Edition Table 5: Linear Dose-Response	252
7.4.5	YY/T 1789.5-2023 Appendix A: Interferent-Spike Screening	255
7.4.6	YY/T 1789.5-2023 Appendix B: Hemoglobin Dose-Response	256
7.4.7	YY/T 1789.5-2023 Appendix C: Patient-Sample Evaluation	259
7.4.8	YY/T 1789.5-2023 Appendix D: Cross-Reactivity Evaluation	261
7.5	Detailed Function Reference	262
7.5.1	Replicate Planning with <code>interference_replicates()</code>	262
7.5.2	Paired Preparation with <code>interference_paired()</code>	263
7.5.3	Dose-Response with <code>interference_dose_response()</code>	263
7.5.4	Patient-Sample Comparison with <code>interference_patient()</code>	264
8	Dilution Recovery, Spike Recovery, and Hook Effect	267
8.1	Main Functions	267
8.1.1	Dilution Recovery	268
8.1.2	Theoretical Spike Concentration	268
8.1.3	Spike Recovery	268
8.1.4	High-Dose Hook Effect	269
8.2	Analysis Examples	269
8.2.1	Dilution Recovery	269
8.2.2	Spike Recovery	271
8.2.3	Hook Effect	272
8.3	Interpretation	274
8.4	Complete Standard Cases	275
8.4.1	CLSI EP34 Table 1: Single-Sample Diluent Suitability	275
8.4.2	CLSI EP34 Table 2: Multiple Original Samples	278
8.4.3	CLSI EP34 Table 3: Basic Spike Recovery	285
8.4.4	CLSI EP34 Table 4: Single-Sample Multiple-Concentration Spiking	287
8.4.5	CLSI EP34 Tables 5-6: Manual Dilution-Procedure Validation	288
8.4.6	CLSI EP34 Tables 7-8: Automatic Dilution-Procedure Validation	291
8.4.7	CLSI EP34 Appendix C: Two Complete Dilution Analyses	293
8.4.8	High-Dose Hook Effect	308
9	Method Comparison and Reference-Material Bias	313
9.1	Detailed Function Reference	313
9.1.1	Workflow and Function Overview	313

9.1.2	Summarizing Replicates with <code>replicate_to_mean()</code>	314
9.1.3	Splitting Data at Prespecified Cutpoints	314
9.1.4	Creating an <code>mcr</code> Object with <code>mcr()</code>	315
9.1.5	Descriptive Statistics with <code>describe()</code>	316
9.1.6	Correlation with <code>correlation()</code>	316
9.1.7	Regression with <code>regression()</code>	316
9.1.8	Bland–Altman Analysis with <code>bland_altman()</code>	317
9.1.9	Outlier Testing with <code>outlier()</code>	317
9.1.10	Bias at Specified Levels with <code>bias()</code>	318
9.1.11	Prediction with <code>predict()</code>	318
9.1.12	Reference-Material Bias with <code>reference_bias()</code>	319
9.1.13	Plotting and Summary	319
9.2	Main Functions	319
9.2.1	Summarize Replicates and Split by Cutpoints	320
9.2.2	Build a Method-Comparison Object	320
9.2.3	Regression Choices	320
9.2.4	Bland–Altman, Outliers, Bias, and Prediction	321
9.2.5	Reference-Material Bias	321
9.3	Analysis Example	321
9.4	Interpretation Checklist	326
9.5	Complete Standard Cases	327
9.5.1	Replicate Summaries and <code>lambda</code>	327
9.5.2	CLSI EP09c Appendix C: Mixed Variance and Regression Models	327
9.5.3	CLSI EP09c Appendix D: Six Data Patterns	332
9.5.4	CLSI EP09c Appendix E: Generalized ESD Outlier Test	334
9.5.5	CLSI EP09c Appendix F: Median-Bias Intervals	335
9.5.6	CLSI EP09c Appendix H: Iteratively Weighted Least Squares	337
9.5.7	CLSI EP09c Appendix I: Passing–Bablok Parameters and Bias Intervals	338
9.5.8	CLSI EP09c Appendix J: Precision-Profile Weighted Deming	338
9.5.9	CLSI EP09c Appendix K2: Regression and Bias Bootstrap	340
9.5.10	YY/T 1789.2–2021 Appendix A: Reference-Material Bias	341
9.5.11	YY/T 1789.2–2021 Appendix B: Patient-Sample Bias and Regression	343
10	Quality Control	353
10.1	Main Functions	353
10.1.1	View Rules with <code>list_westgard()</code>	354
10.1.2	Single-Sequence Control Chart with <code>qc_chart()</code>	355
10.1.3	Two-Level Joint Plot with <code>youden_plot()</code>	356
10.2	Analysis Examples	357
10.2.1	Routine QC with Known Targets	357
10.2.2	Continuous Display with Lot-Change Labels	359
10.2.3	Two-Level Youden Plot	361
11	ROC Curves and Cutoff Determination	363
11.1	Main Functions	363
11.1.1	Convert a Continuous Reference Variable with <code>continuous_to_binary()</code>	364
11.1.2	Create an Analysis Object with <code>roc()</code>	364
11.1.3	Descriptive Statistics with <code>describe()</code>	365
11.1.4	AUC with <code>auc()</code>	366
11.1.5	Cutoff Analysis with <code>cutoff()</code>	366
11.1.6	Plotting and Summary with <code>plot()</code> and <code>summary()</code>	367
11.2	Analysis Examples	367
11.2.1	Teaching Data: Complete Single-Marker Workflow	367
11.2.2	CLSI EP24-A2 Example 1: Empirical ROC and Threshold Intervals	371
11.2.3	CLSI EP24-A2 Appendix D: Single-Curve OxLDL AUC	372
12	ROC Curves and Multivariate Regression	375
12.1	Main Functions	375
12.1.1	Multivariate Logistic Regression with <code>mlr()</code>	376

12.1.2 Paired AUC Comparison with <code>auc_compare()</code>	377
12.1.3 Probability Prediction with <code>predict()</code>	378
12.1.4 ROC Plotting and Summary	379
12.2 Analysis Examples	379
12.2.1 Teaching Data: A Joint Model with Two Markers	379
12.2.2 CLSI EP24-A2 Appendix D: Two Paired ROCs	383
13 Precision Analysis	387
13.1 Main Functions	387
13.2 Analysis Examples	388
13.2.1 EP05 20×2×2 Single-Sample Precision	388
13.2.2 Multiple Samples and a Sadler Profile	392
13.2.3 Multicenter Precision	394
13.2.4 More Complex Nested Models	395
13.2.5 Standard-Data Cross-Checks	396
13.3 Complete Standard Cases	398
13.3.1 CLSI EP05-A3 Appendix A: Glucose 20×2×2	398
13.3.2 CLSI EP05-A3 Appendix B: CA19-9 Multicenter Study	400
13.3.3 CLSI EP15-A3: Ferritin 5×5 Verification	402
13.3.4 WS/T 408–2024: Precision Verification	405
13.3.5 YY/T 1789.1–2021 Appendix A: 25-Hydroxyvitamin D	406
14 Linearity Analysis	409
14.1 Main Functions	409
14.2 Model, Weights, and ADL	411
14.3 Analysis Examples	412
14.3.1 Serial-Dilution Linearity	412
14.3.2 Polynomial Sensitivity Analysis	414
14.3.3 Standard Cross-Checks	417
14.4 Interpretation	421
14.5 Complete Standard Cases	421
14.5.1 CLSI EP06 Second Edition Appendix G: Two-Replicate Study	421
14.5.2 CLSI EP06 Second Edition Appendix H: Two-Panel Study	422
14.5.3 CLSI EP06 Second Edition Appendix I: Assigned Samples	423
14.5.4 WS/T 408–2024: Five-Level Linearity Verification	423
14.5.5 YY/T 1789.4–2022 Appendix A: Linearity Sample 1	424
15 Qualitative Analysis and C5/C95	427
15.1 Main Functions	427
15.1.1 Four-Fold Metrics and Test Formulas	428
15.1.2 C5/C95 Probability Model	429
15.2 Analysis Examples	429
15.2.1 Example 1: Complete Analysis of Raw Paired Data	429
15.2.2 Example 2: Build Directly from TP/FP/TN/FN	433
15.2.3 Example 3: Convert Quantitative Data to Qualitative	434
15.2.4 Example 4: Estimate C5/C95 from an Internal Continuous Response	436
16 Limits of Blank, Detection, and Quantitation	439
16.1 Main Functions	439
16.1.1 Interfaces, Parameters, and Core Statistics	439
16.1.2 Claim Verification and Design Checks	441
16.2 Analysis Examples	441
16.2.1 LoB	441
16.2.2 LoD	442
16.2.3 LoQ	442
16.2.4 Claim Verification	443
16.2.5 Study Design	444
16.2.6 Standard Case: CLSI EP17-A2 Appendix A, Classical LoB and LoD	445
16.2.7 Standard Case: CLSI EP17-A2 Appendix C, Probit LoD	446

16.2.8 Standard Case: CLSI EP17-A2 Appendices E and F, Claim Verification . . .	447
16.2.9 Standard Case: YY/T 1789.3-2022 Appendix A, ProGRP	447
17 Reference Intervals	449
17.1 Main Functions	449
17.2 Analysis Examples	451
17.2.1 Example: Compare All Five Methods	451
17.2.2 Standard Case: CLSI EP28-A3c Appendix B Robust Iteration	454
18 Bottle-to-Bottle Variation	457
18.1 Main Functions	457
18.1.1 ANOVA, Tukey, and Variance Components	458
18.2 Analysis Examples	459
18.2.1 Example 1: 15 Bottles $\times$ 5 Measurements	459
18.2.2 Example 2: Three Reagent Lots Measuring One Control	463
18.2.3 Standard Case: YY/T 1709-2020 Calibrator Homogeneity	464
19 Clinical Sample-Size Calculation	467
19.1 Main Functions	467
19.1.1 Interfaces	467
19.2 Analysis Examples	468
19.2.1 Bland–Altman Agreement	468
19.2.2 Single-Proportion Confidence Interval	469
19.2.3 One-Arm Target-Value Test	469
19.2.4 Standard and Guidance Cross-Checks	469
19.3 Interpretation Checklist	470
19.4 Complete Standard Cases	470
19.4.1 Bland–Altman: PASS Examples 1 and 2	470
19.4.2 Proportion Confidence Intervals: PASS Examples 1–4	471
19.4.3 One-Sample Proportion Target Tests: SAS and PASS Comparison	472
19.4.4 CMDE Appendix Case 1: Target-Value Method with Clinical Acceptance Criteria	474
19.4.5 CMDE Appendix Case 2: Allowable Error Method Requiring CI Precision	476
20 Outlier and Normality Tests	477
20.1 Main Functions	477
20.1.1 Parameters and Formulas for Ordinary Data	477
20.1.2 Statistical Unit for Object-Level Methods	478
20.2 Analysis Examples	479
20.2.1 Test an Ordinary Data Column	479
20.2.2 Outliers in a Method-Comparison Object	480
20.2.3 Outliers and Normality in a Precision Object	481
20.2.4 Standard Case: YY/T 1789.4-2022 Appendix A.2	481
21 Introduction to Algorithm Principles	483
21.1 Replicates, Multiple Samples, and Statistical Units	483
21.1.1 First Decide What One Row Represents	483
21.1.2 Joint or Separate Analysis for Multiple Samples	483
21.2 Outliers and Normality	483
21.2.1 Grubbs Test	483
21.2.2 Generalized ESD	484
21.2.3 Dixon and IQR	484
21.2.4 Automatic Normality Branch	484
21.3 Method Comparison and Reference-Material Bias	484
21.3.1 Direction, Units, and Preprocessing	484
21.3.2 Correlation Options	484
21.3.3 Regression Methods	484
21.3.4 Confidence Intervals and Bland–Altman	485
21.3.5 Reference-Material Bias	485

21.4	Reference-Material Commutability	485
21.5	Qualitative Performance, C5/C95, and ROC	485
21.6	Precision, Bottle Variation, and Reference Intervals	486
21.7	Response-Equation Fitting and Linearity	486
21.8	Interference, Dilution, Spike Recovery, and Hook Effect	486
21.9	Analytical Sensitivity	486
21.10	Quality Control and Stability	486
21.11	Measurement Uncertainty and Traceability	487
21.12	Parameter Choices Change the Estimand	487
21.13	Detailed Algorithm Specification	487
21.13.1	IOLS and WLS Point Estimates	487
21.13.2	Deming Regression and λ	489
21.13.3	Three Weighted-Deming Variants	490
21.13.4	Regression-Parameter Confidence Intervals	491
21.13.5	Passing–Bablok	492
21.13.6	Differences, Percentage Differences, and Outlier Analysis	492
21.13.7	Bland–Altman Axis, Center, and Limits of Agreement	493
21.13.8	Bias at Medical Decision Levels and Prediction Intervals	494
21.13.9	Reference-Material Bias	495
21.14	Reference-Material Commutability	495
21.14.1	EP14-Style Prediction Interval	495
21.14.2	FCC-Style Difference Evaluation	495
21.15	Qualitative Performance, C5/C95, and ROC	496
21.15.1	Four-Fold Proportions and Intervals	496
21.15.2	Likelihood Ratios, Agreement, and McNemar	497
21.15.3	C5/C95	497
21.15.4	Empirical ROC, AUC, and Multivariable Models	498
21.16	Precision, Bottle Variation, and Reference Intervals	498
21.16.1	Variance Components and Precision Profiles	498
21.16.2	Bottle Variation	499
21.16.3	Reference Intervals	499
21.17	Response-Equation Fitting and Linearity	501
21.17.1	Twelve Registered Response Equations	501
21.17.2	Linearity Design Quantities and Regression Comparison	502
21.18	Interference, Dilution, Spike Recovery, and Hook Effect	502
21.18.1	Two-Group Interference and Difference-in-Differences	502
21.18.2	Dilution Recovery	503
21.18.3	Spike Recovery and Hook Effect	503
21.19	Analytical Sensitivity	503
21.19.1	Limits of Blank and Detection	503
21.19.2	Limit of Quantitation and Limit Verification	504
21.20	Quality Control and Stability	504
21.20.1	Levey–Jennings and Westgard	504
21.20.2	Node Bias, Time Regression, and Time to Limit	505
21.20.3	Mean Kinetic Temperature and Arrhenius	505
21.21	Measurement Uncertainty and Traceability	506
21.21.1	Type A and Type B Standardization	506
21.21.2	QC, Homogeneity, Stability, and Characterization	506
21.21.3	First-Order Combination and Effective Degrees of Freedom	507
21.21.4	General Measurement Models and Monte Carlo	507
21.21.5	Stepwise Traceability	508
21.22	Sample Size and Power	508
21.22.1	Bland–Altman Agreement	508
21.22.2	Proportion Intervals and Target Tests	508
22	Standards Mapping	511
22.1	Interpretation Categories	511
22.2	CLSI EP Series	511

22.2.1 Core Statistical Tasks Implemented or Directly Reusable	511
22.2.2 EP Documents with Only Related Tools or Not Yet Implemented	512
22.3 YY/T 1789.1–1789.6	513
22.4 CMDE Guidance Support Level	514
23 R Packages for IVD Data Analysis: A Review	515
23.1 Scope and Inclusion Rules	515
23.2 Classification Framework	515
23.3 Analytical-Performance Evaluation	516
23.3.1 Integrated IVD Tools	516
23.3.2 Method Comparison, Bias, and Agreement	516
23.3.3 Precision, Variance Components, and Repeatability	517
23.3.4 Calibration, Linearity, Dose Response, and Detection Capability	517
23.3.5 Reference Intervals	517
23.3.6 Stability, Interference, Recovery, Lot Differences, and QC	518
23.4 Clinical Diagnostic Performance	518
23.4.1 Binary Metrics, ROC, and Cutoffs	518
23.4.2 Paired Tests, Multiclass, No Gold Standard, and Agreement	518
23.5 Diagnostic Accuracy and General Meta-Analysis	519
23.5.1 Diagnostic-Accuracy Meta-Analysis	519
23.5.2 General Effect Sizes and Prognostic Meta-Analysis	519
23.6 Survival and Event Outcomes	519
23.7 Multivariate Analysis, Chemometrics, and High-Dimensional Biomarkers	519
23.8 Regression, Clinical Prediction, and Utility	520
23.9 Sample Size, Missing Data, and Statistical Support	520
23.10 Data Management, Reporting, and Reproducibility	520
23.11 Complete Review	520
23.11.1 Scope and Inclusion Rules	520
23.12 Analytical-Performance Evaluation	521
23.12.1 Integrated IVD Packages	521
23.12.2 Method Comparison, Bias, and Agreement	521
23.12.3 Precision, Variance Components, and Repeatability	522
23.12.4 Calibration, Linearity, Dose Response, and Detection Capability	522
23.12.5 Reference Intervals	523
23.12.6 Stability, Interference, Recovery, Lot Differences, and QC	523
23.13 Clinical Diagnostic Performance	524
23.13.1 Binary Metrics, ROC, and Cutoffs	524
23.13.2 Paired Tests, Multiclass, No Gold Standard, and Observer Agreement	524
23.14 Diagnostic Accuracy and General Meta-Analysis	525
23.14.1 Diagnostic-Accuracy Meta-Analysis	525
23.14.2 General Effect Sizes, Prognosis, and IPD Meta-Analysis	525
23.15 Survival and Event Outcomes	526
23.16 Multivariate Analysis, Chemometrics, and High-Dimensional Biomarkers	526
23.16.1 Exploratory Multivariate Analysis	526
23.16.2 Chemometrics, Spectra, and Multimarker Quantitation	527
23.16.3 Classification, Prediction, and Machine Learning	527
23.17 Regression, Clinical Prediction, and Clinical Utility	528
23.18 Sample Size, Missing Data, and Statistical Support	528
23.19 Data Management, Reporting, and Reproducibility	529

Welcome

This book is an executable, auditable usage guide for the R package `ivdtools`, intended for IVD practitioners, statisticians, and readers new to R. It covers the workflow from bringing data into a project through exporting results, experimental scenarios, and the statistical algorithms actually used by the package source code.

! Scope of Use

This book is intended for teaching statistical workflows and checking implementations. It is not a substitute for current regulations or standards.

Reader's Guide

- Beginners should read Part 1 and Part 2 in sequence, then enter the function chapters according to their task.
- Experienced analysts can go directly to function signatures, object fields, and boundary behavior.

Getting Help

```
help(mcr, "ivdtools")
```

Visit the project home page to view or download the complete function reference and this user guide.

Version and Citation

```
version <- packageVersion("ivdtools")
version
```

```
[1] '0.2.5'
```

The software may be cited as: hiox-tech (2026). `ivdtools`: Statistical Tools for Evaluation of in Vitro Diagnostic Reagents. R package version 0.2.5, <https://github.com/hiox-tech/ivdtools>.

Project Information

- Project home: <https://github.com/hiox-tech/ivdtools>
- Issue tracker: <https://github.com/hiox-tech/ivdtools/issues>
- License: MIT

Chapter 1

ivdtools Package Overview

`ivdtools` is an R statistical toolkit for in vitro diagnostic (IVD) reagent evaluation. It provides reproducible workflows for data checking, statistical analysis, result summarization, and visualization. The package covers method comparison, ROC analysis, qualitative agreement, precision, linearity, interference, dilution and spike recovery, C5/C95, measurement uncertainty, commutability, reference-material bias, reference intervals, stability, quality control, curve fitting, analytical sensitivity, outlier and normality assessment, and sample-size calculation. Statistical results should be interpreted together with the pre-specified study protocol, applicable standards, laboratory requirements, and clinical or analytical allowable limits.

1.1 Functional Modules

Module	Main entry points	Key features
Method comparison	<code>mcr()</code>	Descriptive statistics, correlation, regression, Bland–Altman, bias, and outliers
Commutability	<code>commutability()</code>	EP14, EP30, and IFCC commutability assessment
ROC	<code>roc()</code>	ROC curves, cutoffs, multivariate logistic regression, and prediction
Qualitative analysis	<code>raw_to_table()</code> , <code>counts_to_table()</code>	Four-fold tables, diagnostic accuracy metrics, Kappa agreement, and McNemar test
Precision	<code>precision()</code>	Variance components, confidence intervals, outliers, normality, and Sadler precision profiles
Linearity	<code>linearity()</code> , <code>linearity_*</code> ()	Study design, HIGH/LOW preparation, linear/polynomial analysis, weights, ADL, and intervals
Interference	<code>interference_*</code> ()	Replicate planning, paired studies, dose response, and patient-sample comparison
Dilution and spike recovery	<code>dilution_recovery()</code> , <code>spike_recovery()</code> , <code>hook_effect()</code>	Dilution recovery, spike recovery, and high-dose hook effect

Module	Main entry points	Key features
C5/C95	<code>c5_c95()</code>	C5/C95 estimation for qualitative reagents from continuous measurements
Measurement uncertainty	<code>uncertainty_*</code>	Type A/B, IQC, bias, homogeneity, stability, combination, and propagation
Reference-material bias	<code>reference_bias()</code>	Bias, measurement-mean uncertainty, and intervals combining reference-material uncertainty
Reference intervals	<code>reference_interval()</code>	EP28 nonparametric, biweight, type 6, parametric, and Huber intervals
Bottle/lot ANOVA	<code>bottle_anova()</code>	One-way or nested ANOVA, Tukey HSD, and compact letter display
Stability	<code>stability_bias()</code> , <code>stability_regression()</code>	Bias, stability regression, time to exceed a limit, stability plans, MKT, and Arrhenius models
Quality control	<code>qc_chart()</code> , <code>youden_plot()</code>	Levey–Jennings charts, Westgard rules, and Youden plots
Calibration and response curves	<code>fit_equation()</code>	Weighted and constrained fitting, and model comparison for linear, exponential, 4PLC, 5PLC, and other response equations; not method-comparison
Analytical sensitivity	<code>sensitivity_lob()</code> / <code>sensitivity_lod()</code> / <code>sensitivity_loq()</code>	linearity assessment Establishment and verification of LoB, LoD, and LoQ
Outliers and distributions	<code>outliers_test()</code> , <code>normal_test()</code>	Grubbs, ESD, Dixon, IQR, and multiple normality tests
Sample size	<code>sample_size_*</code>	Bland–Altman agreement, single-proportion confidence intervals, and one-arm target-value tests

1.2 Design Features

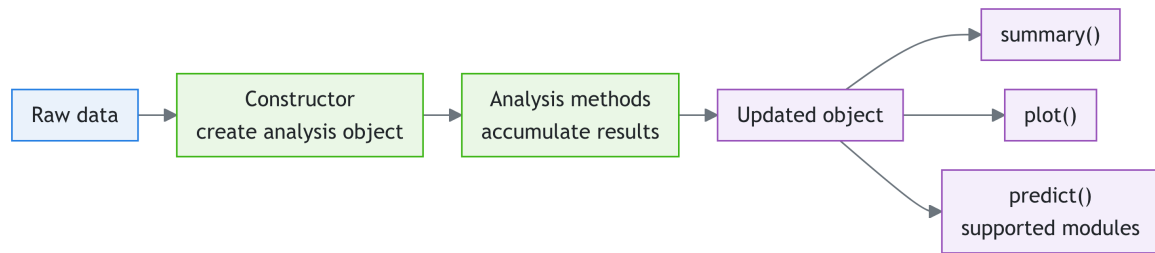
1.2.1 Standalone Workflows

Standalone interfaces complete one task at a time, as in the following examples:

```
normal_test(dat, col = "candidate")
reference_interval(dat, col = "candidate")
```

1.2.2 Progressive Workflows

Modules such as method comparison, ROC, qualitative analysis, and precision organize the analysis process with S3 objects:



Progressive interfaces must be reassigned:

```
obj <- analysis_method(obj)
```

If you call only `analysis_method(obj)` without `obj <-`, the new results will not accumulate in the original object.

1.3 Quick Start

The following code is for demonstrating the analysis process only.

1.3.1 Method Comparison

```
library(ivdtools)

mc <- mcr(
  ivd_mcr_example,
  id = "sid",
  candidate = "test",
  reference = "ref"
)
mc <- describe(mc)
mc <- correlation(mc, method = "pearson")
mc <- regression(mc, method = "deming")
mc <- bland_altman(mc, type = "difference")

summary(mc)
plot(mc, type = "regression")
plot(mc, type = "bland_altman")
```

1.3.2 ROC

```
ro <- roc(
  ivd_roc_example,
  cols = c("x1", "x2"),
  reference = "ref",
  positive = 1
)
ro <- describe(ro)
ro <- auc(ro)
ro <- cutoff(ro)
```

```
ro <- mlr(ro, cols = c("x1", "x2"), name = "combined")

summary(ro)
plot(ro, curves = "all", cutpoint = "Youden")
```

1.3.3 Qualitative Agreement

```
qa <- raw_to_table(
  ivd_qualitative_example,
  candidate = "new",
  reference = "gold",
  id = "id",
  positive = "positive"
)
qa <- describe(qa)
qa <- diagnostics(qa)
qa <- kappa(qa)
qa <- mcnemar(qa)

summary(qa)
```

1.3.4 Bottle-to-Bottle Difference

```
ba <- bottle_anova(
  ivd_bottle_example,
  value ~ bottle
)
print(ba)

posthoc <- tukey(ba)
print(posthoc)
```

1.4 Dependencies

ivdtools primarily depends on ggplot2, ggrepel, VCA, VFP, minpack.lm, nloptr, nls2, nortest, rlang, and ppwdeming.

Chapter 2

Installing and Using the Analysis Environment

This chapter explains how to set up an `ivdtools` analysis environment, create a `qmd` project, import and verify data, run progressive analyses, and save reproducible results.

2.1 Install R

`ivdtools` requires R $\geq 4.1.0$. We recommend using a newer R version that is still supported and keeping the major versions of R and all dependencies consistent across everyone working on a project.

2.1.1 Windows and macOS

Download the installer for your operating system from CRAN: <https://cran.r-project.org/>.

- Windows: install R. A matching version of Rtools is needed only when compiling packages containing native code from source.
- macOS: choose the installer for your processor architecture. Xcode Command Line Tools may be needed when compiling from source.

2.1.2 Debian/Ubuntu

The system repository can install the base R environment:

```
sudo apt update
sudo apt install r-base r-base-dev
```

2.1.3 Fedora

```
sudo dnf install R R-devel
```

R versions in Linux distribution repositories may be old. If a project needs a newer R version, configure the repository by following the instructions for the relevant distribution on CRAN.

2.2 Choose an Editor

We recommend using `.qmd` or `.Rmd` files to record code, text, tables, and figures.

Software	Windows	Linux	macOS	Typical use
RStudio	✓	✓	✓	Ready to use; suitable for beginners and routine analyses
Positron	✓	✓	✓	Suitable for users who want the next-generation Posit IDE
Visual Studio Code	✓	✓	✓	Requires configuration of R, Quarto, and related extensions
Jupyter	✓	✓	✓	Requires Python, Jupyter, and an R kernel
PyCharm	✓	✓	✓	Requires suitable R support and additional configuration
cantor		✓		In development; only recently relatively stable

RStudio and Positron are available from <https://posit.co/downloads/>; Visual Studio Code is available from <https://code.visualstudio.com/Download>.

2.3 Install R Packages

When `ivdttools` has been published to the CRAN mirror you use:

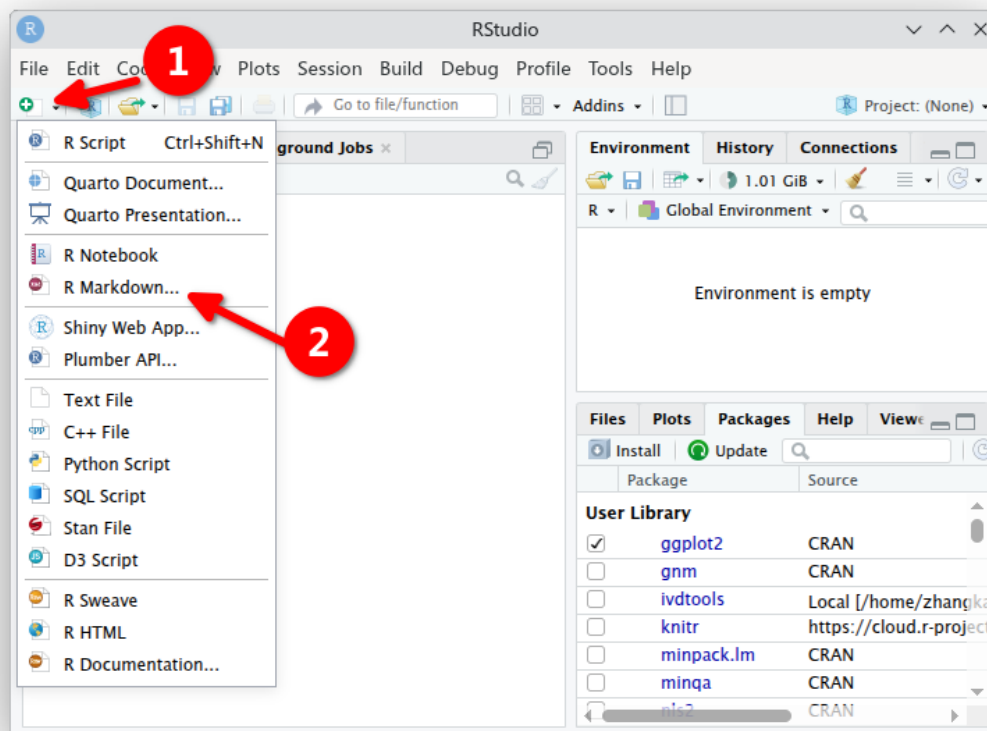
```
install.packages(c("ivdttools", "rmarkdown", "knitr"))
```

Install optional packages for data import as needed:

```
install.packages(c("readr", "readxl"))
```

2.4 Create a Project

In RStudio, choose **File > New File > R Markdown**.



Minimal document template:

```

---
title: "IVD Analysis Report"
author: "Analyst"
date: "Enter report date"
output:
  html_document:
    toc: true
    number_sections: true
---

```

2.5 Import and Verify Data

2.5.1 CSV

Base R:

```

dat <- read.csv(
  "data-raw/method-comparison.csv",
  stringsAsFactors = FALSE,
  check.names = FALSE
)

```

Using readr:

```
dat <- readr::read_csv(
  "data-raw/method-comparison.csv",
  show_col_types = FALSE
)
```

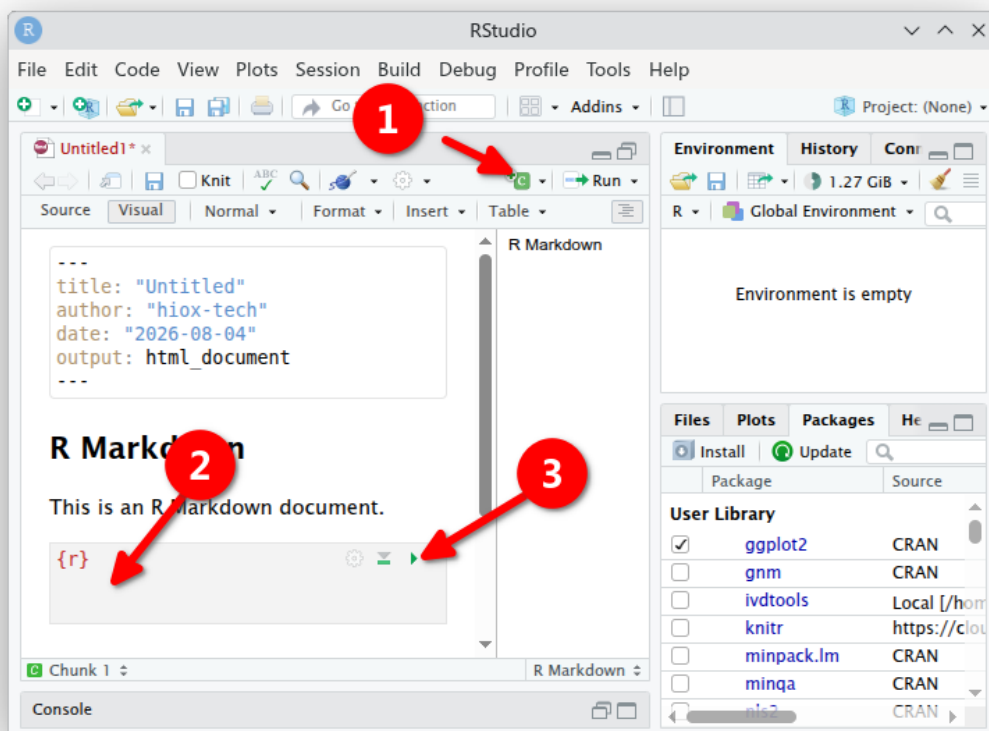
2.5.2 Excel

```
dat <- readxl::read_excel(
  "data-raw/method-comparison.xlsx",
  sheet = "data"
)
```

If an Excel file contains formulas, merged cells, or manual formatting, first confirm that the imported values, column names, and data types agree with the original table.

2.6 Run Code

Insert an R code block in an .Rmd file and use the button in the upper-right corner of the block to run it, or run all code from top to bottom.



For reproducibility, the final report should be rendered from start to finish in a fresh R session rather than relying only on objects already present in the Console.

2.7 Export Results

2.7.1 Tables

After reading the result fields described in the relevant help page, write a table to disk as CSV:

```
# Example: confirm the actual field in the relevant object help page
result_table <- mc$regression
saveRDS(result_table, "tables/regression-result.rds")
```

Prefer saving complex objects as RDS to preserve their classes and attributes. When exchanging results with other software, first organize the object as a flat data frame and then export it as CSV.

2.7.2 Figures

All figures are drawn with ggplot.

```
p <- plot(mc, type = "bland_altman")
ggplot2::ggsave(
  "figures/bland-altman.png",
  plot = p,
  width = 7,
  height = 5,
  dpi = 300
)
```

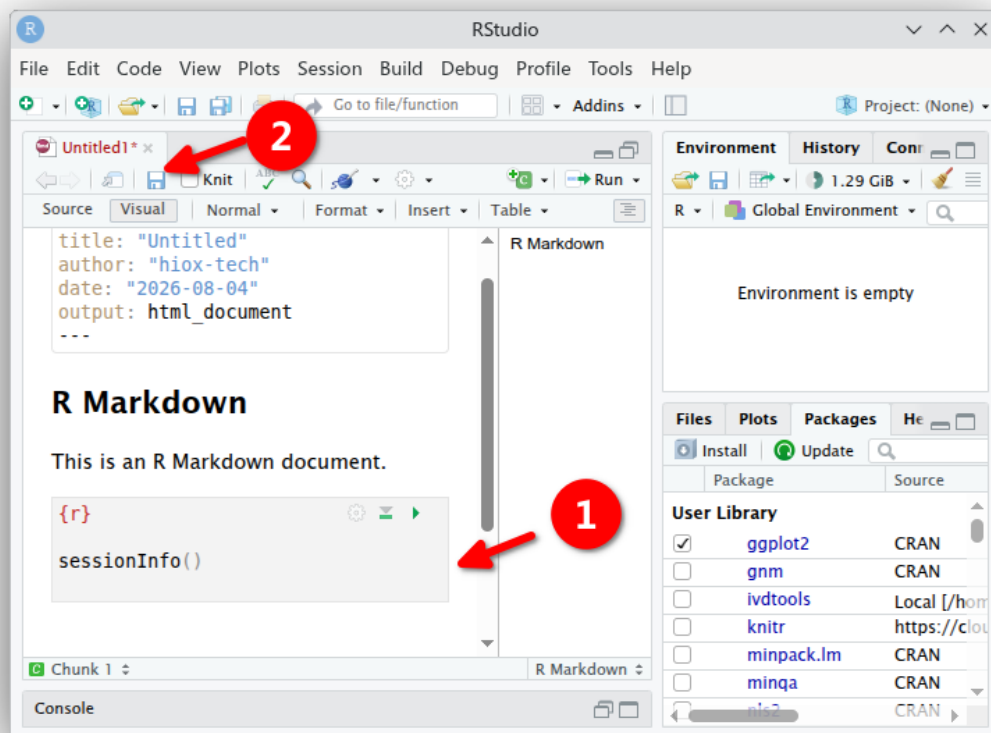
2.8 Render and Save

In RStudio, click **Knit**, or run:

```
rmarkdown::render(
  "report/analysis.Rmd",
  output_format = "html_document",
  clean = TRUE
)
```

Record environment information after completing the analysis:

```
sessionInfo()
```



Chapter 3

Calibration Curve Fitting

This chapter demonstrates how to use `fit_equation()` from `ivdtools` to fit calibration curves and check convergence, residuals, the effect of weights, and forward and inverse predictions. The underlying implementation calls `minpack.lm`, `nloptr`, `nls2`, and `stats`.

```
library(ivdtools)
library(readr)
```

3.1 Main Functions

Function	Purpose
<code>list_equation()</code>	View the built-in equation registry
<code>replicate_to_mean()</code>	Summarize replicate measurements and optionally calculate weights
<code>fit_equation()</code>	Fit a built-in or custom equation
<code>coef()</code>	Extract model parameters
<code>residuals()</code>	Extract observed-minus-fitted residuals
<code>predict()</code>	Predict response values forward or estimate concentrations inversely
<code>plot()</code>	Plot observed points, the fitted curve, and intervals
<code>compare_equation()</code>	Compare candidate equations

3.1.1 View Built-in Equations with `list_equation()`

```
list_equation()
```

ID	Name	Formula	Params	nP	Engine
E01	Linear	$a + b \cdot x$	a, b	2	lm
E02	Quadratic	$a + b \cdot x + c \cdot x^2$	a, b, c	3	lm
E03	ExpoGrowth	$a \cdot \exp(b \cdot x)$	a, b	2	nls
E04	ExpoDecay	$a \cdot \exp(-b \cdot x)$	a, b	2	nls
E05	Power	$a \cdot x^b$	a, b	2	nls
E06	Log	$a + b \cdot \log(x)$	a, b	2	lm
E07	4PLC	$A + (B - A) / (1 + (x / C)^D)$	A, B, C, D	4	nls
E08	5PLC	$A + (B - A) / (1 + (x / C)^D)^E$	A, B, C, D, E	5	nls
E09	Gaussian	$a \cdot \exp(-(x - b)^2 / (2 \cdot c^2))$	a, b, c	3	nls

E10	Michaelis	$V_{\max} * x / (K_m + x)$	$V_{\max}, K_m$	2	nls
E11	Sinusoidal	$a * \sin(b * x + c) + d$	a, b, c, d	4	nls
E12	Cubic	$a + b*x + c*x^2 + d*x^3$	a, b, c, d	4	lm

The function returns a registry data frame containing equation IDs, names, formulas, numbers of parameters, applicable categories, and fitting engines.

3.1.2 Summarize Replicates with `replicate_to_mean()`

If each concentration has replicate measurements, you can fit all observations directly or first summarize them with `replicate_to_mean()`.

```
replicate_to_mean(
  data, x = "Con", y = "RLU", weights = NULL, na.rm = TRUE
)
```

The function groups by `x` and returns summary fields such as the mean, number of replicates, and dispersion at each concentration. `weights` can provide numeric weights or a weighting rule for later fitting. `na.rm = TRUE` excludes missing rows; the number excluded should be recorded. Summary means are suitable for describing each level and should not automatically be treated as the error model for a future single specimen.

3.1.3 Fit a Built-in or Custom Equation with `fit_equation()`

`fit_equation()` accepts an ordinary `data.frame`. `x` is the concentration or relative concentration and `y` is the instrument response; both columns must contain finite numeric values.

`fit_equation()` can use automatic starting values or starting values supplied through `start`; `lower`, `upper`, and `constraints` can set parameter bounds or constraints. `weights` accepts a numeric vector or one of the built-in schemes "equal", "1/y", "1/y^2", "1/x", and "1/x^2".

```
fit_equation(
  eq, data, x = "x", y = "y", start = NULL,
  lower = NULL, upper = NULL, constraints = NULL,
  weights = NULL, ...
)
```

The returned object has S3 class `fit_equation` and contains the selected equation, fitted model, parameters, data, and fitting diagnostics.

3.1.4 Compare Candidate Equations with `compare_equation()`

Main inputs

```
compare_equation(
  data, x = "Con", y = "RLU", eqs = c("4PLC", "5PLC")
)
```

The returned object has S3 class `compare_equation`; its `$table` is sorted by AIC and includes at least Eq, Name, Formula, nPar, AIC, deltaAIC, R2, and RSE. AIC is evidence for comparing candidate models, not a replacement for residuals, plateau behavior, the reporting range, or external validation. If candidate models were not prespecified in the protocol, the comparison should be labeled exploratory.

3.1.5 Get Fitted Parameters and Residuals with `coef()` and `residuals()`

```
coef(fit)
residuals(fit)
```

Here `fit` is the fitted object returned by `fit_equation()`. `coef(fit)` returns a named vector of fitted coefficients; `residuals(fit)` returns observed-minus-fitted residuals corresponding to valid observation rows.

3.1.6 Forward Response Prediction or Inverse Concentration Estimation with `predict()`

```
predict(
  fit,
  newdata,
  interval = "confidence",
  level = 0.95,
  inverse = TRUE
)
```

`newdata` contains the concentrations or responses to predict. `interval` and `level` specify the interval type and confidence level. A confidence interval describes uncertainty in the fitted mean; a prediction interval also includes single-observation error and is usually wider. `inverse` requests inverse estimation of concentration from response.

The function returns a data frame containing input values, point predictions, and the requested lower and upper bounds. For inverse prediction, the point prediction is the estimated concentration.

! Prediction Boundaries

Avoid extrapolating beyond the calibration concentration range. Inverse prediction is highly sensitive to response error on a plateau, and non-monotonic models may have multiple solutions.

3.1.7 Plot the Fitted Curve and Interval with `plot()`

```
plot(fit, interval = "confidence", level = 0.95)
```

Use the figure to check observed points, the fitted curve, intervals, and the calibration range. A formal report should preserve the equation, weights, interval type, parameters, convergence status, and plotting parameters.

3.2 Analysis Examples

3.2.1 Fitting a Linear Calibration Curve

The linear equation is appropriate when the response changes with concentration at an approximately constant rate across the working range. In the `ivdtools` registry it corresponds to E01:

$$y = a + bx$$

Here a is the intercept and b is the slope. This example uses deterministic teaching data in `data/003-校准曲线拟合/Linear.csv` only to demonstrate the workflow. A formal calibration

should define the concentration range, number of replicates, weights, and acceptance criteria in the study protocol.

```
# Read the linear data file
fit_linear_data <- read_csv(
  "./data/003-校准曲线拟合/Linear.csv",
  show_col_types = FALSE
)

# Inspect the data structure
head(fit_linear_data, 6)
```

```
# A tibble: 6 x 3
  Sample   Con   RLU
  <chr>   <dbl> <dbl>
1 c1         0  1185
2 c2         5  3415
3 c3        10  5230
4 c4        20  9810
5 c5        40 18080
6 c6        60 26290
```

```
# Fit the linear calibration curve
fit_linear <- fit_equation(
  "E01",
  data = fit_linear_data,
  x = "Con",
  y = "RLU"
)

# Display fitting results and parameters
print(fit_linear)
```

E01 (Linear)
Formula: $a + b \cdot x$

Call:
stats::lm(formula = frm, data = d, weights = weights)

Residuals:

Min	1Q	Median	3Q	Max
-736.970	-185.656	-38.780	151.657	693.530

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
a	1292.468	183.32973	7.04996	3.4988e-05 ***
b	418.525	1.87989	222.63236	< 2.22e-16 ***

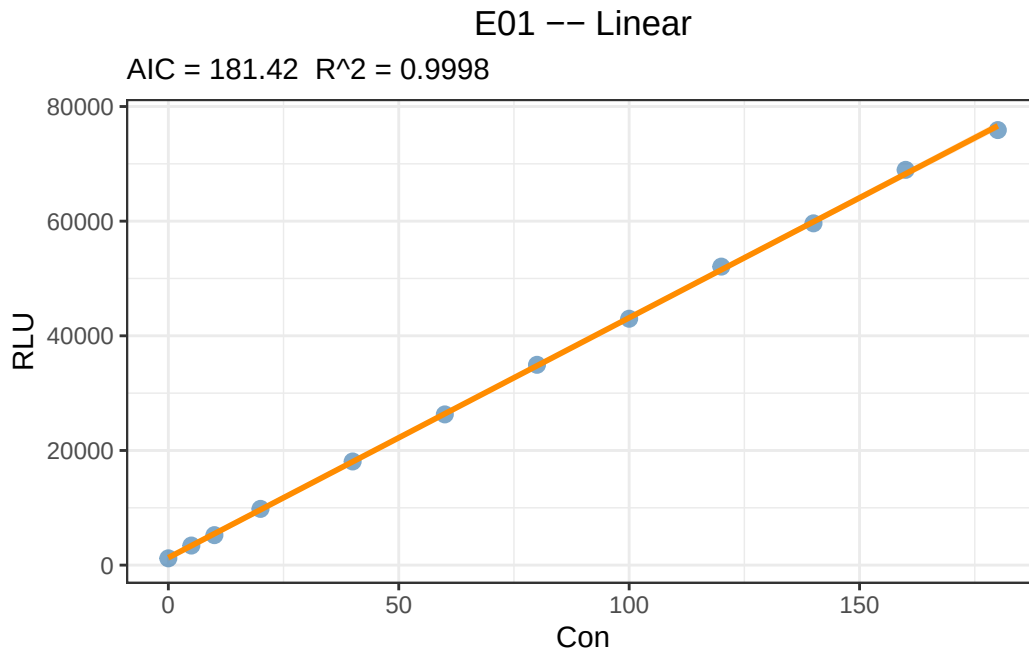
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 395.923 on 10 degrees of freedom
Multiple R-squared: 0.9998, Adjusted R-squared: 0.9998
F-statistic: 49565.2 on 1 and 10 DF, p-value: <2e-16

```
coef(fit_linear)
```

a	b
1292.468	418.525

```
# Plot observations, fitted line, and a 99% confidence interval for the mean
plot(fit_linear, interval = "confidence", level = 0.99)
```



```
# Predict response from concentration
predict(
  fit_linear,
  newdata = data.frame(Con = c(25, 75, 150)),
  interval = "confidence",
  level = 0.95
)
```

x	y	y_ci_lwr	y_ci_upr
25	11755.59	11422.52	12088.66
75	32681.84	32427.13	32936.56
150	64071.22	63670.87	64471.57

```
# Estimate concentration inversely from response
predict(
  fit_linear,
  newdata = data.frame(RLU = c(10000, 30000, 60000)),
  inverse = TRUE
)
```

x	y
20.80529	10000
68.59215	30000
140.27246	60000

The slope represents the average rate at which response changes with concentration, and the intercept is the estimated response at concentration 0. Assess model suitability using residual trends, fitted intervals, back-calculated bias, and prespecified acceptance criteria; a high R^2 alone does not establish complete linearity across the calibration range. Inverse predictions should be restricted to the actual calibration range and should not be extrapolated without justification.

3.2.2 Fitting a 4PLC Calibration Curve for a Sandwich Assay

The four-parameter logistic (4PLC) model is nonlinear:

$$y = A + \frac{B - A}{1 + (x/C)^D}$$

Here A and B are the two asymptotic plateaus, C is the concentration scale at the midpoint of the curve, and D controls the direction and steepness. Interpret the parameters in light of the actual signal direction.

```
# Read the data file
fit_4plc_data <- read_csv("./data/003-校准曲线拟合/4PLC.csv", show_col_types = FALSE)

# Inspect the data structure
head(fit_4plc_data, 6)

# A tibble: 6 x 3
  Sample   Con   RLU
  <chr>   <dbl> <dbl>
1 c1      0     5637
2 c2     0.05  16014
3 c3     0.2   47520
4 c4     0.51 125038
5 c5     1.03 277621
6 c6     5.19 1498023

# Fit the 4PLC curve with 1/y^2 weights
fit_4plc <- fit_equation(
  "4PLC",
  data = fit_4plc_data,
  x = "Con",
  y = "RLU",
  weights = "1/y^2"
)

# Display the fitting results
print(fit_4plc)
```

E07 (4PLC)

Formula: $A + (B - A) / (1 + (x / C)^D)$

Call:

```
fit_equation(eq = "4PLC", data = fit_4plc_data, x = "Con", y = "RLU",
  weights = "1/y^2")
```

Residuals:

Min	1Q	Median	3Q	Max
-1038926.254	-2031.562	491.283	35707.649	809345.965

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
A	5.66828e+03	2.30614e+02	24.57907	4.7016e-08	***
B	2.00185e+08	6.72603e+07	2.97627	0.020624	*
C	4.83741e+02	1.74222e+02	2.77658	0.027433	*
D	-1.08082e+00	1.02400e-02	-105.54858	1.8062e-12	***

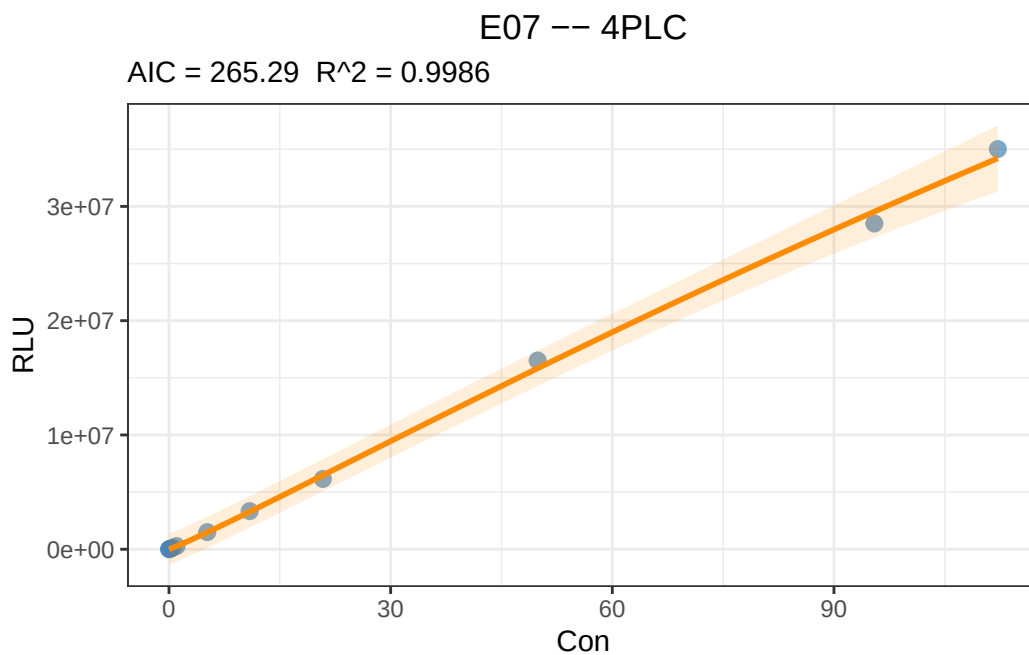
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.042025 on 7 degrees of freedom
 AIC: 265.29 Iterations: 7

```
# Obtain fitted residuals
residuals(fit_4plc)

[1] -3.128012e+01  4.912832e+02 -2.231827e+03 -1.831297e+03  1.304427e+04
[6]  1.466392e+04  5.675138e+04 -3.256634e+05  6.828909e+05 -1.038926e+06
[11]  8.093460e+05
```

```
# Plot the curve
plot(fit_4plc, interval = "prediction", level = 0.95)
```



```
# Predict y from x
predict(
  fit_4plc,
  newdata = data.frame(Con = c(20, 200, 1000)),
)
```

x	y
20	6205093
200	55646993
1000	137475699

```
# Predict x from y
predict(
  fit_4plc,
  newdata = data.frame(RLU = c(10000, 100000, 1000000)),
  inverse = TRUE
)
```

x	y
0.023371	1e+04
0.404396	1e+05
3.589256	1e+06

3.2.3 Constrained Fitting

The following example reuses the linear data above and focuses on how to specify parameter constraints in `fit_equation()`.

When theory says that the response should be 0 at concentration 0, or when the protocol explicitly requires that the intercept not be estimated, the intercept parameter a can be fixed at 0:

```
fit_linear_zero <- fit_equation(
  "E01",
  data = fit_linear_data,
  x = "Con",
  y = "RLU",
  constraints = list(eq = function(p) p["a"])
)

print(fit_linear_zero)
```

E01 (Linear)

Formula: $a + b \cdot x$

Call:

```
fit_equation(eq = "E01", data = fit_linear_data, x = "Con", y = "RLU",
  constraints = list(eq = function(p) p["a"]))
```

Residuals:

Min	1Q	Median	3Q	Max
-1309.733	268.826	616.260	1002.094	1270.563

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
a	0.000000	447.987279	0.000	1
b	428.887404	4.594117	93.356	9.41e-15 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1019.73 on 9 degrees of freedom

AIC: 166.80, R^2 : 0.9988

Convergence: 4 (NLOPT_XTOL_REACHED: Optimization stopped because xtol_rel or xtol_abs (above) was reached.)

Chapter 4

Stability Studies

Stability studies evaluate how measurement results change over time for reagents, calibrators, or QC materials under specified storage, transport, or use conditions. This chapter covers study planning, node bias, trends within and between conditions, time to an allowable limit, Arrhenius accelerated stability, and mean kinetic temperature (MKT).

! Scope of Use

Teaching data and illustrative limits in this chapter are deterministic examples, not universal acceptance criteria. Formal studies must use parameters specified by the protocol, risk assessment, and applicable standards. Decide in advance whether the response is concentration or signal, the time unit, bias scale, change direction, and allowable limit.

```
library(ivdtools)
library(readr)
```

4.1 Main Functions

Function	Main purpose	Key result
<code>stability_plan()</code>	Plan the minimum number of time points	Drift, variability, replicates, expected drift, power
<code>stability_bias()</code>	Calculate bias from the first node and between conditions	Node summaries and limit status
<code>stability_regression()</code>	Regress a single-condition trend or paired test/reference bias	Slope, one-sided limit, node results, diagnostics
<code>stability_time()</code>	Predict when a point estimate or one-sided limit reaches an allowable bias	Predicted time and extrapolation status
<code>arrhenius()</code>	Extrapolate target-temperature shelf life from elevated temperatures	Kinetic order, rate, activation energy, shelf life
<code>mkt()</code>	Calculate equally spaced or time-weighted mean kinetic temperature	Per-probe and combined MKT

`stability_regression()` has two modes: `mode="self"` regresses one condition over time;

`mode="compare"` calculates test-versus-reference bias at matched times and regresses the bias.

4.1.1 Plan a Study

Suppose the protocol specifies an allowable relative drift of 5%, an expected drift of 2.5%, repeatability variability of 1.5%, and a target power of 90%. Compare the minimum number of time points required when 1, 2, or 3 replicates are measured at each time point:

```
plan <- stability_plan(
  allowable_drift = 5,
  variability = 1.5,
  replicates = c(1, 2, 3),
  expected_drift = 2.5,
  power = 0.9,
  mode = "simple",
  bias_type = "relative"
)
print(plan)
```

Stability Study Time-Point Plan

Allowable drift: 5.0000; expected drift: 2.5000

Success target: 90%; method: CLSI EP25 Appendix A Table A2

Replicates	Residual	Allow/Resid	Min nodes
1	1.5000	3.3333	n/a
2	1.0607	4.5455	28
3	0.8660	5.7471	18

n/a -- No feasible entry in EP25 table.

`allowable_drift` must be positive. In simple mode, `variability` is a relative CV/SD; in components mode, it can be a data frame of variance components. `replicates` can contain one or more replicate counts per time point. The returned `stability_plan` object provides time-point/sample-size recommendations for the requested power and replicate counts.

4.1.2 Node Bias

Read the raw data:

```
bias_data <- read_csv(
  "./data/004-稳定性研究/stability-bias-example.csv",
  show_col_types = FALSE
)
bias_result <- stability_bias(
  data = bias_data, condition = "Condition", time = "Day",
  value = "Result", reference_condition = "2-8C", limit = 5,
  bias_type = "relative", time_unit = "d"
)
print(bias_result)
```

Stability Bias Analysis

Conditions: 2 (2-8C, 25C)

Reference: 2-8C

Time unit: d

Node comparisons:20

Acceptance: +/- 5.0000 (relative); 10/10 nodes passed

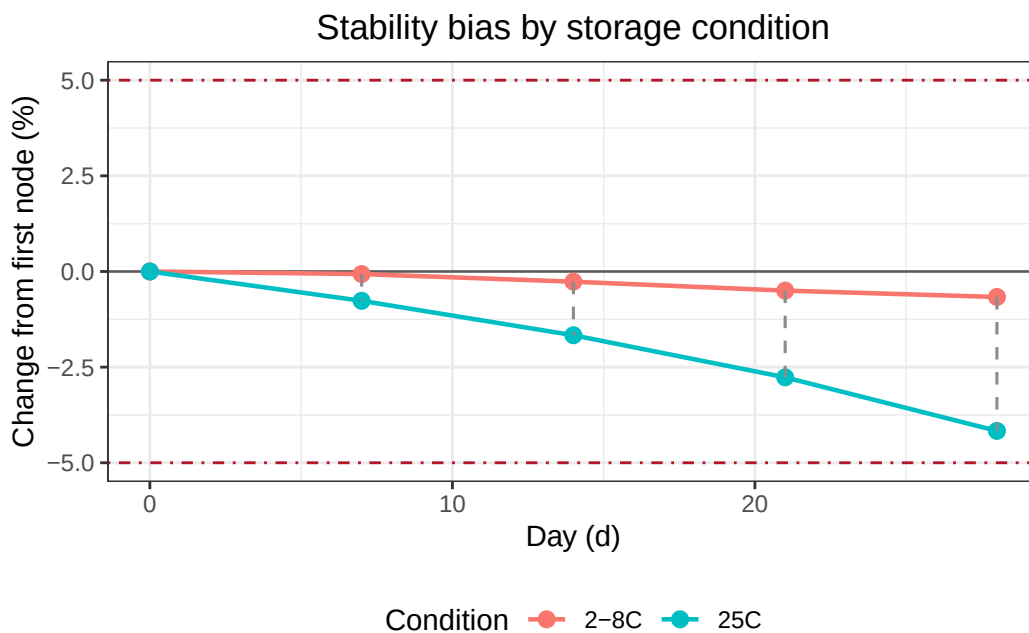
Within-condition bias (vs first node)

Cond	Time	Value	Bias(%)	Status
2-8C	0.0000	100.0000	0.0000	Pass
2-8C	7.0000	99.9333	-0.0667	Pass
2-8C	14.0000	99.7333	-0.2667	Pass
2-8C	21.0000	99.5000	-0.5000	Pass
2-8C	28.0000	99.3333	-0.6667	Pass
25C	0.0000	100.0000	0.0000	Pass
25C	7.0000	99.2333	-0.7667	Pass
25C	14.0000	98.3333	-1.6667	Pass
25C	21.0000	97.2333	-2.7667	Pass
25C	28.0000	95.8333	-4.1667	Pass

Between-condition bias (reference: 2-8C)

Time	Condition	TestVal	Bias(%)
0.0000	25C	100.0000	0.0000
7.0000	25C	99.2333	-0.7005
14.0000	25C	98.3333	-1.4037
21.0000	25C	97.2333	-2.2781
28.0000	25C	95.8333	-3.5235

```
plot(bias_result)
```



The function summarizes the mean by condition and time and returns a `stability_bias` object containing node means, relative/absolute bias, the reference condition, and limit status. When `reference_condition = NULL`, the software selects a reference according to data order; formal analyses should specify it explicitly. Missing or non-finite observations are removed, and the cleaning record must be retained. Within-condition bias uses each condition's first node as its baseline; between-condition bias compares the test and reference conditions at matched time points.

4.1.3 Trend Regression

Read the raw data:

```

regression_data <- read_csv(
  "./data/004-稳定性研究/stability-regression-example.csv",
  show_col_types = FALSE
)
regression_result <- stability_regression(
  data = regression_data, time = "Day", value = "Result",
  condition = "Condition", mode = "compare", test_condition = "25C",
  reference_condition = "2-8C", bias_type = "relative", direction = "decrease",
  conf.level = 0.95, limit = 5, time_unit = "d"
)
print(regression_result)

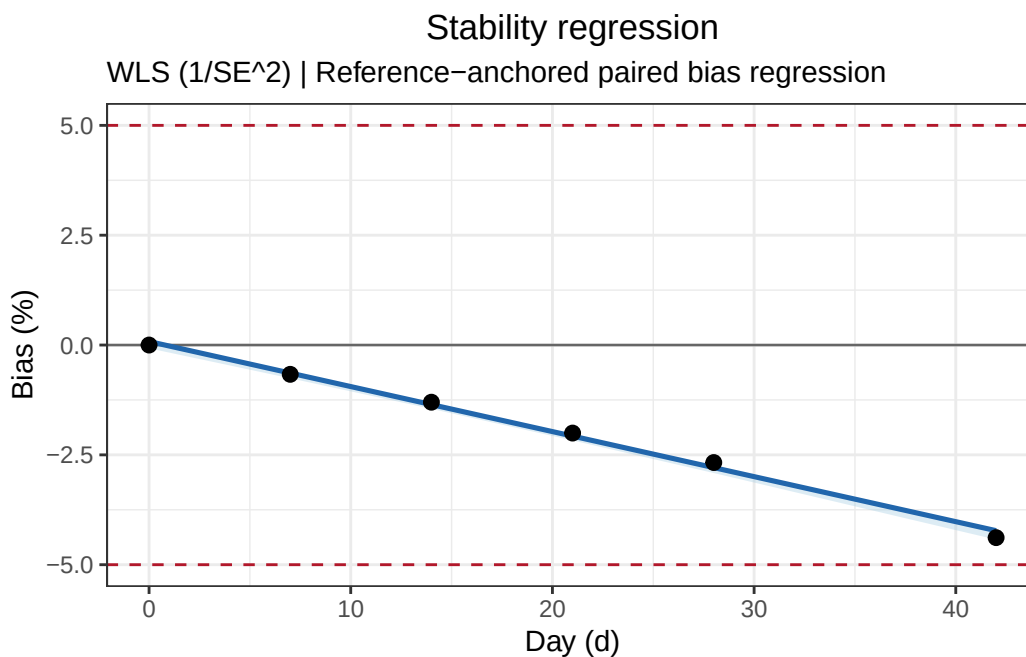
```

Stability Regression

Mode: compare
 Test condition: 25C
 Reference: 2-8C
 Fit: WLS (1/SE²); intercept = 0.0799, slope = -0.1025
 Direction: decrease
 One-sided CI: 95.0%
 R-squared: 0.9956
 Cook distance: row(s) 6 > 1; not removed

Time	ObsBias	Estimate	CI_Limit	Status
0.0000	0.0000	0.0799	-0.0750	
7.0000	-0.6669	-0.6376	-0.7566	
14.0000	-1.3013	-1.3552	-1.4522	
21.0000	-2.0040	-2.0728	-2.1717	
28.0000	-2.6747	-2.7904	-2.9140	
42.0000	-4.3842	-4.2256	-4.4296	

```
plot(regression_result)
```



The one-sided confidence limit evaluates the prespecified least-favorable direction. A large Cook's distance is a diagnostic flag, not an automatic exclusion; review records and per-

form inclusion/exclusion sensitivity analysis when justified.

For a single condition relative to its first node, use `mode="self"` and `test_condition`. The observed first-node bias is fixed at zero, but a fitted regression line need not pass exactly through that point.

At each matched time point, calculate the relative bias of 25C against 2-8C and regress the bias over time. The example prespecifies a decreasing response, so it uses `direction = "decrease"`.

`mode = "self"` uses the first node of one condition as the reference; `mode = "compare"` compares a test condition with a reference condition. With `direction = "auto"`, the direction is inferred from the slope; a formal protocol should preferably specify decrease or increase explicitly. With `limit = NULL`, the model is fitted without an acceptance status.

The returned `stability_regression` object includes the fitted model, slope/intercept, direction, bias scale, confidence limits, observed time range, condition mapping, and exclusions. Use `print()` for a summary and `plot()` for the trend, bias, and confidence limit.

This example identifies a node with a large Cook' s distance, but the function does not delete it automatically. Review the original records, experimental process, and protocol, and perform sensitivity analyses with and without the node only when justified.

4.1.4 Regression Against the First Node `stability_regression()`

`mode = "self"` analyzes the change of one condition over time. The following example uses the regression data but selects only the 25C condition; each node' s `observed_bias` is calculated relative to the mean of the first node:

```
self_regression_result <- stability_regression(
  data = regression_data,
  time = "Day",
  value = "Result",
  condition = "Condition",
  mode = "self",
  test_condition = "25C",
  bias_type = "relative",
  direction = "decrease",
  conf.level = 0.95,
  limit = 5,
  time_unit = "d"
)
print(self_regression_result)
```

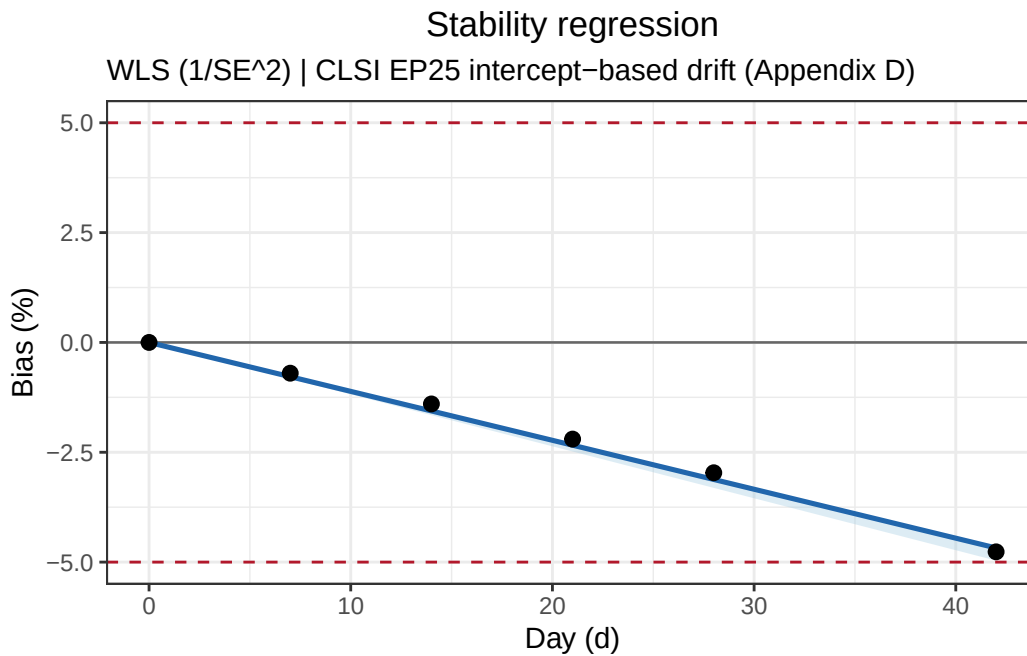
Stability Regression

```
Mode: self
Test condition: 25C
Fit: WLS (1/SE^2); intercept = 100.0849, slope = -0.1115
Direction: decrease
One-sided CI: 95.0%
R-squared: 0.9966
Cook distance: row(s) 1, 6 > 1; not removed
```

Time	ObsBias	Estimate	CI_limit	Status
0.0000	0.0000	-0.0000	-0.0000	
7.0000	-0.7000	-0.7797	-0.8278	
14.0000	-1.4000	-1.5594	-1.6555	
21.0000	-2.2000	-2.3391	-2.4833	
28.0000	-2.9667	-3.1188	-3.3110	

```
42.0000    -4.7667    -4.6782    -4.9665
```

```
plot(self_regression_result)
```



The observed bias at the first node is fixed at 0. The model estimate is fitted jointly from all time points, so the regression intercept is not required to pass exactly through that node.

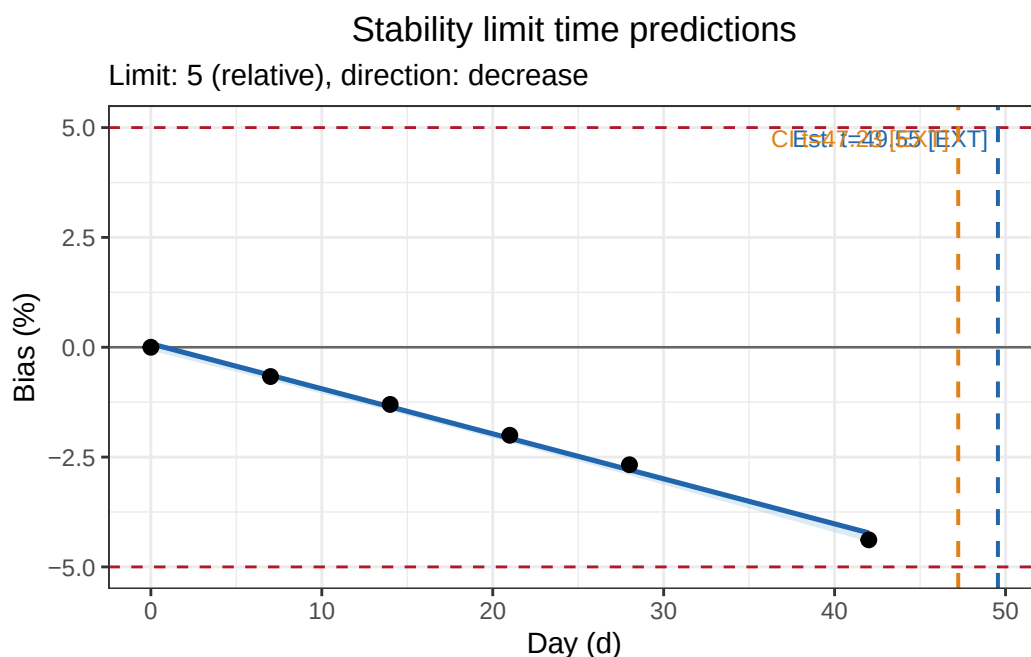
4.1.5 Time to an Allowable Limit

`stability_time()` calculates separately the time when the point estimate and the one-sided confidence limit first reach a bias of -5%:

```
time_result <- stability_time(regression_result, limit = 5, max_time = 90)
print(time_result)
```

```
Stability Limit Time
Limit: 5.0000 (relative); direction: decrease
estimate      49.5548 d [EXTRAPOLATED]
one_sided_ci  47.2316 d [EXTRAPOLATED]
```

```
plot(time_result)
```



If the predicted time lies beyond the longest observed time, the result is marked **EXTRAPOLATED**. Extrapolation assumes that the trend continues and is not equivalent to observed stability evidence.

The input must be a `stability_regression` object. The function evaluates the first time at which the point estimate or one-sided confidence limit reaches the threshold, returns a `stability_time` object, and marks whether the result is beyond the observed range. `max_time` is only a search bound, not a scientifically justified shelf life; substantial extrapolation requires separate justification.

The longest observed time in this example is 42 days, while the predicted limit time is beyond the observed range, so the output is marked **EXTRAPOLATED**. The extrapolation assumes that the linear trend continues outside the observed interval and is not equivalent to measured stability evidence. Report the extrapolation distance and uncertainty clearly, and preferably confirm it with longer real-time observations.

4.1.6 Arrhenius Accelerated Stability

Read and inspect the raw data:

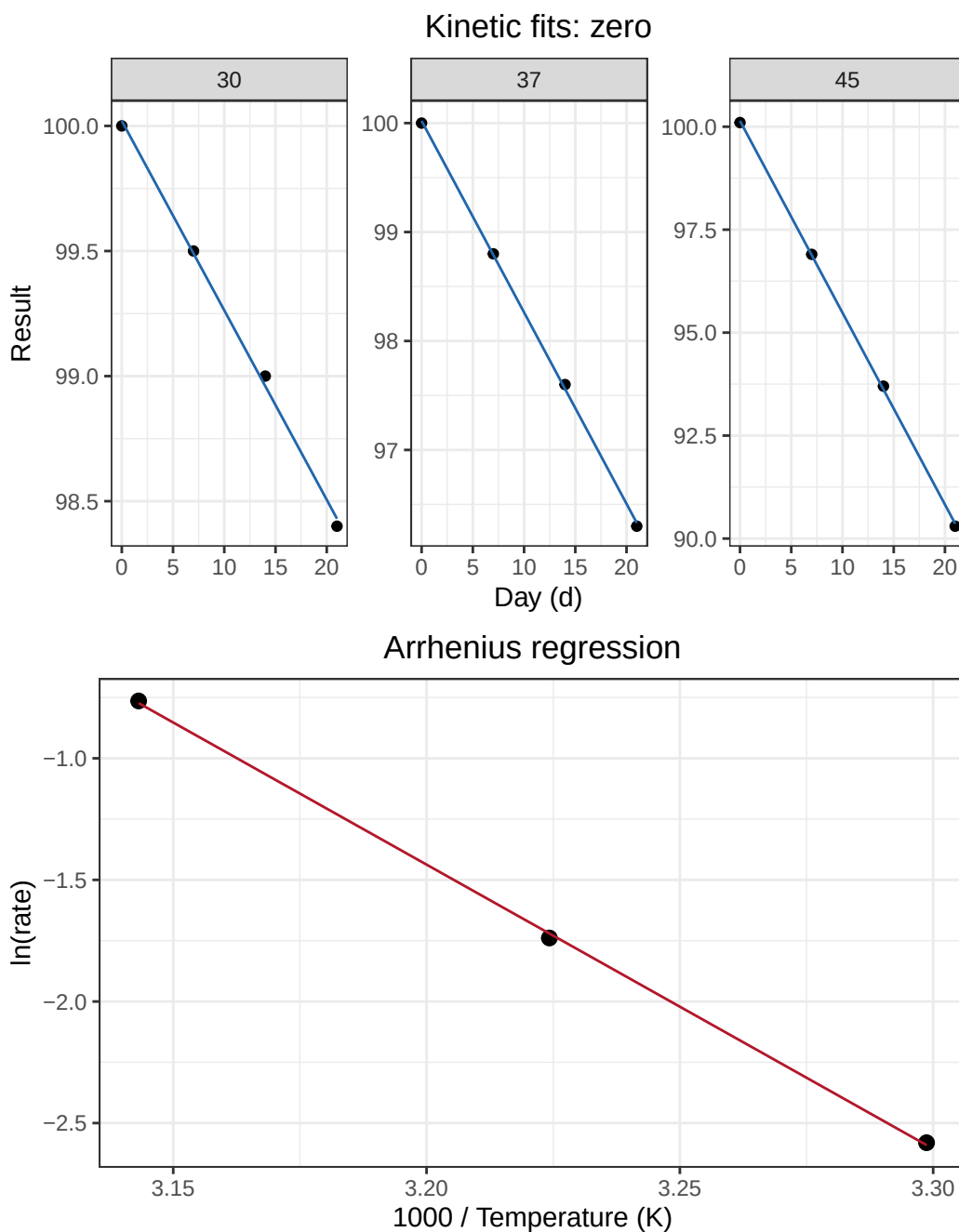
```
arrhenius_data <- read_csv(
  "./data/004-稳定性研究/stability-arrhenius-example.csv",
  show_col_types = FALSE
)
arrhenius_result <- arrhenius(
  data = arrhenius_data, temperature = "Temperature", time = "Day",
  value = "Result", target_temp = 5, order = "auto", direction = "decrease",
  limit = 10, bias_type = "relative", temp_unit = "C", time_unit = "d",
  conf.level = 0.95
)
print(arrhenius_result)
```

Arrhenius Stability Analysis

```
Kinetic order: zero (auto-selected)
Direction: decrease
Activation Ea: 97.159 kJ/mol
Target rate: 0.002346 /d
```

Limit time: 4263.7878 d [1585.3589, 11467.3633]

```
plot(arrhenius_result)
```



`order="auto"` selects from zero- and first-order candidates under the implementation rules. Formal work should justify kinetic order, degradation direction, rate significance, extent of degradation, residuals at each temperature, and the potentially large extrapolation to the target temperature.

4.1.7 Mean Kinetic Temperature

```
mkt_data <- read_csv(
  "./data/004-稳定性研究/stability-mkt-example.csv",
  show_col_types = FALSE
```

```

)
mkt_data$Timestamp <- as.POSIXct(
  mkt_data$Timestamp, format = "%Y-%m-%d %H:%M:%S", tz = "UTC"
)
mkt_result <- mkt(
  data = mkt_data, temp_cols = c("Probe_A", "Probe_B"),
  time = "Timestamp", temp_unit = "C", ea = 83.144
)
print(mkt_result)

```

Mean Kinetic Temperature

Ea: 83.144 kJ/mol

Method: Time-weighted trapezoidal MKT

Probe_A	299.538 K	26.388 C
Probe_B	299.194 K	26.044 C
Combined	299.367 K	26.217 C

MKT is not an arithmetic temperature mean; high temperatures receive greater weight through the Arrhenius relationship. Confirm probe calibration, time ordering, missing intervals, temperature units, exposure duration, and activation energy. MKT summarizes exposure and cannot replace a product-specific stability study.

`mkt()` returns the mean kinetic temperature for each probe and for the combined series. When time is supplied, observations are weighted by their time intervals; otherwise equal spacing is assumed. `ea` is the assumed activation energy in kJ/mol and must be reported.

4.1.8 Split Multiple-Sample Long-Format Data

When different concentration levels, sample types, or lots must be evaluated separately, retain the sample identifier, split the data first, and analyze each sample with the same pre-specified parameters. Read and inspect the multiple-sample data:

```

multiple_samples_data <- read_csv(
  "./data/004-稳定性研究/stability-multiple-samples-example.csv",
  show_col_types = FALSE
)
head(multiple_samples_data, 6)

```

A tibble: 6 x 4

	Sample <chr>	Day <dbl>	Replicate <dbl>	Result <dbl>
1	Sample_A	0	1	100.
2	Sample_A	0	2	99.8
3	Sample_A	0	3	100
4	Sample_A	7	1	99.7
5	Sample_A	7	2	99.5
6	Sample_A	7	3	99.8

Use `split_by_factors()` to build a named list by a classification column. The function keeps the split column, uses the observed sample values as list names, and reports the number of rows in each group and the rows excluded because the grouping value was missing:

```

sample_data_list <- split_by_factors(
  multiple_samples_data,
  variable = "Sample"
)

```

```
sample_data_list
```

Data split by factors

Variable: Sample

Groups: 2

Missing/excluded: 0

Sample_A n = 18

Sample_B n = 18

```
sample_a_data <- sample_data_list[["Sample_A"]]
```

Then run the analysis for each sample.

4.2 Analysis Cases

4.2.1 CLSI EP25-A2 Appendix B: Arrhenius Accelerated Stability

Appendix B provides accelerated-stability data for a control material at 40, 45, 50, and 55 °C. The target storage temperature is 25 °C and the allowable decrease is 20%. The standard first log-transforms the results at each temperature and fits first-order kinetics, then uses the Arrhenius model to extrapolate the reaction rate and shelf life at 25 °C.

Read and inspect the raw data:

```
ep25_b_data <- read_csv(
  "/data/004-稳定性研究/std-ep25-appendix-b.csv",
  show_col_types = FALSE
)
head(ep25_b_data, 6)
```

A tibble: 6 x 3

	Temperature	Day	Result
	<dbl>	<dbl>	<dbl>
1	40	0	1.02
2	40	10	0.956
3	40	21	0.945
4	40	31	0.933
5	40	42	0.908
6	40	52	0.86

Analyze with the first-order model, decreasing direction, 20% allowable change, and 25 °C target temperature specified by Appendix B:

```
ep25_b_result <- arrhenius(
  data = ep25_b_data,
  temperature = "Temperature",
  time = "Day",
  value = "Result",
  target_temp = 25,
  order = "first",
  direction = "decrease",
  limit = 20,
  bias_type = "relative",
  temp_unit = "C",
```



```

time_unit = "d",
conf.level = 0.95
)
print(ep25_b_result)

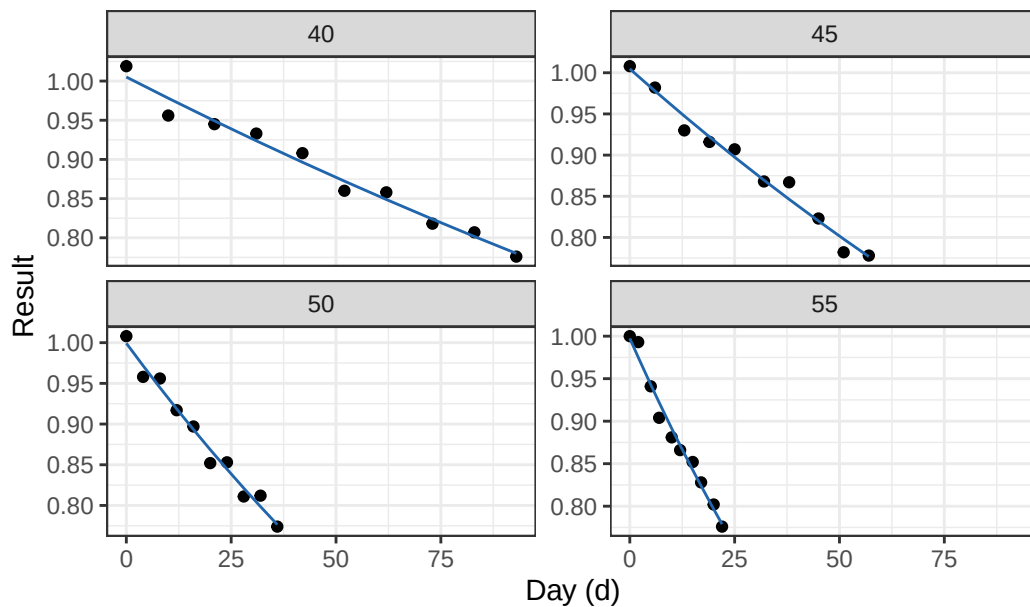
```

Arrhenius Stability Analysis

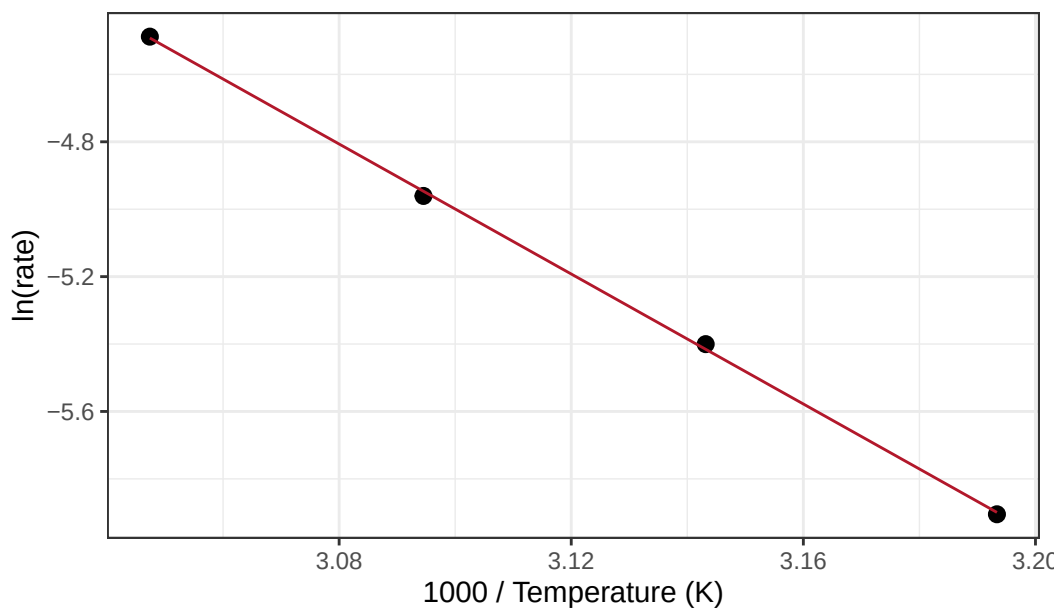
Kinetic order: first
 Direction: decrease
 Activation Ea: 80.160 kJ/mol
 Target rate: 0.000582 /d
 Limit time: 383.1423 d [330.8315, 443.7244]
 WARNING: Observed degradation exceeds 20% at one or more temperatures.

```
plot(ep25_b_result)
```

Kinetic fits: first



Arrhenius regression



Extract the degradation rates at each temperature, the Arrhenius regression coefficients, the target rate at 25 °C, and the shelf life from the `arrhenius()` result, then compare them with the values reported in Appendix B:

Table 4.1: EP25 Appendix B: comparison of `ivdtools` and standard results

appendix	statistic	standard	ivdtools
EP25-A2 Appendix B / 40C	rate	2.7257e-03	0.0027257
EP25-A2 Appendix B / 45C	rate	4.5154e-03	0.0045154
EP25-A2 Appendix B / 50C	rate	7.0087e-03	0.0070087
EP25-A2 Appendix B / 55C	rate	1.1243e-02	0.0112430
EP25-A2 Appendix B / Arrhenius	slope	-9.6320e+03	-9640.9966367
EP25-A2 Appendix B / Arrhenius	intercept	2.4874e+01	24.8877147
EP25-A2 Appendix B / 25C	target_rate	5.8250e-04	0.0005824
EP25-A2 Appendix B / 25C	shelf_life	3.8300e+02	383.1423039

4.2.2 CLSI EP25-A2 Appendix C1: Reagent Stability

Appendix C1 contains single-lot reagent data covering 0–418 days, with a claimed period of 364 days and an allowable drift of 10%. Each of the three series is represented by the mean at each time point; here they are analyzed with `stability_regression()` in self mode, using relative bias and a decreasing direction.

```
ep25_c1 <- read_csv(
  "./data/004-稳定性研究/std-ep25-appendix-c1.csv",
  show_col_types = FALSE
)
head(ep25_c1, 6)
```

```
# A tibble: 6 x 3
  series    day result
  <chr>    <dbl> <dbl>
1 low_panel    0  7.07
2 low_panel   33  6.95
3 low_panel   61  6.99
4 low_panel   95  6.88
5 low_panel  121  6.98
6 low_panel  155  6.83
```

```
ep25_c1_list <- split_by_factors(
  ep25_c1,
  variable = "series"
)
ep25_c1_list
```

Data split by factors

Variable: series

Groups: 3

Missing/excluded: 0

low_panel n = 13

low_control n = 13

high_control n = 13

Analyze the series with `stability_regression()`:

```

ep25_low_panel <- stability_regression(
  ep25_c1_list$low_panel,
  time = "day", value = "result", mode = "self",
  bias_type = "relative", direction = "decrease",
  conf.level = 0.95, limit = 10, time_unit = "d"
)

ep25_low_control <- stability_regression(
  ep25_c1_list$low_control,
  time = "day", value = "result", mode = "self",
  bias_type = "relative", direction = "decrease",
  conf.level = 0.95, limit = 10, time_unit = "d"
)

ep25_high_control <- stability_regression(
  ep25_c1_list$high_control,
  time = "day", value = "result", mode = "self",
  bias_type = "relative", direction = "decrease",
  conf.level = 0.95, limit = 10, time_unit = "d"
)

```

Compare the calculated results with the values reported in the standard:

Table 4.2: EP25 Appendix C1: comparison of ivdtools and standard results

appendix	statistic	standard	ivdtools
EP25-A2 Appendix C1 / low_panel	slope	-0.001567	-0.001567
EP25-A2 Appendix C1 / low_panel	intercept	7.065000	7.065100
EP25-A2 Appendix C1 / low_panel	change_364	-8.070000	-8.074378
EP25-A2 Appendix C1 / low_panel	change_364_upper_CL	-9.260000	-9.262841
EP25-A2 Appendix C1 / low_panel	change_418	-9.270000	-9.272226
EP25-A2 Appendix C1 / low_panel	change_418_upper_CL	-10.640000	-10.636999
EP25-A2 Appendix C1 / low_control	slope	-0.002873	-0.002873
EP25-A2 Appendix C1 / low_control	intercept	25.270000	25.268365
EP25-A2 Appendix C1 / low_control	change_364	-4.140000	-4.139278
EP25-A2 Appendix C1 / low_control	change_364_upper_CL	-5.130000	-5.129081
EP25-A2 Appendix C1 / low_control	change_418	-4.750000	-4.753346
EP25-A2 Appendix C1 / low_control	change_418_upper_CL	-5.890000	-5.889988
EP25-A2 Appendix C1 / high_control	slope	-0.009103	-0.009103
EP25-A2 Appendix C1 / high_control	intercept	99.740000	99.742806
EP25-A2 Appendix C1 / high_control	change_364	-3.320000	-3.321944
EP25-A2 Appendix C1 / high_control	change_364_upper_CL	-4.170000	-4.166660
EP25-A2 Appendix C1 / high_control	change_418	-3.810000	-3.814759
EP25-A2 Appendix C1 / high_control	change_418_upper_CL	-4.780000	-4.784791

4.2.3 CLSI EP25-A2 Appendix C2.1: Calibrator Stability (Reference-Anchored Real-Time Study)

Proportional drift is an analysis definition. After reading the data, calculate it explicitly as `on_test / reference`, then split the data by level:

```

ep25_c2_1_data <- read_csv(
  "./data/004-稳定性研究/std-ep25-appendix-c2-1.csv",
  show_col_types = FALSE
)
head(ep25_c2_1_data, 6)

```

```
# A tibble: 6 x 4
  level month on_test reference
<chr> <dbl> <dbl> <dbl>
1 low      0  1.01      1.00
2 low      1  0.999     1.00
3 low      2  0.992     1.01
4 low      4  0.983     1.00
5 low      5  0.985     1.01
6 low      6  0.97      1.00
```

```
ep25_c2_1_data$proportional_drift <-
  ep25_c2_1_data$on_test / ep25_c2_1_data$reference
ep25_c2_1_list <- split_by_factors(
  ep25_c2_1_data,
  variable = "level"
)
ep25_c2_1_list
```

```
Data split by factors
Variable: level
Groups: 2
Missing/excluded: 0
```

```
low    n = 7
high   n = 7
```

Analyze the two calibrator levels sequentially with `stability_regression()`:

```
ep25_c2_1_low_result <- stability_regression(
  data = ep25_c2_1_list$low,
  time = "month",
  value = "proportional_drift",
  mode = "self",
  bias_type = "relative",
  direction = "decrease",
  limit = 5,
  time_unit = "month",
  conf.level = 0.95
)

ep25_c2_1_high_result <- stability_regression(
  data = ep25_c2_1_list$high,
  time = "month",
  value = "proportional_drift",
  mode = "self",
  bias_type = "relative",
  direction = "decrease",
  limit = 5,
  time_unit = "month",
  conf.level = 0.95
)

ep25_c2_1_low_time <- stability_time(
  ep25_c2_1_low_result,
  max_time = 20
)
```

```
ep25_c2_1_high_time <- stability_time(
  ep25_c2_1_high_result,
  max_time = 20
)

print(ep25_c2_1_low_result)
```

Stability Regression

Mode: self
 Test condition: condition
 Fit: OLS; intercept = 1.0020, slope = -0.0059
 Direction: decrease
 One-sided CI: 95.0%
 R-squared: 0.9566
 Cook distance: row(s) 1 > 1; not removed
 Note: WLS unavailable; OLS used.

Time	ObsBias	Estimate	CI_limit	Status
0.0000	0.0000	-0.0000	-0.0000	
1.0000	-1.2873	-0.5887	-0.6995	
2.0000	-2.1737	-1.1775	-1.3989	
4.0000	-2.8683	-2.3549	-2.7978	
5.0000	-3.2483	-2.9437	-3.4973	
6.0000	-3.8659	-3.5324	-4.1968	
6.5000	-4.3541	-3.8268	-4.5465	

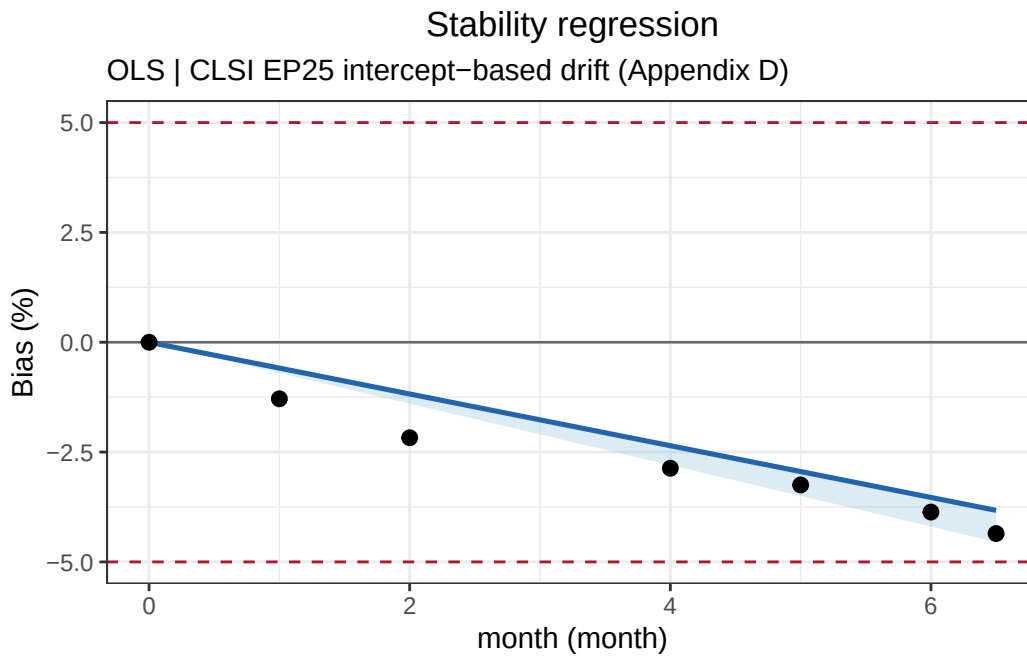
```
print(ep25_c2_1_high_result)
```

Stability Regression

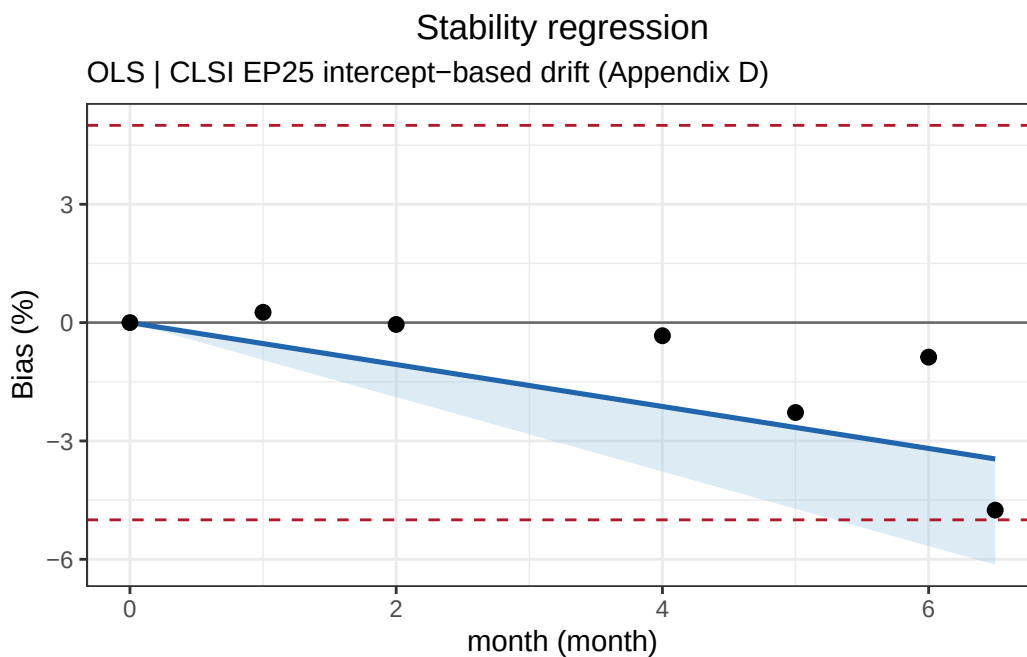
Mode: self
 Test condition: condition
 Fit: OLS; intercept = 1.0078, slope = -0.0054
 Direction: decrease
 One-sided CI: 95.0%
 R-squared: 0.5652
 Cook distance: row(s) 7 > 1; not removed
 Note: WLS unavailable; OLS used.

Time	ObsBias	Estimate	CI_limit	Status
0.0000	0.0000	-0.0000	-0.0000	
1.0000	0.2612	-0.5317	-0.9441	
2.0000	-0.0475	-1.0633	-1.8881	
4.0000	-0.3329	-2.1266	-3.7763	
5.0000	-2.2787	-2.6583	-4.7204	
6.0000	-0.8759	-3.1900	-5.6644	*
6.5000	-4.7539	-3.4558	-6.1365	*

```
plot(ep25_c2_1_low_result)
```



```
plot(ep25_c2_1_high_result)
```



Compare the results with the standard:

Table 4.3: EP25 Appendix C2.1: comparison of ivdtools and standard results

appendix	statistic	standard	ivdtools
EP25-A2 Appendix C2.1 / low_calibrator	change_6	-3.53	-3.532421
EP25-A2 Appendix C2.1 / low_calibrator	change_6_upper_CL	-4.20	-4.196768
EP25-A2 Appendix C2.1 / low_calibrator	change_6.5	-3.83	-3.826789
EP25-A2 Appendix C2.1 / low_calibrator	change_6.5_upper_CL	-4.55	-4.546498
EP25-A2 Appendix C2.1 / low_calibrator	ci_limit_time	7.10	7.148359
EP25-A2 Appendix C2.1 / high_calibrator	change_6	-3.19	-3.189971
EP25-A2 Appendix C2.1 / high_calibrator	change_6_upper_CL	-5.66	-5.664441
EP25-A2 Appendix C2.1 / high_calibrator	change_6.5	-3.46	-3.455801

appendix	statistic	standard	ivdtools
EP25-A2 Appendix C2.1 / high_calibrator	change_6.5_upper_CL	-6.14	-6.136478
EP25-A2 Appendix C2.1 / high_calibrator	ci_limit_time	5.30	5.296198

4.2.4 CLSI EP25-A2 Appendix C2.2: Calibrator Stability (Isochronous Study)

```
ep25_c2_2_data <- read_csv(
  "./data/004-稳定性研究/std-ep25-appendix-c2-2.csv",
  show_col_types = FALSE
)
head(ep25_c2_2_data, 6)
```

```
# A tibble: 6 x 3
  level    day result
<chr>  <dbl>  <dbl>
1 level_1    0   19.9
2 level_1   15   20.2
3 level_1   31   20.2
4 level_1   62   20.3
5 level_1   94   20.7
6 level_1  123   21.1
```

```
ep25_c2_2_list <- split_by_factors(
  ep25_c2_2_data,
  variable = "level"
)
ep25_c2_2_list
```

Data split by factors
Variable: level
Groups: 3
Missing/excluded: 0

```
level_1  n = 9
level_2  n = 9
level_3  n = 9
```

Analyze the three calibrator levels sequentially with `stability_regression()`:

```
ep25_c2_2_level_1_result <- stability_regression(
  data = ep25_c2_2_list$level_1,
  time = "day",
  value = "result",
  mode = "self",
  bias_type = "relative",
  direction = "increase",
  limit = 10,
  time_unit = "d",
  conf.level = 0.95
)

ep25_c2_2_level_2_result <- stability_regression(
  data = ep25_c2_2_list$level_2,
```

```

    time = "day",
    value = "result",
    mode = "self",
    bias_type = "relative",
    direction = "increase",
    limit = 10,
    time_unit = "d",
    conf.level = 0.95
)

ep25_c2_2_level_3_result <- stability_regression(
  data = ep25_c2_2_list$level_3,
  time = "day",
  value = "result",
  mode = "self",
  bias_type = "relative",
  direction = "increase",
  limit = 10,
  time_unit = "d",
  conf.level = 0.95
)

print(ep25_c2_2_level_1_result)

```

Stability Regression

Mode: self
 Test condition: condition
 Fit: OLS; intercept = 19.9291, slope = 0.0088
 Direction: increase
 One-sided CI: 95.0%
 R-squared: 0.9789
 Note: WLS unavailable; OLS used.

Time	ObsBias	Estimate	CI_Limit	Status
0.0000	0.0000	0.0000	0.0000	
15.0000	1.5075	0.6616	0.7341	
31.0000	1.5075	1.3673	1.5171	
62.0000	2.0101	2.7345	3.0343	
94.0000	4.0201	4.1459	4.6003	
123.0000	6.0302	5.4249	6.0196	
154.0000	7.0352	6.7922	7.5367	
180.0000	8.5427	7.9389	8.8092	
210.0000	9.0452	9.2620	10.2774	*

```
print(ep25_c2_2_level_2_result)
```

Stability Regression

Mode: self
 Test condition: condition
 Fit: OLS; intercept = 100.9678, slope = 0.0292
 Direction: increase
 One-sided CI: 95.0%
 R-squared: 0.9426
 Note: WLS unavailable; OLS used.

Time	ObsBias	Estimate	CI_limit	Status
0.0000	0.0000	0.0000	0.0000	
15.0000	1.2859	0.4341	0.5129	
31.0000	0.3956	0.8970	1.0600	
62.0000	0.9891	1.7941	2.1201	
94.0000	2.3739	2.7201	3.2143	
123.0000	2.9674	3.5592	4.2060	
154.0000	4.1543	4.4563	5.2660	
180.0000	5.7369	5.2087	6.1551	
210.0000	6.0336	6.0768	7.1809	

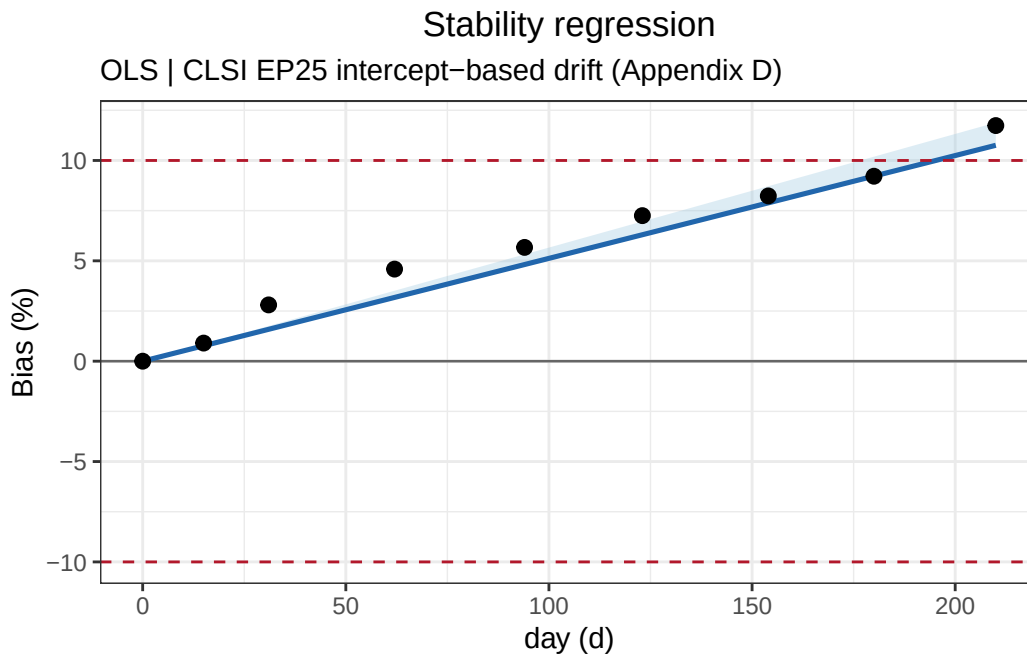
```
print(ep25_c2_2_level_3_result)
```

Stability Regression

Mode: self
 Test condition: condition
 Fit: OLS; intercept = 502.3132, slope = 0.2573
 Direction: increase
 One-sided CI: 95.0%
 R-squared: 0.9808
 Note: WLS unavailable; OLS used.

Time	ObsBias	Estimate	CI_limit	Status
0.0000	0.0000	0.0000	0.0000	
15.0000	0.9014	0.7683	0.8491	
31.0000	2.8045	1.5878	1.7547	
62.0000	4.5873	3.1756	3.5094	
94.0000	5.6691	4.8147	5.3207	
123.0000	7.2516	6.3001	6.9622	
154.0000	8.2332	7.8879	8.7169	
180.0000	9.2147	9.2196	10.1886	*
210.0000	11.7388	10.7562	11.8867	*

```
plot(ep25_c2_2_level_3_result)
```



Compare the results with the standard:

Table 4.4: EP25 Appendix C2.2: comparison of ivdtools and standard results

appendix	statistic	standard	ivdtools
EP25-A2 Appendix C2.2 / level_1	change_180	7.97	7.938892
EP25-A2 Appendix C2.2 / level_1	change_180_upper_CL	8.78	8.809166
EP25-A2 Appendix C2.2 / level_1	change_210	9.30	9.262041
EP25-A2 Appendix C2.2 / level_1	change_210_upper_CL	10.24	10.277360
EP25-A2 Appendix C2.2 / level_2	change_180	5.19	5.208658
EP25-A2 Appendix C2.2 / level_2	change_180_upper_CL	6.11	6.155086
EP25-A2 Appendix C2.2 / level_2	change_210	6.05	6.076767
EP25-A2 Appendix C2.2 / level_2	change_210_upper_CL	7.13	7.180933
EP25-A2 Appendix C2.2 / level_3	change_180	9.19	9.219626
EP25-A2 Appendix C2.2 / level_3	change_180_upper_CL	10.18	10.188639
EP25-A2 Appendix C2.2 / level_3	change_210	10.73	10.756231
EP25-A2 Appendix C2.2 / level_3	change_210_upper_CL	11.87	11.886745

4.2.5 CLSI EP25-A2 Appendix C3.1: Control Stability (Reference-Anchored Real-Time Study)

```
ep25_c3_1_data <- read_csv(
  "./data/004-稳定性研究/std-ep25-appendix-c3-1.csv",
  show_col_types = FALSE
)
head(ep25_c3_1_data, 6)
```

```
# A tibble: 6 x 4
  control month condition result
  <chr>    <dbl> <chr>    <dbl>
1 negative 0 reference 0.23
2 negative 1 reference 0.24
3 negative 2 reference 0.24
4 negative 3 reference 0.23
5 negative 4 reference 0.24
```

6 negative 5 reference 0.25

```
ep25_c3_1_list <- split_by_factors(
  ep25_c3_1_data,
  variable = "control"
)
ep25_c3_1_list
```

Data split by factors

Variable: control

Groups: 2

Missing/excluded: 0

negative n = 18

positive n = 18

Analyze the negative and positive controls separately with `stability_regression()`:

```
ep25_c3_1_negative_result <- stability_regression(
  data = ep25_c3_1_list$negative,
  time = "month",
  value = "result",
  condition = "condition",
  mode = "compare",
  test_condition = "on_test",
  reference_condition = "reference",
  bias_type = "absolute",
  direction = "increase",
  limit = 0.10,
  time_unit = "month",
  conf.level = 0.95
)

ep25_c3_1_positive_result <- stability_regression(
  data = ep25_c3_1_list$positive,
  time = "month",
  value = "result",
  condition = "condition",
  mode = "compare",
  test_condition = "on_test",
  reference_condition = "reference",
  bias_type = "relative",
  direction = "decrease",
  limit = 5,
  time_unit = "month",
  conf.level = 0.95
)

print(ep25_c3_1_negative_result)
```

Stability Regression

Mode: compare

Test condition: on_test

Reference: reference

Fit: OLS; intercept = 0.0004, slope = 0.0103

Direction: increase

One-sided CI: 95.0%
 R-squared: 0.8962
 Note: WLS unavailable; OLS used.

Time	ObsBias	Estimate	CI_limit	Status
0.0000	0.0000	0.0004	0.0121	
1.0000	0.0200	0.0107	0.0204	
2.0000	0.0300	0.0210	0.0290	
3.0000	0.0200	0.0314	0.0381	
4.0000	0.0300	0.0417	0.0480	
5.0000	0.0400	0.0520	0.0588	
6.0000	0.0700	0.0623	0.0705	
7.0000	0.0800	0.0727	0.0826	
7.5000	0.0800	0.0778	0.0888	

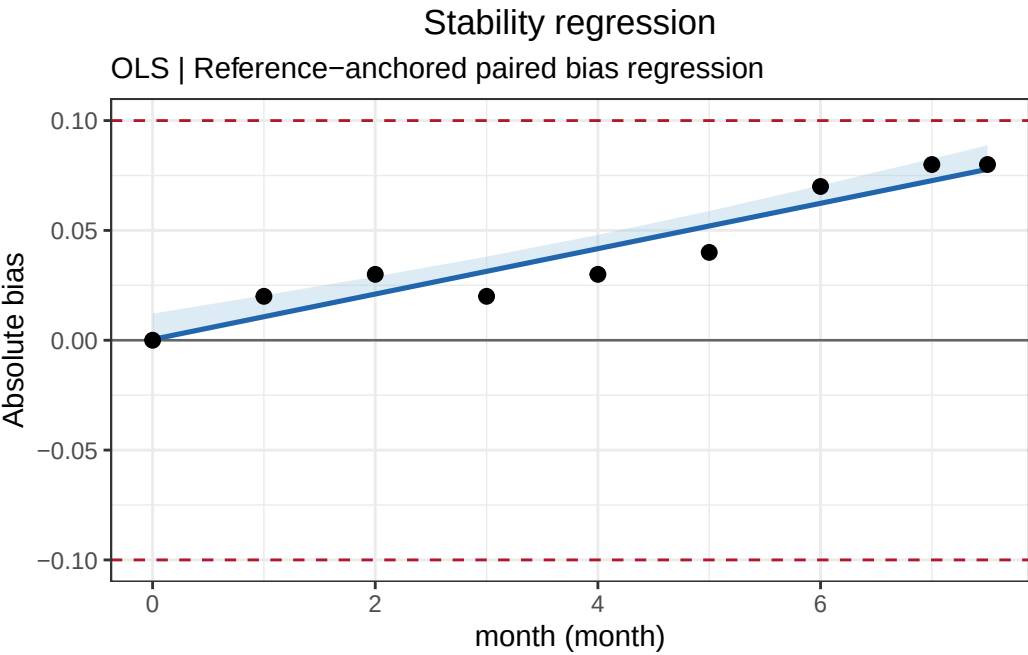
```
print(ep25_c3_1_positive_result)
```

Stability Regression

Mode: compare
 Test condition: on_test
 Reference: reference
 Fit: OLS; intercept = -0.0461, slope = -0.3655
 Direction: decrease
 One-sided CI: 95.0%
 R-squared: 0.7765
 Note: WLS unavailable; OLS used.

Time	ObsBias	Estimate	CI_limit	Status
0.0000	-0.1992	-0.0461	-0.7018	
1.0000	-0.5988	-0.4116	-0.9539	
2.0000	-0.8016	-0.7772	-1.2218	
3.0000	-1.3917	-1.1427	-1.5179	
4.0000	-1.3972	-1.5082	-1.8593	
5.0000	-0.8048	-1.8738	-2.2547	
6.0000	-1.8036	-2.2393	-2.6937	
7.0000	-2.7944	-2.6048	-3.1591	
7.5000	-3.6000	-2.7876	-3.3979	

```
plot(ep25_c3_1_negative_result)
```



Compare the results with the standard:

Table 4.5: EP25 Appendix C3.1: comparison of ivdtools and standard results

appendix	statistic	standard	ivdtools
EP25-A2 Appendix C3.1 / negative_control	change_7	0.07	0.072663
EP25-A2 Appendix C3.1 / negative_control	change_7_upper_CL	0.09	0.082595
EP25-A2 Appendix C3.1 / negative_control	change_7.5	0.08	0.077826
EP25-A2 Appendix C3.1 / negative_control	change_7.5_upper_CL	0.10	0.088761
EP25-A2 Appendix C3.1 / positive_control	change_7	-2.56	-2.604835
EP25-A2 Appendix C3.1 / positive_control	change_7_upper_CL	-3.53	-3.159133
EP25-A2 Appendix C3.1 / positive_control	change_7.5	-2.74	-2.787602
EP25-A2 Appendix C3.1 / positive_control	change_7.5_upper_CL	-3.79	-3.397864

For the negative control, use absolute bias with a 0.10 IU/mL limit; for the positive control, use relative bias with a 5% limit.

4.2.6 CLSI EP25-A2 Appendix C3.2: Control Stability (Isochronous Study)

```
ep25_c3_2_data <- read_csv(  
  "/data/004-稳定性研究/std-ep25-appendix-c3-2.csv",  
  show_col_types = FALSE  
)  
head(ep25_c3_2_data, 6)
```

```
# A tibble: 6 x 3  
  level    day result  
  <chr>  <dbl> <dbl>  
1 level_1    0  1.21  
2 level_1   40  1.17  
3 level_1   80  1.22  
4 level_1  120  1.18  
5 level_1  162  1.15  
6 level_1  203  1.14
```

```
ep25_c3_2_list <- split_by_factors(
  ep25_c3_2_data,
  variable = "level"
)
ep25_c3_2_list
```

Data split by factors

Variable: level

Groups: 3

Missing/excluded: 0

level_1 n = 10

level_2 n = 10

level_3 n = 10

Analyze the three control levels sequentially with `stability_regression()`:

```
ep25_c3_2_level_1_result <- stability_regression(
  data = ep25_c3_2_list$level_1,
  time = "day",
  value = "result",
  mode = "self",
  bias_type = "relative",
  direction = "decrease",
  limit = 10,
  time_unit = "d",
  conf.level = 0.95
)

ep25_c3_2_level_2_result <- stability_regression(
  data = ep25_c3_2_list$level_2,
  time = "day",
  value = "result",
  mode = "self",
  bias_type = "relative",
  direction = "decrease",
  limit = 10,
  time_unit = "d",
  conf.level = 0.95
)

ep25_c3_2_level_3_result <- stability_regression(
  data = ep25_c3_2_list$level_3,
  time = "day",
  value = "result",
  mode = "self",
  bias_type = "relative",
  direction = "decrease",
  limit = 10,
  time_unit = "d",
  conf.level = 0.95
)

print(ep25_c3_2_level_1_result)
```

Stability Regression

Mode: self
 Test condition: condition
 Fit: OLS; intercept = 1.2060, slope = -0.0003
 Direction: decrease
 One-sided CI: 95.0%
 R-squared: 0.7575
 Note: WLS unavailable; OLS used.

Time	ObsBias	Estimate	CI_limit	Status
0.0000	0.0000	-0.0000	-0.0000	
40.0000	-3.3058	-0.9917	-1.3431	
80.0000	0.8264	-1.9833	-2.6861	
120.0000	-2.4793	-2.9750	-4.0292	
162.0000	-4.9587	-4.0162	-5.4394	
203.0000	-5.7851	-5.0327	-6.8160	
255.0000	-8.2645	-6.3219	-8.5620	
305.0000	-10.7438	-7.5614	-10.2408	*
364.0000	-6.6116	-9.0241	-12.2218	*
410.0000	-9.9174	-10.1645	-13.7663	*

```
print(ep25_c3_2_level_2_result)
```

Stability Regression

Mode: self
 Test condition: condition
 Fit: OLS; intercept = 3.4233, slope = -0.0012
 Direction: decrease
 One-sided CI: 95.0%
 R-squared: 0.9402
 Note: WLS unavailable; OLS used.

Time	ObsBias	Estimate	CI_limit	Status
0.0000	0.0000	-0.0000	-0.0000	
40.0000	0.8798	-1.3939	-1.6098	
80.0000	-2.3460	-2.7879	-3.2195	
120.0000	-5.5718	-4.1818	-4.8293	
162.0000	-4.1056	-5.6454	-6.5195	
203.0000	-6.7449	-7.0742	-8.1695	
255.0000	-10.2639	-8.8863	-10.2622	*
305.0000	-11.1437	-10.6287	-12.2744	*
364.0000	-11.7302	-12.6847	-14.6488	*
410.0000	-12.9032	-14.2877	-16.5001	*

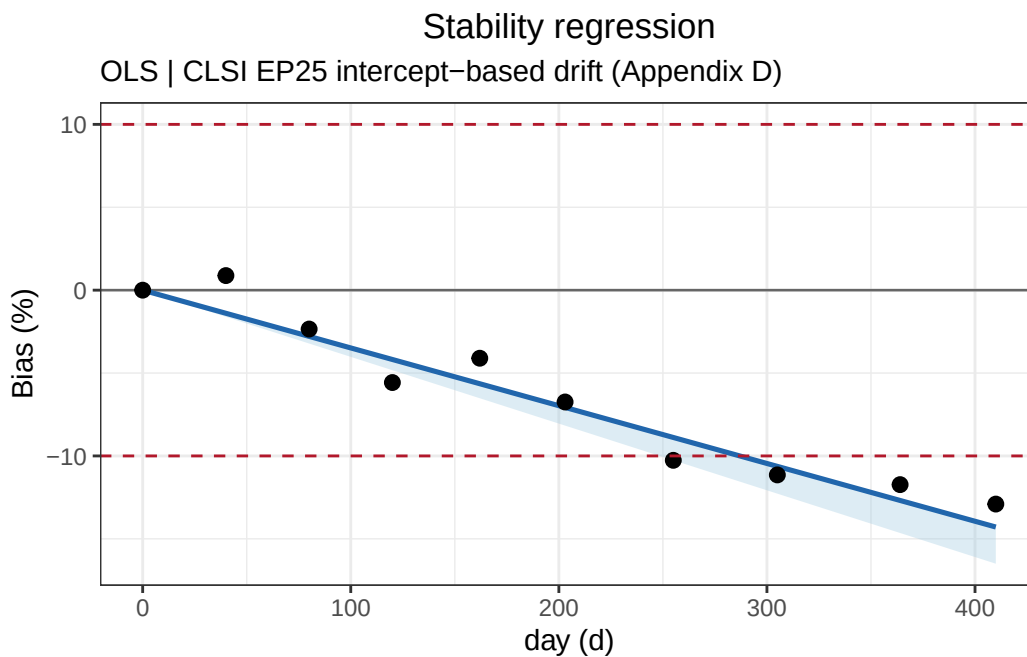
```
print(ep25_c3_2_level_3_result)
```

Stability Regression

Mode: self
 Test condition: condition
 Fit: OLS; intercept = 7.5028, slope = -0.0016
 Direction: decrease
 One-sided CI: 95.0%
 R-squared: 0.9912
 Note: WLS unavailable; OLS used.

Time	ObsBias	Estimate	CI_limit	Status
0.0000	0.0000	-0.0000	-0.0000	
40.0000	-0.9321	-0.8327	-0.8822	
80.0000	-1.5979	-1.6653	-1.7645	
120.0000	-2.6631	-2.4980	-2.6467	
162.0000	-3.1957	-3.3723	-3.5731	
203.0000	-4.5273	-4.2258	-4.4774	
255.0000	-5.8589	-5.3083	-5.6243	
305.0000	-6.7909	-6.3491	-6.7271	
364.0000	-7.3236	-7.5773	-8.0285	
410.0000	-8.3888	-8.5348	-9.0431	

```
plot(ep25_c3_2_level_2_result)
```



Compare the results with the standard:

Table 4.6: EP25 Appendix C3.2: comparison of ivdtools and standard results

appendix	statistic	standard	ivdtools
EP25-A2 Appendix C3.2 / level_1	change_364	-9.00	-9.024133
EP25-A2 Appendix C3.2 / level_1	change_364_upper_CL	-12.19	-12.221806
EP25-A2 Appendix C3.2 / level_1	change_410	-10.14	-10.164545
EP25-A2 Appendix C3.2 / level_1	change_410_upper_CL	-13.73	-13.766320
EP25-A2 Appendix C3.2 / level_2	change_364	-12.68	-12.684724
EP25-A2 Appendix C3.2 / level_2	change_364_upper_CL	-14.66	-14.648829
EP25-A2 Appendix C3.2 / level_2	change_410	-14.28	-14.287738
EP25-A2 Appendix C3.2 / level_2	change_410_upper_CL	-16.52	-16.500055
EP25-A2 Appendix C3.2 / level_3	change_364	-7.58	-7.577276
EP25-A2 Appendix C3.2 / level_3	change_364_upper_CL	-8.05	-8.028464
EP25-A2 Appendix C3.2 / level_3	change_410	-8.54	-8.534844
EP25-A2 Appendix C3.2 / level_3	change_410_upper_CL	-9.07	-9.043050

All three control levels use relative bias, decreasing direction, and a 10% allowable change. The comparison uses node estimates and one-sided confidence limits returned directly by the function.

4.2.7 CLSI EP25-A2 Appendix C4: Transport Conditions

```
ep25_c4_data <- read_csv(
  "../data/004-稳定性研究/std-ep25-appendix-c4.csv",
  show_col_types = FALSE
)
head(ep25_c4_data, 6)
```

```
# A tibble: 6 x 5
  sample condition time replicate result
<chr>   <chr>   <dbl>   <dbl>   <dbl>
1 low_panel unstressed 0         1 7.24
2 low_panel unstressed 0         2 6.98
3 low_panel unstressed 0         3 6.94
4 low_panel unstressed 0         4 6.82
5 low_panel unstressed 0         5 7.09
6 low_panel unstressed 0         6 7.14
```

```
ep25_c4_list <- split_by_factors(
  ep25_c4_data,
  variable = "sample"
)
ep25_c4_list
```

Data split by factors

Variable: sample

Groups: 3

Missing/excluded: 0

low_panel n = 20

low_control n = 20

high_control n = 20

Evaluate the three samples separately with `stability_bias()`:

```
ep25_c4_low_panel_result <- stability_bias(
  data = ep25_c4_list$low_panel,
  condition = "condition",
  time = "time",
  value = "result",
  reference_condition = "unstressed",
  limit = 5,
  bias_type = "relative",
  time_unit = "d"
)

ep25_c4_low_control_result <- stability_bias(
  data = ep25_c4_list$low_control,
  condition = "condition",
  time = "time",
  value = "result",
  reference_condition = "unstressed",
  limit = 5,
  bias_type = "relative",
  time_unit = "d"
)
```

```

ep25_c4_high_control_result <- stability_bias(
  data = ep25_c4_list$high_control,
  condition = "condition",
  time = "time",
  value = "result",
  reference_condition = "unstressed",
  limit = 5,
  bias_type = "relative",
  time_unit = "d"
)

print(ep25_c4_low_panel_result)

```

Stability Bias Analysis

Conditions: 2 (unstressed, stressed)
 Reference: unstressed
 Time unit: d
 Node comparisons: 0
 Acceptance: +/- 5.0000 (relative); 2/2 nodes passed

Within-condition bias (vs first node)

Cond	Time	Value	Bias(%)	Status
unstressed	0.0000	7.0710	0.0000	Pass
stressed	0.0000	7.0700	0.0000	Pass

Between-condition bias (reference: unstressed)

Time	Condition	TestVal	Bias(%)
0.0000	stressed	7.0700	-0.0141

```
print(ep25_c4_low_control_result)
```

Stability Bias Analysis

Conditions: 2 (unstressed, stressed)
 Reference: unstressed
 Time unit: d
 Node comparisons: 0
 Acceptance: +/- 5.0000 (relative); 2/2 nodes passed

Within-condition bias (vs first node)

Cond	Time	Value	Bias(%)	Status
unstressed	0.0000	25.1740	0.0000	Pass
stressed	0.0000	25.1630	0.0000	Pass

Between-condition bias (reference: unstressed)

Time	Condition	TestVal	Bias(%)
0.0000	stressed	25.1630	-0.0437

```
print(ep25_c4_high_control_result)
```

Stability Bias Analysis

Conditions: 2 (unstressed, stressed)
 Reference: unstressed
 Time unit: d
 Node comparisons: 0
 Acceptance: +/- 5.0000 (relative); 2/2 nodes passed

Within-condition bias (vs first node)

Cond	Time	Value	Bias(%)	Status
unstressed	0.0000	99.1610	0.0000	Pass
stressed	0.0000	99.6930	0.0000	Pass

Between-condition bias (reference: unstressed)

Time	Condition	TestVal	Bias(%)
0.0000	stressed	99.6930	0.5365

Compare the results with the standard:

Table 4.7: EP25 Appendix C4: comparison of ivdtools and standard results

appendix	statistic	standard	ivdtools
EP25-A2 Appendix C4 / low_panel	relative_bias	0.0	-0.014142
EP25-A2 Appendix C4 / low_control	relative_bias	0.0	-0.043696
EP25-A2 Appendix C4 / high_control	relative_bias	0.5	0.536501

The transport-condition example uses the reference-condition comparison returned by `stability_bias()`. The standard also reports confidence intervals for replicate measurements, but this function's `between_bias` output does not include that item; therefore the comparison table lists only the relative bias returned directly by the function.

4.2.8 CLSI EP25-A2 Appendix C5: Outlier Evaluation

Read and inspect the standard data:

```

ep25_c5_data <- read_csv(
  "./data/004-稳定性研究/std-ep25-appendix-c5.csv",
  show_col_types = FALSE
)
head(ep25_c5_data, 6)

# A tibble: 6 x 2
  Day Result
<dbl> <dbl>
1     0  12.1
2    31  12.0
3    61  15.0
4    96  11.9
5   120  11.8
6   150  11.8

ep25_c5_without_day_61 <- ep25_c5_data[ep25_c5_data$Day != 61, ]
head(ep25_c5_without_day_61, 6)

# A tibble: 6 x 2
  Day Result
<dbl> <dbl>

```

```

1    0   12.1
2   31   12.0
3   96   11.9
4  120   11.8
5  150   11.8
6  186   11.8

```

Use `stability_regression()` to evaluate the results with and without the outlier:

```

ep25_c5_with_result <- stability_regression(
  data = ep25_c5_data,
  time = "Day",
  value = "Result",
  mode = "self",
  bias_type = "relative",
  direction = "decrease",
  limit = 10,
  time_unit = "d",
  conf.level = 0.95
)

ep25_c5_without_result <- stability_regression(
  data = ep25_c5_without_day_61,
  time = "Day",
  value = "Result",
  mode = "self",
  bias_type = "relative",
  direction = "decrease",
  limit = 10,
  time_unit = "d",
  conf.level = 0.95
)

print(ep25_c5_with_result)

```

Stability Regression

```

Mode:          self
Test condition: condition
Fit:           OLS; intercept = 12.6703, slope = -0.0035
Direction:     decrease
One-sided CI:  95.0%
R-squared:     0.2118
Cook distance: row(s) 3 > 1; not removed
Note:          WLS unavailable; OLS used.

```

Time	ObsBias	Estimate	CI_limit	Status
0.0000	0.0000	-0.0000	-0.0000	
31.0000	-0.7432	-0.8643	-1.7218	
61.0000	23.7820	-1.7008	-3.3880	
96.0000	-2.0644	-2.6766	-5.3320	
120.0000	-2.2296	-3.3457	-6.6650	
150.0000	-2.8076	-4.1822	-8.3313	
186.0000	-2.3947	-5.1859	-10.3308	*
211.0000	-2.6424	-5.8829	-11.7193	*
233.0000	-4.0462	-6.4963	-12.9412	*
270.0000	-3.2205	-7.5279	-14.9963	*

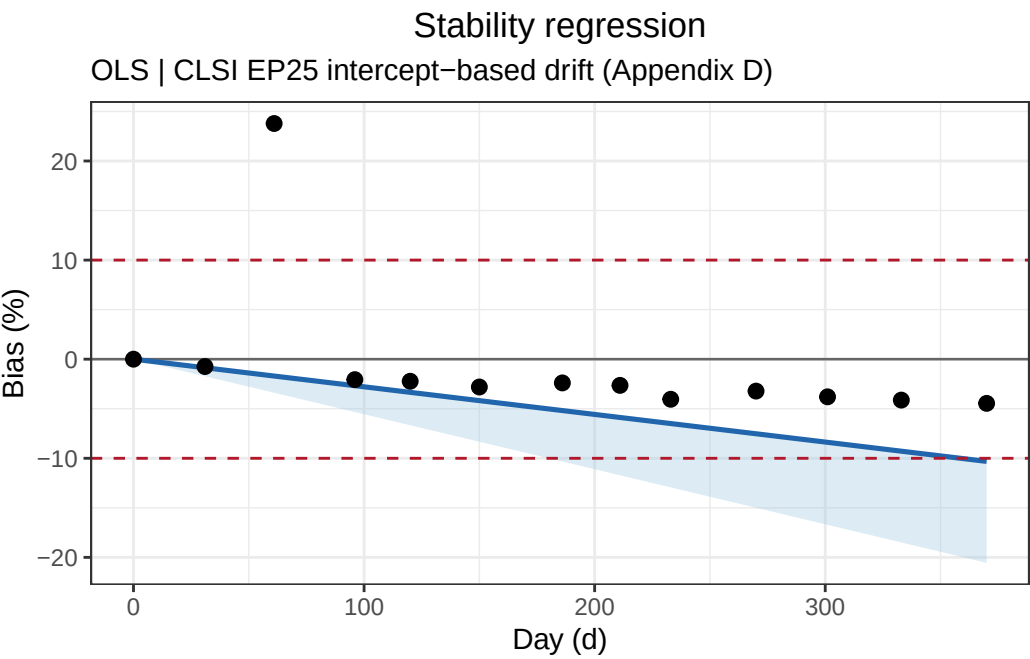
301.0000	-3.7985	-8.3922	-16.7181	*
333.0000	-4.1288	-9.2844	-18.4954	*
370.0000	-4.4591	-10.3160	-20.5504	*

```
print(ep25_c5_without_result)
```

Stability Regression
Mode: self
Test condition: condition
Fit: OLS; intercept = 12.0369, slope = -0.0013
Direction: decrease
One-sided CI: 95.0%
R-squared: 0.8969
Note: WLS unavailable; OLS used.

Time	ObsBias	Estimate	CI_limit	Status
0.0000	0.0000	-0.0000	-0.0000	
31.0000	-0.7432	-0.3428	-0.4080	
96.0000	-2.0644	-1.0615	-1.2634	
120.0000	-2.2296	-1.3269	-1.5793	
150.0000	-2.8076	-1.6586	-1.9741	
186.0000	-2.3947	-2.0567	-2.4479	
211.0000	-2.6424	-2.3331	-2.7769	
233.0000	-4.0462	-2.5764	-3.0665	
270.0000	-3.2205	-2.9855	-3.5534	
301.0000	-3.7985	-3.3283	-3.9614	
333.0000	-4.1288	-3.6821	-4.3826	
370.0000	-4.4591	-4.0912	-4.8695	

```
plot(ep25_c5_with_result)
```



Cook’ s distance:

stability_regression() calculates Cook’ s distance for each time point in the returned object and records observations with Cook’ s distance greater than 1 in diagnostics\$influential:

```

ep25_c5_cook_distance <- data.frame(
  Day = ep25_c5_data$Day,
  Cook_distance = unname(ep25_c5_with_result$diagnostics$cook_distance)
)
ep25_c5_cook_distance

```

```

Day Cook_distance
1    0 1.172637e-01
2   31 7.185586e-02
3   61 1.078556e+00
4   96 2.477807e-02
5  120 1.448582e-02
6  150 9.639484e-03
7  186 2.404885e-03
8  211 1.263118e-03
9  233 4.142662e-03
10 270 1.404264e-06
11 301 3.051755e-04
12 333 3.335777e-03
13 370 1.752518e-02

```

```

ep25_c5_influential <- ep25_c5_cook_distance[
  ep25_c5_cook_distance$Cook_distance > 1,
]
ep25_c5_influential

```

```

Day Cook_distance
3   61      1.078556

```

Compare the results with the standard:

Table 4.8: EP25 Appendix C5: comparison of ivdtools and standard results

appendix	statistic	standard	ivdtools
EP25-A2 Appendix C5 / with_outlier	cook_distance_day_61	1.08	1.078556
EP25-A2 Appendix C5 / with_outlier	change_333	-9.28	-9.284424
EP25-A2 Appendix C5 / with_outlier	change_333_upper_CL	-18.50	-18.495388
EP25-A2 Appendix C5 / with_outlier	change_370	-10.32	-10.316027
EP25-A2 Appendix C5 / with_outlier	change_370_upper_CL	-20.55	-20.550431
EP25-A2 Appendix C5 / without_day_61	change_333	-3.68	-3.682110
EP25-A2 Appendix C5 / without_day_61	change_333_upper_CL	-4.38	-4.382553
EP25-A2 Appendix C5 / without_day_61	change_370	-4.09	-4.091233
EP25-A2 Appendix C5 / without_day_61	change_370_upper_CL	-4.87	-4.869504

Chapter 5

Measurement Uncertainty Analysis

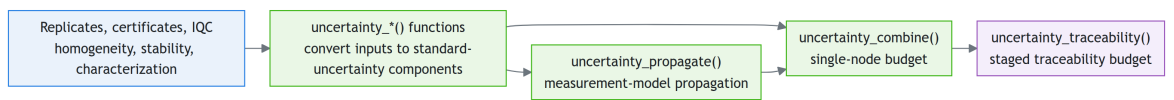
This chapter demonstrates how to define measurement-uncertainty components, propagate a measurement model, combine an uncertainty budget, and aggregate results along a metrological traceability chain. It covers Type A and Type B evaluations, long-term IQC, between-bottle homogeneity, stability, multi-laboratory characterization, and single-node and staged budgets.

! Scope of Use

The ordinary data and standard-case data in this chapter are provided only to verify the analysis workflow. The `uncertainty_*` functions calculate standard, combined, and expanded uncertainty; they do not replace measurand definition, experimental design, clinical acceptance criteria, regulatory assessment, or product-release decisions. Formal reports must state the measurand, unit, whether the result is a single value or a mean, the source/distribution/degrees of freedom of each component, the coverage factor, and the correlation assumptions.

The numerical values in the standard cases follow the cited standard text. Each case compares `ivdtools` results with the corresponding standard result. Differences caused by intermediate rounding, statistical definitions, or approximation methods are explained separately after the comparison table.

Measurement-uncertainty analysis first defines the output quantity and measurement model, then converts information from different sources into standard-uncertainty components:



```
library(ivdtools)
library(readr)
```

5.1 Main Functions

Function	Main purpose
<code>uncertainty_type_a()</code>	Estimate a Type A component from replicate measurements

Function	Main purpose
<code>uncertainty_type_b()</code>	Estimate a Type B component from certificates, limits, or distributions
<code>uncertainty_iqc()</code>	Extract within-run, between-run, and combined components from long-term IQC
<code>uncertainty_homogeneity()</code>	Evaluate between-bottle or between-unit homogeneity
<code>uncertainty_stability()</code>	Estimate a shelf-life stability component from a time trend
<code>uncertainty_characterization()</code>	Estimate an assigned-value component from multi-laboratory results
<code>uncertainty_combine()</code>	Combine an uncertainty budget for one result node
<code>uncertainty_propagate()</code>	Propagate uncertainty through division, multiplication, or another model
<code>uncertainty_traceability()</code>	Accumulate newly introduced components along traceability stages

5.1.1 Type A Components from Replicate Measurements: `uncertainty_type_a()`

`quantity = "single"` uses the sample standard deviation of replicate results to describe a future single result; `quantity = "mean"` uses the standard error to describe the mean of the replicate set. These correspond to different reported quantities.

```
repeatability <- uncertainty_type_a(
  c(99.5, 100.2, 100.0, 99.8, 100.1),
  name = "重复测量",
  quantity = "mean",
  relative = TRUE
)

repeatability
```

Uncertainty component

```
Name      : 重复测量
Estimate   : 99.92
Standard uncertainty : 0.0012419609
Relative uncertainty : 0.0012419609
Degrees of freedom : 4
Evaluation type    : A
Distribution       : normal
Source            : replicate measurements: mean
Relative          : TRUE
```

The returned object includes the estimate, standard uncertainty, relative standard uncertainty, degrees of freedom, evaluation type, distribution, and source. Non-finite values are removed, but formal analyses must retain the raw data and exclusion record.

5.1.2 Type B Components from Certificates or Specifications: `uncertainty_type_b()`

When a certificate gives expanded uncertainty directly, provide the corresponding certificate coverage factor k . When only limits are available, select the distribution using the certificate, specification, or scientific rationale.


```
certificate <- uncertainty_type_b(
  name = "校准品证书",
  estimate = 100,
  uncertainty = 2,
  distribution = "normal",
  k = 2,
  relative = TRUE
)

pipette <- uncertainty_type_b(
  name = "移液器容量",
  estimate = 10,
  lower = 9.9,
  upper = 10.1,
  distribution = "rectangular"
)

certificate
```

Uncertainty component

```
Name      : 校准品证书
Estimate   : 100
Standard uncertainty : 0.01
Relative uncertainty : 0.01
Degrees of freedom : Inf
Evaluation type  : B
Distribution     : normal
Source          : expanded uncertainty
Relative        : TRUE
```

```
pipette
```

Uncertainty component

```
Name      : 移液器容量
Estimate   : 10
Standard uncertainty : 0.057735027
Relative uncertainty : 0.0057735027
Degrees of freedom : Inf
Evaluation type  : B
Distribution     : rectangular
Source          : interval limits
Relative        : FALSE
```

For interval distributions, the standard uncertainty depends on the half-width and the assumed distribution. Normal or Student-*t* intervals also require an explicit coverage probability and degrees of freedom.

5.1.3 Long-Term IQC Components: `uncertainty_iqc()`

When runs and within-run replicates are available, the function uses one-way random-effects ANOVA to separate within-run and between-run variation. If there is no run column, or each run has only one result, it uses the dispersion of the long-term results directly.

```
iqc_data <- data.frame(
  run = rep(1:5, each = 2),
```

```

    result = c(99, 101, 100, 102, 98, 100, 101, 103, 99, 100)
  )

  iqc_component <- uncertainty_iqc(
    iqc_data,
    result = "result",
    run = "run",
    quantity = "single",
    name = "长期 IQC",
    relative = TRUE
  )

  iqc_component

```

IQC uncertainty

```

Method           : one-way ANOVA
Observations      : 10
Runs              : 5
Within-run SD     : 1.3038405
Between-run SD    : 0.77459667
Single-result uncertainty : 1.5165751
Mean uncertainty  : 0.53851648

```

The IQC level should represent the concentration of the reported result. Whether out-of-control periods are excluded must be determined by a prespecified quality procedure. If IQC already contains a calibration or between-lot source, do not add the same source again as an independent component.

5.1.4 Between-Bottle Homogeneity: `uncertainty_homogeneity()`

The data should be in long format, with multiple replicate measurements per packaging unit in the usual design. The function returns mean squares, repeatability SD, the observed between-unit SD, implied heterogeneity, and the final u_{bb} .

```

homogeneity_data <- data.frame(
  bottle = rep(paste0("B", 1:4), each = 2),
  result = c(99, 100, 101, 100, 98, 99, 102, 101)
)

homogeneity_component <- uncertainty_homogeneity(
  homogeneity_data,
  unit = "bottle",
  result = "result",
  name = "瓶间均匀性",
  relative = TRUE
)

homogeneity_component

```

Between-unit homogeneity uncertainty

```

Units           : 4
Results          : 8
Mean             : 100
Effective degrees of freedom : 2
Average replicates : 2
Within-unit mean square : 0.5

```

```

Between-unit mean square      : 3.3333333
Within-unit degrees of freedom : 4
Between-unit degrees of freedom : 3
Repeatability SD             : 0.70710678
Between-unit SD              : 1.1902381
Hidden inhomogeneity         : 0.42044821
Between-unit uncertainty     : 1.1902381
Relative repeatability       : 0.0070710678
Relative between-unit       : 0.011902381
Relative hidden inhomogeneity : 0.0042044821
Relative between-unit uncertainty : 0.011902381

```

The homogeneity result also depends on sampling, measurement order, and minimum sample size. The function does not automatically prove that the sampling plan represents the entire lot.

5.1.5 Stability-Trend Components: `uncertainty_stability()`

The function fits

$$result = intercept + slope \times time$$

The trend uncertainty over the target shelf life is the slope standard error multiplied by shelf life. `shelf_life` and `time` must use the same time unit.

```

stability_data <- data.frame(
  time = rep(0:4, each = 2),
  result = c(100.1, 99.9, 100.0, 99.8, 99.7,
             99.9, 99.6, 99.8, 99.5, 99.7)
)

stability_component <- uncertainty_stability(
  stability_data,
  time = "time",
  result = "result",
  shelf_life = 6,
  name = "6个月长期稳定性",
  relative = TRUE
)

stability_component

```

```

Stability uncertainty
Condition           : all
Observations        : 10
Minimum time        : 0
Maximum time        : 4
Shelf life          : 6
Intercept           : 100
Slope               : -0.1
Mean                : 99.8
Slope SE            : 0.025
Residual SD         : 0.1118034
Degrees of freedom   : 8
Long-term uncertainty : 0.15
Relative long-term uncertainty : 0.001503006

```

Extrapolation far beyond the observed range requires separate justification. Multiple storage conditions often represent different scenarios; do not add components from all conditions together without deciding how they relate to the reported result.

5.1.6 Multi-Laboratory Characterization: `uncertainty_characterization()`

The function first calculates the mean for each laboratory and then combines laboratory means with equal weight. A laboratory with more replicates does not automatically receive more weight.

```
characterization_data <- data.frame(
  laboratory = rep(paste0("Lab", 1:4), each = 3),
  result = c(
    99, 100, 101,
    100, 101, 100,
    98, 99, 100,
    101, 102, 101
  )
)

characterization_component <- uncertainty_characterization(
  characterization_data,
  laboratory = "laboratory",
  result = "result",
  name = "多实验室定值",
  relative = TRUE
)

characterization_component
```

```
Characterization uncertainty
laboratories      : 4
assigned_value    : 100.16667
laboratory_sd     : 0.96225045
standard_uncertainty : 0.48112522
```

5.1.7 Single-Node Combination: `uncertainty_combine()`

`uncertainty_combine()` is appropriate when components are already expressed on the same output scale. `sensitivity` supplies the local effect of an input change on the output; `correlation` supplies the correlation matrix of the inputs. The default is an identity matrix, which assumes independent components.

```
budget <- uncertainty_combine(
  repeatability,
  certificate,
  homogeneity_component$component,
  value = 100,
  coverage = 0.95,
  k = 2
)

budget
```

```
Combined uncertainty budget
```

Scale: absolute
 Combined standard uncertainty: 1.55952
 Expanded uncertainty: 3.11903 ($k = 2$, 95.0%)

Component: 重复测量

Estimate : 99.92
 Standard uncertainty : 0.0012419609
 Relative uncertainty : 0.0012419609
 Degrees of freedom : 4
 Evaluation type : A
 Distribution : normal
 Source : replicate measurements: mean
 Relative : TRUE
 Sensitivity : 1
 Uncertainty on output scale : 0.12419609
 Signed variance contribution : 0.01542467
 Contribution percent : 0.63421424

Component: 校准品证书

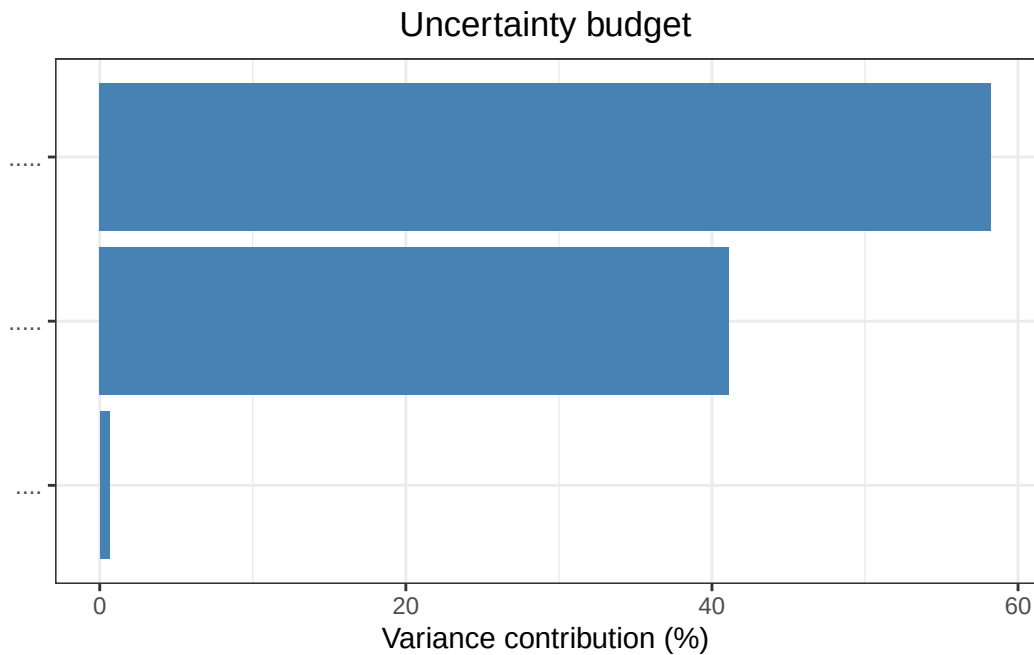
Estimate : 100
 Standard uncertainty : 0.01
 Relative uncertainty : 0.01
 Degrees of freedom : Inf
 Evaluation type : B
 Distribution : normal
 Source : expanded uncertainty
 Relative : TRUE
 Sensitivity : 1
 Uncertainty on output scale : 1
 Signed variance contribution : 1
 Contribution percent : 41.116877

Component: 瓶间均匀性

Estimate : 100
 Standard uncertainty : 0.011902381
 Relative uncertainty : 0.011902381
 Degrees of freedom : 3
 Evaluation type : A
 Distribution : normal
 Source : EP30 homogeneity ANOVA
 Relative : TRUE
 Sensitivity : 1
 Uncertainty on output scale : 1.1902381
 Signed variance contribution : 1.4166667
 Contribution percent : 58.248909

The budget object also stores effective degrees of freedom, coverage factor, covariance matrix, and each component's contribution on the output scale.

`plot(budget)`



5.1.8 Nonlinear Measurement Models: `uncertainty_propagate()`

For ratios, products, powers, logarithms, or other nonlinear measurement models, `uncertainty_propagate()` calculates sensitivities and propagates input uncertainty automatically. Component names must be unique and must match the formal arguments of the measurement-model function.

```
mass <- uncertainty_type_b(
  "mass", estimate = 20, uncertainty = 0.2,
  distribution = "standard"
)
volume <- uncertainty_type_b(
  "volume", estimate = 4, uncertainty = 0.1,
  distribution = "standard"
)

concentration_delta <- uncertainty_propagate(
  model = function(mass, volume) mass / volume,
  components = list(mass, volume),
  method = "delta"
)
concentration_delta
```

Uncertainty propagation

Method: delta

Model estimate: 5

Standard uncertainty: 0.134629

95.0% interval: [4.73613, 5.26387]

```
concentration_mc <- uncertainty_propagate(
  model = function(mass, volume) mass / volume,
  components = list(mass, volume),
  method = "monte_carlo",
  simulations = 10000,
  seed = 20260901
```

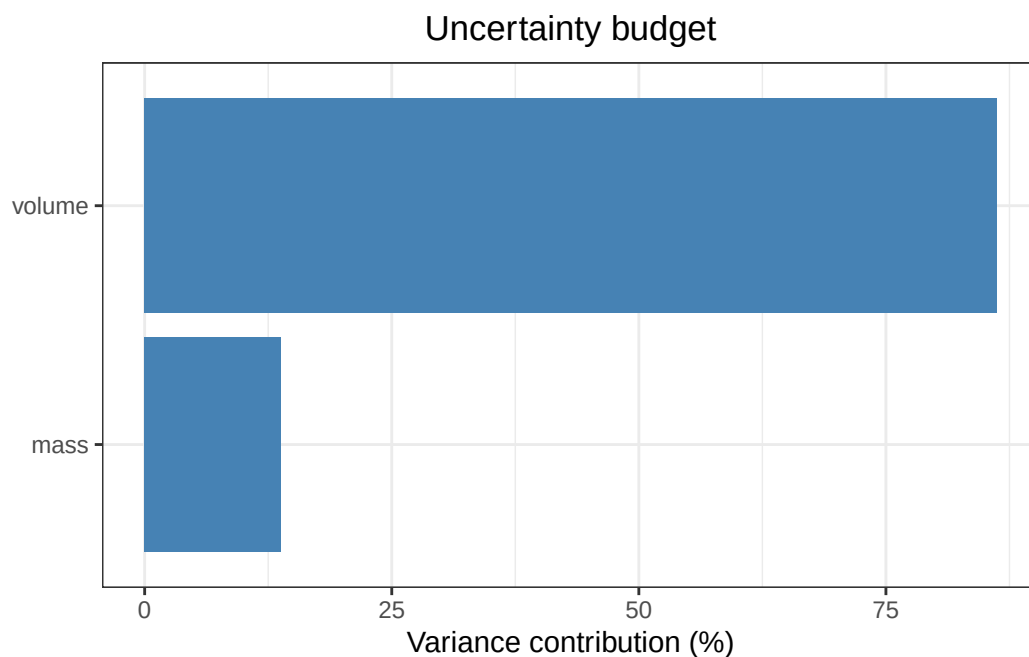
```
)  
concentration_mc
```

Uncertainty propagation

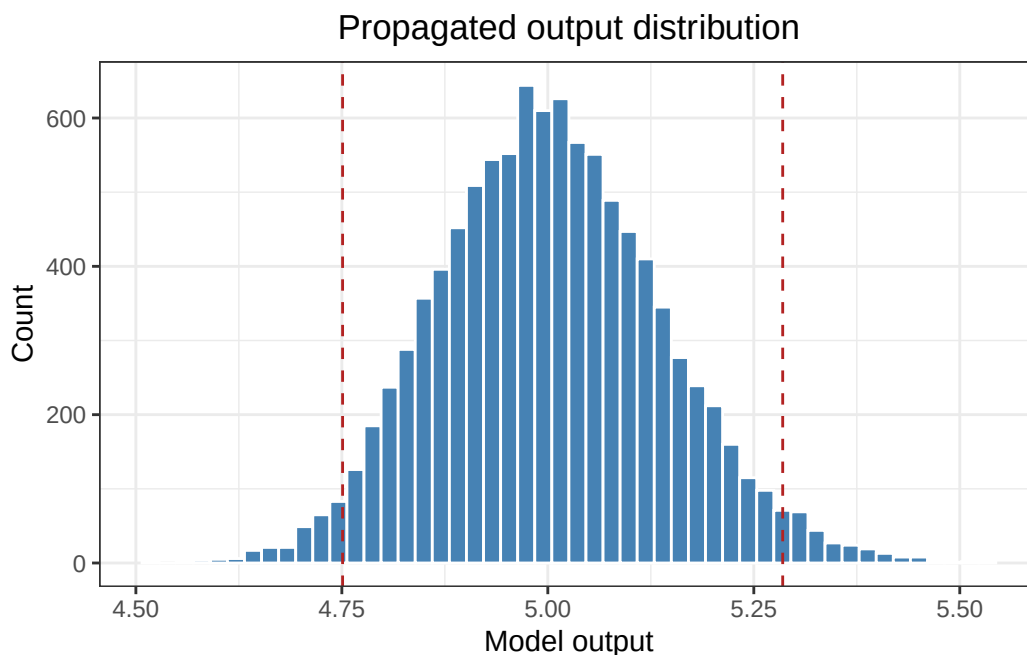
```
Method: monte_carlo  
Model estimate: 5  
Standard uncertainty: 0.13615  
95.0% interval: [4.75089, 5.28508]
```

The delta method is suitable when input uncertainties are small and the model is approximately linear near the estimates. Monte Carlo is useful when the output distribution is markedly skewed or an asymmetric coverage interval is needed. Formal reports should retain the simulation count, random seed, distributions, and correlation structure.

```
plot(concentration_delta, type = "contribution")
```



```
plot(concentration_mc, type = "distribution")
```



5.1.9 Accumulating a Metrological Traceability Chain: `uncertainty_traceability()`

`uncertainty_traceability()` records newly introduced components by stage and returns a cumulative budget for each stage. Stage names should match the calibration-chain nodes used in the quality system.

```
trace <- uncertainty_traceability(
  stages = list(
    reference = list(certificate),
    assignment = list(repeatability),
    release = list(homogeneity_component$component)
  ),
  value = 100,
  coverage = 0.95,
  k = 2
)

trace
```

Traceability-chain uncertainty

Stage: reference

Introduced components	: 1
Cumulative components	: 1
Cumulative standard uncertainty	: 1
Relative standard uncertainty	: 0.01
Coverage factor	: 2
Expanded uncertainty	: 2

Stage: assignment

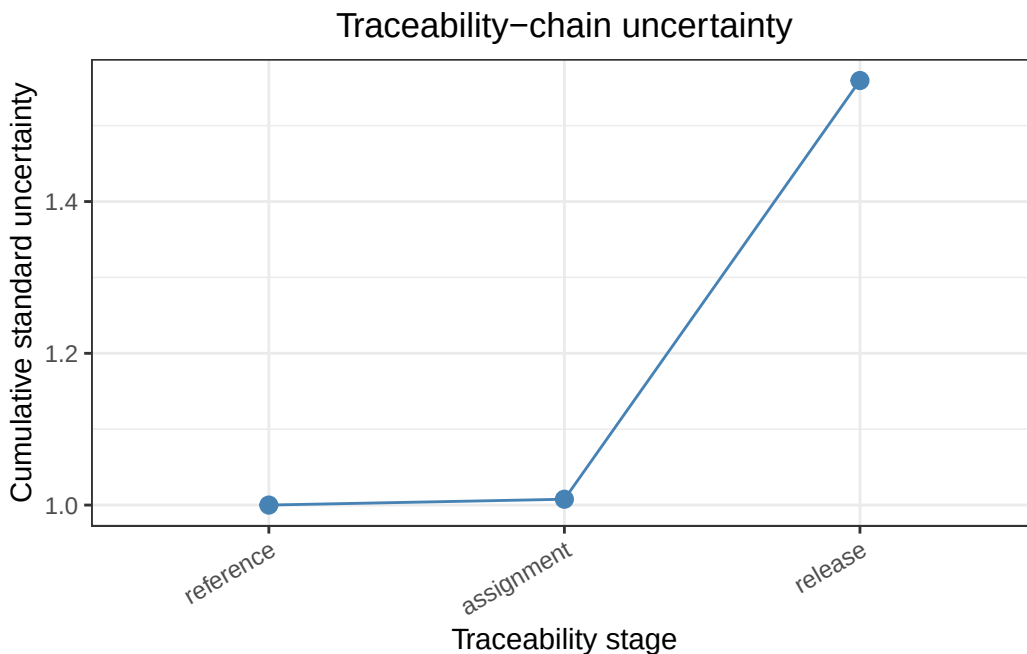
Introduced components	: 1
Cumulative components	: 2
Cumulative standard uncertainty	: 1.0076828
Relative standard uncertainty	: 0.010076828
Coverage factor	: 2
Expanded uncertainty	: 2.0153656

Stage: release

```
Introduced components      : 1
Cumulative components      : 3
Cumulative standard uncertainty : 1.5595164
Relative standard uncertainty : 0.015595164
Coverage factor            : 2
Expanded uncertainty       : 3.1190328
```

`uncertainty_traceability()` currently uses unit sensitivities and independent components at each stage. If a stage contains a complex measurement model or correlated inputs, first model it correctly with `uncertainty_propagate()` or `uncertainty_combine()`, then decide how to pass the node result to the next stage.

```
plot(trace)
```



5.2 Analysis Cases

5.2.1 CLSI EP29-A Appendix A: Converting Type B Limits

CLSI EP29-A Appendix A compares the conversion of the 90–180 limit interval under rectangular, triangular, and normal distributions, and gives weighing-specification, balance-linearity, and repeatability cases. The following recomputes the examples with `uncertainty_type_b()` and `uncertainty_combine()`; the normal case interprets the standard's “ ± 3 standard deviations” as coverage = 0.9973.

```
ep29_a_rectangular <- uncertainty_type_b(
  "EP29 rectangular limits", estimate = 135,
  lower = 90, upper = 180, distribution = "rectangular"
)

ep29_a_triangular <- uncertainty_type_b(
  "EP29 triangular limits", estimate = 135,
  lower = 90, upper = 180, distribution = "triangular"
)
```

```

ep29_a_normal <- uncertainty_type_b(
  "EP29 normal limits", estimate = 135,
  lower = 90, upper = 180, distribution = "normal",
  coverage = 0.9973
)

ep29_a_weighing <- uncertainty_type_b(
  "EP29 weighing specification", estimate = 156,
  lower = 155, upper = 157, distribution = "rectangular"
)

ep29_a_linearity <- lapply(
  1:2,
  function(i) uncertainty_type_b(
    paste0("linearity ", i), estimate = 0,
    uncertainty = 0.02 / sqrt(6), distribution = "standard"
  )
)

ep29_a_repeatability <- lapply(
  1:2,
  function(i) uncertainty_type_b(
    paste0("repeatability ", i), estimate = 0,
    uncertainty = 0.01 / 3, distribution = "standard"
  )
)

ep29_a_balance <- uncertainty_combine(
  ep29_a_linearity[[1]], ep29_a_linearity[[2]],
  ep29_a_repeatability[[1]], ep29_a_repeatability[[2]],
  k = 1
)

ep29_a_tight <- uncertainty_combine(
  uncertainty_type_b(
    "EP29 tight specification", estimate = 156,
    lower = 155.9, upper = 156.1, distribution = "rectangular"
  ),
  uncertainty_type_b(
    "EP29 rounded balance", estimate = 0,
    uncertainty = 0.013, distribution = "standard"
  ),
  k = 1
)

```

Read the standard values from the original Appendix A result-check table and merge the item-by-item comparison:

Table 5.1: EP29-A Appendix A: comparison of ivdtools and standard results

appendix	statistic	standard	ivdtools
A	rectangular_90_180	26.000	25.9807621
A	triangular_90_180	18.400	18.3711731
A	gaussian_90_180	15.000	15.0001150
A	weighing_specification_u	0.577	0.5773503
A	balance_u	0.013	0.0124722
A	tight_combined_u	0.059	0.0591805

`balance_u` uses unrounded linearity and repeatability components. The tight-limit case uses the rounded 0.013 specified in the standard before combining, so the last digits differ because of intermediate rounding.

5.2.2 CLSI EP29-A Appendix B: Cholesterol IQC

Appendix B contains 42 runs with three results per run. The standard identifies run 4 as technically questionable and excludes it, retaining 41 runs and 123 results. The following repeats the same exclusion rule with `uncertainty_iqc()`.

```
ep29_b_data <- read_csv(
  "./data/005-不确定度分析/std-ep29-appendix-b.csv",
  show_col_types = FALSE
)
ep29_b_valid <- subset(ep29_b_data, run != 4L)
head(ep29_b_data, 6)
```

```
# A tibble: 6 x 5
  run reagent_lot technician replicate cholesterol
  <dbl> <chr>          <chr>          <dbl>          <dbl>
1     1 A          AK              1             7.9
2     1 A          AK              2             7.9
3     1 A          AK              3             8.1
4     2 A          AK              1             7.5
5     2 A          AK              2             7.3
6     2 A          AK              3             7.6
```

```
ep29_b_iqc <- uncertainty_iqc(
  ep29_b_valid,
  result = "cholesterol",
  run = "run",
  quantity = "single",
  name = "EP29 cholesterol IQC"
)
```

Extract the ANOVA statistics, combined standard uncertainty, and expanded uncertainty returned by `uncertainty_iqc()`, and compare them with the original EP29-A Appendix B results:

Table 5.2: EP29-A Appendix B: comparison of `ivdtools` and standard results

appendix	statistic	standard	ivdtools
B	lot_A_rep1_mean	7.60000	7.6500000
B	lot_A_rep1_sd	0.33000	0.2438723
B	lot_B_rep1_mean	7.71000	7.7142857
B	lot_B_rep1_sd	0.20000	0.2031889
B	all_rep1_mean	7.66000	7.6571429
B	all_rep1_sd	0.28000	0.2768665
B	valid_rep1_mean	7.68000	7.6829268
B	valid_rep1_sd	0.22000	0.2234977
B	between_ss	6.09000	6.0882927
B	within_ss	0.58000	0.5800000
B	between_ms	0.15200	0.1522073
B	within_ms	0.00707	0.0070732
B	F	21.50000	21.5189655
B	p_value	0.00000	0.0000000
B	within_sd	0.08400	0.0841021
B	between_variance	0.04830	0.0483780

appendix	statistic	standard	ivdtools
B	between_sd	0.22000	0.2199501
B	combined_u	0.23500	0.2354808
B	expanded_U	0.47000	0.4709617
B	grand_mean	7.69000	7.6902439
B	relative_expanded_percent	6.10000	6.1241445

5.2.3 CLSI EP29-A Examples 1–3: Measurement-Model Propagation

EP29-A Examples 1–3 show additive/subtractive, multiplicative, and combined multiplication/division models. They are suitable for checking sensitivity coefficients, correlation, and relative uncertainty handling with `uncertainty_combine()` and `uncertainty_propagate()`.

```

ep29_ex1_v1 <- uncertainty_type_b(
  "V1", estimate = 100, uncertainty = 0.1, distribution = "standard"
)

ep29_ex1_v2 <- uncertainty_type_b(
  "V2", estimate = 90, uncertainty = 0.2, distribution = "standard"
)

ep29_ex1_independent <- uncertainty_combine(
  ep29_ex1_v1, ep29_ex1_v2, value = 190, k = 1
)

ep29_ex1_equal <- uncertainty_combine(
  uncertainty_type_b("V1 equal", estimate = 100, uncertainty = 0.1,
    distribution = "standard"),
  uncertainty_type_b("V2 equal", estimate = 100, uncertainty = 0.1,
    distribution = "standard"),
  value = 200, k = 1
)

ep29_ex1_correlated <- uncertainty_combine(
  uncertainty_type_b("V1 same pipette", estimate = 100, uncertainty = 0.1,
    distribution = "standard"),
  uncertainty_type_b("V2 same pipette", estimate = 100, uncertainty = 0.1,
    distribution = "standard"),
  value = 200,
  correlation = matrix(c(1, 1, 1, 1), nrow = 2),
  k = 1
)

ep29_ex2 <- uncertainty_propagate(
  model = function(protein, volume) protein * volume,
  components = list(
    uncertainty_type_b("protein", estimate = 150, uncertainty = 3,
      distribution = "standard"),
    uncertainty_type_b("volume", estimate = 1.5, uncertainty = 0.015,
      distribution = "standard")
  )
)

ep29_ex3_staged <- uncertainty_propagate(
  model = function(numerator, denominator) numerator / denominator,
  components = list(

```

```

    uncertainty_type_b("numerator", estimate = 330, uncertainty = 11.1,
                      distribution = "standard"),
    uncertainty_type_b("denominator", estimate = 100, uncertainty = 3,
                      distribution = "standard")
  )
)

ep29_ex3_full <- uncertainty_propagate(
  model = function(c1, v1, c2, v2) (c1 * v1 + c2 * v2) / (v1 + v2),
  components = list(
    uncertainty_type_b("c1", estimate = 15, uncertainty = 0.2,
                      distribution = "standard"),
    uncertainty_type_b("v1", estimate = 10, uncertainty = 0.11,
                      distribution = "standard"),
    uncertainty_type_b("c2", estimate = 2, uncertainty = 0.1,
                      distribution = "standard"),
    uncertainty_type_b("v2", estimate = 90, uncertainty = 3,
                      distribution = "standard")
  )
)

```

Table 5.3: EP29-A Examples 1–3: comparison of ivdtools and standard results

appendix	statistic	standard	ivdtools
EP29 Example 1	independent_unequal_u	0.2200	0.2236068
EP29 Example 1	same_pipette_equal_u	0.2000	0.2000000
EP29 Example 1	independent_equal_u	0.1400	0.1414214
EP29 Example 2	relative_u	0.0224	0.0223607
EP29 Example 2	absolute_u	5.0300	5.0311529
EP29 Example 3	independent_staged_u	0.1490	0.1487347
EP29 Example 3	correlated_u	0.1000	0.1009289

When Example 3 propagates the original mixed model directly, the volume quantity appearing in both numerator and denominator naturally generates covariance and produces approximately 0.10 mmol/L. The standard treats the numerator and denominator as independent nodes first and reports approximately 0.149 mmol/L. The dependency structures differ; this is not a display-rounding difference.

5.2.4 CLSI EP29-A Example 4: One-Way ANOVA and IQC Uncertainty

Example 4 provides plasma-creatinine results from 5 runs with 5 replicates per run. Its input structure matches `uncertainty_iqc()` and is suitable for checking within-run and between-run variance and the combined uncertainty for a single result.

```

ep29_ex4_data <- data.frame(
  run = rep(1:5, each = 5),
  replicate = rep(1:5, times = 5),
  creatinine = c(
    140, 140, 140, 141, 140,
    138, 139, 138, 137, 139,
    143, 144, 144, 145, 143,
    143, 143, 142, 143, 142,
    142, 143, 141, 142, 143
  )
)

```

```

ep29_ex4_iqc <- uncertainty_iqc(
  ep29_ex4_data,
  result = "creatinine",
  run = "run",
  quantity = "single",
  name = "EP29 Example 4 P-Creatinine"
)

```

Table 5.4: EP29-A Example 4: comparison of ivdtools and standard results

appendix	statistic	standard	ivdtools
EP29 Example 4	ms_between	24.40	24.4000000
EP29 Example 4	ms_within	0.52	0.5200000
EP29 Example 4	within_sd	0.72	0.7211103
EP29 Example 4	between_sd	2.19	2.1854061
EP29 Example 4	combined_u	2.30	2.3013040
EP29 Example 4	F	46.90	46.9230769

The last digits of the F value differ because the standard displays rounded mean squares, whereas the function uses unrounded mean squares.

5.2.5 CLSI EP29-A Examples 5–6: Bias Estimation and Bias-Correction Uncertainty

Example 5 provides only summary statistics and no raw difference data. Store the summary standard uncertainty given by the standard with `uncertainty_type_b()`, then combine it with `uncertainty_combine()`. Example 6a uses the assigned-value uncertainty of a certified reference material and the uncertainty of the laboratory mean.

```

ep29_ex5_make <- function(n) {
  uncertainty_combine(
    uncertainty_type_b(
      "test method imprecision", estimate = 0,
      uncertainty = 0.71, distribution = "standard"
    ),
    uncertainty_type_b(
      paste0("bias estimate n=", n), estimate = 0,
      uncertainty = 1.2 / sqrt(n), distribution = "standard"
    ),
    k = 1
  )
}
ep29_ex5_n144 <- ep29_ex5_make(144)
ep29_ex5_n16 <- ep29_ex5_make(16)

ep29_ex6a_lab_mean <- uncertainty_type_b(
  "CRM laboratory mean", estimate = 6.04,
  uncertainty = 0.045, distribution = "standard"
)
ep29_ex6a_crm <- uncertainty_type_b(
  "CRM certified value", estimate = 6.3,
  uncertainty = 0.1, k = 2, distribution = "standard"
)
ep29_ex6a_bias <- uncertainty_combine(

```

```

    ep29_ex6a_lab_mean, ep29_ex6a_crm, value = 6.04 - 6.3, k = 1
  )

ep29_ex6b_claim <- uncertainty_combine(
  uncertainty_type_b(
    "future result, literature claim", estimate = 6.04,
    uncertainty = 0.035 * 6.04, distribution = "standard"
  ),
  uncertainty_type_b(
    "bias correction", estimate = 0, uncertainty = 0.068,
    distribution = "standard"
  ),
  k = 1
)

ep29_ex6b_lab <- uncertainty_combine(
  uncertainty_type_b(
    "future result, local ANOVA", estimate = 6.04,
    uncertainty = 0.151, distribution = "standard"
  ),
  uncertainty_type_b(
    "bias correction", estimate = 0, uncertainty = 0.068,
    distribution = "standard"
  ),
  k = 1
)

```

Table 5.5: EP29-A Examples 5–6: comparison of ivdtools and standard results

appendix	statistic	standard	ivdtools
EP29 Example 5	n144_bias_u	0.100	0.1000000
EP29 Example 5	n144_combined_u	0.720	0.7170077
EP29 Example 5	n16_bias_u	0.300	0.3000000
EP29 Example 5	n16_combined_u	0.770	0.7707788
EP29 Example 6a	measured_mean	6.040	6.0400000
EP29 Example 6a	bias	-0.260	-0.2600000
EP29 Example 6a	bias_u	0.068	0.0672681
EP29 Example 6a	bias_expanded	0.140	0.1345362
EP29 Example 6b	claim_combined_u	0.220	0.2220675
EP29 Example 6b	lab_combined_u	0.165	0.1656050

Example 6a includes both the laboratory-mean uncertainty and the assigned-value uncertainty of the certified reference material. Example 6b uses literature precision in one scenario and local-laboratory precision in the other; the two scenarios are not interchangeable.

5.2.6 CLSI EP29-A Example 10: Creatinine-Clearance Uncertainty Propagation

EP29-A Example 10 gives a measurement model for creatinine clearance and the uncertainty of urine creatinine, serum creatinine, urine volume, collection time, and preanalytical volume separately. The following represents the original inputs as Type B components and propagates them with `uncertainty_propagate(method = "delta")`:

```

ep29_u_crea <- uncertainty_type_b(
  "u_crea", estimate = 15, uncertainty = 0.45,
  distribution = "standard"
)

```

```

)
ep29_u_vol <- uncertainty_type_b(
  "u_vol", estimate = 1500, uncertainty = 22.5,
  distribution = "standard"
)
ep29_s_crea <- uncertainty_type_b(
  "s_crea", estimate = 0.115, uncertainty = 0.00575,
  distribution = "standard"
)
ep29_time <- uncertainty_type_b(
  "time", estimate = 1440, uncertainty = 15,
  distribution = "standard"
)
ep29_u_vol_prea <- uncertainty_type_b(
  "u_vol_prea", estimate = 100,
  lower = 0, upper = 200,
  distribution = "rectangular"
)

ep29_creatinine_clearance <- uncertainty_propagate(
  model = function(u_crea, u_vol, s_crea, time, u_vol_prea) {
    (u_vol + u_vol_prea) * u_crea / (s_crea * time)
  },
  components = list(
    ep29_u_crea, ep29_u_vol, ep29_s_crea,
    ep29_time, ep29_u_vol_prea
  ),
  method = "delta"
)

```

Table 5.6: EP29-A Example 10: comparison of ivdtools and standard results

appendix	statistic	standard	ivdtools
EP29 Example 10	estimate	145.0	144.927536
EP29 Example 10	relative_percent	7.1	7.076966

The Kragten numerical approximation in EP29-A Figure 4 gives a relative combined uncertainty of approximately 6.9%, whereas the first-order delta result from `uncertainty_propagate()` is approximately 7.1%. The difference comes from the numerical-approximation path, not from different inputs.

5.2.7 CLSI EP29-A Example 11: Linear-Regression Prediction Equation

Example 11 predicts eAG from A1C using $eAG = \alpha + \beta A1C + \epsilon$. The following propagates uncertainty from the intercept, slope, A1C measurement, and prediction error and calculates three scenarios at A1C = 7% using the parameters in standard Table 4.

```

ep29_ex11_make <- function(relative_alc_u) {
  alc <- 7
  uncertainty_propagate(
    model = function(alpha, beta, A1C, epsilon) {
      alpha + beta * A1C + epsilon
    },
    components = list(
      uncertainty_type_b("alpha", estimate = -46.6, uncertainty = 3.8,

```



```

        distribution = "standard"),
uncertainty_type_b("beta", estimate = 28.7, uncertainty = 0.6,
        distribution = "standard"),
uncertainty_type_b("A1C", estimate = a1c,
        uncertainty = relative_a1c_u * a1c,
        distribution = "standard"),
uncertainty_type_b("epsilon", estimate = 0,
        uncertainty = 2.21 * a1c,
        distribution = "standard")
    )
  )
}
ep29_ex11_results <- lapply(c(0, 0.02, 0.05), ep29_ex11_make)

```

Table 5.7: EP29-A Example 11: comparison with the standard equation

appendix	statistic	standard	ivdtools
EP29 Example 11	A1C_7_U_0pct	30.8	32.94850
EP29 Example 11	A1C_7_U_2pct	31.8	33.91432
EP29 Example 11	A1C_7_U_5pct	36.8	38.59031

The intercept-slope covariance term in standard Table 4 cannot be used to construct a positive-semidefinite correlation matrix because the original parameters are rounded. Therefore the `ivdtools` column uses delta propagation with independent inputs, while the `standard` column is recalculated with the covariance formula in standard Equation 32; the difference is caused by the handling of the correlation term.

5.2.8 YY/T 1709: Calibrator Homogeneity, Stability, and Assigned-Value Uncertainty

5.2.8.1 Appendix A: Uncertainty from Homogeneity

```

yyt1709_homogeneity_data <- read_csv(
  "./data/005-不确定度分析/std-yyt1709-homogeneity.csv",
  show_col_types = FALSE
)
yyt1709_homogeneity <- uncertainty_homogeneity(
  yyt1709_homogeneity_data,
  unit = "bottle",
  result = "result",
  name = "YY/T 1709 瓶间均匀性"
)

```

5.2.8.2 Appendix B: Uncertainty from Stability

The standard calculates a linear trend from the mean at each time point. To maintain the same statistical definition, first aggregate by time point and then pass the summary to `uncertainty_stability()`. If all within-bottle replicates are passed as independent regression rows, the slope standard error uses different information.

```

yyt1709_stability_data <- read_csv(
  "./data/005-不确定度分析/std-yyt1709-stability.csv",
  show_col_types = FALSE
)

```

```

yyt1709_timepoint_means <- function(data) {
  aggregate(result ~ time, data = data, FUN = mean)
}

yyt1709_stability_results <- list(
  长期 = uncertainty_stability(
    yyt1709_timepoint_means(subset(yyt1709_stability_data, study == "长期")),
    time = "time", result = "result", shelf_life = 18,
    name = "长期稳定性"
  ),
  运输室温 = uncertainty_stability(
    yyt1709_timepoint_means(subset(yyt1709_stability_data, study == "运输室温")),
    time = "time", result = "result", shelf_life = 7,
    name = "运输稳定性 (室温)"
  ),
  运输37C = uncertainty_stability(
    yyt1709_timepoint_means(subset(yyt1709_stability_data, study == "运输37C")),
    time = "time", result = "result", shelf_life = 7,
    name = "运输稳定性 (37 C)"
  ),
  复溶 = uncertainty_stability(
    yyt1709_timepoint_means(subset(yyt1709_stability_data, study == "复溶")),
    time = "time", result = "result", shelf_life = 7,
    name = "复溶稳定性"
  )
)

```

5.2.8.3 Appendix C: Combining Assigned-Value Uncertainty

The working-calibrator certificate and reconstitution information are first converted to Type B components. Assigned-value repeatability is estimated with `uncertainty_type_a()`, and `uncertainty_combine()` is then used to combine relative and absolute uncertainty.

```

yyt1709_assignment <- read_csv(
  "./data/005-不确定度分析/std-yyt1709-assignment.csv",
  show_col_types = FALSE
)

yyt1709_assignment_u <- uncertainty_type_a(
  yyt1709_assignment$result,
  name = "赋值重复性",
  quantity = "mean",
  relative = TRUE
)

yyt1709_assigned_mean <- mean(yyt1709_assignment$result)

yyt1709_working_calibration <- uncertainty_type_b(
  "工作校准品",
  estimate = 178.06,
  uncertainty = 6.01,
  distribution = "standard",
  k = 2,
  relative = TRUE
)

yyt1709_reconstitution_1 <- uncertainty_type_b(

```

```

"工作校准品复溶",
estimate = 1,
uncertainty = 0.00003 / (sqrt(3) * 0.99758),
distribution = "standard",
k = 1,
relative = TRUE
)
yyt1709_reconstitution_2 <- uncertainty_type_b(
  "产品校准品复溶",
  estimate = 1,
  uncertainty = 0.01 / sqrt(3),
  distribution = "standard",
  k = 1,
  relative = TRUE
)

yyt1709_reconstitution <- uncertainty_combine(
  yyt1709_reconstitution_1,
  yyt1709_reconstitution_2,
  k = 1
)

yyt1709_characterization_budget <- uncertainty_combine(
  yyt1709_working_calibration,
  yyt1709_assignment_u,
  yyt1709_reconstitution_1,
  yyt1709_reconstitution_2,
  value = yyt1709_assigned_mean,
  coverage = 0.95,
  k = 2
)

```

The supplied uncertainty values are already standard uncertainties, so `distribution = "standard"` tells the function not to perform another distribution conversion. The two `sqrt(3)` terms convert rectangular half-widths into standard uncertainties; 0.99758 in the first term is the actual weighed mass used to convert an absolute weighing error to a relative error, not a distribution parameter.

5.2.8.4 Appendices A–C: Comparison with the Original YY/T 1709 Results

Table 5.8: YY/T 1709 Appendices A–C: comparison of ivdtools and standard results

appendix	statistic	standard	ivdtools
A (均匀性)	SS_between	0.2114405	0.2114405
A (均匀性)	SS_within	0.3556000	0.3556000
A (均匀性)	MS_between	0.0162647	0.0162647
A (均匀性)	MS_within	0.0127000	0.0127000
A (均匀性)	F	1.2806810	1.2806813
A (均匀性)	u_bb	0.0344916	0.0344705
C (定值)	mean	179.7300000	179.7294000
C (定值)	u_REP_rel	0.0012560	0.0012556
C (定值)	u_wcal_rel	0.0168763	0.0168763
C (定值)	u_sol_rel	0.0057735	0.0057735
C (定值)	u_char_rel	0.0178900	0.0178807
C (定值)	u_char	3.2200000	3.2136942
C (赋值确认)	mean	107.7200000	107.7240000
C (赋值确认)	bias	-0.3800000	-0.3760000

appendix	statistic	standard	ivdtools
B (长期)	slope	-0.0137000	-0.0137302
B (长期)	intercept	50.0114000	50.0114286
B (长期)	residual_sd	0.3238000	0.3237945
B (长期)	slope_se	0.0204000	0.0203971
B (长期)	u_stability	0.3672000	0.3671484
B (运输室温)	slope	-0.0487000	-0.0486706
B (运输室温)	intercept	50.3568000	50.3568056
B (运输室温)	residual_sd	0.5451000	0.5297030
B (运输室温)	slope_se	0.0841000	0.0817349
B (运输室温)	u_stability	0.5887000	0.5721446
B (运输37C)	slope	-0.0330000	-0.0329960
B (运输37C)	intercept	50.3936000	50.3936111
B (运输37C)	residual_sd	0.4310000	0.4220273
B (运输37C)	slope_se	0.0665000	0.0651202
B (运输37C)	u_stability	0.4655000	0.4558416
B (复溶)	slope	0.0433000	0.0432540
B (复溶)	intercept	49.6536000	49.6536111
B (复溶)	residual_sd	0.6319000	0.6214551
B (复溶)	slope_se	0.0975000	0.0958926
B (复溶)	u_stability	0.6825000	0.6712482

Chapter 6

Matrix Effects and Commutability

This chapter describes four commutability paths implemented by `commutability()`: CLSI EP14-A3, CLSI EP30-A, IFCC Part 2, and IFCC Part 3. The paths answer different questions and have different statistical units.

Path	Main question	Unit of evaluation	Main result
CLSI EP14-A3	Do processed samples follow the inter-procedure relationship of clinical samples?	Procedure pair $\times$ reference material	Deming prediction interval and classification
CLSI EP30-A	Is a reference material suitable for a set of measurement procedures?	Procedure pair $\times$ reference material	Prediction interval, relative residual, and RMF comparison
IFCC Part 2	Is the difference between reference-material and clinical-sample bias acceptable?	Procedure pair $\times$ reference material	Bias difference, uncertainty, and three-way conclusion
IFCC Part 3	Can recalibration of a reference material reduce inter-procedure differences?	Procedure set	Bias before/after calibration, IMPBR, diagnostics

! Scope of Use

Commutability is a study-design and interpretation problem, not only a regression calculation. The package checks data structure and produces statistical boundaries, but it does not choose materials, reference procedures, clinical acceptability limits, or the final fitness-for-purpose conclusion.

```
library(ivdtools)
library(readr)
```

6.1 Main Function and Input Data

Function	Main purpose
<code>commutability()</code>	Evaluate EP14, EP30, IFCC Part 2, or IFCC Part 3 commutability
<code>print()</code>	Summarize path, procedure pairs, design checks, and main conclusions
<code>summary()</code>	Expand unmet design recommendations and unavailable procedure pairs
<code>plot()</code>	Plot clinical-sample relationships, material positions, and boundaries

Field	Meaning
<code>sample_id</code>	Clinical-sample or reference-material identifier
<code>material_type</code>	Material class; not required for the <code>calibration</code> path
<code>procedure</code>	Measurement-procedure name
<code>result</code>	Single measurement result
<code>position</code>	Position of the material in the IFCC Part 2 run sequence
<code>calibration_stage</code>	before or after for the calibration path

The data must preserve replicate structure, procedure labels, material class, and any run-order information required by the selected path. Do not collapse repeated results without first defining the analysis unit.

6.2 Statistical Paths

For EP14-A3, clinical-sample relationships are fitted with Deming regression and material results are checked against prediction intervals. EP30-A additionally compares relative residuals with reference-material limits. IFCC Part 2 estimates clinical-sample bias around each material and combines bias differences with uncertainty. The supported local-bias models are:

Model	Interpretation
<code>constant_global</code>	One clinical-sample bias estimate across the full concentration range
<code>constant_local</code>	A local constant bias estimate near each reference material
<code>linear_local</code>	A local linear bias estimated from clinical samples on both sides of the material

IFCC Part 3 evaluates calibration validity and compares inter-procedure bias before and after recalibration. Analyze original and log scales separately when the protocol requires it; an apparent change in commutability after transformation is an analysis-scale difference, not an automatic biological conclusion.

6.3 Analysis Examples

6.3.1 EP14-A3 Cholesterol on the Original Scale

```
ep14_c1 <- as.data.frame(read_csv(
  "./data/006-基质效应与互换性/std-ep14-appendix-c1.csv",
  show_col_types = FALSE
))
ep14_fit <- commutability(
  ep14_c1, approach = "ep14", sample_id = "sample_id",
  material_type = "material_type", procedure = "procedure", result = "result"
)
print(ep14_fit)
```

Commutability assessment

Study-design checks: 2 met, 0 unmet. Unmet checks do not invalidate computations; inspect '\$design_checks'.

```
Approach      : EP14
Scale         : raw
Procedure pairs : 1
Reference materials : 5
Procedure pair: A vs B | Reference material: a
  Mean X      : 108.86
  Mean Y      : 110.15
  Predicted Y : 117.19384
  Residual    : -7.0438387
  Prediction SE : 3.9893651
  Lower       : 109.13103
  Upper       : 125.25665
  Extrapolated : FALSE
  Classification : commutable
  procedure_x  : A
  procedure_y  : B
```

```
Procedure pair: A vs B | Reference material: b
  Mean X      : 123.12333
  Mean Y      : 136.95
  Predicted Y : 132.71701
  Residual    : 4.2329936
  Prediction SE : 3.9111051
  Lower       : 124.81237
  Upper       : 140.62164
  Extrapolated : FALSE
  Classification : commutable
  procedure_x  : A
  procedure_y  : B
```

```
Procedure pair: A vs B | Reference material: c
  Mean X      : 146.95
  Mean Y      : 175.36667
  Predicted Y : 158.64821
  Residual    : 16.718461
  Prediction SE : 3.8114445
  Lower       : 150.94499
  Upper       : 166.35142
  Extrapolated : FALSE
  Classification : noncommutable
```

```

procedure_x : A
procedure_y : B

```

Procedure pair: A vs B | Reference material: d_run16

```

Mean X      : 198.44
Mean Y      : 215.98333
Predicted Y : 214.68615
Residual     : 1.2971814
Prediction SE : 3.740138
Lower       : 207.12705
Upper       : 222.24525
Extrapolated : FALSE
Classification : commutable
procedure_x : A
procedure_y : B

```

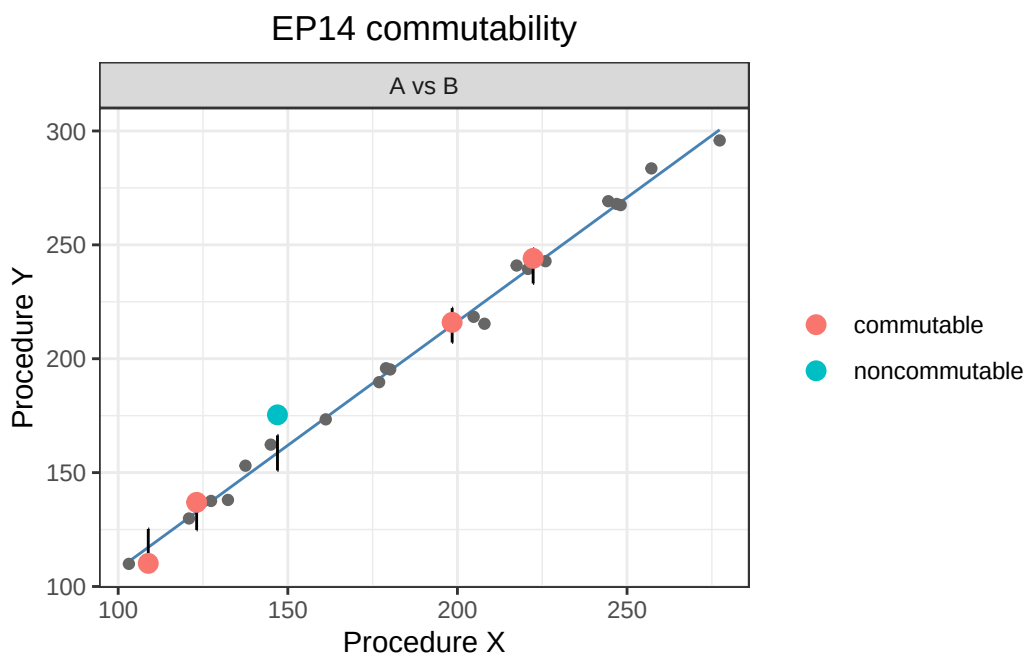
Procedure pair: A vs B | Reference material: d_run07

```

Mean X      : 222.31667
Mean Y      : 244.06667
Predicted Y : 240.67177
Residual     : 3.3948989
Prediction SE : 3.7763512
Lower       : 233.03948
Upper       : 248.30406
Extrapolated : FALSE
Classification : commutable
procedure_x : A
procedure_y : B

```

```
plot(ep14_fit)
```



6.3.2 EP30 Multi-Procedure Reference Materials

```
ep30 <- as.data.frame(read_csv(
  "./data/006-基质效应与互换性/std-ep30-appendix-b.csv",
  show_col_types = FALSE
))
ep30_fit <- commutability(
  ep30, approach = "ep30", sample_id = "specimen",
  material_type = "type", procedure = "method", result = "result"
)
summary(ep30_fit)
```

Commutability assessment

Study-design checks: 180 met, 0 unmet. Unmet checks do not invalidate computations; inspect '\$design_checks'.

```
Approach      : EP30
Scale         : raw
Procedure pairs : 45
Reference materials : 315
Procedure pair: MA vs MB | Reference material: RMA
Mean X       : 12.4
Mean Y       : 14.3
Predicted Y  : 15.576069
Residual     : -1.2760688
Prediction SE : 1.8530797
Lower        : 11.944099
Upper        : 19.208038
Extrapolated : FALSE
Prediction classification : commutable
Sy.x         : 1.1890275
Relative residual : 1.0732038
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x   : MA
procedure_y   : MB
critical_value : 1.959964
rm_set_size   : 1
classification : commutable
```

```
Procedure pair: MA vs MB | Reference material: RMB
Mean X       : 24.1
Mean Y       : 26.9
Predicted Y  : 26.829327
Residual     : 0.070672943
Prediction SE : 1.8454797
Lower        : 23.212253
Upper        : 30.446401
Extrapolated : FALSE
Prediction classification : commutable
Sy.x         : 1.1890275
Relative residual : -0.059437604
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x   : MA
procedure_y   : MB
critical_value : 1.959964
rm_set_size   : 1
classification : commutable
```

Procedure pair: MA vs MB | Reference material: RMC

```

Mean X           : 35.4
Mean Y           : 37
Predicted Y      : 37.697859
Residual         : -0.69785851
Prediction SE     : 1.8688251
Lower            : 34.035029
Upper            : 41.360688
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 1.1890275
Relative residual : 0.58691538
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MA
procedure_y      : MB
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MA vs MB | Reference material: RMD

```

Mean X           : 42.2
Mean Y           : 41.7
Predicted Y      : 44.238214
Residual         : -2.5382137
Prediction SE     : 1.8969998
Lower            : 40.520162
Upper            : 47.956265
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 1.1890275
Relative residual : 2.1346973
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MA
procedure_y      : MB
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MA vs MB | Reference material: RME

```

Mean X           : 15.3
Mean Y           : 19
Predicted Y      : 18.365338
Residual         : 0.63466204
Prediction SE     : 1.8481767
Lower            : 14.742978
Upper            : 21.987698
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 1.1890275
Relative residual : -0.53376567
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MA
procedure_y      : MB
critical_value    : 1.959964

```

```
rm_set_size      : 1
classification    : commutable
```

Procedure pair: MA vs MB | Reference material: RMF

```
Mean X           : 21
Mean Y           : 20
Predicted Y      : 23.847695
Residual         : -3.8476945
Prediction SE    : 1.8443328
Lower           : 20.232869
Upper           : 27.46252
Extrapolated    : FALSE
Prediction classification : noncommutable
Sy.x            : 1.1890275
Relative residual : 3.2360014
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x     : MA
procedure_y     : MB
critical_value   : 1.959964
rm_set_size     : 1
classification   : noncommutable
```

Procedure pair: MA vs MB | Reference material: RMG

```
Mean X           : 17
Mean Y           : 21.1
Predicted Y      : 20.000427
Residual         : 1.0995732
Prediction SE    : 1.8462245
Lower           : 16.381893
Upper           : 23.61896
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.1890275
Relative residual : -0.9247669
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MA
procedure_y     : MB
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable
```

Procedure pair: MA vs MC | Reference material: RMA

```
Mean X           : 12.4
Mean Y           : 11.3
Predicted Y      : 13.351899
Residual         : -2.0518991
Prediction SE    : 1.9589829
Lower           : 9.5123631
Upper           : 17.191435
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.0611977
Relative residual : 1.9335691
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MA
```

```

procedure_y      : MC
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MA vs MC | Reference material: RMB

```

Mean X          : 24.1
Mean Y          : 22.7
Predicted Y     : 25.760137
Residual        : -3.0601371
Prediction SE    : 1.9532717
Lower           : 21.931795
Upper           : 29.588479
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.0611977
Relative residual : 2.8836635
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x     : MA
procedure_y     : MC
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MA vs MC | Reference material: RMC

```

Mean X          : 35.4
Mean Y          : 33.6
Predicted Y     : 37.744162
Residual        : -4.1441618
Prediction SE    : 1.9708366
Lower           : 33.881393
Upper           : 41.60693
Extrapolated    : FALSE
Prediction classification : noncommutable
Sy.x            : 1.0611977
Relative residual : 3.9051742
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x     : MA
procedure_y     : MC
critical_value   : 1.959964
rm_set_size     : 1
classification   : noncommutable

```

Procedure pair: MA vs MC | Reference material: RMD

```

Mean X          : 42.2
Mean Y          : 40.3
Predicted Y     : 44.955787
Residual        : -4.6557873
Prediction SE    : 1.9921183
Lower           : 41.051307
Upper           : 48.860267
Extrapolated    : FALSE
Prediction classification : noncommutable
Sy.x            : 1.0611977
Relative residual : 4.387295
Relative-residual limit : 2

```

```

Relative-residual classification : noncommutable
procedure_x                      : MA
procedure_y                      : MC
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : noncommutable

```

Procedure pair: MA vs MC | Reference material: RME

```

Mean X                          : 15.3
Mean Y                          : 17.3
Predicted Y                     : 16.427445
Residual                       : 0.87255476
Prediction SE                   : 1.9552977
Lower                           : 12.595132
Upper                           : 20.259758
Extrapolated                   : FALSE
Prediction classification       : commutable
Sy.x                           : 1.0611977
Relative residual               : -0.82223584
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                     : MA
procedure_y                     : MC
critical_value                  : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MA vs MC | Reference material: RMF

```

Mean X                          : 21
Mean Y                          : 23.3
Predicted Y                     : 22.472484
Residual                       : 0.82751573
Prediction SE                   : 1.9524105
Lower                           : 18.64583
Upper                           : 26.299138
Extrapolated                   : FALSE
Prediction classification       : commutable
Sy.x                           : 1.0611977
Relative residual               : -0.77979414
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                     : MA
procedure_y                     : MC
critical_value                  : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MA vs MC | Reference material: RMG

```

Mean X                          : 17
Mean Y                          : 19.2
Predicted Y                     : 18.230352
Residual                       : 0.96964839
Prediction SE                   : 1.9538311
Lower                           : 14.400913
Upper                           : 22.05979
Extrapolated                   : FALSE
Prediction classification       : commutable
Sy.x                           : 1.0611977

```

```

Relative residual      : -0.91373022
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MA
procedure_y            : MC
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MA vs MD | Reference material: RMA

```

Mean X                : 12.4
Mean Y                : 11.4
Predicted Y           : 10.746146
Residual              : 0.65385407
Prediction SE         : 1.8882069
Lower                 : 7.0453283
Upper                 : 14.446964
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 1.0324006
Relative residual      : -0.63333369
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MA
procedure_y            : MD
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MA vs MD | Reference material: RMB

```

Mean X                : 24.1
Mean Y                : 19.9
Predicted Y           : 20.509565
Residual              : -0.60956453
Prediction SE         : 1.8825915
Lower                 : 16.819753
Upper                 : 24.199376
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 1.0324006
Relative residual      : 0.59043412
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MA
procedure_y            : MD
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MA vs MD | Reference material: RMC

```

Mean X                : 35.4
Mean Y                : 29.6
Predicted Y           : 29.939191
Residual              : -0.33919104
Prediction SE         : 1.8998608
Lower                 : 26.215532
Upper                 : 33.66285
Extrapolated          : FALSE

```

```

Prediction classification      : commutable
Sy.x                         : 1.0324006
Relative residual             : 0.32854596
Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                  : MA
procedure_y                  : MD
critical_value                : 1.959964
rm_set_size                  : 1
classification                : commutable

```

Procedure pair: MA vs MD | Reference material: RMD

```

Mean X                       : 42.2
Mean Y                       : 34.6
Predicted Y                  : 35.613657
Residual                     : -1.0136566
Prediction SE                 : 1.9207804
Lower                        : 31.848996
Upper                        : 39.378317
Extrapolated                 : FALSE
Prediction classification    : commutable
Sy.x                         : 1.0324006
Relative residual            : 0.98184422
Relative-residual limit     : 2
Relative-residual classification : commutable
procedure_x                  : MA
procedure_y                  : MD
critical_value                : 1.959964
rm_set_size                  : 1
classification                : commutable

```

Procedure pair: MA vs MD | Reference material: RME

```

Mean X                       : 15.3
Mean Y                       : 13.2
Predicted Y                  : 13.166139
Residual                     : 0.033861426
Prediction SE                 : 1.8845835
Lower                        : 9.4724228
Upper                        : 16.859854
Extrapolated                 : FALSE
Prediction classification    : commutable
Sy.x                         : 1.0324006
Relative residual            : -0.032798728
Relative-residual limit     : 2
Relative-residual classification : commutable
procedure_x                  : MA
procedure_y                  : MD
critical_value                : 1.959964
rm_set_size                  : 1
classification                : commutable

```

Procedure pair: MA vs MD | Reference material: RMF

```

Mean X                       : 21
Mean Y                       : 24.9
Predicted Y                  : 17.922676
Residual                     : 6.9773242
Prediction SE                 : 1.8817446
Lower                        : 14.234524

```

```

Upper                : 21.610828
Extrapolated         : FALSE
Prediction classification : noncommutable
Sy.x                 : 1.0324006
Relative residual     : -6.7583496
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x          : MA
procedure_y           : MD
critical_value        : 1.959964
rm_set_size           : 1
classification        : noncommutable

```

Procedure pair: MA vs MD | Reference material: RMG

```

Mean X               : 17
Mean Y               : 14.7
Predicted Y          : 14.584755
Residual              : 0.11524505
Prediction SE         : 1.8831415
Lower                : 10.893865
Upper                : 18.275645
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 1.0324006
Relative residual     : -0.11162823
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MA
procedure_y           : MD
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MA vs ME | Reference material: RMA

```

Mean X               : 12.4
Mean Y               : 12
Predicted Y          : 13.257712
Residual              : -1.2577119
Prediction SE         : 1.9868564
Lower                : 9.3635448
Upper                : 17.151879
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 1.1929761
Relative residual     : 1.0542641
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MA
procedure_y           : ME
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MA vs ME | Reference material: RMB

```

Mean X               : 24.1
Mean Y               : 24.1
Predicted Y          : 26.55133
Residual              : -2.4513305

```


Prediction SE	: 1.9797306
Lower	: 22.67113
Upper	: 30.431531
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.1929761
Relative residual	: 2.0548027
Relative-residual limit	: 2
Relative-residual classification	: noncommutable
procedure_x	: MA
procedure_y	: ME
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MA vs ME | Reference material: RMC

Mean X	: 35.4
Mean Y	: 35.9
Predicted Y	: 39.390466
Residual	: -3.4904664
Prediction SE	: 2.0016311
Lower	: 35.467342
Upper	: 43.313591
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.1929761
Relative residual	: 2.9258478
Relative-residual limit	: 2
Relative-residual classification	: noncommutable
procedure_x	: MA
procedure_y	: ME
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MA vs ME | Reference material: RMD

Mean X	: 42.2
Mean Y	: 43.5
Predicted Y	: 47.116672
Residual	: -3.6166721
Prediction SE	: 2.0281069
Lower	: 43.141656
Upper	: 51.091689
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.1929761
Relative residual	: 3.0316384
Relative-residual limit	: 2
Relative-residual classification	: noncommutable
procedure_x	: MA
procedure_y	: ME
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MA vs ME | Reference material: RME

Mean X	Mean Y
15.3	17.1

```

Predicted Y           : 16.552711
Residual              : 0.54728865
Prediction SE         : 1.9822589
Lower                 : 12.667555
Upper                 : 20.437867
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 1.1929761
Relative residual      : -0.45875912
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MA
procedure_y            : ME
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MA vs ME | Reference material: RMF

```

Mean X                : 21
Mean Y                 : 23.4
Predicted Y           : 23.02909
Residual              : 0.37091035
Prediction SE         : 1.9786556
Lower                 : 19.150996
Upper                 : 26.907183
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 1.1929761
Relative residual      : -0.31091181
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MA
procedure_y            : ME
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MA vs ME | Reference material: RMG

```

Mean X                : 17
Mean Y                 : 19
Predicted Y           : 18.484263
Residual              : 0.51573723
Prediction SE         : 1.9804288
Lower                 : 14.602694
Upper                 : 22.365832
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 1.1929761
Relative residual      : -0.43231146
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MA
procedure_y            : ME
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MA vs MF | Reference material: RMA

```

Mean X           : 12.4
Mean Y           : 9.5
Predicted Y      : 10.725553
Residual         : -1.2255535
Prediction SE    : 1.9315271
Lower            : 6.93983
Upper            : 14.511277
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x            : 0.9474596
Relative residual : 1.2935153
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MA
procedure_y      : MF
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MA vs MF | Reference material: RMB

```

Mean X           : 24.1
Mean Y           : 19
Predicted Y      : 21.312265
Residual         : -2.3122649
Prediction SE    : 1.9269104
Lower            : 17.53559
Upper            : 25.08894
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x            : 0.9474596
Relative residual : 2.4404891
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MA
procedure_y      : MF
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MA vs MF | Reference material: RMC

```

Mean X           : 35.4
Mean Y           : 28.8
Predicted Y      : 31.537037
Residual         : -2.7370374
Prediction SE    : 1.9411168
Lower            : 27.732518
Upper            : 35.341556
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x            : 0.9474596
Relative residual : 2.888817
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MA
procedure_y      : MF
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MA vs MF | Reference material: RMD

Mean X	: 42.2
Mean Y	: 35.1
Predicted Y	: 37.689998
Residual	: -2.5899978
Prediction SE	: 1.9583596
Lower	: 33.851684
Upper	: 41.528312
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 0.9474596
Relative residual	: 2.7336235
Relative-residual limit	: 2
Relative-residual classification	: noncommutable
procedure_x	: MA
procedure_y	: MF
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MA vs MF | Reference material: RME

Mean X	: 15.3
Mean Y	: 14
Predicted Y	: 13.34961
Residual	: 0.65038984
Prediction SE	: 1.9285478
Lower	: 9.5697259
Upper	: 17.129494
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 0.9474596
Relative residual	: -0.68645654
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MA
procedure_y	: MF
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MA vs MF | Reference material: RMF

Mean X	: 21
Mean Y	: 20
Predicted Y	: 18.507239
Residual	: 1.4927612
Prediction SE	: 1.9262144
Lower	: 14.731928
Upper	: 22.28255
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 0.9474596
Relative residual	: -1.5755408
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MA
procedure_y	: MF
critical_value	: 1.959964

```
rm_set_size      : 1
classification   : commutable
```

Procedure pair: MA vs MF | Reference material: RMG

```
Mean X          : 17
Mean Y          : 15.5
Predicted Y     : 14.88785
Residual        : 0.61214973
Prediction SE    : 1.9273625
Lower           : 11.110289
Upper           : 18.665411
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 0.9474596
Relative residual : -0.64609586
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MA
procedure_y     : MF
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable
```

Procedure pair: MA vs MG | Reference material: RMA

```
Mean X          : 12.4
Mean Y          : 9.1
Predicted Y     : 11.511692
Residual        : -2.4116925
Prediction SE    : 1.8485034
Lower           : 7.8886923
Upper           : 15.134693
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 0.85204379
Relative residual : 2.8304795
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x     : MA
procedure_y     : MG
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable
```

Procedure pair: MA vs MG | Reference material: RMB

```
Mean X          : 24.1
Mean Y          : 19.1
Predicted Y     : 22.514905
Residual        : -3.4149045
Prediction SE    : 1.8446068
Lower           : 18.899542
Upper           : 26.130267
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 0.85204379
Relative residual : 4.0078979
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x     : MA
```

```

procedure_y      : MG
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MA vs MG | Reference material: RMC

```

Mean X           : 35.4
Mean Y           : 28.8
Predicted Y      : 33.141938
Residual         : -4.3419384
Prediction SE     : 1.856601
Lower            : 29.503067
Upper            : 36.78081
Extrapolated     : FALSE
Prediction classification : noncommutable
Sy.x             : 0.85204379
Relative residual : 5.09591
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MA
procedure_y      : MG
critical_value    : 1.959964
rm_set_size      : 1
classification    : noncommutable

```

Procedure pair: MA vs MG | Reference material: RMD

```

Mean X           : 42.2
Mean Y           : 35.8
Predicted Y      : 39.536968
Residual         : -3.7369676
Prediction SE     : 1.8711726
Lower            : 35.869537
Upper            : 43.204399
Extrapolated     : FALSE
Prediction classification : noncommutable
Sy.x             : 0.85204379
Relative residual : 4.3858869
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MA
procedure_y      : MG
critical_value    : 1.959964
rm_set_size      : 1
classification    : noncommutable

```

Procedure pair: MA vs MG | Reference material: RME

```

Mean X           : 15.3
Mean Y           : 14.8
Predicted Y      : 14.238984
Residual         : 0.56101565
Prediction SE     : 1.8459887
Lower            : 10.620913
Upper            : 17.857056
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.85204379
Relative residual : -0.65843523
Relative-residual limit : 2

```

```

Relative-residual classification : commutable
procedure_x                     : MA
procedure_y                     : MG
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MA vs MG | Reference material: RMF

```

Mean X                          : 21
Mean Y                          : 20.1
Predicted Y                     : 19.599524
Residual                        : 0.50047644
Prediction SE                   : 1.8440194
Lower                           : 15.985312
Upper                           : 23.213735
Extrapolated                    : FALSE
Prediction classification       : commutable
Sy.x                           : 0.85204379
Relative residual               : -0.58738347
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                     : MA
procedure_y                     : MG
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MA vs MG | Reference material: RMG

```

Mean X                          : 17
Mean Y                          : 16.4
Predicted Y                     : 15.837742
Residual                        : 0.56225834
Prediction SE                   : 1.8449883
Lower                           : 12.221631
Upper                           : 19.453852
Extrapolated                    : FALSE
Prediction classification       : commutable
Sy.x                           : 0.85204379
Relative residual               : -0.65989371
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                     : MA
procedure_y                     : MG
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MA vs MH | Reference material: RMA

```

Mean X                          : 12.4
Mean Y                          : 9.1
Predicted Y                     : 11.512598
Residual                        : -2.4125976
Prediction SE                   : 1.9203196
Lower                           : 7.7488403
Upper                           : 15.276355
Extrapolated                    : FALSE
Prediction classification       : commutable
Sy.x                           : 0.85199356

```

```

Relative residual      : 2.8317087
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : MA
procedure_y            : MH
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MA vs MH | Reference material: RMB

```

Mean X                : 24.1
Mean Y                : 19.1
Predicted Y           : 22.514577
Residual              : -3.4145771
Prediction SE         : 1.9165698
Lower                 : 18.758169
Upper                 : 26.270985
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 0.85199356
Relative residual      : 4.0077499
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : MA
procedure_y            : MH
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MA vs MH | Reference material: RMC

```

Mean X                : 35.4
Mean Y                : 28.8
Predicted Y           : 33.140421
Residual              : -4.3404206
Prediction SE         : 1.9281138
Lower                 : 29.361387
Upper                 : 36.919454
Extrapolated          : FALSE
Prediction classification : noncommutable
Sy.x                  : 0.85199356
Relative residual      : 5.0944289
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : MA
procedure_y            : MH
critical_value         : 1.959964
rm_set_size            : 1
classification         : noncommutable

```

Procedure pair: MA vs MH | Reference material: RMD

```

Mean X                : 42.2
Mean Y                : 35.8
Predicted Y           : 39.534734
Residual              : -3.7347335
Prediction SE         : 1.9421458
Lower                 : 35.728198
Upper                 : 43.341269
Extrapolated          : FALSE

```



```

Prediction classification      : commutable
Sy.x                         : 0.85199356
Relative residual             : 4.3835232
Relative-residual limit      : 2
Relative-residual classification : noncommutable
procedure_x                   : MA
procedure_y                   : MH
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: MA vs MH | Reference material: RME

```

Mean X                       : 15.3
Mean Y                       : 14.8
Predicted Y                  : 14.239584
Residual                     : 0.56041603
Prediction SE                 : 1.9178995
Lower                        : 10.48057
Upper                        : 17.998598
Extrapolated                 : FALSE
Prediction classification    : commutable
Sy.x                         : 0.85199356
Relative residual            : -0.65777027
Relative-residual limit     : 2
Relative-residual classification : commutable
procedure_x                   : MA
procedure_y                   : MH
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: MA vs MH | Reference material: RMF

```

Mean X                       : 21
Mean Y                       : 20.1
Predicted Y                  : 19.599523
Residual                     : 0.50047728
Prediction SE                 : 1.9160046
Lower                        : 15.844223
Upper                        : 23.354823
Extrapolated                 : FALSE
Prediction classification    : commutable
Sy.x                         : 0.85199356
Relative residual            : -0.58741908
Relative-residual limit     : 2
Relative-residual classification : commutable
procedure_x                   : MA
procedure_y                   : MH
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: MA vs MH | Reference material: RMG

```

Mean X                       : 17
Mean Y                       : 16.4
Predicted Y                  : 15.838162
Residual                     : 0.56183781
Prediction SE                 : 1.9169369
Lower                        : 12.081035

```

```

Upper                : 19.595289
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.85199356
Relative residual     : -0.65943903
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MA
procedure_y            : MH
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MA vs MI | Reference material: RMA

```

Mean X                : 12.4
Mean Y                : 9.8
Predicted Y           : 11.822151
Residual              : -2.0221509
Prediction SE          : 1.8406062
Lower                 : 8.2146291
Upper                 : 15.429673
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 0.8461896
Relative residual      : 2.3897137
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x            : MA
procedure_y            : MI
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MA vs MI | Reference material: RMB

```

Mean X                : 24.1
Mean Y                : 19.5
Predicted Y           : 22.685842
Residual              : -3.1858421
Prediction SE          : 1.8367511
Lower                 : 19.085876
Upper                 : 26.285808
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 0.8461896
Relative residual      : 3.764927
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x            : MA
procedure_y            : MI
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MA vs MI | Reference material: RMC

```

Mean X                : 35.4
Mean Y                : 28.6
Predicted Y           : 33.178125
Residual              : -4.578125

```

```

Prediction SE      : 1.8486174
Lower              : 29.554902
Upper              : 36.801349
Extrapolated      : FALSE
Prediction classification : noncommutable
Sy.x              : 0.8461896
Relative residual  : 5.4102828
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x       : MA
procedure_y       : MI
critical_value     : 1.959964
rm_set_size       : 1
classification     : noncommutable

```

Procedure pair: MA vs MI | Reference material: RMD

```

Mean X            : 42.2
Mean Y            : 35
Predicted Y       : 39.492065
Residual          : -4.4920652
Prediction SE      : 1.8630342
Lower             : 35.840585
Upper             : 43.143545
Extrapolated      : FALSE
Prediction classification : noncommutable
Sy.x              : 0.8461896
Relative residual  : 5.3085801
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x       : MA
procedure_y       : MI
critical_value     : 1.959964
rm_set_size       : 1
classification     : noncommutable

```

Procedure pair: MA vs MI | Reference material: RME

```

Mean X            : 15.3
Mean Y            : 15
Predicted Y       : 14.514861
Residual          : 0.48513935
Prediction SE      : 1.8381183
Lower             : 10.912215
Upper             : 18.117506
Extrapolated      : FALSE
Prediction classification : commutable
Sy.x              : 0.8461896
Relative residual  : -0.57332228
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x       : MA
procedure_y       : MI
critical_value     : 1.959964
rm_set_size       : 1
classification     : commutable

```

Procedure pair: MA vs MI | Reference material: RMF

```

Mean X            : 21
Mean Y            : 26

```

```

Predicted Y           : 19.807428
Residual              : 6.1925718
Prediction SE         : 1.83617
Lower                 : 16.208601
Upper                 : 23.406255
Extrapolated         : FALSE
Prediction classification : noncommutable
Sy.x                  : 0.8461896
Relative residual     : -7.3181847
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : MA
procedure_y           : MI
critical_value        : 1.959964
rm_set_size           : 1
classification        : noncommutable

```

Procedure pair: MA vs MI | Reference material: RMG

```

Mean X                : 17
Mean Y                : 16.7
Predicted Y           : 16.093346
Residual              : 0.6066543
Prediction SE         : 1.8371286
Lower                 : 12.49264
Upper                 : 19.694052
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 0.8461896
Relative residual     : -0.71692479
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MA
procedure_y           : MI
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MA vs MJ | Reference material: RMA

```

Mean X                : 12.4
Mean Y                : 8.5
Predicted Y           : 9.3493528
Residual              : -0.84935277
Prediction SE         : 1.8576804
Lower                 : 5.7083662
Upper                 : 12.990339
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 0.84206988
Relative residual     : 1.0086488
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MA
procedure_y           : MJ
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MA vs MJ | Reference material: RMB

```

Mean X           : 24.1
Mean Y           : 17
Predicted Y      : 18.694438
Residual         : -1.694438
Prediction SE     : 1.8538908
Lower            : 15.060879
Upper            : 22.327997
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.84206988
Relative residual : 2.0122297
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MA
procedure_y      : MJ
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MA vs MJ | Reference material: RMC

```

Mean X           : 35.4
Mean Y           : 25.3
Predicted Y      : 27.720033
Residual         : -2.4200332
Prediction SE     : 1.8655562
Lower            : 24.06361
Upper            : 31.376456
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.84206988
Relative residual : 2.8739102
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MA
procedure_y      : MJ
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MA vs MJ | Reference material: RMD

```

Mean X           : 42.2
Mean Y           : 30.6
Predicted Y      : 33.151365
Residual         : -2.5513648
Prediction SE     : 1.8797316
Lower            : 29.467159
Upper            : 36.835571
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.84206988
Relative residual : 3.029873
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MA
procedure_y      : MJ
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MA vs MJ | Reference material: RME

```

Mean X           : 15.3
Mean Y           : 12
Predicted Y      : 11.665656
Residual         : 0.33434405
Prediction SE     : 1.8552347
Lower            : 8.0294628
Upper            : 15.301849
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.84206988
Relative residual : -0.39705024
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MA
procedure_y      : MJ
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MA vs MJ | Reference material: RMF

```

Mean X           : 21
Mean Y           : 21
Predicted Y      : 16.21839
Residual         : 4.7816102
Prediction SE     : 1.8533196
Lower            : 12.58595
Upper            : 19.850829
Extrapolated     : FALSE
Prediction classification : noncommutable
Sy.x             : 0.84206988
Relative residual : -5.6784007
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MA
procedure_y      : MJ
critical_value    : 1.959964
rm_set_size      : 1
classification    : noncommutable

```

Procedure pair: MA vs MJ | Reference material: RMG

```

Mean X           : 17
Mean Y           : 13.4
Predicted Y      : 13.023489
Residual         : 0.37651114
Prediction SE     : 1.8542618
Lower            : 9.3892025
Upper            : 16.657775
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.84206988
Relative residual : -0.44712577
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MA
procedure_y      : MJ
critical_value    : 1.959964

```

```
rm_set_size      : 1
classification    : commutable
```

Procedure pair: MB vs MC | Reference material: RMA

```
Mean X           : 14.3
Mean Y           : 11.3
Predicted Y      : 11.949157
Residual         : -0.64915747
Prediction SE     : 2.1645067
Lower            : 7.7068024
Upper            : 16.191513
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.95212352
Relative residual : 0.68179964
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MB
procedure_y      : MC
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable
```

Procedure pair: MB vs MC | Reference material: RMB

```
Mean X           : 26.9
Mean Y           : 22.7
Predicted Y      : 25.836685
Residual         : -3.1366853
Prediction SE     : 2.1587976
Lower            : 21.60552
Upper            : 30.067851
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.95212352
Relative residual : 3.2944101
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MB
procedure_y      : MC
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable
```

Procedure pair: MB vs MC | Reference material: RMC

```
Mean X           : 37
Mean Y           : 33.6
Predicted Y      : 36.968751
Residual         : -3.3687513
Prediction SE     : 2.1702374
Lower            : 32.715164
Upper            : 41.222338
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.95212352
Relative residual : 3.5381453
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MB
```

```

procedure_y      : MC
critical_value   : 1.959964
rm_set_size      : 1
classification   : commutable

```

Procedure pair: MB vs MC | Reference material: RMD

```

Mean X           : 41.7
Mean Y           : 40.3
Predicted Y      : 42.14902
Residual         : -1.8490197
Prediction SE     : 2.1803678
Lower            : 37.875577
Upper            : 46.422462
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.95212352
Relative residual : 1.9419956
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MB
procedure_y      : MC
critical_value   : 1.959964
rm_set_size      : 1
classification   : commutable

```

Procedure pair: MB vs MC | Reference material: RME

```

Mean X           : 19
Mean Y           : 17.3
Predicted Y      : 17.129426
Residual         : 0.1705742
Prediction SE     : 2.1597819
Lower            : 12.896331
Upper            : 21.362521
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.95212352
Relative residual : -0.17915133
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MB
procedure_y      : MC
critical_value   : 1.959964
rm_set_size      : 1
classification   : commutable

```

Procedure pair: MB vs MC | Reference material: RMF

```

Mean X           : 20
Mean Y           : 23.3
Predicted Y      : 18.231611
Residual         : 5.0683894
Prediction SE     : 2.1591744
Lower            : 13.999707
Upper            : 22.463515
Extrapolated     : FALSE
Prediction classification : noncommutable
Sy.x             : 0.95212352
Relative residual : -5.3232478
Relative-residual limit : 2

```



```

Relative-residual classification : noncommutable
procedure_x                     : MB
procedure_y                     : MC
critical_value                  : 1.959964
rm_set_size                     : 1
classification                  : noncommutable

```

Procedure pair: MB vs MC | Reference material: RMG

```

Mean X                         : 21.1
Mean Y                         : 19.2
Predicted Y                    : 19.444014
Residual                       : -0.24401378
Prediction SE                   : 2.1586676
Lower                          : 15.213103
Upper                          : 23.674925
Extrapolated                   : FALSE
Prediction classification      : commutable
Sy.x                           : 0.95212352
Relative residual              : 0.25628375
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                     : MB
procedure_y                     : MC
critical_value                  : 1.959964
rm_set_size                     : 1
classification                  : commutable

```

Procedure pair: MB vs MD | Reference material: RMA

```

Mean X                         : 14.3
Mean Y                         : 11.4
Predicted Y                    : 9.6426352
Residual                       : 1.7573648
Prediction SE                   : 2.0361834
Lower                          : 5.6517891
Upper                          : 13.633481
Extrapolated                   : FALSE
Prediction classification      : commutable
Sy.x                           : 1.3039134
Relative residual              : -1.3477619
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                     : MB
procedure_y                     : MD
critical_value                  : 1.959964
rm_set_size                     : 1
classification                  : commutable

```

Procedure pair: MB vs MD | Reference material: RMB

```

Mean X                         : 26.9
Mean Y                         : 19.9
Predicted Y                    : 20.569721
Residual                       : -0.66972078
Prediction SE                   : 2.0246944
Lower                          : 16.601393
Upper                          : 24.538049
Extrapolated                   : FALSE
Prediction classification      : commutable
Sy.x                           : 1.3039134

```

```

Relative residual      : 0.51362367
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MB
procedure_y           : MD
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MB vs MD | Reference material: RMC

```

Mean X      : 37
Mean Y      : 29.6
Predicted Y  : 29.328734
Residual     : 0.27126619
Prediction SE : 2.0476812
Lower        : 25.315352
Upper        : 33.342115
Extrapolated : FALSE
Prediction classification : commutable
Sy.x         : 1.3039134
Relative residual : -0.20804003
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x   : MB
procedure_y   : MD
critical_value : 1.959964
rm_set_size   : 1
classification : commutable

```

Procedure pair: MB vs MD | Reference material: RMD

```

Mean X      : 41.7
Mean Y      : 34.6
Predicted Y  : 33.40471
Residual     : 1.1952898
Prediction SE : 2.0679236
Lower        : 29.351654
Upper        : 37.457766
Extrapolated : FALSE
Prediction classification : commutable
Sy.x         : 1.3039134
Relative residual : -0.91669418
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x   : MB
procedure_y   : MD
critical_value : 1.959964
rm_set_size   : 1
classification : commutable

```

Procedure pair: MB vs MD | Reference material: RME

```

Mean X      : 19
Mean Y      : 13.2
Predicted Y  : 13.718612
Residual     : -0.51861158
Prediction SE : 2.0266778
Lower        : 9.7463961
Upper        : 17.690827
Extrapolated : FALSE

```

```

Prediction classification      : commutable
Sy.x                         : 1.3039134
Relative residual             : 0.39773468
Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                  : MB
procedure_y                  : MD
critical_value                : 1.959964
rm_set_size                  : 1
classification                : commutable

```

Procedure pair: MB vs MD | Reference material: RMF

```

Mean X                       : 20
Mean Y                       : 24.9
Predicted Y                  : 14.585841
Residual                     : 10.314159
Prediction SE                 : 2.0254538
Lower                        : 10.616024
Upper                        : 18.555657
Extrapolated                 : FALSE
Prediction classification    : noncommutable
Sy.x                         : 1.3039134
Relative residual            : -7.9101568
Relative-residual limit     : 2
Relative-residual classification : noncommutable
procedure_x                  : MB
procedure_y                  : MD
critical_value                : 1.959964
rm_set_size                  : 1
classification                : noncommutable

```

Procedure pair: MB vs MD | Reference material: RMG

```

Mean X                       : 21.1
Mean Y                       : 14.7
Predicted Y                  : 15.539793
Residual                     : -0.8397925
Prediction SE                 : 2.0244325
Lower                        : 11.571978
Upper                        : 19.507607
Extrapolated                 : FALSE
Prediction classification    : commutable
Sy.x                         : 1.3039134
Relative residual            : 0.64405543
Relative-residual limit     : 2
Relative-residual classification : commutable
procedure_x                  : MB
procedure_y                  : MD
critical_value                : 1.959964
rm_set_size                  : 1
classification                : commutable

```

Procedure pair: MB vs ME | Reference material: RMA

```

Mean X                       : 14.3
Mean Y                       : 12
Predicted Y                  : 11.738151
Residual                     : 0.26184855
Prediction SE                 : 2.2506522
Lower                        : 7.3269543

```

```

Upper                : 16.149349
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 1.7693315
Relative residual     : -0.14799293
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MB
procedure_y           : ME
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MB vs ME | Reference material: RMB

```

Mean X               : 26.9
Mean Y               : 24.1
Predicted Y          : 26.638706
Residual              : -2.5387061
Prediction SE         : 2.2314247
Lower                : 22.265194
Upper                : 31.012218
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 1.7693315
Relative residual     : 1.4348392
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MB
procedure_y           : ME
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MB vs ME | Reference material: RMC

```

Mean X               : 37
Mean Y               : 35.9
Predicted Y          : 38.582802
Residual              : -2.6828016
Prediction SE         : 2.269839
Lower                : 34.133999
Upper                : 43.031604
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 1.7693315
Relative residual     : 1.5162798
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MB
procedure_y           : ME
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MB vs ME | Reference material: RMD

```

Mean X               : 41.7
Mean Y               : 43.5
Predicted Y          : 44.140945
Residual              : -0.64094498

```

Prediction SE	: 2.3034874
Lower	: 39.626193
Upper	: 48.655697
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.7693315
Relative residual	: 0.36225263
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MB
procedure_y	: ME
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MB vs ME | Reference material: RME

Mean X	: 19
Mean Y	: 17.1
Predicted Y	: 17.296295
Residual	: -0.19629486
Prediction SE	: 2.234748
Lower	: 12.916269
Upper	: 21.67632
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.7693315
Relative residual	: 0.11094296
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MB
procedure_y	: ME
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MB vs ME | Reference material: RMF

Mean X	: 20
Mean Y	: 23.4
Predicted Y	: 18.478879
Residual	: 4.9211214
Prediction SE	: 2.2326972
Lower	: 14.102872
Upper	: 22.854885
Extrapolated	: FALSE
Prediction classification	: noncommutable
Sy.x	: 1.7693315
Relative residual	: -2.7813451
Relative-residual limit	: 2
Relative-residual classification	: noncommutable
procedure_x	: MB
procedure_y	: ME
critical_value	: 1.959964
rm_set_size	: 1
classification	: noncommutable

Procedure pair: MB vs ME | Reference material: RMG

Mean X : 21.1
Mean Y : 19

```

Predicted Y           : 19.779721
Residual              : -0.77972065
Prediction SE         : 2.2309857
Lower                 : 15.407069
Upper                 : 24.152372
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 1.7693315
Relative residual     : 0.44068659
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MB
procedure_y           : ME
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MB vs MF | Reference material: RMA

```

Mean X                : 14.3
Mean Y                : 9.5
Predicted Y           : 9.5234244
Residual              : -0.023424432
Prediction SE         : 2.1098187
Lower                 : 5.3882557
Upper                 : 13.658593
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 1.4815125
Relative residual     : 0.01581116
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MB
procedure_y           : MF
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MB vs MF | Reference material: RMB

```

Mean X                : 26.9
Mean Y                : 19
Predicted Y           : 21.379279
Residual              : -2.379279
Prediction SE         : 2.0954497
Lower                 : 17.272273
Upper                 : 25.486285
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 1.4815125
Relative residual     : 1.6059796
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MB
procedure_y           : MF
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MB vs MF | Reference material: RMC

```

Mean X           : 37
Mean Y           : 28.8
Predicted Y      : 30.882781
Residual         : -2.0827814
Prediction SE     : 2.1241822
Lower            : 26.719461
Upper            : 35.046102
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 1.4815125
Relative residual : 1.405848
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MB
procedure_y      : MF
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MB vs MF | Reference material: RMD

```

Mean X           : 41.7
Mean Y           : 35.1
Predicted Y      : 35.305203
Residual         : -0.20520332
Prediction SE     : 2.1494297
Lower            : 31.092398
Upper            : 39.518008
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 1.4815125
Relative residual : 0.13850934
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MB
procedure_y      : MF
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MB vs MF | Reference material: RME

```

Mean X           : 19
Mean Y           : 14
Predicted Y      : 13.945846
Residual         : 0.054153642
Prediction SE     : 2.0979315
Lower            : 9.8339762
Upper            : 18.057716
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 1.4815125
Relative residual : -0.036552943
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MB
procedure_y      : MF
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MB vs MF | Reference material: RMF

Mean X	: 20
Mean Y	: 20
Predicted Y	: 14.886787
Residual	: 5.1132128
Prediction SE	: 2.0963999
Lower	: 10.777919
Upper	: 18.995656
Extrapolated	: FALSE
Prediction classification	: noncommutable
Sy.x	: 1.4815125
Relative residual	: -3.4513463
Relative-residual limit	: 2
Relative-residual classification	: noncommutable
procedure_x	: MB
procedure_y	: MF
critical_value	: 1.959964
rm_set_size	: 1
classification	: noncommutable

Procedure pair: MB vs MF | Reference material: RMG

Mean X	: 21.1
Mean Y	: 15.5
Predicted Y	: 15.921822
Residual	: -0.42182211
Prediction SE	: 2.0951219
Lower	: 11.815459
Upper	: 20.028186
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.4815125
Relative residual	: 0.28472396
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MB
procedure_y	: MF
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MB vs MG | Reference material: RMA

Mean X	: 14.3
Mean Y	: 9.1
Predicted Y	: 10.258893
Residual	: -1.1588934
Prediction SE	: 2.0417754
Lower	: 6.2570871
Upper	: 14.2607
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.3243523
Relative residual	: 0.87506425
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MB
procedure_y	: MG
critical_value	: 1.959964


```
rm_set_size      : 1
classification    : commutable
```

Procedure pair: MB vs MG | Reference material: RMB

```
Mean X           : 26.9
Mean Y           : 19.1
Predicted Y      : 22.585638
Residual         : -3.485638
Prediction SE    : 2.0299559
Lower           : 18.606998
Upper           : 26.564278
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.3243523
Relative residual : 2.6319567
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MB
procedure_y      : MG
critical_value   : 1.959964
rm_set_size      : 1
classification    : commutable
```

Procedure pair: MB vs MG | Reference material: RMC

```
Mean X           : 37
Mean Y           : 28.8
Predicted Y      : 32.4666
Residual         : -3.6665999
Prediction SE    : 2.0536024
Lower           : 28.441613
Upper           : 36.491587
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.3243523
Relative residual : 2.7685985
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MB
procedure_y      : MG
critical_value   : 1.959964
rm_set_size      : 1
classification    : commutable
```

Procedure pair: MB vs MG | Reference material: RMD

```
Mean X           : 41.7
Mean Y           : 35.8
Predicted Y      : 37.064671
Residual         : -1.2646713
Prediction SE    : 2.0744201
Lower           : 32.998883
Upper           : 41.13046
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.3243523
Relative residual : 0.95493567
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MB
```

```

procedure_y      : MG
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MB vs MG | Reference material: RME

```

Mean X          : 19
Mean Y          : 14.8
Predicted Y     : 14.856965
Residual        : -0.056964779
Prediction SE    : 2.0319964
Lower          : 10.874325
Upper          : 18.839605
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x           : 1.3243523
Relative residual : 0.043013311
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MB
procedure_y     : MG
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MB vs MG | Reference material: RMF

```

Mean X          : 20
Mean Y          : 20.1
Predicted Y     : 15.835278
Residual        : 4.2647222
Prediction SE    : 2.0307371
Lower          : 11.855106
Upper          : 19.815449
Extrapolated    : FALSE
Prediction classification : noncommutable
Sy.x           : 1.3243523
Relative residual : -3.2202323
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x     : MB
procedure_y     : MG
critical_value   : 1.959964
rm_set_size     : 1
classification   : noncommutable

```

Procedure pair: MB vs MG | Reference material: RMG

```

Mean X          : 21.1
Mean Y          : 16.4
Predicted Y     : 16.911422
Residual        : -0.51142221
Prediction SE    : 2.0296864
Lower          : 12.93331
Upper          : 20.889534
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x           : 1.3243523
Relative residual : 0.38616779
Relative-residual limit : 2

```

```

Relative-residual classification : commutable
procedure_x                     : MB
procedure_y                     : MG
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MB vs MH | Reference material: RMA

```

Mean X                         : 14.3
Mean Y                         : 9.1
Predicted Y                    : 10.261341
Residual                       : -1.1613407
Prediction SE                   : 2.1068302
Lower                          : 6.1320295
Upper                          : 14.390652
Extrapolated                   : FALSE
Prediction classification      : commutable
Sy.x                           : 1.3241455
Relative residual               : 0.87704921
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                     : MB
procedure_y                     : MH
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MB vs MH | Reference material: RMB

```

Mean X                         : 26.9
Mean Y                         : 19.1
Predicted Y                    : 22.584853
Residual                       : -3.484853
Prediction SE                   : 2.0953837
Lower                          : 18.477976
Upper                          : 26.691729
Extrapolated                   : FALSE
Prediction classification      : commutable
Sy.x                           : 1.3241455
Relative residual               : 2.631775
Relative-residual limit        : 2
Relative-residual classification : noncommutable
procedure_x                     : MB
procedure_y                     : MH
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MB vs MH | Reference material: RMC

```

Mean X                         : 37
Mean Y                         : 28.8
Predicted Y                    : 32.463224
Residual                       : -3.6632239
Prediction SE                   : 2.118288
Lower                          : 28.311456
Upper                          : 36.614992
Extrapolated                   : FALSE
Prediction classification      : commutable
Sy.x                           : 1.3241455

```

```

Relative residual      : 2.7664814
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : MB
procedure_y           : MH
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MB vs MH | Reference material: RMD

```

Mean X      : 41.7
Mean Y      : 35.8
Predicted Y  : 37.06009
Residual     : -1.2600895
Prediction SE : 2.1384655
Lower       : 32.868774
Upper       : 41.251405
Extrapolated : FALSE
Prediction classification : commutable
Sy.x        : 1.3241455
Relative residual : 0.9516247
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x   : MB
procedure_y   : MH
critical_value : 1.959964
rm_set_size   : 1
classification : commutable

```

Procedure pair: MB vs MH | Reference material: RME

```

Mean X      : 19
Mean Y      : 14.8
Predicted Y  : 14.858206
Residual     : -0.058206417
Prediction SE : 2.0973595
Lower       : 10.747457
Upper       : 18.968956
Extrapolated : FALSE
Prediction classification : commutable
Sy.x        : 1.3241455
Relative residual : 0.043957721
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x   : MB
procedure_y   : MH
critical_value : 1.959964
rm_set_size   : 1
classification : commutable

```

Procedure pair: MB vs MH | Reference material: RMF

```

Mean X      : 20
Mean Y      : 20.1
Predicted Y  : 15.836263
Residual     : 4.2637371
Prediction SE : 2.0961401
Lower       : 11.727904
Upper       : 19.944622
Extrapolated : FALSE

```

```

Prediction classification      : noncommutable
Sy.x                         : 1.3241455
Relative residual             : -3.2199914
Relative-residual limit      : 2
Relative-residual classification : noncommutable
procedure_x                   : MB
procedure_y                   : MH
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : noncommutable

```

Procedure pair: MB vs MH | Reference material: RMG

```

Mean X                       : 21.1
Mean Y                       : 16.4
Predicted Y                   : 16.912125
Residual                     : -0.51212512
Prediction SE                 : 2.0951227
Lower                        : 12.80576
Upper                        : 21.01849
Extrapolated                 : FALSE
Prediction classification     : commutable
Sy.x                         : 1.3241455
Relative residual             : 0.38675895
Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                   : MB
procedure_y                   : MH
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: MB vs MI | Reference material: RMA

```

Mean X                       : 14.3
Mean Y                       : 9.8
Predicted Y                   : 10.600736
Residual                     : -0.80073593
Prediction SE                 : 2.0291672
Lower                        : 6.6236413
Upper                        : 14.577831
Extrapolated                 : FALSE
Prediction classification     : commutable
Sy.x                         : 1.328052
Relative residual             : 0.6029402
Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                   : MB
procedure_y                   : MI
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: MB vs MI | Reference material: RMB

```

Mean X                       : 26.9
Mean Y                       : 19.5
Predicted Y                   : 22.750707
Residual                     : -3.2507073
Prediction SE                 : 2.0172357
Lower                        : 18.796998

```

```

Upper                : 26.704417
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 1.328052
Relative residual     : 2.447726
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x          : MB
procedure_y           : MI
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MB vs MI | Reference material: RMC

```

Mean X               : 37
Mean Y               : 28.6
Predicted Y          : 32.48997
Residual              : -3.8899701
Prediction SE         : 2.0411051
Lower                : 28.489478
Upper                : 36.490463
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 1.328052
Relative residual     : 2.9290797
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x          : MB
procedure_y           : MI
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MB vs MI | Reference material: RMD

```

Mean X               : 41.7
Mean Y               : 35
Predicted Y          : 37.022102
Residual              : -2.0221023
Prediction SE         : 2.0621154
Lower                : 32.98043
Upper                : 41.063774
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 1.328052
Relative residual     : 1.5226078
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MB
procedure_y           : MI
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MB vs MI | Reference material: RME

```

Mean X               : 19
Mean Y               : 15
Predicted Y          : 15.132868
Residual              : -0.13286812

```

```

Prediction SE          : 2.0192957
Lower                  : 11.175121
Upper                  : 19.090615
Extrapolated           : FALSE
Prediction classification : commutable
Sy.x                   : 1.328052
Relative residual       : 0.10004738
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x            : MB
procedure_y            : MI
critical_value          : 1.959964
rm_set_size            : 1
classification          : commutable

```

Procedure pair: MB vs MI | Reference material: RMF

```

Mean X                 : 20
Mean Y                 : 26
Predicted Y            : 16.097152
Residual               : 9.9028484
Prediction SE          : 2.0180244
Lower                  : 12.141896
Upper                  : 20.052407
Extrapolated           : FALSE
Prediction classification : noncommutable
Sy.x                   : 1.328052
Relative residual       : -7.4566723
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x            : MB
procedure_y            : MI
critical_value          : 1.959964
rm_set_size            : 1
classification          : noncommutable

```

Procedure pair: MB vs MI | Reference material: RMG

```

Mean X                 : 21.1
Mean Y                 : 16.7
Predicted Y            : 17.157863
Residual               : -0.45786336
Prediction SE          : 2.0169637
Lower                  : 13.204687
Upper                  : 21.11104
Extrapolated           : FALSE
Prediction classification : commutable
Sy.x                   : 1.328052
Relative residual       : 0.34476313
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x            : MB
procedure_y            : MI
critical_value          : 1.959964
rm_set_size            : 1
classification          : commutable

```

Procedure pair: MB vs MJ | Reference material: RMA

```

Mean X                 : 14.3
Mean Y                 : 8.5

```

```

Predicted Y           : 8.2927606
Residual              : 0.20723939
Prediction SE         : 1.9994154
Lower                 : 4.3739784
Upper                 : 12.211543
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 1.2369884
Relative residual     : -0.16753544
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MB
procedure_y           : MJ
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MB vs MJ | Reference material: RMB

```

Mean X                : 26.9
Mean Y                : 17
Predicted Y           : 18.752133
Residual              : -1.7521334
Prediction SE         : 1.9888809
Lower                 : 14.853998
Upper                 : 22.650268
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 1.2369884
Relative residual     : 1.416451
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MB
procedure_y           : MJ
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MB vs MJ | Reference material: RMC

```

Mean X                : 37
Mean Y                : 25.3
Predicted Y           : 27.136234
Residual              : -1.8362338
Prediction SE         : 2.0099621
Lower                 : 23.196781
Upper                 : 31.075687
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 1.2369884
Relative residual     : 1.484439
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MB
procedure_y           : MJ
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MB vs MJ | Reference material: RMD


```

Mean X           : 41.7
Mean Y           : 30.6
Predicted Y      : 31.037746
Residual         : -0.43774585
Prediction SE    : 2.0285391
Lower           : 27.061882
Upper           : 35.013609
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.2369884
Relative residual : 0.35388033
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MB
procedure_y     : MJ
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MB vs MJ | Reference material: RME

```

Mean X           : 19
Mean Y           : 12
Predicted Y      : 12.194273
Residual         : -0.19427268
Prediction SE    : 1.9906992
Lower           : 8.292574
Upper           : 16.095971
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.2369884
Relative residual : 0.15705296
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MB
procedure_y     : MJ
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MB vs MJ | Reference material: RMF

```

Mean X           : 20
Mean Y           : 21
Predicted Y      : 13.024382
Residual         : 7.9756184
Prediction SE    : 1.989577
Lower           : 9.1248823
Upper           : 16.923881
Extrapolated    : FALSE
Prediction classification : noncommutable
Sy.x            : 1.2369884
Relative residual : -6.4476099
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x     : MB
procedure_y     : MJ
critical_value   : 1.959964
rm_set_size     : 1
classification   : noncommutable

```

Procedure pair: MB vs MJ | Reference material: RMG

Mean X	: 21.1
Mean Y	: 13.4
Predicted Y	: 13.937501
Residual	: -0.53750148
Prediction SE	: 1.9886408
Lower	: 10.039837
Upper	: 17.835166
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.2369884
Relative residual	: 0.43452428
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MB
procedure_y	: MJ
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MC vs MD | Reference material: RMA

Mean X	: 11.3
Mean Y	: 11.4
Predicted Y	: 9.1398201
Residual	: 2.2601799
Prediction SE	: 1.9903325
Lower	: 5.2388401
Upper	: 13.0408
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.2844594
Relative residual	: -1.7596351
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MC
procedure_y	: MD
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MC vs MD | Reference material: RMB

Mean X	: 22.7
Mean Y	: 19.9
Predicted Y	: 18.101523
Residual	: 1.798477
Prediction SE	: 1.9760116
Lower	: 14.228611
Upper	: 21.974434
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.2844594
Relative residual	: -1.4001821
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MC
procedure_y	: MD
critical_value	: 1.959964

```
rm_set_size      : 1
classification   : commutable
```

Procedure pair: MC vs MD | Reference material: RMC

```
Mean X           : 33.6
Mean Y           : 29.6
Predicted Y      : 26.670169
Residual         : 2.9298313
Prediction SE    : 1.990261
Lower            : 22.769329
Upper           : 30.571008
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x            : 1.2844594
Relative residual : -2.280984
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MC
procedure_y      : MD
critical_value   : 1.959964
rm_set_size      : 1
classification   : commutable
```

Procedure pair: MC vs MD | Reference material: RMD

```
Mean X           : 40.3
Mean Y           : 34.6
Predicted Y      : 31.937134
Residual         : 2.6628656
Prediction SE    : 2.0123727
Lower            : 27.992956
Upper           : 35.881312
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x            : 1.2844594
Relative residual : -2.0731411
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MC
procedure_y      : MD
critical_value   : 1.959964
rm_set_size      : 1
classification   : commutable
```

Procedure pair: MC vs MD | Reference material: RME

```
Mean X           : 17.3
Mean Y           : 13.2
Predicted Y      : 13.856506
Residual         : -0.65650583
Prediction SE    : 1.9790793
Lower            : 9.9775816
Upper           : 17.73543
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x            : 1.2844594
Relative residual : 0.51111451
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MC
```

```

procedure_y      : MD
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MC vs MD | Reference material: RMF

```

Mean X          : 23.3
Mean Y          : 24.9
Predicted Y     : 18.573192
Residual        : 6.3268085
Prediction SE    : 1.9760858
Lower           : 14.700135
Upper           : 22.446248
Extrapolated    : FALSE
Prediction classification : noncommutable
Sy.x            : 1.2844594
Relative residual : -4.9256586
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x     : MC
procedure_y     : MD
critical_value   : 1.959964
rm_set_size     : 1
classification   : noncommutable

```

Procedure pair: MC vs MD | Reference material: RMG

```

Mean X          : 19.2
Mean Y          : 14.7
Predicted Y     : 15.350123
Residual        : -0.65012297
Prediction SE    : 1.9772339
Lower           : 11.474816
Upper           : 19.22543
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.2844594
Relative residual : 0.50614521
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MC
procedure_y     : MD
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MC vs ME | Reference material: RMA

```

Mean X          : 11.3
Mean Y          : 12
Predicted Y     : 11.05559
Residual        : 0.94440982
Prediction SE    : 2.1638751
Lower           : 6.814473
Upper           : 15.296707
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.5852203
Relative residual : -0.59575935
Relative-residual limit : 2

```

```

Relative-residual classification : commutable
procedure_x                     : MC
procedure_y                     : ME
critical_value                  : 1.959964
rm_set_size                     : 1
classification                  : commutable

```

Procedure pair: MC vs ME | Reference material: RMB

```

Mean X                         : 22.7
Mean Y                         : 24.1
Predicted Y                    : 23.27292
Residual                       : 0.82707984
Prediction SE                   : 2.143768
Lower                          : 19.071212
Upper                          : 27.474628
Extrapolated                   : FALSE
Prediction classification      : commutable
Sy.x                           : 1.5852203
Relative residual              : -0.52174441
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                   : MC
procedure_y                   : ME
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: MC vs ME | Reference material: RMC

```

Mean X                         : 33.6
Mean Y                         : 35.9
Predicted Y                    : 34.954402
Residual                       : 0.94559766
Prediction SE                   : 2.1637748
Lower                          : 30.713482
Upper                          : 39.195323
Extrapolated                   : FALSE
Prediction classification      : commutable
Sy.x                           : 1.5852203
Relative residual              : -0.59650867
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                   : MC
procedure_y                   : ME
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: MC vs ME | Reference material: RMD

```

Mean X                         : 40.3
Mean Y                         : 43.5
Predicted Y                    : 42.134763
Residual                       : 1.3652371
Prediction SE                   : 2.1947388
Lower                          : 37.833154
Upper                          : 46.436372
Extrapolated                   : FALSE
Prediction classification      : commutable
Sy.x                           : 1.5852203

```

```

Relative residual      : -0.86122859
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MC
procedure_y           : ME
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MC vs ME | Reference material: RME

```

Mean X                : 17.3
Mean Y                : 17.1
Predicted Y           : 17.485764
Residual              : -0.38576386
Prediction SE         : 2.1480788
Lower                 : 13.275607
Upper                 : 21.695921
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 1.5852203
Relative residual      : 0.24335031
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MC
procedure_y           : ME
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MC vs ME | Reference material: RMF

```

Mean X                : 23.3
Mean Y                : 23.4
Predicted Y           : 23.915938
Residual              : -0.51593753
Prediction SE         : 2.1438723
Lower                 : 19.714025
Upper                 : 28.11785
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 1.5852203
Relative residual      : 0.3254674
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MC
procedure_y           : ME
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MC vs ME | Reference material: RMG

```

Mean X                : 19.2
Mean Y                : 19
Predicted Y           : 19.521986
Residual              : -0.52198552
Prediction SE         : 2.1454858
Lower                 : 15.316911
Upper                 : 23.72706
Extrapolated          : FALSE

```

```

Prediction classification      : commutable
Sy.x                         : 1.5852203
Relative residual             : 0.32928264
Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                  : MC
procedure_y                   : ME
critical_value                : 1.959964
rm_set_size                   : 1
classification                : commutable

```

Procedure pair: MC vs MF | Reference material: RMA

```

Mean X                       : 11.3
Mean Y                       : 9.5
Predicted Y                  : 8.9794215
Residual                     : 0.5205785
Prediction SE                 : 2.0538816
Lower                        : 4.9538876
Upper                        : 13.004955
Extrapolated                 : FALSE
Prediction classification    : commutable
Sy.x                         : 1.3856472
Relative residual            : -0.37569339
Relative-residual limit     : 2
Relative-residual classification : commutable
procedure_x                  : MC
procedure_y                   : MF
critical_value                : 1.959964
rm_set_size                   : 1
classification                : commutable

```

Procedure pair: MC vs MF | Reference material: RMB

```

Mean X                       : 22.7
Mean Y                       : 19
Predicted Y                  : 18.701259
Residual                     : 0.29874091
Prediction SE                 : 2.0376962
Lower                        : 14.707448
Upper                        : 22.69507
Extrapolated                 : FALSE
Prediction classification    : commutable
Sy.x                         : 1.3856472
Relative residual            : -0.21559666
Relative-residual limit     : 2
Relative-residual classification : commutable
procedure_x                  : MC
procedure_y                   : MF
critical_value                : 1.959964
rm_set_size                   : 1
classification                : commutable

```

Procedure pair: MC vs MF | Reference material: RMC

```

Mean X                       : 33.6
Mean Y                       : 28.8
Predicted Y                  : 27.9967
Residual                     : 0.8032997
Prediction SE                 : 2.0538008
Lower                        : 23.971325

```

```

Upper                : 32.022076
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 1.3856472
Relative residual     : -0.57972887
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MC
procedure_y           : MF
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MC vs MF | Reference material: RMD

```

Mean X               : 40.3
Mean Y               : 35.1
Predicted Y          : 33.710412
Residual              : 1.3895881
Prediction SE         : 2.0787697
Lower                : 29.636098
Upper                : 37.784726
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 1.3856472
Relative residual     : -1.0028441
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MC
procedure_y           : MF
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MC vs MF | Reference material: RME

```

Mean X               : 17.3
Mean Y               : 14
Predicted Y          : 14.096178
Residual              : -0.09617813
Prediction SE         : 2.0411643
Lower                : 10.09557
Upper                : 18.096787
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 1.3856472
Relative residual     : 0.069410257
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MC
procedure_y           : MF
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MC vs MF | Reference material: RMF

```

Mean X               : 23.3
Mean Y               : 20
Predicted Y          : 19.212935
Residual              : 0.78706524

```



```

Prediction SE          : 2.0377801
Lower                  : 15.218959
Upper                  : 23.20691
Extrapolated           : FALSE
Prediction classification : commutable
Sy.x                   : 1.3856472
Relative residual       : -0.56801272
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x             : MC
procedure_y              : MF
critical_value           : 1.959964
rm_set_size              : 1
classification           : commutable

```

Procedure pair: MC vs MF | Reference material: RMG

```

Mean X                  : 19.2
Mean Y                  : 15.5
Predicted Y             : 15.716484
Residual                 : -0.21648439
Prediction SE           : 2.0390781
Lower                   : 11.719965
Upper                   : 19.713004
Extrapolated            : FALSE
Prediction classification : commutable
Sy.x                    : 1.3856472
Relative residual        : 0.15623341
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x              : MC
procedure_y              : MF
critical_value           : 1.959964
rm_set_size              : 1
classification           : commutable

```

Procedure pair: MC vs MG | Reference material: RMA

```

Mean X                  : 11.3
Mean Y                  : 9.1
Predicted Y             : 9.6905532
Residual                 : -0.59055316
Prediction SE           : 1.9783801
Lower                   : 5.8129995
Upper                   : 13.568107
Extrapolated            : FALSE
Prediction classification : commutable
Sy.x                    : 1.1973364
Relative residual        : 0.49322241
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x              : MC
procedure_y              : MG
critical_value           : 1.959964
rm_set_size              : 1
classification           : commutable

```

Procedure pair: MC vs MG | Reference material: RMB

```

Mean X                  : 22.7
Mean Y                  : 19.1

```

```

Predicted Y           : 19.80131
Residual              : -0.70131041
Prediction SE         : 1.9658813
Lower                 : 15.948254
Upper                 : 23.654367
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 1.1973364
Relative residual      : 0.58572544
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MC
procedure_y            : MG
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MC vs MG | Reference material: RMC

```

Mean X                : 33.6
Mean Y                : 28.8
Predicted Y           : 29.468613
Residual              : -0.6686134
Prediction SE         : 1.9783176
Lower                 : 25.591182
Upper                 : 33.346045
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 1.1973364
Relative residual      : 0.55841732
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MC
procedure_y            : MG
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MC vs MG | Reference material: RMD

```

Mean X                : 40.3
Mean Y                : 35.8
Predicted Y           : 35.410901
Residual              : 0.38909944
Prediction SE         : 1.9976373
Lower                 : 31.495603
Upper                 : 39.326198
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 1.1973364
Relative residual      : -0.32497085
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MC
procedure_y            : MG
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MC vs MG | Reference material: RME

```

Mean X           : 17.3
Mean Y           : 14.8
Predicted Y      : 15.012004
Residual         : -0.21200435
Prediction SE    : 1.9685578
Lower           : 11.153702
Upper           : 18.870307
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.1973364
Relative residual : 0.17706331
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MC
procedure_y     : MG
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MC vs MG | Reference material: RMF

```

Mean X           : 23.3
Mean Y           : 20.1
Predicted Y      : 20.333456
Residual         : -0.23345553
Prediction SE    : 1.965946
Lower           : 16.480272
Upper           : 24.186639
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.1973364
Relative residual : 0.19497906
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MC
procedure_y     : MG
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MC vs MG | Reference material: RMG

```

Mean X           : 19.2
Mean Y           : 16.4
Predicted Y      : 16.697131
Residual         : -0.29713055
Prediction SE    : 1.9669477
Lower           : 12.841984
Upper           : 20.552277
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.1973364
Relative residual : 0.24815962
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MC
procedure_y     : MG
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MC vs MH | Reference material: RMA

```

Mean X           : 11.3
Mean Y           : 9.1
Predicted Y      : 9.6927541
Residual         : -0.59275409
Prediction SE    : 2.0455194
Lower           : 5.6836097
Upper           : 13.701898
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.1971996
Relative residual : 0.4951172
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MC
procedure_y     : MH
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MC vs MH | Reference material: RMB

```

Mean X           : 22.7
Mean Y           : 19.1
Predicted Y      : 19.801264
Residual         : -0.70126389
Prediction SE    : 2.0334388
Lower           : 15.815797
Upper           : 23.786731
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.1971996
Relative residual : 0.58575355
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MC
procedure_y     : MH
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MC vs MH | Reference material: RMC

```

Mean X           : 33.6
Mean Y           : 28.8
Predicted Y      : 29.466418
Residual         : -0.66641799
Prediction SE    : 2.045459
Lower           : 25.457392
Upper           : 33.475444
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.1971996
Relative residual : 0.55664737
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MC
procedure_y     : MH
critical_value   : 1.959964

```

```
rm_set_size      : 1
classification   : commutable
```

Procedure pair: MC vs MH | Reference material: RMD

```
Mean X           : 40.3
Mean Y           : 35.8
Predicted Y      : 35.407384
Residual         : 0.39261573
Prediction SE     : 2.0641421
Lower            : 31.36174
Upper            : 39.453028
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 1.1971996
Relative residual : -0.3279451
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MC
procedure_y      : MH
critical_value    : 1.959964
rm_set_size      : 1
classification   : commutable
```

Procedure pair: MC vs MH | Reference material: RME

```
Mean X           : 17.3
Mean Y           : 14.8
Predicted Y      : 15.013022
Residual         : -0.21302241
Prediction SE     : 2.0360253
Lower            : 11.022486
Upper            : 19.003559
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 1.1971996
Relative residual : 0.17793392
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MC
procedure_y      : MH
critical_value    : 1.959964
rm_set_size      : 1
classification   : commutable
```

Procedure pair: MC vs MH | Reference material: RMF

```
Mean X           : 23.3
Mean Y           : 20.1
Predicted Y      : 20.333291
Residual         : -0.23329072
Prediction SE     : 2.0335013
Lower            : 16.347701
Upper            : 24.31888
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 1.1971996
Relative residual : 0.19486369
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MC
```

```

procedure_y      : MH
critical_value   : 1.959964
rm_set_size      : 1
classification   : commutable

```

Procedure pair: MC vs MH | Reference material: RMG

```

Mean X          : 19.2
Mean Y          : 16.4
Predicted Y     : 16.697774
Residual        : -0.29777404
Prediction SE    : 2.0344693
Lower           : 12.710288
Upper           : 20.68526
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.1971996
Relative residual : 0.24872548
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MC
procedure_y     : MH
critical_value   : 1.959964
rm_set_size      : 1
classification   : commutable

```

Procedure pair: MC vs MI | Reference material: RMA

```

Mean X          : 11.3
Mean Y          : 9.8
Predicted Y     : 10.032479
Residual        : -0.23247916
Prediction SE    : 1.9638332
Lower           : 6.1834368
Upper           : 13.881522
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.1089276
Relative residual : 0.20964322
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MC
procedure_y     : MI
critical_value   : 1.959964
rm_set_size      : 1
classification   : commutable

```

Procedure pair: MC vs MI | Reference material: RMB

```

Mean X          : 22.7
Mean Y          : 19.5
Predicted Y     : 20.006479
Residual        : -0.5064793
Prediction SE    : 1.9530739
Lower           : 16.178525
Upper           : 23.834434
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.1089276
Relative residual : 0.45672891
Relative-residual limit : 2

```

```

Relative-residual classification : commutable
procedure_x                      : MC
procedure_y                      : MI
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MC vs MI | Reference material: RMC

```

Mean X                          : 33.6
Mean Y                          : 28.6
Predicted Y                     : 29.543023
Residual                       : -0.9430233
Prediction SE                   : 1.9637794
Lower                           : 25.694086
Upper                           : 33.39196
Extrapolated                   : FALSE
Prediction classification       : commutable
Sy.x                           : 1.1089276
Relative residual               : 0.85039211
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                     : MC
procedure_y                     : MI
critical_value                  : 1.959964
rm_set_size                    : 1
classification                  : commutable

```

Procedure pair: MC vs MI | Reference material: RMD

```

Mean X                          : 40.3
Mean Y                          : 35
Predicted Y                     : 35.404936
Residual                       : -0.40493566
Prediction SE                   : 1.980428
Lower                           : 31.523368
Upper                           : 39.286503
Extrapolated                   : FALSE
Prediction classification       : commutable
Sy.x                           : 1.1089276
Relative residual               : 0.36515969
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                     : MC
procedure_y                     : MI
critical_value                  : 1.959964
rm_set_size                    : 1
classification                  : commutable

```

Procedure pair: MC vs MI | Reference material: RME

```

Mean X                          : 17.3
Mean Y                          : 15
Predicted Y                     : 15.281953
Residual                       : -0.28195292
Prediction SE                   : 1.9553771
Lower                           : 11.449484
Upper                           : 19.114422
Extrapolated                   : FALSE
Prediction classification       : commutable
Sy.x                           : 1.1089276

```

```

Relative residual      : 0.25425728
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MC
procedure_y           : MI
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MC vs MI | Reference material: RMF

```

Mean X      : 23.3
Mean Y      : 26
Predicted Y  : 20.531427
Residual     : 5.4685733
Prediction SE : 1.9531296
Lower        : 16.703363
Upper        : 24.35949
Extrapolated : FALSE
Prediction classification : noncommutable
Sy.x         : 1.1089276
Relative residual : -4.9314069
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x   : MC
procedure_y   : MI
critical_value : 1.959964
rm_set_size   : 1
classification : noncommutable

```

Procedure pair: MC vs MI | Reference material: RMG

```

Mean X      : 19.2
Mean Y      : 16.7
Predicted Y  : 16.944286
Residual     : -0.24428627
Prediction SE : 1.9539915
Lower        : 13.114533
Upper        : 20.774039
Extrapolated : FALSE
Prediction classification : commutable
Sy.x         : 1.1089276
Relative residual : 0.22029055
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x   : MC
procedure_y   : MI
critical_value : 1.959964
rm_set_size   : 1
classification : commutable

```

Procedure pair: MC vs MJ | Reference material: RMA

```

Mean X      : 11.3
Mean Y      : 8.5
Predicted Y  : 7.8093287
Residual     : 0.69067131
Prediction SE : 1.95222
Lower        : 3.9830478
Upper        : 11.63561
Extrapolated : FALSE

```



```

Prediction classification      : commutable
Sy.x                         : 1.1135031
Relative residual             : -0.62026885
Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                  : MC
procedure_y                  : MJ
critical_value                : 1.959964
rm_set_size                   : 1
classification                : commutable

```

Procedure pair: MC vs MJ | Reference material: RMB

```

Mean X                       : 22.7
Mean Y                       : 17
Predicted Y                  : 16.389627
Residual                     : 0.61037277
Prediction SE                 : 1.9412694
Lower                        : 12.584809
Upper                        : 20.194445
Extrapolated                 : FALSE
Prediction classification     : commutable
Sy.x                         : 1.1135031
Relative residual            : -0.54815541
Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                  : MC
procedure_y                  : MJ
critical_value                : 1.959964
rm_set_size                   : 1
classification                : commutable

```

Procedure pair: MC vs MJ | Reference material: RMC

```

Mean X                       : 33.6
Mean Y                       : 25.3
Predicted Y                  : 24.593597
Residual                     : 0.70640311
Prediction SE                 : 1.9521652
Lower                        : 20.767423
Upper                        : 28.41977
Extrapolated                 : FALSE
Prediction classification     : commutable
Sy.x                         : 1.1135031
Relative residual            : -0.63439706
Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                  : MC
procedure_y                  : MJ
critical_value                : 1.959964
rm_set_size                   : 1
classification                : commutable

```

Procedure pair: MC vs MJ | Reference material: RMD

```

Mean X                       : 40.3
Mean Y                       : 30.6
Predicted Y                  : 29.636404
Residual                     : 0.96359607
Prediction SE                 : 1.969107
Lower                        : 25.777025

```

```

Upper                : 33.495783
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 1.1135031
Relative residual     : -0.86537348
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MC
procedure_y           : MJ
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MC vs MJ | Reference material: RME

```

Mean X               : 17.3
Mean Y               : 12
Predicted Y          : 12.325275
Residual              : -0.32527529
Prediction SE         : 1.9436137
Lower                : 8.5158625
Upper                : 16.134688
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 1.1135031
Relative residual     : 0.29211889
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MC
procedure_y           : MJ
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MC vs MJ | Reference material: RMF

```

Mean X               : 23.3
Mean Y               : 21
Predicted Y          : 16.841222
Residual              : 4.1587781
Prediction SE         : 1.9413261
Lower                : 13.036293
Upper                : 20.646151
Extrapolated         : FALSE
Prediction classification : noncommutable
Sy.x                 : 1.1135031
Relative residual     : -3.7348598
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x          : MC
procedure_y           : MJ
critical_value        : 1.959964
rm_set_size           : 1
classification        : noncommutable

```

Procedure pair: MC vs MJ | Reference material: RMG

```

Mean X               : 19.2
Mean Y               : 13.4
Predicted Y          : 13.755325
Residual              : -0.35532505

```

```

Prediction SE          : 1.9422033
Lower                  : 9.9486765
Upper                  : 17.561974
Extrapolated           : FALSE
Prediction classification : commutable
Sy.x                   : 1.1135031
Relative residual       : 0.31910557
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x            : MC
procedure_y            : MJ
critical_value          : 1.959964
rm_set_size            : 1
classification          : commutable

```

Procedure pair: MD vs ME | Reference material: RMA

```

Mean X                  : 11.4
Mean Y                  : 12
Predicted Y             : 14.159035
Residual                 : -2.1590353
Prediction SE           : 2.5863658
Lower                   : 9.0898515
Upper                   : 19.228219
Extrapolated            : FALSE
Prediction classification : commutable
Sy.x                    : 0.83905695
Relative residual        : 2.5731689
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x             : MD
procedure_y             : ME
critical_value          : 1.959964
rm_set_size            : 1
classification          : commutable

```

Procedure pair: MD vs ME | Reference material: RMB

```

Mean X                  : 19.9
Mean Y                  : 24.1
Predicted Y             : 25.717998
Residual                 : -1.6179979
Prediction SE           : 2.5840483
Lower                   : 20.653356
Upper                   : 30.78264
Extrapolated            : FALSE
Prediction classification : commutable
Sy.x                    : 0.83905695
Relative residual        : 1.9283529
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x             : MD
procedure_y             : ME
critical_value          : 1.959964
rm_set_size            : 1
classification          : commutable

```

Procedure pair: MD vs ME | Reference material: RMC

```

Mean X                  : 29.6
Mean Y                  : 35.9

```

```

Predicted Y           : 38.908814
Residual              : -3.008814
Prediction SE         : 2.5920155
Lower                 : 33.828557
Upper                 : 43.989071
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 0.83905695
Relative residual      : 3.5859473
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : MD
procedure_y           : ME
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MD vs ME | Reference material: RMD

```

Mean X                : 34.6
Mean Y                : 43.5
Predicted Y           : 45.708204
Residual              : -2.2082038
Prediction SE         : 2.6005121
Lower                 : 40.611294
Upper                 : 50.805114
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 0.83905695
Relative residual      : 2.6317687
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : MD
procedure_y           : ME
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MD vs ME | Reference material: RME

```

Mean X                : 13.2
Mean Y                : 17.1
Predicted Y           : 16.606816
Residual              : 0.49318443
Prediction SE         : 2.5851498
Lower                 : 11.540015
Upper                 : 21.673616
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 0.83905695
Relative residual      : -0.58778422
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MD
procedure_y           : ME
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MD vs ME | Reference material: RMF

```

Mean X           : 24.9
Mean Y           : 23.4
Predicted Y      : 32.517388
Residual         : -9.1173876
Prediction SE    : 2.5867457
Lower           : 27.447459
Upper           : 37.587316
Extrapolated     : FALSE
Prediction classification : noncommutable
Sy.x            : 0.83905695
Relative residual : 10.866232
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MD
procedure_y      : ME
critical_value    : 1.959964
rm_set_size      : 1
classification    : noncommutable

```

Procedure pair: MD vs ME | Reference material: RMG

```

Mean X           : 14.7
Mean Y           : 19
Predicted Y      : 18.646632
Residual         : 0.3533675
Prediction SE    : 2.5844339
Lower           : 13.581235
Upper           : 23.71203
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x            : 0.83905695
Relative residual : -0.42114841
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MD
procedure_y      : ME
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MD vs MF | Reference material: RMA

```

Mean X           : 11.4
Mean Y           : 9.5
Predicted Y      : 11.436682
Residual         : -1.9366822
Prediction SE    : 2.346992
Lower           : 6.8366624
Upper           : 16.036702
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x            : 1.0101669
Relative residual : 1.9171903
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MD
procedure_y      : MF
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MD vs MF | Reference material: RMB

```

Mean X           : 19.9
Mean Y           : 19
Predicted Y      : 20.650648
Residual         : -1.650648
Prediction SE     : 2.3432735
Lower            : 16.057916
Upper            : 25.24338
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 1.0101669
Relative residual : 1.634035
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MD
procedure_y      : MF
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MD vs MF | Reference material: RMC

```

Mean X           : 29.6
Mean Y           : 28.8
Predicted Y      : 31.165409
Residual         : -2.365409
Prediction SE     : 2.3560467
Lower            : 26.547642
Upper            : 35.783176
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 1.0101669
Relative residual : 2.3416022
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MD
procedure_y      : MF
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MD vs MF | Reference material: RMD

```

Mean X           : 34.6
Mean Y           : 35.1
Predicted Y      : 36.585389
Residual         : -1.4853889
Prediction SE     : 2.369636
Lower            : 31.940988
Upper            : 41.22979
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 1.0101669
Relative residual : 1.4704391
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MD
procedure_y      : MF
critical_value    : 1.959964

```

```

rm_set_size          : 1
classification       : commutable

```

Procedure pair: MD vs MF | Reference material: RME

```

Mean X               : 13.2
Mean Y               : 14
Predicted Y          : 13.387875
Residual             : 0.61212504
Prediction SE        : 2.3450411
Lower               : 8.7916788
Upper               : 17.984071
Extrapolated        : FALSE
Prediction classification : commutable
Sy.x                : 1.0101669
Relative residual    : -0.60596427
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x         : MD
procedure_y         : MF
critical_value       : 1.959964
rm_set_size         : 1
classification       : commutable

```

Procedure pair: MD vs MF | Reference material: RMF

```

Mean X               : 24.9
Mean Y               : 20
Predicted Y          : 26.070628
Residual             : -6.0706279
Prediction SE        : 2.3476013
Lower               : 21.469414
Upper               : 30.671842
Extrapolated        : FALSE
Prediction classification : noncommutable
Sy.x                : 1.0101669
Relative residual    : 6.0095297
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x         : MD
procedure_y         : MF
critical_value       : 1.959964
rm_set_size         : 1
classification       : noncommutable

```

Procedure pair: MD vs MF | Reference material: RMG

```

Mean X               : 14.7
Mean Y               : 15.5
Predicted Y          : 15.013869
Residual             : 0.48613107
Prediction SE        : 2.3438923
Lower               : 10.419924
Upper               : 19.607813
Extrapolated        : FALSE
Prediction classification : commutable
Sy.x                : 1.0101669
Relative residual    : -0.48123837
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x         : MD

```

```

procedure_y      : MF
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MD vs MG | Reference material: RMA

```

Mean X          : 11.4
Mean Y          : 9.1
Predicted Y     : 12.253353
Residual        : -3.1533527
Prediction SE    : 2.3088294
Lower           : 7.7281302
Upper           : 16.778575
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 0.87608703
Relative residual : 3.5993601
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x     : MD
procedure_y     : MG
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MD vs MG | Reference material: RMB

```

Mean X          : 19.9
Mean Y          : 19.1
Predicted Y     : 21.826481
Residual        : -2.7264807
Prediction SE    : 2.3059927
Lower           : 17.306818
Upper           : 26.346143
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 0.87608703
Relative residual : 3.1121117
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x     : MD
procedure_y     : MG
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MD vs MG | Reference material: RMC

```

Mean X          : 29.6
Mean Y          : 28.8
Predicted Y     : 32.751109
Residual        : -3.9511092
Prediction SE    : 2.3157411
Lower           : 28.21234
Upper           : 37.289878
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 0.87608703
Relative residual : 4.5099506
Relative-residual limit : 2

```



```

Relative-residual classification : noncommutable
procedure_x                     : MD
procedure_y                     : MG
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MD vs MG | Reference material: RMD

```

Mean X                          : 34.6
Mean Y                          : 35.8
Predicted Y                     : 38.382361
Residual                       : -2.582361
Prediction SE                   : 2.3261251
Lower                           : 33.82324
Upper                           : 42.941482
Extrapolated                   : FALSE
Prediction classification       : commutable
Sy.x                           : 0.87608703
Relative residual               : 2.9476079
Relative-residual limit        : 2
Relative-residual classification : noncommutable
procedure_x                     : MD
procedure_y                     : MG
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MD vs MG | Reference material: RME

```

Mean X                          : 13.2
Mean Y                          : 14.8
Predicted Y                     : 14.280603
Residual                       : 0.5193967
Prediction SE                   : 2.307341
Lower                           : 9.758298
Upper                           : 18.802909
Extrapolated                   : FALSE
Prediction classification       : commutable
Sy.x                           : 0.87608703
Relative residual               : -0.59285971
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                     : MD
procedure_y                     : MG
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MD vs MG | Reference material: RMF

```

Mean X                          : 24.9
Mean Y                          : 20.1
Predicted Y                     : 27.457733
Residual                       : -7.3577325
Prediction SE                   : 2.3092943
Lower                           : 22.931599
Upper                           : 31.983866
Extrapolated                   : FALSE
Prediction classification       : noncommutable
Sy.x                           : 0.87608703

```

```

Relative residual      : 8.3984037
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : MD
procedure_y           : MG
critical_value         : 1.959964
rm_set_size           : 1
classification         : noncommutable

```

Procedure pair: MD vs MG | Reference material: RMG

```

Mean X                : 14.7
Mean Y                : 16.4
Predicted Y           : 15.969979
Residual              : 0.43002116
Prediction SE         : 2.3064647
Lower                 : 11.449391
Upper                 : 20.490567
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 0.87608703
Relative residual      : -0.49084297
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MD
procedure_y           : MG
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MD vs MH | Reference material: RMA

```

Mean X                : 11.4
Mean Y                : 9.1
Predicted Y           : 12.254145
Residual              : -3.1541448
Prediction SE         : 2.3666331
Lower                 : 7.6156291
Upper                 : 16.89266
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 0.87602552
Relative residual      : 3.600517
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : MD
procedure_y           : MH
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MD vs MH | Reference material: RMB

```

Mean X                : 19.9
Mean Y                : 19.1
Predicted Y           : 21.82624
Residual              : -2.7262395
Prediction SE         : 2.3638664
Lower                 : 17.193147
Upper                 : 26.459332
Extrapolated          : FALSE

```

```

Prediction classification      : commutable
Sy.x                         : 0.87602552
Relative residual             : 3.1120549
Relative-residual limit      : 2
Relative-residual classification : noncommutable
procedure_x                   : MD
procedure_y                   : MH
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: MD vs MH | Reference material: RMC

```

Mean X                       : 29.6
Mean Y                       : 28.8
Predicted Y                  : 32.749689
Residual                     : -3.9496888
Prediction SE                 : 2.373375
Lower                        : 28.097959
Upper                        : 37.401418
Extrapolated                 : FALSE
Prediction classification     : commutable
Sy.x                         : 0.87602552
Relative residual             : 4.5086458
Relative-residual limit      : 2
Relative-residual classification : noncommutable
procedure_x                   : MD
procedure_y                   : MH
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: MD vs MH | Reference material: RMD

```

Mean X                       : 34.6
Mean Y                       : 35.8
Predicted Y                  : 38.380333
Residual                     : -2.5803328
Prediction SE                 : 2.3835058
Lower                        : 33.708747
Upper                        : 43.051918
Extrapolated                 : FALSE
Prediction classification     : commutable
Sy.x                         : 0.87602552
Relative residual             : 2.9454996
Relative-residual limit      : 2
Relative-residual classification : noncommutable
procedure_x                   : MD
procedure_y                   : MH
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: MD vs MH | Reference material: RME

```

Mean X                       : 13.2
Mean Y                       : 14.8
Predicted Y                  : 14.281177
Residual                     : 0.51882339
Prediction SE                 : 2.3651814
Lower                        : 9.6455062

```

```

Upper                : 18.916847
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.87602552
Relative residual     : -0.59224689
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MD
procedure_y          : MH
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MD vs MH | Reference material: RMF

```

Mean X               : 24.9
Mean Y               : 20.1
Predicted Y          : 27.456883
Residual              : -7.3568835
Prediction SE         : 2.3670866
Lower                : 22.817479
Upper                : 32.096288
Extrapolated         : FALSE
Prediction classification : noncommutable
Sy.x                 : 0.87602552
Relative residual     : 8.3980241
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x          : MD
procedure_y          : MH
critical_value        : 1.959964
rm_set_size           : 1
classification        : noncommutable

```

Procedure pair: MD vs MH | Reference material: RMG

```

Mean X               : 14.7
Mean Y               : 16.4
Predicted Y          : 15.97037
Residual              : 0.4296302
Prediction SE         : 2.3643267
Lower                : 11.336375
Upper                : 20.604365
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.87602552
Relative residual     : -0.49043114
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MD
procedure_y          : MH
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MD vs MI | Reference material: RMA

```

Mean X               : 11.4
Mean Y               : 9.8
Predicted Y          : 12.555309
Residual              : -2.7553091

```

Prediction SE	: 2.2970197
Lower	: 8.0532333
Upper	: 17.057385
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.0821696
Relative residual	: 2.5460973
Relative-residual limit	: 2
Relative-residual classification	: noncommutable
procedure_x	: MD
procedure_y	: MI
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MD vs MI | Reference material: RMB

Mean X	: 19.9
Mean Y	: 19.5
Predicted Y	: 22.005873
Residual	: -2.5058727
Prediction SE	: 2.2926602
Lower	: 17.512341
Upper	: 26.499404
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.0821696
Relative residual	: 2.3156007
Relative-residual limit	: 2
Relative-residual classification	: noncommutable
procedure_x	: MD
procedure_y	: MI
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MD vs MI | Reference material: RMC

Mean X	: 29.6
Mean Y	: 28.6
Predicted Y	: 32.790634
Residual	: -4.1906335
Prediction SE	: 2.3076294
Lower	: 28.267763
Upper	: 37.313504
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.0821696
Relative residual	: 3.8724368
Relative-residual limit	: 2
Relative-residual classification	: noncommutable
procedure_x	: MD
procedure_y	: MI
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MD vs MI | Reference material: RMD

Mean X : 34.6
Mean Y : 35

```

Predicted Y           : 38.349789
Residual              : -3.3497886
Prediction SE         : 2.3235373
Lower                 : 33.795739
Upper                 : 42.903838
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 1.0821696
Relative residual     : 3.0954376
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : MD
procedure_y           : MI
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MD vs MI | Reference material: RME

```

Mean X                : 13.2
Mean Y                : 15
Predicted Y           : 14.556605
Residual              : 0.44339505
Prediction SE         : 2.2947327
Lower                 : 10.059011
Upper                 : 19.054198
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 1.0821696
Relative residual     : -0.40972786
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MD
procedure_y           : MI
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MD vs MI | Reference material: RMF

```

Mean X                : 24.9
Mean Y                : 26
Predicted Y           : 27.565028
Residual              : -1.5650278
Prediction SE         : 2.2977339
Lower                 : 23.061552
Upper                 : 32.068503
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 1.0821696
Relative residual     : 1.4461945
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MD
procedure_y           : MI
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MD vs MI | Reference material: RMG

```

Mean X           : 14.7
Mean Y           : 16.7
Predicted Y      : 16.224351
Residual         : 0.47564853
Prediction SE    : 2.2933858
Lower           : 11.729398
Upper           : 20.719305
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x            : 1.0821696
Relative residual : -0.43953232
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MD
procedure_y      : MI
critical_value   : 1.959964
rm_set_size      : 1
classification   : commutable

```

Procedure pair: MD vs MJ | Reference material: RMA

```

Mean X           : 11.4
Mean Y           : 8.5
Predicted Y      : 9.9788234
Residual         : -1.4788234
Prediction SE    : 2.1926862
Lower           : 5.6812375
Upper           : 14.276409
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x            : 0.57964829
Relative residual : 2.5512426
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MD
procedure_y      : MJ
critical_value   : 1.959964
rm_set_size      : 1
classification   : commutable

```

Procedure pair: MD vs MJ | Reference material: RMB

```

Mean X           : 19.9
Mean Y           : 17
Predicted Y      : 18.109886
Residual         : -1.1098856
Prediction SE    : 2.1913825
Lower           : 13.814855
Upper           : 22.404916
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x            : 0.57964829
Relative residual : 1.9147569
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MD
procedure_y      : MJ
critical_value   : 1.959964
rm_set_size      : 1
classification   : commutable

```

Procedure pair: MD vs MJ | Reference material: RMC

Mean X	: 29.6
Mean Y	: 25.3
Predicted Y	: 27.388862
Residual	: -2.0888624
Prediction SE	: 2.1958659
Lower	: 23.085044
Upper	: 31.69268
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 0.57964829
Relative residual	: 3.6036721
Relative-residual limit	: 2
Relative-residual classification	: noncommutable
procedure_x	: MD
procedure_y	: MJ
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MD vs MJ | Reference material: RMD

Mean X	: 34.6
Mean Y	: 30.6
Predicted Y	: 32.17184
Residual	: -1.5718401
Prediction SE	: 2.2006524
Lower	: 27.858641
Upper	: 36.485039
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 0.57964829
Relative residual	: 2.7117135
Relative-residual limit	: 2
Relative-residual classification	: noncommutable
procedure_x	: MD
procedure_y	: MJ
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MD vs MJ | Reference material: RME

Mean X	: 13.2
Mean Y	: 12
Predicted Y	: 11.700695
Residual	: 0.29930459
Prediction SE	: 2.1920021
Lower	: 7.4044503
Upper	: 15.996941
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 0.57964829
Relative residual	: -0.51635551
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MD
procedure_y	: MJ
critical_value	: 1.959964


```
rm_set_size      : 1
classification   : commutable
```

Procedure pair: MD vs MJ | Reference material: RMF

```
Mean X          : 24.9
Mean Y          : 21
Predicted Y     : 22.892863
Residual        : -1.8928633
Prediction SE    : 2.1928999
Lower          : 18.594858
Upper          : 27.190868
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x           : 0.57964829
Relative residual : 3.2655376
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x     : MD
procedure_y     : MJ
critical_value  : 1.959964
rm_set_size     : 1
classification  : commutable
```

Procedure pair: MD vs MJ | Reference material: RMG

```
Mean X          : 14.7
Mean Y          : 13.4
Predicted Y     : 13.135589
Residual        : 0.26441127
Prediction SE    : 2.1915994
Lower          : 8.8401329
Upper          : 17.431045
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x           : 0.57964829
Relative residual : -0.45615812
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MD
procedure_y     : MJ
critical_value  : 1.959964
rm_set_size     : 1
classification  : commutable
```

Procedure pair: ME vs MF | Reference material: RMA

```
Mean X          : 12
Mean Y          : 9.5
Predicted Y     : 9.7214603
Residual        : -0.22146028
Prediction SE    : 1.9560546
Lower          : 5.8876637
Upper          : 13.555257
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x           : 0.77523923
Relative residual : 0.28566702
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : ME
```

```

procedure_y      : MF
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: ME vs MF | Reference material: RMB

```

Mean X           : 24.1
Mean Y           : 19
Predicted Y      : 19.360329
Residual         : -0.36032876
Prediction SE     : 1.951629
Lower            : 15.535206
Upper            : 23.185451
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.77523923
Relative residual : 0.46479686
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : ME
procedure_y      : MF
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: ME vs MF | Reference material: RMC

```

Mean X           : 35.9
Mean Y           : 28.8
Predicted Y      : 28.760217
Residual         : 0.039782979
Prediction SE     : 1.9576878
Lower            : 24.923219
Upper            : 32.597215
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.77523923
Relative residual : -0.051317036
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : ME
procedure_y      : MF
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: ME vs MF | Reference material: RMD

```

Mean X           : 43.5
Mean Y           : 35.1
Predicted Y      : 34.814382
Residual         : 0.28561766
Prediction SE     : 1.9669765
Lower            : 30.959179
Upper            : 38.669585
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.77523923
Relative residual : -0.36842519
Relative-residual limit : 2

```

```

Relative-residual classification : commutable
procedure_x                     : ME
procedure_y                     : MF
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: ME vs MF | Reference material: RME

```

Mean X                          : 17.1
Mean Y                          : 14
Predicted Y                     : 13.784124
Residual                       : 0.21587615
Prediction SE                   : 1.9528767
Lower                          : 9.9565559
Upper                          : 17.611692
Extrapolated                   : FALSE
Prediction classification       : commutable
Sy.x                           : 0.77523923
Relative residual               : -0.27846391
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                     : ME
procedure_y                     : MF
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: ME vs MF | Reference material: RMF

```

Mean X                          : 23.4
Mean Y                          : 20
Predicted Y                     : 18.802708
Residual                       : 1.1972917
Prediction SE                   : 1.9515914
Lower                          : 14.977659
Upper                          : 22.627757
Extrapolated                   : FALSE
Prediction classification       : commutable
Sy.x                           : 0.77523923
Relative residual               : -1.5444158
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                     : ME
procedure_y                     : MF
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: ME vs MF | Reference material: RMG

```

Mean X                          : 19
Mean Y                          : 15.5
Predicted Y                     : 15.297665
Residual                       : 0.20233482
Prediction SE                   : 1.9521813
Lower                          : 11.47146
Upper                          : 19.12387
Extrapolated                   : FALSE
Prediction classification       : commutable
Sy.x                           : 0.77523923

```

```

Relative residual      : -0.26099662
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : ME
procedure_y           : MF
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: ME vs MG | Reference material: RMA

```

Mean X                : 12
Mean Y                : 9.1
Predicted Y           : 10.468498
Residual              : -1.3684984
Prediction SE         : 1.8733729
Lower                 : 6.796755
Upper                 : 14.140242
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 0.52435434
Relative residual      : 2.6098733
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : ME
procedure_y           : MG
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: ME vs MG | Reference material: RMB

```

Mean X                : 24.1
Mean Y                : 19.1
Predicted Y           : 20.486136
Residual              : -1.3861363
Prediction SE         : 1.8712641
Lower                 : 16.818526
Upper                 : 24.153747
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 0.52435434
Relative residual      : 2.6435106
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : ME
procedure_y           : MG
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: ME vs MG | Reference material: RMC

```

Mean X                : 35.9
Mean Y                : 28.8
Predicted Y           : 30.255403
Residual              : -1.455403
Prediction SE         : 1.8741517
Lower                 : 26.582133
Upper                 : 33.928673
Extrapolated          : FALSE

```

```

Prediction classification      : commutable
Sy.x                         : 0.52435434
Relative residual             : 2.7756096
Relative-residual limit      : 2
Relative-residual classification : noncommutable
procedure_x                   : ME
procedure_y                   : MG
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: ME vs MG | Reference material: RMD

```

Mean X                       : 43.5
Mean Y                       : 35.8
Predicted Y                  : 36.547473
Residual                     : -0.74747302
Prediction SE                 : 1.8785874
Lower                        : 32.865509
Upper                        : 40.229437
Extrapolated                 : FALSE
Prediction classification     : commutable
Sy.x                         : 0.52435434
Relative residual            : 1.4255113
Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                   : ME
procedure_y                   : MG
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: ME vs MG | Reference material: RME

```

Mean X                       : 17.1
Mean Y                       : 14.8
Predicted Y                  : 14.690809
Residual                     : 0.10919142
Prediction SE                 : 1.8718584
Lower                        : 11.022034
Upper                        : 18.359584
Extrapolated                 : FALSE
Prediction classification     : commutable
Sy.x                         : 0.52435434
Relative residual            : -0.20823976
Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                   : ME
procedure_y                   : MG
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: ME vs MG | Reference material: RMF

```

Mean X                       : 23.4
Mean Y                       : 20.1
Predicted Y                  : 19.906604
Residual                     : 0.1933965
Prediction SE                 : 1.8712462
Lower                        : 16.239028

```

```

Upper                : 23.574179
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.52435434
Relative residual     : -0.36882788
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : ME
procedure_y           : MG
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: ME vs MG | Reference material: RMG

```

Mean X               : 19
Mean Y               : 16.4
Predicted Y          : 16.263826
Residual             : 0.1361739
Prediction SE         : 1.8715272
Lower                : 12.5957
Upper                : 19.931952
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.52435434
Relative residual     : -0.25969825
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : ME
procedure_y           : MG
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: ME vs MH | Reference material: RMA

```

Mean X               : 12
Mean Y               : 9.1
Predicted Y          : 10.468884
Residual             : -1.3688839
Prediction SE         : 1.9443149
Lower                : 6.6580966
Upper                : 14.279671
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.52434405
Relative residual     : 2.6106597
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x          : ME
procedure_y           : MH
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: ME vs MH | Reference material: RMB

```

Mean X               : 24.1
Mean Y               : 19.1
Predicted Y          : 20.486098
Residual             : -1.3860985

```

```

Prediction SE          : 1.9422834
Lower                  : 16.679293
Upper                  : 24.292904
Extrapolated           : FALSE
Prediction classification : commutable
Sy.x                   : 0.52434405
Relative residual       : 2.6434904
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x             : ME
procedure_y              : MH
critical_value           : 1.959964
rm_set_size              : 1
classification           : commutable

```

Procedure pair: ME vs MH | Reference material: RMC

```

Mean X                  : 35.9
Mean Y                  : 28.8
Predicted Y             : 30.254952
Residual                 : -1.4549524
Prediction SE            : 1.9450653
Lower                   : 26.442694
Upper                   : 34.06721
Extrapolated            : FALSE
Prediction classification : commutable
Sy.x                     : 0.52434405
Relative residual        : 2.7748048
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x              : ME
procedure_y               : MH
critical_value            : 1.959964
rm_set_size               : 1
classification            : commutable

```

Procedure pair: ME vs MH | Reference material: RMD

```

Mean X                  : 43.5
Mean Y                  : 35.8
Predicted Y             : 36.546757
Residual                 : -0.74675661
Prediction SE            : 1.9493392
Lower                   : 32.726122
Upper                   : 40.367391
Extrapolated            : FALSE
Prediction classification : commutable
Sy.x                     : 0.52434405
Relative residual        : 1.4241729
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x              : ME
procedure_y               : MH
critical_value            : 1.959964
rm_set_size               : 1
classification            : commutable

```

Procedure pair: ME vs MH | Reference material: RME

```

Mean X                  : 17.1
Mean Y                  : 14.8

```

```

Predicted Y           : 14.691016
Residual              : 0.10898433
Prediction SE         : 1.9428558
Lower                 : 10.883088
Upper                 : 18.498943
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 0.52434405
Relative residual     : -0.2078489
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : ME
procedure_y           : MH
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: ME vs MH | Reference material: RMF

```

Mean X                : 23.4
Mean Y                : 20.1
Predicted Y           : 19.90659
Residual              : 0.19340979
Prediction SE         : 1.9422661
Lower                 : 16.099819
Upper                 : 23.713362
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 0.52434405
Relative residual     : -0.36886046
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : ME
procedure_y           : MH
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: ME vs MH | Reference material: RMG

```

Mean X                : 19
Mean Y                : 16.4
Predicted Y           : 16.263967
Residual              : 0.13603328
Prediction SE         : 1.9425367
Lower                 : 12.456665
Upper                 : 20.071269
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 0.52434405
Relative residual     : -0.25943515
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : ME
procedure_y           : MH
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: ME vs MI | Reference material: RMA


```

Mean X           : 12
Mean Y           : 9.8
Predicted Y      : 10.787694
Residual         : -0.9876935
Prediction SE    : 1.8675977
Lower           : 7.1272692
Upper           : 14.448118
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 0.63377684
Relative residual : 1.5584247
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : ME
procedure_y     : MI
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: ME vs MI | Reference material: RMB

```

Mean X           : 24.1
Mean Y           : 19.5
Predicted Y      : 20.683239
Residual         : -1.1832387
Prediction SE    : 1.8645064
Lower           : 17.028873
Upper           : 24.337604
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 0.63377684
Relative residual : 1.8669643
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : ME
procedure_y     : MI
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: ME vs MI | Reference material: RMC

```

Mean X           : 35.9
Mean Y           : 28.6
Predicted Y      : 30.33344
Residual         : -1.7334399
Prediction SE    : 1.868739
Lower           : 26.670779
Upper           : 33.996101
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 0.63377684
Relative residual : 2.7350951
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x     : ME
procedure_y     : MI
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: ME vs MI | Reference material: RMD

```

Mean X           : 43.5
Mean Y           : 35
Predicted Y      : 36.548824
Residual         : -1.5488237
Prediction SE     : 1.875235
Lower            : 32.873431
Upper            : 40.224217
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.63377684
Relative residual : 2.4437998
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : ME
procedure_y      : MI
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: ME vs MI | Reference material: RME

```

Mean X           : 17.1
Mean Y           : 15
Predicted Y      : 14.958543
Residual         : 0.041456849
Prediction SE     : 1.8653777
Lower            : 11.30247
Upper            : 18.614616
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.63377684
Relative residual : -0.065412376
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : ME
procedure_y      : MI
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: ME vs MI | Reference material: RMF

```

Mean X           : 23.4
Mean Y           : 26
Predicted Y      : 20.110769
Residual         : 5.8892308
Prediction SE     : 1.8644801
Lower            : 16.456455
Upper            : 23.765083
Extrapolated     : FALSE
Prediction classification : noncommutable
Sy.x             : 0.63377684
Relative residual : -9.2922784
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : ME
procedure_y      : MI
critical_value    : 1.959964

```

```

rm_set_size          : 1
classification       : noncommutable

```

Procedure pair: ME vs MI | Reference material: RMG

```

Mean X              : 19
Mean Y              : 16.7
Predicted Y         : 16.512389
Residual            : 0.1876109
Prediction SE       : 1.864892
Lower               : 12.857268
Upper               : 20.16751
Extrapolated        : FALSE
Prediction classification : commutable
Sy.x                : 0.63377684
Relative residual    : -0.29602044
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x         : ME
procedure_y         : MI
critical_value       : 1.959964
rm_set_size         : 1
classification       : commutable

```

Procedure pair: ME vs MJ | Reference material: RMA

```

Mean X              : 12
Mean Y              : 8.5
Predicted Y         : 8.4554778
Residual            : 0.044522182
Prediction SE       : 1.8695122
Lower               : 4.7913012
Upper               : 12.119654
Extrapolated        : FALSE
Prediction classification : commutable
Sy.x                : 0.021292767
Relative residual    : -2.0909533
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x         : ME
procedure_y         : MJ
critical_value       : 1.959964
rm_set_size         : 1
classification       : commutable

```

Procedure pair: ME vs MJ | Reference material: RMB

```

Mean X              : 24.1
Mean Y              : 17
Predicted Y         : 16.972167
Residual            : 0.027832672
Prediction SE       : 1.8695088
Lower               : 13.307997
Upper               : 20.636337
Extrapolated        : FALSE
Prediction classification : commutable
Sy.x                : 0.021292767
Relative residual    : -1.3071421
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x         : ME

```

```

procedure_y      : MJ
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: ME vs MJ | Reference material: RMC

```

Mean X           : 35.9
Mean Y           : 25.3
Predicted Y      : 25.277699
Residual         : 0.022300753
Prediction SE     : 1.8695135
Lower            : 21.61352
Upper            : 28.941878
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.021292767
Relative residual : -1.0473394
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : ME
procedure_y      : MJ
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: ME vs MJ | Reference material: RMD

```

Mean X           : 43.5
Mean Y           : 30.6
Predicted Y      : 30.627025
Residual         : -0.027024889
Prediction SE     : 1.8695209
Lower            : 26.962831
Upper            : 34.291218
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.021292767
Relative residual : 1.2692051
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : ME
procedure_y      : MJ
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: ME vs MJ | Reference material: RME

```

Mean X           : 17.1
Mean Y           : 12
Predicted Y      : 12.045157
Residual         : -0.045156868
Prediction SE     : 1.8695097
Lower            : 8.3809851
Upper            : 15.709329
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.021292767
Relative residual : 2.1207609
Relative-residual limit : 2

```

```

Relative-residual classification : noncommutable
procedure_x                      : ME
procedure_y                      : MJ
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: ME vs MJ | Reference material: RMF

```

Mean X                          : 23.4
Mean Y                          : 21
Predicted Y                     : 16.479466
Residual                       : 4.5205337
Prediction SE                   : 1.8695087
Lower                           : 12.815296
Upper                           : 20.143636
Extrapolated                   : FALSE
Prediction classification       : noncommutable
Sy.x                           : 0.021292767
Relative residual               : -212.30372
Relative-residual limit        : 2
Relative-residual classification : noncommutable
procedure_x                     : ME
procedure_y                     : MJ
critical_value                  : 1.959964
rm_set_size                     : 1
classification                   : noncommutable

```

Procedure pair: ME vs MJ | Reference material: RMG

```

Mean X                          : 19
Mean Y                          : 13.4
Predicted Y                     : 13.382488
Residual                       : 0.017511721
Prediction SE                   : 1.8695092
Lower                           : 9.7183176
Upper                           : 17.046659
Extrapolated                   : FALSE
Prediction classification       : commutable
Sy.x                           : 0.021292767
Relative residual               : -0.82242581
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                     : ME
procedure_y                     : MJ
critical_value                  : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MF vs MG | Reference material: RMA

```

Mean X                          : 9.5
Mean Y                          : 9.1
Predicted Y                     : 10.240497
Residual                       : -1.140497
Prediction SE                   : 2.2007093
Lower                           : 5.9271861
Upper                           : 14.553808
Extrapolated                   : FALSE
Prediction classification       : commutable
Sy.x                           : 0.64093887

```

```

Relative residual      : 1.7794162
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MF
procedure_y           : MG
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MF vs MG | Reference material: RMB

```

Mean X                : 19
Mean Y                : 19.1
Predicted Y           : 20.111528
Residual              : -1.0115279
Prediction SE         : 2.1978711
Lower                 : 15.80378
Upper                 : 24.419276
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 0.64093887
Relative residual      : 1.5781972
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MF
procedure_y           : MG
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MF vs MG | Reference material: RMC

```

Mean X                : 28.8
Mean Y                : 28.8
Predicted Y           : 30.294276
Residual              : -1.4942757
Prediction SE         : 2.2015904
Lower                 : 25.979238
Upper                 : 34.609314
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 0.64093887
Relative residual      : 2.3313856
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : MF
procedure_y           : MG
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MF vs MG | Reference material: RMD

```

Mean X                : 35.1
Mean Y                : 35.8
Predicted Y           : 36.840328
Residual              : -1.0403277
Prediction SE         : 2.2075326
Lower                 : 32.513643
Upper                 : 41.167012
Extrapolated          : FALSE

```

```

Prediction classification      : commutable
Sy.x                         : 0.64093887
Relative residual             : 1.623131
Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                  : MF
procedure_y                  : MG
critical_value                : 1.959964
rm_set_size                  : 1
classification                : commutable

```

Procedure pair: MF vs MG | Reference material: RME

```

Mean X                       : 14
Mean Y                       : 14.8
Predicted Y                  : 14.916249
Residual                     : -0.1162485
Prediction SE                 : 2.1985744
Lower                        : 10.607122
Upper                        : 19.225375
Extrapolated                 : FALSE
Prediction classification    : commutable
Sy.x                         : 0.64093887
Relative residual            : 0.18137221
Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                  : MF
procedure_y                  : MG
critical_value                : 1.959964
rm_set_size                  : 1
classification                : commutable

```

Procedure pair: MF vs MG | Reference material: RMF

```

Mean X                       : 20
Mean Y                       : 20.1
Predicted Y                  : 21.150584
Residual                     : -1.0505838
Prediction SE                 : 2.1979414
Lower                        : 16.842698
Upper                        : 25.45847
Extrapolated                 : FALSE
Prediction classification    : commutable
Sy.x                         : 0.64093887
Relative residual            : 1.6391327
Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                  : MF
procedure_y                  : MG
critical_value                : 1.959964
rm_set_size                  : 1
classification                : commutable

```

Procedure pair: MF vs MG | Reference material: RMG

```

Mean X                       : 15.5
Mean Y                       : 16.4
Predicted Y                  : 16.474832
Residual                     : -0.074832335
Prediction SE                 : 2.1981788
Lower                        : 12.166481

```

```

Upper                : 20.783184
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.64093887
Relative residual     : 0.11675425
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MF
procedure_y           : MG
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MF vs MH | Reference material: RMA

```

Mean X               : 9.5
Mean Y               : 9.1
Predicted Y          : 10.241081
Residual             : -1.1410813
Prediction SE         : 2.2613564
Lower               : 5.8089041
Upper               : 14.673258
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.64091635
Relative residual     : 1.7803904
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MF
procedure_y           : MH
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MF vs MH | Reference material: RMB

```

Mean X               : 19
Mean Y               : 19.1
Predicted Y          : 20.111495
Residual             : -1.0114955
Prediction SE         : 2.2585948
Lower               : 15.684731
Upper               : 24.53826
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.64091635
Relative residual     : 1.578202
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MF
procedure_y           : MH
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MF vs MH | Reference material: RMC

```

Mean X               : 28.8
Mean Y               : 28.8
Predicted Y          : 30.293607
Residual             : -1.4936069

```



```

Prediction SE          : 2.2622138
Lower                  : 25.859749
Upper                  : 34.727465
Extrapolated           : FALSE
Prediction classification : commutable
Sy.x                   : 0.64091635
Relative residual       : 2.3304241
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x            : MF
procedure_y            : MH
critical_value          : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MF vs MH | Reference material: RMD

```

Mean X                 : 35.1
Mean Y                 : 35.8
Predicted Y            : 36.83925
Residual               : -1.03925
Prediction SE          : 2.2679965
Lower                  : 32.394059
Upper                  : 41.284441
Extrapolated           : FALSE
Prediction classification : commutable
Sy.x                   : 0.64091635
Relative residual       : 1.6215065
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x            : MF
procedure_y            : MH
critical_value          : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MF vs MH | Reference material: RME

```

Mean X                 : 14
Mean Y                 : 14.8
Predicted Y            : 14.916541
Residual               : -0.11654066
Prediction SE          : 2.2592791
Lower                  : 10.488435
Upper                  : 19.344646
Extrapolated           : FALSE
Prediction classification : commutable
Sy.x                   : 0.64091635
Relative residual       : 0.18183443
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x            : MF
procedure_y            : MH
critical_value          : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MF vs MH | Reference material: RMF

```

Mean X                 : 20
Mean Y                 : 20.1

```

```

Predicted Y           : 21.150486
Residual              : -1.0504864
Prediction SE         : 2.2586633
Lower                 : 16.723588
Upper                 : 25.577385
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 0.64091635
Relative residual      : 1.6390383
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MF
procedure_y            : MH
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MF vs MH | Reference material: RMG

```

Mean X                : 15.5
Mean Y                 : 16.4
Predicted Y           : 16.475027
Residual              : -0.075027106
Prediction SE         : 2.2588942
Lower                 : 12.047676
Upper                 : 20.902378
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 0.64091635
Relative residual      : 0.11706224
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MF
procedure_y            : MH
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MF vs MI | Reference material: RMA

```

Mean X                : 9.5
Mean Y                 : 9.8
Predicted Y           : 10.563394
Residual              : -0.76339381
Prediction SE         : 2.1878316
Lower                 : 6.2753227
Upper                 : 14.851465
Extrapolated          : FALSE
Prediction classification : commutable
Sy.x                  : 0.70803357
Relative residual      : 1.0781887
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MF
procedure_y            : MI
critical_value         : 1.959964
rm_set_size            : 1
classification         : commutable

```

Procedure pair: MF vs MI | Reference material: RMB

```

Mean X           : 19
Mean Y           : 19.5
Predicted Y      : 20.313145
Residual         : -0.81314479
Prediction SE    : 2.1843476
Lower           : 16.031902
Upper           : 24.594387
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x            : 0.70803357
Relative residual : 1.1484551
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MF
procedure_y      : MI
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MF vs MI | Reference material: RMC

```

Mean X           : 28.8
Mean Y           : 28.6
Predicted Y      : 30.370783
Residual         : -1.7707826
Prediction SE    : 2.188913
Lower           : 26.080592
Upper           : 34.660973
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x            : 0.70803357
Relative residual : 2.5009868
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MF
procedure_y      : MI
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MF vs MI | Reference material: RMD

```

Mean X           : 35.1
Mean Y           : 35
Predicted Y      : 36.836407
Residual         : -1.836407
Prediction SE    : 2.1962033
Lower           : 32.531928
Upper           : 41.140886
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x            : 0.70803357
Relative residual : 2.5936722
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MF
procedure_y      : MI
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MF vs MI | Reference material: RME

Mean X	: 14
Mean Y	: 15
Predicted Y	: 15.181697
Residual	: -0.18169691
Prediction SE	: 2.185211
Lower	: 10.898762
Upper	: 19.464632
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 0.70803357
Relative residual	: 0.25662188
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MF
procedure_y	: MI
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MF vs MI | Reference material: RMF

Mean X	: 20
Mean Y	: 26
Predicted Y	: 21.339434
Residual	: 4.6605656
Prediction SE	: 2.184434
Lower	: 17.058022
Upper	: 25.620846
Extrapolated	: FALSE
Prediction classification	: noncommutable
Sy.x	: 0.70803357
Relative residual	: -6.5824078
Relative-residual limit	: 2
Relative-residual classification	: noncommutable
procedure_x	: MF
procedure_y	: MI
critical_value	: 1.959964
rm_set_size	: 1
classification	: noncommutable

Procedure pair: MF vs MI | Reference material: RMG

Mean X	: 15.5
Mean Y	: 16.7
Predicted Y	: 16.721131
Residual	: -0.021131271
Prediction SE	: 2.1847254
Lower	: 12.439148
Upper	: 21.003114
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 0.70803357
Relative residual	: 0.029845013
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MF
procedure_y	: MI
critical_value	: 1.959964

```
rm_set_size      : 1
classification    : commutable
```

Procedure pair: MF vs MJ | Reference material: RMA

```
Mean X           : 9.5
Mean Y           : 8.5
Predicted Y      : 8.2663859
Residual         : 0.23361407
Prediction SE     : 2.1170157
Lower            : 4.1171114
Upper            : 12.41566
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.68352665
Relative residual : -0.34177757
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MF
procedure_y      : MJ
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable
```

Procedure pair: MF vs MJ | Reference material: RMB

```
Mean X           : 19
Mean Y           : 17
Predicted Y      : 16.653423
Residual         : 0.346577
Prediction SE     : 2.1136563
Lower            : 12.510733
Upper            : 20.796113
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.68352665
Relative residual : -0.5070424
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MF
procedure_y      : MJ
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable
```

Procedure pair: MF vs MJ | Reference material: RMC

```
Mean X           : 28.8
Mean Y           : 25.3
Predicted Y      : 25.305314
Residual         : -0.0053138853
Prediction SE     : 2.1180584
Lower            : 21.153996
Upper            : 29.456632
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.68352665
Relative residual : 0.0077742182
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MF
```

```

procedure_y      : MJ
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MF vs MJ | Reference material: RMD

```

Mean X          : 35.1
Mean Y          : 30.6
Predicted Y     : 30.867244
Residual        : -0.26724374
Prediction SE    : 2.125088
Lower           : 26.702148
Upper           : 35.03234
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 0.68352665
Relative residual : 0.39097779
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MF
procedure_y     : MJ
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MF vs MJ | Reference material: RME

```

Mean X          : 14
Mean Y          : 12
Predicted Y     : 12.239193
Residual        : -0.23919296
Prediction SE    : 2.1144888
Lower           : 8.094871
Upper           : 16.383515
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 0.68352665
Relative residual : 0.34993948
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MF
procedure_y     : MJ
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MF vs MJ | Reference material: RMF

```

Mean X          : 20
Mean Y          : 21
Predicted Y     : 17.536269
Residual        : 3.463731
Prediction SE    : 2.1137396
Lower           : 13.393416
Upper           : 21.679122
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 0.68352665
Relative residual : -5.067441
Relative-residual limit : 2

```

```

Relative-residual classification : noncommutable
procedure_x                     : MF
procedure_y                     : MJ
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MF vs MJ | Reference material: RMG

```

Mean X                         : 15.5
Mean Y                         : 13.4
Predicted Y                    : 13.563462
Residual                       : -0.16346198
Prediction SE                   : 2.1140206
Lower                          : 9.4200578
Upper                          : 17.706866
Extrapolated                   : FALSE
Prediction classification      : commutable
Sy.x                           : 0.68352665
Relative residual               : 0.23914499
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                     : MF
procedure_y                     : MJ
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MG vs MH | Reference material: RMA

```

Mean X                         : 9.1
Mean Y                         : 9.1
Predicted Y                    : 9.1
Residual                       : 0
Prediction SE                   : 2.1187373
Lower                          : 4.9473513
Upper                          : 13.252649
Extrapolated                   : FALSE
Prediction classification      : commutable
Sy.x                           : 1.7389557e-15
Relative residual               : 0
Relative-residual limit        : 2
Relative-residual classification : commutable
procedure_x                     : MG
procedure_y                     : MH
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MG vs MH | Reference material: RMB

```

Mean X                         : 19.1
Mean Y                         : 19.1
Predicted Y                    : 19.1
Residual                       : 0
Prediction SE                   : 2.1187373
Lower                          : 14.947351
Upper                          : 23.252649
Extrapolated                   : FALSE
Prediction classification      : commutable
Sy.x                           : 1.7389557e-15

```

```

Relative residual      : 0
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MG
procedure_y           : MH
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MG vs MH | Reference material: RMC

```

Mean X      : 28.8
Mean Y      : 28.8
Predicted Y : 28.8
Residual    : 0
Prediction SE : 2.1187373
Lower       : 24.647351
Upper       : 32.952649
Extrapolated : FALSE
Prediction classification : commutable
Sy.x        : 1.7389557e-15
Relative residual : 0
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MG
procedure_y           : MH
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MG vs MH | Reference material: RMD

```

Mean X      : 35.8
Mean Y      : 35.8
Predicted Y : 35.8
Residual    : 7.1054274e-15
Prediction SE : 2.1187373
Lower       : 31.647351
Upper       : 39.952649
Extrapolated : FALSE
Prediction classification : commutable
Sy.x        : 1.7389557e-15
Relative residual : -4.0860313
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : MG
procedure_y           : MH
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MG vs MH | Reference material: RME

```

Mean X      : 14.8
Mean Y      : 14.8
Predicted Y : 14.8
Residual    : 0
Prediction SE : 2.1187373
Lower       : 10.647351
Upper       : 18.952649
Extrapolated : FALSE

```



```

Prediction classification      : commutable
Sy.x                         : 1.7389557e-15
Relative residual             : 0
Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                   : MG
procedure_y                   : MH
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: MG vs MH | Reference material: RMF

```

Mean X                       : 20.1
Mean Y                       : 20.1
Predicted Y                  : 20.1
Residual                     : 3.5527137e-15
Prediction SE                 : 2.1187373
Lower                        : 15.947351
Upper                        : 24.252649
Extrapolated                 : FALSE
Prediction classification     : commutable
Sy.x                         : 1.7389557e-15
Relative residual             : -2.0430157
Relative-residual limit      : 2
Relative-residual classification : noncommutable
procedure_x                   : MG
procedure_y                   : MH
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: MG vs MH | Reference material: RMG

```

Mean X                       : 16.4
Mean Y                       : 16.4
Predicted Y                  : 16.4
Residual                     : 0
Prediction SE                 : 2.1187373
Lower                        : 12.247351
Upper                        : 20.552649
Extrapolated                 : FALSE
Prediction classification     : commutable
Sy.x                         : 1.7389557e-15
Relative residual             : 0
Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                   : MG
procedure_y                   : MH
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: MG vs MI | Reference material: RMA

```

Mean X                       : 9.1
Mean Y                       : 9.8
Predicted Y                  : 9.4307365
Residual                     : 0.3692635
Prediction SE                 : 2.0444861
Lower                        : 5.4236174

```

```

Upper                : 13.437856
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.31374846
Relative residual     : -1.1769412
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MG
procedure_y           : MI
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MG vs MI | Reference material: RMB

```

Mean X               : 19.1
Mean Y               : 19.5
Predicted Y          : 19.313755
Residual              : 0.18624453
Prediction SE         : 2.0435659
Lower                : 15.30844
Upper                : 23.319071
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.31374846
Relative residual     : -0.59361098
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MG
procedure_y           : MI
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MG vs MI | Reference material: RMC

```

Mean X               : 28.8
Mean Y               : 28.6
Predicted Y          : 28.900284
Residual              : -0.30028386
Prediction SE         : 2.0442742
Lower                : 24.89358
Upper                : 32.906988
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.31374846
Relative residual     : 0.95708472
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MG
procedure_y           : MI
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MG vs MI | Reference material: RMD

```

Mean X               : 35.8
Mean Y               : 35
Predicted Y          : 35.818397
Residual              : -0.81839714

```

```
Prediction SE           : 2.045764
Lower                   : 31.808773
Upper                   : 39.828021
Extrapolated           : FALSE
Prediction classification : commutable
Sy.x                   : 0.31374846
Relative residual       : 2.6084499
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x             : MG
procedure_y             : MI
critical_value          : 1.959964
rm_set_size             : 1
classification          : commutable
```

Procedure pair: MG vs MI | Reference material: RME

Mean X	: 14.8
Mean Y	: 15
Predicted Y	: 15.064057
Residual	: -0.064057313
Prediction SE	: 2.0437563
Lower	: 11.058369
Upper	: 19.069746
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 0.31374846
Relative residual	: 0.20416773
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MG
procedure_y	: MI
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MG vs MI | Reference material: RMF

```

Mean X                               : 20.1
Mean Y                               : 26
Predicted Y                          : 20.302057
Residual                             : 5.6979426
Prediction SE                         : 2.0435661
Lower                                : 16.296741
Upper                                : 24.307373
Extrapolated                         : FALSE
Prediction classification             : noncommutable
Sy.x                                 : 0.31374846
Relative residual                     : -18.160862
Relative-residual limit              : 2
Relative-residual classification     : noncommutable
procedure_x                          : MG
procedure_y                          : MI
critical_value                        : 1.959964
rm_set_size                          : 1
classification                        : noncommutable

```

Procedure pair: MG vs MI | Reference material: RMG

Mean X	: 16.4
Mean Y	: 16.7

```

Predicted Y           : 16.64534
Residual              : 0.054659653
Prediction SE         : 2.0436493
Lower                 : 12.639861
Upper                 : 20.650819
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 0.31374846
Relative residual     : -0.17421489
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x           : MG
procedure_y           : MI
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MG vs MJ | Reference material: RMA

```

Mean X                : 9.1
Mean Y                : 8.5
Predicted Y           : 7.2950871
Residual              : 1.2049129
Prediction SE         : 2.0086158
Lower                 : 3.3582725
Upper                 : 11.231902
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 0.4421235
Relative residual     : -2.7252858
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : MG
procedure_y           : MJ
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MG vs MJ | Reference material: RMB

```

Mean X                : 19.1
Mean Y                : 17
Predicted Y           : 15.79386
Residual              : 1.2061396
Prediction SE         : 2.0067537
Lower                 : 11.860695
Upper                 : 19.727025
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                  : 0.4421235
Relative residual     : -2.7280605
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : MG
procedure_y           : MJ
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MG vs MJ | Reference material: RMC

```

Mean X           : 28.8
Mean Y           : 25.3
Predicted Y      : 24.03767
Residual         : 1.2623296
Prediction SE    : 2.0081871
Lower            : 20.101696
Upper            : 27.973645
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.4421235
Relative residual : -2.8551515
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MG
procedure_y      : MJ
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MG vs MJ | Reference material: RMD

```

Mean X           : 35.8
Mean Y           : 30.6
Predicted Y      : 29.986812
Residual         : 0.61318828
Prediction SE    : 2.0112002
Lower            : 26.044932
Upper            : 33.928692
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.4421235
Relative residual : -1.3869163
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MG
procedure_y      : MJ
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MG vs MJ | Reference material: RME

```

Mean X           : 14.8
Mean Y           : 12
Predicted Y      : 12.139388
Residual         : -0.13938785
Prediction SE    : 2.007139
Lower            : 8.2054677
Upper            : 16.073308
Extrapolated     : FALSE
Prediction classification : commutable
Sy.x             : 0.4421235
Relative residual : 0.31526903
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MG
procedure_y      : MJ
critical_value    : 1.959964
rm_set_size      : 1
classification    : commutable

```

Procedure pair: MG vs MJ | Reference material: RMF

Mean X	: 20.1
Mean Y	: 21
Predicted Y	: 16.643738
Residual	: 4.3562623
Prediction SE	: 2.0067539
Lower	: 12.710572
Upper	: 20.576903
Extrapolated	: FALSE
Prediction classification	: noncommutable
Sy.x	: 0.4421235
Relative residual	: -9.853044
Relative-residual limit	: 2
Relative-residual classification	: noncommutable
procedure_x	: MG
procedure_y	: MJ
critical_value	: 1.959964
rm_set_size	: 1
classification	: noncommutable

Procedure pair: MG vs MJ | Reference material: RMG

Mean X	: 16.4
Mean Y	: 13.4
Predicted Y	: 13.499192
Residual	: -0.099191572
Prediction SE	: 2.0069224
Lower	: 9.565696
Upper	: 17.432687
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 0.4421235
Relative residual	: 0.22435263
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MG
procedure_y	: MJ
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MH vs MI | Reference material: RMA

Mean X	: 9.1
Mean Y	: 9.8
Predicted Y	: 9.4305936
Residual	: 0.36940643
Prediction SE	: 2.108231
Lower	: 5.2985368
Upper	: 13.56265
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 0.3137499
Relative residual	: -1.1773914
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MH
procedure_y	: MI
critical_value	: 1.959964

```

rm_set_size          : 1
classification       : commutable

```

Procedure pair: MH vs MI | Reference material: RMB

```

Mean X               : 19.1
Mean Y               : 19.5
Predicted Y          : 19.313749
Residual             : 0.18625124
Prediction SE         : 2.1073386
Lower                : 15.183441
Upper                : 23.444057
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.3137499
Relative residual     : -0.59362963
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MH
procedure_y          : MI
critical_value        : 1.959964
rm_set_size          : 1
classification       : commutable

```

Procedure pair: MH vs MI | Reference material: RMC

```

Mean X               : 28.8
Mean Y               : 28.6
Predicted Y          : 28.900409
Residual             : -0.3004093
Prediction SE         : 2.1080255
Lower                : 24.768755
Upper                : 33.032063
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.3137499
Relative residual     : 0.95748016
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MH
procedure_y          : MI
critical_value        : 1.959964
rm_set_size          : 1
classification       : commutable

```

Procedure pair: MH vs MI | Reference material: RMD

```

Mean X               : 35.8
Mean Y               : 35
Predicted Y          : 35.818618
Residual             : -0.81861794
Prediction SE         : 2.1094702
Lower                : 31.684132
Upper                : 39.953104
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.3137499
Relative residual     : 2.6091417
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x          : MH

```

```

procedure_y      : MI
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MH vs MI | Reference material: RME

```

Mean X          : 14.8
Mean Y          : 15
Predicted Y     : 15.063992
Residual        : -0.063992032
Prediction SE    : 2.1075233
Lower           : 10.933322
Upper           : 19.194662
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 0.3137499
Relative residual : 0.20395873
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x     : MH
procedure_y     : MI
critical_value   : 1.959964
rm_set_size     : 1
classification   : commutable

```

Procedure pair: MH vs MI | Reference material: RMF

```

Mean X          : 20.1
Mean Y          : 26
Predicted Y     : 20.302064
Residual        : 5.6979357
Prediction SE    : 2.1073388
Lower           : 16.171756
Upper           : 24.432372
Extrapolated    : FALSE
Prediction classification : noncommutable
Sy.x            : 0.3137499
Relative residual : -18.160757
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x     : MH
procedure_y     : MI
critical_value   : 1.959964
rm_set_size     : 1
classification   : noncommutable

```

Procedure pair: MH vs MI | Reference material: RMG

```

Mean X          : 16.4
Mean Y          : 16.7
Predicted Y     : 16.645297
Residual        : 0.054703137
Prediction SE    : 2.1074195
Lower           : 12.514831
Upper           : 20.775763
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 0.3137499
Relative residual : -0.17435269
Relative-residual limit : 2

```



```

Relative-residual classification : commutable
procedure_x                     : MH
procedure_y                     : MI
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MH vs MJ | Reference material: RMA

```

Mean X                         : 9.1
Mean Y                         : 8.5
Predicted Y                    : 7.2947327
Residual                       : 1.2052673
Prediction SE                   : 2.0567891
Lower                          : 3.2635002
Upper                          : 11.325965
Extrapolated                   : FALSE
Prediction classification      : commutable
Sy.x                           : 0.44213034
Relative residual              : -2.7260452
Relative-residual limit        : 2
Relative-residual classification : noncommutable
procedure_x                     : MH
procedure_y                     : MJ
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MH vs MJ | Reference material: RMB

```

Mean X                         : 19.1
Mean Y                         : 17
Predicted Y                    : 15.793844
Residual                       : 1.2061563
Prediction SE                   : 2.0549705
Lower                          : 11.766176
Upper                          : 19.821512
Extrapolated                   : FALSE
Prediction classification      : commutable
Sy.x                           : 0.44213034
Relative residual              : -2.7280558
Relative-residual limit        : 2
Relative-residual classification : noncommutable
procedure_x                     : MH
procedure_y                     : MJ
critical_value                   : 1.959964
rm_set_size                     : 1
classification                   : commutable

```

Procedure pair: MH vs MJ | Reference material: RMC

```

Mean X                         : 28.8
Mean Y                         : 25.3
Predicted Y                    : 24.037981
Residual                       : 1.2620186
Prediction SE                   : 2.0563704
Lower                          : 20.007569
Upper                          : 28.068393
Extrapolated                   : FALSE
Prediction classification      : commutable
Sy.x                           : 0.44213034

```

```

Relative residual      : -2.8544039
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x           : MH
procedure_y           : MJ
critical_value         : 1.959964
rm_set_size           : 1
classification         : commutable

```

Procedure pair: MH vs MJ | Reference material: RMD

```

Mean X      : 35.8
Mean Y      : 30.6
Predicted Y  : 29.987359
Residual     : 0.61264085
Prediction SE : 2.0593133
Lower        : 25.951179
Upper        : 34.023539
Extrapolated : FALSE
Prediction classification : commutable
Sy.x         : 0.44213034
Relative residual : -1.3856566
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x   : MH
procedure_y   : MJ
critical_value : 1.959964
rm_set_size   : 1
classification : commutable

```

Procedure pair: MH vs MJ | Reference material: RME

```

Mean X      : 14.8
Mean Y      : 12
Predicted Y  : 12.139226
Residual     : -0.139226
Prediction SE : 2.0553468
Lower        : 8.1108203
Upper        : 16.167632
Extrapolated : FALSE
Prediction classification : commutable
Sy.x         : 0.44213034
Relative residual : 0.31489808
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x   : MH
procedure_y   : MJ
critical_value : 1.959964
rm_set_size   : 1
classification : commutable

```

Procedure pair: MH vs MJ | Reference material: RMF

```

Mean X      : 20.1
Mean Y      : 21
Predicted Y  : 16.643755
Residual     : 4.3562452
Prediction SE : 2.0549708
Lower        : 12.616086
Upper        : 20.671424
Extrapolated : FALSE

```

```

Prediction classification      : noncommutable
Sy.x                         : 0.44213034
Relative residual             : -9.8528527
Relative-residual limit      : 2
Relative-residual classification : noncommutable
procedure_x                   : MH
procedure_y                   : MJ
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : noncommutable

```

Procedure pair: MH vs MJ | Reference material: RMG

```

Mean X                       : 16.4
Mean Y                       : 13.4
Predicted Y                  : 13.499084
Residual                     : -0.09908376
Prediction SE                 : 2.0551352
Lower                        : 9.4710927
Upper                        : 17.527075
Extrapolated                 : FALSE
Prediction classification    : commutable
Sy.x                         : 0.44213034
Relative residual            : 0.22410532
Relative-residual limit     : 2
Relative-residual classification : commutable
procedure_x                   : MH
procedure_y                   : MJ
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: MI vs MJ | Reference material: RMA

```

Mean X                       : 9.8
Mean Y                       : 8.5
Predicted Y                  : 7.6096017
Residual                     : 0.8903983
Prediction SE                 : 2.020301
Lower                        : 3.6498846
Upper                        : 11.569319
Extrapolated                 : FALSE
Prediction classification    : commutable
Sy.x                         : 0.54782818
Relative residual            : -1.625324
Relative-residual limit     : 2
Relative-residual classification : commutable
procedure_x                   : MI
procedure_y                   : MJ
critical_value                 : 1.959964
rm_set_size                   : 1
classification                 : commutable

```

Procedure pair: MI vs MJ | Reference material: RMB

```

Mean X                       : 19.5
Mean Y                       : 17
Predicted Y                  : 15.953928
Residual                     : 1.0460719
Prediction SE                 : 2.0176504
Lower                        : 11.999406

```

```

Upper                : 19.90845
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.54782818
Relative residual     : -1.9094891
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MI
procedure_y           : MJ
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MI vs MJ | Reference material: RMC

```

Mean X               : 28.6
Mean Y               : 25.3
Predicted Y          : 23.782111
Residual              : 1.5178895
Prediction SE         : 2.0197028
Lower                : 19.823566
Upper                : 27.740655
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.54782818
Relative residual     : -2.7707401
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x          : MI
procedure_y           : MJ
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MI vs MJ | Reference material: RMD

```

Mean X               : 35
Mean Y               : 30.6
Predicted Y          : 29.287645
Residual              : 1.3123546
Prediction SE         : 2.0237722
Lower                : 25.321125
Upper                : 33.254166
Extrapolated         : FALSE
Prediction classification : commutable
Sy.x                 : 0.54782818
Relative residual     : -2.3955587
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x          : MI
procedure_y           : MJ
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable

```

Procedure pair: MI vs MJ | Reference material: RME

```

Mean X               : 15
Mean Y               : 12
Predicted Y          : 12.082849
Residual              : -0.082848815

```

```

Prediction SE          : 2.0182595
Lower                  : 8.1271328
Upper                  : 16.038565
Extrapolated           : FALSE
Prediction classification : commutable
Sy.x                   : 0.54782818
Relative residual       : 0.15123139
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x             : MI
procedure_y              : MJ
critical_value           : 1.959964
rm_set_size              : 1
classification           : commutable

```

Procedure pair: MI vs MJ | Reference material: RMF

```

Mean X                  : 26
Mean Y                  : 21
Predicted Y             : 21.545487
Residual                 : -0.54548695
Prediction SE           : 2.0186682
Lower                   : 17.58897
Upper                   : 25.502004
Extrapolated            : FALSE
Prediction classification : commutable
Sy.x                    : 0.54782818
Relative residual        : 0.99572634
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x              : MI
procedure_y               : MJ
critical_value            : 1.959964
rm_set_size               : 1
classification            : commutable

```

Procedure pair: MI vs MJ | Reference material: RMG

```

Mean X                  : 16.7
Mean Y                  : 13.4
Predicted Y             : 13.545257
Residual                 : -0.14525653
Prediction SE           : 2.0179031
Lower                   : 9.5902392
Upper                   : 17.500274
Extrapolated            : FALSE
Prediction classification : commutable
Sy.x                    : 0.54782818
Relative residual        : 0.26514979
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x              : MI
procedure_y               : MJ
critical_value            : 1.959964
rm_set_size               : 1
classification            : commutable

```

EP30 model estimates

Procedure pair: MA vs MB

```

Clinical samples        : 25

```

```

Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range : 3
Equal RM replicates : TRUE
Variance X : 5.6032
Variance Y : 6.8688
Lambda : 1.2258709
Intercept : 3.6495387
Slope : 0.96181694
Intercept variance : 0.24734077
Slope variance : 0.00043809195
Intercept-slope covariance : -0.0091964262
RM variance X : 4.2857143
RM variance Y : 6.0771429
Sy.x : 1.1890275
Critical value : 1.959964
RM set size : 1
procedure_x : MA
procedure_y : MB
clinical_replicates : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates : 3
rm_replicates_min : 3
rm_replicates_max : 3
slope_se : 0.020930646
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MA vs MC

```

Clinical samples : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range : 3
Equal RM replicates : TRUE
Variance X : 5.6032
Variance Y : 6.0064
Lambda : 1.0719589
Intercept : 0.20128784
Slope : 1.0605332
Intercept variance : 0.19669824
Slope variance : 0.00034823454
Intercept-slope covariance : -0.0073101394
RM variance X : 4.2857143
RM variance Y : 6.4857143
Sy.x : 1.0611977
Critical value : 1.959964
RM set size : 1
procedure_x : MA
procedure_y : MC
clinical_replicates : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates : 3
rm_replicates_min : 3
rm_replicates_max : 3
slope_se : 0.018661043
relative_residual_evaluable : TRUE

```

```

relative_residual_limit      : 2

Procedure pair: MA vs MD
Clinical samples              : 25
Clinical replicate range     : 3
Equal clinical replicates    : TRUE
RM replicate range           : 3
Equal RM replicates          : TRUE
Variance X                   : 5.6032
Variance Y                   : 6.544
Lambda                       : 1.1679041
Intercept                    : 0.39859117
Slope                        : 0.83448022
Intercept variance           : 0.18635499
Slope variance               : 0.00033001627
Intercept-slope covariance   : -0.0069277016
RM variance X                : 4.2857143
RM variance Y                : 7.5157143
Sy.x                         : 1.0324006
Critical value                : 1.959964
RM set size                  : 1
procedure_x                  : MA
procedure_y                  : MD
clinical_replicates          : 3
clinical_replicates_min      : 3
clinical_replicates_max      : 3
rm_replicates                : 3
rm_replicates_min            : 3
rm_replicates_max            : 3
slope_se                     : 0.01816635
relative_residual_evaluable  : TRUE
relative_residual_limit      : 2

```

```

Procedure pair: MA vs ME
Clinical samples              : 25
Clinical replicate range     : 3
Equal clinical replicates    : TRUE
RM replicate range           : 3
Equal RM replicates          : TRUE
Variance X                   : 5.6032
Variance Y                   : 5.6408
Lambda                       : 1.0067105
Intercept                    : -0.83125148
Slope                        : 1.1362067
Intercept variance           : 0.24877488
Slope variance               : 0.00044052676
Intercept-slope covariance   : -0.0092475378
RM variance X                : 4.2857143
RM variance Y                : 6.0485714
Sy.x                         : 1.1929761
Critical value                : 1.959964
RM set size                  : 1
procedure_x                  : MA
procedure_y                  : ME
clinical_replicates          : 3
clinical_replicates_min      : 3
clinical_replicates_max      : 3
rm_replicates                : 3

```

```

rm_replicates_min      : 3
rm_replicates_max      : 3
slope_se               : 0.020988729
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MA vs MF

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 5.6032
Variance Y            : 5.3108
Lambda               : 0.94781553
Intercept             : -0.49455087
Slope                 : 0.90484713
Intercept variance     : 0.15681085
Slope variance         : 0.00027762635
Intercept-slope covariance : -0.0058279322
RM variance X          : 4.2857143
RM variance Y          : 7.5185714
Sy.x                  : 0.9474596
Critical value         : 1.959964
RM set size           : 1
procedure_x            : MA
procedure_y            : MF
clinical_replicates     : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates           : 3
rm_replicates_min       : 3
rm_replicates_max       : 3
slope_se               : 0.016662123
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MA vs MG

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 5.6032
Variance Y            : 4.306
Lambda               : 0.76848943
Intercept             : -0.14983142
Slope                 : 0.94044548
Intercept variance     : 0.12671252
Slope variance         : 0.00022428637
Intercept-slope covariance : -0.0047082195
RM variance X          : 4.2857143
RM variance Y          : 6.3271429
Sy.x                  : 0.85204379
Critical value         : 1.959964
RM set size           : 1
procedure_x            : MA
procedure_y            : MG

```



```

clinical_replicates      : 3
clinical_replicates_min  : 3
clinical_replicates_max  : 3
rm_replicates            : 3
rm_replicates_min        : 3
rm_replicates_max        : 3
slope_se                 : 0.014976193
relative_residual_evaluable : TRUE
relative_residual_limit   : 2

```

Procedure pair: MA vs MH

```

Clinical samples          : 25
Clinical replicate range  : 3
Equal clinical replicates : TRUE
RM replicate range        : 3
Equal RM replicates       : TRUE
Variance X                : 5.6032
Variance Y                : 4.6228
Lambda                   : 0.82502856
Intercept                 : -0.14762006
Slope                     : 0.94034013
Intercept variance        : 0.1266871
Slope variance            : 0.00022423613
Intercept-slope covariance : -0.0047071648
RM variance X             : 4.2857143
RM variance Y             : 7.14
Sy.x                     : 0.85199356
Critical value            : 1.959964
RM set size               : 1
procedure_x               : MA
procedure_y               : MH
clinical_replicates       : 3
clinical_replicates_min   : 3
clinical_replicates_max   : 3
rm_replicates             : 3
rm_replicates_min        : 3
rm_replicates_max        : 3
slope_se                  : 0.014974516
relative_residual_evaluable : TRUE
relative_residual_limit    : 2

```

Procedure pair: MA vs MI

```

Clinical samples          : 25
Clinical replicate range  : 3
Equal clinical replicates : TRUE
RM replicate range        : 3
Equal RM replicates       : TRUE
Variance X                : 5.6032
Variance Y                : 8.7928
Lambda                   : 1.5692461
Intercept                 : 0.30849521
Slope                     : 0.92852062
Intercept variance        : 0.12485787
Slope variance            : 0.00022094394
Intercept-slope covariance : -0.0046380552
RM variance X             : 4.2857143
RM variance Y             : 6.3371429
Sy.x                     : 0.8461896

```

```

Critical value      : 1.959964
RM set size        : 1
procedure_x        : MA
procedure_y        : MI
clinical_replicates : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates      : 3
rm_replicates_min  : 3
rm_replicates_max  : 3
slope_se           : 0.014864183
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MA vs MJ

```

Clinical samples    : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range  : 3
Equal RM replicates  : TRUE
Variance X          : 5.6032
Variance Y          : 5.0552
Lambda              : 0.90219874
Intercept           : -0.55484017
Slope               : 0.79872524
Intercept variance   : 0.12382782
Slope variance       : 0.00021921253
Intercept-slope covariance : -0.0046017094
RM variance X        : 4.2857143
RM variance Y        : 7.4885714
Sy.x                : 0.84206988
Critical value       : 1.959964
RM set size          : 1
procedure_x          : MA
procedure_y          : MJ
clinical_replicates  : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates        : 3
rm_replicates_min    : 3
rm_replicates_max    : 3
slope_se             : 0.014805827
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MB vs MC

```

Clinical samples    : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range  : 3
Equal RM replicates  : TRUE
Variance X          : 6.8688
Variance Y          : 6.0064
Lambda              : 0.87444677
Intercept           : -3.8120845
Slope               : 1.1021848
Intercept variance   : 0.20662198
Slope variance       : 0.0003023

```

```

Intercept-slope covariance : -0.0072068319
RM variance X              : 6.0771429
RM variance Y              : 6.4857143
Sy.x                      : 0.95212352
Critical value              : 1.959964
RM set size                 : 1
procedure_x                 : MB
procedure_y                 : MC
clinical_replicates         : 3
clinical_replicates_min    : 3
clinical_replicates_max    : 3
rm_replicates               : 3
rm_replicates_min          : 3
rm_replicates_max          : 3
slope_se                   : 0.017386776
relative_residual_evaluable : TRUE
relative_residual_limit     : 2

```

Procedure pair: MB vs MD

```

Clinical samples           : 25
Clinical replicate range   : 3
Equal clinical replicates  : TRUE
RM replicate range         : 3
Equal RM replicates        : TRUE
Variance X                 : 6.8688
Variance Y                 : 6.544
Lambda                     : 0.95271372
Intercept                  : -2.7587397
Slope                      : 0.86722901
Intercept variance         : 0.39005036
Slope variance             : 0.00057141826
Intercept-slope covariance : -0.013622611
RM variance X              : 6.0771429
RM variance Y              : 7.5157143
Sy.x                      : 1.3039134
Critical value              : 1.959964
RM set size                 : 1
procedure_x                 : MB
procedure_y                 : MD
clinical_replicates         : 3
clinical_replicates_min    : 3
clinical_replicates_max    : 3
rm_replicates               : 3
rm_replicates_min          : 3
rm_replicates_max          : 3
slope_se                   : 0.023904356
relative_residual_evaluable : TRUE
relative_residual_limit     : 2

```

Procedure pair: MB vs ME

```

Clinical samples           : 25
Clinical replicate range   : 3
Equal clinical replicates  : TRUE
RM replicate range         : 3
Equal RM replicates        : TRUE
Variance X                 : 6.8688
Variance Y                 : 5.6408
Lambda                     : 0.82122059

```

```

Intercept          : -5.1727955
Slope              : 1.1825837
Intercept variance : 0.72009785
Slope variance     : 0.001055494
Intercept-slope covariance : -0.025162976
RM variance X      : 6.0771429
RM variance Y      : 6.0485714
Sy.x               : 1.7693315
Critical value      : 1.959964
RM set size        : 1
procedure_x        : MB
procedure_y        : ME
clinical_replicates : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates      : 3
rm_replicates_min  : 3
rm_replicates_max  : 3
slope_se           : 0.032488367
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MB vs MF

```

Clinical samples    : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range  : 3
Equal RM replicates  : TRUE
Variance X          : 6.8688
Variance Y          : 5.3108
Lambda              : 0.77317727
Intercept           : -3.9320295
Slope               : 0.94094084
Intercept variance  : 0.50490079
Slope variance      : 0.00074007333
Intercept-slope covariance : -0.017643348
RM variance X       : 6.0771429
RM variance Y       : 7.5185714
Sy.x                : 1.4815125
Critical value       : 1.959964
RM set size         : 1
procedure_x         : MB
procedure_y         : MF
clinical_replicates : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates       : 3
rm_replicates_min   : 3
rm_replicates_max   : 3
slope_se            : 0.027204289
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MB vs MG

```

Clinical samples    : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range  : 3

```

```

Equal RM replicates      : TRUE
Variance X               : 6.8688
Variance Y               : 4.306
Lambda                   : 0.62689262
Intercept                 : -3.7309834
Slope                    : 0.97831306
Intercept variance       : 0.4023514
Slope variance           : 0.00058943236
Intercept-slope covariance : -0.014052068
RM variance X            : 6.0771429
RM variance Y            : 6.3271429
Sy.x                     : 1.3243523
Critical value           : 1.959964
RM set size              : 1
procedure_x              : MB
procedure_y              : MG
clinical_replicates      : 3
clinical_replicates_min  : 3
clinical_replicates_max  : 3
rm_replicates            : 3
rm_replicates_min       : 3
rm_replicates_max       : 3
slope_se                 : 0.024278228
relative_residual_evaluable : TRUE
relative_residual_limit  : 2

```

Procedure pair: MB vs MH

```

Clinical samples         : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range       : 3
Equal RM replicates      : TRUE
Variance X               : 6.8688
Variance Y               : 4.6228
Lambda                   : 0.67301421
Intercept                 : -3.7248676
Slope                    : 0.97805653
Intercept variance       : 0.40215469
Slope variance           : 0.00058912328
Intercept-slope covariance : -0.014044699
RM variance X            : 6.0771429
RM variance Y            : 7.14
Sy.x                     : 1.3241455
Critical value           : 1.959964
RM set size              : 1
procedure_x              : MB
procedure_y              : MH
clinical_replicates      : 3
clinical_replicates_min  : 3
clinical_replicates_max  : 3
rm_replicates            : 3
rm_replicates_min       : 3
rm_replicates_max       : 3
slope_se                 : 0.024271862
relative_residual_evaluable : TRUE
relative_residual_limit  : 2

```

Procedure pair: MB vs MI

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 6.8688
Variance Y            : 8.7928
Lambda                : 1.2801072
Intercept             : -3.1885173
Slope                 : 0.96428345
Intercept variance    : 0.40379695
Slope variance        : 0.00059131278
Intercept-slope covariance : -0.014096897
RM variance X         : 6.0771429
RM variance Y         : 6.3371429
Sy.x                  : 1.328052
Critical value        : 1.959964
RM set size           : 1
procedure_x           : MB
procedure_y           : MI
clinical_replicates    : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates         : 3
rm_replicates_min     : 3
rm_replicates_max     : 3
slope_se              : 0.024316924
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MB vs MJ

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 6.8688
Variance Y            : 5.0552
Lambda                : 0.73596553
Intercept             : -3.5777974
Slope                 : 0.83010895
Intercept variance    : 0.35121929
Slope variance        : 0.00051458461
Intercept-slope covariance : -0.012267697
RM variance X         : 6.0771429
RM variance Y         : 7.4885714
Sy.x                  : 1.2369884
Critical value        : 1.959964
RM set size           : 1
procedure_x           : MB
procedure_y           : MJ
clinical_replicates    : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates         : 3
rm_replicates_min     : 3
rm_replicates_max     : 3
slope_se              : 0.022684458

```

```
relative_residual_evaluable : TRUE
relative_residual_limit     : 2
```

Procedure pair: MC vs MD

```
Clinical samples           : 25
Clinical replicate range   : 3
Equal clinical replicates  : TRUE
RM replicate range         : 3
Equal RM replicates        : TRUE
Variance X                 : 6.0064
Variance Y                 : 6.544
Lambda                    : 1.0895045
Intercept                  : 0.25672869
Slope                      : 0.78611429
Intercept variance         : 0.29343943
Slope variance             : 0.0004559482
Intercept-slope covariance : -0.01024242
RM variance X              : 6.4857143
RM variance Y              : 7.5157143
Sy.x                      : 1.2844594
Critical value              : 1.959964
RM set size                : 1
procedure_x                : MC
procedure_y                : MD
clinical_replicates        : 3
clinical_replicates_min    : 3
clinical_replicates_max    : 3
rm_replicates              : 3
rm_replicates_min          : 3
rm_replicates_max          : 3
slope_se                   : 0.021352944
relative_residual_evaluable : TRUE
relative_residual_limit     : 2
```

Procedure pair: MC vs ME

```
Clinical samples           : 25
Clinical replicate range   : 3
Equal clinical replicates  : TRUE
RM replicate range         : 3
Equal RM replicates        : TRUE
Variance X                 : 6.0064
Variance Y                 : 5.6408
Lambda                    : 0.93913159
Intercept                  : -1.0545702
Slope                      : 1.0716956
Intercept variance         : 0.44734307
Slope variance             : 0.00069525378
Intercept-slope covariance : -0.015618181
RM variance X              : 6.4857143
RM variance Y              : 6.0485714
Sy.x                      : 1.5852203
Critical value              : 1.959964
RM set size                : 1
procedure_x                : MC
procedure_y                : ME
clinical_replicates        : 3
clinical_replicates_min    : 3
clinical_replicates_max    : 3
```

```

rm_replicates      : 3
rm_replicates_min  : 3
rm_replicates_max  : 3
slope_se           : 0.026367665
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MC vs MF

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 6.0064
Variance Y            : 5.3108
Lambda               : 0.8841902
Intercept             : -0.65713681
Slope                 : 0.85279277
Intercept variance     : 0.34197983
Slope variance         : 0.00053157846
Intercept-slope covariance : -0.011941379
RM variance X          : 6.4857143
RM variance Y          : 7.5185714
Sy.x                  : 1.3856472
Critical value         : 1.959964
RM set size           : 1
procedure_x            : MC
procedure_y            : MF
clinical_replicates    : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates          : 3
rm_replicates_min      : 3
rm_replicates_max      : 3
slope_se               : 0.023055985
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MC vs MG

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 6.0064
Variance Y            : 4.306
Lambda               : 0.71690197
Intercept             : -0.33151324
Slope                 : 0.88690853
Intercept variance     : 0.25474307
Slope variance         : 0.00039571916
Intercept-slope covariance : -0.0088894353
RM variance X          : 6.4857143
RM variance Y          : 6.3271429
Sy.x                  : 1.1973364
Critical value         : 1.959964
RM set size           : 1
procedure_x            : MC

```



```

procedure_y           : MG
clinical_replicates   : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates         : 3
rm_replicates_min     : 3
rm_replicates_max     : 3
slope_se              : 0.019892691
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MC vs MH

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 6.0064
Variance Y            : 4.6228
Lambda               : 0.76964571
Intercept             : -0.32708456
Slope                 : 0.88671139
Intercept variance    : 0.25464172
Slope variance        : 0.00039554326
Intercept-slope covariance : -0.0088854837
RM variance X         : 6.4857143
RM variance Y         : 7.14
Sy.x                  : 1.1971996
Critical value        : 1.959964
RM set size           : 1
procedure_x           : MC
procedure_y           : MH
clinical_replicates   : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates         : 3
rm_replicates_min     : 3
rm_replicates_max     : 3
slope_se              : 0.019888269
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MC vs MI

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 6.0064
Variance Y            : 8.7928
Lambda               : 1.4639052
Intercept             : 0.14597024
Slope                 : 0.87491229
Intercept variance    : 0.21793079
Slope variance        : 0.00033828565
Intercept-slope covariance : -0.0075992489
RM variance X         : 6.4857143
RM variance Y         : 6.3371429

```

```

Sy.x           : 1.1089276
Critical value  : 1.959964
RM set size    : 1
procedure_x    : MC
procedure_y    : MI
clinical_replicates : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates  : 3
rm_replicates_min : 3
rm_replicates_max : 3
slope_se       : 0.018392543
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MC vs MJ

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 6.0064
Variance Y            : 5.0552
Lambda               : 0.84163559
Intercept             : -0.69570407
Slope                 : 0.75265777
Intercept variance     : 0.22031728
Slope variance         : 0.00034224102
Intercept-slope covariance : -0.0076881023
RM variance X          : 6.4857143
RM variance Y          : 7.4885714
Sy.x                  : 1.1135031
Critical value         : 1.959964
RM set size           : 1
procedure_x            : MC
procedure_y            : MJ
clinical_replicates    : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates          : 3
rm_replicates_min      : 3
rm_replicates_max      : 3
slope_se               : 0.018499757
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MD vs ME

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 6.544
Variance Y            : 5.6408
Lambda               : 0.86198044
Intercept             : -1.3435734
Slope                 : 1.359878
Intercept variance     : 0.12687503

```

```

Slope variance           : 0.00031104694
Intercept-slope covariance : -0.005572717
RM variance X            : 7.5157143
RM variance Y            : 6.0485714
Sy.x                     : 0.83905695
Critical value           : 1.959964
RM set size              : 1
procedure_x              : MD
procedure_y              : ME
clinical_replicates      : 3
clinical_replicates_min  : 3
clinical_replicates_max  : 3
rm_replicates            : 3
rm_replicates_min        : 3
rm_replicates_max        : 3
slope_se                 : 0.017636523
relative_residual_evaluable : TRUE
relative_residual_limit   : 2

```

Procedure pair: MD vs MF

```

Clinical samples         : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range       : 3
Equal RM replicates      : TRUE
Variance X               : 6.544
Variance Y               : 5.3108
Lambda                   : 0.81155257
Intercept                : -0.92087196
Slope                    : 1.083996
Intercept variance       : 0.18451044
Slope variance           : 0.00045275179
Intercept-slope covariance : -0.0081115011
RM variance X            : 7.5157143
RM variance Y            : 7.5185714
Sy.x                     : 1.0101669
Critical value           : 1.959964
RM set size              : 1
procedure_x              : MD
procedure_y              : MF
clinical_replicates      : 3
clinical_replicates_min  : 3
clinical_replicates_max  : 3
rm_replicates            : 3
rm_replicates_min        : 3
rm_replicates_max        : 3
slope_se                 : 0.021277965
relative_residual_evaluable : TRUE
relative_residual_limit   : 2

```

Procedure pair: MD vs MG

```

Clinical samples         : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range       : 3
Equal RM replicates      : TRUE
Variance X               : 6.544
Variance Y               : 4.306

```

```

Lambda                : 0.65800733
Intercept              : -0.58590146
Slope                  : 1.1262504
Intercept variance     : 0.13855241
Slope variance         : 0.00033982888
Intercept-slope covariance : -0.0060883743
RM variance X          : 7.5157143
RM variance Y          : 6.3271429
Sy.x                   : 0.87608703
Critical value         : 1.959964
RM set size            : 1
procedure_x            : MD
procedure_y            : MG
clinical_replicates    : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates          : 3
rm_replicates_min      : 3
rm_replicates_max      : 3
slope_se               : 0.018434448
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MD vs MH

```

Clinical samples       : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range     : 3
Equal RM replicates    : TRUE
Variance X             : 6.544
Variance Y             : 4.6228
Lambda                 : 0.70641809
Intercept              : -0.58372347
Slope                  : 1.1261288
Intercept variance     : 0.13852472
Slope variance         : 0.00033975553
Intercept-slope covariance : -0.00608706
RM variance X          : 7.5157143
RM variance Y          : 7.14
Sy.x                   : 0.87602552
Critical value         : 1.959964
RM set size            : 1
procedure_x            : MD
procedure_y            : MH
clinical_replicates    : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates          : 3
rm_replicates_min      : 3
rm_replicates_max      : 3
slope_se               : 0.018432458
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MD vs MI

```

Clinical samples       : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE

```

```

RM replicate range      : 3
Equal RM replicates    : TRUE
Variance X              : 6.544
Variance Y              : 8.7928
Lambda                  : 1.343643
Intercept               : -0.1195644
Slope                   : 1.111831
Intercept variance      : 0.21168945
Slope variance          : 0.00051940298
Intercept-slope covariance : -0.0093056237
RM variance X           : 7.5157143
RM variance Y           : 6.3371429
Sy.x                    : 1.0821696
Critical value           : 1.959964
RM set size             : 1
procedure_x             : MD
procedure_y             : MI
clinical_replicates      : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates           : 3
rm_replicates_min       : 3
rm_replicates_max       : 3
slope_se                : 0.022790414
relative_residual_evaluable : TRUE
relative_residual_limit  : 2

```

Procedure pair: MD vs MJ

```

Clinical samples        : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range      : 3
Equal RM replicates     : TRUE
Variance X              : 6.544
Variance Y              : 5.0552
Lambda                  : 0.77249389
Intercept               : -0.92636579
Slope                   : 0.95659555
Intercept variance      : 0.060523801
Slope variance          : 0.00014836205
Intercept-slope covariance : -0.0026580544
RM variance X           : 7.5157143
RM variance Y           : 7.4885714
Sy.x                    : 0.57964829
Critical value           : 1.959964
RM set size             : 1
procedure_x             : MD
procedure_y             : MJ
clinical_replicates      : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates           : 3
rm_replicates_min       : 3
rm_replicates_max       : 3
slope_se                : 0.012180396
relative_residual_evaluable : TRUE
relative_residual_limit  : 2

```

Procedure pair: ME vs MF

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 5.6408
Variance Y            : 5.3108
Lambda                : 0.94149766
Intercept             : 0.16225188
Slope                 : 0.7966007
Intercept variance     : 0.099273635
Slope variance         : 0.00014378649
Intercept-slope covariance : -0.003309965
RM variance X          : 6.0485714
RM variance Y          : 7.5185714
Sy.x                  : 0.77523923
Critical value         : 1.959964
RM set size            : 1
procedure_x            : ME
procedure_y            : MF
clinical_replicates     : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates          : 3
rm_replicates_min      : 3
rm_replicates_max      : 3
slope_se               : 0.0119911
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: ME vs MG

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 5.6408
Variance Y            : 4.306
Lambda                : 0.7633669
Intercept             : 0.53365093
Slope                 : 0.82790396
Intercept variance     : 0.045349941
Slope variance         : 6.5655065e-05
Intercept-slope covariance : -0.0015113796
RM variance X          : 6.0485714
RM variance Y          : 6.3271429
Sy.x                  : 0.52435434
Critical value         : 1.959964
RM set size            : 1
procedure_x            : ME
procedure_y            : MG
clinical_replicates     : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates          : 3
rm_replicates_min      : 3
rm_replicates_max      : 3

```

```

slope_se           : 0.0081027813
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: ME vs MH

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 5.6408
Variance Y            : 4.6228
Lambda                : 0.81952914
Intercept             : 0.53445619
Slope                 : 0.82786898
Intercept variance     : 0.045346587
Slope variance        : 6.5649517e-05
Intercept-slope covariance : -0.0015112519
RM variance X         : 6.0485714
RM variance Y         : 7.14
Sy.x                  : 0.52434405
Critical value         : 1.959964
RM set size           : 1
procedure_x           : ME
procedure_y           : MH
clinical_replicates    : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates         : 3
rm_replicates_min     : 3
rm_replicates_max     : 3
slope_se              : 0.0081024389
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: ME vs MI

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 5.6408
Variance Y            : 8.7928
Lambda                : 1.558786
Intercept             : 0.97392962
Slope                 : 0.81781366
Intercept variance     : 0.066257176
Slope variance        : 9.5925595e-05
Intercept-slope covariance : -0.0022082072
RM variance X         : 6.0485714
RM variance Y         : 6.3371429
Sy.x                  : 0.63377684
Critical value         : 1.959964
RM set size           : 1
procedure_x           : ME
procedure_y           : MI
clinical_replicates    : 3
clinical_replicates_min : 3

```

```

clinical_replicates_max : 3
rm_replicates           : 3
rm_replicates_min       : 3
rm_replicates_max       : 3
slope_se                : 0.0097941613
relative_residual_evaluable : TRUE
relative_residual_limit  : 2

```

Procedure pair: ME vs MJ

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 5.6408
Variance Y            : 5.0552
Lambda                : 0.89618494
Intercept             : 0.0091741725
Slope                 : 0.70385864
Intercept variance     : 7.4682121e-05
Slope variance         : 1.0807709e-07
Intercept-slope covariance : -2.4879346e-06
RM variance X          : 6.0485714
RM variance Y          : 7.4885714
Sy.x                  : 0.021292767
Critical value         : 1.959964
RM set size            : 1
procedure_x            : ME
procedure_y            : MJ
clinical_replicates     : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates           : 3
rm_replicates_min       : 3
rm_replicates_max       : 3
slope_se               : 0.0003287508
relative_residual_evaluable : TRUE
relative_residual_limit  : 2

```

Procedure pair: MF vs MG

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 5.3108
Variance Y            : 4.306
Lambda                : 0.81080063
Intercept             : 0.36946606
Slope                 : 1.0390559
Intercept variance     : 0.068686655
Slope variance         : 0.00015459995
Intercept-slope covariance : -0.0028600991
RM variance X          : 7.5185714
RM variance Y          : 6.3271429
Sy.x                  : 0.64093887
Critical value         : 1.959964
RM set size            : 1

```



```

procedure_x      : MF
procedure_y      : MG
clinical_replicates : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates    : 3
rm_replicates_min : 3
rm_replicates_max : 3
slope_se         : 0.012433823
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MF vs MH

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 5.3108
Variance Y            : 4.6228
Lambda                : 0.87045266
Intercept             : 0.37066715
Slope                 : 1.038991
Intercept variance    : 0.068678934
Slope variance        : 0.00015458063
Intercept-slope covariance : -0.0028597417
RM variance X         : 7.5185714
RM variance Y         : 7.14
Sy.x                  : 0.64091635
Critical value         : 1.959964
RM set size           : 1
procedure_x           : MF
procedure_y           : MH
clinical_replicates    : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates         : 3
rm_replicates_min     : 3
rm_replicates_max     : 3
slope_se              : 0.012433046
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MF vs MI

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 5.3108
Variance Y            : 8.7928
Lambda                : 1.6556451
Intercept             : 0.81364284
Slope                 : 1.0262896
Intercept variance    : 0.083812322
Slope variance        : 0.00018863976
Intercept-slope covariance : -0.0034898356
RM variance X         : 7.5185714

```

```

RM variance Y           : 6.3371429
Sy.x                   : 0.70803357
Critical value          : 1.959964
RM set size            : 1
procedure_x            : MF
procedure_y            : MI
clinical_replicates    : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates          : 3
rm_replicates_min      : 3
rm_replicates_max      : 3
slope_se               : 0.013734619
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MF vs MJ

```

Clinical samples       : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range     : 3
Equal RM replicates    : TRUE
Variance X             : 5.3108
Variance Y             : 5.0552
Lambda                : 0.95187166
Intercept              : -0.12065115
Slope                  : 0.88284601
Intercept variance     : 0.07817879
Slope variance         : 0.00017600578
Intercept-slope covariance : -0.0032561068
RM variance X          : 7.5185714
RM variance Y          : 7.4885714
Sy.x                   : 0.68352665
Critical value          : 1.959964
RM set size            : 1
procedure_x            : MF
procedure_y            : MJ
clinical_replicates    : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates          : 3
rm_replicates_min      : 3
rm_replicates_max      : 3
slope_se               : 0.013266717
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MG vs MH

```

Clinical samples       : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range     : 3
Equal RM replicates    : TRUE
Variance X             : 4.306
Variance Y             : 4.6228
Lambda                : 1.0735718
Intercept              : 0
Slope                  : 1

```

```

Intercept variance      : 2.8587206e-15
Slope variance         : 5.9666732e-18
Intercept-slope covariance : -1.1689906e-16
RM variance X          : 6.3271429
RM variance Y          : 7.14
Sy.x                   : 1.7389557e-15
Critical value         : 1.959964
RM set size            : 1
procedure_x            : MG
procedure_y            : MH
clinical_replicates    : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates         : 3
rm_replicates_min     : 3
rm_replicates_max     : 3
slope_se              : 2.4426775e-09
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MG vs MI

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X           : 4.306
Variance Y           : 8.7928
Lambda               : 2.0419879
Intercept            : 0.43718925
Slope                : 0.9883019
Intercept variance    : 0.016925594
Slope variance       : 3.4246951e-05
Intercept-slope covariance : -0.00067096627
RM variance X        : 6.3271429
RM variance Y        : 6.3371429
Sy.x                 : 0.31374846
Critical value       : 1.959964
RM set size          : 1
procedure_x          : MG
procedure_y          : MI
clinical_replicates  : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates       : 3
rm_replicates_min   : 3
rm_replicates_max   : 3
slope_se            : 0.0058520895
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MG vs MJ

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X           : 4.306

```

```

Variance Y           : 5.0552
Lambda              : 1.1739898
Intercept           : -0.43879659
Slope               : 0.84987733
Intercept variance  : 0.033635453
Slope variance      : 6.8072223e-05
Intercept-slope covariance : -0.001333671
RM variance X       : 6.3271429
RM variance Y       : 7.4885714
Sy.x                : 0.4421235
Critical value       : 1.959964
RM set size         : 1
procedure_x         : MG
procedure_y         : MJ
clinical_replicates : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates       : 3
rm_replicates_min   : 3
rm_replicates_max   : 3
slope_se            : 0.0082505893
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MH vs MI

```

Clinical samples      : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range    : 3
Equal RM replicates    : TRUE
Variance X            : 4.6228
Variance Y            : 8.7928
Lambda                : 1.9020507
Intercept             : 0.43692235
Slope                 : 0.98831552
Intercept variance    : 0.016925991
Slope variance        : 3.4247895e-05
Intercept-slope covariance : -0.00067098477
RM variance X         : 7.14
RM variance Y         : 6.3371429
Sy.x                  : 0.3137499
Critical value         : 1.959964
RM set size           : 1
procedure_x            : MH
procedure_y            : MI
clinical_replicates    : 3
clinical_replicates_min : 3
clinical_replicates_max : 3
rm_replicates          : 3
rm_replicates_min      : 3
rm_replicates_max      : 3
slope_se              : 0.0058521701
relative_residual_evaluable : TRUE
relative_residual_limit : 2

```

Procedure pair: MH vs MJ

```

Clinical samples      : 25
Clinical replicate range : 3

```

```

Equal clinical replicates : TRUE
RM replicate range       : 3
Equal RM replicates      : TRUE
Variance X               : 4.6228
Variance Y               : 5.0552
Lambda                   : 1.0935364
Intercept                : -0.43945832
Slope                    : 0.8499111
Intercept variance       : 0.033637762
Slope variance           : 6.8077634e-05
Intercept-slope covariance : -0.001333777
RM variance X            : 7.14
RM variance Y            : 7.4885714
Sy.x                     : 0.44213034
Critical value           : 1.959964
RM set size              : 1
procedure_x              : MH
procedure_y              : MJ
clinical_replicates      : 3
clinical_replicates_min  : 3
clinical_replicates_max  : 3
rm_replicates            : 3
rm_replicates_min       : 3
rm_replicates_max       : 3
slope_se                 : 0.0082509172
relative_residual_evaluable : TRUE
relative_residual_limit  : 2

```

Procedure pair: MI vs MJ

```

Clinical samples         : 25
Clinical replicate range : 3
Equal clinical replicates : TRUE
RM replicate range       : 3
Equal RM replicates      : TRUE
Variance X               : 8.7928
Variance Y               : 5.0552
Lambda                   : 0.57492494
Intercept                : -0.82074864
Slope                    : 0.86023983
Intercept variance       : 0.053522283
Slope variance           : 0.00010712641
Intercept-slope covariance : -0.002121103
RM variance X            : 6.3371429
RM variance Y            : 7.4885714
Sy.x                     : 0.54782818
Critical value           : 1.959964
RM set size              : 1
procedure_x              : MI
procedure_y              : MJ
clinical_replicates      : 3
clinical_replicates_min  : 3
clinical_replicates_max  : 3
rm_replicates            : 3
rm_replicates_min       : 3
rm_replicates_max       : 3
slope_se                 : 0.010350189
relative_residual_evaluable : TRUE
relative_residual_limit  : 2

```

6.3.3 IFCC Part 2 Bias Difference and Uncertainty

```
ifcc2 <- as.data.frame(read_csv(
  "/data/006-基质效应与互换性/std-ifcc-part2-example.csv",
  show_col_types = FALSE
))
ifcc2_fit <- commutability(
  ifcc2, approach = "ifcc", sample_id = "sample_id",
  material_type = "material_type", procedure = "procedure", result = "result",
  position = "position", scale = "ln",
  procedure_pairs = list(c("MPx", "MPy")), criterion = 0.12,
  bias_model = "constant_global", coverage_factor = 1.9,
  rm_position = "pooled"
)
print(ifcc2_fit)
```

Commutability assessment

Study-design checks: 4 met, 0 unmet. Unmet checks do not invalidate computations; inspect '\$design_checks'.

```
Approach      : IFCC
Scale         : ln
Procedure pairs : 1
Reference materials : 5
Procedure pair: MPx vs MPy | Reference material: RM1
RM coordinate      : 2.7755556
RM bias            : 0.33925926
Clinical bias      : 0.18095238
Difference in bias  : 0.15830688
RM standard uncertainty : 0.037626363
Clinical standard uncertainty : 0.015516499
Standard uncertainty : 0.040700184
Expanded uncertainty : 0.077330349
Lower              : 0.080976529
Upper              : 0.23563723
Criterion          : 0.12
Clinical samples    : 49
Classification      : inconclusive
procedure_x         : MPx
procedure_y         : MPy
position_mean_sd_x   : 0.067355047
position_mean_sd_y   : 0.050418387
positions_x         : 5
positions_y         : 5
position_pooling     : pooled
```

```
Procedure pair: MPx vs MPy | Reference material: RM2
RM coordinate      : 3.3097037
RM bias            : 0.12903704
Clinical bias      : 0.18095238
Difference in bias  : -0.051915344
RM standard uncertainty : 0.037626363
Clinical standard uncertainty : 0.015516499
Standard uncertainty : 0.040700184
Expanded uncertainty : 0.077330349
Lower              : -0.12924569
Upper              : 0.025415005
Criterion          : 0.12
Clinical samples    : 49
```

```

Classification      : inconclusive
procedure_x         : MPx
procedure_y         : MPy
position_mean_sd_x  : 0.067355047
position_mean_sd_y  : 0.050418387
positions_x         : 5
positions_y         : 5
position_pooling    : pooled

```

Procedure pair: MPx vs MPy | Reference material: RM3

```

RM coordinate       : 5.5979259
RM bias             : 0.16
Clinical bias       : 0.18095238
Difference in bias   : -0.020952381
RM standard uncertainty : 0.037626363
Clinical standard uncertainty : 0.015516499
Standard uncertainty : 0.040700184
Expanded uncertainty : 0.077330349
Lower               : -0.09828273
Upper               : 0.056377968
Criterion           : 0.12
Clinical samples    : 49
Classification      : commutable
procedure_x         : MPx
procedure_y         : MPy
position_mean_sd_x  : 0.067355047
position_mean_sd_y  : 0.050418387
positions_x         : 5
positions_y         : 5
position_pooling    : pooled

```

Procedure pair: MPx vs MPy | Reference material: RM4

```

RM coordinate       : 1.607037
RM bias             : 0.073333333
Clinical bias       : 0.18095238
Difference in bias   : -0.10761905
RM standard uncertainty : 0.037626363
Clinical standard uncertainty : 0.015516499
Standard uncertainty : 0.040700184
Expanded uncertainty : 0.077330349
Lower               : -0.1849494
Upper               : -0.030288699
Criterion           : 0.12
Clinical samples    : 49
Classification      : inconclusive
procedure_x         : MPx
procedure_y         : MPy
position_mean_sd_x  : 0.067355047
position_mean_sd_y  : 0.050418387
positions_x         : 5
positions_y         : 5
position_pooling    : pooled

```

Procedure pair: MPx vs MPy | Reference material: RM5

```

RM coordinate       : 4.4167407
RM bias             : 0.85866667
Clinical bias       : 0.18095238
Difference in bias   : 0.67771429

```

```

RM standard uncertainty      : 0.037626363
Clinical standard uncertainty : 0.015516499
Standard uncertainty         : 0.040700184
Expanded uncertainty         : 0.077330349
Lower                        : 0.60038394
Upper                        : 0.75504463
Criterion                    : 0.12
Clinical samples             : 49
Classification                : noncommutable
procedure_x                  : MPx
procedure_y                  : MPy
position_mean_sd_x           : 0.067355047
position_mean_sd_y           : 0.050418387
positions_x                  : 5
positions_y                  : 5
position_pooling              : pooled

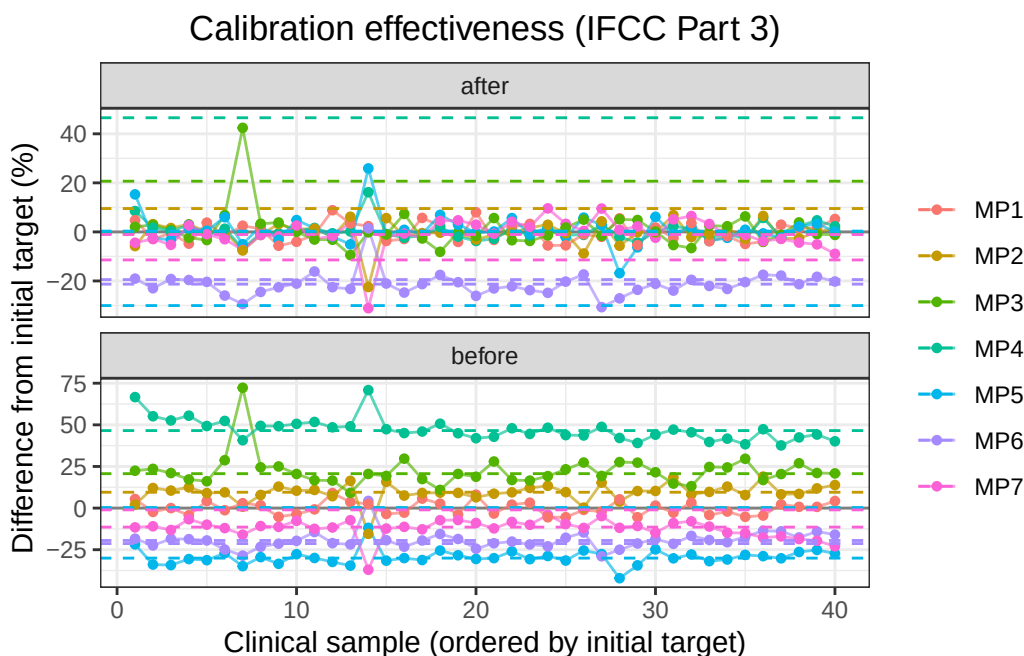
```

6.3.4 IFCC Part 3 Calibration Validity

```

ifcc3 <- as.data.frame(read_csv(
  "./data/006-基质效应与互换性/std-ifcc-part3-example.csv",
  show_col_types = FALSE
))
ifcc3_fit <- commutability(
  ifcc3, approach = "calibration", sample_id = "sample_id",
  procedure = "procedure", result = "result",
  calibration_stage = "calibration_stage", criterion = 6
)
plot(ifcc3_fit)

```



6.4 Interpretation

Prediction-interval classification and bias-difference classification need not agree. A reference material can lie within an EP14/EP30 prediction interval yet have an unacceptable bias difference under IFCC Part 2, or vice versa, because the paths target different estimands

and uncertainty models. The final report should state the path, scale, procedure pair, material definition, repeated-measurement handling, limit, and any procedure pairs that were unavailable or failed design checks.

6.5 Detailed Standard Cases

6.5.1 CLSI EP14-A3 Appendix C Example C1: Cholesterol on the Original Scale

Appendix C Example C1 contains 20 patient samples and 5 processed samples. Both procedures have three replicate measurements for cholesterol in mg/dL. Table C1 labels the last two processed samples as “d”; the data file names them `d_run16` and `d_run07` according to run order only to keep material identities unique. The measurement values are unchanged.

```
ep14_c1_data <- read_csv(
  "./data/006-基质效应与互换性/std-ep14-appendix-c1.csv",
  show_col_types = FALSE
)
head(ep14_c1_data, 6)
```

A tibble: 6 x 7

	sample_id	material_type	run_order	procedure	replicate	result	position
	<chr>	<chr>	<dbl>	<chr>	<dbl>	<dbl>	<lgf>
1	CS01	clinical	21	A	1	206.	NA
2	CS01	clinical	21	A	2	213.	NA
3	CS01	clinical	21	A	3	205.	NA
4	CS01	clinical	21	B	1	217.	NA
5	CS01	clinical	21	B	2	209.	NA
6	CS01	clinical	21	B	3	220.	NA

```
ep14_c1 <- commutability(
  ep14_c1_data,
  sample_id = "sample_id",
  material_type = "material_type",
  procedure = "procedure",
  result = "result",
  approach = "ep14",
  scale = "raw",
  procedure_pairs = list(c("A", "B")),
  conf.level = 0.95
)
ep14_c1
```

Commutability assessment

Study-design checks: 2 met, 0 unmet. Unmet checks do not invalidate computations; inspect '\$design_checks'.

```
Approach      : EP14
Scale         : raw
Procedure pairs : 1
Reference materials : 5
Procedure pair: A vs B | Reference material: a
Mean X       : 108.86
Mean Y       : 110.15
Predicted Y  : 117.19384
Residual     : -7.0438387
Prediction SE : 3.9893651
Lower        : 109.13103
```

Upper : 125.25665
 Extrapolated : FALSE
 Classification : commutable
 procedure_x : A
 procedure_y : B

Procedure pair: A vs B | Reference material: b

Mean X : 123.12333
 Mean Y : 136.95
 Predicted Y : 132.71701
 Residual : 4.2329936
 Prediction SE : 3.9111051
 Lower : 124.81237
 Upper : 140.62164
 Extrapolated : FALSE
 Classification : commutable
 procedure_x : A
 procedure_y : B

Procedure pair: A vs B | Reference material: c

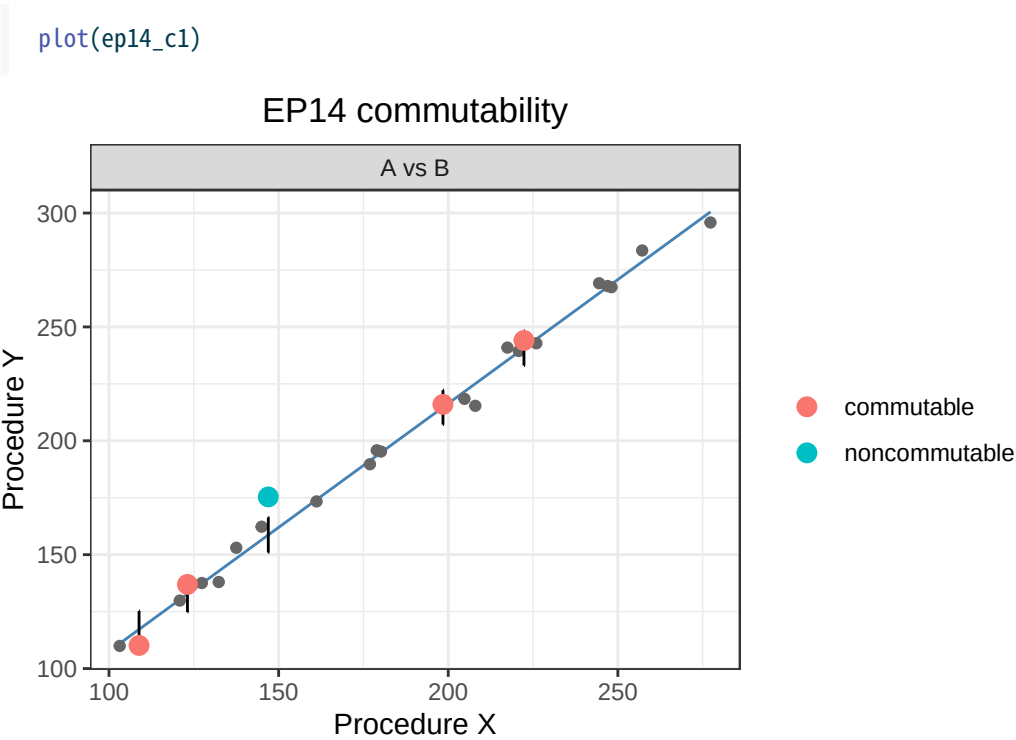
Mean X : 146.95
 Mean Y : 175.36667
 Predicted Y : 158.64821
 Residual : 16.718461
 Prediction SE : 3.8114445
 Lower : 150.94499
 Upper : 166.35142
 Extrapolated : FALSE
 Classification : noncommutable
 procedure_x : A
 procedure_y : B

Procedure pair: A vs B | Reference material: d_run16

Mean X : 198.44
 Mean Y : 215.98333
 Predicted Y : 214.68615
 Residual : 1.2971814
 Prediction SE : 3.740138
 Lower : 207.12705
 Upper : 222.24525
 Extrapolated : FALSE
 Classification : commutable
 procedure_x : A
 procedure_y : B

Procedure pair: A vs B | Reference material: d_run07

Mean X : 222.31667
 Mean Y : 244.06667
 Predicted Y : 240.67177
 Residual : 3.3948989
 Prediction SE : 3.7763512
 Lower : 233.03948
 Upper : 248.30406
 Extrapolated : FALSE
 Classification : commutable
 procedure_x : A
 procedure_y : B



The standard also reports a prediction and prediction interval at $X = 145.0$ mg/dL. The following calculates these quantities at the same reference point from the model, clinical-sample means, and repeatability variances returned by `commutability()`.

Table 6.1: EP14-A3 Example C1: comparison of `ivdtools` and standard results

标准位置	统计量	标准结果	ivdtools结果
EP14 C1	variance_x	15.060	15.062097
EP14 C1	variance_y	22.080	22.077815
EP14 C1	lambda	1.466	1.465786
EP14 C1	slope	1.088	1.088327
EP14 C1	intercept	-1.282	-1.281415
EP14 C1	prediction_at_x_145	156.500	156.525969
EP14 C1	prediction_se_at_x_145	3.820	3.818079
EP14 C1	lower_at_x_145	148.800	148.809344
EP14 C1	upper_at_x_145	164.200	164.242593
EP14 C1	noncommutable_count	1.000	1.000000

Table 6.2: EP14-A3 Example C1: prediction-interval classification of processed samples

rm	mean_x	mean_y	predicted_y	lower	upper	extrapolated	classification
a	108.860	110.150	117.194	109.131	125.257	FALSE	Commutable
b	123.123	136.950	132.717	124.812	140.622	FALSE	Commutable
c	146.950	175.367	158.648	150.945	166.351	FALSE	Non-commutable
d_run16	198.440	215.983	214.686	207.127	222.245	FALSE	Commutable
d_run07	222.317	244.067	240.672	233.039	248.304	FALSE	Commutable

Processed sample c is outside the prediction interval; the other four are inside. No processed sample is extrapolated beyond the concentration range. This conclusion applies only to procedures A–B, the original scale, and the concentration range in this case.

6.5.2 CLSI EP14-A3 Appendix C Example C2: Myoglobin on the log10 Scale

Example C2 contains 20 patient samples and 7 processed samples for myoglobin in ng/mL. Because the dispersion of differences on the original scale increases with concentration, the standard applies a log10 transformation to every replicate result before fitting the Deming relationship.

```
ep14_c2_data <- read_csv(
  "/data/006-基质效应与互换性/std-ep14-appendix-c2.csv",
  show_col_types = FALSE
)
head(ep14_c2_data, 6)
```

```
# A tibble: 6 x 7
  sample_id material_type run_order procedure replicate result position
  <chr>      <chr>          <dbl> <chr>          <dbl> <dbl> <lgl>
1 CS01      clinical          6 A              1  81.2 NA
2 CS01      clinical          6 A              2  77.0 NA
3 CS01      clinical          6 A              3  76.4 NA
4 CS01      clinical          6 B              1  80.7 NA
5 CS01      clinical          6 B              2  82.2 NA
6 CS01      clinical          6 B              3  81.3 NA
```

```
ep14_c2 <- commutability(
  ep14_c2_data,
  sample_id = "sample_id",
  material_type = "material_type",
  procedure = "procedure",
  result = "result",
  approach = "ep14",
  scale = "log10",
  procedure_pairs = list(c("A", "B")),
  conf.level = 0.95
)
summary(ep14_c2)
```

Commutability assessment

Study-design checks: 2 met, 0 unmet. Unmet checks do not invalidate computations; inspect '\$design_checks'.

```
Approach      : EP14
Scale         : log10
Procedure pairs : 1
Reference materials : 7
Procedure pair: A vs B | Reference material: a
  Mean X      : 2.8940948
  Mean Y      : 2.8397634
  Predicted Y  : 2.8935335
  Residual     : -0.053770073
  Prediction SE : 0.020229796
  Lower       : 2.8526475
  Upper       : 2.9344194
  Extrapolated : FALSE
  Classification : noncommutable
  procedure_x   : A
  procedure_y   : B
```

```
Procedure pair: A vs B | Reference material: b
```

Mean X : 2.6008874
 Mean Y : 2.5519334
 Predicted Y : 2.5995654
 Residual : -0.047631984
 Prediction SE : 0.01996673
 Lower : 2.5592112
 Upper : 2.6399197
 Extrapolated : FALSE
 Classification : noncommutable
 procedure_x : A
 procedure_y : B

Procedure pair: A vs B | Reference material: c

Mean X : 3.1955496
 Mean Y : 3.2183062
 Predicted Y : 3.1957703
 Residual : 0.022535935
 Prediction SE : 0.021016093
 Lower : 3.1532952
 Upper : 3.2382454
 Extrapolated : FALSE
 Classification : commutable
 procedure_x : A
 procedure_y : B

Procedure pair: A vs B | Reference material: d

Mean X : 2.0161898
 Mean Y : 1.96499
 Predicted Y : 2.0133509
 Residual : -0.048360943
 Prediction SE : 0.020959923
 Lower : 1.9709893
 Upper : 2.0557125
 Extrapolated : FALSE
 Classification : noncommutable
 procedure_x : A
 procedure_y : B

Procedure pair: A vs B | Reference material: e

Mean X : 3.4766983
 Mean Y : 3.4720027
 Predicted Y : 3.4776484
 Residual : -0.0056457436
 Prediction SE : 0.022172569
 Lower : 3.432836
 Upper : 3.5224608
 Extrapolated : FALSE
 Classification : commutable
 procedure_x : A
 procedure_y : B

Procedure pair: A vs B | Reference material: f

Mean X : 2.3100027
 Mean Y : 2.2570492
 Predicted Y : 2.307926
 Residual : -0.050876857
 Prediction SE : 0.020214197
 Lower : 2.2670716

```

Upper      : 2.3487804
Extrapolated : FALSE
Classification : noncommutable
procedure_x  : A
procedure_y  : B

```

Procedure pair: A vs B | Reference material: g

```

Mean X      : 1.7036698
Mean Y      : 1.6448325
Predicted Y : 1.7000201
Residual    : -0.05518756
Prediction SE : 0.022243376
Lower       : 1.6550646
Upper       : 1.7449756
Extrapolated : FALSE
Classification : noncommutable
procedure_x  : A
procedure_y  : B

```

EP14 model estimates

Procedure pair: A vs B

```

Clinical samples : 20
Variance X      : 0.00056562731
Variance Y      : 0.00057048831
Lambda         : 1.008594
Intercept      : -0.008069495
Slope          : 1.0025943
Slope SE       : 0.010966691
DF             : 40
procedure_x     : A
procedure_y     : B

```

Table 6.3: EP14-A3 Example C2: comparison of ivdtools and standard results

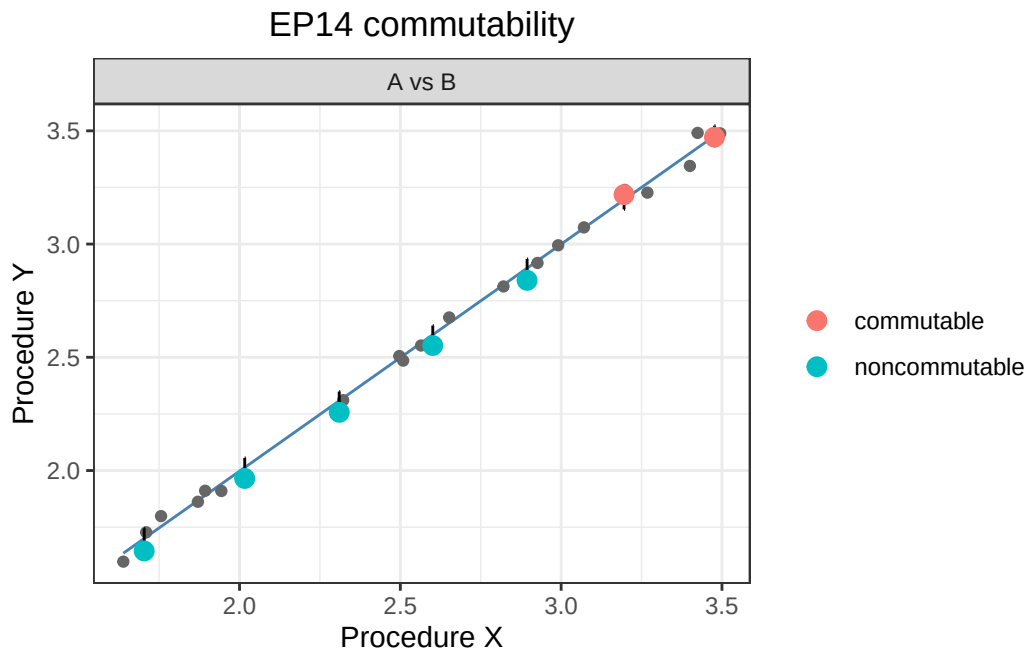
标准位置	统计量	标准结果	ivdtools结果
EP14 C2	variance_x	0.000566	0.0005656
EP14 C2	variance_y	0.000571	0.0005705
EP14 C2	lambda	1.009000	1.0085940
EP14 C2	slope	1.002600	1.0025943
EP14 C2	intercept	-0.008100	-0.0080695
EP14 C2	prediction_at_x_1.64	1.635000	1.6361851
EP14 C2	prediction_se_at_x_1.64	0.022600	0.0225598
EP14 C2	lower_at_x_1.64	1.589000	1.5905901
EP14 C2	upper_at_x_1.64	1.681000	1.6817802
EP14 C2	noncommutable_count	5.000000	5.0000000

Table 6.4: EP14-A3 Example C2: classification on the log10 scale

rm	mean_x	mean_y	predicted_y	lower	upper	extrapolated	classification
a	2.8941	2.8398	2.8935	2.8526	2.9344	FALSE	Non-commutable
b	2.6009	2.5519	2.5996	2.5592	2.6399	FALSE	Non-commutable
c	3.1955	3.2183	3.1958	3.1533	3.2382	FALSE	Commutable
d	2.0162	1.9650	2.0134	1.9710	2.0557	FALSE	Non-commutable

rm	mean_x	mean_y	predicted_y	lower	upper	extrapolated	classification
e	3.4767	3.4720	3.4776	3.4328	3.5225	FALSE	Commutable
f	2.3100	2.2570	2.3079	2.2671	2.3488	FALSE	Non-commutable
g	1.7037	1.6448	1.7000	1.6551	1.7450	FALSE	Non-commutable

```
plot(ep14_c2)
```



Processed samples a, b, d, f, and g are noncommutable, while c and e are commutable, giving five noncommutable materials in total. No processed sample is extrapolated.

6.5.3 CLSI EP30-A Appendix B: Multi-Procedure Reference-Material Data

EP30-A Appendix B provides 25 clinical samples, 7 candidate reference materials, 10 measurement procedures, and three replicate measurements per combination.

```
ep30_raw <- read_csv(
  "./data/006-基质效应与互换性/std-ep30-appendix-b.csv",
  show_col_types = FALSE
)
```

6.5.3.1 MA-MB and MA-MD Regression Results

Use `approach = "ep30"` directly. The seven candidate reference materials are used independently in this example, so `rm_set_size = 1`; the function does not automatically treat the total number of materials as the size of a correlated set.

```
ep30_result <- commutability(
  ep30_raw,
  sample_id = "specimen",
  material_type = "type",
)
```

```

procedure = "method",
result = "result",
approach = "ep30",
scale = "raw",
procedure_pairs = list(c("MA", "MB"), c("MA", "MD")),
conf.level = 0.95,
rm_set_size = 1,
relative_residual_limit = 2
)
ep30_result

```

Commutability assessment

Study-design checks: 8 met, 0 unmet. Unmet checks do not invalidate computations; inspect '\$design_checks'.

```

Approach      : EP30
Scale         : raw
Procedure pairs : 2
Reference materials : 14
Procedure pair: MA vs MB | Reference material: RMA
  Mean X      : 12.4
  Mean Y      : 14.3
  Predicted Y  : 15.576069
  Residual     : -1.2760688
  Prediction SE : 1.8530797
  Lower       : 11.944099
  Upper       : 19.208038
  Extrapolated : FALSE
  Prediction classification : commutable
  Sy.x        : 1.1890275
  Relative residual : 1.0732038
  Relative-residual limit : 2
  Relative-residual classification : commutable
  procedure_x   : MA
  procedure_y   : MB
  critical_value : 1.959964
  rm_set_size   : 1
  classification : commutable

```

```

Procedure pair: MA vs MB | Reference material: RMB
  Mean X      : 24.1
  Mean Y      : 26.9
  Predicted Y  : 26.829327
  Residual     : 0.070672943
  Prediction SE : 1.8454797
  Lower       : 23.212253
  Upper       : 30.446401
  Extrapolated : FALSE
  Prediction classification : commutable
  Sy.x        : 1.1890275
  Relative residual : -0.059437604
  Relative-residual limit : 2
  Relative-residual classification : commutable
  procedure_x   : MA
  procedure_y   : MB
  critical_value : 1.959964
  rm_set_size   : 1
  classification : commutable

```


Procedure pair: MA vs MB | Reference material: RMC

Mean X	: 35.4
Mean Y	: 37
Predicted Y	: 37.697859
Residual	: -0.69785851
Prediction SE	: 1.8688251
Lower	: 34.035029
Upper	: 41.360688
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.1890275
Relative residual	: 0.58691538
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MA
procedure_y	: MB
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MA vs MB | Reference material: RMD

Mean X	: 42.2
Mean Y	: 41.7
Predicted Y	: 44.238214
Residual	: -2.5382137
Prediction SE	: 1.8969998
Lower	: 40.520162
Upper	: 47.956265
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.1890275
Relative residual	: 2.1346973
Relative-residual limit	: 2
Relative-residual classification	: noncommutable
procedure_x	: MA
procedure_y	: MB
critical_value	: 1.959964
rm_set_size	: 1
classification	: commutable

Procedure pair: MA vs MB | Reference material: RME

Mean X	: 15.3
Mean Y	: 19
Predicted Y	: 18.365338
Residual	: 0.63466204
Prediction SE	: 1.8481767
Lower	: 14.742978
Upper	: 21.987698
Extrapolated	: FALSE
Prediction classification	: commutable
Sy.x	: 1.1890275
Relative residual	: -0.53376567
Relative-residual limit	: 2
Relative-residual classification	: commutable
procedure_x	: MA
procedure_y	: MB
critical_value	: 1.959964
rm_set_size	: 1

classification : commutable

Procedure pair: MA vs MB | Reference material: RMF

Mean X : 21
 Mean Y : 20
 Predicted Y : 23.847695
 Residual : -3.8476945
 Prediction SE : 1.8443328
 Lower : 20.232869
 Upper : 27.46252
 Extrapolated : FALSE
 Prediction classification : noncommutable
 Sy.x : 1.1890275
 Relative residual : 3.2360014
 Relative-residual limit : 2
 Relative-residual classification : noncommutable
 procedure_x : MA
 procedure_y : MB
 critical_value : 1.959964
 rm_set_size : 1
 classification : noncommutable

Procedure pair: MA vs MB | Reference material: RMG

Mean X : 17
 Mean Y : 21.1
 Predicted Y : 20.000427
 Residual : 1.0995732
 Prediction SE : 1.8462245
 Lower : 16.381893
 Upper : 23.61896
 Extrapolated : FALSE
 Prediction classification : commutable
 Sy.x : 1.1890275
 Relative residual : -0.9247669
 Relative-residual limit : 2
 Relative-residual classification : commutable
 procedure_x : MA
 procedure_y : MB
 critical_value : 1.959964
 rm_set_size : 1
 classification : commutable

Procedure pair: MA vs MD | Reference material: RMA

Mean X : 12.4
 Mean Y : 11.4
 Predicted Y : 10.746146
 Residual : 0.65385407
 Prediction SE : 1.8882069
 Lower : 7.0453283
 Upper : 14.446964
 Extrapolated : FALSE
 Prediction classification : commutable
 Sy.x : 1.0324006
 Relative residual : -0.63333369
 Relative-residual limit : 2
 Relative-residual classification : commutable
 procedure_x : MA
 procedure_y : MD

```
critical_value      : 1.959964
rm_set_size        : 1
classification      : commutable
```

Procedure pair: MA vs MD | Reference material: RMB

```
Mean X              : 24.1
Mean Y              : 19.9
Predicted Y         : 20.509565
Residual            : -0.60956453
Prediction SE       : 1.8825915
Lower               : 16.819753
Upper               : 24.199376
Extrapolated        : FALSE
Prediction classification : commutable
Sy.x                : 1.0324006
Relative residual    : 0.59043412
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MA
procedure_y           : MD
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable
```

Procedure pair: MA vs MD | Reference material: RMC

```
Mean X              : 35.4
Mean Y              : 29.6
Predicted Y         : 29.939191
Residual            : -0.33919104
Prediction SE       : 1.8998608
Lower               : 26.215532
Upper               : 33.66285
Extrapolated        : FALSE
Prediction classification : commutable
Sy.x                : 1.0324006
Relative residual    : 0.32854596
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x          : MA
procedure_y           : MD
critical_value        : 1.959964
rm_set_size           : 1
classification        : commutable
```

Procedure pair: MA vs MD | Reference material: RMD

```
Mean X              : 42.2
Mean Y              : 34.6
Predicted Y         : 35.613657
Residual            : -1.0136566
Prediction SE       : 1.9207804
Lower               : 31.848996
Upper               : 39.378317
Extrapolated        : FALSE
Prediction classification : commutable
Sy.x                : 1.0324006
Relative residual    : 0.98184422
Relative-residual limit : 2
Relative-residual classification : commutable
```

```

procedure_x      : MA
procedure_y      : MD
critical_value   : 1.959964
rm_set_size      : 1
classification   : commutable

```

Procedure pair: MA vs MD | Reference material: RME

```

Mean X          : 15.3
Mean Y          : 13.2
Predicted Y     : 13.166139
Residual        : 0.033861426
Prediction SE    : 1.8845835
Lower           : 9.4724228
Upper           : 16.859854
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.0324006
Relative residual : -0.032798728
Relative-residual limit : 2
Relative-residual classification : commutable
procedure_x      : MA
procedure_y      : MD
critical_value   : 1.959964
rm_set_size      : 1
classification   : commutable

```

Procedure pair: MA vs MD | Reference material: RMF

```

Mean X          : 21
Mean Y          : 24.9
Predicted Y     : 17.922676
Residual        : 6.9773242
Prediction SE    : 1.8817446
Lower           : 14.234524
Upper           : 21.610828
Extrapolated    : FALSE
Prediction classification : noncommutable
Sy.x            : 1.0324006
Relative residual : -6.7583496
Relative-residual limit : 2
Relative-residual classification : noncommutable
procedure_x      : MA
procedure_y      : MD
critical_value   : 1.959964
rm_set_size      : 1
classification   : noncommutable

```

Procedure pair: MA vs MD | Reference material: RMG

```

Mean X          : 17
Mean Y          : 14.7
Predicted Y     : 14.584755
Residual        : 0.11524505
Prediction SE    : 1.8831415
Lower           : 10.893865
Upper           : 18.275645
Extrapolated    : FALSE
Prediction classification : commutable
Sy.x            : 1.0324006
Relative residual : -0.11162823

```

```

Relative-residual limit      : 2
Relative-residual classification : commutable
procedure_x                  : MA
procedure_y                  : MD
critical_value                : 1.959964
rm_set_size                   : 1
classification                : commutable

```

6.5.3.2 Reference-Material Predictions and Relative Residuals

The relative residual in EP30 Table 1 is the distance from the reference material to the regression line divided by the clinical-sample regression standard error. `commutability()` returns the sign used in the table, `predicted value – measured value`, and uses $n - 1$ as the denominator degrees of freedom for $S_{y.x}$ as in the example. Prediction intervals are calculated separately using Appendix C16–C20.

Table 6.5: EP30-A Appendix B Table 1: regression parameters and relative residuals

标准位置	统计量	标准结果	ivdtools结果
EP30 MA-MB	intercept	3.660	3.6495
EP30 MA-MB	slope	0.961	0.9618
EP30 MA-MB	syx	1.188	1.1890
EP30 MA-MB	RMA	1.088	1.0732
EP30 MA-MB	RMB	-0.117	-0.0594
EP30 MA-MB	RMC	0.601	0.5869
EP30 MA-MB	RMD	2.164	2.1347
EP30 MA-MB	RME	-0.523	-0.5338
EP30 MA-MB	RMF	3.241	3.2360
EP30 MA-MB	RMG	-0.923	-0.9248
EP30 MA-MD	intercept	0.396	0.3986
EP30 MA-MD	slope	0.834	0.8345
EP30 MA-MD	syx	1.033	1.0324
EP30 MA-MD	RMA	-0.603	-0.6333
EP30 MA-MD	RMB	0.518	0.5904
EP30 MA-MD	RMC	0.326	0.3285
EP30 MA-MD	RMD	0.997	0.9818
EP30 MA-MD	RME	-0.021	-0.0328
EP30 MA-MD	RMF	-6.758	-6.7583
EP30 MA-MD	RMG	-0.066	-0.1116

Under the $\pm 2S_{y.x}$ rule, RMF in MA–MD is noncommutable; RMD and RMF in MA–MB exceed the limits. RMD remains commutable under the 95% prediction-interval method for MA–MB, showing that materials near a boundary can receive different classifications under different evaluation rules.

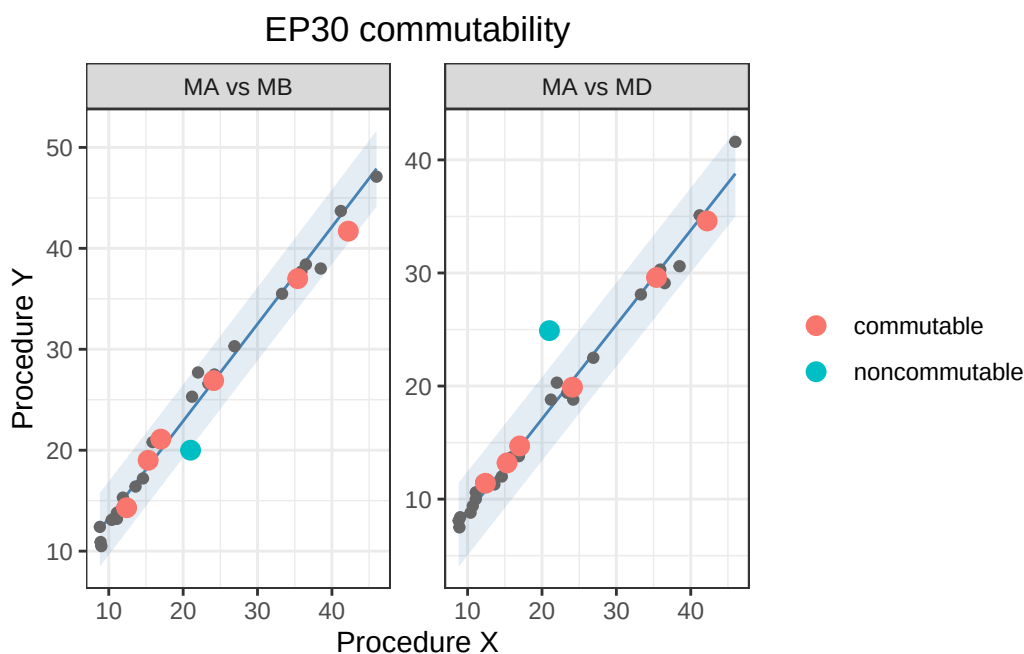
Prediction intervals and relative residuals classify RMD in MA–MB differently: the former calls it commutable, while the latter calls it noncommutable. This is a boundary difference caused by the evaluation rule; do not merge the two columns into an undefined “final classification.”

Table 6.6: EP30-A Appendix B: prediction-interval and relative-residual classifications

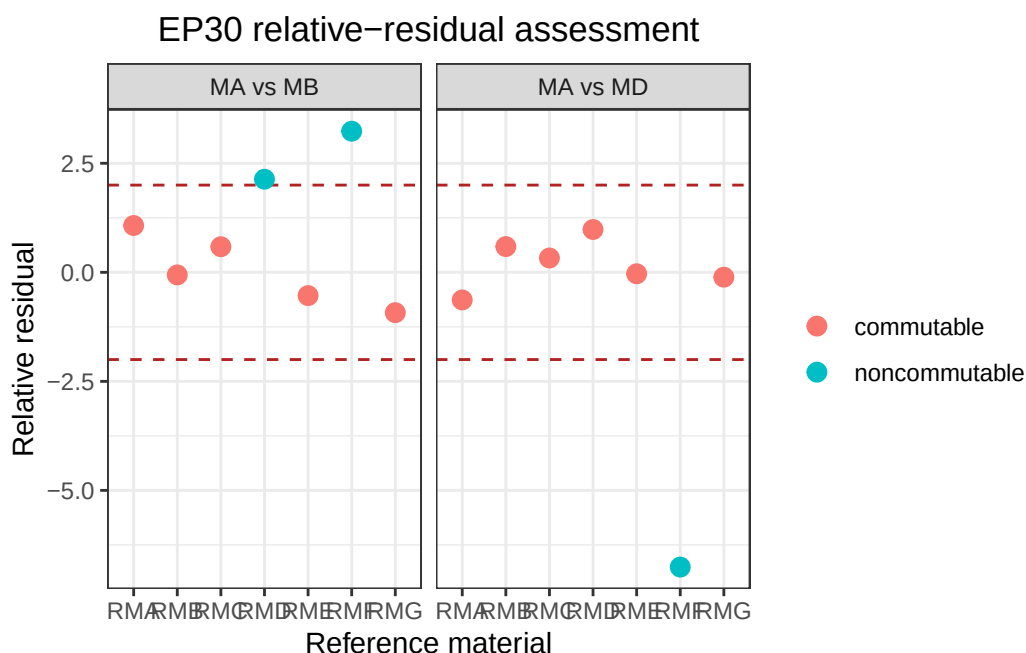
程序对	参考物质	预测值	下限	上限	区间判定	相对残差	残差判定
MA vs MB	RMA	15.576	11.944	19.208	Commutable	1.073	Commutable
MA vs MB	RMB	26.829	23.212	30.446	Commutable	-0.059	Commutable

程序对	参考物质	预测值	下限	上限	区间判定	相对残差	残差判定
MA vs MB	RMC	37.698	34.035	41.361	Commutable	0.587	Commutable
MA vs MB	RMD	44.238	40.520	47.956	Commutable	2.135	Non-commutable
MA vs MB	RME	18.365	14.743	21.988	Commutable	-0.534	Commutable
MA vs MB	RMF	23.848	20.233	27.463	Non-commutable	3.236	Non-commutable
MA vs MB	RMG	20.000	16.382	23.619	Commutable	-0.925	Commutable
MA vs MD	RMA	10.746	7.045	14.447	Commutable	-0.633	Commutable
MA vs MD	RMB	20.510	16.820	24.199	Commutable	0.590	Commutable
MA vs MD	RMC	29.939	26.216	33.663	Commutable	0.329	Commutable
MA vs MD	RMD	35.614	31.849	39.378	Commutable	0.982	Commutable
MA vs MD	RME	13.166	9.472	16.860	Commutable	-0.033	Commutable
MA vs MD	RMF	17.923	14.235	21.611	Non-commutable	-6.758	Non-commutable
MA vs MD	RMG	14.585	10.894	18.276	Commutable	-0.112	Commutable

```
plot(ep30_result)
```



```
plot(ep30_result, method = "relative_residual")
```



6.5.4 IFCC Part 2 Official Supplement: Bias Difference and Uncertainty

The original IFCC Part 2 workbook contains 50 clinical samples and 5 reference materials. Following the supplement's exclusions, all results for CS21 and the third replicate of CS6 on MPx are removed, leaving 49 clinical samples. Each reference material has five positions and three replicates per procedure. The analysis uses the natural-log scale, criterion $C = 0.12$, and coverage factor $k = 1.9$.

```
ifcc_data <- read_csv(
  "./data/006-基质效应与互换性/std-ifcc-part2-example.csv",
  show_col_types = FALSE
)
table(ifcc_data$material_type, ifcc_data$procedure)
```

```
      MPx MPy
clinical 146 147
rm        75  75
```

```
ifcc_result <- commutability(
  ifcc_data,
  sample_id = "sample_id",
  material_type = "material_type",
  procedure = "procedure",
  result = "result",
  position = "position",
  approach = "ifcc",
  scale = "ln",
  procedure_pairs = list(c("MPx", "MPy")),
  criterion = 0.12,
  bias_model = "constant_global",
  coverage_factor = 1.9,
  rm_position = "pooled"
)
ifcc_result
```

Commutability assessment

Study-design checks: 4 met, 0 unmet. Unmet checks do not invalidate computations; inspect '\$design_checks'.

```

Approach      : IFCC
Scale         : ln
Procedure pairs : 1
Reference materials : 5
Procedure pair: MPx vs MPy | Reference material: RM1
  RM coordinate      : 2.7755556
  RM bias            : 0.33925926
  Clinical bias      : 0.18095238
  Difference in bias  : 0.15830688
  RM standard uncertainty : 0.037626363
  Clinical standard uncertainty : 0.015516499
  Standard uncertainty : 0.040700184
  Expanded uncertainty : 0.077330349
  Lower              : 0.080976529
  Upper              : 0.23563723
  Criterion           : 0.12
  Clinical samples    : 49
  Classification      : inconclusive
  procedure_x         : MPx
  procedure_y         : MPy
  position_mean_sd_x  : 0.067355047
  position_mean_sd_y  : 0.050418387
  positions_x         : 5
  positions_y         : 5
  position_pooling    : pooled

```

```

Procedure pair: MPx vs MPy | Reference material: RM2
  RM coordinate      : 3.3097037
  RM bias            : 0.12903704
  Clinical bias      : 0.18095238
  Difference in bias  : -0.051915344
  RM standard uncertainty : 0.037626363
  Clinical standard uncertainty : 0.015516499
  Standard uncertainty : 0.040700184
  Expanded uncertainty : 0.077330349
  Lower              : -0.12924569
  Upper              : 0.025415005
  Criterion           : 0.12
  Clinical samples    : 49
  Classification      : inconclusive
  procedure_x         : MPx
  procedure_y         : MPy
  position_mean_sd_x  : 0.067355047
  position_mean_sd_y  : 0.050418387
  positions_x         : 5
  positions_y         : 5
  position_pooling    : pooled

```

```

Procedure pair: MPx vs MPy | Reference material: RM3
  RM coordinate      : 5.5979259
  RM bias            : 0.16
  Clinical bias      : 0.18095238
  Difference in bias  : -0.020952381
  RM standard uncertainty : 0.037626363
  Clinical standard uncertainty : 0.015516499

```


Standard uncertainty	: 0.040700184
Expanded uncertainty	: 0.077330349
Lower	: -0.09828273
Upper	: 0.056377968
Criterion	: 0.12
Clinical samples	: 49
Classification	: commutable
procedure_x	: MPx
procedure_y	: MPy
position_mean_sd_x	: 0.067355047
position_mean_sd_y	: 0.050418387
positions_x	: 5
positions_y	: 5
position_pooling	: pooled

Procedure pair: MPx vs MPy | Reference material: RM4

RM coordinate	: 1.607037
RM bias	: 0.073333333
Clinical bias	: 0.18095238
Difference in bias	: -0.10761905
RM standard uncertainty	: 0.037626363
Clinical standard uncertainty	: 0.015516499
Standard uncertainty	: 0.040700184
Expanded uncertainty	: 0.077330349
Lower	: -0.1849494
Upper	: -0.030288699
Criterion	: 0.12
Clinical samples	: 49
Classification	: inconclusive
procedure_x	: MPx
procedure_y	: MPy
position_mean_sd_x	: 0.067355047
position_mean_sd_y	: 0.050418387
positions_x	: 5
positions_y	: 5
position_pooling	: pooled

Procedure pair: MPx vs MPy | Reference material: RM5

RM coordinate	: 4.4167407
RM bias	: 0.85866667
Clinical bias	: 0.18095238
Difference in bias	: 0.67771429
RM standard uncertainty	: 0.037626363
Clinical standard uncertainty	: 0.015516499
Standard uncertainty	: 0.040700184
Expanded uncertainty	: 0.077330349
Lower	: 0.60038394
Upper	: 0.75504463
Criterion	: 0.12
Clinical samples	: 49
Classification	: noncommutable
procedure_x	: MPx
procedure_y	: MPy
position_mean_sd_x	: 0.067355047
position_mean_sd_y	: 0.050418387
positions_x	: 5
positions_y	: 5
position_pooling	: pooled

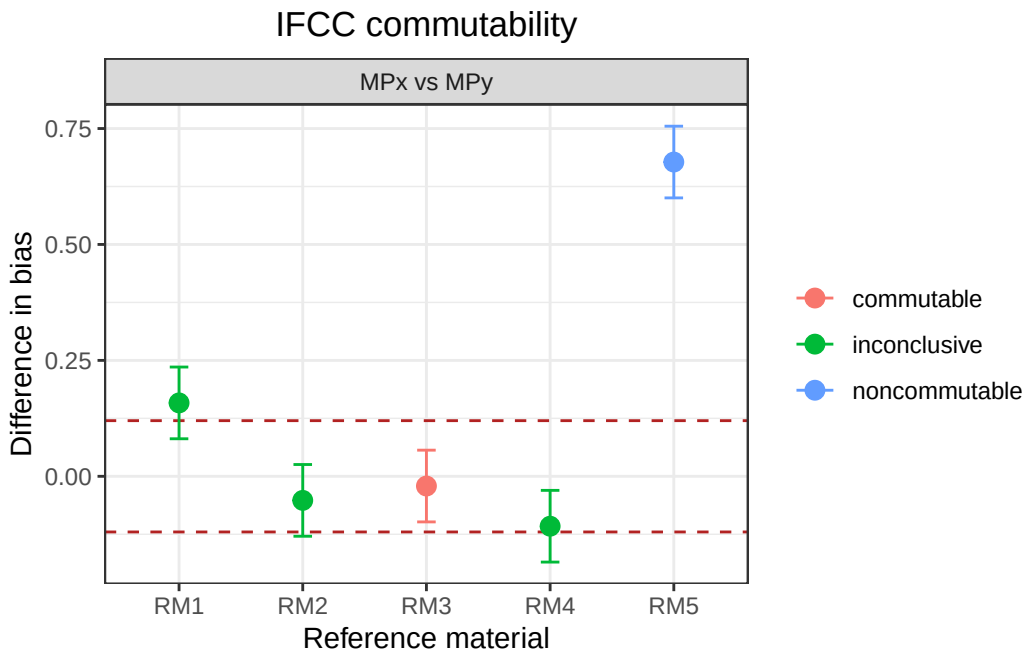
Table 6.7: IFCC Part 2: material-level comparison of ivdtools and the official workbook

标准位置	统计量	标准结果	ivdtools结果
IFCC Part 2 / RM1	偏倚差 D_RM	0.1583069	0.1583069
IFCC Part 2 / RM1	扩展不确定度 U(D_RM)	0.0773303	0.0773303
IFCC Part 2 / RM1	区间下限	0.0809765	0.0809765
IFCC Part 2 / RM1	区间上限	0.2356372	0.2356372
IFCC Part 2 / RM1	分类	Inconclusive	Inconclusive
IFCC Part 2 / RM2	偏倚差 D_RM	-0.0519153	-0.0519153
IFCC Part 2 / RM2	扩展不确定度 U(D_RM)	0.0773303	0.0773303
IFCC Part 2 / RM2	区间下限	-0.1292457	-0.1292457
IFCC Part 2 / RM2	区间上限	0.0254150	0.0254150
IFCC Part 2 / RM2	分类	Inconclusive	Inconclusive
IFCC Part 2 / RM3	偏倚差 D_RM	-0.0209524	-0.0209524
IFCC Part 2 / RM3	扩展不确定度 U(D_RM)	0.0773303	0.0773303
IFCC Part 2 / RM3	区间下限	-0.0982827	-0.0982827
IFCC Part 2 / RM3	区间上限	0.0563780	0.0563780
IFCC Part 2 / RM3	分类	Commutable	Commutable
IFCC Part 2 / RM4	偏倚差 D_RM	-0.1076190	-0.1076190
IFCC Part 2 / RM4	扩展不确定度 U(D_RM)	0.0773303	0.0773303
IFCC Part 2 / RM4	区间下限	-0.1849494	-0.1849494
IFCC Part 2 / RM4	区间上限	-0.0302887	-0.0302887
IFCC Part 2 / RM4	分类	Inconclusive	Inconclusive
IFCC Part 2 / RM5	偏倚差 D_RM	0.6777143	0.6777143
IFCC Part 2 / RM5	扩展不确定度 U(D_RM)	0.0773303	0.0773303
IFCC Part 2 / RM5	区间下限	0.6003839	0.6003839
IFCC Part 2 / RM5	区间上限	0.7550446	0.7550446
IFCC Part 2 / RM5	分类	Non-commutable	Non-commutable

Table 6.8: IFCC Part 2: uncertainty components compared with the official workbook

标准位置	统计量	标准结果	ivdtools结果
IFCC Part 2 / uncertainty components	临床样品平均偏倚	0.18095238	0.18095238
IFCC Part 2 / uncertainty components	MPx 合并位置均值 SD	0.06735505	0.06735505
IFCC Part 2 / uncertainty components	MPy 合并位置均值 SD	0.05041839	0.05041839
IFCC Part 2 / uncertainty components	参考物质标准不确定度	0.03762636	0.03762636
IFCC Part 2 / uncertainty components	临床样品标准不确定度	0.01551650	0.01551650
IFCC Part 2 / uncertainty components	偏倚差标准不确定度	0.04070018	0.04070018
IFCC Part 2 / uncertainty components	偏倚差扩展不确定度	0.07733035	0.07733035

```
plot(ifcc_result)
```



The classification of RM3 depends on pooling the position-mean variance across reference materials as specified by the official method. If position variance is estimated separately for each material, both the algorithm and classification definition change; formal reports must record the `rm_position` setting.

6.5.5 IFCC Part 3 Official Supplement: Calibration-Validity Assessment

The official IFCC Part 3 case contains 40 clinical samples and 7 measurement procedures. Table S1 contains results from the manufacturer's original calibration scheme. For each clinical sample, the initial target is the mean of the middle five values after removing the largest and smallest of the seven procedure results. Table S2 contains results after recalibration with two candidate reference-material levels but still uses the initial targets from Table S1; targets must not be redefined from the post-calibration results.

```
ifcc_part3_data <- read_csv(
  "./data/006-基质效应与互换性/std-ifcc-part3-example.csv",
  show_col_types = FALSE
)
table(
  ifcc_part3_data$calibration_stage,
  ifcc_part3_data$procedure
)
```

	MP1	MP2	MP3	MP4	MP5	MP6	MP7
after	40	40	40	40	40	40	40
before	40	40	40	40	40	40	40

First evaluate all seven procedures. `criterion = 6` is the prespecified IMPBR percentage criterion in the paper example; `w3_reference_procedure = "MP1"` applies the MP1 SD/W(3) ratio to adjust each procedure's W(3), as in the original analysis.

```
ifcc_part3_result <- commutability(
  ifcc_part3_data,
  sample_id = "sample_id",
  procedure = "procedure",
  result = "result",
```

```

    approach = "calibration",
    calibration_stage = "calibration_stage",
    criterion = 6,
    w3_reference_procedure = "MP1"
  )

ifcc_part3_excluding_mp6 <- commutability(
  ifcc_part3_data,
  sample_id = "sample_id",
  procedure = "procedure",
  result = "result",
  approach = "calibration",
  calibration_stage = "calibration_stage",
  criterion = 6,
  exclude_procedures = "MP6",
  w3_reference_procedure = "MP1"
)
ifcc_part3_excluding_mp6

```

Commutability assessment

Study-design checks: 4 met, 0 unmet. Unmet checks do not invalidate computations; inspect '\$design_checks'.

```

Approach      : CALIBRATION (IFCC PART 3)
Scale         : raw
Procedures    : 7
Clinical samples : 40
IMPBR criterion (%) : 6
Stage: before | Procedure: MP1
Clinical samples      : 40
Median bias (%)      : -0.38910506
Bias SD (%)          : 4.0149558
Adjusted W(3) (%)    : 4.0149558
Included after investigation : TRUE
w3_percent           : 11.18415
trend_intercept      : 1.288401
trend_slope          : -0.019331579
trend_p_value        : 0.54012483

```

```

Stage: before | Procedure: MP2
Clinical samples      : 40
Median bias (%)      : 9.5282484
Bias SD (%)          : 5.7222879
Adjusted W(3) (%)    : 5.5726509
Included after investigation : TRUE
w3_percent           : 15.5233
trend_intercept      : 5.3027017
trend_slope          : 0.060976356
trend_p_value        : 0.17087519

```

```

Stage: before | Procedure: MP3
Clinical samples      : 40
Median bias (%)      : 20.680902
Bias SD (%)          : 9.5434313
Adjusted W(3) (%)    : 5.9037259
Included after investigation : TRUE
w3_percent           : 16.44555
trend_intercept      : 24.218286
trend_slope          : -0.029099244

```

```

trend_p_value          : 0.69847835

Stage: before | Procedure: MP4
Clinical samples       : 40
Median bias (%)        : 46.552662
Bias SD (%)            : 6.7290862
Adjusted W(3) (%)      : 5.8855004
Included after investigation : TRUE
w3_percent             : 16.394781
trend_intercept        : 61.15902
trend_slope            : -0.20881729
trend_p_value          : 7.623953e-06

```

```

Stage: before | Procedure: MP5
Clinical samples       : 40
Median bias (%)        : -30.046827
Bias SD (%)            : 4.5852825
Adjusted W(3) (%)      : 3.4941653
Included after investigation : TRUE
w3_percent             : 9.7334245
trend_intercept        : -30.931755
trend_slope            : 0.021966809
trend_p_value          : 0.54217093

```

```

Stage: before | Procedure: MP6
Clinical samples       : 40
Median bias (%)        : -19.447183
Bias SD (%)            : 5.2950133
Adjusted W(3) (%)      : 3.9604755
Included after investigation : FALSE
w3_percent             : 11.032389
trend_intercept        : -22.796747
trend_slope            : 0.052621097
trend_p_value          : 0.20230744

```

```

Stage: before | Procedure: MP7
Clinical samples       : 40
Median bias (%)        : -11.427474
Bias SD (%)            : 5.6481377
Adjusted W(3) (%)      : 4.6290488
Included after investigation : TRUE
w3_percent             : 12.894781
trend_intercept        : -6.5515631
trend_slope            : -0.085122896
trend_p_value          : 0.049743566

```

```

Stage: after | Procedure: MP1
Clinical samples       : 40
Median bias (%)        : -0.73741264
Bias SD (%)            : 4.0413178
Adjusted W(3) (%)      : 4.0413178
Included after investigation : TRUE
w3_percent             : 11.18415
trend_intercept        : -0.026587184
trend_slope            : 0.00031214807
trend_p_value          : 0.99217849

```

```

Stage: after | Procedure: MP2

```

Clinical samples : 40
 Median bias (%) : -0.2908762
 Bias SD (%) : 4.9916593
 Adjusted W(3) (%) : 4.9666072
 Included after investigation : TRUE
 w3_percent : 13.744843
 trend_intercept : -1.641973
 trend_slope : 0.015950627
 trend_p_value : 0.68473953

Stage: after | Procedure: MP3

Clinical samples : 40
 Median bias (%) : -0.17136128
 Bias SD (%) : 7.8422761
 Adjusted W(3) (%) : 4.8633169
 Included after investigation : TRUE
 w3_percent : 13.458993
 trend_intercept : 4.3152226
 trend_slope : -0.048689788
 trend_p_value : 0.42878141

Stage: after | Procedure: MP4

Clinical samples : 40
 Median bias (%) : -0.28195489
 Bias SD (%) : 3.7935062
 Adjusted W(3) (%) : 3.3947339
 Included after investigation : TRUE
 w3_percent : 9.3947606
 trend_intercept : 0.83067063
 trend_slope : -0.0060932695
 trend_p_value : 0.8384197

Stage: after | Procedure: MP5

Clinical samples : 40
 Median bias (%) : 0.3245325
 Bias SD (%) : 6.3376024
 Adjusted W(3) (%) : 4.3522181
 Included after investigation : TRUE
 w3_percent : 12.044551
 trend_intercept : 4.4394388
 trend_slope : -0.051202332
 trend_p_value : 0.30174764

Stage: after | Procedure: MP6

Clinical samples : 40
 Median bias (%) : -21.313277
 Bias SD (%) : 4.8895959
 Adjusted W(3) (%) : 3.5271045
 Included after investigation : FALSE
 w3_percent : 9.7610897
 trend_intercept : -21.87256
 trend_slope : 0.0074119906
 trend_p_value : 0.8473998

Stage: after | Procedure: MP7

Clinical samples : 40
 Median bias (%) : -1.0730622
 Bias SD (%) : 6.4531842

```

Adjusted W(3) (%)      : 4.9939824
Included after investigation : TRUE
w3_percent            : 13.820603
trend_intercept       : -1.6436293
trend_slope           : 0.013117511
trend_p_value         : 0.79632873
Stage: before | Assessment set: all
Procedures            : 7
IMPBR (%)             : 76.599489
Minimum procedure     : MP5
Maximum procedure     : MP4
Criterion (%)         : 6
Excluded procedures   :
Classification        : screening
min_bias_percent      : -30.046827
max_bias_percent      : 46.552662

Stage: before | Assessment set: specified_exclusions
Procedures            : 6
IMPBR (%)             : 76.599489
Minimum procedure     : MP5
Maximum procedure     : MP4
Criterion (%)         : 6
Excluded procedures   : MP6
Classification        : screening
min_bias_percent      : -30.046827
max_bias_percent      : 46.552662

Stage: after | Assessment set: all
Procedures            : 7
IMPBR (%)             : 21.63781
Minimum procedure     : MP6
Maximum procedure     : MP5
Criterion (%)         : 6
Excluded procedures   :
Classification        : requires_investigation
min_bias_percent      : -21.313277
max_bias_percent      : 0.3245325

Stage: after | Assessment set: specified_exclusions
Procedures            : 6
IMPBR (%)             : 1.3975947
Minimum procedure     : MP7
Maximum procedure     : MP5
Criterion (%)         : 6
Excluded procedures   : MP6
Classification        : commutable
min_bias_percent      : -1.0730622
max_bias_percent      : 0.3245325

```

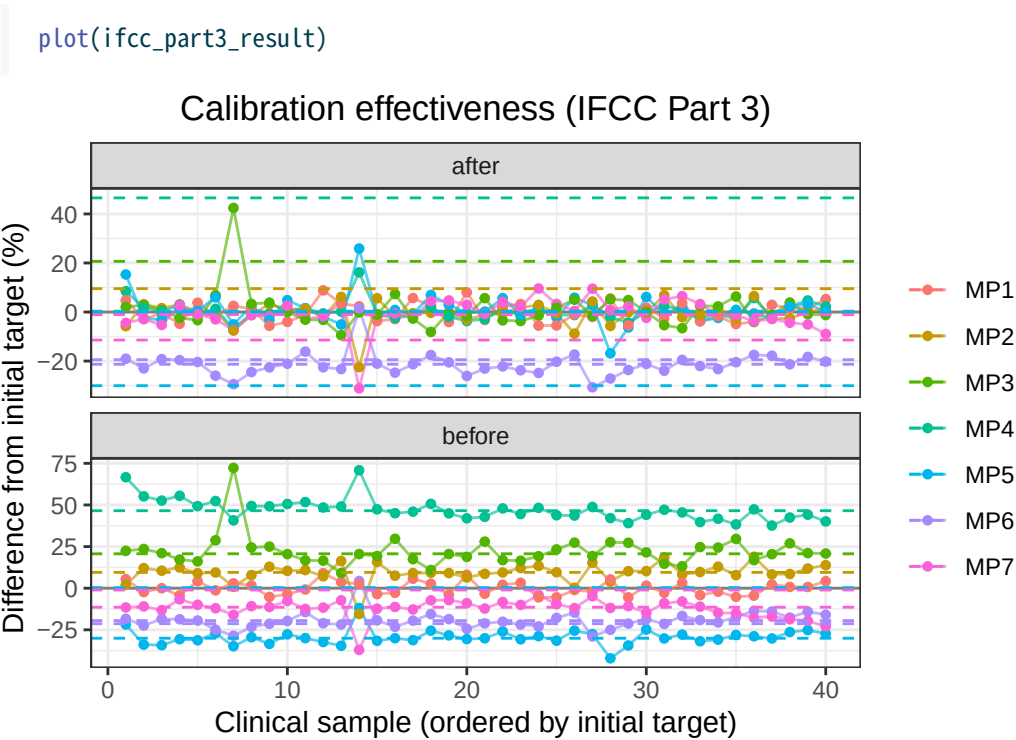
Table 6.9: IFCC Part 3: comparison with official Tables 1 and 2

标准位置	统计量	标准结果	ivdtools结果
IFCC Part 3 / before / MP1	median_bias_percent	-0.40	-0.3891
IFCC Part 3 / before / MP1	sd_bias_percent	4.01	4.0150
IFCC Part 3 / before / MP1	adjusted_w3_percent	4.01	4.0150
IFCC Part 3 / before / MP2	median_bias_percent	9.50	9.5282
IFCC Part 3 / before / MP2	sd_bias_percent	5.72	5.7223

标准位置	统计量	标准结果	ivdtools结果
IFCC Part 3 / before / MP2	adjusted_w3_percent	5.57	5.5727
IFCC Part 3 / before / MP3	median_bias_percent	20.70	20.6809
IFCC Part 3 / before / MP3	sd_bias_percent	9.54	9.5434
IFCC Part 3 / before / MP3	adjusted_w3_percent	5.90	5.9037
IFCC Part 3 / before / MP4	median_bias_percent	46.60	46.5527
IFCC Part 3 / before / MP4	sd_bias_percent	6.73	6.7291
IFCC Part 3 / before / MP4	adjusted_w3_percent	5.89	5.8855
IFCC Part 3 / before / MP5	median_bias_percent	-30.00	-30.0468
IFCC Part 3 / before / MP5	sd_bias_percent	4.59	4.5853
IFCC Part 3 / before / MP5	adjusted_w3_percent	3.49	3.4942
IFCC Part 3 / before / MP6	median_bias_percent	-19.40	-19.4472
IFCC Part 3 / before / MP6	sd_bias_percent	5.30	5.2950
IFCC Part 3 / before / MP6	adjusted_w3_percent	3.96	3.9605
IFCC Part 3 / before / MP7	median_bias_percent	-11.40	-11.4275
IFCC Part 3 / before / MP7	sd_bias_percent	5.65	5.6481
IFCC Part 3 / before / MP7	adjusted_w3_percent	4.63	4.6290
IFCC Part 3 / after / MP1	median_bias_percent	-0.80	-0.7374
IFCC Part 3 / after / MP1	sd_bias_percent	4.05	4.0413
IFCC Part 3 / after / MP1	adjusted_w3_percent	4.05	4.0413
IFCC Part 3 / after / MP2	median_bias_percent	-0.30	-0.2909
IFCC Part 3 / after / MP2	sd_bias_percent	4.99	4.9917
IFCC Part 3 / after / MP2	adjusted_w3_percent	4.98	4.9666
IFCC Part 3 / after / MP3	median_bias_percent	-0.20	-0.1714
IFCC Part 3 / after / MP3	sd_bias_percent	7.83	7.8423
IFCC Part 3 / after / MP3	adjusted_w3_percent	4.95	4.8633
IFCC Part 3 / after / MP4	median_bias_percent	-0.30	-0.2820
IFCC Part 3 / after / MP4	sd_bias_percent	3.80	3.7935
IFCC Part 3 / after / MP4	adjusted_w3_percent	3.45	3.3947
IFCC Part 3 / after / MP5	median_bias_percent	0.40	0.3245
IFCC Part 3 / after / MP5	sd_bias_percent	6.34	6.3376
IFCC Part 3 / after / MP5	adjusted_w3_percent	4.39	4.3522
IFCC Part 3 / after / MP6	median_bias_percent	-21.30	-21.3133
IFCC Part 3 / after / MP6	sd_bias_percent	4.89	4.8896
IFCC Part 3 / after / MP6	adjusted_w3_percent	3.57	3.5271
IFCC Part 3 / after / MP7	median_bias_percent	-1.10	-1.0731
IFCC Part 3 / after / MP7	sd_bias_percent	6.45	6.4532
IFCC Part 3 / after / MP7	adjusted_w3_percent	5.06	4.9940
IFCC Part 3 / before / ALL	impbr_percent	76.60	76.5995
IFCC Part 3 / after / ALL	impbr_percent	21.60	21.6378
IFCC Part 3 / after / EXCLUDE_MP6	impbr_percent	1.50	1.3976

For the complete procedure set, IMPBR decreases from approximately 76.6% before calibration to approximately 21.6% after calibration, but remains above 6%, so the function returns “requires investigation.” It does not automatically classify the candidate reference material as noncommutable for all procedures or remove a procedure. The original investigation identified the candidate reference material as unsuitable for MP6; after explicitly setting `exclude_procedures = "MP6"`, the IMPBR for the remaining six procedures is approximately 1.4% based on the values listed in the supplement, meeting the 6% criterion.

The paper reports an IMPBR of 1.5% after excluding MP6, whereas recalculation from the published Table S1 and S2 values gives approximately 1.4%. The unrounded internal values used in the paper were not published with the supplement, so it is not possible to determine whether the approximately 0.1 percentage-point difference is caused only by rounding; both values are retained as reported.



leave_one_out shows sensitivity results after omitting each procedure and is intended only to identify an investigation direction. Formal conclusions require support from the calibration model, precision, and interference investigations; the MP6 exclusion in this case is the conclusion of the original investigation, not a procedure selected automatically by the software to minimize IMPBR.

6.5.6 Real-Application Comparison: POCT Glucose EQA Materials

Wang et al. reported a real-application study in 2020: three EQA materials at three concentration levels were measured together with 33 individual clinical samples on five POCT glucose meters and one Hitachi 7600 analyzer. The study used both the CLSI EP30-A Deming-regression 95% prediction interval and the IFCC bias-difference method. After exclusions, the IFCC analysis used 30 clinical samples, a 10% criterion, and a coverage factor of 1.9.

EQA material	CLSI prediction-interval result	IFCC bias-difference result
In-house human whole blood (HWB)	All three levels were commutable on all 5/5 POCT meters	The three levels were commutable on 3/5, 4/5, and 3/5 meters; two meters were commutable at all three levels
Commercial control material (CCM)	The three levels were commutable on 3/5, 2/5, and 4/5 meters	All 5 meters were inconclusive or noncommutable
Pooled human serum (PHS)	The three levels were commutable on 4/5, 3/5, and 5/5 meters	All 5 meters were inconclusive or noncommutable

Both paths showed that in-house human whole blood performed better than commercial control material and pooled human serum, while the IFCC conclusion was more conservative. The investigators suggested that inconclusive results could be related to insufficient replicate or clinical-sample counts, poorer measurement precision, and larger sample-specific differences. This case illustrates that prediction intervals and bias differences ad-

dress different questions and handle uncertainty differently; the same material can receive different classifications for different procedure pairs, concentrations, and decision paths.

The paper does not provide enough individual measurements to reconstruct the complete numerical analysis here. This is therefore a real-application comparison of study design and published results, not numerical validation evidence for `ivdttools`. See Wang et al., 2020, Journal of Clinical Laboratory Analysis.

Chapter 7

Interference Studies

Interference studies ask whether a potential interferent changes a measurement, how the effect changes with interferent concentration, and whether patient samples containing the factor show a larger method difference or dispersion. These are different analysis paths and should not be collapsed into one summary number.

Path	Main question	Main result
Paired preparation screen	Is the test sample sufficiently different from the solvent control at specified analyte/interferent levels?	Absolute/relative effect and confidence interval
Dose response	How does the effect change with interferent concentration?	Baseline-relative effect, monotonicity, interpolation
Patient-sample comparison	Do patient samples with the factor show method differences or increased dispersion?	Per-sample difference, group summary, difference–interferent regression

! Scope of Use

Interference conclusions require prespecified interferents, concentrations, replicates, direction, and allowable effect. A statistical difference is not automatically clinically or analytically important. Keep the preparation design, solvent control, patient grouping, and method-difference direction explicit.

```
library(ivdtools)
library(readr)
```

7.1 Main Functions

Function	Purpose	Return value
<code>interference_replicates()</code>	Calculate required test and control replicates	<code>interference_replicates</code> data frame
<code>interference_paired()</code>	Analyze paired-preparation interference	<code>interference_paired</code> object

Function	Purpose	Return value
<code>interference_dose_response()</code>	Analyze interferent dose response	<code>interference_dose_response</code> object
<code>predict()</code>	Forward or inverse prediction from a dose-response result	Prediction data frame
<code>interference_patient()</code>	Analyze method differences in patient samples	<code>interference_patient</code> object

7.1.1 Replicate Planning

```
interference_replicates(
  sd, detectable_difference, alpha = 0.05, power = 0.90,
  alternative = c("two.sided", "one.sided"), min_replicates = 5,
  label = NULL
)
```

`sd` is expected repeatability SD, or repeatability CV on a relative scale. `detectable_difference` uses the same scale and compatible length. `alpha` and `power` must be finite values between 0 and 1; `alternative` determines one- or two-sided planning. The result is a design quantity, not an acceptance decision.

7.1.2 Paired-Preparation Studies

```
interference_paired(
  data, result, design = c("ep07", "analyte_spike", "blank"),
  by = NULL, condition = NULL, test_level = NULL, control_level = NULL,
  matrix = NULL, normal_level = NULL, interferent_level = NULL,
  spike = NULL, unspiked_level = NULL, spiked_level = NULL,
  conf.level = 0.95
)
```

The data are long format, one replicate per row. For `ep07`, identify test and solvent-control conditions; for `analyte_spike`, identify normal/interferent matrix and unspiked/spiked analyte; for `blank`, identify the relevant condition. The returned object contains paired effects, relative effects, standard errors, intervals, and group summaries.

7.1.3 Dose–Response Studies

```
interference_dose_response(
  data, result, concentration, by = NULL, baseline,
  method = c("point_to_point", "linear"), conf.level = 0.95
)
```

`baseline` must be an observed interferent concentration in every stratum. `point_to_point` interpolates adjacent concentration means; `linear` fits a straight line to individual results. The object preserves the baseline-relative effect, model coefficients, and predictions.

7.1.4 Patient-Sample Comparison

```
interference_patient(
  data, id, group, method, result,
  test_group, control_group, evaluated_method, comparative_method,
  interferent_cols = NULL, conf.level = 0.95
)
```

The difference is defined as evaluated method minus comparative method. Optional interferent columns are fitted one at a time; group summaries and regression intervals use the specified confidence level.

7.2 Analysis Examples

7.2.1 EP07 Replicate Planning and Paired Screening

```
ep07_paired <- as.data.frame(read_csv(
  "./data/007-干扰研究/std-ep07-table3-paired.csv",
  show_col_types = FALSE
))
ep07_fit <- interference_paired(
  ep07_paired, result = "result", design = "ep07",
  by = "level", condition = "condition",
  test_level = "test", control_level = "control", conf.level = 0.95
)
print(ep07_fit)
```

Interference paired-difference analysis

Design: ep07

Confidence level: 95%

level: low

stratum	: level=low
design	: ep07
n_test	: 5
n_control	: 5
test_mean	: 0.3
control_mean	: 0.5
absolute_difference	: -0.2
absolute_se	: 0.033166248
df	: 8
absolute_ci_lower	: -0.2764815
absolute_ci_upper	: -0.1235185
percent_difference	: -40
percent_se	: 6.6332496
percent_ci_lower	: -55.296301
percent_ci_upper	: -24.703699

level: high

stratum	: level=high
design	: ep07
n_test	: 5
n_control	: 5
test_mean	: 4.55
control_mean	: 5
absolute_difference	: -0.45
absolute_se	: 0.096436508

```

df                : 8
absolute_ci_lower : -0.67238299
absolute_ci_upper : -0.22761701
percent_difference : -9
percent_se        : 1.9287302
percent_ci_lower  : -13.44766
percent_ci_upper  : -4.5523403

```

7.2.2 EP07 Dose Response

```

ep07_dose <- as.data.frame(read_csv(
  "/data/007-干扰研究/std-ep07-table4-dose-response.csv",
  show_col_types = FALSE
))
ep07_dose_fit <- interference_dose_response(
  ep07_dose, result = "result", concentration = "concentration",
  baseline = 0, method = "point_to_point"
)
print(ep07_dose_fit)

```

Interference dose-response analysis

Method: point_to_point

Baseline concentration: 0

Stratum: all | Concentration: 0

```

Observations      : 5
Mean              : 200.02
SD                : 1.7753873
SE                : 0.79397733
Absolute difference : 0
Percent difference : 0
CV percent        : 0.88760488

```

Stratum: all | Concentration: 10

```

Observations      : 5
Mean              : 223.9
SD                : 4.645966
SE                : 2.0777392
Absolute difference : 23.88
Percent difference : 11.938806
CV percent        : 2.0750183

```

Stratum: all | Concentration: 20

```

Observations      : 5
Mean              : 240.08
SD                : 3.3536547
SE                : 1.4998
Absolute difference : 40.06
Percent difference : 20.027997
CV percent        : 1.3968905

```

Stratum: all | Concentration: 30

```

Observations      : 5
Mean              : 245.72
SD                : 4.7892588
SE                : 2.1418217
Absolute difference : 45.7
Percent difference : 22.847715

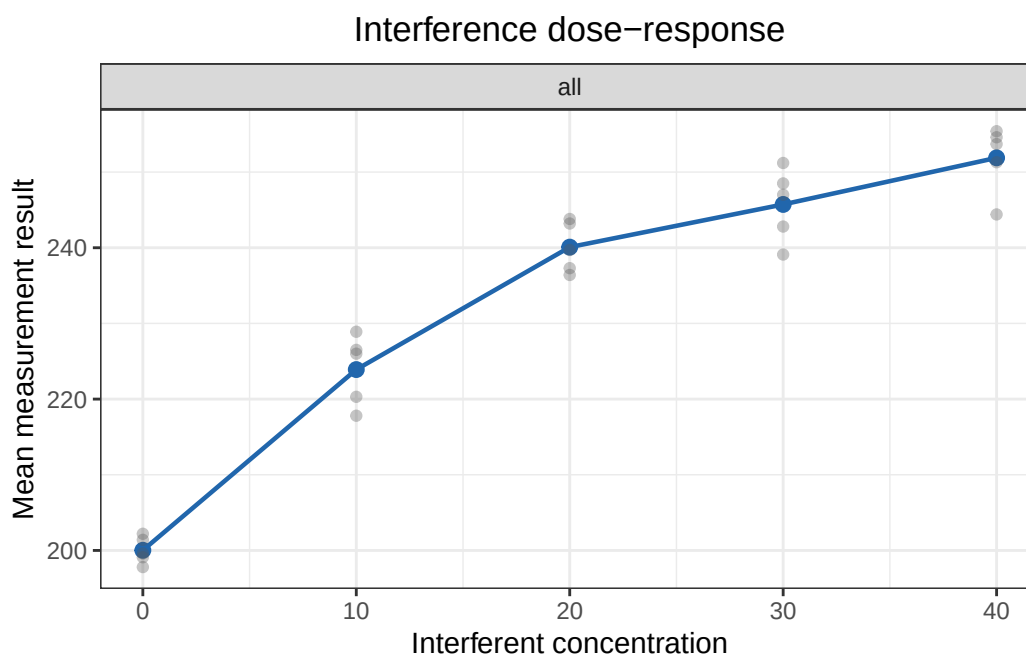
```

CV percent : 1.9490716

Stratum: all | Concentration: 40

Observations : 5
 Mean : 251.88
 SD : 4.4549972
 SE : 1.9923353
 Absolute difference : 51.86
 Percent difference : 25.927407
 CV percent : 1.7686983

```
plot(ep07_dose_fit)
```



7.2.3 Patient-Sample Comparison

```
patient_data <- as.data.frame(read_csv(
  "./data/007-干扰研究/std-yyt1789-5-appendix-c-patient.csv",
  show_col_types = FALSE
))
patient_fit <- interference_patient(
  patient_data, id = "id", group = "group", method = "method",
  result = "result", test_group = "test", control_group = "control",
  evaluated_method = "evaluated", comparative_method = "comparative"
)
print(patient_fit)
```

Patient-specimen interference evaluation

Evaluated - comparative: evaluated - comparative

Group: test

Observations : 20
 Mean difference : -1.1835
 SD difference : 2.3930541
 SE difference : 0.53510317
 Median difference : -0.49
 25th percentile : -1.3425
 75th percentile : -0.32

```

CI lower          : -2.3034838
CI upper          : -0.063516183
Mean percent difference : -17.916419
SD percent difference  : 18.774944
Percent CI lower     : -26.703363
Percent CI upper     : -9.1294749

```

```

Group: control
Observations      : 20
Mean difference    : -1.039
SD difference      : 2.1614978
SE difference      : 0.4833256
Median difference  : -0.415
25th percentile   : -0.835
75th percentile   : -0.075
CI lower          : -2.0506121
CI upper          : -0.027387902
Mean percent difference : -11.712959
SD percent difference  : 24.21947
Percent CI lower     : -23.04802
Percent CI upper     : -0.37789805

```

7.3 Interpretation and Reporting

Report the design, interferent and concentration, control preparation, replicate count, effect definition, confidence method, and allowable effect. A dose-response line can describe an association without proving mechanism. A patient-sample regression can be confounded by disease severity, sample matrix, or method range. EP07 and YY/T 1789.5 standard examples in the supplied data directory can be cross-checked with the same functions, but standard-specific acceptance limits and study conclusions remain external to the package.

7.4 Full Standard Cases

7.4.1 CLSI EP07 Third Edition Section 5.1: Replicate-Count Design

Section 5.1.1 of EP07 uses a two-sided example with repeatability SD 1.0 mg/dL, 95% confidence, 95% power, and a detectable effect of 1.5 mg/dL. Section 5.1.4 assumes a repeatability CV of 5% and detects effects of 7% and 10% with 95% confidence and 90% power. SD and difference, or CV and percentage difference, are on matching scales, so the same formula can be used.

```

ep07_replicates_511 <- interference_replicates(
  sd = 1,
  detectable_difference = 1.5,
  alpha = 0.05,
  power = 0.95,
  label = "5.1.1 two-sided"
)
ep07_replicates_514 <- interference_replicates(
  sd = c(5, 5),
  detectable_difference = c(7, 10),
  alpha = 0.05,
  power = 0.90,
  label = c("5.1.4 7%", "5.1.4 10%")
)
rbind(ep07_replicates_511, ep07_replicates_514)

```


EP07 paired-difference replicate design

Label: 5.1.1 two-sided

SD : 1
 Detectable difference : 1.5
 Difference / SD : 1.5
 Alpha : 0.05
 Power : 0.95
 Alternative : two.sided
 Z alpha : 1.959964
 Z power : 1.6448536
 Calculated replicates : 11.550853
 Replicates per sample : 12
 Minimum replicates : 5

Label: 5.1.4 7%

SD : 5
 Detectable difference : 7
 Difference / SD : 1.4
 Alpha : 0.05
 Power : 0.9
 Alternative : two.sided
 Z alpha : 1.959964
 Z power : 1.2815516
 Calculated replicates : 10.72186
 Replicates per sample : 11
 Minimum replicates : 5

Label: 5.1.4 10%

SD : 5
 Detectable difference : 10
 Difference / SD : 2
 Alpha : 0.05
 Power : 0.9
 Alternative : two.sided
 Z alpha : 1.959964
 Z power : 1.2815516
 Calculated replicates : 5.2537115
 Replicates per sample : 6
 Minimum replicates : 5

Table 7.1: EP07 Section 5.1: replicate-count comparison

Standard position	Standard calculated	ivdtools calculated	Standard replicates	ivdtools replicates
5.1.1 two-sided	11.550853	11.550853	12	12
5.1.4 7%	10.721860	10.721860	11	11
5.1.4 10%	5.253712	5.253712	6	6

The replicate count must be satisfied separately for every test sample and control sample; the two groups must not be added together and interpreted as one total sample size.

7.4.2 CLSI EP07 Third Edition Table 3: Paired Screening

EP07 Table 3 compares five test replicates with five solvent-control replicates at analyte levels 0.50 and 5.00 mg/dL and an interferent concentration of 500 mg/dL. The effect direction

is consistently test – control; use the absolute difference at the low level and the percentage difference relative to the control mean at the high level.

```
ep07_table3 <- read_csv(
  "./data/007-干扰研究/std-ep07-table3-paired.csv",
  show_col_types = FALSE
)
head(ep07_table3, 6)
```

A tibble: 6 x 5

	level	condition	replicate	result	unit
	<chr>	<chr>	<dbl>	<dbl>	<chr>
1	low	control	1	0.45	mg/dL
2	low	control	2	0.56	mg/dL
3	low	control	3	0.48	mg/dL
4	low	control	4	0.54	mg/dL
5	low	control	5	0.47	mg/dL
6	low	test	1	0.24	mg/dL

```
ep07_table3_result <- interference_paired(
  as.data.frame(ep07_table3),
  result = "result",
  design = "ep07",
  condition = "condition",
  test_level = "test",
  control_level = "control",
  by = "level",
  conf.level = 0.95
)
ep07_table3_result
```

Interference paired-difference analysis

Design: ep07

Confidence level: 95%

Level: low

```
stratum      : level=low
design        : ep07
n_test       : 5
n_control    : 5
test_mean    : 0.3
control_mean : 0.5
absolute_difference : -0.2
absolute_se   : 0.033166248
df           : 8
absolute_ci_lower : -0.2764815
absolute_ci_upper : -0.1235185
percent_difference : -40
percent_se     : 6.6332496
percent_ci_lower : -55.296301
percent_ci_upper : -24.703699
```

Level: high

```
stratum      : level=high
design        : ep07
n_test       : 5
n_control    : 5
test_mean    : 4.55
control_mean : 5
```

```

absolute_difference : -0.45
absolute_se         : 0.096436508
df                  : 8
absolute_ci_lower   : -0.67238299
absolute_ci_upper   : -0.22761701
percent_difference   : -9
percent_se          : 1.9287302
percent_ci_lower    : -13.44766
percent_ci_upper    : -4.5523403

```

The recalculated low-level control/test means are 0.50/0.30 mg/dL, with an absolute effect of -0.20 mg/dL and a 95% CI of -0.28 to -0.12 mg/dL. The high-level means are 5.00/4.55 mg/dL, with a relative effect of -9.0% and a 95% CI of -13.4% to -4.6% in ascending order. The standard prints the high-level interval in reverse order; only the display order is changed here.

Table 7.2: EP07 Third Edition Table 3: comparison of ivdtools and standard results

level	statistic	standard	ivdtools
low	control_mean	0.500	0.5000
low	test_mean	0.300	0.3000
low	control_sd	0.047	0.0474
low	test_sd	0.057	0.0570
low	difference	-0.200	-0.2000
low	ci_lower	-0.280	-0.2765
low	ci_upper	-0.120	-0.1235
high	control_mean	5.000	5.0000
high	test_mean	4.550	4.5500
high	control_sd	0.141	0.1409
high	test_sd	0.163	0.1632
high	difference_percent	-9.000	-9.0000
high	ci_lower_percent	-13.400	-13.4477
high	ci_upper_percent	-4.600	-4.5523

Under the case' s prespecified limits, the low-level point estimate of -0.20 mg/dL exceeds ± 0.10 mg/dL, while the high-level point estimate of -9.0% is within $\pm 10\%$; this screening case therefore reports interference only at the low level. If equivalence is prespecified, the entire confidence interval should instead be required to lie within the allowable difference.

7.4.3 CLSI EP07 Third Edition Table 4: Curved Dose–Response and Point-to-Point Interpolation

EP07 Table 4 presents simulated dose–response data: interferent concentrations of 0, 10, 20, 30, and 40 mg/dL, with five measurements at each concentration. The analyte results are expressed in nmol/L. The means increase monotonically with interferent concentration but gradually flatten at high concentrations, so the main text uses a point-to-point method between adjacent means rather than summarizing the full range with one straight line.

```

ep07_table4 <- read_csv(
  "../data/007-干扰研究/std-ep07-table4-dose-response.csv",
  show_col_types = FALSE
)
head(ep07_table4, 6)

```

A tibble: 6 x 6

```

sample replicate concentration result concentration_unit result_unit
<dbl>    <dbl>          <dbl>  <dbl> <chr>              <chr>

```

1	1	1	0	201. mg/dL	nmol/L
2	1	2	0	199. mg/dL	nmol/L
3	1	3	0	198. mg/dL	nmol/L
4	1	4	0	200. mg/dL	nmol/L
5	1	5	0	202. mg/dL	nmol/L
6	2	1	10	218. mg/dL	nmol/L

```

ep07_table4_result <- interference_dose_response(
  ep07_table4,
  result = "result",
  concentration = "concentration",
  baseline = 0,
  method = "point_to_point"
)

ep07_table4_result

```

Interference dose-response analysis

Method: point_to_point

Baseline concentration: 0

Stratum: all | Concentration: 0

```

Observations      : 5
Mean              : 200.02
SD                : 1.7753873
SE                : 0.79397733
Absolute difference : 0
Percent difference : 0
CV percent        : 0.88760488

```

Stratum: all | Concentration: 10

```

Observations      : 5
Mean              : 223.9
SD                : 4.645966
SE                : 2.0777392
Absolute difference : 23.88
Percent difference : 11.938806
CV percent        : 2.0750183

```

Stratum: all | Concentration: 20

```

Observations      : 5
Mean              : 240.08
SD                : 3.3536547
SE                : 1.4998
Absolute difference : 40.06
Percent difference : 20.027997
CV percent        : 1.3968905

```

Stratum: all | Concentration: 30

```

Observations      : 5
Mean              : 245.72
SD                : 4.7892588
SE                : 2.1418217
Absolute difference : 45.7
Percent difference : 22.847715
CV percent        : 1.9490716

```

Stratum: all | Concentration: 40

```

Observations      : 5

```

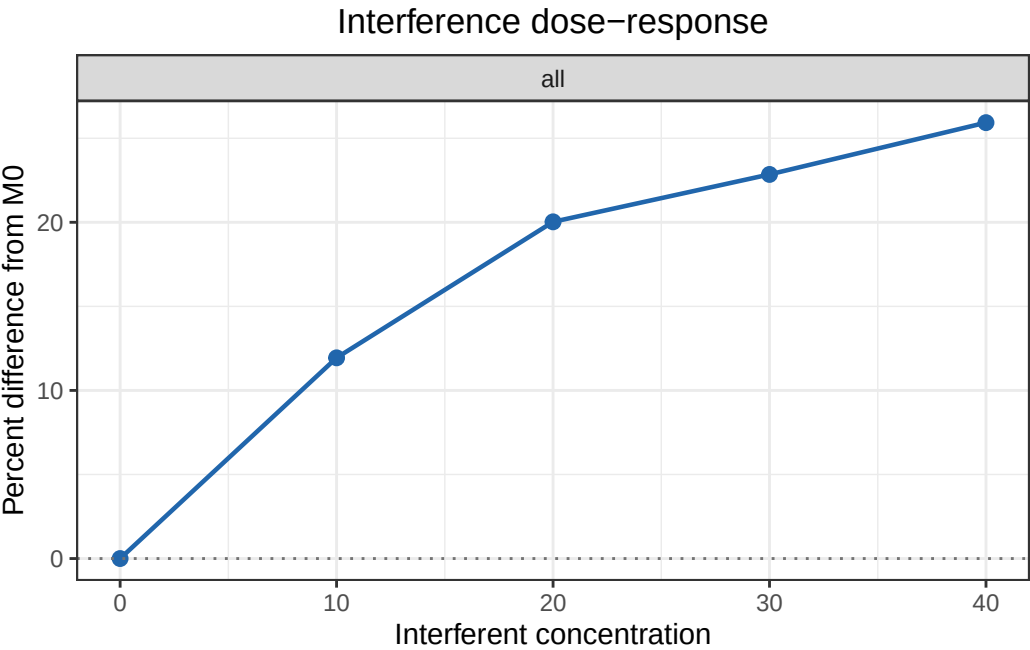
Mean : 251.88
SD : 4.4549972
SE : 1.9923353
Absolute difference : 51.86
Percent difference : 25.927407
CV percent : 1.7686983

Using the mean at 0 mg/dL, $M_0 = 200.02$ nmol/L, as the baseline, calculate the absolute and relative effects at each concentration. The main text also asks at which interferent concentration the mean effect reaches 10%. This can be obtained by reverse interpolation between the measured 0 and 10 mg/dL nodes:

```
ep07_table4_limit <- predict(
  ep07_table4_result,
  target = 10,
  metric = "percent"
)
ep07_table4_limit
```

```
stratum direction metric input estimate within_observed_range
1 all inverse percent 10 8.376047 TRUE
```

```
plot(ep07_table4_result, type = "percent")
```



Compare the recalculated values with EP07 Table 4 and Section 6.2.4.3:

Table 7.3: EP07 Third Edition Table 4: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
0	mean	200.020	200.02000
10	mean	223.900	223.90000
20	mean	240.080	240.08000
30	mean	245.720	245.72000
40	mean	251.880	251.88000
0	cv_percent	0.900	0.88760
10	cv_percent	2.100	2.07502
20	cv_percent	1.400	1.39689

Standard location	Statistic	Standard	ivdtools
30	cv_percent	1.900	1.94907
40	cv_percent	1.800	1.76870
10	absolute_difference	23.880	23.88000
20	absolute_difference	40.060	40.06000
30	absolute_difference	45.700	45.70000
40	absolute_difference	51.860	51.86000
10	percent_difference	11.940	11.93881
20	percent_difference	20.030	20.02800
30	percent_difference	22.850	22.84772
40	percent_difference	25.930	25.92741
all	interference_at_10_percent	8.375	8.37605

Reverse interpolation gives 8.376 mg/dL. The main text reports 8.375 mg/dL when calculated from the rounded 11.94% value in the table, and recommends reporting at most one decimal place after considering study uncertainty, or approximately 8.4 mg/dL. The estimate lies within the measured range but remains an estimate between two nodes; for a critical claim, samples should be prepared and tested near this concentration.

EP07 Section 6.2.5.4 also uses these data to illustrate the limitation of a quadratic curve. Following the main text, fit a quadratic polynomial to all 25 individual results:

```
ep07_table4_quadratic <- lm(
  result ~ concentration + I(concentration^2),
  data = ep07_table4
)
ep07_table4_quadratic_coef <- coef(ep07_table4_quadratic)
```

	term	estimate
1	(Intercept)	200.64342857
2	concentration	2.56911429
3	I(concentration^2)	-0.03284286

Table 7.4: Quadratic-curve recalculation for EP07 Section 6.2.5.4

Standard location	Statistic	Standard	ivdtools
EP07 6.2.5.4	intercept	200.640000	200.643429
EP07 6.2.5.4	linear_term	2.569100	2.569114
EP07 6.2.5.4	quadratic_term	-0.032843	-0.032843
EP07 6.2.5.4	turning_point	39.100000	39.112223

The quadratic model suggests that the effect turns downward after approximately 39.1 mg/dL. This is mainly a shape imposed by the function, not a mechanism demonstrated by the experiment. The quadratic fit is therefore retained here to reproduce the warning in the standard text, while interpretation continues to use the point-to-point result.

7.4.4 CLSI EP07 Third Edition Table 5: Linear Dose–Response

Table 5 uses the same five interferent concentrations and five measurements per concentration, but the data follow an approximately linear trajectory. In this case, method = “linear” fits the analyte result against interferent concentration using all 25 individual results, rather than regressing only the five concentration means.

```
ep07_table5 <- read_csv(
  "/data/007-干扰研究/std-ep07-table5-dose-response.csv",
```

```

      show_col_types = FALSE
    )
    head(ep07_table5, 6)

# A tibble: 6 x 6
  sample replicate concentration result concentration_unit result_unit
  <dbl>     <dbl>         <dbl> <dbl> <chr>              <chr>
1     1         1             0  200. mg/dL          nmol/L
2     1         2             0  200. mg/dL          nmol/L
3     1         3             0  203. mg/dL          nmol/L
4     1         4             0  198. mg/dL          nmol/L
5     1         5             0  206. mg/dL          nmol/L
6     2         1            10  212. mg/dL          nmol/L

ep07_table5_result <- interference_dose_response(
  ep07_table5,
  result = "result",
  concentration = "concentration",
  baseline = 0,
  method = "linear"
)
ep07_table5_result

```

Interference dose-response analysis

Method: linear

Baseline concentration: 0

Stratum: all | Concentration: 0

```

Observations      : 5
Mean              : 201.44
SD                : 3.1101447
SE                : 1.390899
Absolute difference : 0
Percent difference : 0
CV percent        : 1.5439559

```

Stratum: all | Concentration: 10

```

Observations      : 5
Mean              : 212.28
SD                : 2.2785961
SE                : 1.0190191
Absolute difference : 10.84
Percent difference : 5.381255
CV percent        : 1.0733918

```

Stratum: all | Concentration: 20

```

Observations      : 5
Mean              : 226.56
SD                : 5.2122932
SE                : 2.3310084
Absolute difference : 25.12
Percent difference : 12.470214
CV percent        : 2.3006237

```

Stratum: all | Concentration: 30

```

Observations      : 5
Mean              : 239.2
SD                : 2.9597297

```

```

SE           : 1.3236314
Absolute difference : 37.76
Percent difference : 18.745036
CV percent    : 1.2373452

```

```
Stratum: all | Concentration: 40
```

```

Observations : 5
Mean         : 249.6
SD           : 2.9732137
SE           : 1.3296616
Absolute difference : 48.16
Percent difference : 23.907863
CV percent    : 1.1911914

```

```
plot(ep07_table5_result, type = "absolute")
```

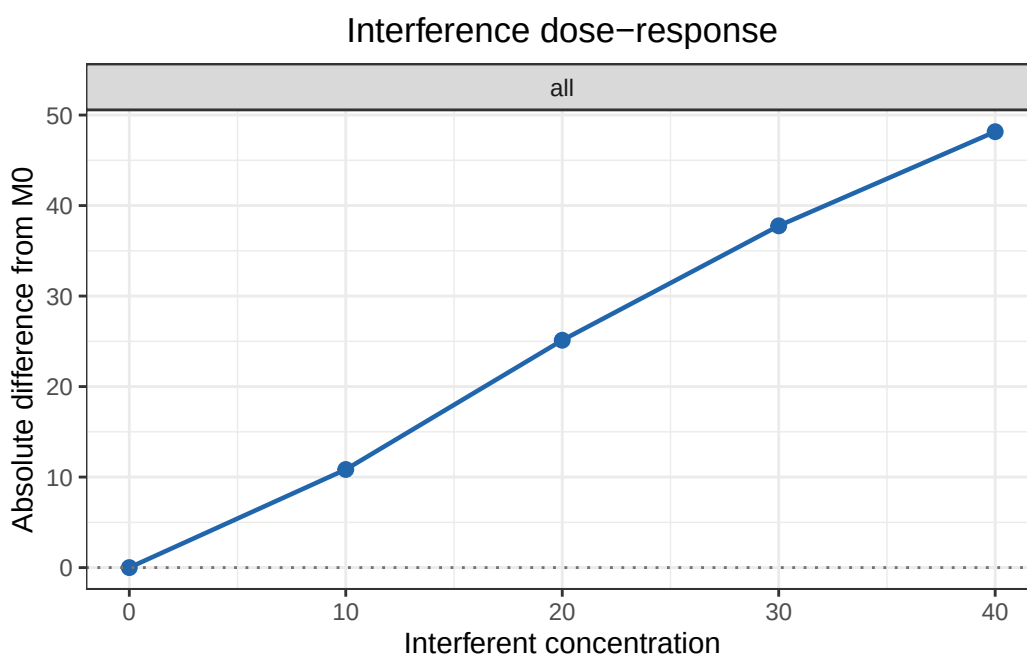


Table 7.5: EP07 Third Edition Table 5: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
0	mean	201.4400	201.44000
10	mean	212.2800	212.28000
20	mean	226.5600	226.56000
30	mean	239.2000	239.20000
40	mean	249.6000	249.60000
0	cv_percent	1.5000	1.54396
10	cv_percent	1.1000	1.07339
20	cv_percent	2.3000	2.30062
30	cv_percent	1.2000	1.23735
40	cv_percent	1.2000	1.19119
all	intercept	201.1700	201.16800
all	slope	1.2324	1.23240
all	difference_intercept	-0.2720	-0.27200

Using the five replicates at 0 mg/dL gives $M_0 = 201.44$ nmol/L. Regression of all individual results gives $Y = 1.2324X + 201.168$; the main text writes the intercept as 201.17 nmol/L. If

the results are first converted to absolute effects using M_0 , the equation is $d = 1.2324X - 0.272$. The regression intercept and the measured baseline mean are two different estimates of the no-interferent result and should not be mixed. The linear method is appropriate only when the scatter is approximately linear across the measured range; a high goodness of fit does not justify ignoring local curvature or extrapolating beyond the observed range.

7.4.5 YY/T 1789.5–2023 Appendix A: Interferent-Spike Screening

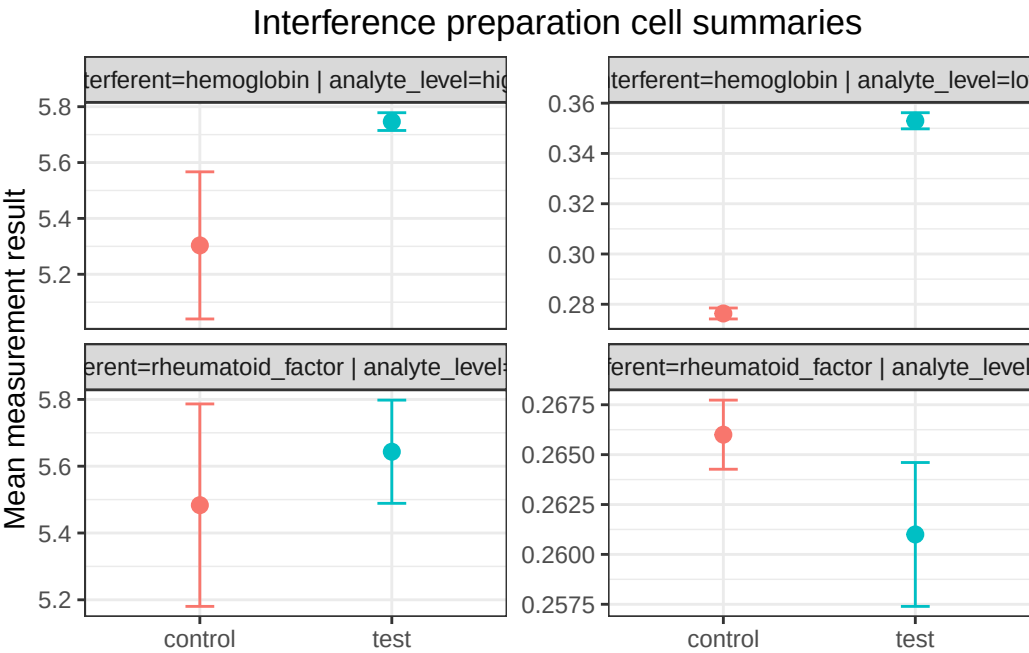
Appendix A uses low- and high-level thyroid-stimulating hormone (TSH) samples, at 0.3 and 5.0 $\mu\text{IU/mL}$, respectively. It screens 2,000 mg/dL hemoglobin and 2,000 IU/mL rheumatoid factor. Test preparations receive interferent stock solution, while control preparations receive the same volume of physiological saline; each preparation is measured three times.

```
yyt1789_a_data <- read_csv(
  "./data/007-干扰研究/std-yyt1789-5-appendix-a-paired.csv",
  show_col_types = FALSE
)
head(yyt1789_a_data, 6)
```

```
# A tibble: 6 x 7
  interferent analyte_level condition interferent_concentration replicate result
  <chr>        <chr>        <chr>                <dbl>         <dbl> <dbl>
1 hemoglobin  low          control                0             1  0.272
2 hemoglobin  low          control                0             2  0.279
3 hemoglobin  low          control                0             3  0.278
4 hemoglobin  low          test                  2000          1  0.347
5 hemoglobin  low          test                  2000          2  0.358
6 hemoglobin  low          test                  2000          3  0.354
# i 1 more variable: unit <chr>
```

```
yyt1789_a <- interference_paired(
  yyt1789_a_data,
  result = "result",
  design = "ep07",
  condition = "condition",
  test_level = "test",
  control_level = "control",
  by = c("interferent", "analyte_level"),
  conf.level = 0.95
)
```

```
plot(yyt1789_a)
```



Compare the results with the standard text:

Table 7.6: YY/T 1789.5 Appendix A: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
hemoglobin_low	control_mean	0.276	0.2763
hemoglobin_low	test_mean	0.353	0.3530
hemoglobin_low	difference	0.077	0.0767
hemoglobin_high	control_mean	5.303	5.3033
hemoglobin_high	test_mean	5.747	5.7467
hemoglobin_high	difference	0.443	0.4433
rheumatoid_factor_low	control_mean	0.266	0.2660
rheumatoid_factor_low	test_mean	0.261	0.2610
rheumatoid_factor_low	difference	-0.005	-0.0050
rheumatoid_factor_high	control_mean	5.483	5.4833
rheumatoid_factor_high	test_mean	5.643	5.6433
rheumatoid_factor_high	difference	0.160	0.1600

The hemoglobin effects at the low and high levels are 0.0767 and 0.4433 $\mu\text{IU/mL}$, respectively. The critical bias values B_c in Table A.3 are 0.024 and 0.432 $\mu\text{IU/mL}$, so both levels proceed to further dose–response evaluation under the statistical rule in the standard text. The rheumatoid-factor effects are -0.0050 and 0.1600 $\mu\text{IU/mL}$, within the case’ s technical limits of ± 0.03 and ± 0.5 $\mu\text{IU/mL}$, respectively.

The confidence intervals from the paired-interference analysis use the observed SD of the preparation units. Table A.3 of YY/T 1789.5 instead uses the repeatability SD from a prior precision study and a standard-specific formula for B_c . This case therefore compares preparation-unit means and effects with the standard table without misrepresenting the software interval as the standard’ s B_c decision.

7.4.6 YY/T 1789.5–2023 Appendix B: Hemoglobin Dose–Response

Appendix B prepares low- and high-level TSH samples at five hemoglobin concentrations: 0, 500, 1,000, 1,500, and 2,000 mg/dL, with three measurements at each concentration. The standard uses the mean at 0 mg/dL as the control and uses point-to-point interpolation to determine the interferent concentration at which the allowable bias is reached.

```

yyt1789_b_data <- read_csv(
  "./data/007-干扰研究/std-yyt1789-5-appendix-b-dose.csv",
  show_col_types = FALSE
)
head(yyt1789_b_data, 6)

# A tibble: 6 x 7
  interferent analyte_level concentration replicate result result_unit
  <chr>        <chr>          <dbl>      <dbl>   <dbl> <chr>
1 hemoglobin  low                0          1  0.282 uIU/mL
2 hemoglobin  low                0          2  0.29  uIU/mL
3 hemoglobin  low                0          3  0.263 uIU/mL
4 hemoglobin  low               500          1  0.274 uIU/mL
5 hemoglobin  low               500          2  0.307 uIU/mL
6 hemoglobin  low               500          3  0.31  uIU/mL
# i 1 more variable: concentration_unit <chr>

```

```

yyt1789_b <- interference_dose_response(
  yyt1789_b_data,
  result = "result",
  concentration = "concentration",
  by = "analyte_level",
  baseline = 0,
  method = "point_to_point"
)

```

The allowable bias for the low-level sample is ± 0.03 $\mu\text{IU/mL}$. Reverse-interpolate the absolute effect at 0.03 $\mu\text{IU/mL}$:

```

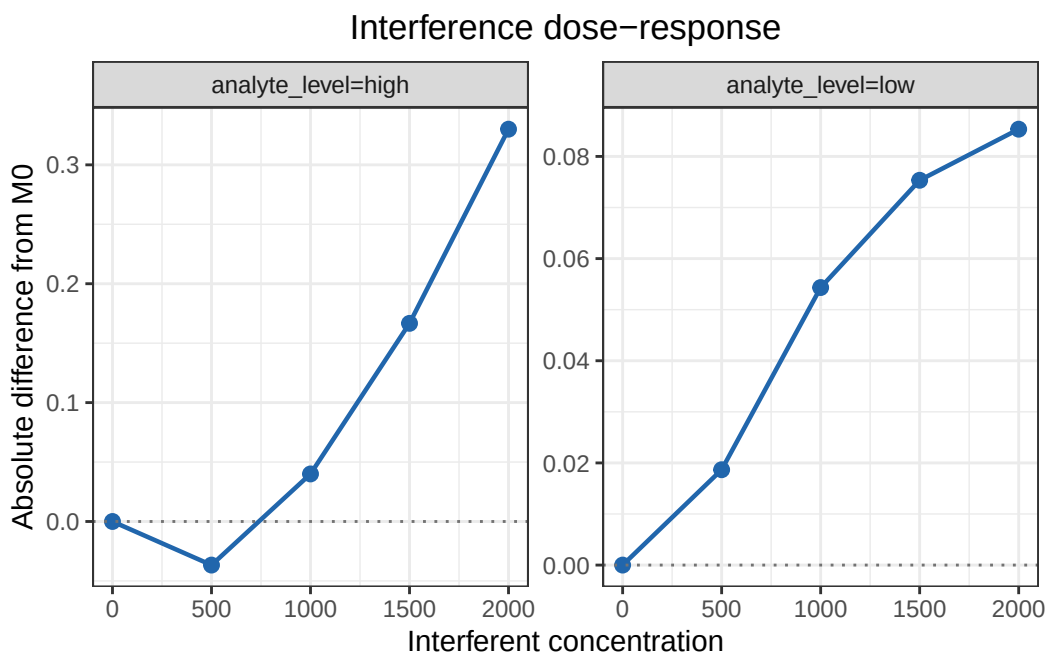
yyt1789_b_limit <- predict(
  yyt1789_b,
  target = 0.03,
  metric = "absolute",
  stratum = "analyte_level=low"
)

```

```

plot(yyt1789_b, type = "absolute")

```



Compare the results with the standard text:

Table 7.7: YY/T 1789.5 Appendix B: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
low_0	mean	0.278	0.2783
low_0	difference	0.000	0.0000
low_500	mean	0.297	0.2970
low_500	difference	0.019	0.0187
low_1000	mean	0.333	0.3327
low_1000	difference	0.054	0.0543
low_1500	mean	0.354	0.3537
low_1500	difference	0.075	0.0753
low_2000	mean	0.364	0.3637
low_2000	difference	0.085	0.0853
high_0	mean	5.273	5.2733
high_0	difference	0.000	0.0000
high_500	mean	5.237	5.2367
high_500	difference	-0.037	-0.0367
high_1000	mean	5.313	5.3133
high_1000	difference	0.040	0.0400
high_1500	mean	5.440	5.4400
high_1500	difference	0.167	0.1667
high_2000	mean	5.603	5.6033
high_2000	difference	0.330	0.3300
low	interference_concentration	658.880	658.8785

The low-level effect crosses $0.03 \mu\text{IU/mL}$ between 500 and 1,000 mg/dL, giving 658.88 mg/dL by point-to-point reverse interpolation. At the high level, the absolute effect remains within $\pm 0.5 \mu\text{IU/mL}$ at all measured concentrations. The case therefore reports 658.88 mg/dL as the hemoglobin concentration resisted by the evaluated system, taking the lower result of the two analyte levels. This is an interpolation between measured nodes, not a result measured directly at 658.88 mg/dL.

7.4.7 YY/T 1789.5–2023 Appendix C: Patient-Sample Evaluation

Appendix C uses EDTA-anticoagulated plasma as the test group and serum as the control group, with 20 patient samples in each group. Every sample is measured by both the evaluated measurement procedure and a comparative procedure claimed for serum and EDTA plasma. The source Table C.1 lists one procedure result per sample, rather than single-replicate records, so the input has one row per sample-by-procedure combination.

```
yyt1789_c_data <- read_csv(
  "/data/007-干扰研究/std-yyt1789-5-appendix-c-patient.csv",
  show_col_types = FALSE
)
head(yyt1789_c_data, 6)
```

```
# A tibble: 6 x 5
  id   group method    result unit
<chr> <chr>  <chr>    <dbl> <chr>
1 C01  control comparative 23.3  uIU/mL
2 C01  control evaluated 23.5  uIU/mL
3 C02  control comparative 8     uIU/mL
4 C02  control evaluated 2.73  uIU/mL
5 C03  control comparative 5.35  uIU/mL
6 C03  control evaluated 4.47  uIU/mL
```

```
yyt1789_c <- interference_patient(
  yyt1789_c_data,
  id = "id",
  group = "group",
  method = "method",
  result = "result",
  test_group = "test",
  control_group = "control",
  evaluated_method = "evaluated",
  comparative_method = "comparative",
  conf.level = 0.95
)
yyt1789_c
```

Patient-specimen interference evaluation

Evaluated - comparative: evaluated - comparative

Group: test

Observations	: 20
Mean difference	: -1.1835
SD difference	: 2.3930541
SE difference	: 0.53510317
Median difference	: -0.49
25th percentile	: -1.3425
75th percentile	: -0.32
CI lower	: -2.3034838
CI upper	: -0.063516183
Mean percent difference	: -17.916419
SD percent difference	: 18.774944
Percent CI lower	: -26.703363
Percent CI upper	: -9.1294749

Group: control

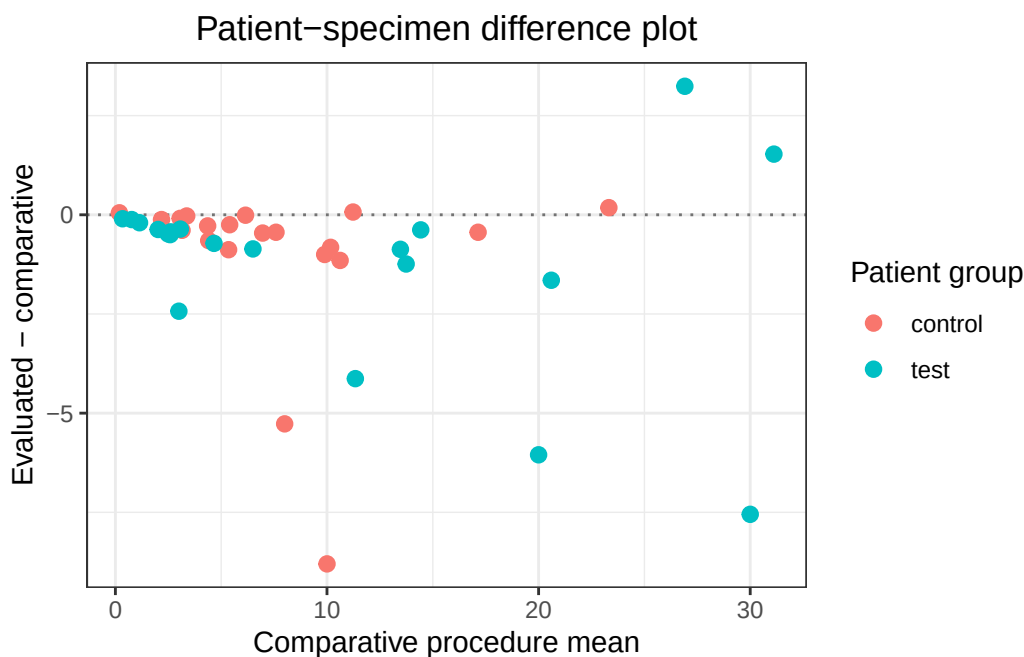
Observations	: 20
Mean difference	: -1.039

```

SD difference      : 2.1614978
SE difference      : 0.4833256
Median difference  : -0.415
25th percentile   : -0.835
75th percentile   : -0.075
CI lower           : -2.0506121
CI upper           : -0.027387902
Mean percent difference : -11.712959
SD percent difference : 24.21947
Percent CI lower   : -23.04802
Percent CI upper   : -0.37789805

```

```
plot(yyt1789_c, type = "difference")
```



Compare the results with the standard text:

Table 7.8: YY/T 1789.5 Appendix C: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
control	mean_difference	-1.04	-1.0390
control	sd_difference	2.16	2.1615
control	ci_lower	-2.05	-2.0506
control	ci_upper	-0.03	-0.0274
test	mean_difference	-1.18	-1.1835
test	sd_difference	2.39	2.3931
test	ci_lower	-2.30	-2.3035
test	ci_upper	-0.06	-0.0635

The mean difference for the control group is $-1.039 \mu\text{IU/mL}$, with a 95% CI of -2.051 to $-0.027 \mu\text{IU/mL}$. For the test group, the mean is $-1.184 \mu\text{IU/mL}$, with a 95% CI of -2.303 to $-0.064 \mu\text{IU/mL}$. The intervals overlap, so Appendix C concludes that the EDTA anticoagulant does not interfere with this TSH measurement. Both groups show negative bias relative to the comparative procedure; the decision is based on overlap between the two group intervals, not on whether either interval contains zero.

7.4.8 YY/T 1789.5–2023 Appendix D: Cross-Reactivity Evaluation

Appendix D adds 200,000 $\mu\text{IU/mL}$ luteinizing hormone (LH) to low- and high-level TSH base samples and uses controls with the same volume of diluent. First obtain the test-minus-control effect with paired interference analysis, then calculate the standard-defined cross-reactivity rate:

$$\text{Cross-reactivity rate} = \frac{\bar{x}_T - \bar{x}_C}{C_{\text{interferent}}} \times 100\%.$$

```
yyt1789_d_data <- read_csv(
  "./data/007-干扰研究/std-yyt1789-5-appendix-d-cross-reactivity.csv",
  show_col_types = FALSE
)
head(yyt1789_d_data)
```

```
# A tibble: 6 x 8
  interferent analyte_level condition interferent_concentration replicate result
  <chr>        <chr>        <chr>                <dbl>      <dbl> <dbl>
1 LH          low          control                0          1 0.281
2 LH          low          control                0          2 0.283
3 LH          low          control                0          3 0.282
4 LH          low          test                200000     1 0.312
5 LH          low          test                200000     2 0.305
6 LH          low          test                200000     3 0.298
# i 2 more variables: result_unit <chr>, concentration_unit <chr>
```

```
yyt1789_d <- interference_paired(
  yyt1789_d_data,
  result = "result",
  design = "ep07",
  condition = "condition",
  test_level = "test",
  control_level = "control",
  by = "analyte_level",
  conf.level = 0.95
)
yyt1789_d
```

Interference paired-difference analysis

Design: ep07

Confidence level: 95%

analyte_level: low

```
stratum      : analyte_level=low
design        : ep07
n_test       : 3
n_control    : 3
test_mean    : 0.305
control_mean : 0.282
absolute_difference : 0.023
absolute_se  : 0.0040824829
df           : 4
absolute_ci_lower : 0.01166521
absolute_ci_upper : 0.03433479
percent_difference : 8.1560284
percent_se    : 1.447689
percent_ci_lower : 4.1365994
```

```

percent_ci_upper : 12.175457

analyte_level: high
  stratum      : analyte_level=high
  design       : ep07
  n_test       : 3
  n_control    : 3
  test_mean    : 5.2366667
  control_mean : 5.04
  absolute_difference : 0.19666667
  absolute_se   : 0.04772607
  df           : 4
  absolute_ci_lower : 0.064157853
  absolute_ci_upper : 0.32917548
  percent_difference : 3.9021164
  percent_se     : 0.94694584
  percent_ci_lower : 1.2729733
  percent_ci_upper : 6.5312595

```

```

yyt1789_d_effects <- yyt1789_d$effects
yyt1789_d_effects$cross_reactivity_percent <-
  100 * yyt1789_d_effects$absolute_difference / 200000

```

Compare the results with the standard text:

Table 7.9: YY/T 1789.5 Appendix D: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
low	control_mean	0.28200	0.282000
low	test_mean	0.30500	0.305000
low	cross_reactivity_percent	0.00001	0.000012
high	control_mean	5.04000	5.040000
high	test_mean	5.24000	5.236667
high	cross_reactivity_percent	0.00010	0.000098

The cross-reactivity rates for the low- and high-level samples are approximately 0.00001% and 0.00010%, respectively, both within the case' s prespecified range of -0.1% to 0.1%. This conversion is appropriate only when the unit relationship between analyte result and interferent concentration has a clear scientific meaning. The report must state both units and the interferent concentration used in the experiment.

7.5 Detailed Function Reference

7.5.1 Replicate Planning with `interference_replicates()`

```

interference_replicates(
  sd,
  detectable_difference,
  alpha = 0.05,
  power = 0.90,
  alternative = c("two.sided", "one.sided"),
  min_replicates = 5L,
  label = NULL
)

```


sd is expected repeatability SD; for relative-scale planning it is repeatability CV. detectable_difference uses the same scale and a compatible length. alpha is the type-I error probability, power is the target power, alternative selects a two- or one-sided design, min_replicates is the minimum number of test and control replicates per preparation, and label optionally names each calculation.

The result is an `interference_replicates` data frame with one row per design. It includes the inputs, `difference_over_sd`, `z_alpha`, `z_power`, `calculated_replicates` before rounding, and the final `replicates_per_sample`. The final count applies separately to each test and control preparation.

7.5.2 Paired Preparation with `interference_paired()`

```
interference_paired(
  data,
  result,
  design = c("ep07", "analyte_spike", "blank"),
  by = NULL,
  condition = NULL,
  test_level = "test",
  control_level = "control",
  matrix = NULL,
  normal_level = "normal",
  interferent_level = "interferent",
  spike = NULL,
  unspiked_level = "unspiked",
  spiked_level = "spiked",
  conf.level = 0.95
)
```

`data` is a long-format data frame with one replicate result per row. `result` names the numeric measurement column. `design` can be `ep07`, `analyte_spike`, or `blank`. `by` optionally supplies stratification columns such as `interferent` and `analyte_level`. `condition` identifies test/control conditions for `ep07` and optionally `blank` designs. `test_level` and `control_level` identify those values. `matrix` identifies the matrix column for `analyte_spike`; `normal_level` and `interferent_level` identify normal and interferent matrices; `spike` identifies `analyte_spike` status, and `unspiked_level` and `spiked_level` identify the two states. `conf.level` is the effect-interval confidence level.

For `ep07`, use `condition` with `test_level` and `control_level`. For `analyte_spike`, use `matrix` and `spike` settings. For `blank`, use `condition` to distinguish interferent blank and control, or compare with a theoretical zero when no control records exist.

The returned `interference_paired` object contains effects by stratum, including sample sizes, test/control means, absolute and relative effects, SEs, and intervals. `cell_summary` contains preparation-unit `n`, mean, SD, and SE; `issues` records data problems; `excluded_rows` stores omitted input rows; `analysis_data` stores analyzed rows. `plot()` returns preparation-unit and effect figures.

7.5.3 Dose–Response with `interference_dose_response()`

```
interference_dose_response(
  data,
  result,
  concentration,
  by = NULL,
  baseline = 0,
```

```

    method = c("point_to_point", "linear"),
    conf.level = 0.95
  )

predict(
  object,
  newdata = NULL,
  target = NULL,
  metric = c("result", "absolute", "percent"),
  stratum = NULL
)

plot(x, type = c("response", "absolute", "percent"))

```

data is a long-format dose-response data frame. result and concentration name the numeric result and interferent concentration columns. by optionally defines strata. baseline is the measured interferent concentration used for the baseline mean and must occur in every stratum. point_to_point interpolates between adjacent concentration means; linear fits a line to individual results. conf.level is the regression-coefficient confidence level.

predict() takes a dose-response object. Supply newdata to predict a result or effect at interferent concentrations, or target to reverse-predict the interferent concentration at a specified result/effect; exactly one of newdata and target is required. metric selects result, absolute effect, or percentage effect; stratum selects one or more strata. The function does not extrapolate and returns missing values outside the observed range. plot() selects result, absolute-effect, or percentage-effect figures.

The returned interference_dose_response object contains a summary of n, mean, SD, SE, CV, absolute effect, and percentage effect at each concentration; diagnostics for baseline, level count, replicate range, and monotonicity; coefficients and models for the linear method; and issues, excluded_rows, and analysis_data. predict() returns stratum, direction, metric, input, estimate, and within_observed_range.

7.5.4 Patient-Sample Comparison with interference_patient()

```

interference_patient(
  data,
  id,
  group,
  method,
  result,
  test_group,
  control_group,
  evaluated_method,
  comparative_method,
  interferent_cols = NULL,
  conf.level = 0.95
)

plot(
  x,
  type = c("difference", "interferent"),
  interferent = NULL
)

```

data is a long-format patient-sample result data frame. id names the sample identifier, group the patient test/control group, method the measurement procedure, and result the numeric result. test_group and control_group identify the two patient groups. evaluated_method

and `comparative_method` define the difference direction as `evaluated` minus `comparative`. `interferent_cols` optionally names numeric interferent columns; each is fitted in a separate univariate model. `conf.level` controls group summaries and regression coefficients.

For `plot()`, `type = "difference"` plots method difference against the comparative-procedure mean; `type = "interferent"` plots difference against a selected interferent concentration. When several interferent columns exist, `interferent` selects the column to plot.

The returned `interference_patient` object contains `method_summary` for sample-by-method `n`, mean, and SD; `specimen_results` for complete method pairs and absolute/relative differences; `group_summary` for group-level differences and intervals; regressions and models for each interferent; and `issues`, `incomplete_ids`, and `excluded_rows`. `plot()` returns the corresponding figure.

Chapter 8

Dilution Recovery, Spike Recovery, and Hook Effect

This chapter covers three related but distinct analyses: dilution recovery, spike recovery, and high-dose hook effect. Each uses a different expected-value calculation and requires its own acceptance limits.

Path	Main input	Main result
Dilution recovery	Sample ID, result, dilution factor, optional assignment	Target, dilution-corrected concentration, per-sample and across-sample recovery
Spike recovery	Spiked and solvent-control results, theoretical added concentration	Observed increment and recovery
High-dose hook effect	Known concentration, raw response, ULoQ, optional series	Concentration summaries, hook-effect boundary, or lower bound when no hook is observed

! Scope of Use

Recovery and hook-effect calculations do not by themselves establish an extended measuring interval. Interpret them together with precision, LoQ, specimen commutability, carryover, and prespecified acceptance limits. Use raw instrument response for hook-effect analysis; reported concentrations may already be truncated.

```
library(ivdtools)
library(readr)
```

8.1 Main Functions

Function	Purpose	Return value
<code>dilution_recovery()</code>	Calculate dilution targets, corrected concentrations, and recovery	<code>dilution_recovery</code> object

Function	Purpose	Return value
<code>spike_concentration()</code>	Calculate theoretical post-spike concentration by mass balance	Data frame
<code>spike_recovery()</code>	Calculate recovery relative to solvent control	<code>spike_recovery</code> object
<code>hook_effect()</code>	Summarize high-concentration response and identify hook boundaries	<code>hook_effect</code> object

8.1.1 Dilution Recovery

```
dilution_recovery(
  data, sample, result, dilution_factor, assigned = NULL,
  diluent_value = 0, conf.level = 0.95
)
```

`dilution_factor` is 1 for undiluted material and 2 for a 1:2 dilution. If `assigned` is omitted, the mean of each sample's undiluted results is used as the original concentration. The target after dilution is

$$C_{target} = \frac{C_{assigned}}{d},$$

and the corrected concentration is $dC_{measured}$ when the diluent contains no analyte. Report the dilution-level and across-sample recovery, not only a pooled mean.

8.1.2 Theoretical Spike Concentration

```
spike_concentration(
  sample_concentration, sample_volume,
  stock_concentration, spike_volume
)
```

By mass balance,

$$C_{final} = \frac{C_s V_s + C_{stock} V_{spike}}{V_s + V_{spike}}.$$

8.1.3 Spike Recovery

```
spike_recovery(
  data, sample, result, condition,
  analyte_level, solvent_level, added_concentration = NULL,
  sample_volume = NULL, stock_concentration = NULL, spike_volume = NULL,
  conf.level = 0.95
)
```

The observed increment is the spiked result minus the solvent-control result. Recovery is the observed increment divided by the theoretical added concentration. Supply `added_concentration` directly or let the function calculate it from volumes and stock concentration.

8.1.4 High-Dose Hook Effect

```
hook_effect(
  data, concentration, response, series = NULL, uloq,
  uloq_response = NULL, conf.level = 0.95
)
```

The input must contain known concentrations and raw responses. `series` identifies independent hook-effect series. If `uloq_response` is omitted, every series must contain a record at `concentration == uloq`. The function summarizes response at each concentration, compares it with the ULoQ reference, and reports an observed hook boundary or the lower bound supported when no hook is observed.

8.2 Analysis Examples

8.2.1 Dilution Recovery

```
dilution <- as.data.frame(read_csv(
  "/data/008-稀释回收与HOOK/std-ep34-table1-single-sample.csv",
  show_col_types = FALSE
))
dilution_fit <- dilution_recovery(
  dilution, sample = "sample", result = "observed",
  dilution_factor = "dilution_factor", assigned = NULL
)
print(dilution_fit)
```

Dilution recovery

Samples: 1

Dilution factors: 1, 2, 4, 10, 20

Confidence level: 95.0%

Dilution factor: 1

```
Specimen content : 100
Samples          : 1
Mean recovery    : 100
SD recovery      : NA
SE recovery      : NA
Degrees of freedom : NA
CI lower         : NA
CI upper         : NA
```

Dilution factor: 2

```
Specimen content : 50
Samples          : 1
Mean recovery    : 95.583142
SD recovery      : NA
SE recovery      : NA
Degrees of freedom : NA
CI lower         : NA
CI upper         : NA
```

Dilution factor: 4

```
Specimen content : 25
Samples          : 1
```

Mean recovery : 101.46518
 SD recovery : NA
 SE recovery : NA
 Degrees of freedom : NA
 CI lower : NA
 CI upper : NA

Dilution factor: 10

Specimen content : 10
 Samples : 1
 Mean recovery : 105.81278
 SD recovery : NA
 SE recovery : NA
 Degrees of freedom : NA
 CI lower : NA
 CI upper : NA

Dilution factor: 20

Specimen content : 5
 Samples : 1
 Mean recovery : 99.312696
 SD recovery : NA
 SE recovery : NA
 Degrees of freedom : NA
 CI lower : NA
 CI upper : NA

Sample results

Sample: EP34-Table1 | Dilution factor: 1

Replicates : 3
 Mean result : 625.63333
 SD result : 3.1214313
 CV result : 0.49892343
 Specimen content : 100
 Assigned value : 625.63333
 Target concentration : 625.63333
 Corrected concentration : 625.63333
 Recovery : 100

Sample: EP34-Table1 | Dilution factor: 2

Replicates : 3
 Mean result : 299
 SD result : 6.2793312
 CV result : 2.1001108
 Specimen content : 50
 Assigned value : 625.63333
 Target concentration : 312.81667
 Corrected concentration : 598
 Recovery : 95.583142

Sample: EP34-Table1 | Dilution factor: 4

Replicates : 3
 Mean result : 158.7
 SD result : 2.8213472
 CV result : 1.7777865
 Specimen content : 25
 Assigned value : 625.63333

Target concentration : 156.40833
 Corrected concentration : 634.8
 Recovery : 101.46518

Sample: EP34-Table1 | Dilution factor: 10

Replicates : 3
 Mean result : 66.2
 SD result : 3.6345564
 CV result : 5.4902664
 Specimen content : 10
 Assigned value : 625.63333
 Target concentration : 62.563333
 Corrected concentration : 662
 Recovery : 105.81278

Sample: EP34-Table1 | Dilution factor: 20

Replicates : 3
 Mean result : 31.066667
 SD result : 2.7934447
 CV result : 8.9917748
 Specimen content : 5
 Assigned value : 625.63333
 Target concentration : 31.281667
 Corrected concentration : 621.33333
 Recovery : 99.312696

For multiple original samples, retain the sample identifier and use `std-ep34-table2-multiple-samples.csv` or the Appendix C files to assess both within-sample and across-sample recovery.

8.2.2 Spike Recovery

```
spike <- as.data.frame(read_csv(
  "/data/008-稀释回收与H00K/std-ep34-table3-spike-recovery.csv",
  show_col_types = FALSE
))
spike_fit <- spike_recovery(
  spike, sample = "sample", result = "result", condition = "condition",
  analyte_level = "analyte", solvent_level = "solvent",
  added_concentration = 3.64
)
print(spike_fit)
```

Spike recovery

Samples: 3
 Confidence level: 95.0%

Samples : 3
 Mean recovery : 90.659341
 SD of recovery : 2.7472527
 SE of recovery : 1.5861271
 Degrees of freedom : 2
 CI lower : 83.834787
 CI upper : 97.483895

Sample results

Sample: 1

```

Analyte replicates : 2
Analyte mean      : 13.1
Analyte SD       : 0.28284271
Solvent replicates : 2
Solvent mean     : 9.8
Solvent SD       : 0.14142136
Observed increase : 3.3
Added concentration : 3.64
Recovery         : 90.659341

```

Sample: 2

```

Analyte replicates : 2
Analyte mean      : 12.7
Analyte SD       : 0.14142136
Solvent replicates : 2
Solvent mean     : 9.5
Solvent SD       : 0
Observed increase : 3.2
Added concentration : 3.64
Recovery         : 87.912088

```

Sample: 3

```

Analyte replicates : 2
Analyte mean      : 12.3
Analyte SD       : 0.14142136
Solvent replicates : 2
Solvent mean     : 8.9
Solvent SD       : 0.28284271
Observed increase : 3.4
Added concentration : 3.64
Recovery         : 93.406593

```

8.2.3 Hook Effect

```

hook_data <- expand.grid(
  concentration = c(1, 2, 3, 4, 5, 6, 7, 8),
  replicate = 1:3,
  KEEP.OUT.ATTRS = FALSE
)
hook_mean <- c(100000, 150000, 180000, 178000, 183000, 160000, 110000, 80000)
hook_data$response <-
  rep(hook_mean, times = 3) + rep(c(-300, 0, 300), each = 8)
hook_fit <- hook_effect(
  hook_data, concentration = "concentration", response = "response",
  uloq = 1.5, uloq_response = 130000
)
print(hook_fit)

```

High-dose hook-effect experiment

ULoQ concentration: 1.5

ULoQ response: supplied

Series: 1

Series: all

```

ULoQ concentration      : 1.5
ULoQ response           : 130000

```

Hook concentration : 6
Hook response : 160000
First in-range concentration : 7
First in-range response : 110000
Highest tested concentration : 8
Hook lower bound : NA

Concentration-level results

Series: all | Concentration: 1

Replicates : 3
Mean response : 1e+05
SD response : 300
CV response : 0.3
SE response : 173.20508
CI lower : 99254.759
CI upper : 100745.24

Series: all | Concentration: 2

Replicates : 3
Mean response : 150000
SD response : 300
CV response : 0.2
SE response : 173.20508
CI lower : 149254.76
CI upper : 150745.24

Series: all | Concentration: 3

Replicates : 3
Mean response : 180000
SD response : 300
CV response : 0.16666667
SE response : 173.20508
CI lower : 179254.76
CI upper : 180745.24

Series: all | Concentration: 4

Replicates : 3
Mean response : 178000
SD response : 300
CV response : 0.16853933
SE response : 173.20508
CI lower : 177254.76
CI upper : 178745.24

Series: all | Concentration: 5

Replicates : 3
Mean response : 183000
SD response : 300
CV response : 0.16393443
SE response : 173.20508
CI lower : 182254.76
CI upper : 183745.24

Series: all | Concentration: 6

Replicates : 3
Mean response : 160000
SD response : 300

CV response : 0.1875
 SE response : 173.20508
 CI lower : 159254.76
 CI upper : 160745.24

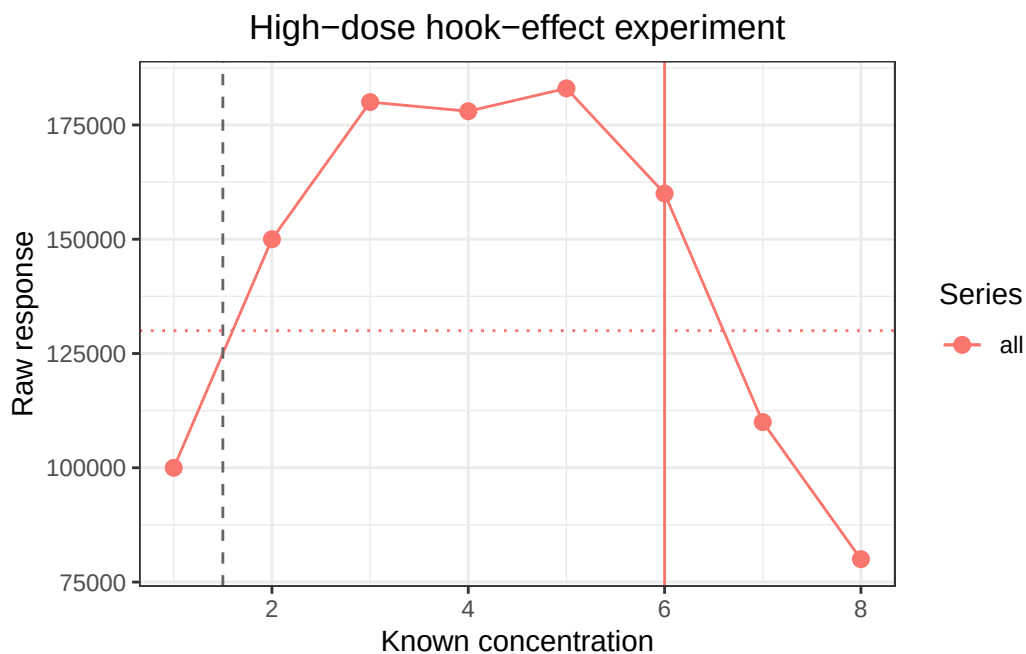
Series: all | Concentration: 7

Replicates : 3
 Mean response : 110000
 SD response : 300
 CV response : 0.27272727
 SE response : 173.20508
 CI lower : 109254.76
 CI upper : 110745.24

Series: all | Concentration: 8

Replicates : 3
 Mean response : 80000
 SD response : 300
 CV response : 0.375
 SE response : 173.20508
 CI lower : 79254.759
 CI upper : 80745.241

`plot(hook_fit)`



8.3 Interpretation

Recovery near 100% does not prove accuracy across the entire interval; bias can vary with concentration and matrix. A lack of observed hook effect means only that the tested concentrations and responses support a lower bound. The EP34 Tables 1–8 and Appendix C files supplied with the project can be used for reproducible cross-checks, but the standard's final measuring-interval decision still requires all relevant evidence and prespecified limits.

8.4 Complete Standard Cases

8.4.1 CLSI EP34 Table 1: Single-Sample Diluent Suitability

EP34 Table 1 measures one neat sample and 1/2, 1/4, 1/10, and 1/20 dilutions three times each. Because no independent assignment is provided, `dilution_recovery()` uses the mean of the neat results as the original-sample concentration.

```
ep34_table1 <- read_csv(
  "./data/008-稀释回收与H00K/std-ep34-table1-single-sample.csv",
  show_col_types = FALSE
)

head(ep34_table1, 6)
```

```
# A tibble: 6 x 6
  sample      dilution fraction replicate observed dilution_factor
  <chr>      <chr>      <dbl>      <dbl>      <dbl>      <dbl>
1 EP34-Table1 neat          1          1      624.          1
2 EP34-Table1 1/2          0.5          1     293.          2
3 EP34-Table1 1/4          0.25          1     156.          4
4 EP34-Table1 1/10         0.1          1      63.1         10
5 EP34-Table1 1/20         0.05          1      34.1         20
6 EP34-Table1 neat          1          2     629.          1
```

```
ep34_table1_result <- dilution_recovery(
  ep34_table1,
  sample = "sample",
  result = "observed",
  dilution_factor = "dilution_factor",
  conf.level = 0.95
)
ep34_table1_result
```

Dilution recovery

Samples: 1

Dilution factors: 1, 2, 4, 10, 20

Confidence level: 95.0%

Dilution factor: 1

```
Specimen content : 100
Samples          : 1
Mean recovery    : 100
SD recovery      : NA
SE recovery      : NA
Degrees of freedom : NA
CI lower         : NA
CI upper         : NA
```

Dilution factor: 2

```
Specimen content : 50
Samples          : 1
Mean recovery    : 95.583142
SD recovery      : NA
SE recovery      : NA
Degrees of freedom : NA
CI lower         : NA
```

CI upper : NA

Dilution factor: 4

Specimen content : 25
 Samples : 1
 Mean recovery : 101.46518
 SD recovery : NA
 SE recovery : NA
 Degrees of freedom : NA
 CI lower : NA
 CI upper : NA

Dilution factor: 10

Specimen content : 10
 Samples : 1
 Mean recovery : 105.81278
 SD recovery : NA
 SE recovery : NA
 Degrees of freedom : NA
 CI lower : NA
 CI upper : NA

Dilution factor: 20

Specimen content : 5
 Samples : 1
 Mean recovery : 99.312696
 SD recovery : NA
 SE recovery : NA
 Degrees of freedom : NA
 CI lower : NA
 CI upper : NA

Sample results

Sample: EP34-Table1 | Dilution factor: 1

Replicates : 3
 Mean result : 625.63333
 SD result : 3.1214313
 CV result : 0.49892343
 Specimen content : 100
 Assigned value : 625.63333
 Target concentration : 625.63333
 Corrected concentration : 625.63333
 Recovery : 100

Sample: EP34-Table1 | Dilution factor: 2

Replicates : 3
 Mean result : 299
 SD result : 6.2793312
 CV result : 2.1001108
 Specimen content : 50
 Assigned value : 625.63333
 Target concentration : 312.81667
 Corrected concentration : 598
 Recovery : 95.583142

Sample: EP34-Table1 | Dilution factor: 4

Replicates : 3

Mean result : 158.7
 SD result : 2.8213472
 CV result : 1.7777865
 Specimen content : 25
 Assigned value : 625.63333
 Target concentration : 156.40833
 Corrected concentration : 634.8
 Recovery : 101.46518

Sample: EP34-Table1 | Dilution factor: 10

Replicates : 3
 Mean result : 66.2
 SD result : 3.6345564
 CV result : 5.4902664
 Specimen content : 10
 Assigned value : 625.63333
 Target concentration : 62.563333
 Corrected concentration : 662
 Recovery : 105.81278

Sample: EP34-Table1 | Dilution factor: 20

Replicates : 3
 Mean result : 31.066667
 SD result : 2.7934447
 CV result : 8.9917748
 Specimen content : 5
 Assigned value : 625.63333
 Target concentration : 31.281667
 Corrected concentration : 621.33333
 Recovery : 99.312696

The SD, SE, and confidence intervals printed in the summary are calculated from recovery values across different samples at a given dilution, rather than from the three replicate measurements. They are therefore NA for this single-sample case.

```
plot(ep34_table1_result, type = "recovery")
```

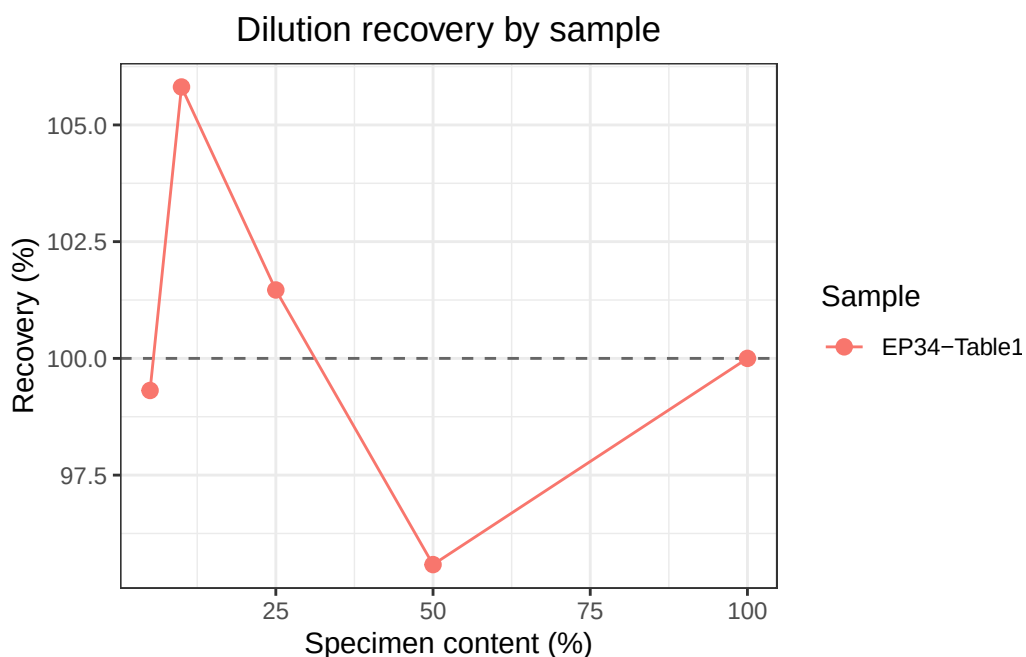


Table 8.1: EP34 Table 1: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
neat	recovery	100.00	100.000
1/2	recovery	95.59	95.583
1/4	recovery	101.47	101.465
1/10	recovery	105.75	105.813
1/20	recovery	99.36	99.313

The software uses unrounded means from the three replicates, whereas the standard table first displays observed means and targets to one decimal place before calculating recovery. The resulting difference is no more than 0.07 percentage points. Under the example's pre-specified 90%–110% range, all five point estimates are within the limits; the range is not built into the function.

8.4.2 CLSI EP34 Table 2: Multiple Original Samples

Table 2 gives observed means for five original samples at the same five dilution levels. The source does not publish all replicate results behind those means, so the input remains at summary granularity, with one row per sample-by-dilution level.

```
ep34_table2 <- read_csv(
  "/data/008-稀释回收与HOOK/std-ep34-table2-multiple-samples.csv",
  show_col_types = FALSE
)

head(ep34_table2, 6)
```

A tibble: 6 x 6

	sample	dilution	fraction	observed_mean	assigned	dilution_factor
	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	neat	1	506	506	1
2	1	1/2	0.5	264.	506	2
3	1	1/4	0.25	120.	506	4
4	1	1/10	0.1	46	506	10
5	1	1/20	0.05	27.2	506	20
6	2	neat	1	598	598	1

```
ep34_table2_result <- dilution_recovery(
  ep34_table2,
  sample = "sample",
  result = "observed_mean",
  dilution_factor = "dilution_factor",
  assigned = "assigned",
  conf.level = 0.95
)

ep34_table2_result
```

Dilution recovery

Samples: 5

Dilution factors: 1, 2, 4, 10, 20

Confidence level: 95.0%

Dilution factor: 1

Specimen content : 100

Samples : 5
 Mean recovery : 100
 SD recovery : 0
 SE recovery : 0
 Degrees of freedom : 4
 CI lower : 100
 CI upper : 100

Dilution factor: 2

Specimen content : 50
 Samples : 5
 Mean recovery : 100.53079
 SD recovery : 3.5222063
 SE recovery : 1.5751786
 Degrees of freedom : 4
 CI lower : 96.157389
 CI upper : 104.90418

Dilution factor: 4

Specimen content : 25
 Samples : 5
 Mean recovery : 102.74826
 SD recovery : 5.3015808
 SE recovery : 2.370939
 Degrees of freedom : 4
 CI lower : 96.165478
 CI upper : 109.33104

Dilution factor: 10

Specimen content : 10
 Samples : 5
 Mean recovery : 101.81449
 SD recovery : 8.2466472
 SE recovery : 3.6880127
 Degrees of freedom : 4
 CI lower : 91.574928
 CI upper : 112.05406

Dilution factor: 20

Specimen content : 5
 Samples : 5
 Mean recovery : 99.785763
 SD recovery : 8.8098638
 SE recovery : 3.9398909
 Degrees of freedom : 4
 CI lower : 88.846872
 CI upper : 110.72465

Sample results

Sample: 1 | Dilution factor: 1

Replicates : 1
 Mean result : 506
 SD result : NA
 CV result : NA
 Specimen content : 100
 Assigned value : 506
 Target concentration : 506

Corrected concentration : 506
Recovery : 100

Sample: 2 | Dilution factor: 1
Replicates : 1
Mean result : 598
SD result : NA
CV result : NA
Specimen content : 100
Assigned value : 598
Target concentration : 598
Corrected concentration : 598
Recovery : 100

Sample: 3 | Dilution factor: 1
Replicates : 1
Mean result : 570.4
SD result : NA
CV result : NA
Specimen content : 100
Assigned value : 570.4
Target concentration : 570.4
Corrected concentration : 570.4
Recovery : 100

Sample: 4 | Dilution factor: 1
Replicates : 1
Mean result : 625.6
SD result : NA
CV result : NA
Specimen content : 100
Assigned value : 625.6
Target concentration : 625.6
Corrected concentration : 625.6
Recovery : 100

Sample: 5 | Dilution factor: 1
Replicates : 1
Mean result : 561.2
SD result : NA
CV result : NA
Specimen content : 100
Assigned value : 561.2
Target concentration : 561.2
Corrected concentration : 561.2
Recovery : 100

Sample: 1 | Dilution factor: 2
Replicates : 1
Mean result : 264.5
SD result : NA
CV result : NA
Specimen content : 50
Assigned value : 506
Target concentration : 253
Corrected concentration : 529
Recovery : 104.54545

Sample: 2 | Dilution factor: 2
Replicates : 1
Mean result : 294.4
SD result : NA
CV result : NA
Specimen content : 50
Assigned value : 598
Target concentration : 299
Corrected concentration : 588.8
Recovery : 98.461538

Sample: 3 | Dilution factor: 2
Replicates : 1
Mean result : 292.1
SD result : NA
CV result : NA
Specimen content : 50
Assigned value : 570.4
Target concentration : 285.2
Corrected concentration : 584.2
Recovery : 102.41935

Sample: 4 | Dilution factor: 2
Replicates : 1
Mean result : 299
SD result : NA
CV result : NA
Specimen content : 50
Assigned value : 625.6
Target concentration : 312.8
Corrected concentration : 598
Recovery : 95.588235

Sample: 5 | Dilution factor: 2
Replicates : 1
Mean result : 285.2
SD result : NA
CV result : NA
Specimen content : 50
Assigned value : 561.2
Target concentration : 280.6
Corrected concentration : 570.4
Recovery : 101.63934

Sample: 1 | Dilution factor: 4
Replicates : 1
Mean result : 119.6
SD result : NA
CV result : NA
Specimen content : 25
Assigned value : 506
Target concentration : 126.5
Corrected concentration : 478.4
Recovery : 94.545455

Sample: 2 | Dilution factor: 4
Replicates : 1
Mean result : 154.1

SD result : NA
CV result : NA
Specimen content : 25
Assigned value : 598
Target concentration : 149.5
Corrected concentration : 616.4
Recovery : 103.07692

Sample: 3 | Dilution factor: 4

Replicates : 1
Mean result : 151.8
SD result : NA
CV result : NA
Specimen content : 25
Assigned value : 570.4
Target concentration : 142.6
Corrected concentration : 607.2
Recovery : 106.45161

Sample: 4 | Dilution factor: 4

Replicates : 1
Mean result : 158.7
SD result : NA
CV result : NA
Specimen content : 25
Assigned value : 625.6
Target concentration : 156.4
Corrected concentration : 634.8
Recovery : 101.47059

Sample: 5 | Dilution factor: 4

Replicates : 1
Mean result : 151.8
SD result : NA
CV result : NA
Specimen content : 25
Assigned value : 561.2
Target concentration : 140.3
Corrected concentration : 607.2
Recovery : 108.19672

Sample: 1 | Dilution factor: 10

Replicates : 1
Mean result : 46
SD result : NA
CV result : NA
Specimen content : 10
Assigned value : 506
Target concentration : 50.6
Corrected concentration : 460
Recovery : 90.909091

Sample: 2 | Dilution factor: 10

Replicates : 1
Mean result : 65.3
SD result : NA
CV result : NA
Specimen content : 10

Assigned value : 598
Target concentration : 59.8
Corrected concentration : 653
Recovery : 109.19732

Sample: 3 | Dilution factor: 10
Replicates : 1
Mean result : 61.6
SD result : NA
CV result : NA
Specimen content : 10
Assigned value : 570.4
Target concentration : 57.04
Corrected concentration : 616
Recovery : 107.99439

Sample: 4 | Dilution factor: 10
Replicates : 1
Mean result : 66.2
SD result : NA
CV result : NA
Specimen content : 10
Assigned value : 625.6
Target concentration : 62.56
Corrected concentration : 662
Recovery : 105.81841

Sample: 5 | Dilution factor: 10
Replicates : 1
Mean result : 53.4
SD result : NA
CV result : NA
Specimen content : 10
Assigned value : 561.2
Target concentration : 56.12
Corrected concentration : 534
Recovery : 95.153243

Sample: 1 | Dilution factor: 20
Replicates : 1
Mean result : 27.2
SD result : NA
CV result : NA
Specimen content : 5
Assigned value : 506
Target concentration : 25.3
Corrected concentration : 544
Recovery : 107.50988

Sample: 2 | Dilution factor: 20
Replicates : 1
Mean result : 27
SD result : NA
CV result : NA
Specimen content : 5
Assigned value : 598
Target concentration : 29.9
Corrected concentration : 540

Recovery : 90.301003

Sample: 3 | Dilution factor: 20

Replicates : 1
Mean result : 31.3
SD result : NA
CV result : NA
Specimen content : 5
Assigned value : 570.4
Target concentration : 28.52
Corrected concentration : 626
Recovery : 109.74755

Sample: 4 | Dilution factor: 20

Replicates : 1
Mean result : 31.1
SD result : NA
CV result : NA
Specimen content : 5
Assigned value : 625.6
Target concentration : 31.28
Corrected concentration : 622
Recovery : 99.424552

Sample: 5 | Dilution factor: 20

Replicates : 1
Mean result : 25.8
SD result : NA
CV result : NA
Specimen content : 5
Assigned value : 561.2
Target concentration : 28.06
Corrected concentration : 516
Recovery : 91.94583

```
plot(ep34_table2_result, type = "summary")
```

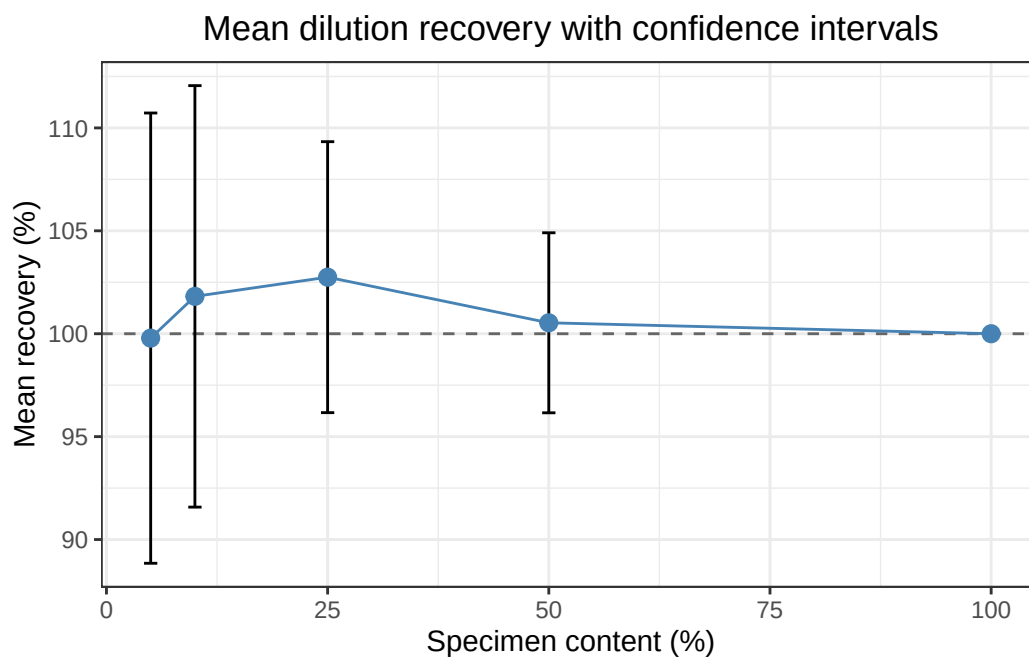


Table 8.2: EP34 Table 2: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
neat	mean_recovery	100.00	100.000
neat	lower_ci	100.00	100.000
neat	upper_ci	100.00	100.000
1/2	mean_recovery	100.53	100.531
1/2	lower_ci	96.16	96.157
1/2	upper_ci	104.90	104.904
1/4	mean_recovery	102.75	102.748
1/4	lower_ci	96.17	96.165
1/4	upper_ci	109.33	109.331
1/10	mean_recovery	101.82	101.814
1/10	lower_ci	91.58	91.575
1/10	upper_ci	112.06	112.054
1/20	mean_recovery	99.76	99.786
1/20	lower_ci	88.76	88.847
1/20	upper_ci	110.77	110.725

The 95% CIs for the 1/2 and 1/4 dilutions lie within the example' s 90%–110% range; the intervals for 1/10 and 1/20 cross that range. Small differences from the source arise from display rounding of sample targets and intermediate statistics.

8.4.3 CLSI EP34 Table 3: Basic Spike Recovery

Table 3 adds 0.1 mL of a 40 mg/dL calcium spike solution, or an equal volume of pure solvent, to three serum samples; each condition is measured twice. The source displays the theoretical added concentration in the final mixture as 3.64 mg/dL and uses that displayed value for recovery.

```
ep34_table3_added <- spike_concentration(
  sample_concentration = 0,
  sample_volume = 1,
  stock_concentration = 40,
  spike_volume = 0.1
)
ep34_table3_added
```

```
Spike concentration
Sample concentration: 0
Stock concentration : 40
Added concentration : 3.6363636
Final concentration : 3.6363636
Sample volume      : 1
Spike volume       : 0.1
Total volume       : 1.1
Spike fraction      : 0.090909091
```

```
ep34_table3 <- read_csv(
  "/data/008-稀释回收与H00K/std-ep34-table3-spike-recovery.csv",
  show_col_types = FALSE
)
head(ep34_table3, 6)
```

```
# A tibble: 6 x 4
```

	sample	condition	replicate	result
	<dbl>	<chr>	<dbl>	<dbl>
1	1	solvent	1	9.7
2	1	solvent	2	9.9
3	1	analyte	1	12.9
4	1	analyte	2	13.3
5	2	solvent	1	9.5
6	2	solvent	2	9.5

```
ep34_table3_result <- spike_recovery(
  ep34_table3,
  sample = "sample",
  result = "result",
  condition = "condition",
  analyte_level = "analyte",
  solvent_level = "solvent",
  added_concentration = 3.64,
  conf.level = 0.95
)
ep34_table3_result
```

Spike recovery

Samples: 3

Confidence level: 95.0%

```
Samples      : 3
Mean recovery : 90.659341
SD of recovery : 2.7472527
SE of recovery : 1.5861271
Degrees of freedom : 2
CI lower      : 83.834787
CI upper      : 97.483895
```

Sample results

Sample: 1

```
Analyte replicates : 2
Analyte mean       : 13.1
Analyte SD         : 0.28284271
Solvent replicates : 2
Solvent mean       : 9.8
Solvent SD         : 0.14142136
Observed increase  : 3.3
Added concentration : 3.64
Recovery           : 90.659341
```

Sample: 2

```
Analyte replicates : 2
Analyte mean       : 12.7
Analyte SD         : 0.14142136
Solvent replicates : 2
Solvent mean       : 9.5
Solvent SD         : 0
Observed increase  : 3.2
Added concentration : 3.64
Recovery           : 87.912088
```


Sample: 3
Analyte replicates : 2
Analyte mean : 12.3
Analyte SD : 0.14142136
Solvent replicates : 2
Solvent mean : 8.9
Solvent SD : 0.28284271
Observed increase : 3.4
Added concentration : 3.64
Recovery : 93.406593

```
plot(ep34_table3_result)
```

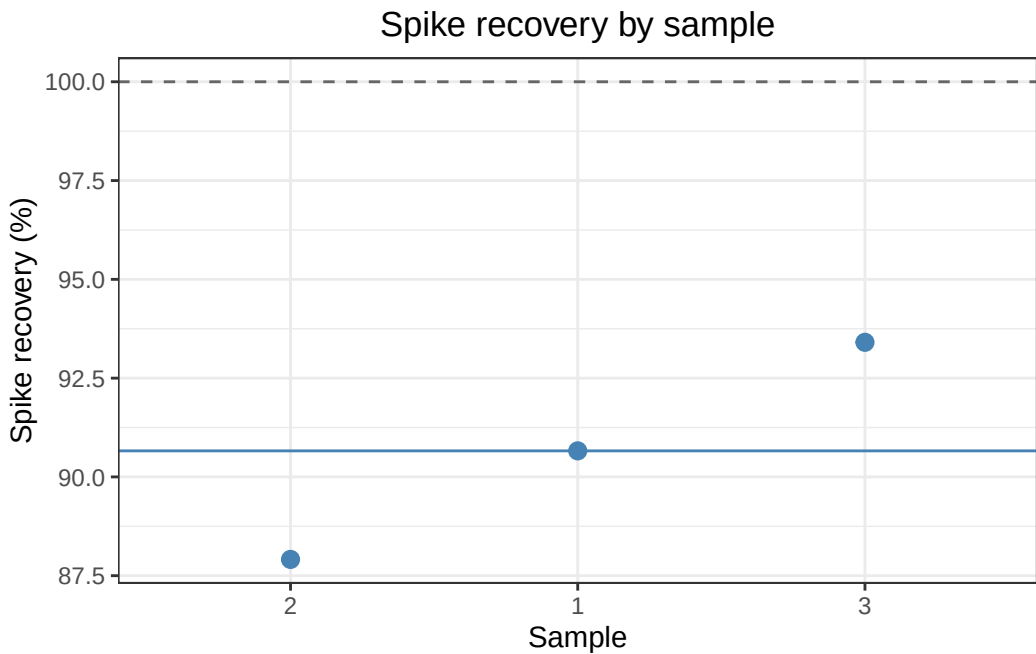


Table 8.3: EP34 Table 3: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
1	recovery	90.7	90.659
2	recovery	88.0	87.912
3	recovery	93.4	93.407
all	mean_recovery	90.7	90.659

The recovery values for the three samples are approximately 90.7%, 87.9%, and 93.4%, with a across-sample mean of approximately 90.7%. The standard table displays one decimal place, and its intermediate rounding is not completely uniform; the verification tolerance is therefore 0.1 percentage points. Formal analyses should use unrounded theoretical additions and measurement results.

8.4.4 CLSI EP34 Table 4: Single-Sample Multiple-Concentration Spiking

Table 4 mixes 100 μ L of a patient sample at 1,000 ng/mL with 900 μ L of spike solutions at different concentrations. `spike_concentration()` recalculates the mass-balance concentration; because the source does not provide matched solvent-control replicates, this case does not call `spike_recovery()`.

```

ep34_table4 <- read_csv(
  "/data/008-稀释回收与HOOK/std-ep34-table4-single-spike.csv",
  show_col_types = FALSE
)
ep34_table4_balance <- spike_concentration(
  sample_concentration = 1000,
  sample_volume = 0.1,
  stock_concentration = ep34_table4$spike_concentration,
  spike_volume = 0.9
)
ep34_table4$computed_ivdtools <- ep34_table4_balance$final_concentration
ep34_table4$recovery_ivdtools <-
  100 * ep34_table4$measured_concentration / ep34_table4$computed_ivdtools
ep34_table4

# A tibble: 5 x 6
  spike_concentration computed_concentration measured_concentration
      <dbl>              <dbl>              <dbl>
1         0             100                94
2        10             109                99
3       100             190               181
4      1000            1000               917
5     10000            9100              9024
# i 3 more variables: standard_recovery <dbl>, computed_ivdtools <dbl>,
#   recovery_ivdtools <dbl>

```

Table 8.4: EP34 Table 4: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
0	computed_concentration	100	100.000
0	recovery	94	94.000
10	computed_concentration	109	109.000
10	recovery	91	90.826
100	computed_concentration	190	190.000
100	recovery	95	95.263
1000	computed_concentration	1000	1000.000
1000	recovery	92	91.700
10000	computed_concentration	9100	9100.000
10000	recovery	99	99.165

The sample prepared from the 10,000 ng/mL spike solution has a theoretical concentration of 9,100 ng/mL, above the example AMI, and therefore requires measurement through an established EMI.

8.4.5 CLSI EP34 Tables 5–6: Manual Dilution-Procedure Validation

This case uses five patient samples assigned by a reference measurement procedure. Two operators independently prepare two 1/20 dilution samples each, and each preparation is measured twice, giving eight dilution results per original sample.

```

ep34_table5 <- read_csv(
  "/data/008-稀释回收与HOOK/std-ep34-table5-manual-validation.csv",
  show_col_types = FALSE
)
head(ep34_table5, 6)

```

```
# A tibble: 6 x 7
  sample assigned scientist preparation replicate result dilution_factor
  <dbl>   <dbl>   <dbl>       <dbl>     <dbl>   <dbl>       <dbl>
1     1     140.     1         1         1     6.99         20
2     1     140.     1         1         2     6.51         20
3     1     140.     1         2         1     6.37         20
4     1     140.     1         2         2     6.12         20
5     1     140.     2         1         1     6.53         20
6     1     140.     2         1         2     6.74         20
```

```
ep34_table5_result <- dilution_recovery(
  ep34_table5,
  sample = "sample",
  result = "result",
  dilution_factor = "dilution_factor",
  assigned = "assigned",
  conf.level = 0.95
)
ep34_table5_result
```

Dilution recovery

Samples: 5
Dilution factors: 20
Confidence level: 95.0%

Dilution factor: 20

Specimen content : 5
Samples : 5
Mean recovery : 98.761501
SD recovery : 4.0632964
SE recovery : 1.8171614
Degrees of freedom : 4
CI lower : 93.716252
CI upper : 103.80675

Sample results

Sample: 1 | Dilution factor: 20

Replicates : 8
Mean result : 6.65625
SD result : 0.32845689
CV result : 4.9345636
Specimen content : 5
Assigned value : 139.68
Target concentration : 6.984
Corrected concentration : 133.125
Recovery : 95.307131

Sample: 2 | Dilution factor: 20

Replicates : 8
Mean result : 15.23875
SD result : 0.67979908
CV result : 4.4609898
Specimen content : 5
Assigned value : 292.04
Target concentration : 14.602
Corrected concentration : 304.775

Recovery : 104.3607

Sample: 3 | Dilution factor: 20

Replicates : 8
Mean result : 31.73875
SD result : 2.1664879
CV result : 6.8260025
Specimen content : 5
Assigned value : 647.32
Target concentration : 32.366
Corrected concentration : 634.775
Recovery : 98.06201

Sample: 4 | Dilution factor: 20

Replicates : 8
Mean result : 53.6675
SD result : 1.979623
CV result : 3.6886812
Specimen content : 5
Assigned value : 1132.45
Target concentration : 56.6225
Corrected concentration : 1073.35
Recovery : 94.781227

Sample: 5 | Dilution factor: 20

Replicates : 8
Mean result : 92.41375
SD result : 2.4947198
CV result : 2.6995115
Specimen content : 5
Assigned value : 1824.62
Target concentration : 91.231
Corrected concentration : 1848.275
Recovery : 101.29643

```
plot(ep34_table5_result, type = "observed")
```

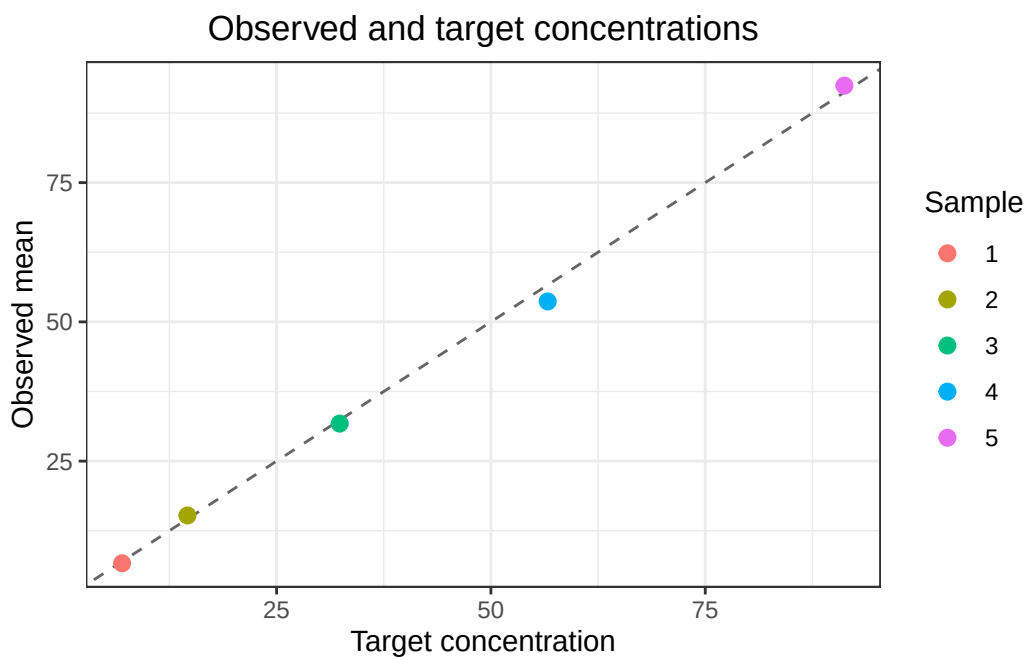


Table 8.5: EP34 Tables 5–6: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
1	mean	6.66	6.656
1	corrected_concentration	133.13	133.125
1	recovery	95.30	95.307
2	mean	15.24	15.239
2	corrected_concentration	304.78	304.775
2	recovery	104.40	104.361
3	mean	31.74	31.739
3	corrected_concentration	634.78	634.775
3	recovery	98.10	98.062
4	mean	53.67	53.667
4	corrected_concentration	1073.35	1073.350
4	recovery	94.80	94.781
5	mean	92.41	92.414
5	corrected_concentration	1848.28	1848.275
5	recovery	101.30	101.296

The dilution recoveries for the five samples range from 94.8% to 104.4%, all within the example's prespecified 90%–110% range. Here, assigned is the value assigned by the reference measurement procedure.

8.4.6 CLSI EP34 Tables 7–8: Automatic Dilution-Procedure Validation

The AMI in this case is 5–40 pg/mL, and the proposed EMI is 40–80 pg/mL. Five samples characterized by spike recovery are measured four times on each of two instruments after an automatic 1/2 dilution.

```
ep34_table7 <- read_csv(
  "./data/008-稀释回收与H00K/std-ep34-table7-auto-validation.csv",
  show_col_types = FALSE
)
head(ep34_table7, 6)
```

A tibble: 6 x 6

	sample	assigned	instrument	replicate	result	dilution_factor
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	44.9	1	1	20.3	2
2	1	44.9	1	2	26.0	2
3	1	44.9	1	3	24.3	2
4	1	44.9	1	4	25.0	2
5	1	44.9	2	1	22.5	2
6	1	44.9	2	2	21.3	2

```
ep34_table7_result <- dilution_recovery(
  ep34_table7,
  sample = "sample",
  result = "result",
  dilution_factor = "dilution_factor",
  assigned = "assigned",
  conf.level = 0.95
)
ep34_table7_result
```

Dilution recovery

Samples: 5
Dilution factors: 2
Confidence level: 95.0%

Dilution factor: 2

Specimen content : 50
Samples : 5
Mean recovery : 104.55585
SD recovery : 6.022583
SE recovery : 2.693381
Degrees of freedom : 4
CI lower : 97.077824
CI upper : 112.03387

Sample results

Sample: 1 | Dilution factor: 2

Replicates : 8
Mean result : 23.6775
SD result : 2.1002432
CV result : 8.8702067
Specimen content : 50
Assigned value : 44.92
Target concentration : 22.46
Corrected concentration : 47.355
Recovery : 105.42075

Sample: 2 | Dilution factor: 2

Replicates : 8
Mean result : 26.44
SD result : 1.9634299
CV result : 7.4259831
Specimen content : 50
Assigned value : 53.78
Target concentration : 26.89
Corrected concentration : 52.88
Recovery : 98.326515

Sample: 3 | Dilution factor: 2

Replicates : 8
Mean result : 35.44125
SD result : 1.9058139
CV result : 5.3773892
Specimen content : 50
Assigned value : 62.32
Target concentration : 31.16
Corrected concentration : 70.8825
Recovery : 113.73957

Sample: 4 | Dilution factor: 2

Replicates : 8
Mean result : 35.975
SD result : 1.3997653
CV result : 3.8909389
Specimen content : 50
Assigned value : 68.34
Target concentration : 34.17

```
Corrected concentration : 71.95
Recovery                : 105.28241

Sample: 5 | Dilution factor: 2
Replicates              : 8
Mean result             : 37.50875
SD result               : 1.5492066
CV result               : 4.1302539
Specimen content        : 50
Assigned value          : 75.01
Target concentration    : 37.505
Corrected concentration : 75.0175
Recovery                : 100.01
```

Table 8.6: EP34 Tables 7–8: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
1	mean	23.68	23.678
1	corrected_concentration	47.36	47.355
1	recovery	105.40	105.421
2	mean	26.44	26.440
2	corrected_concentration	52.88	52.880
2	recovery	98.30	98.327
3	mean	35.44	35.441
3	corrected_concentration	70.88	70.882
3	recovery	113.70	113.740
4	mean	35.98	35.975
4	corrected_concentration	71.95	71.950
4	recovery	105.30	105.282
5	mean	37.51	37.509
5	corrected_concentration	75.02	75.018
5	recovery	100.00	100.010

The recoveries for the five samples range from 98.3% to 113.7%, all within the example’s prespecified 85%–115% range.

8.4.7 CLSI EP34 Appendix C: Two Complete Dilution Analyses

Appendix C Tables C1 and C2 publish only the observed mean for each sample at each dilution level, not the underlying individual results. This section follows the source granularity and recalculates across-sample recovery for analytes B and C separately.

Table C1: Analyte B

```
ep34_c1 <- read_csv(
  "/data/008-稀释回收与H00K/std-ep34-appendix-c1.csv",
  show_col_types = FALSE
)
head(ep34_c1, 6)

# A tibble: 6 x 5
  sample dilution dilution_factor assigned observed_mean
  <dbl> <chr>          <dbl>    <dbl>         <dbl>
1     1 neat              1    220.         220.
2     1 1/2              2    220.         115.
3     1 1/4              4    220.          58.2
4     1 1/10             10    220.          25.6
```

5	1 1/20	20	220.	15.7
6	2 neat	1	260.	260.

```

ep34_c1_result <- dilution_recovery(
  ep34_c1,
  sample = "sample",
  result = "observed_mean",
  dilution_factor = "dilution_factor",
  assigned = "assigned",
  conf.level = 0.95
)
ep34_c1_result

```

Dilution recovery

Samples: 5

Dilution factors: 1, 2, 4, 10, 20

Confidence level: 95.0%

Dilution factor: 1

```

Specimen content : 100
Samples          : 5
Mean recovery    : 100
SD recovery      : 0
SE recovery      : 0
Degrees of freedom : 4
CI lower         : 100
CI upper         : 100

```

Dilution factor: 2

```

Specimen content : 50
Samples          : 5
Mean recovery    : 100.49583
SD recovery      : 3.5627839
SE recovery      : 1.5933254
Degrees of freedom : 4
CI lower         : 96.072052
CI upper         : 104.91961

```

Dilution factor: 4

```

Specimen content : 25
Samples          : 5
Mean recovery    : 104.98711
SD recovery      : 2.8223671
SE recovery      : 1.2622009
Degrees of freedom : 4
CI lower         : 101.48268
CI upper         : 108.49154

```

Dilution factor: 10

```

Specimen content : 10
Samples          : 5
Mean recovery    : 115.13247
SD recovery      : 3.1142636
SE recovery      : 1.392741
Degrees of freedom : 4
CI lower         : 111.2656
CI upper         : 118.99934

```


Dilution factor: 20

Specimen content : 5
Samples : 5
Mean recovery : 135.48746
SD recovery : 8.945831
SE recovery : 4.0006973
Degrees of freedom : 4
CI lower : 124.37974
CI upper : 146.59517

Sample results

Sample: 1 | Dilution factor: 1

Replicates : 1
Mean result : 220.3
SD result : NA
CV result : NA
Specimen content : 100
Assigned value : 220.3
Target concentration : 220.3
Corrected concentration : 220.3
Recovery : 100

Sample: 2 | Dilution factor: 1

Replicates : 1
Mean result : 260.1
SD result : NA
CV result : NA
Specimen content : 100
Assigned value : 260.1
Target concentration : 260.1
Corrected concentration : 260.1
Recovery : 100

Sample: 3 | Dilution factor: 1

Replicates : 1
Mean result : 248
SD result : NA
CV result : NA
Specimen content : 100
Assigned value : 248
Target concentration : 248
Corrected concentration : 248
Recovery : 100

Sample: 4 | Dilution factor: 1

Replicates : 1
Mean result : 272.2
SD result : NA
CV result : NA
Specimen content : 100
Assigned value : 272.2
Target concentration : 272.2
Corrected concentration : 272.2
Recovery : 100

Sample: 5 | Dilution factor: 1

Replicates : 1
Mean result : 244
SD result : NA
CV result : NA
Specimen content : 100
Assigned value : 244
Target concentration : 244
Corrected concentration : 244
Recovery : 100

Sample: 1 | Dilution factor: 2

Replicates : 1
Mean result : 114.8
SD result : NA
CV result : NA
Specimen content : 50
Assigned value : 220.3
Target concentration : 110.15
Corrected concentration : 229.6
Recovery : 104.22152

Sample: 2 | Dilution factor: 2

Replicates : 1
Mean result : 128.2
SD result : NA
CV result : NA
Specimen content : 50
Assigned value : 260.1
Target concentration : 130.05
Corrected concentration : 256.4
Recovery : 98.57747

Sample: 3 | Dilution factor: 2

Replicates : 1
Mean result : 127.3
SD result : NA
CV result : NA
Specimen content : 50
Assigned value : 248
Target concentration : 124
Corrected concentration : 254.6
Recovery : 102.66129

Sample: 4 | Dilution factor: 2

Replicates : 1
Mean result : 129.7
SD result : NA
CV result : NA
Specimen content : 50
Assigned value : 272.2
Target concentration : 136.1
Corrected concentration : 259.4
Recovery : 95.297575

Sample: 5 | Dilution factor: 2

Replicates : 1
Mean result : 124.1
SD result : NA

CV result : NA
Specimen content : 50
Assigned value : 244
Target concentration : 122
Corrected concentration : 248.2
Recovery : 101.72131

Sample: 1 | Dilution factor: 4
Replicates : 1
Mean result : 58.2
SD result : NA
CV result : NA
Specimen content : 25
Assigned value : 220.3
Target concentration : 55.075
Corrected concentration : 232.8
Recovery : 105.67408

Sample: 2 | Dilution factor: 4
Replicates : 1
Mean result : 67.1
SD result : NA
CV result : NA
Specimen content : 25
Assigned value : 260.1
Target concentration : 65.025
Corrected concentration : 268.4
Recovery : 103.19108

Sample: 3 | Dilution factor: 4
Replicates : 1
Mean result : 66.3
SD result : NA
CV result : NA
Specimen content : 25
Assigned value : 248
Target concentration : 62
Corrected concentration : 265.2
Recovery : 106.93548

Sample: 4 | Dilution factor: 4
Replicates : 1
Mean result : 68.8
SD result : NA
CV result : NA
Specimen content : 25
Assigned value : 272.2
Target concentration : 68.05
Corrected concentration : 275.2
Recovery : 101.10213

Sample: 5 | Dilution factor: 4
Replicates : 1
Mean result : 65.9
SD result : NA
CV result : NA
Specimen content : 25
Assigned value : 244

Target concentration : 61
Corrected concentration : 263.6
Recovery : 108.03279

Sample: 1 | Dilution factor: 10
Replicates : 1
Mean result : 25.6
SD result : NA
CV result : NA
Specimen content : 10
Assigned value : 220.3
Target concentration : 22.03
Corrected concentration : 256
Recovery : 116.20517

Sample: 2 | Dilution factor: 10
Replicates : 1
Mean result : 28.8
SD result : NA
CV result : NA
Specimen content : 10
Assigned value : 260.1
Target concentration : 26.01
Corrected concentration : 288
Recovery : 110.72664

Sample: 3 | Dilution factor: 10
Replicates : 1
Mean result : 29.6
SD result : NA
CV result : NA
Specimen content : 10
Assigned value : 248
Target concentration : 24.8
Corrected concentration : 296
Recovery : 119.35484

Sample: 4 | Dilution factor: 10
Replicates : 1
Mean result : 31.2
SD result : NA
CV result : NA
Specimen content : 10
Assigned value : 272.2
Target concentration : 27.22
Corrected concentration : 312
Recovery : 114.6216

Sample: 5 | Dilution factor: 10
Replicates : 1
Mean result : 28
SD result : NA
CV result : NA
Specimen content : 10
Assigned value : 244
Target concentration : 24.4
Corrected concentration : 280
Recovery : 114.7541

Sample: 1 | Dilution factor: 20
 Replicates : 1
 Mean result : 15.7
 SD result : NA
 CV result : NA
 Specimen content : 5
 Assigned value : 220.3
 Target concentration : 11.015
 Corrected concentration : 314
 Recovery : 142.53291

Sample: 2 | Dilution factor: 20
 Replicates : 1
 Mean result : 17.4
 SD result : NA
 CV result : NA
 Specimen content : 5
 Assigned value : 260.1
 Target concentration : 13.005
 Corrected concentration : 348
 Recovery : 133.79469

Sample: 3 | Dilution factor: 20
 Replicates : 1
 Mean result : 15.6
 SD result : NA
 CV result : NA
 Specimen content : 5
 Assigned value : 248
 Target concentration : 12.4
 Corrected concentration : 312
 Recovery : 125.80645

Sample: 4 | Dilution factor: 20
 Replicates : 1
 Mean result : 17.5
 SD result : NA
 CV result : NA
 Specimen content : 5
 Assigned value : 272.2
 Target concentration : 13.61
 Corrected concentration : 350
 Recovery : 128.58193

Sample: 5 | Dilution factor: 20
 Replicates : 1
 Mean result : 17.9
 SD result : NA
 CV result : NA
 Specimen content : 5
 Assigned value : 244
 Target concentration : 12.2
 Corrected concentration : 358
 Recovery : 146.72131

```
plot(ep34_c1_result, type = "summary")
```

Mean dilution recovery with confidence intervals

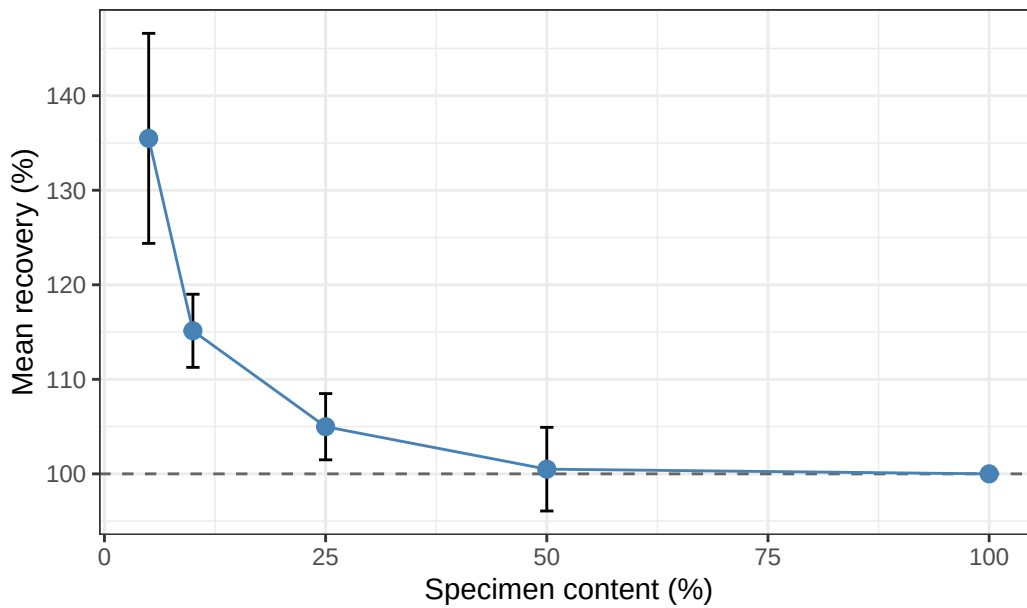


Table C2: Analyte C

```
ep34_c2 <- read_csv(
  "/data/008-稀释回收与HOOK/std-ep34-appendix-c2.csv",
  show_col_types = FALSE
)
head(ep34_c2, 6)
```

A tibble: 6 x 5

	sample	dilution	dilution_factor	assigned	observed_mean
	<dbl>	<chr>	<dbl>	<dbl>	<dbl>
1	1	neat	1	1492.	1492.
2	1	1/2	2	1492.	751.
3	1	1/10	10	1492.	145.
4	1	1/50	50	1492.	28.7
5	1	1/200	200	1492.	7.5
6	2	neat	1	1386.	1386.

```
ep34_c2_result <- dilution_recovery(
  ep34_c2,
  sample = "sample",
  result = "observed_mean",
  dilution_factor = "dilution_factor",
  assigned = "assigned",
  conf.level = 0.95
)
ep34_c2_result
```

Dilution recovery

Samples: 6

Dilution factors: 1, 2, 10, 50, 200

Confidence level: 95.0%

Dilution factor: 1

Specimen content : 100

Samples : 6

Mean recovery : 100
 SD recovery : 0
 SE recovery : 0
 Degrees of freedom : 5
 CI lower : 100
 CI upper : 100

Dilution factor: 2

Specimen content : 50
 Samples : 6
 Mean recovery : 101.33362
 SD recovery : 3.2217773
 SE recovery : 1.3152851
 Degrees of freedom : 5
 CI lower : 97.952571
 CI upper : 104.71467

Dilution factor: 10

Specimen content : 10
 Samples : 6
 Mean recovery : 101.55253
 SD recovery : 5.9523994
 SE recovery : 2.4300569
 Degrees of freedom : 5
 CI lower : 95.305866
 CI upper : 107.79919

Dilution factor: 50

Specimen content : 2
 Samples : 6
 Mean recovery : 102.34874
 SD recovery : 5.2080713
 SE recovery : 2.1261862
 Degrees of freedom : 5
 CI lower : 96.883208
 CI upper : 107.81428

Dilution factor: 200

Specimen content : 0.5
 Samples : 6
 Mean recovery : 102.55094
 SD recovery : 5.5420679
 SE recovery : 2.2625397
 Degrees of freedom : 5
 CI lower : 96.734901
 CI upper : 108.36699

Sample results

Sample: 1 | Dilution factor: 1

Replicates : 1
 Mean result : 1492.1
 SD result : NA
 CV result : NA
 Specimen content : 100
 Assigned value : 1492.1
 Target concentration : 1492.1
 Corrected concentration : 1492.1

Recovery : 100

Sample: 2 | Dilution factor: 1

Replicates : 1
Mean result : 1386.5
SD result : NA
CV result : NA
Specimen content : 100
Assigned value : 1386.5
Target concentration : 1386.5
Corrected concentration : 1386.5
Recovery : 100

Sample: 3 | Dilution factor: 1

Replicates : 1
Mean result : 1290.8
SD result : NA
CV result : NA
Specimen content : 100
Assigned value : 1290.8
Target concentration : 1290.8
Corrected concentration : 1290.8
Recovery : 100

Sample: 4 | Dilution factor: 1

Replicates : 1
Mean result : 1451.6
SD result : NA
CV result : NA
Specimen content : 100
Assigned value : 1451.6
Target concentration : 1451.6
Corrected concentration : 1451.6
Recovery : 100

Sample: 5 | Dilution factor: 1

Replicates : 1
Mean result : 1209.8
SD result : NA
CV result : NA
Specimen content : 100
Assigned value : 1209.8
Target concentration : 1209.8
Corrected concentration : 1209.8
Recovery : 100

Sample: 6 | Dilution factor: 1

Replicates : 1
Mean result : 1326.4
SD result : NA
CV result : NA
Specimen content : 100
Assigned value : 1326.4
Target concentration : 1326.4
Corrected concentration : 1326.4
Recovery : 100

Sample: 1 | Dilution factor: 2

Replicates : 1
Mean result : 750.9
SD result : NA
CV result : NA
Specimen content : 50
Assigned value : 1492.1
Target concentration : 746.05
Corrected concentration : 1501.8
Recovery : 100.65009

Sample: 2 | Dilution factor: 2
Replicates : 1
Mean result : 680.9
SD result : NA
CV result : NA
Specimen content : 50
Assigned value : 1386.5
Target concentration : 693.25
Corrected concentration : 1361.8
Recovery : 98.218536

Sample: 3 | Dilution factor: 2
Replicates : 1
Mean result : 648.3
SD result : NA
CV result : NA
Specimen content : 50
Assigned value : 1290.8
Target concentration : 645.4
Corrected concentration : 1296.6
Recovery : 100.44933

Sample: 4 | Dilution factor: 2
Replicates : 1
Mean result : 745.2
SD result : NA
CV result : NA
Specimen content : 50
Assigned value : 1451.6
Target concentration : 725.8
Corrected concentration : 1490.4
Recovery : 102.67291

Sample: 5 | Dilution factor: 2
Replicates : 1
Mean result : 598.3
SD result : NA
CV result : NA
Specimen content : 50
Assigned value : 1209.8
Target concentration : 604.9
Corrected concentration : 1196.6
Recovery : 98.908911

Sample: 6 | Dilution factor: 2
Replicates : 1
Mean result : 710.3
SD result : NA

CV result : NA
Specimen content : 50
Assigned value : 1326.4
Target concentration : 663.2
Corrected concentration : 1420.6
Recovery : 107.10193

Sample: 1 | Dilution factor: 10
Replicates : 1
Mean result : 145.4
SD result : NA
CV result : NA
Specimen content : 10
Assigned value : 1492.1
Target concentration : 149.21
Corrected concentration : 1454
Recovery : 97.446552

Sample: 2 | Dilution factor: 10
Replicates : 1
Mean result : 135.8
SD result : NA
CV result : NA
Specimen content : 10
Assigned value : 1386.5
Target concentration : 138.65
Corrected concentration : 1358
Recovery : 97.944464

Sample: 3 | Dilution factor: 10
Replicates : 1
Mean result : 135.7
SD result : NA
CV result : NA
Specimen content : 10
Assigned value : 1290.8
Target concentration : 129.08
Corrected concentration : 1357
Recovery : 105.1286

Sample: 4 | Dilution factor: 10
Replicates : 1
Mean result : 136.5
SD result : NA
CV result : NA
Specimen content : 10
Assigned value : 1451.6
Target concentration : 145.16
Corrected concentration : 1365
Recovery : 94.034169

Sample: 5 | Dilution factor: 10
Replicates : 1
Mean result : 127.2
SD result : NA
CV result : NA
Specimen content : 10
Assigned value : 1209.8

Target concentration : 120.98
Corrected concentration : 1272
Recovery : 105.14135

Sample: 6 | Dilution factor: 10
Replicates : 1
Mean result : 145.4
SD result : NA
CV result : NA
Specimen content : 10
Assigned value : 1326.4
Target concentration : 132.64
Corrected concentration : 1454
Recovery : 109.62002

Sample: 1 | Dilution factor: 50
Replicates : 1
Mean result : 28.7
SD result : NA
CV result : NA
Specimen content : 2
Assigned value : 1492.1
Target concentration : 29.842
Corrected concentration : 1435
Recovery : 96.173179

Sample: 2 | Dilution factor: 50
Replicates : 1
Mean result : 28.5
SD result : NA
CV result : NA
Specimen content : 2
Assigned value : 1386.5
Target concentration : 27.73
Corrected concentration : 1425
Recovery : 102.77678

Sample: 3 | Dilution factor: 50
Replicates : 1
Mean result : 26.8
SD result : NA
CV result : NA
Specimen content : 2
Assigned value : 1290.8
Target concentration : 25.816
Corrected concentration : 1340
Recovery : 103.81159

Sample: 4 | Dilution factor: 50
Replicates : 1
Mean result : 27.9
SD result : NA
CV result : NA
Specimen content : 2
Assigned value : 1451.6
Target concentration : 29.032
Corrected concentration : 1395
Recovery : 96.100854

Sample: 5 | Dilution factor: 50

Replicates : 1
Mean result : 25.9
SD result : NA
CV result : NA
Specimen content : 2
Assigned value : 1209.8
Target concentration : 24.196
Corrected concentration : 1295
Recovery : 107.04249

Sample: 6 | Dilution factor: 50

Replicates : 1
Mean result : 28.7
SD result : NA
CV result : NA
Specimen content : 2
Assigned value : 1326.4
Target concentration : 26.528
Corrected concentration : 1435
Recovery : 108.18758

Sample: 1 | Dilution factor: 200

Replicates : 1
Mean result : 7.5
SD result : NA
CV result : NA
Specimen content : 0.5
Assigned value : 1492.1
Target concentration : 7.4605
Corrected concentration : 1500
Recovery : 100.52946

Sample: 2 | Dilution factor: 200

Replicates : 1
Mean result : 7.2
SD result : NA
CV result : NA
Specimen content : 0.5
Assigned value : 1386.5
Target concentration : 6.9325
Corrected concentration : 1440
Recovery : 103.85864

Sample: 3 | Dilution factor: 200

Replicates : 1
Mean result : 6.4
SD result : NA
CV result : NA
Specimen content : 0.5
Assigned value : 1290.8
Target concentration : 6.454
Corrected concentration : 1280
Recovery : 99.16331

Sample: 4 | Dilution factor: 200

Replicates : 1

Mean result : 7.1
SD result : NA
CV result : NA
Specimen content : 0.5
Assigned value : 1451.6
Target concentration : 7.258
Corrected concentration : 1420
Recovery : 97.823092

Sample: 5 | Dilution factor: 200
Replicates : 1
Mean result : 6.1
SD result : NA
CV result : NA
Specimen content : 0.5
Assigned value : 1209.8
Target concentration : 6.049
Corrected concentration : 1220
Recovery : 100.84311

Sample: 6 | Dilution factor: 200
Replicates : 1
Mean result : 7.5
SD result : NA
CV result : NA
Specimen content : 0.5
Assigned value : 1326.4
Target concentration : 6.632
Corrected concentration : 1500
Recovery : 113.08806

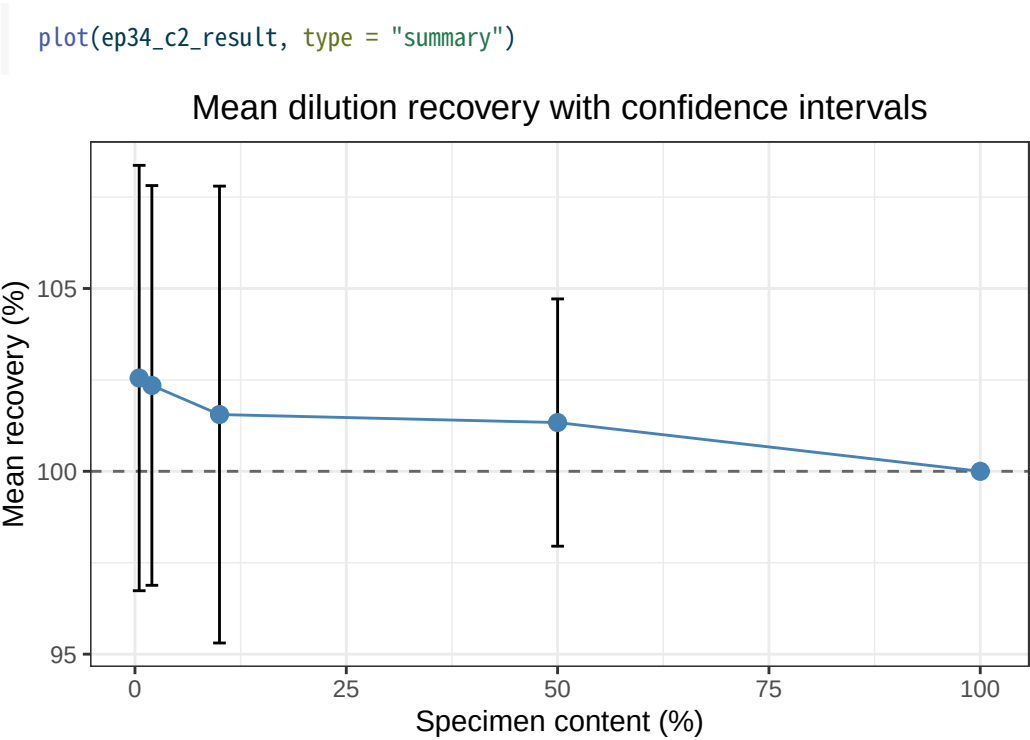


Table 8.7: EP34 Appendix C: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
appendix_c1 / neat	mean_recovery	100.00	100.000
appendix_c1 / neat	lower_ci	100.00	100.000
appendix_c1 / neat	upper_ci	100.00	100.000
appendix_c1 / 1/2	mean_recovery	100.50	100.496
appendix_c1 / 1/2	lower_ci	96.06	96.072
appendix_c1 / 1/2	upper_ci	104.89	104.920
appendix_c1 / 1/4	mean_recovery	104.99	104.987
appendix_c1 / 1/4	lower_ci	101.45	101.483
appendix_c1 / 1/4	upper_ci	108.50	108.492
appendix_c1 / 1/10	mean_recovery	115.13	115.132
appendix_c1 / 1/10	lower_ci	111.33	111.266
appendix_c1 / 1/10	upper_ci	119.05	118.999
appendix_c1 / 1/20	mean_recovery	135.49	135.487
appendix_c1 / 1/20	lower_ci	124.43	124.380
appendix_c1 / 1/20	upper_ci	146.69	146.595
appendix_c2 / neat	mean_recovery	100.00	100.000
appendix_c2 / neat	lower_ci	100.00	100.000
appendix_c2 / neat	upper_ci	100.00	100.000
appendix_c2 / 1/2	mean_recovery	101.33	101.334
appendix_c2 / 1/2	lower_ci	97.95	97.953
appendix_c2 / 1/2	upper_ci	104.71	104.715
appendix_c2 / 1/10	mean_recovery	101.55	101.553
appendix_c2 / 1/10	lower_ci	95.31	95.306
appendix_c2 / 1/10	upper_ci	107.80	107.799
appendix_c2 / 1/50	mean_recovery	102.35	102.349
appendix_c2 / 1/50	lower_ci	96.88	96.883
appendix_c2 / 1/50	upper_ci	107.81	107.814
appendix_c2 / 1/200	mean_recovery	102.71	102.551
appendix_c2 / 1/200	lower_ci	96.82	96.735
appendix_c2 / 1/200	upper_ci	108.60	108.367

Recovery in Table C1 rises clearly as dilution increases; the 1/10 and 1/20 dilutions do not meet the example requirement. Table C2 remains relatively stable even at 1/200. At the highest dilution in Table C2, display rounding of the published one-decimal observed means produces a difference of up to approximately 0.23 percentage points in the software-recalculated summary.

8.4.8 High-Dose Hook Effect

The EP34 hook-effect figure provides an illustrative trajectory rather than raw responses at each concentration. The following teaching data demonstrate `hook_effect()`; these values are not from the standard text.

```
hook_data <- expand.grid(
  concentration = c(1, 2, 3, 4, 5, 6, 7, 8),
  replicate = 1:3,
  KEEP.OUT.ATTRS = FALSE
)
hook_mean <- c(100000, 150000, 180000, 178000, 183000, 160000, 110000, 80000)
hook_data$response <-
  rep(hook_mean, times = 3) + rep(c(-300, 0, 300), each = 8)
head(hook_data, 6)
```

	concentration	replicate	response
1	1	1	99700
2	2	1	149700
3	3	1	179700
4	4	1	177700
5	5	1	182700
6	6	1	159700

```
hook_result <- hook_effect(
  hook_data,
  concentration = "concentration",
  response = "response",
  uloq = 1.5,
  uloq_response = 130000,
  conf.level = 0.95
)
hook_result
```

High-dose hook-effect experiment

ULoQ concentration: 1.5

ULoQ response: supplied

Series: 1

Series: all

ULoQ concentration	: 1.5
ULoQ response	: 130000
Hook concentration	: 6
Hook response	: 160000
First in-range concentration	: 7
First in-range response	: 110000
Highest tested concentration	: 8
Hook lower bound	: NA

Concentration-level results

Series: all | Concentration: 1

Replicates	: 3
Mean response	: 1e+05
SD response	: 300
CV response	: 0.3
SE response	: 173.20508
CI lower	: 99254.759
CI upper	: 100745.24

Series: all | Concentration: 2

Replicates	: 3
Mean response	: 150000
SD response	: 300
CV response	: 0.2
SE response	: 173.20508
CI lower	: 149254.76
CI upper	: 150745.24

Series: all | Concentration: 3

Replicates	: 3
Mean response	: 180000
SD response	: 300

CV response : 0.16666667
SE response : 173.20508
CI lower : 179254.76
CI upper : 180745.24

Series: all | Concentration: 4

Replicates : 3
Mean response : 178000
SD response : 300
CV response : 0.16853933
SE response : 173.20508
CI lower : 177254.76
CI upper : 178745.24

Series: all | Concentration: 5

Replicates : 3
Mean response : 183000
SD response : 300
CV response : 0.16393443
SE response : 173.20508
CI lower : 182254.76
CI upper : 183745.24

Series: all | Concentration: 6

Replicates : 3
Mean response : 160000
SD response : 300
CV response : 0.1875
SE response : 173.20508
CI lower : 159254.76
CI upper : 160745.24

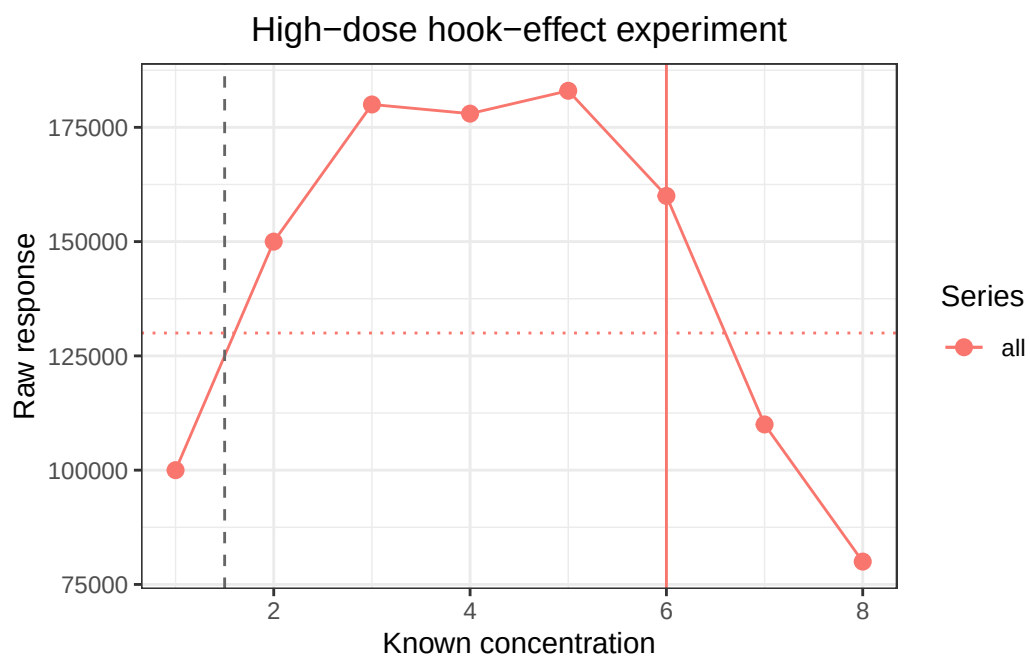
Series: all | Concentration: 7

Replicates : 3
Mean response : 110000
SD response : 300
CV response : 0.27272727
SE response : 173.20508
CI lower : 109254.76
CI upper : 110745.24

Series: all | Concentration: 8

Replicates : 3
Mean response : 80000
SD response : 300
CV response : 0.375
SE response : 173.20508
CI lower : 79254.759
CI upper : 80745.241

`plot(hook_result)`



In this example, the measured concentration 7 is the first point that returns to the ULoQ response range. The function therefore reports the preceding measured concentration, 6, as `hook_concentration`. This is a boundary determined by the spacing of tested concentrations, not an interpolated turning point.

Chapter 9

Method Comparison and Reference-Material Bias

Method comparison evaluates agreement between candidate and reference procedures on the same samples, corresponding to CLSI EP09. Reference-material bias asks a related but distinct question: how far are repeated measurements of a reference material from its assigned value, including uncertainty in the assigned value?

Analysis path	Main input	Main result
Patient-sample method comparison	Sample ID, comparison and candidate results, optional weights	Difference, regression, agreement limits, outliers, bias at specified levels
Reference-material bias	Replicates, assigned value, and assigned-value uncertainty	Mean, bias, relative bias, and bias interval

! Scope of Use

Correlation is not agreement, and a statistically significant bias is not automatically clinically important. Choose the regression and error model before seeing results, preserve the candidate-minus-reference direction, and compare bias and agreement limits with prespecified allowable limits at relevant decision levels.

```
library(ivdtools)
library(readr)
```

9.1 Detailed Function Reference

9.1.1 Workflow and Function Overview

Function	Purpose	Return value
<code>replicate_to_mean()</code>	Summarize one or more response columns by sample with mean, SD, and n	Summary data frame

Function	Purpose	Return value
<code>split_by_cutpoints()</code>	Split analysis data at prespecified numeric boundaries	Named data-frame list
<code>mcr()</code>	Create a method-comparison object	mcr object
<code>describe()</code>	Add descriptive statistics	Updated mcr object
<code>correlation()</code>	Add correlation analysis	Updated mcr object
<code>regression()</code>	Fit OLS, WLS, Deming, weighted Deming, or Passing–Bablok regression	Updated mcr object
<code>bland_altman()</code>	Estimate center difference, center interval, and limits of agreement	Updated mcr object
<code>outlier()</code>	Test paired-difference outliers	Updated mcr object
<code>bias()</code>	Calculate bias at specified levels	Updated mcr object
<code>predict()</code>	Forward or inverse prediction	Prediction data frame
<code>reference_bias()</code>	Analyze reference-material bias and intervals	reference_bias object
<code>plot()</code>	Plot scatter, regression, agreement, or bias	ggplot object
<code>summary()</code>	Summarize completed analyses	Original mcr object, invisibly

9.1.2 Summarizing Replicates with `replicate_to_mean()`

```
replicate_to_mean(data, x, y, weights = NULL, na.rm = TRUE)

summary_data <- replicate_to_mean(
  raw_data, "sample", c("candidate", "reference")
)
```

`x` is the sample-ID column and `y` can contain one or more numeric result columns. With one column, the function returns `x`, `y_mean`, `y_sd`, and `y_n` to preserve the original interface. With multiple columns it returns one set of mean, SD, and `n` columns per variable. The multiple-column mode does not create one common weight because each method has its own precision information.

9.1.3 Splitting Data at Prespecified Cutpoints

Segmentation is data preparation and is not stored in an mcr object. One mcr object always represents one defined data set, so descriptive statistics, correlation, outlier testing, regression, Bland–Altman analysis, bias, and plots use the same sample range.

```
mcr_data_list <- split_by_cutpoints(
  method_data,
  variable = "reference",
  cutpoints = c(100, 500)
)

mcr_data_list
```

The console summary has the following form:

Data split by cutpoints

Variable: reference

Cutpoints: 100, 500

segment_1	$[-\text{Inf}, 100)$	n = 35
segment_2	$[100, 500)$	n = 62
segment_3	$[500, \text{Inf})$	n = 28

The return value is a named data-frame list; each segment retains all original columns and row order:

```
mcr_data_list$segment_1
mcr_data_list$segment_2
mcr_data_list$segment_3
```

Use `lapply()` to create comparison objects for each segment and then run the complete workflow:

```
mcr_results <- lapply(
  mcr_data_list,
  function(d) mcr(
    d,
    id = "sample",
    candidate = "candidate",
    reference = "reference"
  )
)

mcr_results$segment_1 |>
  describe() |>
  correlation() |>
  regression(method = "deming") |>
  bland_altman()
```

cutpoints must be finite, strictly increasing, and unique. Intervals are left-closed and right-open, so a cutpoint belongs to the next segment. Empty segments are retained by default to reveal prespecified ranges with no observations; `drop_empty = TRUE` removes them. Nonfinite segmentation values error by default; `na_action = "drop"` excludes them and records the count. labels can provide unique list names.

9.1.4 Creating an mcr Object with `mcr()`

```
mcr(
  data,
  id,
  candidate,
  reference,
  weights = NULL,
  candidate_sd = NULL,
  reference_sd = NULL,
  candidate_n = NULL,
  reference_n = NULL
)
```

data must contain one pair of sample-level means per row; duplicate IDs error. `id` is the sample-ID column. `candidate` is the evaluated procedure, the regression Y axis, and the first term in the difference. `reference` is the comparison procedure, the regression X axis, and

the second term. `weights` is an optional finite nonnegative weight column. `candidate_sd` and `reference_sd` are optional within-sample SD columns and must be supplied together for `lambda = "replicates"`. `candidate_n` and `reference_n` are optional replicate-count columns and must also be supplied together; if omitted, equal replicate counts are assumed.

The returned `mcr` object preserves the input data and column names, complete paired values, the input row numbers in `complete_idx`, and the number of complete pairs `n`. Later results are stored in `describe`, `correlation`, `regression`, `outlier`, `bland_altman`, and `bias` slots. Split first with `split_by_cutpoints()` when separate ranges are required.

9.1.5 Descriptive Statistics with `describe()`

```
describe(x, cols = NULL, digits = 4L)
```

`x` is an `mcr` object. `cols` selects additional columns; when omitted, all columns other than ID and the two measurement columns are used, and a minus-prefixed name can exclude a column. `digits` controls display precision.

The returned object contains complete-pair count, duplicate-ID information, summaries of additional columns, and descriptive statistics for the candidate, reference, and paired difference.

9.1.6 Correlation with `correlation()`

```
correlation(
  x,
  method = c("pearson", "spearman", "kendall")
)
```

`method` selects Pearson, Spearman, or Kendall. The correlation slot contains coefficient, test statistic, degrees of freedom, P value, interval, and `n`. Correlation describes common variation and is not agreement.

9.1.7 Regression with `regression()`

```
regression(
  x,
  method = c("ols", "wls", "deming", "wdeming", "pb"),
  conf.level = 0.95,
  lambda = 1,
  weights = NULL,
  variance_models = NULL,
  profile_components = c(reference = "total", candidate = "total"),
  ci_method = c(
    "auto", "analytic", "jackknife", "rank",
    "bootstrap_percentile", "bootstrap_standard"
  ),
  bootstrap_replicates = 5000L,
  seed = NULL,
  mdl = NULL,
  weight_iterations = 4L,
  epsilon = NULL
)
```

`method` selects `ols`, `wls`, `deming`, `wdeming`, or `pb`. `conf.level` is the regression-parameter confidence level. `lambda` is candidate-Y variance divided by reference-X variance, as a pos-

itive number or “replicates” ; the latter uses SD and n supplied to `mcr()`. `weights` supports fixed weights, residual, constant_cv, and precision_profile. For `precision_profile`, `variance_models` is a list with reference and candidate functions or precision objects that have already been profiled, and `profile_components` selects the component on each side.

`ci_method = “auto”` uses analytic intervals for OLS/WLS, complete-refit jackknife for Deming variants, and rank intervals for Passing–Bablok. The explicit bootstrap methods are the K2 percentile and standard methods. At least 5,000 bootstrap replicates are required. `seed` controls reproducibility without changing the caller’s random-number state. `mdl` supplies medical decision levels; when bootstrap is run during regression, bias draws for those levels are retained. `weight_iterations` controls residual-weight updates, and `epsilon` controls convergence in iterative weighted Deming.

The returned regression slot stores method, intercept, slope, SEs, parameter intervals, intercept-slope covariance, residuals, regression SE, n, input row numbers, confidence level, and, when applicable, R-squared, variance ratio, and weight specification. Iterative weighted Deming also stores convergence, latent concentrations, and final weights. Bootstrap stores the seed, requested/valid/failed counts, and parameter draws. Passing–Bablok follows EP09c Appendix I Algorithm I for vertical pairs, slope -1, and K shifts.

9.1.8 Bland–Altman Analysis with `bland_altman()`

```
bland_altman(
  x,
  type = c("difference", "ratio", "percent"),
  x_axis = c("mean", "candidate", "reference"),
  percent_denominator = c("reference", "mean", "candidate"),
  agree.level = 0.95,
  conf.level = 0.95,
  method = c("parametric", "nonparametric", "hodges_lehmann"),
  median_ci_method = c("order_statistic", "interpolated"),
  bootstrap_replicates = 5000L,
  seed = NULL
)
```

`type` selects absolute difference, ratio, or percentage difference. `x_axis` uses the two-method mean, candidate, or reference as horizontal coordinate. `percent_denominator` independently selects reference, mean, or candidate. `agree.level` is the LoA coverage proportion, while `conf.level` is confidence in the center and LoA intervals. The parametric method uses the mean difference, nonparametric uses the ordinary median, and `hodges_lehmann` uses the HL pseudomedian. `median_ci_method` selects the conservative order-statistic interval or Appendix F continuous-rank interpolation. `bootstrap_replicates` and `seed` control non-parametric LoA resampling.

The `bland_altman` slot contains center value, center_estimand, center_ci, method-specific center interval, SD, LoA and their intervals, percent denominator, source row numbers, plotting coordinate, and labels. Order-statistic intervals record actual coverage; interpolated intervals record continuous rank and nominal coverage.

9.1.9 Outlier Testing with `outlier()`

```
outlier(
  x,
  method = c("grubbs", "esd", "dixon", "iqr"),
  type = c("difference", "percent"),
  percent_denominator = c("reference", "mean", "candidate"),
  alpha = 0.05,
```

```

    coef = 1.5,
    ...
)

```

method selects Grubbs, generalized ESD, Dixon, or IQR. type tests absolute or percent differences, and percent_denominator uses the same definition as bland_altman(). alpha controls the Grubbs, ESD, or Dixon tests; coef controls IQR fences. Additional arguments are passed to the selected test, including r for the maximum number of generalized-ESD candidates.

The returned outlier slot stores method, difference scale, number of outliers, input row numbers, values, test statistics, and critical values. Detection does not authorize deletion.

9.1.10 Bias at Specified Levels with bias()

```

bias(
  x,
  mdl,
  level = 0.95,
  interval = c("", "confidence", "prediction", "both"),
  ci_method = c("auto", "analytic", "jackknife", "rank",
               "bootstrap_percentile", "bootstrap_standard"),
  bootstrap_replicates = 5000L,
  seed = NULL
)

```

x must be an mcr object with regression completed. mdl contains one or more reference-scale levels. interval selects confidence, prediction, or both; level is the bias-interval confidence level. ci_method, bootstrap_replicates, and seed control parameter uncertainty. Passing-Bablok under auto uses percentile bootstrap, not jackknife.

The returned bias slot defines bias as fitted candidate minus mdl. After prespecified data segmentation, mdl should lie within that segment to avoid accidental extrapolation.

9.1.11 Prediction with predict()

```

predict(
  object,
  newdata = NULL,
  reference = NULL,
  candidate = NULL,
  inverse = FALSE,
  interval = c("none", "confidence", "prediction", "both"),
  conf.level = 0.95,
  ci_method = c("auto", "analytic", "jackknife", "rank",
               "bootstrap_percentile", "bootstrap_standard"),
  bootstrap_replicates = 5000L,
  seed = NULL
)

```

With inverse = FALSE, predict candidate from reference; with inverse = TRUE, algebraically solve for reference from candidate. newdata supplies new values; otherwise the original reference or candidate column is used. The function returns input, predicted value, and requested intervals. Prediction intervals are an ivdtools extension and do not represent EP09c parameter intervals. Inverse prediction is an algebraic back-calculation and does not return an interval.

9.1.12 Reference-Material Bias with `reference_bias()`

```
reference_bias(
  data,
  result,
  reference,
  reference_uncertainty,
  level = NULL,
  k = 2
)
```

`data` is long-format replicate data. `result` is the measurement column. `reference` is a single assigned value, a vector with one value per level, or a column. `reference_uncertainty` follows the same rules. `level` optionally identifies material/concentration strata; assignment and uncertainty must be constant within each stratum. `k` expands the standard uncertainty of the measurement mean into an interval component and defaults to 2.

The returned `reference_bias` object contains, for each level, total and valid counts, excluded count, mean, SD, standard and expanded uncertainty of the mean, assigned value and its uncertainty, absolute and relative bias, half-width, and interval limits. It also records the input columns, `k`, and counts.

9.1.13 Plotting and Summary

```
plot(
  x,
  type = c("scatter", "regression", "bland_altman", "bias"),
  interval = c("none", "confidence", "prediction", "both"),
  conf.level = NULL,
  mdl = NULL
)
```

`type` selects scatter, regression, Bland–Altman, or bias plots. `interval` and `conf.level` control regression/bias bands; `mdl` marks medical decision levels. The function invisibly returns a ggplot object. Plots use the same data range as their corresponding analysis slots.

```
summary(object)
```

`summary` prints analyses already completed in the `mcr` object, does not run missing steps, and invisibly returns the original object.

9.2 Main Functions

Function	Purpose	Return value
<code>replicate_to_mean()</code>	Summarize one or more response columns by sample	Summary data frame
<code>split_by_cutpoints()</code>	Split data at prespecified numerical boundaries	Named data-frame list
<code>mcr()</code>	Create a method-comparison object	<code>mcr</code> object
<code>describe()</code>	Add descriptive statistics	Updated object
<code>correlation()</code>	Add Pearson, Spearman, or Kendall correlation	Updated object

Function	Purpose	Return value
<code>regression()</code>	Fit OLS, WLS, Deming, weighted Deming, or Passing–Bablok	Updated object
<code>bland_altman()</code>	Estimate center difference and limits of agreement	Updated object
<code>outlier()</code>	Test paired-difference outliers	Updated object
<code>bias()</code>	Estimate bias at medical decision levels	Updated object
<code>predict()</code>	Forward or inverse prediction	Prediction data frame
<code>reference_bias()</code>	Analyze reference-material bias and intervals	<code>reference_bias</code> object
<code>plot()</code> / <code>summary()</code>	Plot and summarize completed analyses	Figure or updated object

9.2.1 Summarize Replicates and Split by Cutpoints

```
replicate_to_mean(data, x = "sample", y = c("candidate", "reference"))
split_by_cutpoints(data, col = "reference", cutpoints = c(30, 80, 150))
```

The resulting data frame should contain one row per sample and one paired mean for each method. Define cutpoints before analysis and do not repeatedly split to find favorable results.

9.2.2 Build a Method-Comparison Object

```
mcr(data, id, candidate, reference, weights = NULL,
     candidate_sd = NULL, reference_sd = NULL,
     candidate_n = NULL, reference_n = NULL)
```

Each row must be one sample-level pair; duplicate IDs are errors. The candidate is the regression Y axis and the first term in candidate-minus-reference differences; the reference is the X axis. Sample-level SD and replicate count columns can support `lambda="replicates"`.

9.2.3 Regression Choices

```
regression(
  x, method = c("ols", "wls", "deming", "wdeming", "pb"),
  conf.level = 0.95, lambda = 1,
  weights = NULL, variance_models = NULL,
  profile_components = c("total"), ci_method = "auto",
  bootstrap_replicates = 5000, seed = NULL, mdl = NULL
)
```

OLS treats X error as negligible; Deming accounts for error in both methods; WLS and weighted Deming address concentration-dependent variance; Passing–Bablok uses rank-based slopes without a normal-error model. `lambda` is the candidate/reference variance ratio and should come from independent precision evidence, not be tuned to match a desired slope. `ci_method="auto"` uses analytic intervals for OLS/WLS, jackknife for Deming variants, and rank intervals for Passing–Bablok.

9.2.4 Bland–Altman, Outliers, Bias, and Prediction

```

bland_altman(x, type = c("difference", "ratio", "percent"),
             x_axis = c("mean", "candidate", "reference"))
outlier(x, method = c("grubbs", "esd", "dixon", "iqr"), type = "difference")
bias(x, mdl = c(30, 80, 150), interval = "both")
predict(x, candidate = c(40, 100))
predict(x, reference = c(50, 120), inverse = TRUE)

```

The Bland–Altman center and limits of agreement describe individual differences, not a test of equality of means. Investigate candidate outliers in the original records and report sensitivity analyses; detection does not authorize automatic deletion.

9.2.5 Reference-Material Bias

```

reference_bias(
  data,
  result,
  reference,
  reference_uncertainty,
  level = NULL,
  k = 2
)

```

The function combines the expanded uncertainty of the measured mean with the reference-material uncertainty. It reports results only; state the bias direction and the coverage factor before interpreting the interval.

9.3 Analysis Example

```

mc_dat <- as.data.frame(read_csv(
  "./data/009-方法学比较与参考物偏倚/method-comparison.csv",
  show_col_types = FALSE
))
mc <- mcr(mc_dat, id = "id", candidate = "candidate", reference = "reference")
mc <- describe(mc)

```

Descriptive Statistics

Variable	n	Mean	SD	Min	Q1	Median	Q3	Max	NA
reference (X)	40	105.0000	56.9555	10.0000	57.4750	105.0000	152.5250	200.0000	
candidate (Y)	40	112.4977	59.2513	11.5900	63.1075	112.5750	159.5925	209.9000	
Diff (Y-X)	40	7.4977	3.9583	1.2000	5.2725	6.6100	9.7950	22.1000	

batch

Level	Count	Percent
B1	20	50.0%
B2	20	50.0%

sample_type

Level	Count	Percent
plasma	13	32.5%

serum	14	35.0%
urine	13	32.5%

```
mc <- correlation(mc, method = "pearson")
```

Correlation

Method: pearson
 $r = 0.9985$, 95% CI [0.9971, 0.9992]
 p-value: $< 2.2e-16$
 n = 40

```
mc <- regression(mc, method = "deming")
```

Regression

Method: Deming
 CI method: jackknife
 Intercept = 3.2588 (SE = 1.4495)
 Slope = 1.0404 (SE = 0.0100)
 Intercept 95% CI: [0.3245, 6.1930]
 Slope 95% CI: [1.0202, 1.0606]
 H_0 : slope = 1 $\rightarrow$ t = 4.0474, p = 0.0002453
 slope significantly different from 1 (CI excludes 1)
 Sigma = 3.3319
 n = 40

```
mc <- bland_altman(mc, type = "difference")
```

Bland-Altman

Type: difference
 Y-axis: candidate - reference
 X-axis: Mean of candidate and reference
 Method: parametric
 n = 40

Mean diff: 7.4977
 95% CI for mean diff: [6.2318, 8.7637]
 SD diff: 3.9583
 95% LoA: [-0.2603, 15.2558]

95% CI for LoA:
 Lower LoA: [-2.4419, 1.9213]
 Upper LoA: [13.0742, 17.4374]

```
mc <- outlier(mc, method = "grubbs", type = "difference")
```

Outlier

Method: grubbs
 Data: Difference (candidate - reference)
 Alpha: 0.05

1 outlier(s) found:

Row	Value	Statistic	Critical

```
7                22.1000      3.6891  3.0361
```

```
mc <- bias(mc, mdl = c(30, 80, 150), interval = "both")
```

Bias

```
Method: Deming
Interval: both (95%)
MDL points: 3
Bias range: [4.4699, 9.3145]
```

mdl	bias	ci_lower	ci_upper	pi_lower	pi_upper
30	4.469895	2.109892	6.829897	-2.6761742	11.61596
80	6.488465	5.027934	7.948996	-0.4129728	13.38990
150	9.314463	8.450122	10.178804	2.5141854	16.11474

```
summary(mc)
```

Method Comparison Regression (mcr)

Method Comparison

```
Data class:      data.frame
Total rows:      40
Complete pairs:  40
ID variable:     id
Reference (X):   reference
Candidate (Y):   candidate
Duplicate IDs:   none
```

Analysis slots:

```
[x] Describe
[x] Correlation (pearson)
[x] Regression (Deming)
[x] Outlier (grubbs)
[x] Bland-Altman (difference)
[x] Bias
```

Descriptive Statistics

Variable	n	Mean	SD	Min	Q1	Median	Q3	Max	NA
reference (X)	40	105.0000	56.9555	10.0000	57.4750	105.0000	152.5250	200.0000	
candidate (Y)	40	112.4977	59.2513	11.5900	63.1075	112.5750	159.5925	209.9000	
Diff (Y-X)	40	7.4977	3.9583	1.2000	5.2725	6.6100	9.7950	22.1000	

batch

Level	Count	Percent
B1	20	50.0%
B2	20	50.0%

sample_type

Level	Count	Percent
plasma	13	32.5%
serum	14	35.0%

urine 13 32.5%

Correlation

Method: pearson
 $r = 0.9985$, 95% CI [0.9971, 0.9992]
 p-value: < 2.2e-16
 n = 40

Regression

Method: Deming
 CI method: jackknife
 Intercept = 3.2588 (SE = 1.4495)
 Slope = 1.0404 (SE = 0.0100)
 Intercept 95% CI: [0.3245, 6.1930]
 Slope 95% CI: [1.0202, 1.0606]
 H_0 : slope = 1 $\rightarrow$ t = 4.0474, p = 0.0002453
 slope significantly different from 1 (CI excludes 1)
 Sigma = 3.3319
 n = 40

Outlier

Method: grubbs
 Data: Difference (candidate - reference)
 Alpha: 0.05

1 outlier(s) found:

Row	Value	Statistic	Critical
7	22.1000	3.6891	3.0361

Bland-Altman

Type: difference
 Y-axis: candidate - reference
 X-axis: Mean of candidate and reference
 Method: parametric
 n = 40

Mean diff: 7.4977
 95% CI for mean diff: [6.2318, 8.7637]
 SD diff: 3.9583
 95% LoA: [-0.2603, 15.2558]

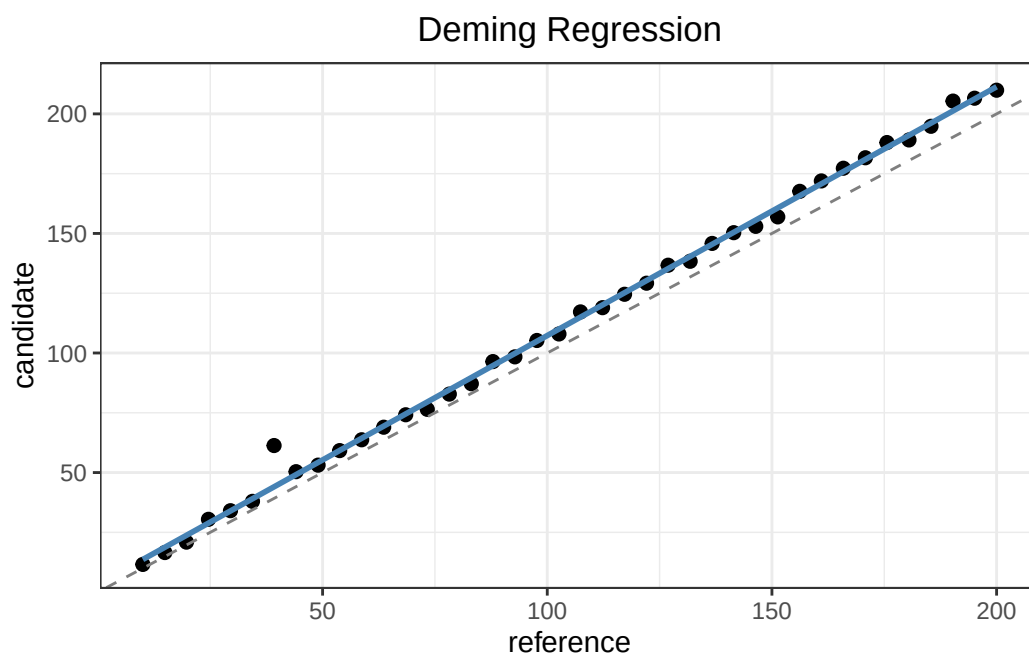
95% CI for LoA:
 Lower LoA: [-2.4419, 1.9213]
 Upper LoA: [13.0742, 17.4374]

Bias

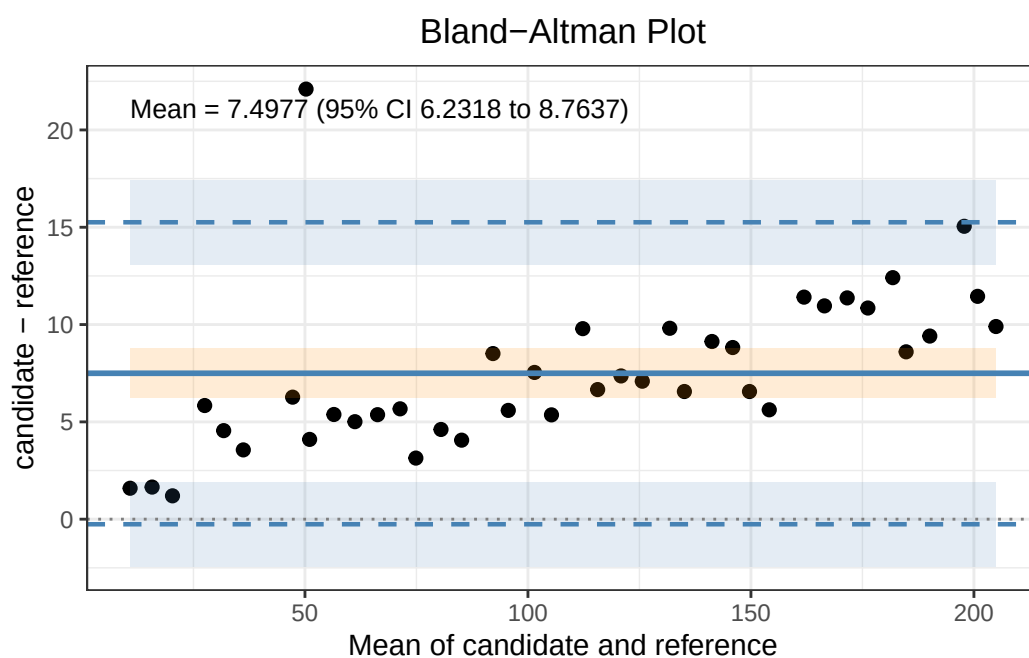
Method: Deming
 Interval: both (95%)
 MDL points: 3
 Bias range: [4.4699, 9.3145]

mdl	bias	ci_lower	ci_upper	pi_lower	pi_upper
30	4.469895	2.109892	6.829897	-2.6761742	11.61596
80	6.488465	5.027934	7.948996	-0.4129728	13.38990
150	9.314463	8.450122	10.178804	2.5141854	16.11474

```
plot(mc, type = "regression")
```



```
plot(mc, type = "bland_altman")
```



For subgroup analyses, split by lot or sample type using prespecified boundaries and repeat the same workflow on every subgroup. Do not let a pooled mean conceal subgroup-specific bias.

```
rm_data <- as.data.frame(read_csv(
  "./data/009-方法学比较与参考物偏倚/std-wst408-reference-material.csv",
  show_col_types = FALSE
))
rm_fit <- reference_bias(
  rm_data,
```

```

    result = "result",
    reference = "assigned",
    reference_uncertainty = "expanded_uncertainty",
    level = "sample",
    k = 2
)
print(rm_fit)

```

Reference Material Bias

Standard: YY/T 1789.2-2021, Section 5.3

Data: rm_data

Result: result

Levels: 2

Records: 20 valid, 0 excluded

Coverage factor: k = 2

Measurement and uncertainty

Level: GBW09193

n	: 10
Mean	: 1.528
SD	: 0.016193277
Reference	: 1.49
U_ref	: 0.04
u_X	: 0.0051207638
U_X	: 0.010241528

Level: GBW09195

n	: 10
Mean	: 1.05
SD	: 0.014142136
Reference	: 1.02
U_ref	: 0.04
u_X	: 0.004472136
U_X	: 0.0089442719

Bias interval

Level: GBW09193

Bias	: 0.038
Bias (%)	: 2.5503356
Interval half-width	: 0.0412903
Lower	: -0.0032903002
Upper	: 0.0792903

Level: GBW09195

Bias	: 0.03
Bias (%)	: 2.9411765
Interval half-width	: 0.040987803
Lower	: -0.010987803
Upper	: 0.070987803

Interval: Bias +/- sqrt($U_X^2 + U_{ref}^2$)

9.4 Interpretation Checklist

1. Confirm candidate/reference direction and measurement units.
2. Use correlation only descriptively; interpret regression and Bland–Altman together.

3. Choose OLS, WLS, Deming, weighted Deming, or Passing–Bablok from the error structure.
4. Investigate, document, and sensitivity-test outliers.
5. Report bias at clinically relevant levels and compare it with prespecified limits.
6. Distinguish patient-sample agreement from reference-material bias and commutability.

9.5 Complete Standard Cases

9.5.1 Replicate Summaries and lambda

`mcr()` does not accept raw replicate rows with the same ID. First summarize both methods, then pass the mean columns as X/Y and the SD and n columns as replicate precision information. The two methods may have different replicate counts, but each count must be constant across samples; otherwise a single scalar lambda is not appropriate and a precision profile should be used.

```
replicate_raw <- data.frame(
  sample = rep(1:8, each = 3),
  reference = rep(seq(10, 80, 10), each = 3) + rep(c(-1, 0, 1), 8),
  candidate = rep(2 + 1.05 * seq(10, 80, 10), each = 3) +
    rep(c(-0.5, 0, 0.5), 8)
)
replicate_summary <- replicate_to_mean(
  replicate_raw, "sample", c("candidate", "reference")
)
replicate_mcr <- mcr(
  replicate_summary,
  id = "x", candidate = "candidate_mean", reference = "reference_mean",
  candidate_sd = "candidate_sd", reference_sd = "reference_sd",
  candidate_n = "candidate_n", reference_n = "reference_n"
)
replicate_fit <- regression(
  replicate_mcr, method = "deming", lambda = "replicates"
)
```

Regression

```
Method: Deming
CI method: jackknife
Lambda: 0.2500 (from summarized SD data)
Intercept = 2.0000 (SE = 0.0000)
Slope      = 1.0500 (SE = 0.0000)
Intercept 95% CI: [2.0000, 2.0000]
Slope     95% CI: [1.0500, 1.0500]
H0: slope = 1 -> t = 170220065980364.3125, p = < 2.2e-16
slope significantly different from 1 (CI excludes 1)
Sigma      = 0.0000
n = 8
```

9.5.2 CLSI EP09c Appendix C: Mixed Variance and Regression Models

Appendix C compares two reagent lots from the same measurement procedure using 79 samples. For samples 1–40, it uses the absolute difference $Y - X$; for samples 41–79, it uses the percentage difference $Y - X$ relative to the mean of the two lots.

```

ep09_c <- read_csv(
  "/data/009-方法学比较与参考物偏倚/std-ep09-appendix-c.csv",
  show_col_types = FALSE
)
ep09_c_diff <- mcr(
  as.data.frame(ep09_c),
  id = "id", candidate = "mp_y", reference = "mp_x"
)
ep09_c_diff <- describe(ep09_c_diff)

```

Descriptive Statistics

Variable	n	Mean	SD	Min	Q1	Median	Q3	Max	NA
mp_x (X)	79	8.7969	16.2611	0.0010	0.1835	1.7730	11.5160	91.2350	
mp_y (Y)	79	9.0292	17.4625	0.0010	0.1995	1.8330	11.5980	99.8020	
Diff (Y-X)	79	0.2323	1.9178	-2.4890	-0.0515	0.0060	0.0760	13.3600	

unit

Level	Count	Percent
ug/L	79	100.0%

Call `bland_altman()` separately for the low- and high-concentration groups prespecified in Appendix C. These groups are the samples numbered 1–40 and 41–79 in the study design. Their reference results overlap slightly at the boundary and cannot be reconstructed without loss by a single numeric cutpoint, so the data are split directly according to the study design rather than with `split_by_cutpoints()`. The low-concentration group uses absolute differences; for the high-concentration group, EP09c sets `percent_denominator = "mean"` to calculate $Y - X$ relative to the mean of the two lots. `x_axis` controls the plotting position, while the percentage denominator controls the bias definition; they are separate parameters.

```

ep09_c_low_mcr <- mcr(
  as.data.frame(ep09_c[1:40, ]),
  id = "id", candidate = "mp_y", reference = "mp_x"
)
ep09_c_low_mcr <- bland_altman(
  ep09_c_low_mcr, type = "difference", x_axis = "mean"
)

```

Bland-Altman

Type: difference
 Y-axis: mp_y - mp_x
 X-axis: Mean of mp_y and mp_x
 Method: parametric
 n = 40

Mean diff: 0.0204
 95% CI for mean diff: [-0.0101, 0.0509]
 SD diff: 0.0954
 95% LoA: [-0.1666, 0.2074]

95% CI for LoA:
 Lower LoA: [-0.2192, -0.1140]
 Upper LoA: [0.1548, 0.2599]

```

ep09_c_high_mcr <- mcr(
  as.data.frame(ep09_c[41:79, ]),
  id = "id", candidate = "mp_y", reference = "mp_x"
)
ep09_c_high_mcr <- bland_altman(
  ep09_c_high_mcr,
  type = "percent", x_axis = "mean", percent_denominator = "mean"
)

```

Bland-Altman

Type: percent
 Y-axis: $(mp_y - mp_x) / \text{mean}(mp_y, mp_x) * 100$
 X-axis: Mean of mp_y and mp_x
 Percent base: mean
 Method: parametric
 n = 39

Mean diff: 0.4303
 95% CI for mean diff: [-1.8286, 2.6892]
 SD diff: 6.9685
 95% LoA: [-13.2277, 14.0883]

95% CI for LoA:

Lower LoA: [-17.1215, -9.3339]
 Upper LoA: [10.1945, 17.9821]

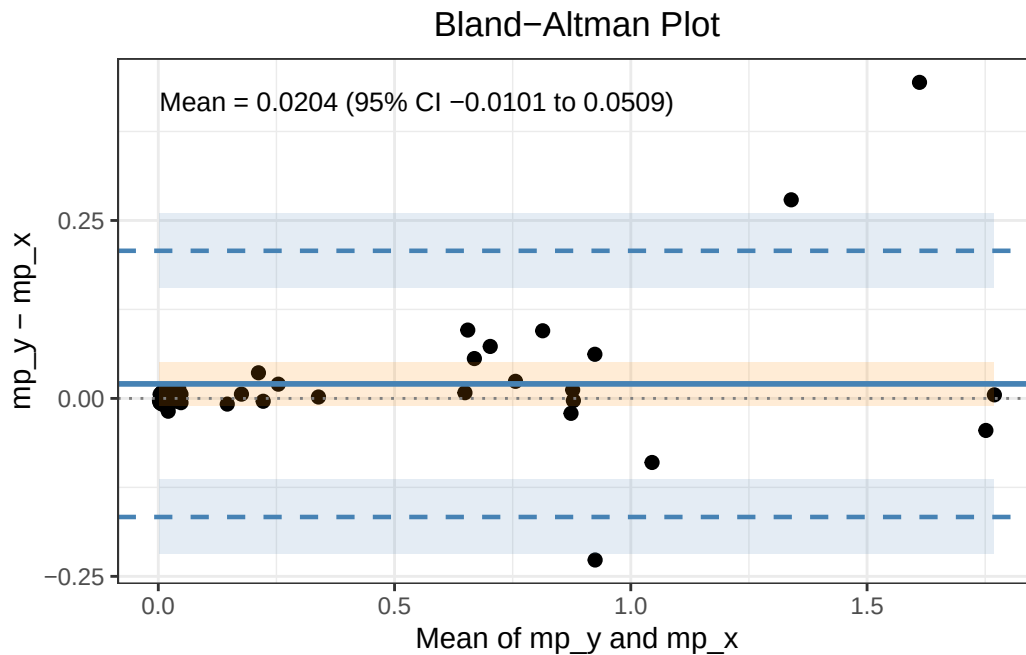
Table 9.1: EP09c Appendix C: segmented bias comparison

Standard location	Statistic	Standard	ivdtools
low_absolute	estimate	0.020	0.0204
low_absolute	lower	-0.010	-0.0101
low_absolute	upper	0.051	0.0509
high_percent	estimate	0.430	0.4303
high_percent	lower	-1.830	-1.8286
high_percent	upper	2.690	2.6892

```

plot(ep09_c_low_mcr, type = "bland_altman")

```



The orange band is the 95% CI for the mean difference; the blue bands are the 95% CIs for the lower and upper limits of agreement.

Fit the five models shown in standard Figures C(f)–C(j). `weights = "constant_cv"` uses $1/x^2$ for WLS and the Linnet iterative constant-CV algorithm for weighted Deming; the latter refits the complete model after each jackknife deletion.

```
ep09_c_mcr <- mcr(
  as.data.frame(ep09_c),
  id = "id", candidate = "mp_y", reference = "mp_x"
)
ep09_c_ols <- regression(ep09_c_mcr, method = "ols")
```

Regression

```
Method: OLS
CI method: analytic
Intercept = -0.3804 (SE = 0.1995)
Slope      = 1.0697 (SE = 0.0108)
Intercept 95% CI: [-0.7777, 0.0169]
Slope      95% CI: [1.0481, 1.0912]
H0: slope = 1 -> t = 6.4219, p = 1.011e-08
slope significantly different from 1 (CI excludes 1)
R-squared = 0.9921
Sigma     = 1.5576
n = 79
```

```
ep09_c_wls <- regression(
  ep09_c_mcr, method = "wls", weights = "constant_cv"
)
```

Regression

```
Method: WLS
Weights: constant_cv
CI method: analytic
Intercept = 0.0054 (SE = 0.0004)
```

```

Slope      = 0.9238 (SE = 0.0521)
Intercept 95% CI: [0.0046, 0.0062]
Slope      95% CI: [0.8201, 1.0274]
H0: slope = 1 -> t = -1.4642, p = 0.1472
slope not significantly different from 1 (CI contains 1)
R-squared = 0.8035
Sigma      = 0.4456
n = 79

```

```
ep09_c_deming <- regression(ep09_c_mcr, method = "deming")
```

Regression

```

Method: Deming
CI method: jackknife
Intercept = -0.4202 (SE = 0.1792)
Slope      = 1.0742 (SE = 0.0367)
Intercept 95% CI: [-0.7771, -0.0634]
Slope      95% CI: [1.0012, 1.1472]
H0: slope = 1 -> t = 2.0230, p = 0.04654
slope significantly different from 1 (CI excludes 1)
Sigma      = 1.5594
n = 79

```

```

ep09_c_wdeming <- regression(
  ep09_c_mcr, method = "wdeming", weights = "constant_cv"
)

```

Regression

```

Method: Constant-CV Deming
Weights: constant_cv
CI method: jackknife
Intercept = -0.0023 (SE = 0.0019)
Slope      = 1.0372 (SE = 0.0264)
Intercept 95% CI: [-0.0061, 0.0015]
Slope      95% CI: [0.9846, 1.0899]
H0: slope = 1 -> t = 1.4074, p = 0.1633
slope not significantly different from 1 (CI contains 1)
Sigma      = 0.0056
n = 79

```

```
ep09_c_pb <- regression(ep09_c_mcr, method = "pb")
```

Regression

```

Method: Passing-Bablok
CI method: rank
Intercept = 0.0055 (SE = 0.0083)
Slope      = 1.0028 (SE = 0.0132)
Intercept 95% CI: [-0.0059, 0.0089]
Slope      95% CI: [0.9830, 1.0162]
H0: slope = 1 -> t = 0.2140, p = 0.8311
slope not significantly different from 1 (CI contains 1)
Sigma      = 1.9142
n = 79

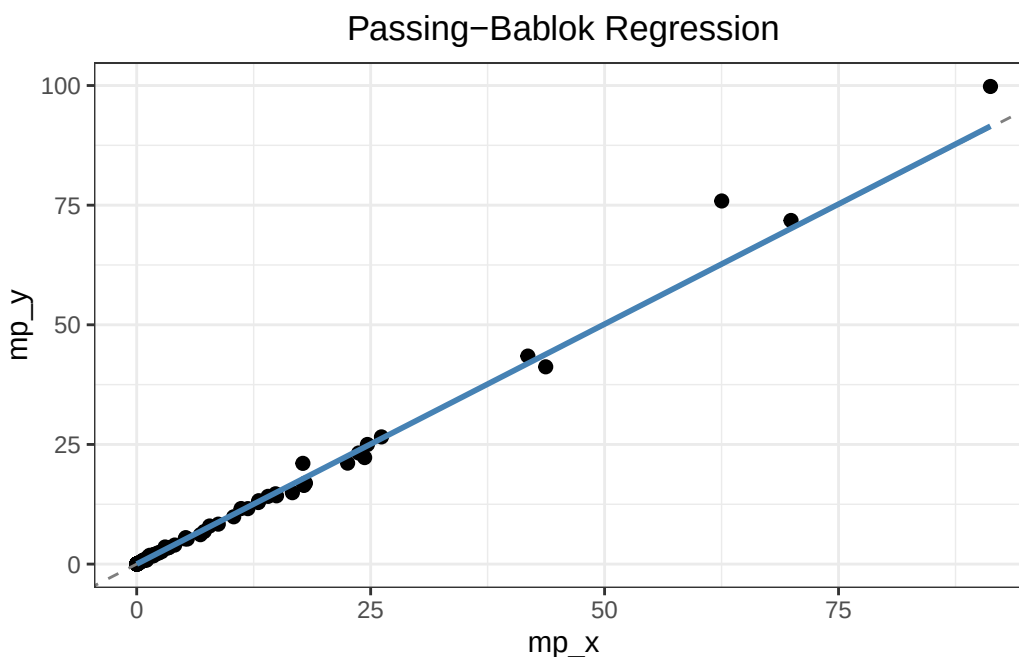
```

Table 9.2: EP09c Appendix C: regression parameters

Standard location	Statistic	Standard	ivdtools
OLS	intercept	-0.3800	-0.3804
OLS	slope	1.0700	1.0697
WLS	intercept	0.0100	0.0054
WLS	slope	0.9200	0.9238
Deming	intercept	-0.4200	-0.4202
Deming	slope	1.0700	1.0742
Weighted Deming	intercept	0.0000	-0.0023
Weighted Deming	slope	1.0400	1.0372
Passing-Bablok	intercept	0.0055	0.0055
Passing-Bablok	slope	1.0028	1.0028

The constant-CV fit stores convergence, latent_concentration, and the final weights; internally it converts ppwdeming's X/Y variance ratio to this package's common Y/X lambda. Passing-Bablok follows Appendix I Algorithm I: it retains signed infinite slopes for vertical pairs, excludes slopes of -1, and applies the K shift.

```
plot(ep09_c_pb, type = "regression")
```



9.5.3 CLSI EP09c Appendix D: Six Data Patterns

The six Appendix D data sets each contain 40 paired results and illustrate how constant SD, constant CV, concentration distribution, and outliers affect difference plots and regression. The standard does not publish one uniform set of regression parameters for all six data sets, so the following OLS parameters are used only to verify the data structure.

```
ep09_d <- read_csv(
  "/data/009-方法学比较与参考物偏倚/std-ep09-appendix-d.csv",
  show_col_types = FALSE
)
ep09_d_pattern <- c(
  D1 = "constant SD", D2 = "constant CV", D3 = "constant CV",
```

```

D4 = "constant CV with outliers", D5 = "constant SD with outliers",
D6 = "constant SD"
)

# Demonstrate the workflow on D4, which contains outliers, then use the
# same call for D1--D6 in the hidden summary below.
ep09_d4 <- as.data.frame(ep09_d[ep09_d$dataset == "D4", ])
ep09_d4_mcr <- mcr(
  ep09_d4, id = "sample",
  candidate = "candidate", reference = "comparative"
)
ep09_d4_mcr <- describe(ep09_d4_mcr)

```

Descriptive Statistics

Variable	n	Mean	SD	Min	Q1	Median	Q3	Max	NA
comparative (X)	40	203.8413	226.8297	0.8810	31.6217	104.2190	282.3255	785.5660	
candidate (Y)	40	224.5340	235.1985	1.0780	37.6600	140.9205	347.1503	801.2270	
Diff (Y-X)	40	20.6928	107.4543	-205.3500	-2.0363	4.9985	18.7690	585.1470	

dataset

Level	Count	Percent
D4	40	100.0%

```

ep09_d4_mcr <- regression(ep09_d4_mcr, method = "ols")

```

Regression

```

Method: OLS
CI method: analytic
Intercept = 35.9058 (SE = 22.9826)
Slope      = 0.9254 (SE = 0.0759)
Intercept 95% CI: [-10.6200, 82.4316]
Slope      95% CI: [0.7717, 1.0790]
H0: slope = 1 -> t = -0.9834, p = 0.3316
slope not significantly different from 1 (CI contains 1)
R-squared = 0.7965
Sigma     = 107.4996
n = 40

```

```

plot(ep09_d4_mcr, type = "regression")

```

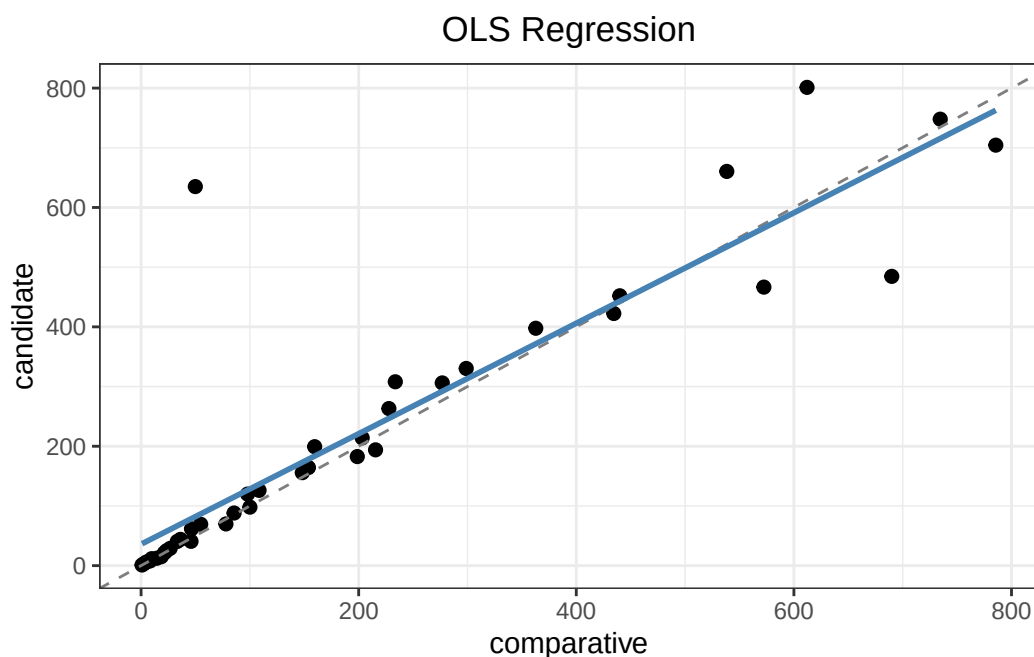


Table 9.3: EP09c Appendix D: data structure and OLS diagnostics

	Dataset	Data pattern	Sample count	Intercept	Slope	Largest-residual sample
D1	D1	constant SD	40	9.1854	0.9959	29
D2	D2	constant CV	40	12.4287	0.9697	36
D3	D3	constant CV	40	7.3639	0.8942	39
D4	D4	constant CV with outliers	40	35.9058	0.9254	14
D5	D5	constant SD with outliers	40	0.5373	0.9594	3
D6	D6	constant SD	40	0.2171	1.0024	18

All six sets contain 40 complete pairs. Model choice should still be based on the error structure shown in the difference plot, not only on the correlation coefficient.

9.5.4 CLSI EP09c Appendix E: Generalized ESD Outlier Test

Appendix E uses the 100 relative biases from Appendix F to demonstrate generalized ESD with $\alpha = 0.01$ and a maximum of five outliers. Relative bias is defined as $100 * (Y - X) / X$.

```
ep09_f <- read_csv(
  "/data/009-方法学比较与参考物偏倚/std-ep09-appendix-f.csv",
  show_col_types = FALSE
)
ep09_f_mcr <- mcr(
  as.data.frame(ep09_f),
  id = "id", candidate = "y", reference = "x"
)
ep09_e_result <- outlier(
  ep09_f_mcr, method = "esd", type = "percent", alpha = 0.01, r = 5
)
```

Outlier


```
Method: esd
Data:   Percent diff ((y - x) / x * 100)
Alpha:  0.01
```

```
1 outlier(s) found:
```

Row	Value	Statistic	Critical
3	-55.7047	6.0896	3.7540

9.5.5 CLSI EP09c Appendix F: Median-Bias Intervals

Appendix F provides 100 paired results. The ordinary median demonstrates both the default conservative order-statistic interval and the continuous-rank interpolation interval; method = "hodges_lehmann" returns the Walsh-average pseudomedian and a Wilcoxon/Tukey interval without continuity correction. Nonparametric limits of agreement remain empirical quantiles with percentile-bootstrap intervals.

```
ep09_f_ba <- bland_altman(
  ep09_f_mcr,
  type = "percent", x_axis = "reference",
  percent_denominator = "reference",
  method = "nonparametric", bootstrap_replicates = 1000, seed = 2026
)
```

Bland-Altman

```
Type:      percent
Y-axis:    (y - x) / x * 100
X-axis:    x
Percent base: reference
Method:    nonparametric
n = 100
```

```
Median diff:      -0.3345
95% requested CI for median diff (actual 96.5%): [-2.0202, 1.5873]
SD diff:          9.1499
95% LoA: [-12.6938, 16.2952]
```

95% CI for LoA:

```
Lower LoA: [-35.4958, -9.2515]
Upper LoA: [11.6452, 22.4088]
```

```
ep09_f_interpolated <- bland_altman(
  ep09_f_mcr,
  type = "percent", x_axis = "reference",
  percent_denominator = "reference",
  method = "nonparametric", median_ci_method = "interpolated",
  bootstrap_replicates = 1000, seed = 2026
)
```

Bland-Altman

```
Type:      percent
Y-axis:    (y - x) / x * 100
X-axis:    x
Percent base: reference
Method:    nonparametric
n = 100
```

```

Median diff:      -0.3345
95% requested CI for median diff (actual 95.0%): [-1.9970, 1.1783]
SD diff:          9.1499
95% LoA: [-12.6938, 16.2952]

95% CI for LoA:
  Lower LoA: [-35.4958, -9.2515]
  Upper LoA: [11.6452, 22.4088]

```

```

ep09_f_hl <- bland_altman(
  ep09_f_mcr,
  type = "percent", x_axis = "reference",
  percent_denominator = "reference",
  method = "hodges_lehmann",
  bootstrap_replicates = 1000, seed = 2026
)

```

Bland-Altman

```

Type:      percent
Y-axis:    (y - x) / x * 100
X-axis:    x
Percent base: reference
Method:    hodges_lehmann
n = 100

```

```

HL pseudomedian: 0.0460
95% CI for hl pseudomedian: [-1.4034, 1.6075]
SD diff:          9.1499
95% LoA: [-12.6938, 16.2952]

```

```

95% CI for LoA:
  Lower LoA: [-35.4958, -9.2515]
  Upper LoA: [11.6452, 22.4088]

```

Table 9.4: EP09c Appendix F: median-bias comparison

Standard location	Statistic	Standard	ivdtools
EP09c Appendix F	median	-0.335	-0.3345
EP09c Appendix F	position_40	-2.020	-2.0202
EP09c Appendix F	position_41	-1.990	-1.9871
EP09c Appendix F	position_60	1.000	1.0031
EP09c Appendix F	position_61	1.590	1.5873
EP09c Appendix F	interpolated_lower	-1.997	-1.9970
EP09c Appendix F	interpolated_upper	1.152	1.1783
EP09c Appendix F	HL_pseudomedian	-0.050	0.0460
EP09c Appendix F	HL_lower	-1.410	-1.4034
EP09c Appendix F	HL_upper	1.610	1.6075

The default `median_diff_ci` uses positions 40 and 61 to form a conservative order-statistic interval, with actual coverage of approximately 96.4%. The interpolated ranks are 40.7002 and 60.2998, giving -1.9970% to 1.1783% by linear interpolation of adjacent order statistics; the upper limit calculated from the public data differs from the 1.152% in the main text. Exact recalculation from Table F2 gives a Hodges–Lehmann pseudomedian of $+0.05\%$, whereas the standard text reports -0.05% ; the interval remains approximately -1.41% to 1.61% .

9.5.6 CLSI EP09c Appendix H: Iteratively Weighted Least Squares

Appendix H compares 120 platelet results from two analyzers. weights = “residual” first fits OLS, then estimates a linear SD function from absolute residuals and uses the inverse squared fitted SD as weights; weight_iterations = 4 updates the weights for four rounds as recommended by the standard.

```
ep09_h <- read_csv(
  "/data/009-方法学比较与参考物偏倚/std-ep09-appendix-h.csv",
  show_col_types = FALSE
)
ep09_h_mcr <- mcr(
  as.data.frame(ep09_h),
  id = "sample", candidate = "candidate", reference = "comparative"
)
ep09_h_wls <- regression(
  ep09_h_mcr,
  method = "wls", weights = "residual", weight_iterations = 4
)
```

Regression

```
Method: WLS
Weights: residual
Weight updates: 4
CI method: analytic
Intercept = 3.0202 (SE = 1.0724)
Slope      = 1.0209 (SE = 0.0070)
Intercept 95% CI: [0.8966, 5.1439]
Slope     95% CI: [1.0071, 1.0347]
H0: slope = 1 -> t = 2.9977, p = 0.003318
slope significantly different from 1 (CI excludes 1)
R-squared = 0.9945
Sigma     = 1.2216
n = 120
```

Regression

```
Method: WLS
Weights: residual
Weight updates: 4
CI method: analytic
Intercept = 3.0202 (SE = 1.0724)
Slope      = 1.0209 (SE = 0.0070)
Intercept 95% CI: [0.8966, 5.1439]
Slope     95% CI: [1.0071, 1.0347]
H0: slope = 1 -> t = 2.9977, p = 0.003318
slope significantly different from 1 (CI excludes 1)
R-squared = 0.9945
Sigma     = 1.2216
n = 120
```

Table 9.5: EP09c Appendix H: WLS comparison

Standard location	Statistic	Standard	ivdtools
EP09c Appendix H	intercept	3.013	3.0202
EP09c Appendix H	intercept_se	1.073	1.0724
EP09c Appendix H	intercept_lower	0.889	0.8966

Standard location	Statistic	Standard	ivdtools
EP09c Appendix H	intercept_upper	5.138	5.1439
EP09c Appendix H	slope	1.021	1.0209
EP09c Appendix H	slope_se	0.007	0.0070
EP09c Appendix H	slope_lower	1.007	1.0071
EP09c Appendix H	slope_upper	1.035	1.0347
EP09c Appendix H	regression_se	1.222	1.2216

The recalculated intercept is 3.020 versus 3.013 in the standard. Small interval differences may arise from unreported input precision or the standard's recalculation process; they should not be interpreted as a different WLS formula.

9.5.7 CLSI EP09c Appendix I: Passing–Bablok Parameters and Bias Intervals

`ci_method = "auto"` selects the rank interval from Appendix I for Passing–Bablok regression parameters. Bias at a specified medical decision level is a joint function of the regression parameters.

```
ep09_i_bias <- bias(
  ep09_c_pb, mdl = 1,
  interval = "confidence",
  bootstrap_replicates = 5000, seed = 202609
)
```

Bias

```
Method: Passing-Bablok
Interval: confidence (95%)
MDL points: 1
Bias range: [0.0083, 0.0083]
```

```
mdl      bias      ci_lower  ci_upper
-----
1 0.008343138 -0.01802986 0.02214245
```

The bias interval is obtained by resampling complete (X, Y) pairs; the point estimate remains the original Passing–Bablok fit. The returned object's `ci_method` attribute is `bootstrap_percentile`.

9.5.8 CLSI EP09c Appendix J: Precision-Profile Weighted Deming

Appendix J requires the variance function for each method to be incorporated into the errors-in-variables regression. `variance_models` can receive functions that return variances or precision objects that have already been processed by `variance()` and `profile()`. Values outside the range of means in the precision study generate a warning; predicted nonpositive or nonfinite variances generate an error.

```
ep09_j_precision_data <- read_csv(
  "./data/013-精密度分析/precision-profile.csv",
  show_col_types = FALSE
)
ep09_j_precision <- precision(
  as.data.frame(ep09_j_precision_data), value ~ day, by = "sample"
)
```

```
ep09_j_precision <- variance(ep09_j_precision)
```

Variance components

sample	component	VC	%Total	SD	CV[%]	Method
L1	day	0.6376	63.0	0.7985	3.9446	anova
L1	error	0.3750	37.0	0.6124	3.0250	anova
L2	day	5.7139	75.1	2.3904	4.7584	anova
L2	error	1.8937	24.9	1.3761	2.7393	anova
L3	day	3.3913	15.7	1.8415	1.8088	anova
L3	error	18.1466	84.3	4.2599	4.1840	anova
L4	day	30.5549	54.6	5.5276	2.7529	anova
L4	error	25.4525	45.4	5.0451	2.5126	anova
L5	day	0.0000	0.0	0.0000	0.0000	anova
L5	error	171.7986	100.0	13.1072	3.3003	anova

```
ep09_j_precision <- profile(ep09_j_precision)
```

Precision profiles -- Sadler

total: $\sigma^2 = 0.0000 + 0.0086 * u^J$ (AIC = 56.45, $R^2 = 0.9847$)

day: $\sigma^2 = 0.0000 + 0.0095 * u^J$ (AIC = 43.46, $R^2 = 0.8919$)

error: $\sigma^2 = 0.0010 * u^2$ (AIC = 55.96, $R^2 = 0.9806$)

```
# Generate teaching data to demonstrate the analysis.
ep09_j_data <- data.frame(
  sample = 1:8,
  reference = seq(25, 350, length.out = 8)
)
ep09_j_data$candidate <- 1 + 1.03 * ep09_j_data$reference +
  c(-1.0, 0.8, -1.2, 1.1, -0.7, 0.9, -0.6, 0.7)

ep09_j_mcr <- mcr(
  ep09_j_data, "sample", "candidate", "reference"
)
ep09_j_profile_fit <- regression(
  ep09_j_mcr, method = "wdeming", weights = "precision_profile",
  variance_models = list(
    reference = ep09_j_precision,
    candidate = ep09_j_precision
  )
)
```

Regression

Method: Precision-profile Deming

Weights: precision_profile

CI method: jackknife

Intercept = 0.0131 (SE = 1.3393)

Slope = 1.0360 (SE = 0.0074)

Intercept 95% CI: [-3.2639, 3.2902]

Slope 95% CI: [1.0178, 1.0542]

H0: slope = 1 -> t = 4.8363, p = 0.002892

slope significantly different from 1 (CI excludes 1)

Sigma = 1.0955

n = 8

9.5.9 CLSI EP09c Appendix K2: Regression and Bias Bootstrap

The percentile method takes parameter-sample quantiles directly. The standard method constructs intervals from the bootstrap standard error, intercept-slope covariance, and $t(n-2)$. Each replicate resamples complete sample pairs and refits the selected model.

```
ep09_k_percentile <- regression(
  ep09_c_mcr, method = "ols",
  ci_method = "bootstrap_percentile",
  bootstrap_replicates = 5000, seed = 202610, mdl = c(0.5, 2)
)
```

Regression

```
Method: OLS
CI method: bootstrap_percentile
Bootstrap: 5000 effective / 5000 requested (0 failed)
Intercept = -0.3804 (SE = 0.1738)
Slope      = 1.0697 (SE = 0.0358)
Intercept 95% CI: [-0.6645, 0.0425]
Slope      95% CI: [0.9822, 1.1270]
H0: slope = 1 -> t = 1.9441, p = 0.05554
slope not significantly different from 1 (CI contains 1)
R-squared = 0.9921
Sigma     = 1.5576
n = 79
```

```
ep09_k_standard <- regression(
  ep09_c_mcr, method = "ols",
  ci_method = "bootstrap_standard",
  bootstrap_replicates = 5000, seed = 202610, mdl = c(0.5, 2)
)
```

Regression

```
Method: OLS
CI method: bootstrap_standard
Bootstrap: 5000 effective / 5000 requested (0 failed)
Intercept = -0.3804 (SE = 0.1738)
Slope      = 1.0697 (SE = 0.0358)
Intercept 95% CI: [-0.7265, -0.0343]
Slope      95% CI: [0.9983, 1.1410]
H0: slope = 1 -> t = 1.9441, p = 0.05554
slope not significantly different from 1 (CI contains 1)
R-squared = 0.9921
Sigma     = 1.5576
n = 79
```

Table 9.6: Appendix K2: two bootstrap parameter intervals

method	parameter	estimate	lower	upper
percentile	intercept	-0.38040	-0.66451	0.04253
percentile	slope	1.06965	0.98218	1.12699
standard	intercept	-0.38040	-0.72652	-0.03428
standard	slope	1.06965	0.99831	1.14099

`regression$bootstrap` records the random seed, requested, valid, and failed replicate

counts, and `parameter_samples` stores the joint parameter draws. When `mdl` is supplied during regression, bias draws are stored as well. `bias()` and `predict()` reuse draws from the same method. Prediction intervals are an `ivdtools` extension and should not be called `EP09c` regression-parameter or bias-confidence intervals.

9.5.10 YY/T 1789.2–2021 Appendix A: Reference-Material Bias

Appendix A uses national serum-cholesterol reference materials at three levels, with six replicate measurements at each level under repeatability conditions. `reference_bias()` combines expanded uncertainty of the measurement mean with reference-material uncertainty to form the bias interval.

```
yyt_a <- read_csv(
  "./data/009-方法学比较与参考物偏倚/std-yyt1789-2-appendix-a.csv",
  show_col_types = FALSE
)
yyt_a_result <- reference_bias(
  as.data.frame(yyt_a),
  result = "result",
  reference = "assigned",
  reference_uncertainty = "reference_uncertainty",
  level = "level",
  k = 2
)
yyt_a_result
```

Reference Material Bias

Standard: YY/T 1789.2-2021, Section 5.3

Data: `as.data.frame(yyt_a)`

Result: `result`

Levels: 3

Records: 18 valid, 0 excluded

Coverage factor: `k = 2`

Measurement and uncertainty

Level: L1

```
n      : 6
Mean   : 192.35
SD     : 0.60909769
Reference : 197.6
U_ref  : 2.5
u_X    : 0.24866309
U_X    : 0.49732618
```

Level: L2

```
n      : 6
Mean   : 149.8
SD     : 0.35777088
Reference : 152.4
U_ref  : 1.9
u_X    : 0.14605935
U_X    : 0.2921187
```

Level: L3

```
n      : 6
Mean   : 119.5
SD     : 0.42426407
```

Reference : 122.1
 U_ref : 1.6
 u_X : 0.17320508
 U_X : 0.34641016

Bias interval

Level: L1

Bias : -5.25
 Bias (%) : -2.6568826
 Interval half-width : 2.5489867
 Lower : -7.7989867
 Upper : -2.7010133

Level: L2

Bias : -2.6
 Bias (%) : -1.7060367
 Interval half-width : 1.922325
 Lower : -4.522325
 Upper : -0.67767502

Level: L3

Bias : -2.6
 Bias (%) : -2.1294021
 Interval half-width : 1.6370706
 Lower : -4.2370706
 Upper : -0.96292945

Interval: Bias +/- sqrt($U_X^2 + U_{ref}^2$)

Table 9.7: YY/T 1789.2 Appendix A: reference-material bias

Standard location	Statistic	Standard	ivdtools
Appendix A L1	mean	192.35	192.3500
Appendix A L1	sd	0.61	0.6091
Appendix A L1	standard_uncertainty	0.25	0.2487
Appendix A L1	expanded_uncertainty	0.50	0.4973
Appendix A L1	bias	-5.20	-5.2500
Appendix A L1	lower	-7.70	-7.7990
Appendix A L1	upper	-2.70	-2.7010
Appendix A L2	mean	149.80	149.8000
Appendix A L2	sd	0.36	0.3578
Appendix A L2	standard_uncertainty	0.15	0.1461
Appendix A L2	expanded_uncertainty	0.30	0.2921
Appendix A L2	bias	-2.60	-2.6000
Appendix A L2	lower	-4.50	-4.5223
Appendix A L2	upper	-0.70	-0.6777
Appendix A L3	mean	119.50	119.5000
Appendix A L3	sd	0.42	0.4243
Appendix A L3	standard_uncertainty	0.17	0.1732
Appendix A L3	expanded_uncertainty	0.34	0.3464
Appendix A L3	bias	-2.60	-2.6000
Appendix A L3	lower	-4.20	-4.2371
Appendix A L3	upper	-1.00	-0.9629

The means, SDs, uncertainties, and bias intervals at all three levels can be recalculated directly from the 18 raw results. At Level 1, the exact bias is -5.25 mg/dL while the standard displays -5.2 mg/dL; the exact lower bound of the combined interval is -7.80 mg/dL while

the standard displays -7.7 mg/dL. These are differences caused by the standard's stepwise display rounding.

9.5.11 YY/T 1789.2–2021 Appendix B: Patient-Sample Bias and Regression

Appendix B provides complete paired results for 120 patient samples. The comparative procedure is the X axis and the evaluated product is the Y axis. The standard analyzes the 75 samples with comparative results below 100 mg/dL using absolute differences and the 45 samples at or above 100 mg/dL using relative differences.

```
yyt_b <- read_csv(
  "/data/009-方法学比较与参考物偏倚/std-yyt1789-2-appendix-b.csv",
  show_col_types = FALSE
)
head(yyt_b, 6)
```

```
# A tibble: 6 x 3
  sample comparative evaluated
  <dbl>         <dbl>         <dbl>
1     1           21           23
2     2          128          130
3     3           43           40
4     4          125          128
5     5          165          163
6     6          223          228
```

```
yyt_b_data_list <- split_by_cutpoints(
  as.data.frame(yyt_b),
  variable = "comparative",
  cutpoints = 100,
  labels = c("low", "high")
)
yyt_b_data_list
```

Data split by cutpoints
Variable: comparative
Cutpoints: 100

low	$[-\text{Inf}, 100)$	n = 75
high	$[100, \text{Inf})$	n = 45

```
yyt_b_mcr <- lapply(
  yyt_b_data_list,
  \(d) mcr(
    d,
    id = "sample", candidate = "evaluated", reference = "comparative"
  )
)
yyt_b_mcr$low <- describe(yyt_b_mcr$low)
```

Descriptive Statistics

Variable	n	Mean	SD	Min	Q1	Median	Q3	Max	NA
comparative (X)	75	60.4533	24.8647	12.0000	37.5000	65.0000	81.5000	98.0000	

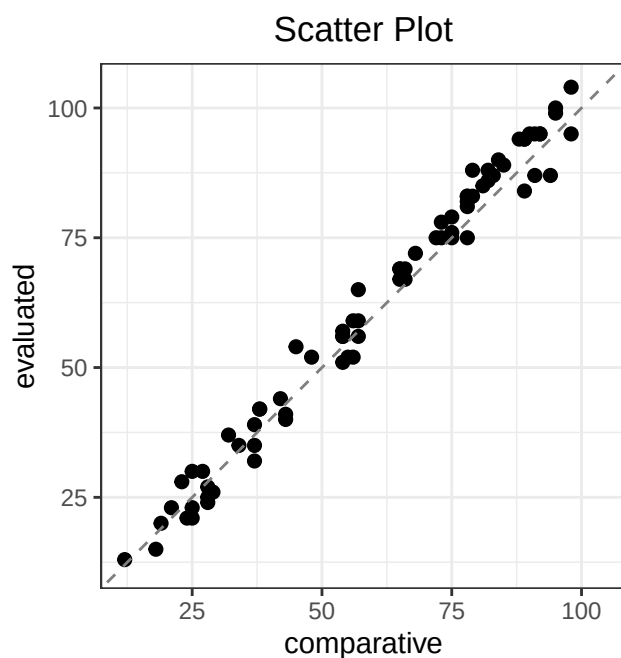
evaluated (Y)	75	62.2800	26.1847	13.0000	39.5000	67.0000	85.5000	104.0000
Diff (Y-X)	75	1.8267	3.6331	-7.0000	-1.5000	3.0000	4.0000	9.0000

```
yyt_b_mcr$high <- describe(yyt_b_mcr$high)
```

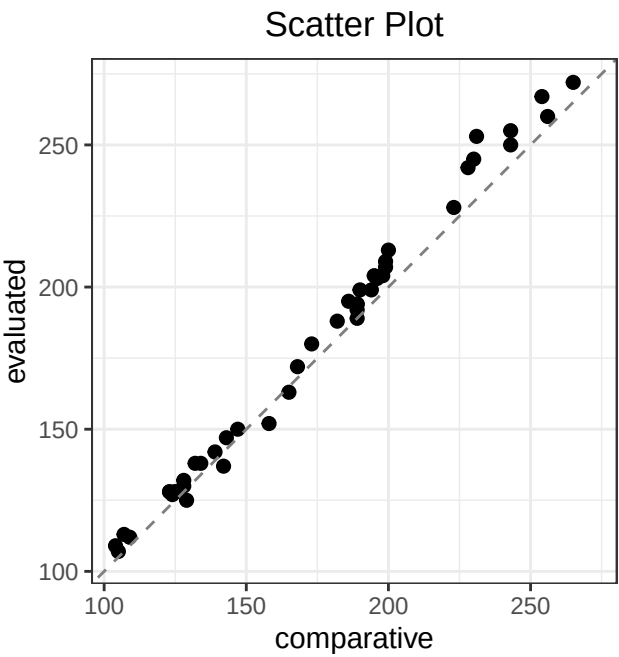
Descriptive Statistics

Variable	n	Mean	SD	Min	Q1	Median	Q3	Max	NA
comparative (X)	45	173.3111	45.8782	104.0000	129.0000	182.0000	199.0000	265.0000	
evaluated (Y)	45	178.8444	48.9554	107.0000	132.0000	188.0000	207.0000	272.0000	
Diff (Y-X)	45	5.5333	5.1416	-6.0000	3.0000	5.0000	7.0000	22.0000	

```
plot(yyt_b_mcr$low, type = "scatter")
```



```
plot(yyt_b_mcr$high, type = "scatter")
```



9.5.11.1 Generalized ESD Outlier Test

Difference analysis requires evaluated product minus comparative procedure, so the outlier objects pass the evaluated product as candidate.

```
yyt_b_low_mcr <- yyt_b_mcr$low
yyt_b_high_mcr <- yyt_b_mcr$high
yyt_b_low_outlier <- outlier(
  yyt_b_low_mcr, method = "esd", type = "difference", alpha = 0.05, r = 3
)
```

Outlier
Method: esd
Data: Difference (evaluated - comparative)
Alpha: 0.05

No outliers detected.

```
yyt_b_high_outlier <- outlier(
  yyt_b_high_mcr, method = "esd", type = "percent", alpha = 0.05, r = 3
)
```

Outlier
Method: esd
Data: Percent diff ((evaluated - comparative) / comparative * 100)
Alpha: 0.05

No outliers detected.

Table 9.8: YY/T 1789.2 Appendix B: generalized ESD conclusions

interval	difference_scale	maximum_tests	detected_outliers
<100 mg/dL	absolute difference	3	0
>=100 mg/dL	relative difference	3	0

9.5.11.2 Segmented Bias

Call `bland_altman()` separately on the two prespecified `mcr` objects to obtain absolute differences in the low range and relative differences in the high range.

```
yyt_b_low_ba <- bland_altman(
  yyt_b_low_mcr, type = "difference", x_axis = "mean"
)
```

Bland-Altman

Type: difference
 Y-axis: evaluated - comparative
 X-axis: Mean of evaluated and comparative
 Method: parametric
 n = 75

Mean diff: 1.8267
 95% CI for mean diff: [0.9908, 2.6626]
 SD diff: 3.6331
 95% LoA: [-5.2940, 8.9474]

95% CI for LoA:
 Lower LoA: [-6.7289, -3.8591]
 Upper LoA: [7.5125, 10.3823]

```
yyt_b_high_ba <- bland_altman(
  yyt_b_high_mcr, type = "percent", x_axis = "mean",
  percent_denominator = "reference"
)
```

Bland-Altman

Type: percent
 Y-axis: (evaluated - comparative) / comparative * 100
 X-axis: Mean of evaluated and comparative
 Percent base: reference
 Method: parametric
 n = 45

Mean diff: 3.0020
 95% CI for mean diff: [2.2441, 3.7600]
 SD diff: 2.5229
 95% LoA: [-1.9427, 7.9468]

95% CI for LoA:
 Lower LoA: [-3.2477, -0.6377]
 Upper LoA: [6.6418, 9.2518]

Table 9.9: YY/T 1789.2 Appendix B: segmented bias comparison

Standard location	Statistic	Standard	ivdtools
Appendix B low	mean_bias	1.8267	1.8267
Appendix B low	sd_bias	3.6331	3.6331
Appendix B low	lower	1.0000	0.9908
Appendix B low	upper	2.7000	2.6626
Appendix B high	mean_relative_bias	3.0000	3.0020
Appendix B high	sd_relative_bias	2.5230	2.5229

Standard location	Statistic	Standard	ivdtools
Appendix B high	lower	2.6000	2.2441
Appendix B high	upper	4.0000	3.7600

The confidence intervals differ somewhat because the statistical definitions are not identical.

9.5.11.3 Four Regression Models

The regression is evaluated-product-on-comparative-procedure, so the evaluated product is passed as candidate (Y) and the comparative procedure as reference (X). WLS uses weights = “residual” and iterates the standard’s absolute-residual SD function four times.

```
yyt_b_reg_mcr <- mcr(
  as.data.frame(yyt_b),
  id = "sample", candidate = "evaluated", reference = "comparative"
)
yyt_b_ols <- regression(yyt_b_reg_mcr, method = "ols")
```

Regression

```
Method: OLS
CI method: analytic
Intercept = -0.8416 (SE = 0.6630)
Slope      = 1.0395 (SE = 0.0055)
Intercept 95% CI: [-2.1545, 0.4713]
Slope      95% CI: [1.0287, 1.0503]
H0: slope = 1 -> t = 7.2214, p = 5.448e-11
slope significantly different from 1 (CI excludes 1)
R-squared = 0.9967
Sigma      = 3.8533
n = 120
```

```
yyt_b_deming <- regression(yyt_b_reg_mcr, method = "deming")
```

Regression

```
Method: Deming
CI method: jackknife
Intercept = -1.0230 (SE = 0.6625)
Slope      = 1.0413 (SE = 0.0066)
Intercept 95% CI: [-2.3348, 0.2889]
Slope      95% CI: [1.0283, 1.0542]
H0: slope = 1 -> t = 6.2964, p = 5.403e-09
slope significantly different from 1 (CI excludes 1)
Sigma      = 3.8550
n = 120
```

```
yyt_b_pb <- regression(yyt_b_reg_mcr, method = "pb")
```

Regression

```
Method: Passing-Bablok
CI method: rank
Intercept = 0.0250 (SE = 0.7022)
Slope      = 1.0375 (SE = 0.0075)
```

```

Intercept 95% CI: [-0.8786, 1.0124]
Slope      95% CI: [1.0248, 1.0485]
H0: slope = 1 -> t = 4.9790, p = 2.204e-06
slope significantly different from 1 (CI excludes 1)
Sigma      = 3.9129
n = 120

```

```

yyt_b_wls <- regression(
  yyt_b_reg_mcr,
  method = "wls", weights = "residual", weight_iterations = 4
)

```

Regression

```

Method: WLS
Weights: residual
Weight updates: 4
CI method: analytic
Intercept = -0.6980 (SE = 0.6193)
Slope      = 1.0380 (SE = 0.0058)
Intercept 95% CI: [-1.9243, 0.5283]
Slope      95% CI: [1.0266, 1.0495]
H0: slope = 1 -> t = 6.5761, p = 1.383e-09
slope significantly different from 1 (CI excludes 1)
R-squared = 0.9963
Sigma      = 1.2988
n = 120

```

Table 9.10: YY/T 1789.2 Appendix B: regression parameters

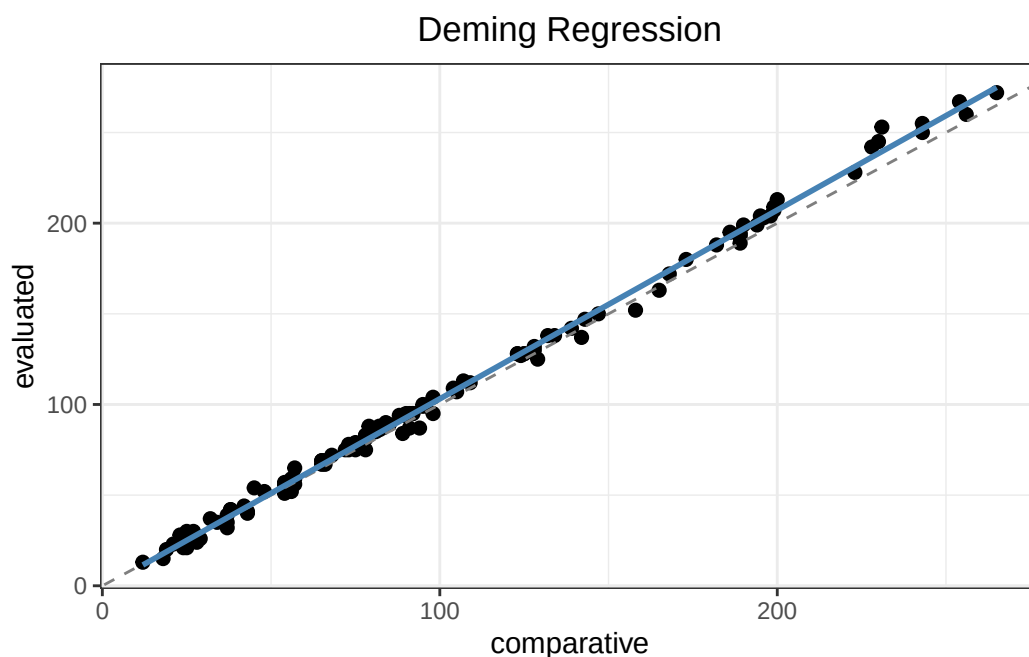
Standard location	Statistic	Standard	ivdtools
Appendix B OLS	intercept	-0.842	-0.8416
Appendix B OLS	slope	1.039	1.0395
Appendix B WLS	intercept	-0.700	-0.6980
Appendix B WLS	slope	1.038	1.0380
Appendix B Deming	intercept	-1.023	-1.0230
Appendix B Deming	slope	1.041	1.0413
Appendix B Passing-Bablok	intercept	0.025	0.0250
Appendix B Passing-Bablok	slope	1.038	1.0375

The WLS intercept differs in the final displayed digits, possibly because of iteration stopping or display precision. The comparison retains both the standard and function values.

```

plot(yyt_b_deming, type = "regression")

```



9.5.11.4 Bias at 125 mg/dL

The standard direction is evaluated product minus comparative procedure, namely intercept + (slope - 1) × 125, which is also `bias()`'s default direction.

```
yyt_b_ols_bias <- bias(
  yyt_b_ols, mdl = 125, level = 0.95, interval = "confidence"
)
```

Bias

Method: OLS
Interval: confidence (95%)
MDL points: 1
Bias range: [4.0943, 4.0943]

mdl	bias	ci_lower	ci_upper
125	4.094271	3.357293	4.831248

```
yyt_b_wls_bias <- bias(
  yyt_b_wls, mdl = 125, level = 0.95, interval = "confidence"
)
```

Bias

Method: WLS
Interval: confidence (95%)
MDL points: 1
Bias range: [4.0575, 4.0575]

mdl	bias	ci_lower	ci_upper
125	4.057517	3.271421	4.843613

```
yyt_b_deming_bias <- bias(
  yyt_b_deming, mdl = 125, level = 0.95, interval = "confidence"
)
```

Bias

Method: Deming
 Interval: confidence (95%)
 MDL points: 1
 Bias range: [4.1335, 4.1335]

```
mdl    bias ci_lower ci_upper
-----
125 4.133481 3.299802 4.96716
```

Table 9.11: YY/T 1789.2 Appendix B: bias at 125 mg/dL

Standard location	Statistic	Standard	ivdtools
Appendix B OLS MDL	bias	4.1	4.0943
Appendix B OLS MDL	lower	3.4	3.3573
Appendix B OLS MDL	upper	4.8	4.8312
Appendix B WLS MDL	bias	4.1	4.0575
Appendix B WLS MDL	lower	3.5	3.2714
Appendix B WLS MDL	upper	4.7	4.8436
Appendix B Deming MDL	bias	4.1	4.1335
Appendix B Deming MDL	lower	3.4	3.2998
Appendix B Deming MDL	upper	4.9	4.9672

The WLS and Deming differences concern confidence-interval construction. The chapter retains the EP09c defaults and does not alter confidence level, random seed, or weight iterations to force agreement with a published value.

The WLS difference comes from whether the weighted-regression residual scale is applied a second time. YY/T 1789.2 Equation (8) uses

$$SE\{\widehat{B}(x_c)\} = \sqrt{\frac{1}{\sum_i w_i} + \frac{(x_c - \bar{x}_w)^2}{SSX_w}}.$$

It does not multiply by the weighted residual scale $S_{y \cdot x}$. Here the degrees of freedom are 118, $t_{0.975, 118} = 1.980272$, the bias estimate is 4.057517 mg/dL, and the Equation (8) SE is 0.305646 mg/dL. The half-width is 0.605262 mg/dL, giving 3.4523–4.6628 mg/dL, displayed as 3.5–4.7 mg/dL.

The ivdtools WLS analytic interval agrees with EP09c Appendix H H17–H18 and multiplies that SE by $S_{y \cdot x}$. Here $S_{y \cdot x} = 1.298770$, giving adjusted SE 0.396964 mg/dL, half-width 0.786096 mg/dL, and interval 3.2714–4.8436 mg/dL. The half-width ratio is exactly 1.298770, so this is a formula choice about residual-scale correction, not random variation or decimal rounding. If weights are known absolute inverse variances from an independent precision study, the Equation (8) expression has a clear model basis; with weights iteratively estimated from the same residuals, residual-scale correction is more defensible. The standard also uses $S_{y \cdot x}$ for WLS regression-parameter SEs but not for Equation (8), so the two scale conventions differ.

For Deming, the bias SE uses intercept, slope, and covariance:

$$SE\{\widehat{B}(x_c)\} = \sqrt{\hat{\sigma}_a^2 + x_c^2 \hat{\sigma}_b^2 + 2x_c \hat{\sigma}_{ab}}.$$

YY/T 1789.2 uses its analytic covariance formula, whereas `ci_method = "auto"` in `ivdtools` follows EP09c Appendix K1: delete each sample, refit Deming, and calculate the covariance from 120 leave-one-out parameter sets. Here the bias estimate is 4.133481 mg/dL and the jackknife SE is 0.420992 mg/dL; the *t* multiplier gives a half-width of 0.833679 mg/dL and interval 3.2998–4.9672 mg/dL. YY/T 1789.2 publishes 3.4–4.9 mg/dL. Both intervals have explicit assumptions and should be reported as methodological differences, not calculation errors or simple rounding differences.

Chapter 10

Quality Control

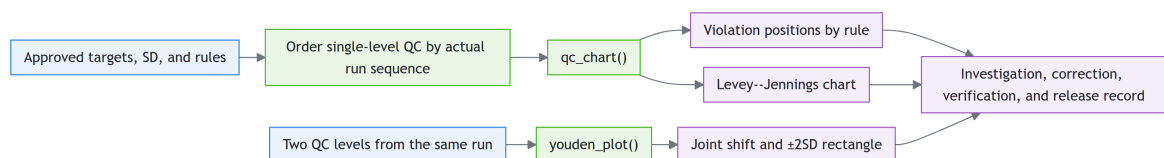
This chapter demonstrates how to use `ivdtools` to monitor continuous quality-control results. Current functionality includes listing implemented Westgard rules, creating a Levey–Jennings chart and locating rule violations, and using a Youden plot to inspect two QC levels jointly.

! Scope of Use

The routine QC, lot-change labels, and Youden data in this chapter are deterministic teaching data. They are not original standard data from CLSI EP23, EP26, or EP33 and are not intended to establish general acceptance criteria.

`qc_chart()` analyzes one numeric sequence at a time. Rules are scanned in the current row order of the data frame; `run` supplies only the x-axis for the plot and does not sort the data. `group` is used only for faceting and does not reset continuous rules between groups. The current severity labels mark only 1-3s as `out_of_control`; all other rule violations are labeled `warning`. Thus `n_warn` and `n_oc` are software display categories and are not equivalent to a laboratory multirule release strategy.

The statistical unit and workflow for QC analysis are:



```
library(ivdtools)
library(readr)
```

10.1 Main Functions

Function	Main purpose	Main results
<code>list_westgard()</code>	View rule names recognized by <code>qc_chart()</code>	Named vector of rule descriptions
<code>qc_chart()</code>	Standardize one QC sequence, scan rules, and create a Levey–Jennings chart	Targets, SD, positions by rule, unique violation points, and plotting data

Function	Main purpose	Main results
<code>print()</code>	Print a summary of a QC or Youden object	Targets, sample size, rules, or outside-rectangle positions
<code>plot()</code>	Plot a result object	Levey–Jennings or Youden plot
<code>youden_plot()</code>	Jointly inspect two QC levels from the same run	Pearson correlation, positions outside the $\pm 2SD$ rectangle, and plotting data

10.1.1 View Rules with `list_westgard()`

Usage

```
list_westgard()
```

Westgard QC Rules

```
-----
1-2s    1 point exceeds +/-2s (warning)
1-3s    1 point exceeds +/-3s (out of control)
2-2s    2 consecutive points exceed +/-2s (same side)
R-4s    2 consecutive points differ by at least 4s
4-1s    4 consecutive points exceed +/-1s (same side)
8x      8 consecutive points on same side of mean
10x     10 consecutive points on same side of mean
-----
```

Exact definitions in the current source code

Let the target mean be μ_0 and the target standard deviation be $\sigma_0 > 0$. The standardized deviation of result i is

$$z_i = \frac{x_i - \mu_0}{\sigma_0}.$$

Rule	ivdtools condition	Points marked
1-2s	$ z_i > 2$	Current point
1-3s	$ z_i > 3$	Current point
2-2s	Both adjacent points are > 2 , or both are < -2	The two points in the adjacent window
R-4s	Each adjacent point has $ z \geq 2$ and $ z_i - z_{i+1} \geq 4$	The two points in the adjacent window
4-1s	Four consecutive points are all > 1 , or all < -1	The four points in the window
8x	Eight consecutive points are all > 0 , or all < 0	The eight points in the window
10x	Ten consecutive points are all > 0 , or all < 0	The ten points in the window

Interpret inequality boundaries exactly as shown: 1-2s, 1-3s, 2-2s, and 4-1s use strict “greater than”; a point exactly at 2s, 3s, or 1s does not trigger the rule. The single-point threshold and two-point range for R-4s use “greater than or equal to.” A point may occur in several rules; `n_violations` counts unique positions across all rules.

Boundary of Multilevel Westgard Rules

Classical multirule schemes can combine rules across QC levels in the same run or across consecutive runs at the same level. The current `qc_chart()` scans only the vector supplied to it; R-4s and the other window rules are implemented over adjacent rows, and the function does not automatically recognize two levels from the same run. Multi-level data must not simply be stacked into one column and treated as a standard multi-level Westgard decision. `youden_plot()` can help inspect two levels, but it does not apply cross-level Westgard rejection rules.

10.1.2 Single-Sequence Control Chart with `qc_chart()`

Usage

```
qc_chart(
  data,
  value,
  rules = "1-2s,1-3s,2-2s,R-4s,4-1s,10x",
  mean = NULL,
  sd = NULL,
  group = NULL,
  run = NULL
)

print(x)
plot(x)
```

Arguments

Argument	Description
<code>data</code>	Data frame; each row is one QC result in the sequence.
<code>value</code>	Name of the numeric QC result column.
<code>rules</code>	Comma-separated string; only names listed by <code>list_westgard()</code> are allowed. 8x is not included by default.
<code>mean</code>	Target mean; if <code>NULL</code> , estimated from the current complete data. Routine monitoring should normally pass an approved target explicitly.
<code>sd</code>	Positive target SD; if <code>NULL</code> , estimated from the current complete data.
<code>group</code>	Optional grouping column used only for plot facets; targets are not estimated separately and rule windows are not reset.
<code>run</code>	Optional x-axis column; if <code>NULL</code> , uses 1, ..., <i>n</i> . It does not change scan order.

If different control materials, lots, or instruments need their own targets and independent rule resets, first split the data into a named list with `split_by_factors(data, variable="...")`, then call `qc_chart()` separately for each data frame. A `group` facet cannot replace independent analyses.

Calculations and returned values

Rows with missing `value` are removed before calculation; corresponding complete rows of `run` and `group` are retained. If targets are not supplied, the sample mean and sample SD of all complete results are used:

$$\hat{\mu} = \frac{1}{n} \sum_i x_i, \quad \hat{\sigma} = \sqrt{\frac{\sum_i (x_i - \hat{\mu})^2}{n - 1}}.$$

Estimating targets from the same monitoring sequence can absorb an ongoing shift and widen the control limits. It is therefore suitable only for exploratory or establishment phases and should not be re-estimated on a rolling basis for each routine decision.

Main returned fields are:

Field	Meaning
<code>target_mean</code> , <code>target_sd</code>	Targets actually used for standardization and control limits
<code>rule_vec</code> <code>violations</code>	Rules actually scanned and their order Named list; each element contains positions in the sequence after missing rows are removed
<code>n_violations</code>	Number of unique positions hit by any rule
<code>n_warn</code>	Number of unique points hit but not by 1-3s
<code>n_oc</code>	Number of unique points hit by 1-3s
<code>plot_data</code>	Complete values, x-axis, group, displayed severity, and rule labels
<code>n_total</code> , <code>n_complete</code> , <code>n_miss</code>	Original, complete, and missing record counts

`plot()` draws the target mean and the $\pm 1s$, $\pm 2s$, and $\pm 3s$ lines and labels points hit by rules. Positions in `violations` refer to the cleaned sequence and may not equal original data-frame row numbers; when missing values occur, trace records through `plot_data$run` or the original run identifier.

10.1.3 Two-Level Joint Plot with `youden_plot()`

Usage

```

youden_plot(
  data,
  sample1,
  sample2,
  mean1 = NULL,
  mean2 = NULL,
  sd1 = NULL,
  sd2 = NULL
)

print(x)
plot(x)

```

Arguments

Argument	Description
<code>data</code>	Wide-format data frame; each row must represent two levels from the same run.
<code>sample1, sample2</code>	Names of the two numeric QC-level columns.
<code>mean1, mean2</code>	Target means; if omitted, estimated from complete paired data.
<code>sd1, sd2</code>	Positive target SDs; if omitted, estimated from complete paired data.

Only rows with both levels nonmissing are retained. Standardized coordinates are

$$z_{1i} = \frac{x_{1i} - \mu_1}{\sigma_1}, \quad z_{2i} = \frac{x_{2i} - \mu_2}{\sigma_2}.$$

The source code defines a point as inside the rectangle when

$$|z_{1i}| \leq 2 \quad \text{and} \quad |z_{2i}| \leq 2.$$

If either level strictly exceeds $\pm 2s$, the point is included in `outside_rows`. Boundary points remain inside. The object also calculates the Pearson correlation between the two original result columns:

$$r = \frac{\sum_i (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)}{\sqrt{\sum_i (x_{1i} - \bar{x}_1)^2 \sum_i (x_{2i} - \bar{x}_2)^2}}.$$

Main fields include `mean1`, `mean2`, `sd1`, `sd2`, `correlation`, `n_outside`, `outside_rows`, and `plot_data`. `outside_rows` contains original data-frame row numbers, unlike the cleaned positions returned by `qc_chart()`.

The current plot displays both levels on their original measurement scales and overlays $\pm 1s$ and $\pm 2s$ rectangles. To determine whether the two levels shifted in the same direction, compare the directions of (z_1, z_2) relative to the center. Do not interpret an original-scale $y = x$ reference line, with different targets or units, as meaning that the two levels should have equal numerical values. `correlation` and the outside-rectangle flag are descriptive results; there is no error ellipse, joint probability test, or automatic out-of-control conclusion.

Return value

`youden_plot()` returns two-level coordinates, ellipse/limit data, and outlier classifications, and invisibly returns a `ggplot` object.

10.2 Analysis Examples

10.2.1 Routine QC with Known Targets

Read 30 teaching results ordered by run:

```
qc_daily <- read_csv(
  "./data/010-质量控制/qc-daily.csv",
  show_col_types = FALSE
)
qc_daily <- as.data.frame(qc_daily)
dim(qc_daily)
```

```
[1] 30 2
```

```
head(qc_daily)
```

```
run value
1  1 103.24
2  2 103.06
3  3  94.52
4  4  99.51
5  5  99.14
6  6  95.62
```

The protocol specifies a target mean of 100 and target SD of 3, so pass them explicitly rather than re-estimating them from the monitoring data:

```
qc_daily_fit <- qc_chart(
  qc_daily,
  value = "value",
  rules = "1-2s,1-3s,2-2s,R-4s,4-1s,10x",
  mean = 100,
  sd = 3,
  run = "run"
)
print(qc_daily_fit)
```

QC Control Chart

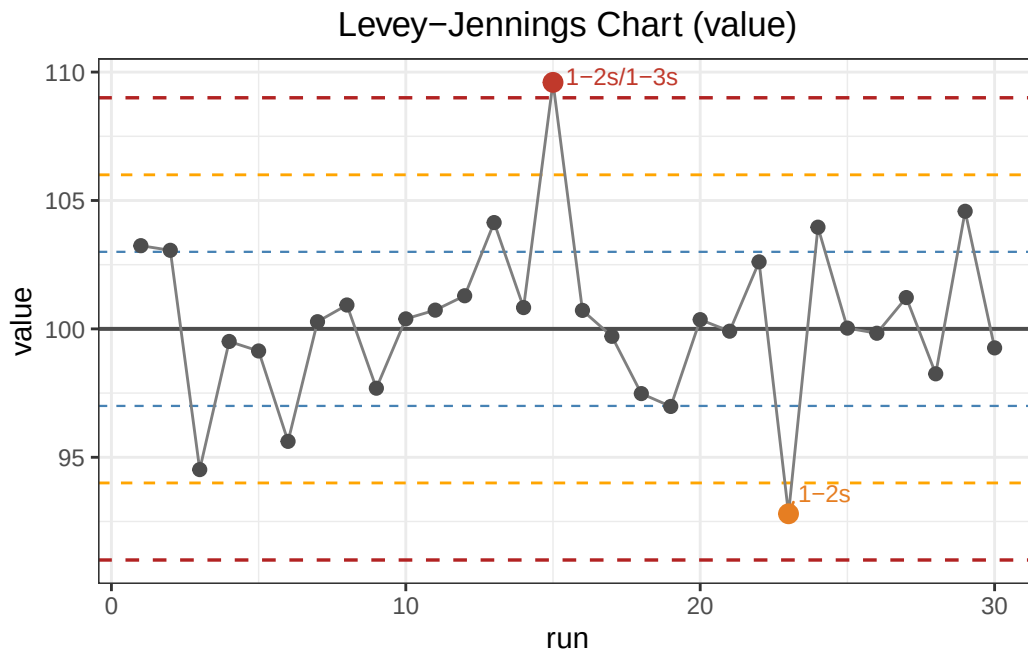
```
Value column:  value
Mean (target): 100.0000
SD (target):   3.0000
N (total):     30 | Complete: 30 | Missing: 0
Rules applied: 1-2s, 1-3s, 2-2s, R-4s, 4-1s, 10x
```

Violations:

```
-----
1-2s : points 15, 23
1-3s : points 15
-> 2 violation(s): 1 warning(s), 1 out of control
```

Point 15 meets both 1-2s and 1-3s and is counted as one `out_of_control` point; point 23 meets only 1-2s and is counted as one warning point. Therefore `n_violations=2`, `n_warn=1`, and `n_oc=1`.

```
plot(qc_daily_fit)
```

10.2.1.1 Known Targets versus In-Data Estimates

If mean and sd are omitted, the function re-estimates the center and dispersion from these 30 results:

```
qc_daily_estimated <- qc_chart(
  qc_daily,
  value = "value",
  run = "run"
)
qc_daily_estimated
```

QC Control Chart

```
Value column:  value
Mean (target): 100.2890
SD (target):   3.2314
N (total):     30 | Complete: 30 | Missing: 0
Rules applied: 1-2s, 1-3s, 2-2s, R-4s, 4-1s, 10x
```

Violations:

```
-----
1-2s : points 15, 23
-> 2 violation(s): 2 warning(s), 0 out of control
```

The in-data estimates are a mean of 100.2890 and an SD of 3.2314. Point 15 no longer strictly exceeds the new $+3s$ limit, so its display classification changes from `out_of_control` to `warning`. This is why routine monitoring should not update targets and SDs on a rolling basis using the current lot's data.

10.2.2 Continuous Display with Lot-Change Labels

The first, middle, and last 10 points of this teaching data are labeled L1, L2, and L3, respectively; L2 has an artificial overall shift:

```
qc_lot <- read_csv(
  "./data/010-质量控制/qc-lot-change.csv",
  show_col_types = FALSE
)
qc_lot <- as.data.frame(qc_lot)
```

```
qc_lot_fit <- qc_chart(
  qc_lot,
  value = "value",
  mean = 100,
  sd = 3,
  run = "run",
  group = "lot"
)
print(qc_lot_fit)
```

QC Control Chart

Value column: value
 Mean (target): 100.0000
 SD (target): 3.0000
 N (total): 30 | Complete: 30 | Missing: 0
 Rules applied: 1-2s, 1-3s, 2-2s, R-4s, 4-1s, 10x

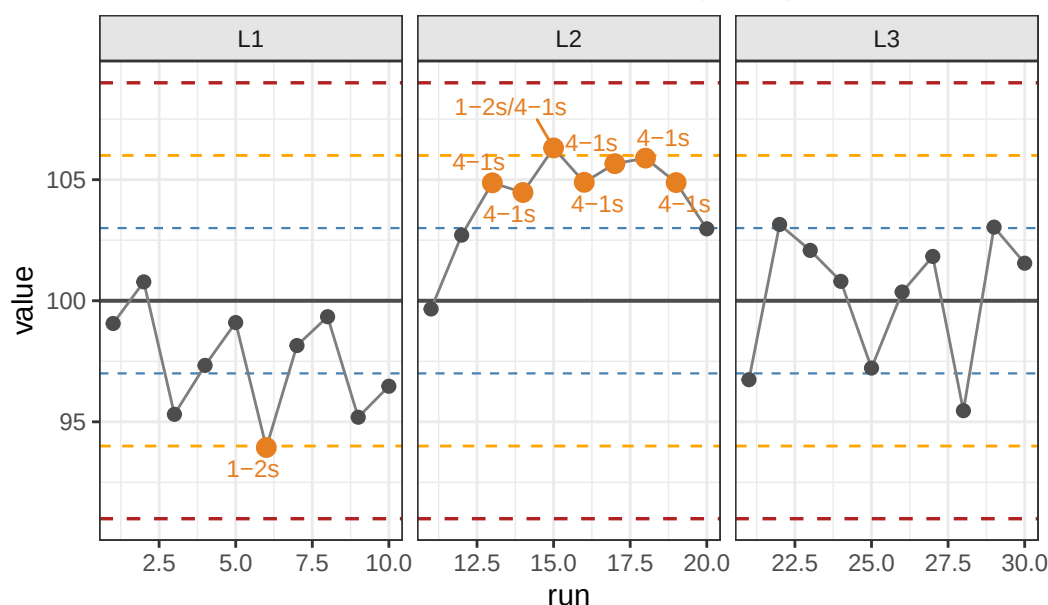
Violations:

 1-2s : points 6, 15
 4-1s : points 13-19
 -> 8 violation(s): 8 warning(s), 0 out of control

1-2s hits points 6 and 15; several overlapping 4-1s windows mark points 13–19. After combining the two rule types, there are 8 unique points, all classified as **warning** by the current severity implementation. Rule windows are scanned continuously across L1/L2/L3 boundaries; plot facets do not change these positions.

```
plot(qc_lot_fit)
```

Levey–Jennings Chart (value)



10.2.3 Two-Level Youden Plot

Each row contains two QC levels from the same run:

```
youden_data <- read_csv(
  "./data/010-质量控制/qc-youden.csv",
  show_col_types = FALSE
)
youden_data <- as.data.frame(youden_data)
head(youden_data)
```

```
run   low   high
1    1 77.86 209.36
2    2 82.19 207.10
3    3 75.50 189.00
4    4 80.40 195.36
5    5 80.64 198.23
6    6 78.22 200.14
```

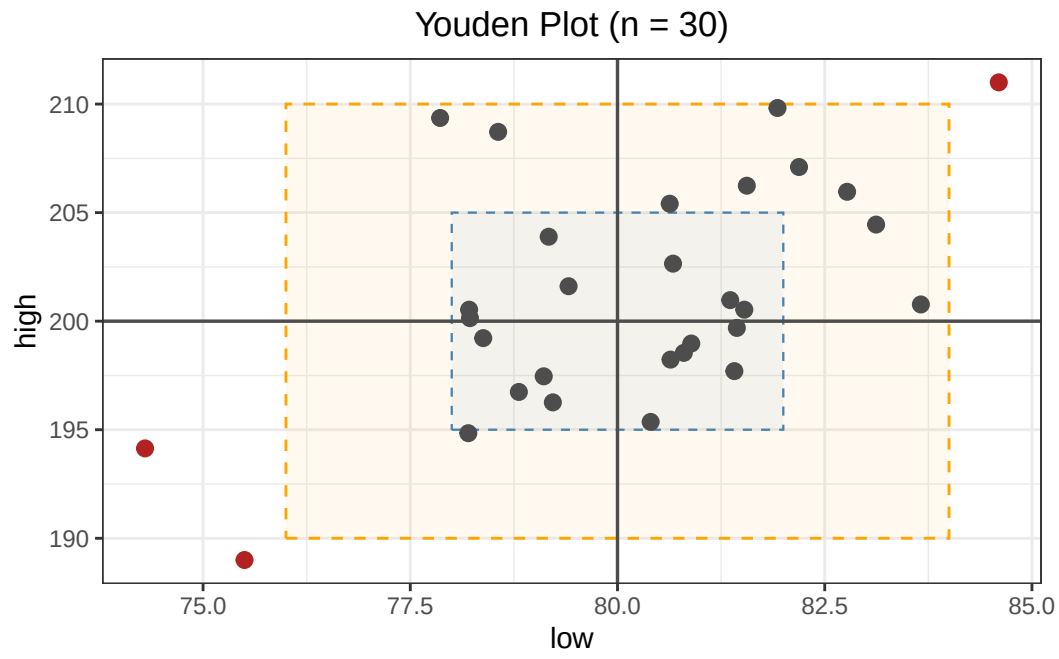
```
youden_fit <- youden_plot(
  youden_data,
  sample1 = "low",
  sample2 = "high",
  mean1 = 80,
  mean2 = 200,
  sd1 = 2,
  sd2 = 5
)
print(youden_fit)
```

Youden Plot

```
Sample 1 (low):
  Mean:  80.0000
   SD:   2.0000
Sample 2 (high):
  Mean:  200.0000
   SD:   5.0000
Correlation: 0.5428
N (total):  30 | Complete: 30 | Missing: 0
Points outside +/-2SD: 3 (rows: 3, 17, 19)
```

The Pearson correlation in this example is 0.5428; original rows 3, 17, and 19 lie outside the $\pm 2s$ rectangle. Correlation does not determine whether a process is out of control, and an outside-rectangle flag is only a localization aid. Use each run's two standardized deviations together with single-level control charts, instrument events, and QC records to distinguish a shared systematic shift from a problem affecting one level alone.

```
plot(youden_fit)
```



Chapter 11

ROC Curves and Cutoff Determination

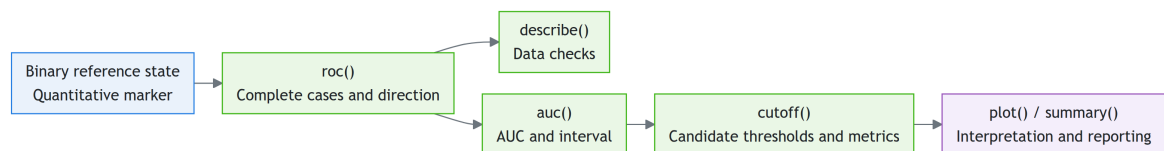
This chapter demonstrates how to use `ivdtools` to evaluate the ability of a quantitative marker to distinguish a target state and, after confirming the classification direction, inspect candidate cutoffs. An ROC curve describes the trade-off between sensitivity and specificity across thresholds, AUC summarizes the complete empirical ROC, and a cutoff represents a particular decision point. These answer different questions.

! Scope of Use

ROC conclusions assume that the reference state is correct, the study population is representative, and the positive class is defined correctly. `ivdtools` currently provides an empirical ROC, trapezoidal AUC, a Hanley–McNeil standard error for a single curve, and two within-sample cutoff criteria. It does not provide partial AUC, intervals at fixed sensitivity or specificity, paired ROC tests, resampling uncertainty for cutoffs, sample-size design, or external validation.

The function automatically selects the `geq` or `leq` direction by comparing marker means in the positive and negative groups; there is no separate direction argument. A formal analysis must verify that the automatic direction agrees with the prespecified clinical meaning. An in-sample “optimal” cutoff is subject to selection bias and cannot be interpreted directly as a validated threshold in an independent population or as a recommended standard cutoff.

The single-marker ROC workflow is:



```
library(ivdtools)
library(readr)
```

11.1 Main Functions

Function	Main purpose	Main result
<code>continuous_to_binary()</code>	Convert a continuous reference variable to 0/1 using a supplied threshold	Original data and a new <code><variable>_binary</code> column
<code>roc()</code>	Create an ROC object and determine common complete cases and direction	Sample information, reference encoding, marker values, and direction
<code>describe()</code>	Describe the raw variables in the object	Numeric summaries and categorical frequencies
<code>auc()</code>	Calculate empirical ROC, AUC, standard error, and 95% CI	<code>auc_list</code> and <code>auc_table</code>
<code>cutoff()</code>	Calculate diagnostic metrics at candidate thresholds and two optimization criteria	<code>cutoff_table</code>
<code>plot()</code>	Plot ROC or cutoff–metric curves	ggplot figure
<code>summary()</code>	Summarize analysis slots already run	ROC object summary

11.1.1 Convert a Continuous Reference Variable with `continuous_to_binary()`

```
continuous_to_binary(data, cols, cutoff)
```

Argument	Description
<code>data</code>	Data frame.
<code>cols</code>	One or more names of numeric columns to convert.
<code>cutoff</code>	A finite numeric value. A length-one value is recycled for all columns; otherwise it must have the same length as <code>cols</code> .

For variable j , the function applies

$$B_{ij} = I(X_{ij} \geq c_j).$$

The original columns are retained and `<variable>_binary` is added. Equality at the cutoff is always assigned to 1; the function does not support a `<=` direction. Use it only when this binary definition has already been established by the protocol or an external reference. Do not repeatedly search for a reference-variable threshold in the same data to improve ROC performance.

The function returns a data frame with the original columns and the new `<variable>_binary` columns.

11.1.2 Create an Analysis Object with `roc()`

```
roc(data, cols, reference, id = NULL, positive = NULL)
```

Argument	Description
<code>data</code>	Data frame.
<code>cols</code>	One or more numeric marker-column names.
<code>reference</code>	Name of one reference-state column; it may be numeric 0/1, a two-level factor, or a two-value character vector.
<code>id</code>	Optional subject identifier; used only to report repeated IDs and does not deduplicate automatically.
<code>positive</code>	Positive level for a factor or character reference. If omitted, the last level after sorting is used. For numeric 0/1, 1 is positive when omitted.

`roc()` requires the `reference` and all `cols` to be finite and complete. Define

$$R_i = I(Y_i \text{ is complete}) \prod_{j=1}^p I(X_{ij} \text{ is finite and complete}),$$

and include only subjects with $R_i = 1$ in the object. Even if `auc()` is later called for only a subset of `cols`, the analysis sample remains the common complete-case set determined by all markers when the object was created. Create separate objects if each marker should use its own complete cases.

For marker X_j , the source compares group means:

$$\bar{X}_{j,+} \geq \bar{X}_{j,-} \Rightarrow \hat{Y} = I(X_j \geq c) \quad (\text{geq}),$$

$$\bar{X}_{j,+} < \bar{X}_{j,-} \Rightarrow \hat{Y} = I(X_j \leq c) \quad (\text{leq}).$$

This is a computational direction, not a validation of clinical direction. An extreme value can affect a group mean and select the wrong direction; inspect `col_direction` and compare it with the protocol.

Main fields include `n`, `n_pos`, `n_neg`, `n_total`, `cols_values`, `reference_binary`, `col_direction`, and `print`, followed by `describe`, `auc_list`, `auc_table`, `cutoff_table`, and `mlr_results` as later analyses are added.

The returned object has class `roc`, stores common complete cases, reference states, marker values, and automatic directions, and is updated by subsequent analyses.

11.1.3 Descriptive Statistics with `describe()`

The function returns a descriptive-statistics data frame. Numeric columns contain sample size, mean, SD, quantiles, and range; categorical columns contain frequencies and percentages.

```
describe(x, cols = NULL, digits = 4L)
```

When `cols=NULL`, the function describes all columns in the original data except an explicitly supplied ID, not only ROC markers. Numeric columns provide n , mean, SD, minimum, quantiles, and maximum; factors, characters, and the reference column provide frequencies and percentages. The current implementation does not summarize marker distributions separately by positive and negative group, so group differences require a separate table or plot.

11.1.4 AUC with `auc()`

```
auc(x, cols = NULL)
```

The empirical ROC is calculated at every distinct observed value and at the two $\pm\infty$ thresholds, and the AUC is integrated by trapezoids over false-positive rate, $FPR = 1 - Sp$. Equivalently, it is the rank statistic

$$\hat{A} = \frac{1}{n_+ n_-} \sum_{i:Y_i=1} \sum_{j:Y_j=0} \left[I(X_i > X_j) + \frac{1}{2} I(X_i = X_j) \right],$$

with the order reversed when low values represent the positive class. AUC estimates the probability that a randomly selected positive and negative subject are correctly ordered by the selected direction; it is not the accuracy at a particular cutoff.

The current standard error uses the Hanley–McNeil approximation. Let

$$Q_1 = \frac{\hat{A}}{2 - \hat{A}}, \quad Q_2 = \frac{2\hat{A}^2}{1 + \hat{A}},$$

then

$$SE(\hat{A}) = \sqrt{\frac{\hat{A}(1 - \hat{A})(n_+ - 1)(Q_1 - \hat{A}^2)(n_- - 1)(Q_2 - \hat{A}^2)}{n_+ n_-}}.$$

The 95% CI is $\hat{A} \pm 1.96 SE(\hat{A})$, truncated to $[0, 1]$. When AUC is exactly 0 or 1, the current implementation returns NA for the standard error and interval rather than an exact or resampling interval.

The updated `roc` object stores ROC points, AUC, standard error, and interval for each marker in `auc_list`, with a summary table in `auc_table`.

11.1.5 Cutoff Analysis with `cutoff()`

```
cutoff(x, cols = NULL)
```

Run `auc()` first. For the `geq` direction, predicted positive means $X_i \geq c$; for `leq`, it means $X_i \leq c$. At each candidate threshold, calculate

$$\begin{aligned} Se &= \frac{TP}{TP + FN}, & Sp &= \frac{TN}{TN + FP}, \\ PPV &= \frac{TP}{TP + FP}, & NPV &= \frac{TN}{TN + FN}, & Accuracy &= \frac{TP + TN}{N}. \end{aligned}$$

precision in the table means PPV. The two automatic criteria are the maximum of

$$J(c) = Se(c) + Sp(c) - 1$$

(Youden) and the minimum of

$$d(c) = \sqrt{[1 - Se(c)]^2 + [1 - Sp(c)]^2}$$

(Corner). The candidate set consists of observed values and two infinite endpoints; the endpoints complete the ROC but are not reported as optimal finite cutoffs. If several finite thresholds tie, the source selects the first in its internal order: descending for `geq` and ascending for `leq`. Thus the automatic result also has a deterministic tie-breaking rule.

⚠ Observed-Value Cutoffs versus Open-Interval Cutoffs

Manual EP24 tables often place a cutoff in the open interval between adjacent observations. `ivdtools` uses observed values as thresholds and assigns equality to the positive side. A report must therefore state the decision rule, such as “results ≥ 9.5 are positive,” rather than only writing “cutoff 9.5”; open-interval endpoints and implemented thresholds must not be conflated.

The updated `roc` object stores candidate thresholds, confusion matrices, diagnostic metrics, and the two selection criteria in `cutoff_table`.

11.1.6 Plotting and Summary with `plot()` and `summary()`

```
plot(x, curves = NULL, cutpoint = NULL,
     type = c("roc", "cutoff"))
summary(x)
```

- `curves=NULL` selects all original markers by default; `curves="all"` selects all original markers and saved MLR models. A character vector such as `c("x1", "combined")` can mix both.
- `type="roc"` plots sensitivity against $1 - \text{Sp}$. An original marker requires `auc()` first. `cutpoint` can be "Youden", "Corner", or "all" and requires `cutoff()` first; it labels only the selected original marker.
- `type="cutoff"` plots sensitivity and specificity against finite thresholds for one original marker. Only one original marker may be specified in `curves`; MLR model probabilities are not currently supported.
- `summary()` summarizes analyses already present in the object; it does not automatically run `describe()`, `auc()`, or `cutoff()`.

`plot()` invisibly returns a `ggplot` object; `summary()` prints and invisibly returns the input `roc` object.

11.2 Analysis Examples

11.2.1 Teaching Data: Complete Single-Marker Workflow

```
roc_dat <- as.data.frame(read_csv(
  "./data/011-ROC曲线与cutoff/roc-example.csv",
  show_col_types = FALSE
))
head(roc_dat, 6)
```

	sid	ref	x1
1	1	0	7.594080
2	2	0	8.312513
3	3	0	5.885744
4	4	0	5.585100
5	5	0	9.568483
6	6	0	2.364555

```
roc_fit <- roc(
  roc_dat, cols = "x1", reference = "ref", id = "sid"
)
print(roc_fit)
```

ROC Analysis

```
Data class:      data.frame
Total rows:      100
Complete cases:  100
Reference:        ref (binary)
Reference coding: binary (0/1)
Positive / Neg:   50 / 50
Evaluation cols:  1 (x1)
Duplicate IDs:    none
```

Analysis slots:

```
[ ] Describe
[ ] AUC
[ ] Paired AUC comparison
[ ] Cutoff
[ ] MLR
```

```
roc_fit <- describe(roc_fit, cols = c("x1", "ref"))
```

Descriptive Statistics

Variable	n	Mean	SD	Min	Q1	Median	Q3	Max	NA
x1	100	9.2576	4.4744	-0.3894	5.8224	8.5472	12.9591	18.7295	

ref		
Level	Count	Percent
0	50	50.0%
1	50	50.0%

```
roc_fit <- auc(roc_fit)
```

AUC Table

Column	AUC	95% CI	Direction
x1	0.9420	[0.8940, 0.9900]	geq

The AUC in this example is 0.9420, with a Hanley–McNeil 95% CI of 0.8937–0.9903; the direction is `geq`. These results describe the apparent discrimination of the current 100 teaching cases.

```
roc_fit <- cutoff(roc_fit)
```

Optimal Cutoff Analysis

```
Column: x1
Youden / Corner: cutoff = 8.7617 (sens = 0.9000, spec = 0.9200, Youden = 0.8200)
```

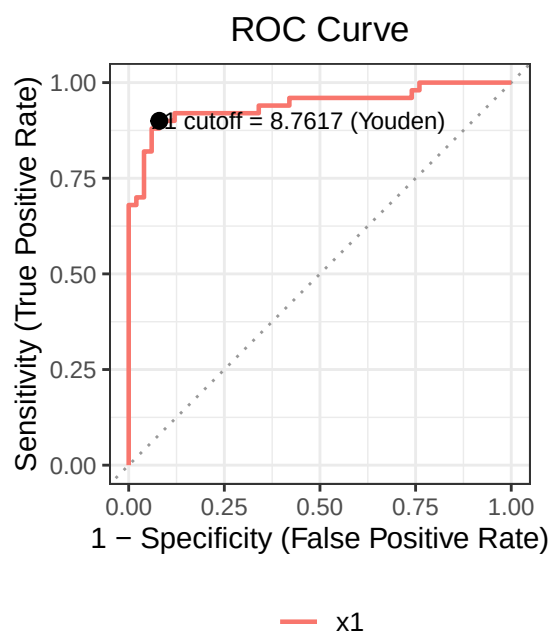
Full cutoff table:

Cutoff	Sens	Spec	TP	FP	TN	FN	Youden
Inf	0.0000	1.0000	0	0	50	50	0.0000
18.7295	0.0200	1.0000	1	0	50	49	0.0200
18.5675	0.0400	1.0000	2	0	50	48	0.0400
18.2248	0.0600	1.0000	3	0	50	47	0.0600
17.7837	0.0800	1.0000	4	0	50	46	0.0800
16.4453	0.1000	1.0000	5	0	50	45	0.1000
15.8893	0.1200	1.0000	6	0	50	44	0.1200
15.8087	0.1400	1.0000	7	0	50	43	0.1400
15.6058	0.1600	1.0000	8	0	50	42	0.1600
15.5003	0.1800	1.0000	9	0	50	41	0.1800
15.2855	0.2000	1.0000	10	0	50	40	0.2000
15.0168	0.2200	1.0000	11	0	50	39	0.2200
14.8639	0.2400	1.0000	12	0	50	38	0.2400
14.6980	0.2600	1.0000	13	0	50	37	0.2600
14.4757	0.2800	1.0000	14	0	50	36	0.2800
14.4469	0.3000	1.0000	15	0	50	35	0.3000
14.4042	0.3200	1.0000	16	0	50	34	0.3200
14.1952	0.3400	1.0000	17	0	50	33	0.3400
14.1938	0.3600	1.0000	18	0	50	32	0.3600
13.9382	0.3800	1.0000	19	0	50	31	0.3800
13.7845	0.4000	1.0000	20	0	50	30	0.4000
13.3718	0.4200	1.0000	21	0	50	29	0.4200
13.0699	0.4400	1.0000	22	0	50	28	0.4400
12.9970	0.4600	1.0000	23	0	50	27	0.4600
12.9854	0.4800	1.0000	24	0	50	26	0.4800
12.9659	0.5000	1.0000	25	0	50	25	0.5000
12.9569	0.5200	1.0000	26	0	50	24	0.5200
12.7812	0.5400	1.0000	27	0	50	23	0.5400
12.7534	0.5600	1.0000	28	0	50	22	0.5600
12.5227	0.5800	1.0000	29	0	50	21	0.5800
12.3440	0.6000	1.0000	30	0	50	20	0.6000
12.2962	0.6200	1.0000	31	0	50	19	0.6200
12.1394	0.6400	1.0000	32	0	50	18	0.6400
12.1354	0.6600	1.0000	33	0	50	17	0.6600
12.0393	0.6800	1.0000	34	0	50	16	0.6800
11.7418	0.6800	0.9800	34	1	49	16	0.6600
11.6642	0.7000	0.9800	35	1	49	15	0.6800
11.5082	0.7000	0.9600	35	2	48	15	0.6600
11.4038	0.7200	0.9600	36	2	48	14	0.6800
11.2133	0.7400	0.9600	37	2	48	13	0.7000
11.0696	0.7600	0.9600	38	2	48	12	0.7200
10.3902	0.7800	0.9600	39	2	48	11	0.7400
10.1719	0.8000	0.9600	40	2	48	10	0.7600
9.8836	0.8200	0.9600	41	2	48	9	0.7800
9.5685	0.8200	0.9400	41	3	47	9	0.7600
9.3526	0.8400	0.9400	42	3	47	8	0.7800
9.1944	0.8600	0.9400	43	3	47	7	0.8000
9.1032	0.8800	0.9400	44	3	47	6	0.8200
8.8558	0.8800	0.9200	44	4	46	6	0.8000
8.7617	0.9000	0.9200	45	4	46	5	0.8200
8.5538	0.9000	0.9000	45	5	45	5	0.8000
8.5406	0.9000	0.8800	45	6	44	5	0.7800
8.5296	0.9200	0.8800	46	6	44	4	0.8000
8.5259	0.9200	0.8600	46	7	43	4	0.7800
8.3125	0.9200	0.8400	46	8	42	4	0.7600
8.2985	0.9200	0.8200	46	9	41	4	0.7400

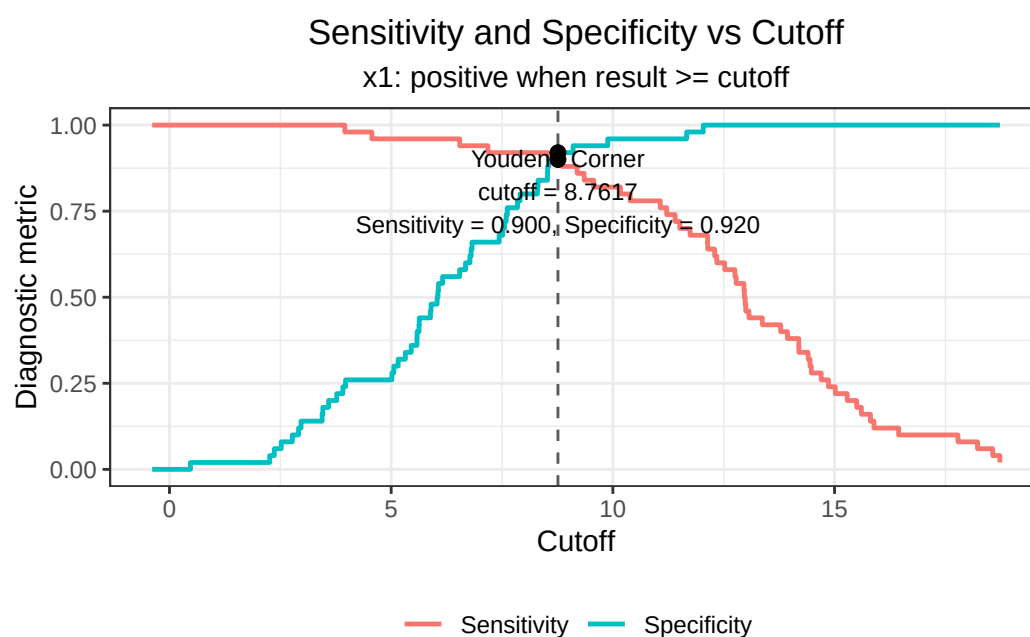
7.9078	0.9200	0.8000	46	10	40	4	0.7200
7.8493	0.9200	0.7800	46	11	39	4	0.7000
7.6187	0.9200	0.7600	46	12	38	4	0.6800
7.5941	0.9200	0.7400	46	13	37	4	0.6600
7.5556	0.9200	0.7200	46	14	36	4	0.6400
7.5150	0.9200	0.7000	46	15	35	4	0.6200
7.4355	0.9200	0.6800	46	16	34	4	0.6000
7.1837	0.9200	0.6600	46	17	33	4	0.5800
6.8235	0.9400	0.6600	47	17	33	3	0.6000
6.8029	0.9400	0.6400	47	18	32	3	0.5800
6.7765	0.9400	0.6200	47	19	31	3	0.5600
6.6770	0.9400	0.6000	47	20	30	3	0.5400
6.5446	0.9400	0.5800	47	21	29	3	0.5200
6.5440	0.9600	0.5800	48	21	29	2	0.5400
6.1603	0.9600	0.5600	48	22	28	2	0.5200
6.0665	0.9600	0.5400	48	23	27	2	0.5000
6.0554	0.9600	0.5200	48	24	26	2	0.4800
6.0361	0.9600	0.5000	48	25	25	2	0.4600
5.9000	0.9600	0.4800	48	26	24	2	0.4400
5.8857	0.9600	0.4600	48	27	23	2	0.4200
5.6326	0.9600	0.4400	48	28	22	2	0.4000
5.6281	0.9600	0.4200	48	29	21	2	0.3800
5.5851	0.9600	0.4000	48	30	20	2	0.3600
5.5802	0.9600	0.3800	48	31	19	2	0.3400
5.4514	0.9600	0.3600	48	32	18	2	0.3200
5.3207	0.9600	0.3400	48	33	17	2	0.3000
5.1582	0.9600	0.3200	48	34	16	2	0.2800
5.0599	0.9600	0.3000	48	35	15	2	0.2600
5.0167	0.9600	0.2800	48	36	14	2	0.2400
4.5654	0.9600	0.2600	48	37	13	2	0.2200
3.9706	0.9800	0.2600	49	37	13	1	0.2400
3.9552	0.9800	0.2400	49	38	12	1	0.2200
3.9103	1.0000	0.2400	50	38	12	0	0.2400
3.7719	1.0000	0.2200	50	39	11	0	0.2200
3.5881	1.0000	0.2000	50	40	10	0	0.2000
3.4583	1.0000	0.1800	50	41	9	0	0.1800
3.4422	1.0000	0.1600	50	42	8	0	0.1600
2.9673	1.0000	0.1400	50	43	7	0	0.1400
2.9111	1.0000	0.1200	50	44	6	0	0.1200
2.7698	1.0000	0.1000	50	45	5	0	0.1000
2.5203	1.0000	0.0800	50	46	4	0	0.0800
2.3646	1.0000	0.0600	50	47	3	0	0.0600
2.2596	1.0000	0.0400	50	48	2	0	0.0400
0.4733	1.0000	0.0200	50	49	1	0	0.0200
-0.3894	1.0000	0.0000	50	50	0	0	0.0000
-Inf	1.0000	0.0000	50	50	0	0	0.0000

Both criteria select 8.76. Because the direction is `geq`, this means $x_1 \geq 8.76$ is classified as positive. In the current sample, sensitivity is 0.90, specificity is 0.92, and the Youden index is 0.82. PPV and NPV also depend on the positive proportion in the study data and cannot be extrapolated directly to populations with a different prevalence.

```
plot(roc_fit, cutpoint = "Youden")
```



```
plot(roc_fit, curves = "x1", type = "cutoff", cutpoint = "all")
```



11.2.2 CLSI EP24-A2 Example 1: Empirical ROC and Threshold Intervals

EP24 Table 1 gives 8 subjects and 7 distinct measurement values; there are 4 positive and 4 negative cases, with one cross-group tie at 15.1 ng/mL.

```
ep24_example1 <- as.data.frame(read_csv(
  "./data/011-ROC曲线与cutoff/std-ep24-example1.csv",
  show_col_types = FALSE
))
ep24_example1_fit <- roc(
  ep24_example1, cols = "assay",
  reference = "status", id = "id"
)
```

```
ep24_example1_fit <- auc(ep24_example1_fit)
```

AUC Table

Column	AUC	95% CI	Direction
assay	0.7812	[0.4390, 1.0000]	geq

```
ep24_example1_fit <- cutoff(ep24_example1_fit)
```

Optimal Cutoff Analysis

Column: assay

Youden / Corner: cutoff = 9.5000 (sens = 0.7500, spec = 0.7500, Youden = 0.5000)

Full cutoff table:

Cutoff	Sens	Spec	TP	FP	TN	FN	Youden
Inf	0.0000	1.0000	0	0	4	4	0.0000
24.8000	0.2500	1.0000	1	0	4	3	0.2500
15.1000	0.5000	0.7500	2	1	3	2	0.2500
9.5000	0.7500	0.7500	3	1	3	1	0.5000
7.0000	0.7500	0.5000	3	2	2	1	0.2500
6.4000	1.0000	0.5000	4	2	2	0	0.5000
2.1000	1.0000	0.2500	4	3	1	0	0.2500
1.6000	1.0000	0.0000	4	4	0	0	0.0000
-Inf	1.0000	0.0000	4	4	0	0	0.0000

Table 11.1: EP24-A2 Example 1: empirical ROC and cutoff conventions

Standard position	Statistic	Standard result	ivdtools result
EP24-A2 Example 1	AUC	0.78	0.7812
EP24-A2 Example 1	Youden cutoff	NA	9.5000
EP24-A2 Example 1	Sensitivity	0.75	0.7500
EP24-A2 Example 1	Specificity	0.75	0.7500

EP24 Table 2 places the decision point between adjacent observations, whereas `ivdtools` uses the observed value 9.5 with a “ $\geq$ ” rule. This is a difference in threshold notation.

The Youden index also reaches the same maximum across 2.1–6.4. EP24 uses this example to show that the maximum sum and the closest point to the upper-left corner can yield two cutoffs. When Youden values tie, the current software takes 9.5 first under the descending `geq` candidate order and labels that row `Youden / Corner`. This is a difference in threshold representation and tie handling, not an error in sensitivity, specificity, or AUC.

11.2.3 CLSI EP24-A2 Appendix D: Single-Curve OxLDL AUC

Appendix D contains 50 subjects, including 28 diagnostic positives and 22 negatives. This section checks only the published OxLDL single-curve AUC, Hanley–McNeil standard error, and normal-approximation interval.

```
ep24 <- as.data.frame(read_csv(
  "/data/公共/std-ep24-appendix-d.csv",
  show_col_types = FALSE
))
ep24_oxldl <- roc(
```

```

ep24, cols = "oxldl",
reference = "diagnosis", id = "id"
)
ep24_oxldl <- auc(ep24_oxldl)

```

AUC Table

Column	AUC	95% CI	Direction
oxldl	0.7995	[0.6777, 0.9213]	geq

Table 11.2: EP24-A2 Appendix D: OxLDL single-curve comparison

Standard position	Statistic	Standard result	ivdtools result
OxLDL	AUC	0.800	0.799513
OxLDL	SE	0.062	0.062140
OxLDL	CI_lower	0.680	0.677718
OxLDL	CI_upper	0.920	0.921308

Appendix D does not recommend an OxLDL cutoff. The following automatically selected result is therefore only a teaching-derived result for the current data:

```
ep24_oxldl <- cutoff(ep24_oxldl)
```

Optimal Cutoff Analysis

Column: oxldl

Youden / Corner: cutoff = 71.0000 (sens = 0.6429, spec = 0.8636, Youden = 0.5065)

Full cutoff table:

Cutoff	Sens	Spec	TP	FP	TN	FN	Youden
Inf	0.0000	1.0000	0	0	22	28	0.0000
143.0000	0.0357	1.0000	1	0	22	27	0.0357
116.0000	0.0714	1.0000	2	0	22	26	0.0714
96.0000	0.1071	1.0000	3	0	22	25	0.1071
95.0000	0.1429	1.0000	4	0	22	24	0.1429
88.0000	0.1786	1.0000	5	0	22	23	0.1786
86.0000	0.2143	1.0000	6	0	22	22	0.2143
83.0000	0.2500	1.0000	7	0	22	21	0.2500
82.0000	0.2857	1.0000	8	0	22	20	0.2857
78.0000	0.3571	1.0000	10	0	22	18	0.3571
77.0000	0.4286	0.9545	12	1	21	16	0.3831
76.0000	0.5357	0.9545	15	1	21	13	0.4903
74.0000	0.5357	0.9091	15	2	20	13	0.4448
73.0000	0.5714	0.8636	16	3	19	12	0.4351
72.0000	0.6071	0.8636	17	3	19	11	0.4708
71.0000	0.6429	0.8636	18	3	19	10	0.5065
70.0000	0.6429	0.8182	18	4	18	10	0.4610
66.0000	0.7143	0.7273	20	6	16	8	0.4416
64.0000	0.7500	0.6818	21	7	15	7	0.4318
63.0000	0.7500	0.6364	21	8	14	7	0.3864
62.0000	0.7500	0.5909	21	9	13	7	0.3409
61.0000	0.7500	0.5455	21	10	12	7	0.2955
60.0000	0.8571	0.5000	24	11	11	4	0.3571
59.0000	0.8571	0.4545	24	12	10	4	0.3117

57.0000	0.8929	0.4545	25	12	10	3	0.3474
54.0000	0.8929	0.3636	25	14	8	3	0.2565
53.0000	0.9286	0.3636	26	14	8	2	0.2922
52.0000	0.9286	0.3182	26	15	7	2	0.2468
51.0000	0.9286	0.2727	26	16	6	2	0.2013
48.0000	0.9286	0.2273	26	17	5	2	0.1558
45.0000	0.9643	0.2273	27	17	5	1	0.1916
44.0000	0.9643	0.1818	27	18	4	1	0.1461
42.0000	0.9643	0.0909	27	20	2	1	0.0552
41.0000	1.0000	0.0909	28	20	2	0	0.0909
37.0000	1.0000	0.0455	28	21	1	0	0.0455
22.0000	1.0000	0.0000	28	22	0	0	0.0000
-Inf	1.0000	0.0000	28	22	0	0	0.0000

Chapter 12

ROC Curves and Multivariate Regression

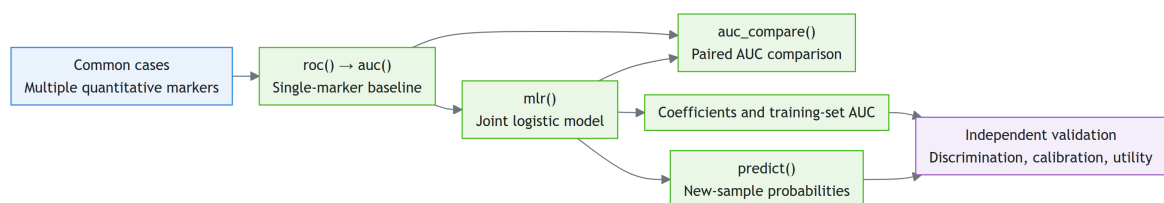
Building on single-marker ROC analysis, this chapter demonstrates how to fit a binomial logistic regression with multiple markers, calculate joint predicted probabilities, plot a training-set ROC, and predict new samples. A joint model must be evaluated for discrimination, calibration, clinical utility, and transportability; an increase in AUC is only one aspect.

! Scope of Use

`mlr()` fits an unpenalized, main-effects logistic regression for the supplied variables. The current implementation does not perform variable selection, automatically construct nonlinear terms, search for interactions, impute missing values, diagnose collinearity, perform internal validation, assess calibration, create decision curves, conduct external validation, or update models.

The joint AUC reported by `mlr()` uses the same data that fitted the model and therefore represents apparent performance. Its Hanley–McNeil interval also does not correct for optimism from model fitting and variable selection. Intervals from `predict()` are confidence intervals for the mean predicted probability due to model-coefficient uncertainty, not prediction intervals for future individual outcomes.

The multiple-marker workflow is:



```
library(ivdtools)
library(readr)
```

12.1 Main Functions

Function	Main purpose	Main result
<code>roc()</code>	Build a common complete-case object from multiple markers	Marker values, reference state, and single-marker directions
<code>auc()</code>	Establish a single-marker performance baseline	AUC, Hanley–McNeil SE, and CI for each marker
<code>mlr()</code>	Fit a named multivariate logistic model	Model, coefficients, predicted probabilities, and training ROC/AUC
<code>auc_compare()</code>	Compare two ROC curves from the same subjects	AUC difference, SE, CI, z, and P value
<code>predict()</code>	Predict training data or new data with a saved model	Probability, SE, CI, and fixed 0.5 classification
<code>plot()</code>	Overlay original-marker and joint-model ROCs	ggplot figure
<code>summary()</code>	Summarize all analyses already run	Single-marker and all named-model results

The complete arguments, direction rules, AUC, and interval formulas for `roc()` and `auc()` are given in the previous chapter. This chapter focuses on the joint-model interface.

12.1.1 Multivariate Logistic Regression with `mlr()`

```
mlr(x, cols = NULL, name = NULL, ...)
```

Argument	Description
<code>x</code>	A <code>roc</code> object on which <code>auc()</code> has already been run.
<code>cols</code>	Marker columns to enter the model; <code>NULL</code> uses all markers in the object.
<code>name</code>	Unique model name; if omitted, names are generated as <code>MLR01</code> , <code>MLR02</code> , and so on.
<code>...</code>	Other arguments passed to <code>stats::glm()</code> , such as control options; the formula and binomial family are fixed by the function.

Let $Y_i \in \{0, 1\}$. After selecting p markers, the current function fits

$$\eta_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij}, \quad \pi_i = P(Y_i = 1 \mid \mathbf{x}_i) = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}.$$

Coefficients are estimated by binomial maximum likelihood. Holding other variables constant, a one-unit increase in x_j multiplies the odds of a positive outcome by

$$\text{OR}_j = \exp(\beta_j).$$

This is not a risk ratio and does not represent a causal effect. If variables are measured in very small or very large units, rescale them before modeling according to the prespecified plan so that coefficients have interpretable units.

The formula is fixed as `reference_binary ~ x1 + x2 + ...`; squared terms, splines, and interactions are not added automatically. `roc()` has already established common complete

cases using all initial cols, so a submodel from `mlr(cols=...)` still uses the same analysis set. This helps compare nested marker combinations but is not a substitute for missing-data handling.

After fitting, an empirical ROC is constructed from the fitted probabilities $\hat{\pi}_i$ in the same cases, with direction fixed at `geq` so that larger probabilities indicate a greater tendency to be positive. The joint AUC SE and 95% CI continue to use the single-curve Hanley–McNeil formula from the previous chapter.

Each model is stored in `mlr_results[[name]]`. Main fields are:

Field	Meaning
<code>formula, cols</code>	Actual formula and predictors
<code>fit, fit_summary</code>	glm object and its summary
<code>predicted</code>	Fitted probabilities in the training set
<code>auc, auc_se, ci_lower, ci_upper</code>	Training-set AUC and approximate interval
<code>roc_curve</code>	Complete cutoff metric table for the training probabilities

The updated roc object is returned and the new model is stored in `mlr_results[[name]]`.

An AUC Increase Does Not Validate a Model

The single-marker and joint-model ROCs come from the same subjects and are correlated; use `auc_compare()` for a paired comparison. However, the joint probabilities are fitted from the same outcome data. EP24 and DeLong treat input scores as fixed and do not correct for model fitting, variable selection, or overfitting optimism. Therefore a paired P value cannot replace internal validation, independent validation, calibration, or assessment of clinical net benefit.

12.1.2 Paired AUC Comparison with `auc_compare()`

```
auc_compare(
  x,
  curves,
  method = c("ep24", "deLong"),
  conf.level = 0.95,
  rating_method = c("pearson", "kendall"),
  name = NULL
)
```

Argument	Description
<code>x</code>	The same roc object with the required <code>auc()</code> or <code>mlr()</code> analysis completed.
<code>curves</code>	Exactly two curve names: original markers, named MLR models, or one of each. The difference is first minus second.
<code>method</code>	Default "ep24", the EP24-A2 Hanley–McNeil paired approximation; "deLong" uses nonparametric paired covariance.
<code>conf.level</code>	Confidence level for the AUC-difference interval.

Argument	Description
<code>rating_method</code>	Within-group score-correlation method for EP24; Pearson is the default for continuous laboratory results and Kendall can be used for ordinal ratings. DeLong does not use this argument.
<code>name</code>	Optional result name; if omitted, it is generated from the two curve names and method.

The function accepts curves only from the same `roc` object, ensuring fully paired positive and negative states and complete cases. The result is stored in `auc_comparisons[[name]]` and contains both AUCs, their SEs, the difference, difference SE, confidence interval, z statistic, P value, and method-specific correlation or covariance information. `summary()` prints all saved comparisons.

The EP24 method first calculates the correlation between the two scores within the positive and negative groups, averages the two correlations, and combines that value with the average AUC to linearly interpolate the correlation between the two AUC estimates from EP24-A2 Table 5. If the coordinates fall outside the table range, the nearest boundary is used and the result and console output are marked explicitly. DeLong directly estimates the covariance of the two AUCs from paired rank contributions in the positive and negative groups and does not use the EP24 lookup approximation.

12.1.3 Probability Prediction with `predict()`

```
predict(
  object,
  mlr = NULL,
  newdata = NULL,
  column_map = NULL
)
```

Argument	Description
<code>object</code>	A <code>roc</code> object containing at least one <code>mlr()</code> result.
<code>mlr</code>	Model name. It may be omitted when the object contains one model and is required when it contains multiple models.
<code>newdata</code>	New data frame; <code>NULL</code> returns fitted results for the common complete-case training data.
<code>column_map</code>	Named character vector where names are new-data columns and values are training-model columns, such as <code>c(new_x1="x1")</code> .

For a new-sample design vector $\mathbf{x}_0 = (1, x_{01}, \dots, x_{0p})^\top$, the link-scale prediction and its standard error are

$$\hat{\eta}_0 = \mathbf{x}_0^\top \hat{\beta}, \quad \text{SE}(\hat{\eta}_0) = \sqrt{\mathbf{x}_0^\top \widehat{\text{Cov}}(\hat{\beta}) \mathbf{x}_0}.$$

The function first forms, on the logit scale,

$$\hat{\eta}_0 \pm z_{0.975} \text{SE}(\hat{\eta}_0),$$

and transforms the bounds back to the probability scale with logit^{-1} to obtain `ci_lower` and `ci_upper`. The probability-scale `pred_se` uses the delta method:

$$\text{SE}(\hat{\pi}_0) \approx \text{SE}(\hat{\eta}_0) \hat{\pi}_0 (1 - \hat{\pi}_0).$$

`pred_class` is fixed at $I(\hat{\pi}_0 \geq 0.5)$; the current interface cannot change this probability threshold. If a study protocol uses another decision threshold, classify `pred_prob` separately according to that prespecified rule and record it explicitly.

The function returns a data frame containing input row identifiers, `pred_prob`, `pred_se`, `ci_lower`, `ci_upper`, and `pred_class`.

12.1.4 ROC Plotting and Summary

`plot()` invisibly returns a `ggplot` object; `summary()` prints and invisibly returns the input `roc` object.

```
plot(x, curves = NULL, cutpoint = NULL,
     type = c("roc", "cutoff"))
summary(x)
```

`curves` uses the same curve namespace as `auc_compare()`: it can contain original marker names or MLR model names. `plot(x, curves="all")` draws all original markers and saved models; `plot(x, curves=c("x1", "combined"))` can mix them, while specifying only "combined" draws only that model. `curves=NULL` retains the default of plotting all original markers. `type="cutoff"` supports only one original marker in `curves`; cutoff plots for multivariate model probabilities are not currently supported. `cutpoint` labels only the selected original marker. `summary()` prints all saved models and `auc_compare()` results.

12.2 Analysis Examples

12.2.1 Teaching Data: A Joint Model with Two Markers

```
mlr_dat <- as.data.frame(read_csv(
  "./data/012-ROC曲线与多元回归/roc-multimarker.csv",
  show_col_types = FALSE
))
head(mlr_dat)
```

	sid	ref	x1	x2
1	1	0	5.510540	8.298655
2	2	0	3.188811	12.942968
3	3	0	8.126533	12.351778
4	4	0	6.159571	16.105370
5	5	0	11.118763	11.867919
6	6	0	2.736035	19.585368

First calculate the AUC of both original markers on the same common-case set:

```
mlr_fit <- roc(
  mlr_dat, cols = c("x1", "x2"),
  reference = "ref", id = "sid"
)
```

```
mlr_fit <- auc(mlr_fit)
```

AUC Table

Column	AUC	95% CI	Direction
x1	0.9219	[0.8712, 0.9727]	geq
x2	0.8322	[0.7587, 0.9057]	geq

```
mlr_fit
```

ROC Analysis

```
Data class:      data.frame
Total rows:      120
Complete cases:  120
Reference:       ref (binary)
Reference coding: binary (0/1)
Positive / Neg:  60 / 60
Evaluation cols: 2 (x1, x2)
Duplicate IDs:   none
```

Analysis slots:

```
[ ] Describe
[x] AUC
[ ] Paired AUC comparison
[ ] Cutoff
[ ] MLR
```

The AUCs of x1 and x2 are 0.9219 and 0.8322, respectively; both directions are `geq`. Next fit the prespecified main-effects model with two markers:

```
mlr_fit <- mlr(
  mlr_fit, cols = c("x1", "x2"), name = "combined"
)
```

Multivariate Logistic Regression (combined)

```
-----
Formula: reference_binary ~ x1 + x2
n = 120 (positive = 60, negative = 60)
AUC = 0.9642 [0.9298, 0.9985]
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-13.6586717	2.7569949	-4.954188	7.263286e-07
x1	0.7638502	0.1549103	4.930919	8.184356e-07
x2	0.3949617	0.1064673	3.709700	2.075049e-04

* Note: AUC is evaluated on the same data used for fitting; the estimate may be optimistic.

The fitted equation is

$$\text{logit}(\hat{\pi}) = -13.6587 + 0.76385x_1 + 0.39496x_2.$$

Holding the other marker constant, a one-unit increase in x1 and x2 multiplies the adjusted odds by approximately $\exp(0.76385) = 2.146$ and $\exp(0.39496) = 1.484$, respectively. The

exponentiated intercept usually has no practical interpretation because $x_1 = x_2 = 0$ may lie outside the data range.

The joint model has a training-set AUC of 0.9642 and a 95% CI of 0.9298–0.9985. Compare the joint model with each original marker using EP24 paired comparisons:

```
mlr_fit <- auc_compare(
  mlr_fit, curves = c("combined", "x1"), method = "ep24"
)
```

Paired AUC Comparison

Curves: combined - x1
 Method: EP24-A2 Hanley-McNeil paired approximation
 Subjects: 120 (positive = 60, negative = 60)
 AUC combined: 0.964167
 AUC x1: 0.921944
 Difference: 0.042222
 SE difference: 0.022111
 95.0% CI: [-0.001115, 0.085559]
 z = 1.909555
 p-value: 0.0561906
 Mean within-group rating correlation: 0.660442
 EP24 area correlation: 0.538934
 Note: fitted training probabilities are treated as fixed scores;
 model optimism and fitting uncertainty are not corrected.

```
mlr_fit <- auc_compare(
  mlr_fit, curves = c("combined", "x2"), method = "ep24"
)
```

Paired AUC Comparison

Curves: combined - x2
 Method: EP24-A2 Hanley-McNeil paired approximation
 Subjects: 120 (positive = 60, negative = 60)
 AUC combined: 0.964167
 AUC x2: 0.832222
 Difference: 0.131944
 SE difference: 0.035277
 95.0% CI: [0.062803, 0.201085]
 z = 3.740275
 p-value: 0.000183819
 Mean within-group rating correlation: 0.445998
 EP24 area correlation: 0.357226
 Note: fitted training probabilities are treated as fixed scores;
 model optimism and fitting uncertainty are not corrected.

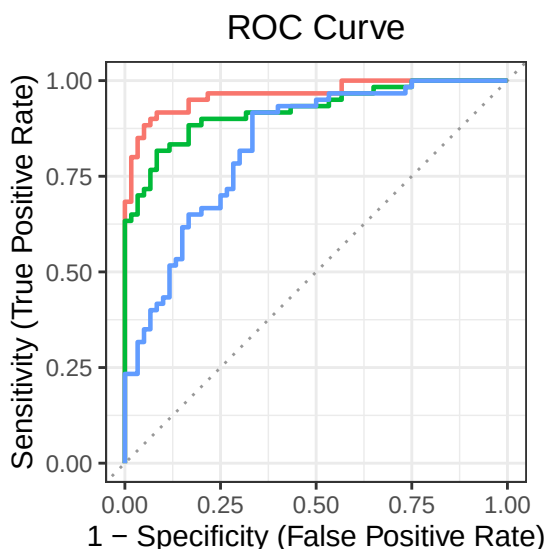
Table 12.1: Paired AUC comparison of the joint training model and single markers

comparison	auc_difference	SE	CI_lower	CI_upper	z	P_value
combined - x1	0.04222	0.02211	-0.00111	0.08556	1.90955	0.05619
combined - x2	0.13194	0.03528	0.06280	0.20109	3.74028	0.00018

These comparisons address correlation between curves caused by the same subjects. However, the joint probabilities are still fitted from these 120 cases; the paired test does not cor-

rect for model-fitting uncertainty or optimism and does not evaluate a validation set. The result cannot be described as confirmed clinical benefit.

```
plot(mlr_fit, curves = "all")
```



— combined (AUC=0.9642) — x1 — x2

Predict new samples and map column names:

```
new_patients <- data.frame(
  x1 = c(7, 12),
  x2 = c(16, 21)
)
new_predictions <- predict(
  mlr_fit, mlr = "combined", newdata = new_patients
)
new_predictions
```

	x1	x2	pred_prob	pred_se	ci_lower	ci_upper	pred_class
1	7	16	0.1200103	0.05313595	0.04840777	0.2677265	0
2	12	21	0.9781557	0.01751040	0.89984790	0.9955390	1

The predicted probabilities for the two cases are 0.1200 and 0.9782, displayed as 0 and 1 using the built-in 0.5 threshold. The interval reflects parameter uncertainty for the fitted mean probability; it does not include the future individual's binomial outcome randomness, model error, or population shift.

When new data use different column names, the mapping direction is “new column name = training column name” :

```
renamed_patients <- data.frame(
  marker_a = c(7, 12),
  marker_b = c(16, 21)
)
mapped_predictions <- predict(
  mlr_fit,
  mlr = "combined",
  newdata = renamed_patients,
  column_map = c(marker_a = "x1", marker_b = "x2")
)
```


)

12.2.2 CLSI EP24-A2 Appendix D: Two Paired ROCs

OxLDL and LDL in EP24 Appendix D were measured in the same 50 subjects, so their AUCs are correlated. First calculate both empirical AUCs and single-curve Hanley–McNeil intervals in one roc object, then call `auc_compare()`; common complete cases preserve the pairing.

```
ep24_pair <- as.data.frame(read_csv(
  "./data/公共/std-ep24-appendix-d.csv",
  show_col_types = FALSE
))
ep24_pair_fit <- roc(
  ep24_pair, cols = c("oxldl", "ldl"),
  reference = "diagnosis", id = "id"
)
ep24_pair_fit <- auc(ep24_pair_fit)
```

AUC Table

Column	AUC	95% CI	Direction
oxldl	0.7995	[0.6777, 0.9213]	geq
ldl	0.5617	[0.4012, 0.7222]	geq

```
ep24_pair_fit
```

ROC Analysis

```
Data class:      data.frame
Total rows:      50
Complete cases:  50
Reference:       diagnosis (binary)
Reference coding: binary (0/1)
Positive / Neg:  28 / 22
Evaluation cols: 2 (oxldl, ldl)
Duplicate IDs:   none
```

Analysis slots:

```
[ ] Describe
[x] AUC
[ ] Paired AUC comparison
[ ] Cutoff
[ ] MLR
```

The default EP24 method uses standard equation (6):

$$SE(\hat{A}_1 - \hat{A}_2) = \sqrt{SE_1^2 + SE_2^2 - 2r SE_1 SE_2},$$

where r is the AUC correlation interpolated from EP24 Table 5 using the average within-group score correlation and average AUC. DeLong does not use this lookup approximation; it estimates the covariance from paired rank contributions for each positive and negative subject.

Define

$$\phi(x, y) = I(x > y) + \frac{1}{2}I(x = y).$$

For marker k , with m positives and n negatives, construct

$$V_{10,i}^{(k)} = \frac{1}{n} \sum_{j=1}^n \phi(X_{ki}, Y_{kj}), \quad V_{01,j}^{(k)} = \frac{1}{m} \sum_{i=1}^m \phi(X_{ki}, Y_{kj}).$$

The DeLong covariance estimate for the two-marker AUC vector is

$$\widehat{\Sigma} = \frac{\text{Cov}(\mathbf{V}_{10})}{m} + \frac{\text{Cov}(\mathbf{V}_{01})}{n}.$$

Let $\mathbf{c} = (1, -1)^\top$. The standard error of $\widehat{\Delta} = \widehat{A}_{\text{OxLDL}} - \widehat{A}_{\text{LDL}}$ is

$$\text{SE}(\widehat{\Delta}) = \sqrt{\mathbf{c}^\top \widehat{\Sigma} \mathbf{c}}.$$

```
ep24_pair_fit <- auc_compare(
  ep24_pair_fit,
  curves = c("oxldl", "ldl"),
  method = "ep24"
)
```

Paired AUC Comparison

Curves: oxldl - ldl
 Method: EP24-A2 Hanley-McNeil paired approximation
 Subjects: 50 (positive = 28, negative = 22)
 AUC oxldl: 0.799513
 AUC ldl: 0.561688
 Difference: 0.237825
 SE difference: 0.075261
 95.0% CI: [0.090315, 0.385334]
 z = 3.159979
 p-value: 0.0015778
 Mean within-group rating correlation: 0.511818
 EP24 area correlation: 0.481818
 Table lookup used bounded coordinates: correlation 0.5118, AUC 0.7000

Compare live values with the EP24 published values in the check table:

Table 12.2: EP24-A2 Appendix D: single-curve and paired-difference comparison

Standard position	Statistic	Standard result	ivdtools result
OxLDL	AUC	0.8000	0.799513
OxLDL	SE	0.0620	0.062140
OxLDL	CI_lower	0.6800	0.677718
OxLDL	CI_upper	0.9200	0.921308
LDL	AUC	0.5600	0.561688
LDL	SE	0.0820	0.081895
LDL	CI_lower	0.4000	0.401174
LDL	CI_upper	0.7200	0.722203
Difference	AUC_difference	0.2400	0.237825
Difference	SE	0.0750	0.075261

Standard position	Statistic	Standard result	ivdtools result
Difference	CI_lower	0.0900	NA
Difference	CI_upper	0.3900	NA
Difference	z	3.1600	3.159979
Difference	p_value	0.0016	0.001578

Chapter 13

Precision Analysis

Precision evaluates agreement among results obtained from the same measurement system under repeatable conditions, usually expressed as SD or CV%. CLSI EP05 separates within-run, between-run, between-day, and related variance components and reports their estimates and confidence intervals for comparison with acceptance limits.

This chapter demonstrates how to use `precision()` to build a variance-component analysis, perform outlier and normality checks, estimate variance components and intervals, fit a Sadler precision profile, and rearrange results into EP05-style tables.



```
library(ivdtools)
library(readr)
```

13.1 Main Functions

Function	Main purpose	Key input or output
<code>precision()</code>	Create a precision variance-component object	Nested formula, <code>by</code> , balance checks
<code>outlier()</code>	Detect outliers by sample	Grubbs or IQR
<code>normal()</code>	Test normality by sample	Shapiro–Wilk and related methods
<code>variance() / vc()</code>	Estimate variance components	VC, % total, SD, CV%, <code>NegVC</code>
<code>ci()</code>	Confidence intervals for components and total SD/CV	Satterthwaite approximation
<code>profile()</code>	Fit a Sadler precision profile	Ten candidate models, AIC selection
<code>report()</code>	Generate EP05-structured tables	Explicit <code>day</code> , <code>run</code> , and <code>site</code> mapping
<code>list_sadler()</code>	View the ten Sadler models	Model formulas and types
<code>summary() / plot()</code>	Summaries and figures	<code>plot(type=...)</code>

After creating an object, analysis results accumulate through `outlier()`, `normal()`, `variance()`, `ci()`, and `profile()`. **Reassign the object after every call.** Only completed analyses appear in `summary()`, and some plots require a preceding analysis.

```
precision(data, form, by = NULL)
variance(x, digits = 4, method = c("auto", "anova", "reml"), NegVC = FALSE)
ci(x, digits = 4)
profile(object, model.no = 1:10)
report(x, day, run = NULL, site = NULL,
       table = c("simple", "detailed", "CI"), digits = 4)
```

`precision()` ultimately calls `VCA::anovaVCA()` and requires an ordinary `data.frame`; convert `readr::read_csv()` tibbles before use and convert design factors such as `day`, `lot`, and `replicate` to factors.

For a two-level nested design,

$$Y_{drk} = \mu + D_d + R_{r(d)} + \varepsilon_{drk},$$

with independent random terms of mean zero,

$$\sigma_T^2 = \sigma_D^2 + \sigma_{R(D)}^2 + \sigma_e^2, \quad SD_T = \sqrt{\sigma_T^2}, \quad CV_T = 100 SD_T / \bar{Y}.$$

The right side of `form` must represent the actual random hierarchy; the replicate layer is usually the residual error. `by` fits different samples or concentrations separately and does not combine them into one common variance. With `method="auto"`, `VCA::isBalanced()` selects `anovaVCA()` for balanced data and `remlVCA()` otherwise; estimates and degrees of freedom can differ, so preserve the actual `Method` in reports.

`variance()` stores numeric values in `results[[sample]]$vc`, `sd_comp`, and `cv_comp`; formatted display tables are separate. Use the numeric components for later calculations. Run `ci()` after `variance()`; it uses `VCA::VCAinference()` for two-sided SD/CV intervals. `profile()` fits $SD = f(\bar{Y}; \theta)$ across at least three valid levels and selects a Sadler/VFP candidate by AIC; extrapolation outside observed concentrations requires caution. `report()` maps components to EP05 terminology and can calculate within-laboratory and reproducibility intervals from component covariances and Satterthwaite degrees of freedom.

13.2 Analysis Examples

13.2.1 EP05 20×2×2 Single-Sample Precision

Twenty days, two runs per day, and two replicates per run produce 80 results:

```
ep05 <- read_csv("./data/013-精密度分析/precision-ep05.csv", show_col_types = FALSE)
ep05 <- as.data.frame(ep05)
ep05$day <- factor(ep05$day)
ep05$run <- factor(ep05$run)
ep05$rep <- factor(ep05$rep)
head(ep05, 6)
```

	day	run	rep	value
1	1	1	1	96.94
2	1	1	2	93.48
3	1	2	1	99.28
4	1	2	2	101.54
5	2	1	1	102.96
6	2	1	2	102.85

```
p <- precision(ep05, value ~ day/run)
p <- outlier(p, method = "grubbs")
```

Outlier detection -- Grubbs (alpha = 0.05)
No outliers detected.

```
p <- normal(p, method = "shapiro")
```

Normality test -- Shapiro-Wilk

Sample	n	Method	Statistic	p-value
all	80	Shapiro-Wilk	W=0.9868	0.5858

```
p <- variance(p)
```

Variance components

sample	component	VC	%Total	SD	CV[%]	Method
all	day	18.8358	42.6	4.3400	4.3788	anova
all	day:run	17.4343	39.4	4.1754	4.2128	anova
all	error	7.9482	18.0	2.8193	2.8445	anova

```
p <- ci(p)
```

Confidence intervals (SD)

sample	component	estimate	lower	upper
all	total	6.6497	5.4288	8.5842
all	day	4.3400	0.0000	6.2255
all	day:run	4.1754	2.0129	5.5513
all	error	2.8193	2.3147	3.6073

Confidence intervals (%CV)

sample	component	estimate	lower	upper
all	total	6.7091	5.4773	8.6609
all	day	4.3788	0.0000	6.2812
all	day:run	4.2128	2.0309	5.6009
all	error	2.8445	2.3353	3.6395

```
summary(p)
```

Precision analysis -- summary

Precision analysis object

```
Data class      : data.frame
Total rows      : 80
Formula         : value ~ day/run
Samples         : 1
```

Factor levels

```
day      : 1 (4), 2 (4), 3 (4), 4 (4), 5 (4), 6 (4), 7 (4), 8 (4), 9 (4), 10 (4), 11 (4), 12 (4), 13 (4), 14 (4), 15
run      : 1 (40), 2 (40)
rep      : 1 (40), 2 (40)
```

Analysis status

```
[x] outlier
[x] normal
[x] variance
[x] ci
[ ] profile
```

Outlier detection -- Grubbs (alpha = 0.05)

No outliers detected.

Normality test -- Shapiro-Wilk

Sample	n	Method	Statistic	p-value
all	80	Shapiro-Wilk	W=0.9868	0.5858

Variance components

sample	component	VC	%Total	SD	CV[%]	Method
all	day	18.8358	42.6	4.3400	4.3788	anova
all	day:run	17.4343	39.4	4.1754	4.2128	anova
all	error	7.9482	18.0	2.8193	2.8445	anova

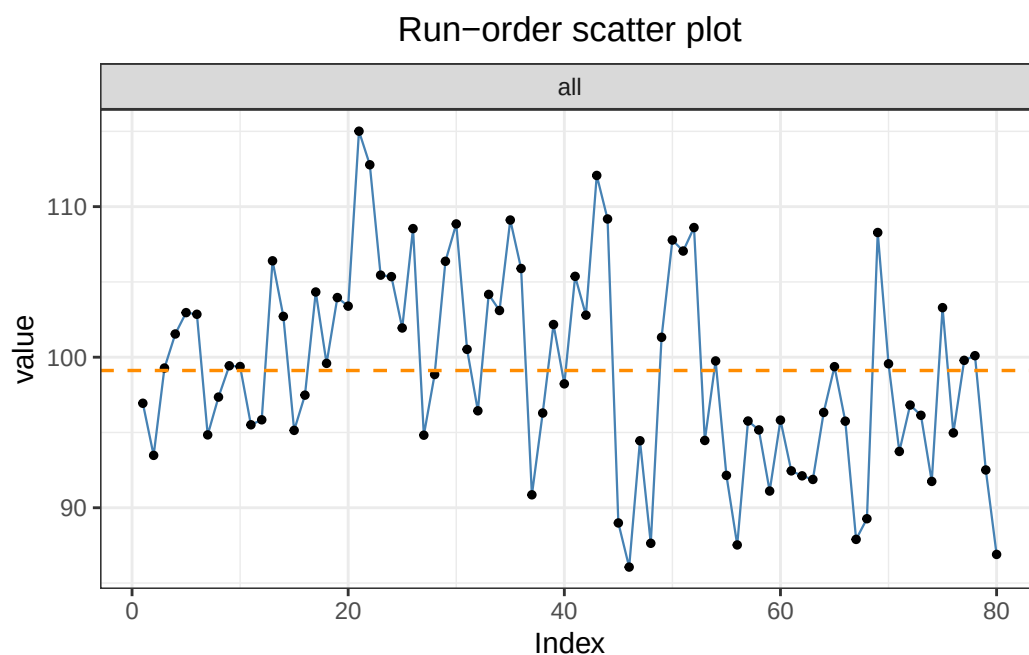
Confidence intervals (SD)

sample	component	estimate	lower	upper
all	total	6.6497	5.4288	8.5842
all	day	4.3400	0.0000	6.2255
all	day:run	4.1754	2.0129	5.5513
all	error	2.8193	2.3147	3.6073

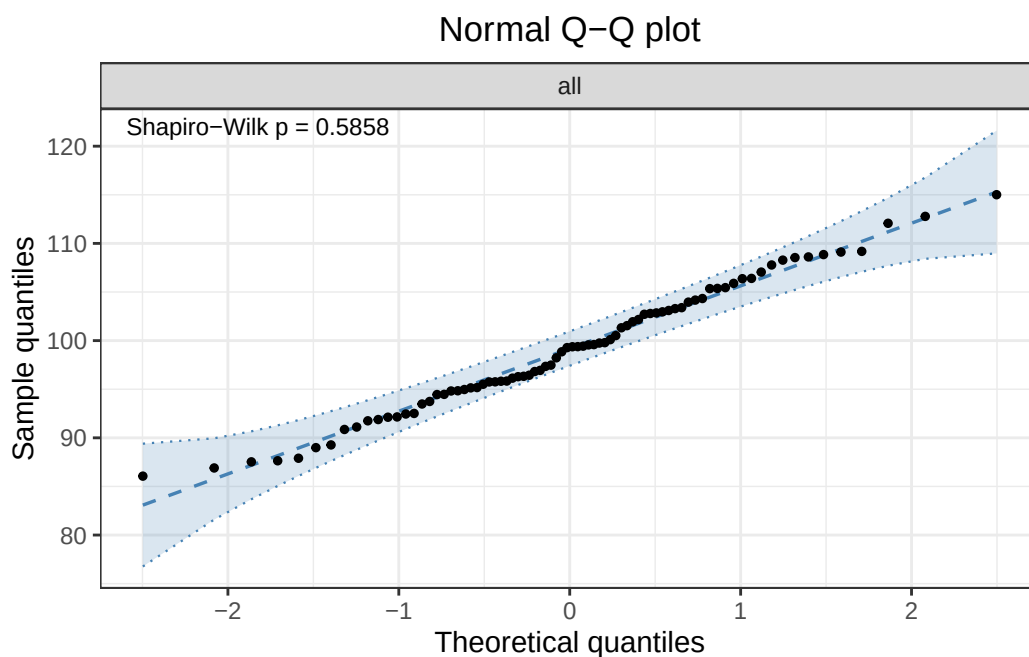
Confidence intervals (%CV)

sample	component	estimate	lower	upper
all	total	6.7091	5.4773	8.6609
all	day	4.3788	0.0000	6.2812
all	day:run	4.2128	2.0309	5.6009
all	error	2.8445	2.3353	3.6395

```
plot(p, type = "dot")
```

```
plot(p, type = "qq")
```



value ~ day/run means day, nested run within day, and replicate residual error. outlier() accepts "grubbs" or "iqr"; normal(method="auto") selects by sample size. NegVC is supplied to variance(), not to precision():

```
p <- variance(p, NegVC = TRUE)
```

13.2.2 Multiple Samples and a Sadler Profile

```
profile_data <- as.data.frame(read_csv(
  "/data/013-精密度分析/precision-profile.csv", show_col_types = FALSE
))
profile_data$sample <- factor(profile_data$sample)
profile_data$day <- factor(profile_data$day)
profile_data$rep <- factor(profile_data$rep)

p2 <- precision(profile_data, value ~ day, by = "sample")
p2 <- variance(p2)
```

Variance components

sample	component	VC	%Total	SD	CV[%]	Method

L1	day	0.6376	63.0	0.7985	3.9446	anova
L1	error	0.3750	37.0	0.6124	3.0250	anova
L2	day	5.7139	75.1	2.3904	4.7584	anova
L2	error	1.8937	24.9	1.3761	2.7393	anova
L3	day	3.3913	15.7	1.8415	1.8088	anova
L3	error	18.1466	84.3	4.2599	4.1840	anova
L4	day	30.5549	54.6	5.5276	2.7529	anova
L4	error	25.4525	45.4	5.0451	2.5126	anova
L5	day	0.0000	0.0	0.0000	0.0000	anova
L5	error	171.7986	100.0	13.1072	3.3003	anova

```
p2 <- ci(p2)
```

Confidence intervals (SD)

sample	component	estimate	lower	upper

L1	total	1.0063	0.6007	2.9368
L1	day	0.7985	0.0000	1.4267
L1	error	0.6124	0.4391	1.0108
L2	total	2.7582	1.5714	9.9889
L2	day	2.3904	0.0000	4.2023
L2	error	1.3761	0.9868	2.2716
L3	total	4.6409	3.2872	7.8819
L3	day	1.8415	0.0000	4.1779
L3	error	4.2599	3.0547	7.0319
L4	total	7.4838	4.6166	19.1652
L4	day	5.5276	0.0000	10.0268
L4	error	5.0451	3.6177	8.3280
L5	total	13.1072	9.5629	20.8255
L5	day	0.0000	NA	NA
L5	error	13.1072	9.3990	21.6365

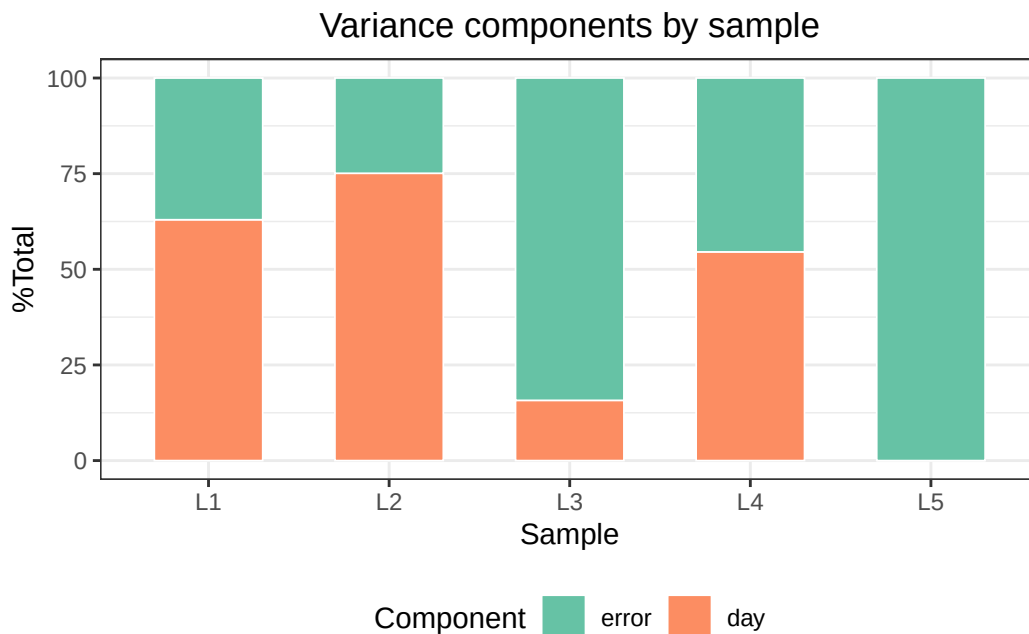
Confidence intervals (%CV)

sample	component	estimate	lower	upper

L1	total	4.9710	2.9673	14.5079
L1	day	3.9446	0.0000	7.0480
L1	error	3.0250	2.1692	4.9936
L2	total	5.4905	3.1281	19.8843
L2	day	4.7584	0.0000	8.3652
L2	error	2.7393	1.9643	4.5219
L3	total	4.5583	3.2287	7.7416

L3	day	1.8088	0.0000	4.1035
L3	error	4.1840	3.0003	6.9067
L4	total	3.7271	2.2992	9.5447
L4	day	2.7529	0.0000	4.9936
L4	error	2.5126	1.8017	4.1476
L5	total	3.3003	2.4078	5.2436
L5	day	0.0000	NA	NA
L5	error	3.3003	2.3666	5.4478

```
plot(p2, type = "var")
```



```
list_sadler()
```

Sadler Precision Profile Models

- 1 Constant variance $\sigma^2 = \beta_1$
- 2 Constant CV $\sigma^2 = \beta_1 * u^2$
- 3 Mixed (const + prop var) $\sigma^2 = \beta_1 + \beta_2 * u^2$
- 4 Constrained power (const exponent) $\sigma^2 = (\beta_1 + \beta_2 * u)^2$
- 5 Alternative constrained power $\sigma^2 = \beta_1 + \beta_2 * u^K$
- 6 Unconstrained power (with min) $\sigma^2 = \beta_1 + \beta_2 * u + \beta_3 * u^J$
- 7 Alternative unconstrained power $\sigma^2 = \beta_1 + \beta_2 * u^J$
- 8 Unconstrained power (Sadler default) $\sigma^2 = (\beta_1 + \beta_2 * u)^J$
- 9 CLSI EP17-like $\sigma^2 = \beta_1 * u^J$
- 10 CLSI EP17 exact (log-log) $CV = \beta_1 * u^J$

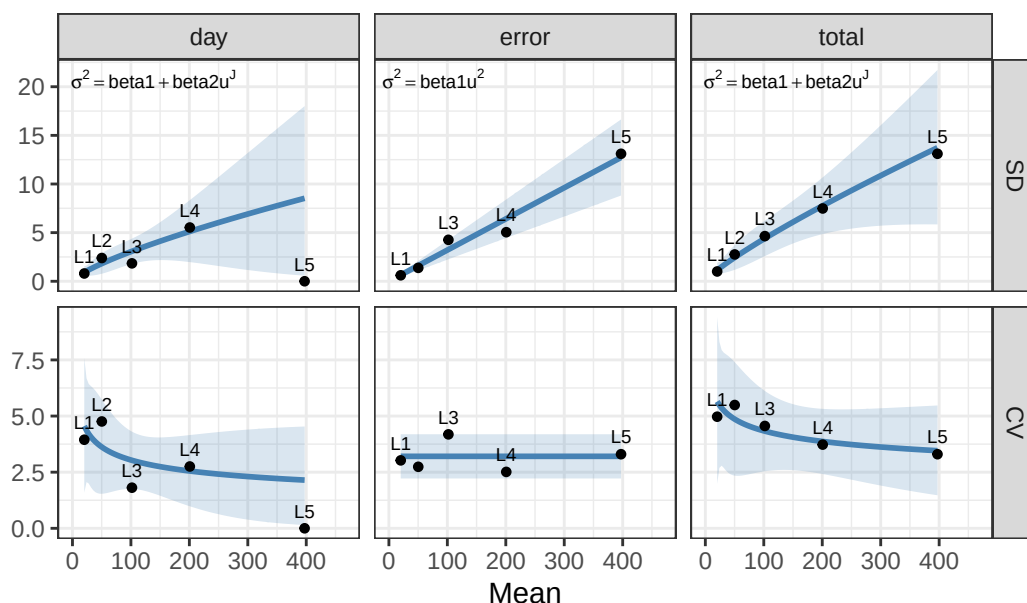
```
p2 <- profile(p2, model.no = 1:10)
```

Precision profiles -- Sadler

total: $\sigma^2 = 0.0000 + 0.0086 * u^J$ (AIC = 56.45, $R^2 = 0.9847$)
 day: $\sigma^2 = 0.0000 + 0.0095 * u^J$ (AIC = 43.46, $R^2 = 0.8919$)
 error: $\sigma^2 = 0.0010 * u^2$ (AIC = 55.96, $R^2 = 0.9806$)

```
plot(p2, type = "profile")
```

Precision profiles



Each variance component needs at least three finite-SD concentrations for profile fitting. The profile can predict SD/CV at arbitrary concentrations for applications such as functional sensitivity, but predictions outside the observed range are extrapolations.

13.2.3 Multicenter Precision

```
sites_data <- as.data.frame(read_csv(
  "./data/013-精密度分析/precision-sites.csv", show_col_types = FALSE
))
p3 <- precision(sites_data, value ~ site/day/run, by = "sample")
p3 <- variance(p3)
```

Variance components

sample	component	VC	%Total	SD	CV[%]	Method

S1	site	0.2746	10.4	0.5241	1.0550	anova
S1	site:day	0.0000	0.0	0.0000	0.0000	anova
S1	site:day:run	0.3553	13.4	0.5960	1.1998	anova
S1	error	2.0231	76.3	1.4223	2.8632	anova
S2	site	0.7117	8.6	0.8436	0.8359	anova
S2	site:day	0.1018	1.2	0.3191	0.3162	anova
S2	site:day:run	0.9106	11.0	0.9543	0.9456	anova
S2	error	6.5670	79.2	2.5626	2.5392	anova
S3	site	22.0901	42.6	4.7000	2.3517	anova
S3	site:day	2.6252	5.1	1.6202	0.8107	anova
S3	site:day:run	0.0000	0.0	0.0000	0.0000	anova
S3	error	27.1915	52.4	5.2145	2.6091	anova

```
knitr::kable(report(p3, run = "run", day = "day", site = "site"))
```

Sample Description	Mean Value	Repeatability SD	%CV Repeatability	Reproducibility SD	%CV Reproducibility
S1	49.6766	1.4223	2.8632	1.6288	3.2788
S2	100.9203	2.5626	2.5392	2.8794	2.8532
S3	199.8592	5.2145	2.6091	7.2046	3.6049

To analyze each site separately, use multiple grouping columns while retaining all runs within each subset:

```
p3s <- precision(sites_data, value ~ day/run, by = c("site", "sample"))
p3s <- variance(p3s)
```

Variance components

sample	component	VC	%Total	SD	CV[%]	Method

A.S1	day	0.0000	0.0	0.0000	0.0000	anova
A.S1	day:run	0.1646	6.8	0.4057	0.8231	anova
A.S1	error	2.2677	93.2	1.5059	3.0553	anova
B.S1	day	0.0000	0.0	0.0000	0.0000	anova
B.S1	day:run	0.5459	23.5	0.7389	1.4758	anova
B.S1	error	1.7784	76.5	1.3335	2.6637	anova
A.S2	day	0.3241	4.3	0.5693	0.5602	anova
A.S2	day:run	0.1523	2.0	0.3903	0.3840	anova
A.S2	error	7.0483	93.7	2.6549	2.6124	anova
B.S2	day	0.0000	0.0	0.0000	0.0000	anova
B.S2	day:run	1.6690	21.5	1.2919	1.2891	anova
B.S2	error	6.0856	78.5	2.4669	2.4616	anova
A.S3	day	4.4871	15.6	2.1183	1.0420	anova
A.S3	day:run	1.7006	5.9	1.3041	0.6415	anova
A.S3	error	22.5797	78.5	4.7518	2.3375	anova
B.S3	day	0.7633	2.3	0.8737	0.4448	anova
B.S3	day:run	0.0000	0.0	0.0000	0.0000	anova
B.S3	error	31.8033	97.7	5.6394	2.8709	anova

```
knitr::kable(report(p3s, run = "run", day = "day"))
```

Sample De- scription	Mean Value	Repeatability SD	%CV Re- peatability	Within-Laboratory Precision SD	%CV Within-Laboratory Precision
A.S1	49.2884	1.5059	3.0553	1.5596	3.1642
B.S1	50.0647	1.3335	2.6637	1.5246	3.0452
A.S2	101.6249	2.6549	2.6124	2.7431	2.6993
B.S2	100.2158	2.4669	2.4616	2.7847	2.7787
A.S3	203.2864	4.7518	2.3375	5.3635	2.6384
B.S3	196.4320	5.6394	2.8709	5.7067	2.9052

13.2.4 More Complex Nested Models

precision() can represent other VCA hierarchies, for example operator, day, and run:

```
custom_data <- as.data.frame(read_csv(
  "./data/013-精密度分析/precision-custom.csv", show_col_types = FALSE
))
p4 <- precision(custom_data, value ~ operator/day/run)
p4 <- variance(p4)
```

Variance components

sample	component	VC	%Total	SD	CV[%]	Method

all	operator	2.2914	11.8	1.5137	1.4898	anova
all	operator:day	6.3159	32.4	2.5132	2.4734	anova
all	operator:day:run	3.2463	16.7	1.8017	1.7733	anova

```
all          error 7.6322   39.2 2.7627 2.7190  anova
```

13.2.5 Standard-Data Cross-Checks

The supplied EP05-A3 Appendix A and B files can be analyzed using the same pattern. Appendix A is a balanced day/run design; Appendix B is a multicenter site/day design with samples grouped by `by = "sample"`:

```
ep05_a <- as.data.frame(read_csv(
  "./data/013-精密度分析/std-ep05-appendix-a-glucose.csv", show_col_types = FALSE
))
ep05_a$day <- factor(ep05_a$day)
ep05_a$run <- factor(ep05_a$run)
ep05_a_precision <- precision(ep05_a, result ~ day/run)
ep05_a_precision <- variance(ep05_a_precision, NegVC = FALSE)
```

Variance components

sample	component	VC	%Total	SD	CV[%]	Method
all	day	1.9586	15.1	1.3995	0.5731	anova
all	day:run	3.0750	23.8	1.7536	0.7181	anova
all	error	7.9000	61.1	2.8107	1.1510	anova

```
ep05_a_precision <- ci(ep05_a_precision)
```

Confidence intervals (SD)

sample	component	estimate	lower	upper
all	total	3.5963	3.0696	4.3430
all	day	1.3995	0.0000	2.4622
all	day:run	1.7536	0.0000	2.7858
all	error	2.8107	2.3076	3.5963

Confidence intervals (%CV)

sample	component	estimate	lower	upper
all	total	1.4727	1.2570	1.7785
all	day	0.5731	0.0000	1.0083
all	day:run	0.7181	0.0000	1.1408
all	error	1.1510	0.9450	1.4727

```
knitr::kable(report(ep05_a_precision, day = "day", run = "run", digits = 6))
```

Sample De- scription	Mean Value	Repeatability SD	%CV Re- peatability	Within-Laboratory Precision SD	%CV
					Within-Laboratory Precision
all	244.2	2.810694	1.15098	3.596325	1.472697

```
ep05_b <- as.data.frame(read_csv(
  "./data/013-精密度分析/std-ep05-appendix-b-ca19-9.csv", show_col_types = FALSE
))
ep05_b_precision <- precision(ep05_b, result ~ site/day, by = "sample")
ep05_b_precision <- variance(ep05_b_precision)
```

Variance components

sample	component	VC	%Total	SD	CV[%]	Method

P1	site	0.3843	35.4	0.6199	5.1312	anova
P1	site:day	0.1778	16.4	0.4216	3.4899	anova
P1	error	0.5248	48.3	0.7244	5.9963	anova
P2	site	1.6189	47.9	1.2724	3.0597	anova
P2	site:day	0.1232	3.6	0.3509	0.8439	anova
P2	error	1.6348	48.4	1.2786	3.0747	anova
P5	site	24.9068	29.3	4.9907	1.3165	anova
P5	site:day	3.1861	3.7	1.7850	0.4709	anova
P5	error	56.9669	67.0	7.5476	1.9910	anova
Q3	site	3.1742	60.4	1.7816	3.1959	anova
Q3	site:day	0.5232	10.0	0.7233	1.2975	anova
Q3	error	1.5599	29.7	1.2490	2.2404	anova
Q4	site	30.0735	75.7	5.4839	3.3104	anova
Q4	site:day	1.8663	4.7	1.3661	0.8247	anova
Q4	error	7.8128	19.7	2.7951	1.6873	anova
Q6	site	164.1097	68.1	12.8105	3.0922	anova
Q6	site:day	3.0208	1.3	1.7380	0.4195	anova
Q6	error	73.9590	30.7	8.5999	2.0758	anova

```
ep05_b_precision <- ci(ep05_b_precision)
```

Confidence intervals (SD)

sample	component	estimate	lower	upper

P1	total	1.0425	0.7415	1.7535
P1	site	0.6199	0.0000	1.1178
P1	site:day	0.4216	0.0000	0.6380
P1	error	0.7244	0.6148	0.8819
P2	total	1.8376	1.2313	3.5995
P2	site	1.2724	0.0000	2.2291
P2	site:day	0.3509	0.0000	0.7084
P2	error	1.2786	1.0852	1.5566
P5	total	9.2228	6.9060	13.8849
P5	site	4.9907	0.0000	8.9156
P5	site:day	1.7850	0.0000	3.9426
P5	error	7.5476	6.4058	9.1888
Q3	total	2.2929	1.4259	5.7003
Q3	site	1.7816	0.0000	3.1184
Q3	site:day	0.7233	0.0000	1.0958
Q3	error	1.2490	1.0600	1.5205
Q4	total	6.3050	3.6468	21.1983
Q4	site	5.4839	0.0000	9.5060
Q4	site:day	1.3661	0.0000	2.1602
Q4	error	2.7951	2.3723	3.4029
Q6	total	15.5271	9.3516	43.6651
Q6	site	12.8105	0.0000	22.1981
Q6	site:day	1.7380	0.0000	4.2690
Q6	error	8.5999	7.2989	10.4699

Confidence intervals (%CV)

sample	component	estimate	lower	upper

P1	total	8.6292	6.1376	14.5139
P1	site	5.1312	0.0000	9.2524
P1	site:day	3.4899	0.0000	5.2812

```

P1    error    5.9963 5.0891 7.3001
P2    total    4.4191 2.9611 8.6559
P2    site     3.0597 0.0000 5.3606
P2    site:day 0.8439 0.0000 1.7036
P2    error    3.0747 2.6095 3.7433
P5    total    2.4329 1.8217 3.6627
P5    site     1.3165 0.0000 2.3518
P5    site:day 0.4709 0.0000 1.0400
P5    error    1.9910 1.6898 2.4239
Q3    total    4.1130 2.5578 10.2254
Q3    site     3.1959 0.0000 5.5938
Q3    site:day 1.2975 0.0000 1.9656
Q3    error    2.2404 1.9015 2.7276
Q4    total    3.8061 2.2014 12.7966
Q4    site     3.3104 0.0000 5.7384
Q4    site:day 0.8247 0.0000 1.3040
Q4    error    1.6873 1.4320 2.0542
Q6    total    3.7479 2.2573 10.5398
Q6    site     3.0922 0.0000 5.3581
Q6    site:day 0.4195 0.0000 1.0305
Q6    error    2.0758 1.7618 2.5272

```

```
knitr::kable(report(ep05_b_precision, day = "day", site = "site", table = "detailed"))
```

Sample De- scrip- tion	Mean Value	N	Repeatability SD	%CV				Within- Laboratory			%CV		%CV	
				Repeatability SD	Between-Run SD	%CV	Between-Day SD	%CV	Between-Site SD	%CV	Precision SD	Within- Laboratory Precision SD	Reproducibility SD	Reproducibility
P1	12.0873	13	0.7244	5.9963	NA	NA	0.4216	6.4899	0.6195	5.1312	0.8382	6.9379	1.0425	8.6292
P2	41.5875	10	1.2786	3.0747	NA	NA	0.3509	0.8439	1.2724	3.0597	1.3259	3.1884	1.8376	4.4191
P5	379.0907	10	7.5476	1.9910	NA	NA	1.7850	0.4709	4.9907	1.3165	7.7558	2.0459	9.2228	2.4329
Q3	55.7467	10	1.2490	2.2404	NA	NA	0.7233	1.2975	1.7816	3.1959	1.4433	2.5890	2.2929	4.1130
Q4	165.6562	10	2.7951	1.6873	NA	NA	1.3661	0.8247	5.4838	3.3104	3.1111	1.8781	6.3050	3.8061
Q6	414.2867	10	8.5999	2.0758	NA	NA	1.7380	0.4195	12.8105	3.0922	28.7738	2.1178	15.5273	3.7479

Standard examples can still differ because of exclusion rules, displayed intermediate values, ANOVA versus REML, and the exact degrees-of-freedom convention. Disclose the path used rather than treating similar rounded values as identical algorithms.

13.3 Complete Standard Cases

13.3.1 CLSI EP05-A3 Appendix A: Glucose $20 \times 2 \times 2$

Appendix A uses one patient-plasma pool measured in two runs per day on 20 days, with two replicates per run, for 80 glucose results. The data are complete and balanced; the nested structure is day/run, with replicate measurements forming the residual layer.

```

ep05_a <- as.data.frame(read_csv(
  "/data/013-精密度分析/std-ep05-appendix-a-glucose.csv",
  show_col_types = FALSE
))
ep05_a$day <- factor(ep05_a$day)
ep05_a$run <- factor(ep05_a$run)
ep05_a$replicate <- factor(ep05_a$replicate)

```



```
ep05_a_precision <- precision(
  ep05_a, form = result ~ day/run
)
ep05_a_precision <- variance(ep05_a_precision, NegVC = FALSE)
```

Variance components

sample	component	VC	%Total	SD	CV[%]	Method
all	day	1.9586	15.1	1.3995	0.5731	anova
all	day:run	3.0750	23.8	1.7536	0.7181	anova
all	error	7.9000	61.1	2.8107	1.1510	anova

```
ep05_a_precision <- ci(ep05_a_precision)
```

Confidence intervals (SD)

sample	component	estimate	lower	upper
all	total	3.5963	3.0696	4.3430
all	day	1.3995	0.0000	2.4622
all	day:run	1.7536	0.0000	2.7858
all	error	2.8107	2.3076	3.5963

Confidence intervals (%CV)

sample	component	estimate	lower	upper
all	total	1.4727	1.2570	1.7785
all	day	0.5731	0.0000	1.0083
all	day:run	0.7181	0.0000	1.1408
all	error	1.1510	0.9450	1.4727

```
knitr::kable(report(
  ep05_a_precision,
  day = "day", run = "run", digits = 6
))
```

Sample De- scription	Mean Value	Repeatability SD	%CV Re- peatability	Within-Laboratory Precision SD	%CV Within-Laboratory Precision
all	244.2	2.810694	1.15098	3.596325	1.472697

Table 13.1: EP05-A3 Appendix A: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
EP05-A3 Appendix A	mean	244.20	244.2000
EP05-A3 Appendix A	SS_day	415.80	415.8000
EP05-A3 Appendix A	SS_run_day	281.00	281.0000
EP05-A3 Appendix A	SS_error	316.00	316.0000
EP05-A3 Appendix A	MS_day	21.88	21.8842
EP05-A3 Appendix A	MS_run_day	14.05	14.0500
EP05-A3 Appendix A	MS_error	7.90	7.9000
EP05-A3 Appendix A	V_day	1.96	1.9586
EP05-A3 Appendix A	V_run	3.08	3.0750
EP05-A3 Appendix A	V_error	7.90	7.9000
EP05-A3 Appendix A	SD_repeatability	2.81	2.8107

Standard location	Statistic	Standard	ivdtools
EP05-A3 Appendix A	SD_within_lab	3.60	3.5963
EP05-A3 Appendix A	CV_repeatability	1.20	1.1510
EP05-A3 Appendix A	CV_within_lab	1.50	1.4727

13.3.2 CLSI EP05-A3 Appendix B: CA19-9 Multicenter Study

Appendix B contains three sites, six samples, five days per site, and five replicates per day, for 450 results. The sites use different instruments and operators but the same reagent and calibrator lots. The nested model is site/day, with replicate measurements as the residual layer.

```
ep05_b <- as.data.frame(read_csv(
  "/data/013-精密度分析/std-ep05-appendix-b-ca19-9.csv",
  show_col_types = FALSE
))
ep05_b$sample <- factor(ep05_b$sample)
ep05_b$site <- factor(ep05_b$site)
ep05_b$day <- factor(ep05_b$day)
ep05_b$replicate <- factor(ep05_b$replicate)

ep05_b_precision <- precision(
  ep05_b, form = result ~ site/day, by = "sample"
)
ep05_b_precision <- variance(ep05_b_precision)
```

Variance components

sample	component	VC	%Total	SD	CV[%]	Method

P1	site	0.3843	35.4	0.6199	5.1312	anova
P1	site:day	0.1778	16.4	0.4216	3.4899	anova
P1	error	0.5248	48.3	0.7244	5.9963	anova
P2	site	1.6189	47.9	1.2724	3.0597	anova
P2	site:day	0.1232	3.6	0.3509	0.8439	anova
P2	error	1.6348	48.4	1.2786	3.0747	anova
P5	site	24.9068	29.3	4.9907	1.3165	anova
P5	site:day	3.1861	3.7	1.7850	0.4709	anova
P5	error	56.9669	67.0	7.5476	1.9910	anova
Q3	site	3.1742	60.4	1.7816	3.1959	anova
Q3	site:day	0.5232	10.0	0.7233	1.2975	anova
Q3	error	1.5599	29.7	1.2490	2.2404	anova
Q4	site	30.0735	75.7	5.4839	3.3104	anova
Q4	site:day	1.8663	4.7	1.3661	0.8247	anova
Q4	error	7.8128	19.7	2.7951	1.6873	anova
Q6	site	164.1097	68.1	12.8105	3.0922	anova
Q6	site:day	3.0208	1.3	1.7380	0.4195	anova
Q6	error	73.9590	30.7	8.5999	2.0758	anova

```
ep05_b_precision <- ci(ep05_b_precision)
```

Confidence intervals (SD)

sample	component	estimate	lower	upper

P1	total	1.0425	0.7415	1.7535
P1	site	0.6199	0.0000	1.1178
P1	site:day	0.4216	0.0000	0.6380

P1	error	0.7244	0.6148	0.8819
P2	total	1.8376	1.2313	3.5995
P2	site	1.2724	0.0000	2.2291
P2	site:day	0.3509	0.0000	0.7084
P2	error	1.2786	1.0852	1.5566
P5	total	9.2228	6.9060	13.8849
P5	site	4.9907	0.0000	8.9156
P5	site:day	1.7850	0.0000	3.9426
P5	error	7.5476	6.4058	9.1888
Q3	total	2.2929	1.4259	5.7003
Q3	site	1.7816	0.0000	3.1184
Q3	site:day	0.7233	0.0000	1.0958
Q3	error	1.2490	1.0600	1.5205
Q4	total	6.3050	3.6468	21.1983
Q4	site	5.4839	0.0000	9.5060
Q4	site:day	1.3661	0.0000	2.1602
Q4	error	2.7951	2.3723	3.4029
Q6	total	15.5271	9.3516	43.6651
Q6	site	12.8105	0.0000	22.1981
Q6	site:day	1.7380	0.0000	4.2690
Q6	error	8.5999	7.2989	10.4699

Confidence intervals (%CV)

sample	component	estimate	lower	upper
--------	-----------	----------	-------	-------

P1	total	8.6292	6.1376	14.5139
P1	site	5.1312	0.0000	9.2524
P1	site:day	3.4899	0.0000	5.2812
P1	error	5.9963	5.0891	7.3001
P2	total	4.4191	2.9611	8.6559
P2	site	3.0597	0.0000	5.3606
P2	site:day	0.8439	0.0000	1.7036
P2	error	3.0747	2.6095	3.7433
P5	total	2.4329	1.8217	3.6627
P5	site	1.3165	0.0000	2.3518
P5	site:day	0.4709	0.0000	1.0400
P5	error	1.9910	1.6898	2.4239
Q3	total	4.1130	2.5578	10.2254
Q3	site	3.1959	0.0000	5.5938
Q3	site:day	1.2975	0.0000	1.9656
Q3	error	2.2404	1.9015	2.7276
Q4	total	3.8061	2.2014	12.7966
Q4	site	3.3104	0.0000	5.7384
Q4	site:day	0.8247	0.0000	1.3040
Q4	error	1.6873	1.4320	2.0542
Q6	total	3.7479	2.2573	10.5398
Q6	site	3.0922	0.0000	5.3581
Q6	site:day	0.4195	0.0000	1.0305
Q6	error	2.0758	1.7618	2.5272

```

ep05_b_report <- report(
  ep05_b_precision,
  day = "day", site = "site", table = "detailed", digits = 6
)
ep05_b_report_ci <- report(
  ep05_b_precision,
  day = "day", site = "site", table = "CI", digits = 6
)

```

Table 13.2: EP05-A3 Appendix B: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
EP05-A3 Appendix B	P1_mean	12.100	12.0813
EP05-A3 Appendix B	P1_SD_R	0.724	0.7244
EP05-A3 Appendix B	P1_SD_WL	0.838	0.8382
EP05-A3 Appendix B	P1_SD_REP	1.040	1.0425
EP05-A3 Appendix B	P1_CV_R	6.000	5.9963
EP05-A3 Appendix B	P1_CV_WL	6.900	6.9379
EP05-A3 Appendix B	P1_CV_REP	8.600	8.6292
EP05-A3 Appendix B	P2_mean	41.600	41.5840
EP05-A3 Appendix B	P2_SD_R	1.280	1.2786
EP05-A3 Appendix B	P2_SD_WL	1.330	1.3259
EP05-A3 Appendix B	P2_SD_REP	1.840	1.8376
EP05-A3 Appendix B	P2_CV_R	3.100	3.0747
EP05-A3 Appendix B	P2_CV_WL	3.200	3.1884
EP05-A3 Appendix B	P2_CV_REP	4.400	4.4191
EP05-A3 Appendix B	P5_mean	379.000	379.0907
EP05-A3 Appendix B	P5_SD_R	7.550	7.5476
EP05-A3 Appendix B	P5_SD_WL	7.760	7.7558
EP05-A3 Appendix B	P5_SD_REP	9.220	9.2228
EP05-A3 Appendix B	P5_CV_R	2.000	1.9910
EP05-A3 Appendix B	P5_CV_WL	2.000	2.0459
EP05-A3 Appendix B	P5_CV_REP	2.400	2.4329
EP05-A3 Appendix B	Q3_mean	55.700	55.7467
EP05-A3 Appendix B	Q3_SD_R	1.250	1.2490
EP05-A3 Appendix B	Q3_SD_WL	1.440	1.4433
EP05-A3 Appendix B	Q3_SD_REP	2.290	2.2929
EP05-A3 Appendix B	Q3_CV_R	2.200	2.2404
EP05-A3 Appendix B	Q3_CV_WL	2.600	2.5890
EP05-A3 Appendix B	Q3_CV_REP	4.100	4.1130
EP05-A3 Appendix B	Q4_mean	166.000	165.6560
EP05-A3 Appendix B	Q4_SD_R	2.800	2.7951
EP05-A3 Appendix B	Q4_SD_WL	3.110	3.1111
EP05-A3 Appendix B	Q4_SD_REP	6.300	6.3050
EP05-A3 Appendix B	Q4_CV_R	1.700	1.6873
EP05-A3 Appendix B	Q4_CV_WL	1.900	1.8781
EP05-A3 Appendix B	Q4_CV_REP	3.800	3.8061
EP05-A3 Appendix B	Q6_mean	414.000	414.2867
EP05-A3 Appendix B	Q6_SD_R	8.600	8.5999
EP05-A3 Appendix B	Q6_SD_WL	8.770	8.7738
EP05-A3 Appendix B	Q6_SD_REP	15.500	15.5271
EP05-A3 Appendix B	Q6_CV_R	2.100	2.0758
EP05-A3 Appendix B	Q6_CV_WL	2.100	2.1178
EP05-A3 Appendix B	Q6_CV_REP	3.700	3.7479

13.3.3 CLSI EP15-A3: Ferritin 5×5 Verification

The complete EP15 case contains three serum ferritin levels, measured for five days with five replicates per day. Keep the raw data intact, then reproduce the standard case's pre-specified Grubbs exclusion of 30.2 µg/L from Level 1, Day 1, Replicate 3. This exclusion reproduces the standard case and is not selected retrospectively by the code.

```

ep15_ferritin_raw <- as.data.frame(read_csv(
  "./data/013-精密度分析/std-ep15-ferritin.csv", show_col_types = FALSE
))
ep15_ferritin_raw$sample <- factor(ep15_ferritin_raw$sample)
ep15_ferritin_raw$run <- factor(ep15_ferritin_raw$run)
ep15_ferritin_raw$replicate <- factor(ep15_ferritin_raw$replicate)

ep15_exclude <- with(
  ep15_ferritin_raw,
  sample == 1L & run == 1L & replicate == 3L
)
ep15_ferritin_raw[ep15_exclude, ]

run replicate sample result
7 1 3 1 30.2

# Exclude the prespecified result.
ep15_ferritin <- ep15_ferritin_raw[!ep15_exclude, ]

```

After exclusion, Level 1 has 24 results, with four replicates on Day 1; Levels 2 and 3 each retain 25 results. Split by sample and fit one result ~ run model for each level. run represents the between-run component, while daily replicates form the residual error component.

```

# Separate levels to compare ANOVA and REML explicitly.
ep15_ferritin_list <- split(ep15_ferritin, ep15_ferritin$sample)
ep15_level_1 <- droplevels(ep15_ferritin_list[["1"]])
ep15_level_2 <- droplevels(ep15_ferritin_list[["2"]])
ep15_level_3 <- droplevels(ep15_ferritin_list[["3"]])

# Level 1
ep15_level_1_fit <- precision(ep15_level_1, result ~ run)
# The unbalanced design automatically switches to REML.
ep15_level_1_fit_r <- variance(
  ep15_level_1_fit, NegVC = FALSE
)

```

Variance components

sample	component	VC	%Total	SD	CV[%]	Method
all	run	0.2820	27.5	0.5311	2.0816	reml
all	error	0.7423	72.5	0.8616	3.3770	reml

```

# Use ANOVA explicitly to match the standard case.
ep15_level_1_fit <- variance(
  ep15_level_1_fit, method = "anova", NegVC = FALSE
)

```

Variance components

sample	component	VC	%Total	SD	CV[%]	Method
all	run	0.2804	27.4	0.5295	2.0756	anova
all	error	0.7414	72.6	0.8610	3.3749	anova

```

ep15_level_1_fit <- ci(ep15_level_1_fit, digits = 6)

```

Confidence intervals (SD)

sample	component	estimate	lower	upper
--------	-----------	----------	-------	-------

```
-----
all    total 1.010837 0.752556 1.539604
all    run  0.529550 0.000000 0.944544
all    error 0.861028 0.654803 1.257592
```

Confidence intervals (%CV)

```
sample component estimate    lower    upper
-----
all    total 3.962125 2.949754 6.034706
all    run  2.075649 0.000000 3.702279
all    error 3.374924 2.566597 4.929319
```

```
# Level 2
ep15_level_2_fit <- precision(ep15_level_2, result ~ run)
ep15_level_2_fit <- variance(
  ep15_level_2_fit, method = "anova", NegVC = FALSE
)
```

Variance components

```
sample component    VC %Total    SD  CV[%] Method
-----
all    run 2.5400    44.6 1.5937 1.1374 anova
all    error 3.1600    55.4 1.7776 1.2687 anova
```

```
ep15_level_2_fit <- ci(ep15_level_2_fit, digits = 6)
```

Confidence intervals (SD)

```
sample component estimate    lower    upper
-----
all    total 2.387467 1.701094 3.999257
all    run  1.593738 0.000000 2.636950
all    error 1.777639 1.359999 2.567034
```

Confidence intervals (%CV)

```
sample component estimate    lower    upper
-----
all    total 1.703873 1.214026 2.854166
all    run  1.137409 0.000000 1.881922
all    error 1.268655 0.970596 1.832025
```

```
# Level 3
ep15_level_3_fit <- precision(ep15_level_3, result ~ run)
ep15_level_3_fit <- variance(
  ep15_level_3_fit, method = "anova", NegVC = FALSE
)
```

Variance components

```
sample component    VC %Total    SD  CV[%] Method
-----
all    run 102.6080    47.5 10.1296 1.6262 anova
all    error 113.5200    52.5 10.6546 1.7105 anova
```

```
ep15_level_3_fit <- ci(ep15_level_3_fit, digits = 6)
```

Confidence intervals (SD)

```
sample component estimate    lower    upper
-----
```

```

all    total 14.701292 10.382650 25.141336
all    run  10.129561  0.000000 16.638736
all    error 10.654576  8.151381 15.385949

```

Confidence intervals (%CV)

```

sample component estimate    lower    upper
-----
all    total 2.360213 1.666878 4.036305
all    run  1.626246 0.000000 2.671259
all    error 1.710534 1.308660 2.470130

```

Extract the total mean, run variance component, repeatability SD, and total within-laboratory SD from each object and compare them with the displayed standard values:

Table 13.3: EP15-A3 ferritin: ivdtools compared with the standard text

Sample	Statistic	Standard	ivdtools
Level 1	overall mean	25.5000	25.5125
Level 1	between-run variance	0.2805	0.2804
Level 1	repeatability SD	0.8600	0.8610
Level 1	within-laboratory SD	1.0100	1.0108
Level 2	overall mean	140.1000	140.1200
Level 2	between-run variance	2.5400	2.5400
Level 2	repeatability SD	1.7800	1.7776
Level 2	within-laboratory SD	2.4000	2.3875
Level 3	overall mean	623.0000	622.8800
Level 3	between-run variance	102.6100	102.6080
Level 3	repeatability SD	10.7000	10.6546
Level 3	within-laboratory SD	14.7000	14.7013

The recalculated results for all three levels agree with or are close to the displayed standard values. Level 1 is unbalanced after exclusion, so small differences in variance components arise from unbalanced-data calculation details and intermediate rounding. For Level 2, the exact within-laboratory SD is 2.3875 µg/L, while the standard displays 2.40 µg/L after rounding its variance components. The standard case passes all three levels under its stated PI and UVL criteria; that conclusion applies only to the stated claims and limits.

13.3.4 WS/T 408–2024: Precision Verification

WS/T 408 Appendix A.2 uses two HDL-C levels, five batches per level, and three replicates per batch. After import, verify that each sample and batch contains three results.

```

wst408_precision <- as.data.frame(read_csv(
  "./data/013-精密度分析/std-wst408-precision.csv",
  show_col_types = FALSE
))
wst408_precision$sample <- factor(wst408_precision$sample)
wst408_precision$batch <- factor(wst408_precision$batch)
wst408_precision$replicate <- factor(wst408_precision$replicate)

wst408_precision_fit <- precision(
  wst408_precision,
  result ~ batch,
  by = "sample"
)
wst408_precision_fit <- variance(

```

```
wst408_precision_fit,
method = "anova",
NegVC = FALSE
)
```

Variance components

sample	component	VC	%Total	SD	CV[%]	Method
样品1	batch	0.0001	10.9	0.0080	0.7637	anova
样品1	error	0.0005	89.1	0.0228	2.1884	anova
样品2	batch	0.0001	12.1	0.0091	0.5892	anova
样品2	error	0.0006	87.9	0.0246	1.5898	anova

```
wst408_precision_fit <- ci(wst408_precision_fit, digits = 6)
```

Confidence intervals (SD)

sample	component	estimate	lower	upper
样品1	total	0.024152	0.017523	0.038843
样品1	batch	0.007958	0.000000	0.020611
样品1	error	0.022804	0.015933	0.040019
样品2	total	0.026268	0.019035	0.042360
样品2	batch	0.009129	0.000000	0.022737
样品2	error	0.024631	0.017210	0.043225

Confidence intervals (%CV)

sample	component	estimate	lower	upper
样品1	total	2.317879	1.681694	3.727688
样品1	batch	0.763745	0.000000	1.978014
样品1	error	2.188437	1.529098	3.840561
样品2	total	1.695429	1.228567	2.734049
样品2	batch	0.589202	0.000000	1.467529
样品2	error	1.589755	1.110789	2.789914

Table 13.4: WS/T 408 Appendix A.2: ivdtools compared with the standard text

Sample	Statistic	Standard	ivdtools
Level 1	overall mean	1.0420	1.0420
Level 1	s_WR	0.0228	0.0228
Level 1	s_WL	0.0242	0.0242
Level 1	CV_WL (%)	NA	2.3179
Level 2	overall mean	1.5493	1.5493
Level 2	s_WR	0.0246	0.0246
Level 2	s_WL	0.0263	0.0263
Level 2	CV_WL (%)	NA	1.6954

13.3.5 YY/T 1789.1–2021 Appendix A: 25-Hydroxyvitamin D

Appendix A provides complete 20-day $\times$ 2-run $\times$ 2-replicate data, for 80 results.

```
yyt1789_1 <- read.csv(
  "/data/013-精密度分析/std-yyt1789-1-appendix-a.csv"
)
yyt1789_1$day <- factor(yyt1789_1$day)
yyt1789_1$run <- factor(yyt1789_1$run)
```



```
yyt1789_1_fit <- precision(yyt1789_1, result ~ day/run)
yyt1789_1_fit <- variance(
  yyt1789_1_fit, method = "anova"
)
```

Variance components

sample	component	VC	%Total	SD	CV[%]	Method
all	day	0.1769	36.2	0.4206	2.4324	anova
all	day:run	0.0648	13.2	0.2545	1.4721	anova
all	error	0.2475	50.6	0.4975	2.8773	anova

```
yyt1789_1_fit <- ci(yyt1789_1_fit)
```

Confidence intervals (SD)

sample	component	estimate	lower	upper
all	total	0.6994	0.5861	0.8674
all	day	0.4206	0.0000	0.5991
all	day:run	0.2545	0.0000	0.4400
all	error	0.4975	0.4084	0.6365

Confidence intervals (%CV)

sample	component	estimate	lower	upper
all	total	4.0450	3.3897	5.0170
all	day	2.4324	0.0000	3.4650
all	day:run	1.4721	0.0000	2.5448
all	error	2.8773	2.3623	3.6815

Table 13.5: YY/T 1789.1–2021 Appendix A: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
YY/T 1789.1 Appendix A	mean	17.290	17.2898
YY/T 1789.1 Appendix A	repeatability SD	0.497	0.4975
YY/T 1789.1 Appendix A	within-laboratory SD	0.699	0.6994
YY/T 1789.1 Appendix A	repeatability CV (%)	2.900	2.8773
YY/T 1789.1 Appendix A	within-laboratory CV (%)	4.000	4.0450
YY/T 1789.1 Appendix A	repeatability CI lower	0.408	0.4084
YY/T 1789.1 Appendix A	repeatability CI upper	0.637	0.6365
YY/T 1789.1 Appendix A	within-laboratory CI lower	0.586	0.5861
YY/T 1789.1 Appendix A	within-laboratory CI upper	0.867	0.8674

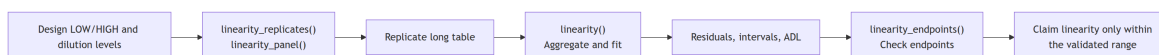
Chapter 14

Linearity Analysis

Linearity evaluates whether measured results have an acceptable linear relationship with analyte concentration or target value across the working range. This chapter uses the dedicated `linearity()` workflow to summarize levels, apply weights, calculate straight-line residuals, compare allowable deviation from linearity (ADL), and perform exploratory polynomial analysis.

! Scope of Use

The core of CLSI EP06, 2nd edition, is comparison of straight-line residuals with a pre-specified ADL. A high correlation or a nonsignificant quadratic term alone does not establish linearity. `method="polynomial"` reproduces the superseded EP6-A approach and should be labeled exploratory. The functions do not replace professional judgment about sample preparation, commutability, endpoint coverage, interference, or the reporting range.



`library(ivdtools)`
`library(readr)`

14.1 Main Functions

Function	Main purpose	Key input or output
<code>linearity_replicates()</code>	Calculate replicates per level from precision and ADL	<code>imprecision</code> , <code>adl</code> , <code>family_error</code>
<code>linearity_panel()</code>	Design a HIGH/LOW preparation series	<code>proportion_high</code> , <code>high</code> , <code>low</code> , <code>total_volume</code>
<code>linearity_endpoints()</code>	Calculate laboratory-confirmed LOW/HIGH targets	<code>LLoQ/ULoQ</code> and low/high-end CV
<code>linearity()</code>	EP06 parameterized linearity analysis	<code>method</code> , <code>weights</code> , <code>conf.level</code> , <code>adl_*</code>
<code>plot()</code>	Plot fit, residuals, precision, or model differences	<code>type</code>

```
linearity_replicates(imprecision = 4, adl = 5, levels = 8, family_error = 0.10)
```

```
Type: relative
Imprecision      : 4
ADL              : 5
True deviation   : 0
Effective ADL    : 5
Level probability : 0.98691628
Z               : 2.4814824
Calculated replicates : 3.940963
Replicates       : 4
Recommend four   : FALSE
```

```
linearity_panel(proportion_high = seq(0, 1, length.out = 6), high = 1000, low = 10, total_volume = 1)
```

```
Sample: S1
Proportion high      : 0
Proportion low       : 1
Expected             : 10
Relative concentration : 0.01
Total volume         : 1
High volume          : 0
Low volume           : 1
```

```
Sample: S2
Proportion high      : 0.2
Proportion low       : 0.8
Expected             : 208
Relative concentration : 0.208
Total volume         : 1
High volume          : 0.2
Low volume           : 0.8
```

```
Sample: S3
Proportion high      : 0.4
Proportion low       : 0.6
Expected             : 406
Relative concentration : 0.406
Total volume         : 1
High volume          : 0.4
Low volume           : 0.6
```

```
Sample: S4
Proportion high      : 0.6
Proportion low       : 0.4
Expected             : 604
Relative concentration : 0.604
Total volume         : 1
High volume          : 0.6
Low volume           : 0.4
```

```
Sample: S5
Proportion high      : 0.8
Proportion low       : 0.2
Expected             : 802
Relative concentration : 0.802
Total volume         : 1
```

```

High volume      : 0.8
Low volume       : 0.2

```

```

Sample: S6
Proportion high  : 1
Proportion low   : 0
Expected         : 1000
Relative concentration : 1
Total volume     : 1
High volume      : 1
Low volume       : 0

```

```
linearity_endpoints(lloq = 10, uloq = 1000, cv_high = 5, cv_at_lloq = 20, cv_at_k_lloq = 15)
```

CLSI EP06-Ed2 target concentrations

LOW target: 14.14116

HIGH target: 924.0072

The replicate-count function reports a design quantity only. Final level and replicate counts must satisfy the protocol, applicable standard, and available samples. LOW/HIGH calculations use laboratory-confirmed target concentrations and are not a substitute for a manufacturer's validation process.

14.2 Model, Weights, and ADL

```

linearity_replicates(
  imprecision, adl, type = c("relative", "absolute"),
  level_probability = 0.99, levels = NULL, family_error = NULL,
  true_deviation = 0, min_replicates = 2
)
linearity_panel(proportion_high, high, low = 0, total_volume = NULL, sample = NULL)
linearity(
  data, sample, result, x, method = c("linear", "polynomial"),
  intercept = TRUE,
  weights = c("equal", "level_variance", "pooled", "precision_profile"),
  variance_group = NULL, conf.level = NULL,
  multiplicity = c("familywise", "none"), adl = NULL,
  adl_abs = NULL, adl_rel = NULL
)

```

If the per-level central coverage is P_L , repeatability is s , expected true deviation is δ_0 , and effective ADL is $A_{\text{eff}} = A - \delta_0$, the replicate count is

$$n = \left\lceil \left(\frac{z_{(1+P_L)/2s}}{A_{\text{eff}}} \right)^2 \right\rceil.$$

For familywise error α_F over L levels, $P_L = (1 - \alpha_F)^{1/L}$. Relative inputs use percentage units: 5 means 5%.

With high-sample proportion p_h ,

$$x_{\text{expected}} = p_h x_H + (1 - p_h) x_L.$$

`linearity(method="linear")` first summarizes each level as $\bar{y}_j, s_j, n_j$ and fits $\bar{y}_j = \beta_0 + \beta_1 x_j + e_j$.

weights	Level-mean weight	Meaning
equal	1	Equal weight
level_variance	n_j/s_j^2	Use the variance at each level
pooled	$n_j/\hat{\sigma}_g^2$	Use groups supplied in <code>variance_group</code> ; no automatic grouping
precision_profile	$1/\hat{\sigma}_j^2$	Use variance predicted from an SD–mean profile

The residual is $r_j = \bar{y}_j - \hat{y}_j$, and ADL is exceeded when $|r_j| > ADL_j$. When `weights="pooled"`, groups must be explicit, contiguous after sorting by x , and contain at least two levels. `percent_base` determines the denominator of relative residuals. `multiplicity="familywise"` converts overall coverage to per-level coverage; "none" uses the overall coverage at every level.

14.3 Analysis Examples

14.3.1 Serial-Dilution Linearity

```
dilution <- read_csv("./data/014-线性分析/linearity-dilution.csv", show_col_types = FALSE)
dilution <- as.data.frame(dilution)
head(dilution, 6)
```

```
conc replicate signal
1  50         1  48.6
2  50         2  49.2
3  50         3  50.0
4 100         1 101.3
5 100         2  99.7
6 100         3 101.4
```

```
fit_lin <- linearity(
  dilution, sample = "conc", result = "signal", x = "conc",
  method = "linear", weights = "equal", conf.level = 0.95,
  multiplicity = "familywise", adl_rel = 5
)
print(fit_lin)
```

Linearity regression analysis

Analysis settings

```
Method: linear (EP06-Ed2)
Observations / levels: 24 / 8
Intercept: estimated
Weights: equal
Fit unit: level_mean
```

Model comparison

```
Degree: 1
Observations      : 8
Parameters        : 2
```

```

Residual degrees of freedom : 6
RSS                          : 146.65863
Weighted RSS                 : 146.65863
SY.X                         : 4.9439969
Weighted SY.X                : 4.9439969
R-squared                    : 0.99999571
Adjusted R-squared           : 0.999995

```

Regression coefficients

```

      Estimate Std. Error   t value Pr(>|t|)
a 0.975290473 2.205878722   0.44213 0.67388
b 0.998893935 0.000844269 1183.14720 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Sample fit information

Sample	Expected	Predicted	Observed mean	Weight	Observed SD	n
50	50	50.920	49.2667	1	0.702377	3
100	100	100.865	100.8000	1	0.953939	3
200	200	200.754	202.6667	1	4.452340	3
400	400	400.533	397.8333	1	7.396170	3
800	800	800.090	798.8667	1	10.261741	3
1600	1600	1599.206	1596.9000	1	13.753545	3
3200	3200	3197.436	3207.8000	1	20.940869	3
6400	6400	6393.896	6389.5667	1	19.923437	3

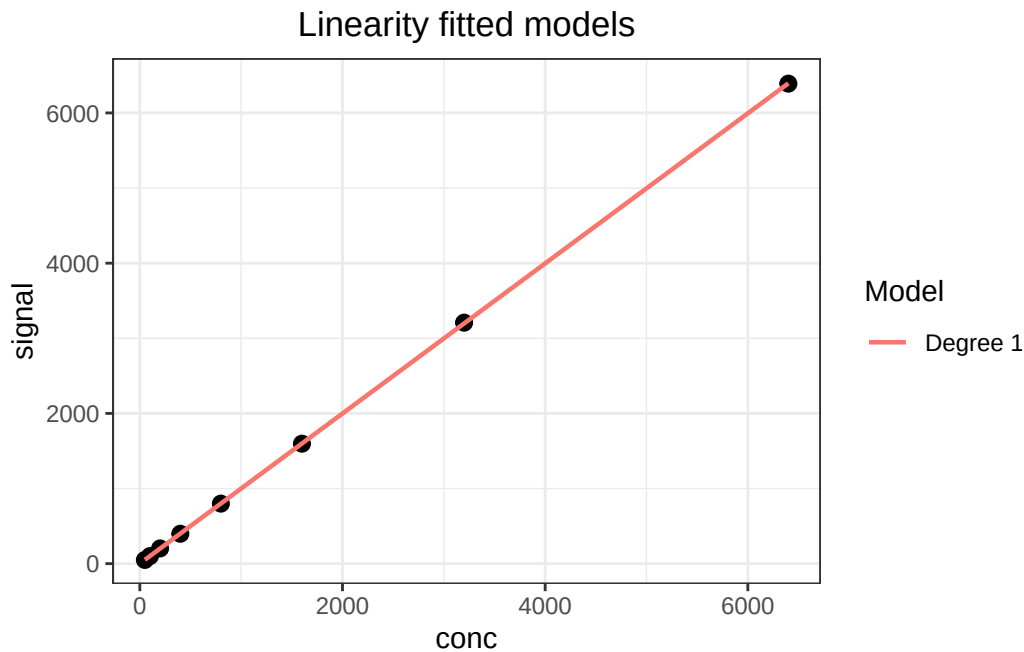
Sample bias information

Sample	Residual	Recovery bias	Residual (%)	Recovery bias (%)	ADL	Decision
50	-1.653321	-0.733333	-3.2468990	-1.466667	2.54600	Within ADL
100	-0.064684	0.800000	-0.0641295	0.800000	5.04323	Within ADL
200	1.912589	2.666667	0.9527025	1.333333	10.03770	Within ADL
400	-2.699531	-2.166667	-0.6739850	-0.541667	20.02664	Within ADL
800	-1.223772	-1.133333	-0.1529542	-0.141667	40.00452	Within ADL
1600	-2.305587	-3.100000	-0.1441708	-0.193750	79.96028	Within ADL
3200	10.364116	7.800000	0.3241384	0.243750	159.87179	Within ADL
6400	-4.329810	-10.433333	-0.0677179	-0.163021	319.69482	Within ADL

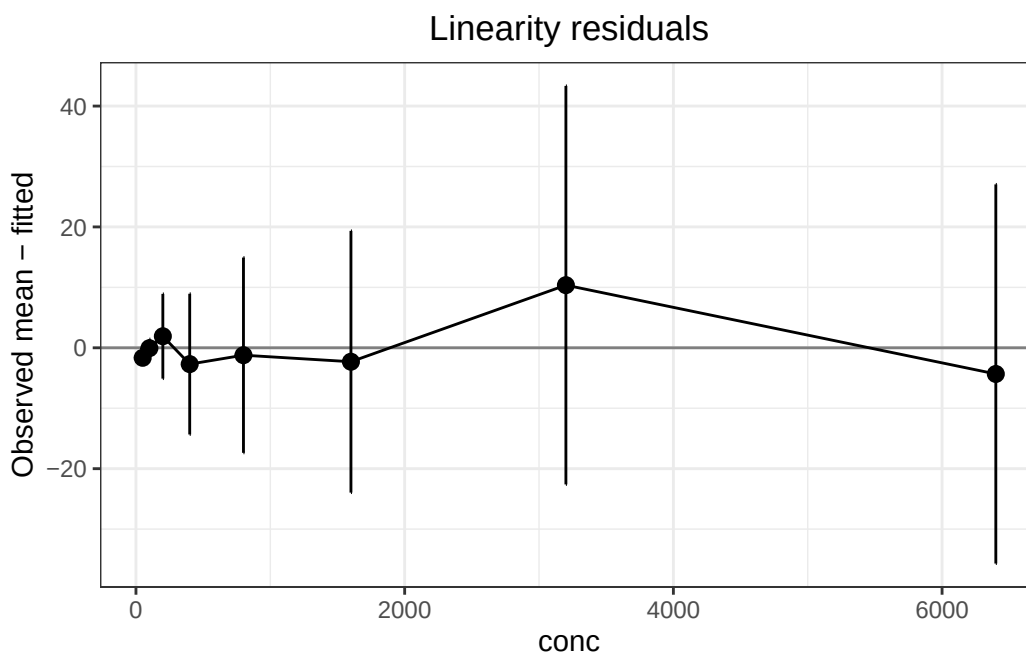
Definitions

Residual = observed mean - predicted value.
 Recovery bias = observed mean - expected value.
 Recovery bias (%) = 100 * recovery bias / expected value.
 Residual (%) denominator: predicted.
 ADL decision metric = residual.

```
plot(fit_lin, type = "fit")
```



```
plot(fit_lin, type = "residual")
```



`adl_exceeded` is a numeric comparison with the ADL supplied by the caller, not a complete product-acceptance decision. Choose weights according to the design and heteroscedasticity structure, and compare residuals, back-calculated bias, and prespecified criteria.

14.3.2 Polynomial Sensitivity Analysis

```
poly_data <- read_csv("./data/014-线性分析/linearity-polynomial.csv", show_col_types = FALSE)
poly_data <- as.data.frame(poly_data)
head(poly_data, 6)
```

	conc	replicate	signal
1	75.00	1	75.0
2	75.00	2	75.0

3	75.00	3	74.4
4	483.75	1	454.1
5	483.75	2	455.3
6	483.75	3	456.0

```
fit_poly <- linearity(
  poly_data, sample = "conc", result = "signal", x = "conc",
  method = "polynomial", weights = "level_variance", max_degree = 3,
  poly_alpha = 0.05, percent_base = "predicted"
)
print(fit_poly)
```

Linearity regression analysis

Analysis settings

Method: polynomial (EP6-A, exploratory)
 Observations / levels: 15 / 5
 Intercept: estimated
 Weights: level_variance
 Fit unit: observation
 Selected degree: 3

Model comparison

Degree: 1

Observations	: 15
Parameters	: 2
Residual degrees of freedom	: 13
RSS	: 52631.917
Weighted RSS	: 550.93015
SY.X	: 63.62868
Weighted SY.X	: 6.5099341
R-squared	: 0.99902344
Adjusted R-squared	: 0.99894832

Degree: 2

Observations	: 15
Parameters	: 3
Residual degrees of freedom	: 12
RSS	: 10241.163
Weighted RSS	: 30.175825
SY.X	: 29.213529
Weighted SY.X	: 1.5857655
R-squared	: 0.99994651
Adjusted R-squared	: 0.9999376

Degree: 3

Observations	: 15
Parameters	: 4
Residual degrees of freedom	: 11
RSS	: 4360.5146
Weighted RSS	: 15.881083
SY.X	: 19.910058
Weighted SY.X	: 1.2015552
R-squared	: 0.99997185
Adjusted R-squared	: 0.99996417
Selected	: Yes

Regression coefficients (degree 1)

```

      Estimate Std. Error   t value Pr(>|t|)
a 3.26682147 1.60149874   2.03985 0.062224 .
b 0.94635339 0.00820623 115.32135 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Regression coefficients (degree 2)

```

      Estimate Std. Error   t value   Pr(>|t|)
a 7.07723e+00 4.71485e-01  15.0105 3.8555e-09 ***
b 8.98365e-01 3.88796e-03 231.0636 < 2.22e-16 ***
c 5.87581e-05 4.08310e-06  14.3905 6.2393e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Regression coefficients (degree 3, selected)

```

      Estimate Std. Error   t value   Pr(>|t|)
a 1.08167e+01 1.24094e+00  8.71655 2.8637e-06 ***
b 8.38991e-01 1.90976e-02 43.93179 1.0374e-13 ***
c 1.93013e-04 4.27784e-05  4.51193 0.0008837 ***
d -5.97186e-08 1.89786e-08 -3.14662 0.0092980 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Sample fit information

Sample	Expected	Predicted	Observed mean	Weight	Observed SD	n
75	75.00	74.8015	74.800	8.33333333	0.346410	3
483.75	483.75	455.0861	455.133	1.08303249	0.960902	3
892.5	892.50	870.9067	865.967	0.01552474	8.025792	3
1301.25	1301.25	1297.7933	1328.467	0.00166688	24.493332	3
1710	1710.00	1711.2760	1710.433	0.01516914	8.119319	3

Sample bias information

Sample	Residual	Recovery bias	Residual (%)	Recovery bias (%)	DL	ADL	Decision
75	-0.00153386	-0.200000	-0.00206599	-0.2666667	0.558208	NA	Not assessed
483.75	0.04720887	-28.616667	0.01023909	-5.9155900	-5.979149	NA	Not assessed
892.5	-4.94005897	-26.533333	-0.58263161	-2.9729225	23.019504	NA	Not assessed
1301.25	30.67332481	27.216667	2.48425504	2.0915786	63.084173	NA	Not assessed
1710	-0.84264426	0.433333	-0.05196596	0.0253411	89.744861	NA	Not assessed

Definitions

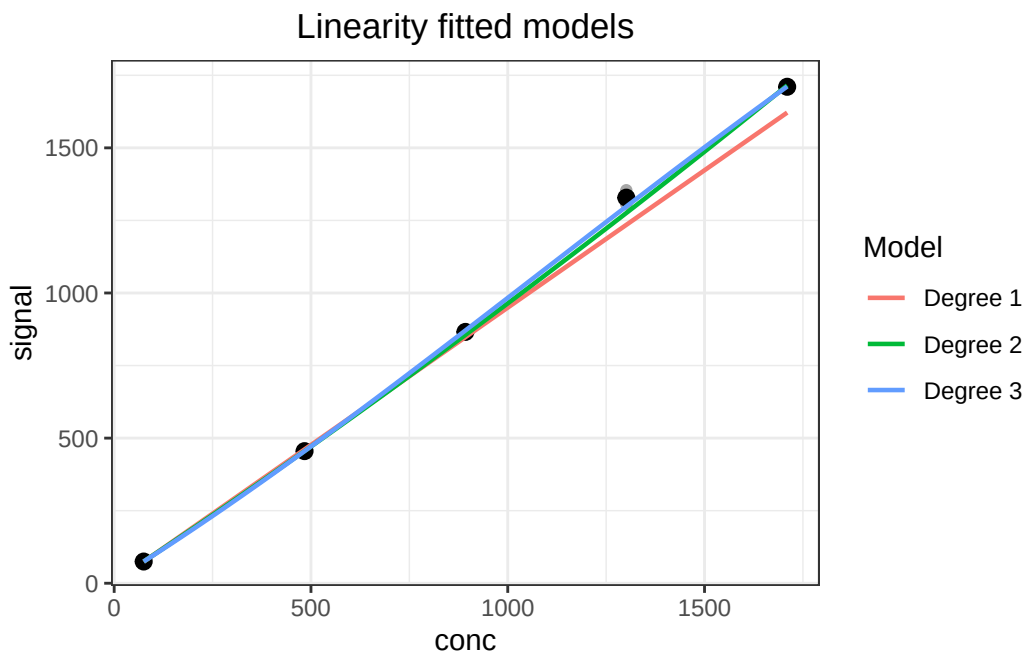
Residual = observed mean - predicted value.
 Recovery bias = observed mean - expected value.
 Recovery bias (%) = 100 * recovery bias / expected value.
 Residual (%) denominator: predicted.
 ADL decision metric = deviation from linearity (DL).

Warnings / interpretation notes

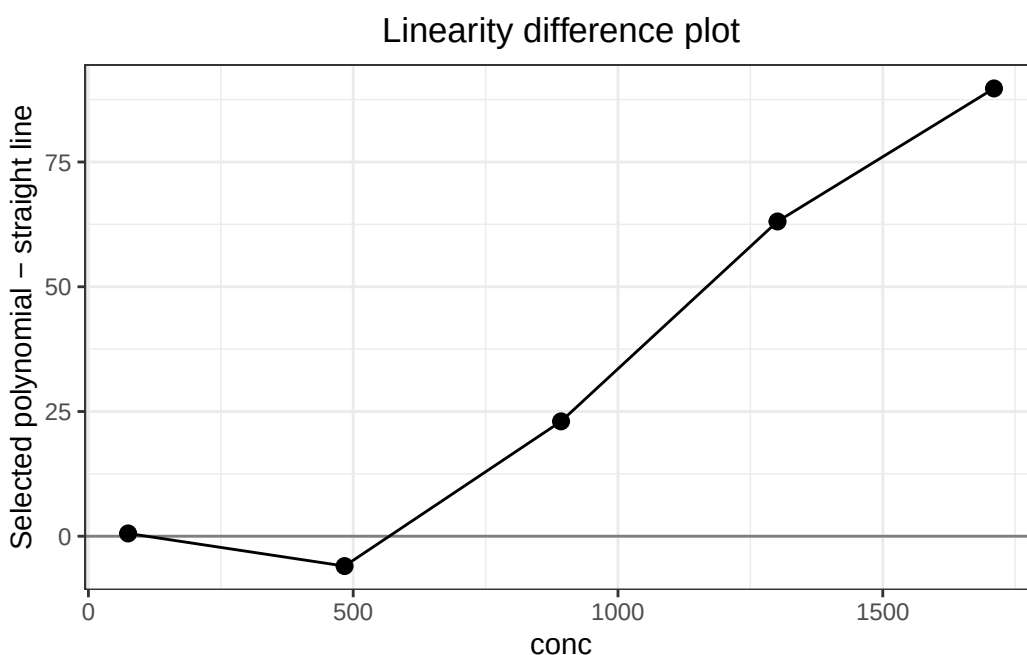
1 duplicate sample/result/x row(s) were retained.

Polynomial regression follows superseded EP6-A and is exploratory under EP06-Ed2.

```
plot(fit_poly, type = "fit")
```



```
plot(fit_poly, type = "difference")
```



14.3.3 Standard Cross-Checks

The package can reproduce the principal calculations in CLSI EP06 Appendices G-I and the YY/T 1789.4 Appendix A.5 linearity example using the supplied CSV files. The following pattern illustrates the EP06 Appendix G weighting and the reporting convention:

```
ep06_g <- as.data.frame(read_csv(
  "/data/014-线性分析/std-ep06-appendix-g.csv", show_col_types = FALSE
```

```

))
ep06_g_ivd <- linearity(
  ep06_g, sample = "level", result = "result", x = "expected",
  method = "linear", intercept = FALSE, weights = "pooled",
  variance_group = "variance_group"
)
ep06_g_ivd

```

Linearity regression analysis

Analysis settings

Method: linear (EP06-Ed2)
 Observations / levels: 20 / 10
 Intercept: fixed at zero
 Weights: pooled
 Variance group: variance_group
 Fit unit: level_mean

Model comparison

Degree: 1
 Observations : 10
 Parameters : 1
 Residual degrees of freedom : 9
 RSS : 2010.7972
 Weighted RSS : 83.183953
 SY.X : 14.947304
 Weighted SY.X : 3.0401746
 R-squared : 0.99911786
 Adjusted R-squared : 0.99901984

Regression coefficients

	Estimate	Std. Error	t value	Pr(> t)
b	1.0677279	0.0105755	100.962	4.6544e-15 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Pooled variance groups

Variance group: low

Levels : 3
 Degrees of freedom : 3
 Minimum x : 52.9
 Maximum x : 158.7
 Pooled variance : 1.21
 Pooled SD : 1.1
 Level-mean weight : 1.6528926
 Minimum weight : 1.6528926
 Maximum weight : 1.6528926

Variance group: middle

Levels : 4
 Degrees of freedom : 4
 Minimum x : 211.6
 Maximum x : 370.3
 Pooled variance : 55.255

Pooled SD : 7.4333707
 Level-mean weight : 0.036195819
 Minimum weight : 0.036195819
 Maximum weight : 0.036195819

Variance group: high
 Levels : 3
 Degrees of freedom : 3
 Minimum x : 423.2
 Maximum x : 529
 Pooled variance : 267.72167
 Pooled SD : 16.362202
 Level-mean weight : 0.0074704451
 Minimum weight : 0.0074704451
 Maximum weight : 0.0074704451

Sample fit information

Sample	Expected	Predicted	Observed mean	Weight	Observed SD	n
10	52.9	56.4828	51.25	1.65289256	0.353553	2
9	105.8	112.9656	111.30	1.65289256	1.838478	2
8	158.7	169.4484	171.95	1.65289256	0.353553	2
7	211.6	225.9312	233.15	0.03619582	5.586144	2
6	264.5	282.4140	292.05	0.03619582	6.293250	2
5	317.4	338.8968	350.15	0.03619582	4.879037	2
4	370.3	395.3796	391.45	0.03619582	11.242998	2
3	423.2	451.8624	468.20	0.00747045	20.364675	2
2	476.1	508.3453	496.65	0.00747045	18.879751	2
1	529.0	564.8281	529.00	0.00747045	5.656854	2

Sample bias information

Sample	Residual	Recovery bias	Residual (%)	Recovery bias (%)	ADL	Decision
10	-5.23281	-1.65	-9.26442	-3.11909	NA	Not assessed
9	-1.66561	5.50	-1.47444	5.19849	NA	Not assessed
8	2.50158	13.25	1.47631	8.34909	NA	Not assessed
7	7.21878	21.55	3.19512	10.18431	NA	Not assessed
6	9.63597	27.55	3.41200	10.41588	NA	Not assessed
5	11.25317	32.75	3.32053	10.31821	NA	Not assessed
4	-3.92964	21.15	-0.99389	5.71159	NA	Not assessed
3	16.33755	45.00	3.61560	10.63327	NA	Not assessed
2	-11.69525	20.55	-2.30065	4.31632	NA	Not assessed
1	-35.82806	0.00	-6.34318	0.00000	NA	Not assessed

Definitions

Residual = observed mean - predicted value.
 Recovery bias = observed mean - expected value.
 Recovery bias (%) = 100 * recovery bias / expected value.
 Residual (%) denominator: predicted.
 ADL decision metric = residual.

The ten levels are explicitly grouped as 1–3, 4–7, and 8–10. The user supplies these groups from the repeatability-SD–concentration assessment and protocol; the function does not select a favorable grouping.

For Appendix H, use `weights = "level_variance"` and fit the combined panel and each panel separately. For Appendix I, use the published summary data and weights $10/s^2$ rather than manufacturing pseudo-raw replicates. The standard's higher-order model-selection and

ADL-PctBnd lookup are not identical to the package' s exploratory polynomial mode; report this algorithm difference explicitly.

```
yyt1789_4 <- read.csv("./data/014-线性分析/std-ep06-appendix-g.csv")
yyt1789_4_fit <- linearity(
  yyt1789_4, sample = "level", result = "result", x = "expected",
  method = "linear", weights = "equal", adl_rel = 10
)
yyt1789_4_fit
```

Linearity regression analysis

Analysis settings

Method: linear (EP06-Ed2)
 Observations / levels: 20 / 10
 Intercept: estimated
 Weights: equal
 Fit unit: level_mean

Model comparison

Degree: 1
 Observations : 10
 Parameters : 2
 Residual degrees of freedom : 8
 RSS : 1709.9271
 Weighted RSS : 1709.9271
 SY.X : 14.61988
 Weighted SY.X : 14.61988
 R-squared : 0.99309927
 Adjusted R-squared : 0.99223668

Regression coefficients

	Estimate	Std. Error	t value	Pr(> t)
a	9.1333333	9.9872792	0.9145	0.38719
b	1.0324168	0.0304271	33.9308	6.2178e-10 ***

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Sample fit information

Sample	Expected	Predicted	Observed mean	Weight	Observed SD	n
10	52.9	63.7482	51.25	1	0.353553	2
9	105.8	118.3630	111.30	1	1.838478	2
8	158.7	172.9779	171.95	1	0.353553	2
7	211.6	227.5927	233.15	1	5.586144	2
6	264.5	282.2076	292.05	1	6.293250	2
5	317.4	336.8224	350.15	1	4.879037	2
4	370.3	391.4373	391.45	1	11.242998	2
3	423.2	446.0521	468.20	1	20.364675	2
2	476.1	500.6670	496.65	1	18.879751	2
1	529.0	555.2818	529.00	1	5.656854	2

Sample bias information

Sample	Residual	Recovery bias	Residual (%)	Recovery bias (%)	ADL	Decision
--------	----------	---------------	--------------	-------------------	-----	----------

10	-12.4981818	-1.65	-19.60555025	-3.11909	6.37482	Exceeds ADL
9	-7.0630303	5.50	-5.96726046	5.19849	11.83630	Within ADL
8	-1.0278788	13.25	-0.59422557	8.34909	17.29779	Within ADL
7	5.5572727	21.55	2.44176200	10.18431	22.75927	Within ADL
6	9.8424242	27.55	3.48765416	10.41588	28.22076	Within ADL
5	13.3275758	32.75	3.95685524	10.31821	33.68224	Within ADL
4	0.0127273	21.15	0.00325142	5.71159	39.14373	Within ADL
3	22.1478788	45.00	4.96531184	10.63327	44.60521	Within ADL
2	-4.0169697	20.55	-0.80232369	4.31632	50.06670	Within ADL
1	-26.2818182	0.00	-4.73305938	0.00000	55.52818	Within ADL

Definitions

Residual = observed mean - predicted value.

Recovery bias = observed mean - expected value.

Recovery bias (%) = 100 * recovery bias / expected value.

Residual (%) denominator: predicted.

ADL decision metric = residual.

14.4 Interpretation

1. A successful fit does not establish model suitability; evaluate residuals, bias, acceptance limits, and the protocol.
2. Examine residuals on both original and relative scales, especially when values span orders of magnitude.
3. Select weights using the observed variance structure or protocol requirements.
4. Conclusions apply only within the validated range; avoid unsupported extrapolation.
5. Preserve raw data, code, model parameters, convergence information, package version, and session information.

14.5 Complete Standard Cases

14.5.1 CLSI EP06 Second Edition Appendix G: Two-Replicate Study

Appendix G prepares ten levels by mixing HIGH with blank at relative concentrations of 0.1–1.0, with two measurements per level. The prespecified linear interval is 55–500 U/L and the allowable deviation from linearity is $\pm 10\%$. Because there are only two replicates per level, the standard pools adjacent-level variances into three groups and uses $2/\overline{s^2}$ as the no-intercept WLS weight. The grouping column is written directly into the CSV according to standard Table G3, and expected values are already calculated from the HIGH concentration and preparation proportions.

```
ep06_g <- as.data.frame(read_csv(
  "./data/014-线性分析/std-ep06-appendix-g.csv",
  show_col_types = FALSE
))

ep06_g_ivd <- linearity(
  ep06_g, sample = "level", result = "result", x = "expected",
  method = "linear", intercept = FALSE,
  weights = "pooled", variance_group = "variance_group"
)
```

Table 14.1: EP06 Second Edition Appendix G: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
EP06 Ed2 Appendix G	WLS_slope	1.0677	1.0677
EP06 Ed2 Appendix G	max_abs_deviation_percent	9.3000	9.2644

variance_group explicitly assigns levels 1–3, 4–7, and 8–10 to three groups. The grouping is supplied by the analyst from the repeatability-SD versus concentration plot and the pre-specified plan; the function does not select the most favorable grouping automatically.

14.5.2 CLSI EP06 Second Edition Appendix H: Two-Panel Study

Appendix H prepares two panels, HIGH1 and HIGH2, separately with blank, at ten levels per panel and five measurements per level, for 100 results. HIGH1 covers the high end and HIGH2 the low end; together they cover the case' s prespecified 20–1,000 U/L range. Each level receives weight $5/s^2$ in a no-intercept WLS fit. The CSV includes sample_id and expected values determined by panel, level, and preparation proportion, so it can enter the analysis directly.

```
ep06_h <- as.data.frame(read_csv(
  "./data/014-线性分析/std-ep06-appendix-h.csv",
  show_col_types = FALSE
))

ep06_h_fit <- function(z) {
  fit <- linearity(
    z, sample = "sample_id", result = "result", x = "expected",
    method = "linear", intercept = FALSE, weights = "level_variance"
  )
  data.frame(
    slope = fit$coefficients$estimate,
    max_abs_deviation_percent = max(abs(fit$level_summary$percent_residual_linear))
  )
}
ep06_h_combined <- ep06_h_fit(ep06_h)
ep06_h_separate <- do.call(rbind, lapply(
  split(ep06_h, ep06_h$panel), ep06_h_fit
))
```

Table 14.2: EP06 Second Edition Appendix H: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
EP06 Ed2 Appendix H	combined WLS slope	0.9686	0.9686
EP06 Ed2 Appendix H	combined maximum deviation (%)	14.5000	14.4759
EP06 Ed2 Appendix H	HIGH1 WLS slope	0.9496	0.9496
EP06 Ed2 Appendix H	HIGH1 maximum deviation (%)	12.8000	12.7612
EP06 Ed2 Appendix H	HIGH2 WLS slope	1.0013	1.0013
EP06 Ed2 Appendix H	HIGH2 maximum deviation (%)	9.2000	9.2134

The maximum deviation in all three analyses is below the case' s prespecified $\pm 15\%$, but the overlap, range coverage, and weighting structure of the two panels are prerequisites for that conclusion.

14.5.3 CLSI EP06 Second Edition Appendix I: Assigned Samples

Appendix I provides true values, ten-measurement means, and SDs for nine spiked samples but does not publish the 90 individual measurements. This case therefore retains the published summary granularity and does not create pseudo-raw observations from the means and SDs. The weight is $10/s^2$, the fitted model is a no-intercept WLS line, and the prespecified allowable deviation is $\pm 2\%$.

```
ep06_i <- as.data.frame(read_csv(
  "/data/014-线性分析/std-ep06-appendix-i-summary.csv",
  show_col_types = FALSE
))
ep06_i$weight <- ep06_i$n / ep06_i$sd^2
ep06_i_ivd <- linearity(
  ep06_i, sample = "sample", result = "measured_mean", x = "true_value",
  method = "linear", intercept = FALSE, weights = ep06_i$weight
)
```

Table 14.3: EP06 Second Edition Appendix I: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
EP06 Ed2 Appendix I	WLS_slope	1.0045	1.0045
EP06 Ed2 Appendix I	max_abs_deviation_percent	1.7000	1.7212

The recalculated maximum absolute deviation is within the case's prespecified $\pm 2\%$. Recovery bias and linearity deviation answer different questions: the former is relative to the true value, while the latter is relative to the best-fit line, so they cannot replace one another.

14.5.4 WS/T 408–2024: Five-Level Linearity Verification

The HDL-C case in Appendix A.4 contains five levels with three measurements per level. The standard focuses on the residual SD of the straight-line regression, $s_{y|x}$, the within-run SD s_{WR} , their F ratio, and the nonlinear SD s_{NL} .

```
wst408_linearity <- read_csv(
  "/data/014-线性分析/std-wst408-linearity.csv",
  show_col_types = FALSE
)
wst408_linearity_ivd <- linearity(
  as.data.frame(wst408_linearity), sample = "level", result = "measured",
  x = "expected", method = "linear", conf.level = 0.95
)
```

Table 14.4: WS/T 408 Appendix A.4: ivdtools compared with the standard text

Standard location	Statistic	Standard	ivdtools
WS/T 408 Appendix A.4	s_yx	0.0318	0.03277
WS/T 408 Appendix A.4	s_WR	0.0186	NA
WS/T 408 Appendix A.4	F	2.9130	NA
WS/T 408 Appendix A.4	s_NL	0.0258	NA

linearity() fits the line to the five level means and returns $s_{y|x} \approx 0.0328$; the standard companion table gives 0.0318. The s_{WR} , F, and s_{NL} quantities required by the standard are

not public results in the current object, so the table marks them explicitly as NA.

14.5.5 YY/T 1789.4–2022 Appendix A: Linearity Sample 1

Appendix A.5 provides 14 dilution levels for AFP linearity sample 1, with four results per level. The standard selects a model using significance of higher-order polynomial terms and then evaluates level-specific relative bias from the final line against a $\pm 10\%$ allowable error.

```
yyt1789_4 <- read.csv(
  "./data/014-线性分析/std-yyt1789-4-appendix-a-sample1.csv"
)
yyt1789_4_linear_p <- linearity(
  yyt1789_4, sample = "level", result = "result", x = "dilution",
  method = "polynomial", weights = "equal", adl_rel = 10
)
yyt1789_4_linear <- linearity(
  yyt1789_4, sample = "level", result = "result", x = "dilution",
  method = "linear", weights = "equal", adl_rel = 10
)
```

Table 14.5: YY/T 1789.4–2022 Appendix A.5: final line and relative bias

Standard location	Statistic	Standard	ivdtools
YY/T 1789.4 Appendix A.5 / level 1	final-line prediction	4.28	4.284
YY/T 1789.4 Appendix A.5 / level 1	relative bias (%)	6.07	5.974
YY/T 1789.4 Appendix A.5 / level 2	final-line prediction	19.40	19.401
YY/T 1789.4 Appendix A.5 / level 2	relative bias (%)	5.05	5.046
YY/T 1789.4 Appendix A.5 / level 3	final-line prediction	34.52	34.518
YY/T 1789.4 Appendix A.5 / level 3	relative bias (%)	4.46	4.453
YY/T 1789.4 Appendix A.5 / level 4	final-line prediction	64.75	64.752
YY/T 1789.4 Appendix A.5 / level 4	relative bias (%)	-3.69	-3.697
YY/T 1789.4 Appendix A.5 / level 5	final-line prediction	125.22	125.219
YY/T 1789.4 Appendix A.5 / level 5	relative bias (%)	7.25	7.250
YY/T 1789.4 Appendix A.5 / level 6	final-line prediction	246.16	246.154
YY/T 1789.4 Appendix A.5 / level 6	relative bias (%)	0.47	0.467
YY/T 1789.4 Appendix A.5 / level 7	final-line prediction	367.10	367.089
YY/T 1789.4 Appendix A.5 / level 7	relative bias (%)	3.77	3.773
YY/T 1789.4 Appendix A.5 / level 8	final-line prediction	488.04	488.025
YY/T 1789.4 Appendix A.5 / level 8	relative bias (%)	-4.04	-4.043
YY/T 1789.4 Appendix A.5 / level 9	final-line prediction	608.98	608.960
YY/T 1789.4 Appendix A.5 / level 9	relative bias (%)	-3.05	-3.051
YY/T 1789.4 Appendix A.5 / level 10	final-line prediction	729.92	729.895
YY/T 1789.4 Appendix A.5 / level 10	relative bias (%)	3.59	3.598
YY/T 1789.4 Appendix A.5 / level 11	final-line prediction	850.86	850.830
YY/T 1789.4 Appendix A.5 / level 11	relative bias (%)	-3.99	-3.990
YY/T 1789.4 Appendix A.5 / level 12	final-line prediction	971.80	971.765
YY/T 1789.4 Appendix A.5 / level 12	relative bias (%)	2.60	2.607
YY/T 1789.4 Appendix A.5 / level 13	final-line prediction	1092.74	1092.700
YY/T 1789.4 Appendix A.5 / level 13	relative bias (%)	-3.20	-3.198
YY/T 1789.4 Appendix A.5 / level 14	final-line prediction	1213.68	1213.635
YY/T 1789.4 Appendix A.5 / level 14	relative bias (%)	2.56	2.566

The model-selection procedures are not identical: the standard fits polynomials to level means and reports regression standard errors, whereas `linearity(method = "polynomial")` currently fits individual observations and labels this mode as the legacy EP6-A exploratory

workflow; the degrees of freedom and regression standard errors differ. This case therefore calls the package's linear method a reproduction of the final line only and does not present its polynomial selection as YY/T 1789.4's higher-order significance test.

The ADL-PctBnd critical table in standard Appendix A.3 also combines regression imprecision, number of levels, and replicate count. `adl_rel = 10` is a level-wise numerical limit and does not implement the PctBnd lookup rule. The two similarly named ADL concepts use different algorithms and must be distinguished in reporting.

Chapter 15

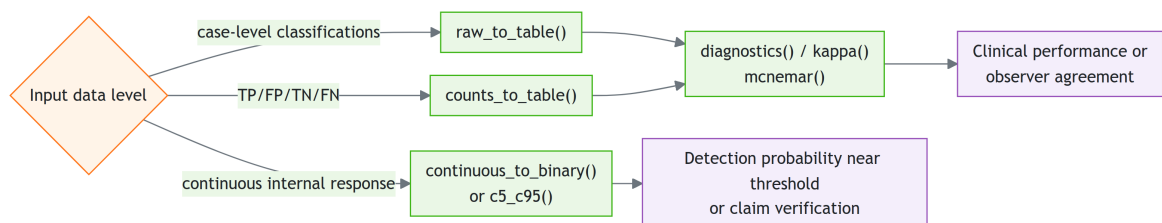
Qualitative Analysis and C5/C95

This chapter addresses three different questions: whether binary results from a candidate and reference method agree, how strong the agreement between positive/negative classifications is, and where a continuous response reaches a specified detection probability near a critical concentration. The first two questions use paired four-fold tables; the last uses `c5_c95()`.

! Scope of Use

Before analysis, fix the positive direction, the reference method or clinical-state definition, the rule for combining replicate measurements, and the confidence-interval method. If a subject has multiple results, define a subject-level classification according to the protocol first; do not treat replicates as independent samples to inflate the sample size.

Workflow: from classification results to qualitative-performance conclusions



Qualitative analysis evaluates agreement between candidate and reference positive/negative classifications, corresponding to CLSI EP12. `ivdtools` provides `raw_to_table()` to build a four-fold table from raw paired data, `counts_to_table()` to build one directly from TP/FP/TN/FN counts, and three analysis methods: `diagnostics()` (sensitivity, specificity, PPV, NPV, likelihood ratios, and so on), `kappa()` (Kappa and PABAK), and `mcnemar()` (paired-difference test). Use `continuous_to_binary()` when quantitative results must be converted to qualitative classifications.

```
library(ivdtools)
library(readr)
```

15.1 Main Functions

Function	Main purpose	Key inputs or outputs
<code>raw_to_table()</code>	Build a four-fold table from raw paired data	<code>candidate</code> , <code>reference</code> , <code>positive</code> , <code>na.rm</code>
<code>counts_to_table()</code>	Build a four-fold table from TP/FP/TN/FN	<code>candidate/reference</code> labels
<code>continuous_to_binary()</code>	Convert quantitative results to qualitative classes	<code>cutoff</code> ($\geq$ cutoff is 1)
<code>describe()</code>	Describe raw data	Available only for <code>raw_to_table()</code> sources
<code>diagnostics()</code>	Diagnostic accuracy metrics	<code>ci.method</code> , <code>prevalence</code>
<code>kappa()</code>	Kappa / PABAK	<code>prevalence switch</code>
<code>mcnemar()</code>	Paired-difference test	Exact test when discordant pairs <25
<code>c5_c95()</code>	C5/C95 from an internal continuous response	<code>cutoff</code> , <code>direction</code> , <code>precision profile</code>

Choose the function according to the data level. Use `counts_to_table()` when TP/FP/TN/FN are already available; use `raw_to_table()` for paired subject-level classifications; with only continuous signal and reference concentrations, first confirm that the replicate/precision relationship required by `c5_c95()` can be established. `diagnostics()`, `kappa()`, and `mcnemar()` all operate on the same four-fold table, so denominators must remain consistent in the report.

15.1.1 Four-Fold Metrics and Test Formulas

```
raw_to_table(data, candidate, reference, id = NULL,
             na.rm = TRUE, positive = NULL)
counts_to_table(tp, fp, tn, fn,
               candidate = "candidate", reference = "reference")
diagnostics(object, conf.level = 0.95,
            ci.method = "wilson", prevalence = NULL)
kappa(x, prevalence = NULL, conf.level = 0.95)
mcnemar(x, conf.level = 0.95)
```

Let the four-fold table be $a = TP$, $b = FP$, $c = FN$, $d = TN$. Main metrics are

$$Se = \frac{a}{a + c}, \quad Sp = \frac{d}{b + d}, \quad PPV = \frac{a}{a + b}, \quad NPV = \frac{d}{c + d},$$

$$LR^+ = \frac{Se}{1 - Sp}, \quad LR^- = \frac{1 - Se}{Sp}.$$

`diagnostics()` can use Wilson, Wald, continuity-corrected Wald/Wilson, Agresti-Coull, Jeffreys, or Clopper-Pearson proportion intervals. When a target-population prevalence π is supplied, PPV/NPV are recalculated by Bayes' theorem; the remaining metrics still come from the original table. When a cell is zero, log-approximation intervals for LR or OR may be NA; the function does not automatically apply a 0.5 correction.

Cohen's Kappa is

$$\kappa = \frac{P_o - P_e}{1 - P_e}, \quad P_o = \frac{a + d}{n},$$

where P_e is calculated from the marginal proportions of the two classifications. Passing `prevalence` to `kappa()` switches the current implementation to $PABAK = 2P_o - 1$; this numeric argument triggers PABAK and records its label but does not enter the formula. McNe-

mar uses only discordant pairs b, c : when $b + c < 25$ it performs a two-sided exact binomial test; otherwise it uses the continuity-corrected statistic

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}.$$

raw_to_table() and counts_to_table() return four-fold-table objects; continuous_to_binary() returns a data frame with binary columns; describe() returns a descriptive table; and diagnostics(), kappa(), and mcnemar() return statistics, intervals, and test results.

15.1.2 C5/C95 Probability Model

```
c5_c95(data, result = NULL, sample = NULL, cutoff,
        direction = c("geq", "leq"),
        probabilities = c(0.05, 0.95), component = NULL,
        model.no = 1:10, K = 2, search_range = NULL)
```

For an internal continuous-response mean μ and precision profile $\sigma(\mu)$, when `direction="geq"`,

$$P(\text{positive} \mid \mu) = 1 - \Phi\left(\frac{c - \mu}{\sigma(\mu)}\right).$$

C5/C95 are the values of μ at which this probability equals 0.05 and 0.95, respectively. Input may be raw replicate data or a precision object after `variance()` has been run. If there are multiple roots, the function selects the root closest to the cutoff; missing roots, multiple roots, and roots outside the observed-mean range are recorded in `issues` and should not be ignored.

`c5_c95()` returns C5/C95, the fitted model, and `issues`.

15.2 Analysis Examples

15.2.1 Example 1: Complete Analysis of Raw Paired Data

```
qual <- read_csv("../data/015-定性分析与C5C95/qualitative-raw.csv", show_col_types = FALSE)
qual <- as.data.frame(qual)

qa <- raw_to_table(
  qual,
  candidate = "new",
  reference = "gold",
  id = "id",
  positive = "positive"
)
qa
```

Fourfold Table

Source	: raw data
Total n	: 60
Duplicate IDs	: none
Candidate	: new
Reference	: gold

2x2 Contingency Table

		gold		
		positive	negative	Sum
new	positive	23	1	24
	negative	1	35	36
	sum	24	36	60

`raw_to_table()` requires exactly two levels in both candidate and reference columns; `positive` specifies the positive level for factor/character mapping; `na.rm` controls whether missing rows are removed. `id` is optional and is used to detect duplicate samples.

```
qa <- describe(qa)
```

Data Summary

```
Rows: 60
Columns: 5
ID: 60 unique, no duplicates
```

```
new
  negative      36 ( 60.0%)
  positive      24 ( 40.0%)
gold
  negative      36 ( 60.0%)
  positive      24 ( 40.0%)
age
Mean (SD):      51.05 (13.45)
Median (Q1-Q3): 48.90 (42.15 - 57.28)
Range:          24.30 - 90.10
gender
  F              32 ( 53.3%)
  M              28 ( 46.7%)
```

`describe()` is available only for `raw_to_table()` sources and reports row count, duplicate IDs, column summaries, and missingness.

```
qa <- diagnostics(qa, ci.method = "wilson")
```

Diagnostic Performance Summary

```
Sensitivity      0.958 (0.798, 0.993)
Specificity      0.972 (0.858, 0.995)
PPV              0.958 (0.798, 0.993)
NPV              0.972 (0.858, 0.995)
Accuracy         0.967 (0.886, 0.991)
Prevalence       0.400 (0.286, 0.526)
LR+              34.500 (4.986, 238.724)
LR-              0.043 (0.006, 0.292)
Odds Ratio       805.000 (47.921, 13522.837)
```

```
# PPV & NPV are based on data prevalence of 0.400.
```

```
# Confidence intervals for proportions use 'wilson' method.
```

Sensitivity is 0.958 and specificity is 0.972 in this example. If target-population prevalence is known, use prevalence to recalculate PPV/NPV:


```
qa <- diagnostics(qa, prevalence = 0.4)
```

Diagnostic Performance Summary

Sensitivity	0.958	(0.798, 0.993)
Specificity	0.972	(0.858, 0.995)
PPV	0.958	(0.790, 0.993)
NPV	0.972	(0.864, 0.995)
Accuracy	0.967	(0.886, 0.991)
Prevalence	0.400	
LR+	34.500	(4.986, 238.724)
LR-	0.043	(0.006, 0.292)
Odds Ratio	805.000	(47.921, 13522.837)

PPV & NPV are based on user-specified prevalence of 0.400.

Confidence intervals for proportions use 'wilson' method.

diagnostics() supports seven proportion confidence-interval methods (default "wilson"). With prevalence, PPV/NPV use Bayes' theorem; otherwise they use the observed prevalence.

```
qa <- kappa(qa)
```

Cohen's Kappa

new vs gold
n = 60

Kappa	0.9306
SE	0.0483
95% CI	(0.8359, 1.0000)
Observed agreement	0.9667
Expected agreement	0.5200

```
qa <- kappa(qa, prevalence = 0.5)
```

PABAK

new vs gold
n = 60

Kappa	0.9333
SE	0.0463
95% CI	(0.8425, 1.0000)
Observed agreement	0.9667
Expected agreement	0.5000

PABAK (prevalence = 0.500): $2 \cdot p_{\text{obs}} - 1$, assumes expected agreement = 0.5

kappa() returns Cohen's Kappa; with prevalence it instead computes PABAK ($2 \cdot p_{\text{obs}} - 1$). These handle prevalence bias differently, so report the method.

```
qa <- mcnemar(qa)
```

McNemar Test

new vs gold
Discordant pairs: b (pos, neg) = 1, c (neg, pos) = 1
n (b + c) = 2

Method: Exact binomial test
P-value: 1.0000
Conclusion: Not significant at 95% confidence level

Discordant pairs satisfy $b + c = 2 < 25$, so the function automatically uses the exact binomial test.

```
summary(qa)
```

Qualitative Analysis - summary

Fourfold Table

Source : raw data
Total n : 60
Duplicate IDs : none
Candidate : new
Reference : gold

2x2 Contingency Table

		gold		
		positive	negative	Sum
new	positive	23	1	24
	negative	1	35	36
	sum	24	36	60

Analyses performed:

☒ Describe
☒ Diagnostics
☒ Kappa
☒ McNemar

Data Summary

Rows: 60
Columns: 5
ID: 60 unique, no duplicates

new
negative 36 (60.0%)
positive 24 (40.0%)
gold
negative 36 (60.0%)
positive 24 (40.0%)
age
Mean (SD): 51.05 (13.45)
Median (Q1-Q3): 48.90 (42.15 - 57.28)
Range: 24.30 - 90.10
gender
F 32 (53.3%)
M 28 (46.7%)

Diagnostic Performance Summary

Sensitivity 0.958 (0.798, 0.993)

```

Specificity      0.972 (0.858, 0.995)
PPV              0.958 (0.790, 0.993)
NPV              0.972 (0.864, 0.995)
Accuracy         0.967 (0.886, 0.991)
Prevalence       0.400
LR+              34.500 (4.986, 238.724)
LR-              0.043 (0.006, 0.292)
Odds Ratio       805.000 (47.921, 13522.837)

```

PPV & NPV are based on user-specified prevalence of 0.400.

Confidence intervals for proportions use 'wilson' method.

PABAK

```

new vs gold
n = 60

```

```

Kappa           0.9333
SE              0.0463
95% CI          (0.8425, 1.0000)
Observed agreement 0.9667
Expected agreement 0.5000
# PABAK (prevalence = 0.500): 2 * p_obs - 1, assumes expected agreement = 0.5

```

McNemar Test

```

new vs gold
Discordant pairs: b (pos, neg) = 1, c (neg, pos) = 1
n (b + c)         = 2

```

Method: Exact binomial test

P-value: 1.0000

Conclusion: Not significant at 95% confidence level

15.2.2 Example 2: Build Directly from TP/FP/TN/FN

When counts are already available, raw data are not needed:

```

tab <- counts_to_table(
  tp = 80, fp = 5, tn = 120, fn = 3,
  candidate = "new method",
  reference = "gold standard"
)
tab

```

Fourfold Table

```

Source      : counts
Total n     : 208
Candidate   : new method
Reference    : gold standard

```

2x2 Contingency Table

```

-----
                        gold standard
                        positive  negative  Sum
-----
new method  positive  80         5        85
            negative  3         120       123
            sum       83        125       208
-----

```

```
-----
```

```
tab <- diagnostics(tab)
```

Diagnostic Performance Summary

Sensitivity	0.964	(0.899, 0.988)
Specificity	0.960	(0.910, 0.983)
PPV	0.941	(0.870, 0.975)
NPV	0.976	(0.931, 0.992)
Accuracy	0.962	(0.926, 0.980)
Prevalence	0.399	(0.335, 0.467)
LR+	24.096	(10.198, 56.934)
LR-	0.038	(0.012, 0.114)
Odds Ratio	640.000	(148.773, 2753.180)

PPV & NPV are based on data prevalence of 0.399.

Confidence intervals for proportions use 'wilson' method.

```
tab <- kappa(tab)
```

Cohen's Kappa

new method vs gold standard
n = 208

Kappa	0.9201
SE	0.0277
95% CI	(0.8659, 0.9744)
Observed agreement	0.9615
Expected agreement	0.5184

```
tab <- mcnemar(tab)
```

McNemar Test

new method vs gold standard
Discordant pairs: b (pos, neg) = 5, c (neg, pos) = 3
n (b + c) = 8

Method: Exact binomial test
P-value: 0.7266
Conclusion: Not significant at 95% confidence level

```
describe(tab) # count sources have no raw data; describe() is unavailable
```

Objects built by `counts_to_table()` have no raw data, so `describe()` is unavailable; `diagnostics()`, `kappa()`, and `mcnemar()` work normally.

15.2.3 Example 3: Convert Quantitative Data to Qualitative

The candidate method is the quantitative result result and the reference is the qualitative variable gold:

```
cont <- read_csv("./data/015-定性分析与C5C95/qualitative-continuous.csv", show_col_types = FALSE)
cont <- as.data.frame(cont)
```

```
head(cont, 6)
```

```

  id    result    gold
1 S001 10.111087 positive
2 S002 13.057209 positive
3 S003  1.534680 negative
4 S004  7.782787 negative
5 S005  4.982424 negative
6 S006  9.443399 negative

```

Convert to binary at cutoff 8:

```
cont_bin <- continuous_to_binary(cont, cols = "result", cutoff = 8)
head(cont_bin, 6)
```

```

  id    result    gold result_binary
1 S001 10.111087 positive            1
2 S002 13.057209 positive            1
3 S003  1.534680 negative            0
4 S004  7.782787 negative            0
5 S005  4.982424 negative            0
6 S006  9.443399 negative            1

```

`continuous_to_binary()` creates `result_binary` ($\geq$ cutoff is 1, otherwise 0). To satisfy the positive mapping requirement of `raw_to_table()`, relabel it consistently with the reference:

```
cont_bin$result_binary_lab <- ifelse(cont_bin$result_binary == 1, "positive", "negative")
```

```

qa3 <- raw_to_table(
  cont_bin,
  candidate = "result_binary_lab",
  reference = "gold",
  id = "id",
  positive = "positive"
)
qa3 <- diagnostics(qa3)

```

Diagnostic Performance Summary

Sensitivity	0.952	(0.773, 0.992)
Specificity	0.897	(0.736, 0.964)
PPV	0.870	(0.679, 0.955)
NPV	0.963	(0.817, 0.993)
Accuracy	0.920	(0.812, 0.968)
Prevalence	0.420	(0.294, 0.558)
LR+	9.206	(3.140, 26.994)
LR-	0.053	(0.008, 0.361)
Odds Ratio	173.333	(16.746, 1794.100)

PPV & NPV are based on data prevalence of 0.420.

Confidence intervals for proportions use 'wilson' method.

The level supplied to `positive` must exist in both columns, so relabel the binary column as positive/negative first. If `positive` is omitted, the function may reverse the positive/negative direction according to sorted levels and flip TP/FP; always check.

15.2.4 Example 4: Estimate C5/C95 from an Internal Continuous Response

When a qualitative classification is based on an internal continuous signal and a fixed cutoff, use replicate measurements to establish an SD–mean precision profile and then find the mean response at the specified positive probabilities. Here `result >= 40` defines the target event:

```
head(c5_data, 6)
```

```
level response
1    L1 35.14000
2    L1 35.29636
3    L1 35.45273
4    L1 35.60909
5    L1 35.76545
6    L1 35.92182
```

```
c5_result <- c5_c95(
  c5_data,
  result = "response",
  sample = "level",
  cutoff = 40,
  direction = "geq",
  probabilities = c(0.05, 0.95),
  model.no = 1
)
c5_result
```

C5/C95 estimation from an internal continuous response

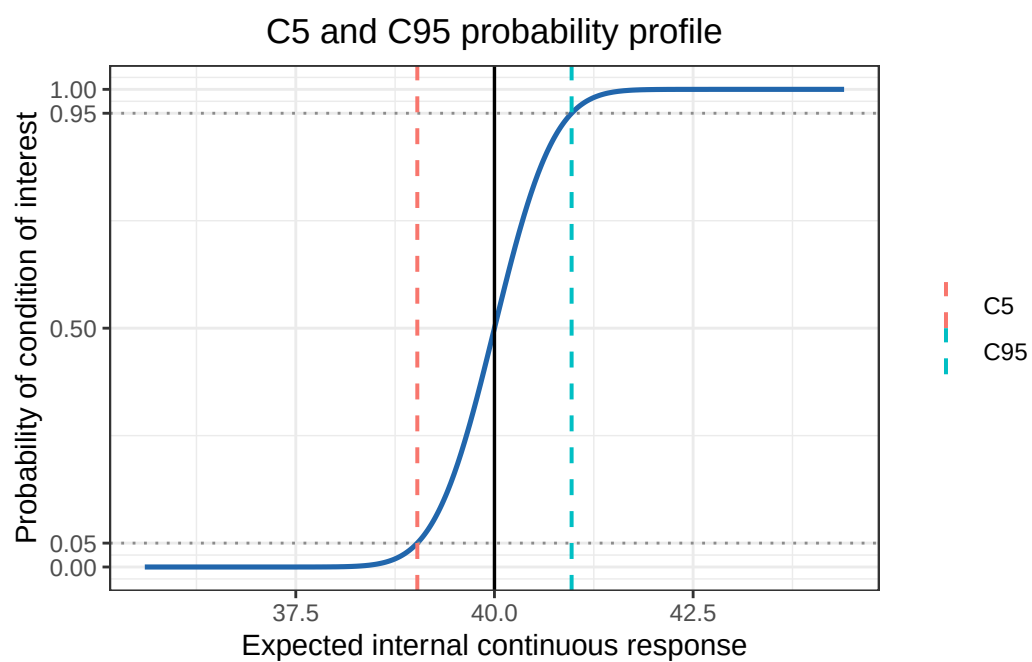
Source: raw data

Precision component: repeatability

Event: response >= cutoff

cutoff	label	probability	mean_response	SD	actual_probability	extrapolated
40	C5	0.05	39.0291	0.5903	0.05	FALSE
40	C95	0.95	40.9709	0.5903	0.95	FALSE

```
plot(c5_result, type = "probability")
```



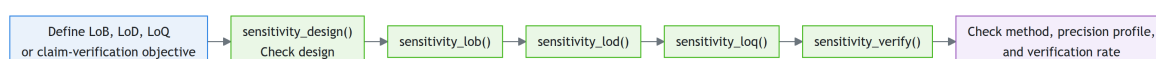
direction = "geq" means a response greater than or equal to the cutoff is the target event; "leq" means less than or equal to the cutoff. If a root lies outside the observed mean range, the object marks extrapolation. The precision-profile model and its coverage range should be included in the interpretation.

Chapter 16

Limits of Blank, Detection, and Quantitation

This chapter explains how to turn low-concentration experimental data into auditable sensitivity conclusions for the limit of blank (LoB), limit of detection (LoD), limit of quantitation (LoQ), and verification of those claims. The metrics answer different questions: LoB describes the upper tail of the blank distribution, LoD describes detection capability, and LoQ must additionally specify a precision or total-error requirement.

Workflow: establish or verify LoB, LoD, and LoQ



`library(ivdtools)`

16.1 Main Functions

The analytical-sensitivity module uses raw long-format data as its main input and separates limit establishment, claim verification, and study design:

Function	Purpose
<code>sensitivity_lob()</code>	Establish nonparametric, parametric, or zero LoB
<code>sensitivity_lod()</code>	Classical, nonparametric, precision-profile, or Probit LoD
<code>sensitivity_loq()</code>	Westgard, RMS, precision-based, or custom-target LoQ
<code>sensitivity_verify()</code>	Verify an existing LoB, LoD, or LoQ claim
<code>sensitivity_design()</code>	View minimum design requirements and actual data counts

16.1.1 Interfaces, Parameters, and Core Statistics

```
sensitivity_lob(  
  data, result, sample, lot = NULL,  
  method = c("nonparametric", "parametric", "zero"),  
  alpha = 0.05, lot_rule = c("ep17", "separate", "combined"),  
  exclude = NULL
```

```

)
sensitivity_lod(
  data, result = NULL, sample = NULL, lob, lot = NULL,
  concentration = NULL, detected = NULL,
  method = c("classical", "nonparametric", "profile", "probit"),
  beta = 0.05, profile_model = c("linear", "quadratic", "sadler", "vfp")
)
sensitivity_loq(
  data, result, sample, assigned, lot = NULL,
  method = c("westgard", "rms", "precision", "custom"),
  goal, goal_scale = c("relative", "absolute"),
  lod = NULL, k = 2, profile = FALSE
)

```

Nonparametric LoB is the $(1 - \alpha)$ rank quantile of blank results; the parametric method uses

$$LoB = \bar{x}_B + c_p s_B,$$

where c_p applies a finite-sample correction based on α and blank degrees of freedom $B - K$. `method="zero"` fixes $LoB = 0$ and reports the proportion of blank results above zero. For classical LoD, SDs from multiple low-level samples are pooled by degrees of freedom:

$$s_p = \sqrt{\frac{\sum_j (n_j - 1) s_j^2}{\sum_j (n_j - 1)}},$$

and combined with LoB, β , and a finite-sample correction. The precision-profile method solves for concentration-dependent SD; the Probit method fits the detection-probability-concentration relationship. Their data structures and assumptions differ.

At each LoQ level, bias is $b = \bar{x} - x_a$. The four error quantities are:

method	Quantity compared with goal
westgard	$ b + ks$
rms	$\sqrt{s^2 + b^2}$
precision	s
custom	Value returned by <code>accuracy_function()</code>

When `goal_scale="relative"`, the quantity is divided by $|x_a|$. A candidate LoQ is the lowest observed mean or fitted-profile intersection satisfying the target. Multiple lots are handled by the selected `lot_rule`, and the most conservative finite candidate is reported. When LoD is supplied,

$$LoQ_{\text{reported}} = \max(LoQ_{\text{candidate}}, LoD).$$

`sensitivity_lob()`, `sensitivity_lod()`, and `sensitivity_loq()` return the corresponding limit, method, core statistics, and diagnostics.

16.1.2 Claim Verification and Design Checks

```
sensitivity_verify(
  data, result, sample, limit = c("lob", "lod", "loq"),
  claim, lob = NULL, assigned = NULL, allowable_error = NULL,
  error_scale = c("relative", "absolute"), alpha = 0.05
)
sensitivity_design(method = "all", data = NULL, ...)
```

LoB verification counts $x \leq \text{claim}$, LoD verification counts $x \geq \text{LoB}$, and LoQ verification checks whether results fall within the allowable-error window around the assigned value. The function returns the observed success proportion, a one-sided Wilson score interval, and EP17 Table 1 boundaries, but intentionally does not generate a pass field. `sensitivity_design()` gives minimum design requirements and can summarize actual data counts; it shows gaps but does not replace protocol decisions.

`sensitivity_verify()` returns the observed success proportion, Wilson interval, and boundary information. `sensitivity_design()` returns minimum requirements and an actual-count table.

16.2 Analysis Examples

16.2.1 LoB

Each row of the raw data is one measurement:

```
# Simulate data
set.seed(17)
blank <- expand_grid(
  lot = c("A", "B"), sample = paste0("blank", 1:4), replicate = 1:15
)
blank$result <- rnorm(nrow(blank), 0.2, 0.1)
head(blank, 6)
```

	lot	sample	replicate	result
1	A	blank1	1	0.09849913
2	B	blank1	1	0.19203633
3	A	blank2	1	0.17670130
4	B	blank2	1	0.11827321
5	A	blank3	1	0.27720908
6	B	blank3	1	0.18343881

```
# Establish LoB
lob_np <- sensitivity_lob(
  blank, result = "result", sample = "sample", lot = "lot",
  method = "nonparametric"
)
lob_np
```

Analytical sensitivity: LOB
 Method: nonparametric
 Estimate: 0.378828

EP17 lot rules are used by default for all establishment functions: 2–3 lots are calculated separately and the maximum is reported; 4 or more lots are combined. Negative measurement results are not automatically truncated to 0.

16.2.2 LoD

Classical LoD uses the pooled SD of low-level samples:

```
# Simulate data
low <- expand.grid(
  lot = c("A", "B"), sample = paste0("low", 1:4), replicate = 1:15
)
low$result <- rnorm(nrow(low), 1.2, 0.2)
head(low, 6)
```

	lot	sample	replicate	result
1	A	low1	1	1.4763513
2	B	low1	1	1.2897865
3	A	low2	1	1.1695380
4	B	low2	1	1.2518588
5	A	low3	1	0.8777197
6	B	low3	1	0.7982088

```
# Establish LoD
lod_classical <- sensitivity_lod(
  low, result = "result", sample = "sample", lot = "lot",
  lob = lob_np, method = "classical"
)
lod_classical
```

Analytical sensitivity: LOD

Method: classical

Estimate: 0.771614

method = "profile" supports linear, quadratic, sadler, and vfp precision profiles; method = "probit" supports raw 0/1 detection results or count/total summaries. Cochran statistics and fit measures are diagnostics only; the program does not choose the analysis path for the user.

16.2.3 LoQ

The LoQ accuracy target must be prespecified. A relative target is expressed as a proportion, so 21.6% is written 0.216:

```
# Simulate data
loq_data <- expand.grid(
  lot = c("A", "B"), sample = paste0("pool", 1:4), replicate = 1:9
)
loq_data$assigned <- rep(rep(c(30, 35, 40, 45), each = 9), 2)
loq_data$result <- rnorm(nrow(loq_data), loq_data$assigned,
  0.06 * loq_data$assigned)
head(loq_data, 6)
```

	lot	sample	replicate	assigned	result
1	A	pool1	1	30	29.70842
2	B	pool1	1	30	28.62287
3	A	pool2	1	30	29.82049
4	B	pool2	1	30	30.07043
5	A	pool3	1	30	28.84843
6	B	pool3	1	30	28.98056

```
# Establish LoQ
loq_te <- sensitivity_loq(
  loq_data, result = "result", sample = "sample", assigned = "assigned",
  lot = "lot", method = "westgard", goal = 0.216,
  goal_scale = "relative", lod = lod_classical
)
loq_te
```

Analytical sensitivity: LOQ

Method: westgard

Estimate: 29.4924

Candidate LoQ: 29.4924

Reported LoQ : 29.4924

The result stores both `candidate_loq` and `reported_loq`, so the LoD constraint does not overwrite the original candidate value.

16.2.4 Claim Verification

limit can specify only one claim type at a time:

```
# Simulate data
verification <- data.frame(
  sample = rep(c("blank1", "blank2"), each = 10),
  result = c(rep(0, 19), 1.2)
)
head(verification, 6)
```

```
sample result
1 blank1      0
2 blank1      0
3 blank1      0
4 blank1      0
5 blank1      0
6 blank1      0
```

```
# Verify LoB
v_lob <- sensitivity_verify(
  verification, result = "result", sample = "sample",
  limit = "lob", claim = 1
)
v_lob
```

Analytical sensitivity: VERIFICATION

Method: lob

Limit : lob

Claim : 1

LoB : 1

Total results : 20

Consistent results : 19

Observed proportion : 0.95

Score CI lower : 0.80399276

Score CI upper : 0.98876518

EP17 boundary : 0.85

Difference from boundary : 0.1

- LoB: count result <= claim.
- LoD: count result >= lob; the LoD claim is not a detection decision threshold.

- LoQ: check whether the result falls within the allowable-error window around each sample's assigned value.

The result contains the observed proportion, one-sided Wilson interval, EP17 Table 1 boundaries, and differences. It does not contain a pass or verified field.

16.2.5 Study Design

```
sensitivity_design("all")
```

Analytical sensitivity study design

Method: classical

```
Reagent lots      : 2
Instruments       : 1
Days              : 3
Blank samples     : 4
Low samples       : 4
Concentration levels : NA
Replicates per lot : 60
Replicate scope   : blank and low results separately
```

Method: profile

```
Reagent lots      : 2
Instruments       : 1
Days              : 5
Blank samples     : 0
Low samples       : 5
Concentration levels : 5
Replicates per lot : 40
Replicate scope   : per sample
```

Method: probit

```
Reagent lots      : 2
Instruments       : NA
Days              : NA
Blank samples     : 0
Low samples       : 3
Concentration levels : 5
Replicates per lot : 20
Replicate scope   : per sample and concentration level
```

Method: loq

```
Reagent lots      : 2
Instruments       : 1
Days              : 3
Blank samples     : 0
Low samples       : 4
Concentration levels : 1
Replicates per lot : 36
Replicate scope   : total low results
```

Method: verification

```
Reagent lots      : 1
Instruments       : 1
Days              : 3
Blank samples     : 2
Low samples       : 0
Concentration levels : 1
```

```
Replicates per lot : 20
Replicate scope   : total verification results
```

```
sensitivity_design("verification", verification,
                  result = "result", sample = "sample", limit = "lob")
```

Analytical sensitivity study design

```
Method           : verification
Reagent lots     : 1
Instruments      : 1
Days             : 3
Blank samples    : 2
Low samples      : 0
Concentration levels : 1
Replicates per lot : 20
Replicate scope  : total verification results
```

Observed study counts

```
Observations     : 20
Reagent lots     : NA
Days             : NA
Samples          : 2
Concentration levels : NA
```

The design object reports minimum requirements alongside actual counts but does not make a pass/fail decision. Exclusions must be supplied explicitly through `exclude` and are preserved in the result object.

16.2.6 Standard Case: CLSI EP17-A2 Appendix A, Classical LoB and LoD

Two reagent lots each have 60 blank and 60 low-level results. Calculate each lot under the EP17 lot rule and report the larger value.

```
ep17_blank <- read.csv("../data/016-分析灵敏度/std-ep17-appendix-a-blank.csv", check.names = FALSE)
ep17_low <- read.csv("../data/016-分析灵敏度/std-ep17-appendix-a-low.csv", check.names = FALSE)

ep17_lob_np <- sensitivity_lob(
  ep17_blank, result = "result", sample = "sample", lot = "lot",
  method = "nonparametric"
)
ep17_lod_np <- sensitivity_lod(
  ep17_low, result = "result", sample = "sample", lot = "lot",
  lob = ep17_lob_np, method = "classical"
)
ep17_lob_p <- sensitivity_lob(
  ep17_blank, result = "result", sample = "sample", lot = "lot",
  method = "parametric"
)
ep17_lod_p <- sensitivity_lod(
  ep17_low, result = "result", sample = "sample", lot = "lot",
  lob = ep17_lob_p, method = "classical"
)
```

Table 16.1: EP17-A2 Appendix A: comparison with the standard

Standard position	Statistic	Standard result	ivdtools result
EP17-A2 Appendix A	nonparametric LoB lot 1	7.6	7.6000

Standard position	Statistic	Standard result	ivdtools result
EP17-A2 Appendix A	nonparametric LoB lot 2	9.4	9.4000
EP17-A2 Appendix A	nonparametric LoB report	9.4	9.4000
EP17-A2 Appendix A	classical LoD lot 1	14.5	14.5309
EP17-A2 Appendix A	classical LoD lot 2	13.8	13.8343
EP17-A2 Appendix A	classical LoD report	14.5	14.5309
EP17-A2 Appendix A	parametric LoB report	8.8	8.7956
EP17-A2 Appendix A	parametric LoD report	13.9	13.9266

Negative measurement results are retained as in the source table and are not truncated to 0.

16.2.7 Standard Case: CLSI EP17-A2 Appendix C, Probit LoD

All 66 zero-concentration results from three lots are negative, supporting $\text{LoB} = 0$ in this example. Because zero concentration cannot be log-transformed, only positive-concentration rows are sent to the Probit model; zero-concentration counts remain in the input table for LoB checking.

```
ep17_probit <- read.csv("./data/016-分析灵敏度/std-ep17-appendix-c-probit.csv", check.names = FALSE)
ep17_probit_fit_data <- ep17_probit[ep17_probit$concentration > 0, ]

ep17_c_lod <- sensitivity_lod(
  ep17_probit_fit_data,
  result = "positive", sample = "sample", lob = 0, lot = "lot",
  concentration = "concentration", detected = "positive", total = "total",
  method = "probit", concentration_scale = "log10"
)
plot(ep17_c_lod)
```

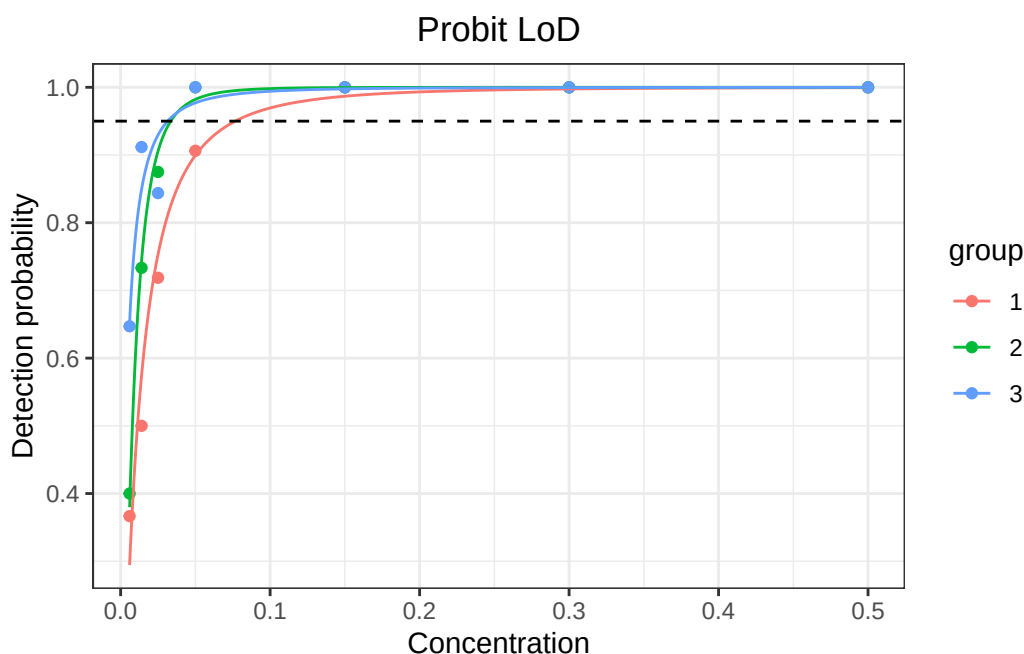


Table 16.2: EP17-A2 Appendix C: comparison with the standard

Standard position	Statistic	Standard result	ivdtools result
EP17-A2 Appendix C	Probit LoD lot 1	0.077	0.0766
EP17-A2 Appendix C	Probit LoD lot 2	0.033	0.0335

Standard position	Statistic	Standard result	ivdtools result
EP17-A2 Appendix C	Probit LoD lot 3	0.031	0.0314

The three LoDs are 0.077, 0.033, and 0.031 CFU/mL; the maximum, 0.077 CFU/mL, is reported. Neither goodness-of-fit test indicates lack of fit, but this appendix uses one sample, a deliberate simplification that falls below the minimum three-sample design and must not be treated as a compliant study template.

16.2.8 Standard Case: CLSI EP17-A2 Appendices E and F, Claim Verification

```
ep17_ef <- read.csv("./data/016-分析灵敏度/std-ep17-appendix-e-verification.csv", check.names = FALSE)
ep17_f <- read.csv("./data/016-分析灵敏度/std-ep17-appendix-f-loq-verification.csv", check.names = FALSE)

ep17_verify_lob <- sensitivity_verify(
  ep17_ef[ep17_ef$type == "blank", ],
  result = "result", sample = "sample", limit = "lob", claim = 20
)
ep17_verify_lod <- sensitivity_verify(
  ep17_ef[ep17_ef$type == "positive", ],
  result = "result", sample = "sample", limit = "lod", claim = 45,
  lob = 20
)
ep17_verify_loq <- sensitivity_verify(
  ep17_f, result = "result", sample = "sample", limit = "loq", claim = 50,
  assigned = "target", allowable_error = 0.15
)
```

Table 16.3: EP17-A2 Appendices E-F: comparison with the standard

Standard position	Statistic	Standard result	ivdtools result
EP17-A2 Appendices E-F	LoB consistency (%)	95.8	95.8333
EP17-A2 Appendices E-F	LoD consistency (%)	91.7	91.6667
EP17-A2 Appendices E-F	LoQ consistency (%)	91.1	91.1111

16.2.9 Standard Case: YY/T 1789.3-2022 Appendix A, ProGRP

Appendix A gives 60 blank and 60 low-concentration results for each of two lots and can be analyzed directly with `sensitivity_lob()` and `sensitivity_lod()`. Negative blanks are retained as reported and are not truncated to zero.

```
yyt1789_3_blank <- read.csv(
  "./data/016-分析灵敏度/std-yyt1789-3-appendix-a-blank.csv"
)
yyt1789_3_low <- read.csv(
  "./data/016-分析灵敏度/std-yyt1789-3-appendix-a-low.csv"
)

yyt1789_3_lob_np <- sensitivity_lob(
  yyt1789_3_blank, result = "result", sample = "sample", lot = "lot",
  method = "nonparametric"
)
yyt1789_3_lob_p <- sensitivity_lob(
```

```

yyt1789_3_blank, result = "result", sample = "sample", lot = "lot",
method = "parametric"
)
yyt1789_3_lod_classical <- sensitivity_lod(
  yyt1789_3_low, result = "result", sample = "sample", lot = "lot",
  lob = yyt1789_3_lob_np, method = "classical"
)
yyt1789_3_lod_np <- sensitivity_lod(
  yyt1789_3_low, result = "result", sample = "sample", lot = "lot",
  lob = yyt1789_3_lob_np, method = "nonparametric"
)

```

Table 16.4: YY/T 1789.3-2022 Appendix A: comparison with the standard

Standard position	Statistic	Standard result	ivdtools result
YY/T 1789.3 Appendix A	nonparametric LoB lot 1	0.240	0.2450
YY/T 1789.3 Appendix A	nonparametric LoB lot 2	0.250	0.2500
YY/T 1789.3 Appendix A	reported LoB	0.250	0.2500
YY/T 1789.3 Appendix A	parametric LoB lot 1	0.174	0.1758
YY/T 1789.3 Appendix A	parametric LoB lot 2	0.176	0.1779
YY/T 1789.3 Appendix A	classical LoD lot 1	0.350	0.3570
YY/T 1789.3 Appendix A	classical LoD lot 2	0.360	0.3673
YY/T 1789.3 Appendix A	nonparametric LoD lot 1	1.075	1.0750
YY/T 1789.3 Appendix A	nonparametric LoD lot 2	1.130	1.1300

Three intermediate algorithm differences must be disclosed:

- The standard takes the 57th order statistic for nonparametric LoB, whereas the package uses continuous rank 57.5 and interpolates between positions 57 and 58. Lot 1 is therefore 0.245 rather than 0.24; after taking the larger of the two lots, the reported value is still exactly 0.25.
- Standard Table A.6 uses displayed pooled SDs of 0.065/0.066 and coefficient 1.653, producing classical LoDs of 0.35/0.36. The package calculates pooled SD from raw results and uses its own finite-sample coefficient 1.6486, producing approximately 0.357/0.367. The methods are similar but not identical; close final rounding does not establish complete reproduction.
- Parametric LoB also differs because of the finite-sample coefficient convention, yielding approximately 0.1758/0.1779 rather than the standard's 0.174/0.176.

Chapter 17

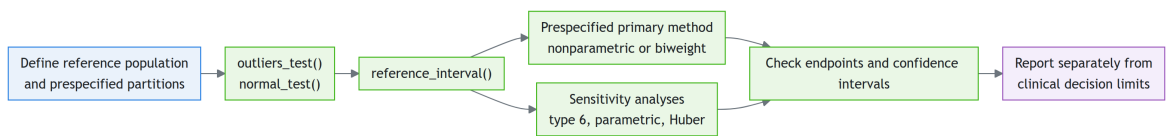
Reference Intervals

This chapter introduces reference-interval establishment, method selection, and result reporting. A reference interval describes the distribution range of measurements in a target reference population. It is not a disease-diagnostic cutoff and does not automatically apply to populations with different age, sex, region, or measurement system; population definitions and inclusion/exclusion criteria must be reported with the statistical method.

! Scope of Use

reference_interval() calculates a statistical interval for a defined sample set. It does not recruit reference individuals, determine partitions, justify sample size, investigate causes of outliers, or validate transfer across systems. Detection by an outlier test is not an automatic reason for exclusion, and a normality test does not replace a distribution plot. Reference coverage interval and endpoint confidence level ci are different parameters.

Workflow: establish and interpret a reference interval



A reference interval (RI) is defined by measurements in a reference sample population and is used to interpret individual results, corresponding to CLSI EP28-A3c. `ivdtools` uses the standard simple nonparametric method by default and implements the Appendix B biweight method. It also provides R type-6 quantiles with normal-approximation rank intervals, normal/log-normal parametric intervals, and Huber M robust intervals as explicitly named extensions. These methods differ in sensitivity to sample size, distribution shape, and outliers. Perform outlier and normality assessment before establishment; presenting multiple methods is a sensitivity comparison, not a post hoc choice of the most favorable result.

```
library(ivdtools)
library(readr)
```

17.1 Main Functions

Function	Main purpose	Key inputs or outputs
<code>outliers_test()</code>	Outlier detection	Grubbs / ESD / Dixon / IQR

Function	Main purpose	Key inputs or outputs
<code>normal_test()</code>	Normality test and plot	<code>method="auto"</code> selects by sample size
<code>reference_interval()</code>	Reference intervals by five methods	<code>method="all"</code> , <code>interval</code> , <code>ci</code> , bootstrap settings
<code>plot.reference_interval()</code>	Plot intervals by method	Jitter plus lower/upper limits

```
reference_interval(
  data, col, interval = 0.95, ci = 0.95,
  method = "nonparametric",
  id = NULL,
  bootstrap_replicates = 1999,
  seed = NULL
)
```

Let the central coverage be γ . The two-tail probability is

$$\alpha = (1 - \gamma)/2.$$

The five methods are defined as follows:

Method	Reference limits	Endpoint interval
<code>nonparametric</code> (default)	EP28 $r = p(n + 1)$ with linear interpolation at noninteger ranks	Order-statistic interval from binomial inversion, with actual coverage reported
<code>biweight</code>	EP28 Appendix B T_{bi} , $s_{bi}[205.6]$, and $S_T[3.7]$	Percentile bootstrap
<code>type6</code>	R <code>quantile(type=6)</code> at α and $1 - \alpha$	Retained normal-approximation rank positions
<code>parametric</code>	$x \pm z_{1-\alpha} s$	t approximation for the SE of reference limits
<code>huber</code>	Huber location $T \pm z_{1-\alpha} S$	Percentile bootstrap

The parametric method first performs Shapiro–Wilk. If normality is rejected and all values are positive, it tries a natural-log transformation; only if normality is not rejected after transformation does it calculate on the log scale and back-transform. If both original and log scales reject normality, the function warns and continues on the original scale. Biweight and Huber both start from the median and MAD, but use different weights, scales, and reference-limit formulas; they must not be treated as one generic “robust method.” `bootstrap_replicates` must be at least 100. Setting `seed` makes the bootstrap reproducible, and the function restores the caller’s random-number state.

Complete finite values in `col` enter the analysis; `id` is used only to report duplicate rows and does not automatically retain one row per person. The object stores `nonparametric`, `biweight`, `type6`, `parametric`, and `huber` results along with cleaned values. Showing all five methods is a sensitivity comparison, not a license to choose the most favorable method after seeing results.

`outliers_test()` and `normal_test()` return the method, statistic, P value, and indices. `reference_interval()` returns reference limits, endpoint confidence intervals, actual coverage, and cleaned data. `plot.reference_interval()` returns a `ggplot` object.

17.2 Analysis Examples

17.2.1 Example: Compare All Five Methods

This example performs prespecified outlier and normality checks and then calls `reference_interval(method = "all")` to output all five methods. The deterministic teaching data have $n = 120$, mild right skew, and outliers.

```
ri_data <- read_csv("./data/017-参考区间/reference-interval.csv", show_col_types = FALSE)
ri_data <- as.data.frame(ri_data)
```

EP28 recommends reviewing outliers before establishing an RI. With $n = 120$, Dixon is not suitable (it is limited to $n \leq 30$), so use Grubbs and generalized ESD:

```
outliers_test(ri_data, col = "value", method = "grubbs")
```

Grubbs Outliers Test

Data: ri_data
Column: value (n = 120, missing = 0)
Parameters: alpha = 0.05
Outliers: 1

Index	Value	G stat	Critical
88	183.0000	4.0104	3.4451

```
outliers_test(ri_data, col = "value", method = "esd", r = 3)
```

Generalized ESD Outliers Test

Data: ri_data
Column: value (n = 120, missing = 0)
Parameters: alpha = 0.05, r = 3
Outliers: 2

Index	Value	R stat	Critical
88	183.0000	4.0104	3.4451
57	178.0000	4.0576	3.4424

`outliers_test()` accepts "grubbs", "esd", "dixon", and "iqr"; ESD uses r for the maximum number of detections, Dixon uses type ("both", "min", or "max"), and IQR uses `coef` (default 1.5).

Detecting an outlier does not mean it should be deleted. Review original records, the experimental process, and protocol, and perform inclusion/exclusion sensitivity analyses when justified.

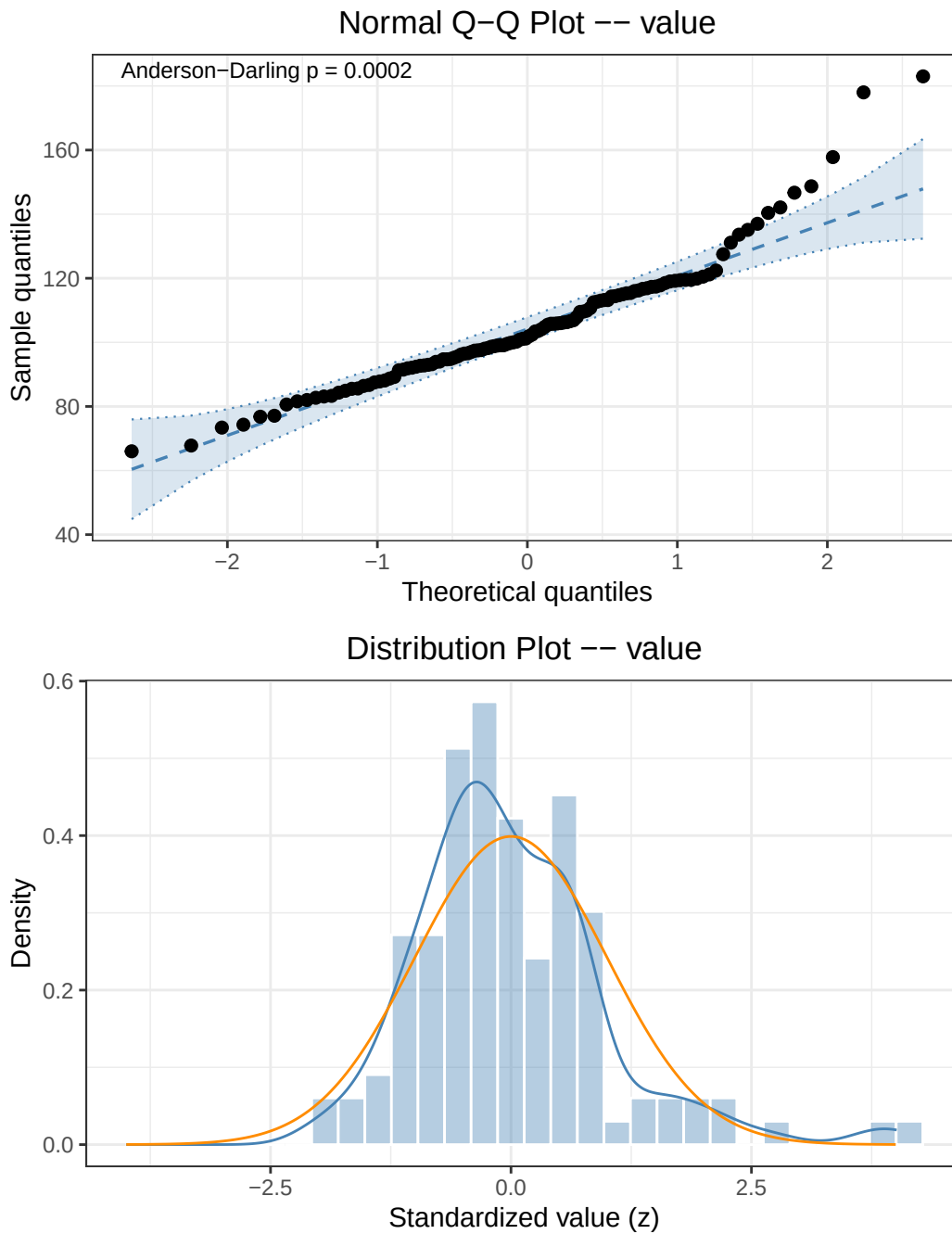
```
nt <- normal_test(ri_data, col = "value", method = "auto")
nt
```

Normality Test

Data: ri_data
Column: value (n = 120, missing = 0)
Method: Anderson-Darling

A = 1.7577, p = 0.0002

```
plot(nt, type = c("qq", "his"))
```



With $n = 120$, `method="auto"` selects the Anderson–Darling test. A small P value suggests non-normal data; the right-skewed distribution and outliers can be checked in the Q–Q plot and histogram.

```
set.seed(1728)
ri <- reference_interval(
  ri_data,
  col = "value",
  interval = 0.95,
  ci = 0.95,
  method = "all",
  id = "id",
  bootstrap_replicates = 1999,
  seed = 1728
)
```

```
print(ri)
```

Reference Interval Analysis

```
Column:          value
Reference interval: 95% (alpha = 0.025)
Confidence level:  0.95
ID variable:      id
Complete cases:    120 / 120
```

No missing values.

No duplicate IDs found.

Nonparametric (CLSI EP28-A3c; exact binomial rank CI)

```
Lower = 73.4225 [66.0000, 81.6000]
Upper = 157.5725 [137.0000, 183.0000]
Point ranks: 3.0250 and 117.9750
CI ranks: lower 1-8; upper 113-120
Actual rank coverage: lower 0.9412; upper 0.9412
Warning: requested rank-CI coverage is not fully attainable at this sample size.
```

Biweight (CLSI EP28-A3c Appendix B; 6 iterations)

```
Lower = 63.2548 [54.4269, 71.4132]
Upper = 141.7011 [133.3354, 149.8618]
Tbi = 102.477938; sbi[205.6] = 19.748969; ST[3.7] = 1.536778
Bootstrap valid: 1999 / 1999
```

Type 6 (R quantile type 6; normal-approximation rank CI)

```
Lower = 73.4225 [66.0000, 80.6000]
Upper = 157.5725 [137.0000, 183.0000]
```

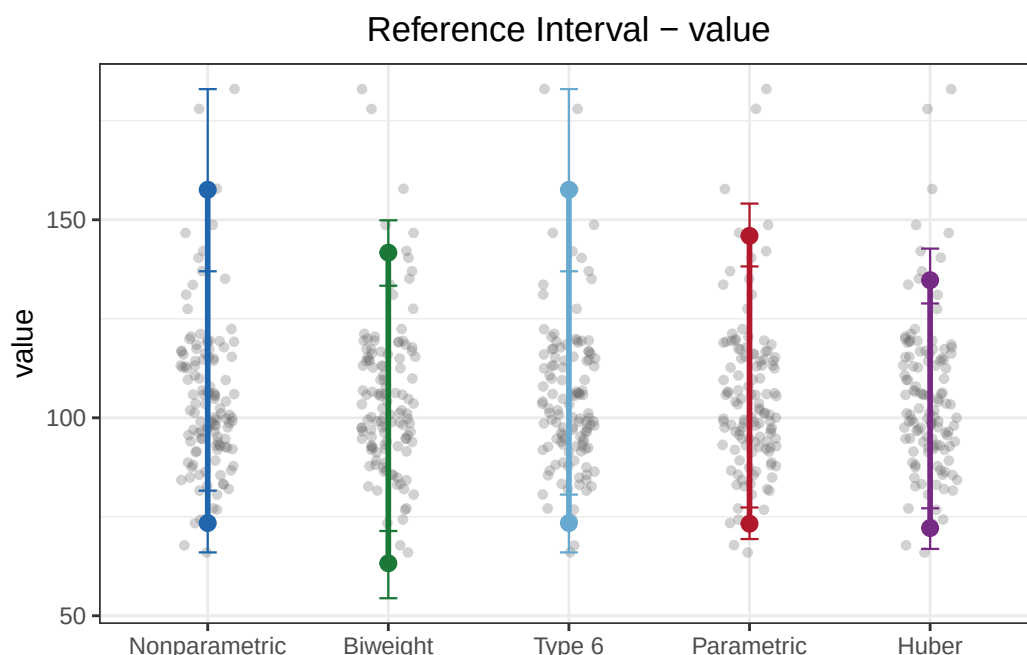
Parametric (log-transformed, Shapiro-Wilk p = 0.0661)

```
Lower = 73.2539 [69.3807, 77.3434]
Upper = 145.9407 [138.2241, 154.0880]
```

Huber (Huber M-estimation (k=1.5, 16 iter) + MAD)

```
Lower = 72.1573 [66.8868, 77.1452]
Upper = 134.7365 [128.8693, 142.7196]
Bootstrap valid: 1999 / 1999
```

```
plot(ri)
```



In `reference_interval()`, `interval` is reference-interval coverage (default 0.95, corresponding to 2.5%–97.5%) and `ci` is endpoint confidence level; they have different meanings. `method` can be any combination of "nonparametric", "biweight", "type6", "parametric", and "huber", or the single value "all". `id` is used for duplicate-sample detection.

The five methods in this example are:

- **Nonparametric** (73.4–157.6): EP28 point estimates make no distributional assumption and endpoint CIs use binomial ranks.
- **Biweight** (63.3–141.7): uses the two scale components from EP28 Appendix B; this teaching data set with outliers does not satisfy the prerequisite of investigating anomalies first.
- **Type 6** (73.4–157.6): point estimates match the nonparametric method, but endpoint CIs use retained normal-approximation ranks.
- **Parametric** (73.3–145.9): after detecting non-normality, the function applies a log transformation (`transformed=TRUE`) and back-transforms.
- **Huber** (72.2–134.7): based on Huber M estimation and MAD.

Differences among methods reflect different rank intervals, distributional assumptions, and robust-scale definitions.

17.2.2 Standard Case: CLSI EP28-A3c Appendix B Robust Iteration

Appendix B demonstrates the biweight calculation with 20 female calcium observations. This small sample is for algorithm demonstration only; the standard explicitly does not recommend it for establishing a formal RI. `method="biweight"` directly returns the location, scale components, reference limits, and bootstrap endpoint intervals from Appendix B.

```
ep28_b <- as.data.frame(read_csv(
  "./data/017-参考区间/std-ep28-appendix-b.csv",
  show_col_types = FALSE
))
ep28_biweight <- reference_interval(
  ep28_b, col = "calcium", interval = 0.95,
  ci = 0.90, method = "biweight",
  bootstrap_replicates = 1999, seed = 2803
)
ep28_biweight
```


Reference Interval Analysis

Column: calcium
 Reference interval: 95% (alpha = 0.025)
 Confidence level: 0.9
 ID variable: --
 Complete cases: 20 / 20

No missing values.

Note: 'id' not provided, skip duplicate detection.

Biweight (CLSI EP28-A3c Appendix B; 6 iterations)

Lower = 9.0495 [8.8339, 9.2781]
 Upper = 10.1994 [9.9725, 10.4052]
 Tbi = 9.624439; sbi[205.6] = 0.270427; ST[3.7] = 0.048187
 Bootstrap valid: 1989 / 1999

Table 17.1: EP28 Appendix B: biweight intermediate values and reference limits

Standard position	Statistic	Standard result	ivdtools result
EP28-A3c Appendix B	Tbi	9.62440	9.62444
EP28-A3c Appendix B	sbi[205.6]	0.27043	0.27043
EP28-A3c Appendix B	ST[3.7]	0.04816	0.04819
EP28-A3c Appendix B	lower	9.05000	9.04951
EP28-A3c Appendix B	upper	10.20000	10.19937

The unrounded S_T differs from the displayed standard value by approximately 2.7×10^{-5} , a rounding difference in an intermediate standard value, and is retained transparently in the table. A formal RI still requires a sample that meets the protocol; the standard explicitly states that 20 observations are for algorithm demonstration and that robust methods also require prior investigation of justified outliers.

Chapter 18

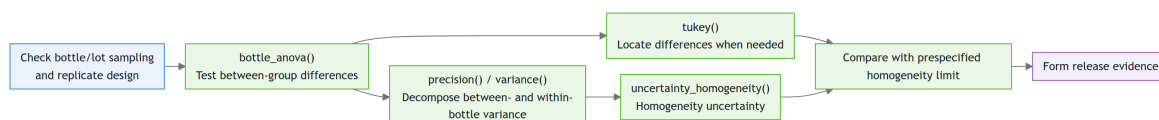
Bottle-to-Bottle Variation

This chapter uses bottle-to-bottle and lot-to-lot comparisons to assess material homogeneity and demonstrates how to convert homogeneity results into an uncertainty component. A significance test only indicates whether a difference was observed; it does not by itself indicate practical impact. The final conclusion must also consider repeatability, target values, and prespecified acceptance criteria.

! Scope of Use

The F test from `bottle_anova()` asks whether all group means are identical; it does not ask whether the difference meets a product-homogeneity limit. `tukey()` locates pairwise differences only when the overall model and its assumptions are appropriate. Homogeneity uncertainty or bottle SD must be estimated from a variance model consistent with the sampling, measurement order, and replicate design, and then compared with a prespecified technical limit; release or rejection must not rely only on a P value.

Workflow: from bottle differences to homogeneity uncertainty



Bottle-to-bottle variability evaluates differences among bottles (or lots) of the same product. The analysis generally has two steps: use `bottle_anova()` to test whether bottle or lot means differ statistically, then use variance-component analysis to separate between-bottle and within-bottle variation and assess whether the magnitude meets prespecified precision limits. This chapter demonstrates 15 bottles $\times$ 5 measurements and three lots measured with one control material.

The examples are deterministic teaching data. Statistical significance does not mean that a product is unacceptable; decisions must use acceptance limits specified in advance.

```
library(ivdtools)
library(readr)
```

18.1 Main Functions

Function	Main purpose	Key inputs or outputs
<code>bottle_anova()</code>	One-way or nested ANOVA	<code>value ~ batch</code> or <code>value ~ batch/vial</code>

Function	Main purpose	Key inputs or outputs
<code>tukey()</code>	Tukey HSD pairwise comparisons and compact letters	<code>term</code> specifies the effect
<code>precision()</code> / <code>variance()</code>	Decompose between-bottle and within-bottle variance	One-way <code>value ~ bottle</code>

18.1.1 ANOVA, Tukey, and Variance Components

```

bottle_anova(data, formula, conf.level = 0.95)
tukey(x, term = NULL, conf.level = NULL)

precision(data, value ~ bottle)
variance(x, method = c("auto", "anova", "reml"))
uncertainty_homogeneity(data, unit = "bottle", result = "value", ...)

```

The one-way model is

$$Y_{ij} = \mu + B_i + \varepsilon_{ij},$$

with the global null hypothesis that all bottle means are equal. The classical F statistic is

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}.$$

`bottle_anova()` removes rows with missing or nonfinite responses or grouping values, converts right-hand variables to factors, and fits with `aov()`. A nested formula such as `value ~ lot/vial` expands to `lot` and `lot:vial`. The object also reports group-mean t intervals, the Bartlett homogeneity test, and a Shapiro–Wilk test on model residuals. These assumption checks do not automatically select a model.

`tukey()` performs Tukey HSD for the specified effect and returns differences, simultaneous intervals, adjusted P values, and compact letters. In a complex model, verify the default effect and supply `term` explicitly when necessary. Variance-component analysis estimates

$$\sigma_{\text{total}}^2 = \sigma_B^2 + \sigma_e^2,$$

so between-bottle SD is $\sqrt{\sigma_B^2}$; this is a different scale from the ANOVA P value. Homogeneity conclusions should focus on the prespecified maximum between-bottle variation or its uncertainty contribution.

`bottle_anova()` returns the ANOVA model and between-group diagnostics; `tukey()` returns pairwise comparisons; `precision()/variance()` return variance components; and `uncertainty_homogeneity()` returns a standard uncertainty component for homogeneity.

18.2 Analysis Examples

18.2.1 Example 1: 15 Bottles $\times$ 5 Measurements

18.2.1.1 Read and Check Data

```
bottle <- read_csv("./data/018-瓶间差/bottle-difference.csv", show_col_types = FALSE)
bottle <- as.data.frame(bottle)
bottle$bottle <- factor(bottle$bottle)
head(bottle, 6)
```

```
bottle replicate value
1    B01          1  97.52
2    B01          2 100.53
3    B01          3  98.81
4    B01          4  97.55
5    B01          5  97.85
6    B02          1 100.92
```

18.2.1.2 Step 1: Test for Significant Differences

The one-way model `value ~ bottle` tests whether the 15 bottle means are equal:

```
ba <- bottle_anova(bottle, value ~ bottle)
print(ba)
```

Bottle ANOVA Analysis

```
Response:      value
Group variable: bottle (15 groups)
Complete cases: 75 / 75
Confidence level: 0.95
```

Descriptive Statistics:

bottle	n	Mean	SD
B01	5	98.4520	1.2739
B02	5	101.7140	0.9360
B03	5	99.8060	0.8153
B04	5	102.7980	0.4992
B05	5	98.7520	0.8662
B06	5	100.0960	1.4971
B07	5	100.7180	0.8466
B08	5	100.5000	1.3179
B09	5	99.8160	1.0246
B10	5	101.6400	0.8034
B11	5	99.3300	0.9360
B12	5	98.4700	1.0884
B13	5	99.7440	1.1004
B14	5	100.9040	0.4803
B15	5	99.7920	1.4477

ANOVA Table:

Term	Df	Sum Sq	Mean Sq	F-value	p-value	Sig
bottle	14	107.7233	7.6945	7.1460	0.0000	***

Residuals 60 64.6057 1.0768

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Assumption Checks:

Bartlett test: K-squared = 9.9325, df = 14, p-value = 0.7671 (OK)

Shapiro-Wilk: W = 0.9828, p-value = 0.4011 (OK)

Post-hoc: use `tukey()` for pairwise comparisons

The ANOVA table provides the between-bottle F test; assumptions reports Bartlett's test of variance homogeneity and the Shapiro-Wilk normality test.

18.2.1.3 Pairwise Comparisons

```
posthoc <- tukey(ba)
posthoc
```

Tukey HSD Post-hoc Test

Response: value

Complete cases: 75 / 75

Confidence level: 0.95

Term: bottle

Comparison	Diff	lwr	upr	p-value	Sig
B02-B01	3.2620	0.9412	5.5828	0.0005	***
B03-B01	1.3540	-0.9668	3.6748	0.7526	
B04-B01	4.3460	2.0252	6.6668	0.0000	***
B05-B01	0.3000	-2.0208	2.6208	1.0000	
B06-B01	1.6440	-0.6768	3.9648	0.4526	
B07-B01	2.2660	-0.0548	4.5868	0.0624	
B08-B01	2.0480	-0.2728	4.3688	0.1408	
B09-B01	1.3640	-0.9568	3.6848	0.7431	
B10-B01	3.1880	0.8672	5.5088	0.0008	***
B11-B01	0.8780	-1.4428	3.1988	0.9896	
B12-B01	0.0180	-2.3028	2.3388	1.0000	
B13-B01	1.2920	-1.0288	3.6128	0.8078	
B14-B01	2.4520	0.1312	4.7728	0.0287	*
B15-B01	1.3400	-0.9808	3.6608	0.7656	
B03-B02	-1.9080	-4.2288	0.4128	0.2229	
B04-B02	1.0840	-1.2368	3.4048	0.9391	
B05-B02	-2.9620	-5.2828	-0.6412	0.0026	**
B06-B02	-1.6180	-3.9388	0.7028	0.4794	
B07-B02	-0.9960	-3.3168	1.3248	0.9686	
B08-B02	-1.2140	-3.5348	1.1068	0.8674	
B09-B02	-1.8980	-4.2188	0.4228	0.2298	
B10-B02	-0.0740	-2.3948	2.2468	1.0000	
B11-B02	-2.3840	-4.7048	-0.0632	0.0384	*
B12-B02	-3.2440	-5.5648	-0.9232	0.0006	***
B13-B02	-1.9700	-4.2908	0.3508	0.1830	
B14-B02	-0.8100	-3.1308	1.5108	0.9952	
B15-B02	-1.9220	-4.2428	0.3988	0.2134	
B04-B03	2.9920	0.6712	5.3128	0.0022	**
B05-B03	-1.0540	-3.3748	1.2668	0.9507	
B06-B03	0.2900	-2.0308	2.6108	1.0000	

B07-B03	0.9120	-1.4088	3.2328	0.9852	
B08-B03	0.6940	-1.6268	3.0148	0.9990	
B09-B03	0.0100	-2.3108	2.3308	1.0000	
B10-B03	1.8340	-0.4868	4.1548	0.2778	
B11-B03	-0.4760	-2.7968	1.8448	1.0000	
B12-B03	-1.3360	-3.6568	0.9848	0.7693	
B13-B03	-0.0620	-2.3828	2.2588	1.0000	
B14-B03	1.0980	-1.2228	3.4188	0.9330	
B15-B03	-0.0140	-2.3348	2.3068	1.0000	
B05-B04	-4.0460	-6.3668	-1.7252	0.0000	***
B06-B04	-2.7020	-5.0228	-0.3812	0.0092	**
B07-B04	-2.0800	-4.4008	0.2408	0.1258	
B08-B04	-2.2980	-4.6188	0.0228	0.0549	
B09-B04	-2.9820	-5.3028	-0.6612	0.0023	**
B10-B04	-1.1580	-3.4788	1.1628	0.9026	
B11-B04	-3.4680	-5.7888	-1.1472	0.0002	***
B12-B04	-4.3280	-6.6488	-2.0072	0.0000	***
B13-B04	-3.0540	-5.3748	-0.7332	0.0016	**
B14-B04	-1.8940	-4.2148	0.4268	0.2326	
B15-B04	-3.0060	-5.3268	-0.6852	0.0020	**
B06-B05	1.3440	-0.9768	3.6648	0.7619	
B07-B05	1.9660	-0.3548	4.2868	0.1854	
B08-B05	1.7480	-0.5728	4.0688	0.3515	
B09-B05	1.0640	-1.2568	3.3848	0.9470	
B10-B05	2.8880	0.5672	5.2088	0.0037	**
B11-B05	0.5780	-1.7428	2.8988	0.9999	
B12-B05	-0.2820	-2.6028	2.0388	1.0000	
B13-B05	0.9920	-1.3288	3.3128	0.9696	
B14-B05	2.1520	-0.1688	4.4728	0.0968	
B15-B05	1.0400	-1.2808	3.3608	0.9556	
B07-B06	0.6220	-1.6988	2.9428	0.9997	
B08-B06	0.4040	-1.9168	2.7248	1.0000	
B09-B06	-0.2800	-2.6008	2.0408	1.0000	
B10-B06	1.5440	-0.7768	3.8648	0.5576	
B11-B06	-0.7660	-3.0868	1.5548	0.9972	
B12-B06	-1.6260	-3.9468	0.6948	0.4711	
B13-B06	-0.3520	-2.6728	1.9688	1.0000	
B14-B06	0.8080	-1.5128	3.1288	0.9953	
B15-B06	-0.3040	-2.6248	2.0168	1.0000	
B08-B07	-0.2180	-2.5388	2.1028	1.0000	
B09-B07	-0.9020	-3.2228	1.4188	0.9866	
B10-B07	0.9220	-1.3988	3.2428	0.9837	
B11-B07	-1.3880	-3.7088	0.9328	0.7199	
B12-B07	-2.2480	-4.5688	0.0728	0.0670	
B13-B07	-0.9740	-3.2948	1.3468	0.9739	
B14-B07	0.1860	-2.1348	2.5068	1.0000	
B15-B07	-0.9260	-3.2468	1.3948	0.9831	
B09-B08	-0.6840	-3.0048	1.6368	0.9992	
B10-B08	1.1400	-1.1808	3.4608	0.9125	
B11-B08	-1.1700	-3.4908	1.1508	0.8956	
B12-B08	-2.0300	-4.3508	0.2908	0.1498	
B13-B08	-0.7560	-3.0768	1.5648	0.9976	
B14-B08	0.4040	-1.9168	2.7248	1.0000	
B15-B08	-0.7080	-3.0288	1.6128	0.9988	
B10-B09	1.8240	-0.4968	4.1448	0.2859	
B11-B09	-0.4860	-2.8068	1.8348	1.0000	
B12-B09	-1.3460	-3.6668	0.9748	0.7601	
B13-B09	-0.0720	-2.3928	2.2488	1.0000	

B14-B09	1.0880	-1.2328	3.4088	0.9374
B15-B09	-0.0240	-2.3448	2.2968	1.0000
B11-B10	-2.3100	-4.6308	0.0108	0.0523
B12-B10	-3.1700	-5.4908	-0.8492	0.0009 ***
B13-B10	-1.8960	-4.2168	0.4248	0.2312
B14-B10	-0.7360	-3.0568	1.5848	0.9982
B15-B10	-1.8480	-4.1688	0.4728	0.2668
B12-B11	-0.8600	-3.1808	1.4608	0.9914
B13-B11	0.4140	-1.9068	2.7348	1.0000
B14-B11	1.5740	-0.7468	3.8948	0.5257
B15-B11	0.4620	-1.8588	2.7828	1.0000
B13-B12	1.2740	-1.0468	3.5948	0.8226
B14-B12	2.4340	0.1132	4.7548	0.0311 *
B15-B12	1.3220	-0.9988	3.6428	0.7819
B14-B13	1.1600	-1.1608	3.4808	0.9015
B15-B13	0.0480	-2.2728	2.3688	1.0000
B15-B14	-1.1120	-3.4328	1.2088	0.9266

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Compact Letter Display:

B04: a
 B02: a
 B10: a
 B14: a
 B07: ab
 B08: ab
 B06: b
 B09: b
 B03: b
 B15: b
 B13: b
 B11: b
 B05: b
 B12: b
 B01: b

Tukey HSD returns all pairwise comparisons and compact letter display. Bottles with different letters have significantly different means. With many bottles, some significant differences are expected; the key question is whether the **magnitude** is acceptable.

18.2.1.4 Step 2: Does the Magnitude Meet the Requirement?

Use `precision(value ~ bottle)` to split total variation into between-bottle and within-bottle components:

```
bp <- precision(bottle, value ~ bottle)
bp <- variance(bp)
```

Variance components

sample	component	VC	%Total	SD	CV[%]	Method

all	bottle	1.3236	55.1	1.1505	1.1485	anova
all	error	1.0768	44.9	1.0377	1.0359	anova

Between-bottle SD is approximately 1.15 (CV approximately 1.15%) and within-bottle SD is approximately 1.04 (CV approximately 1.04%). If the protocol specifies a total precision limit of 2% CV, the between-bottle difference is within the illustrative limit. **Note:** this limit is an

example; a formal study must use the limit specified by the protocol or standard and compare both total CV (including between- and within-bottle components) and the individual components.

```
bp <- ci(bp)
```

Confidence intervals (SD)

sample	component	estimate	lower	upper
all	total	1.5493	1.2449	2.0519
all	bottle	1.1505	0.4254	1.5704
all	error	1.0377	0.8807	1.2633

Confidence intervals (%CV)

sample	component	estimate	lower	upper
all	total	1.5467	1.2428	2.0485
all	bottle	1.1485	0.4246	1.5678
all	error	1.0359	0.8792	1.2612

18.2.2 Example 2: Three Reagent Lots Measuring One Control

The three reagent lots jointly form one between-lot model and therefore must not be split by lot. If a raw long table also contains multiple products or analytes, first use `split_by_factors(data, variable="product")` to split independent evaluation projects; retain all lots within each subset and call `bottle_anova()` or `precision()` separately.

18.2.2.1 Read Data

Three reagent lots (L1, L2, L3) each measure the same control 15 times:

```
lot_data <- read_csv("./data/018-瓶间差/reagent-lot.csv", show_col_types = FALSE)
lot_data <- as.data.frame(lot_data)
lot_data$lot <- factor(lot_data$lot)
```

18.2.2.2 Test Lot-to-Lot Differences

```
ba_lot <- bottle_anova(lot_data, value ~ lot)
print(ba_lot)
```

Bottle ANOVA Analysis

Response: value
 Group variable: lot (3 groups)
 Complete cases: 45 / 45
 Confidence level: 0.95

Descriptive Statistics:

lot	n	Mean	SD
L1	15	100.8173	1.1626
L2	15	101.2760	1.4271
L3	15	100.5673	0.7829

ANOVA Table:

```
-----
Term                Df    Sum Sq   Mean Sq   F-value   p-value   Sig
-----
lot                  2     3.8754    1.9377    1.4528    0.2454
Residuals           42    56.0183    1.3338
-----
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Assumption Checks:

Bartlett test: K-squared = 4.6245, df = 2, p-value = 0.0990 (OK)

Shapiro-Wilk: W = 0.9817, p-value = 0.6870 (OK)

Post-hoc: use `tukey()` for pairwise comparisons

```
tukey(ba_lot)
```

Tukey HSD Post-hoc Test

Response: value
Complete cases: 45 / 45
Confidence level: 0.95

Term: lot

```
-----
Comparison          Diff      lwr      upr    p-value   Sig
-----
L2-L1                0.4587  -0.5659   1.4832   0.5269
L3-L1               -0.2500  -1.2745   0.7745   0.8247
L3-L2               -0.7087  -1.7332   0.3159   0.2245
-----
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Compact Letter Display:

```
-----
L2: a
L1: a
L3: a
```

Compact letters show which lots have significantly different means and the direction of the differences. Even a statistically significant lot-mean difference must be evaluated against a prespecified acceptable bias, such as the target range for a control; a P value alone is insufficient.

18.2.3 Standard Case: YY/T 1709-2020 Calibrator Homogeneity

Appendix A samples 14 bottles of a CRP calibrator from one lot and measures each three times, using one-way ANOVA to separate between- and within-bottle variation.

```
yyt1709_homogeneity <- read_csv(
  "./data/018-瓶间差/std-yyt1709-homogeneity.csv",
  show_col_types = FALSE
)
yyt1709_homogeneity$bottle <- factor(yyt1709_homogeneity$bottle)

yyt1709_fit <- ivdtools::bottle_anova(
  yyt1709_homogeneity, result ~ bottle, conf.level = 0.95
)
```

```
yyt1709_fit
```

Bottle ANOVA Analysis

Response: result
 Group variable: bottle (14 groups)
 Complete cases: 42 / 42
 Confidence level: 0.95

Descriptive Statistics:

bottle	n	Mean	SD
1	3	7.1433	0.1415
2	3	7.2333	0.1301
3	3	7.0433	0.1210
4	3	7.1333	0.1106
5	3	7.0233	0.0723
6	3	7.1133	0.0987
7	3	7.2533	0.1069
8	3	7.2033	0.0462
9	3	7.2500	0.0872
10	3	7.1767	0.1115
11	3	7.0767	0.1234
12	3	7.1433	0.1115
13	3	7.1633	0.1563
14	3	7.0800	0.1153

ANOVA Table:

Term	Df	Sum Sq	Mean Sq	F-value	p-value	Sig
bottle	13	0.2114	0.0163	1.2807	0.2809	
Residuals	28	0.3556	0.0127			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Assumption Checks:

Bartlett test: K-squared = 3.1688, df = 13, p-value = 0.9973 (OK)
 Shapiro-Wilk: W = 0.9532, p-value = 0.0839 (OK)

Post-hoc: use `tukey()` for pairwise comparisons

```
yyt1709_u <- ivdtools::uncertainty_homogeneity(
  yyt1709_homogeneity, unit = "bottle", result = "result"
)
yyt1709_u
```

Between-unit homogeneity uncertainty

Units : 14
 Results : 42
 Mean : 7.1454762
 Effective degrees of freedom : 3
 Average replicates : 3
 Within-unit mean square : 0.0127
 Between-unit mean square : 0.016264652

Within-unit degrees of freedom : 28
 Between-unit degrees of freedom : 13
 Repeatability SD : 0.11269428
 Between-unit SD : 0.034470529
 Hidden inhomogeneity : 0.033636378
 Between-unit uncertainty : 0.034470529
 Relative repeatability : 0.015771416
 Relative between-unit : 0.0048241052
 Relative hidden inhomogeneity : 0.0047073669
 Relative between-unit uncertainty : 0.0048241052

Table 18.1: YY/T 1709 Appendix A: comparison with the standard

Standard position	Statistic	Standard result	ivdtools result
YY/T 1709 Appendix A	SS_between	0.211440	0.211440
YY/T 1709 Appendix A	SS_within	0.355600	0.355600
YY/T 1709 Appendix A	MS_between	0.016265	0.016265
YY/T 1709 Appendix A	MS_within	0.012700	0.012700
YY/T 1709 Appendix A	F	1.280681	1.280681
YY/T 1709 Appendix A	u_bb	0.034492	0.034471

Chapter 19

Clinical Sample-Size Calculation

Sample size is determined by effect size, variability, confidence level, power, and study design; it is not a fixed constant. This chapter covers the three sample-size problems implemented by `ivdtools`: Bland–Altman agreement precision/power, precision of a single-proportion confidence interval, and a one-arm test of a proportion against a target.

! Scope of Use

The functions do not add attrition, invalid samples, minimum numbers per stratum, design effects, clustering, or multiplicity adjustments. `delta` in a Bland–Altman calculation is the clinically acceptable limit, not the observed LoA. Confidence-interval methods and one- or two-sided directions must be prespecified. Preserve all assumptions and sensitivity analyses, not only the final integer.

`library(ivdtools)`

19.1 Main Functions

Function	Main purpose	Solving direction
<code>sample_size_bland_altman()</code>	Bland–Altman agreement evaluation	Solve for <code>n</code> from target power
<code>sample_size_proportion_ci()</code>	Single-proportion confidence interval	Solve for <code>n</code> from the target half-width
<code>sample_size_proportion()</code>	One-arm target-value test	Solve for <code>n</code> from the target alpha and power

All three functions are parameter-driven and do not require raw data. They return structured results; the sample size is in `result$n`, and `print()` shows target and achieved quantities. Round-up, attrition, invalid samples, and minimum stratum counts must be handled separately.

19.1.1 Interfaces

```
sample_size_bland_altman(  
  power = 0.80, mu, sd, delta,  
  conf.level = 0.95, agree.level = 0.95,  
  method = "lu", n.min = 3, n.max = 1e5
```

```

)
sample_size_proportion_ci(
  p, half_width, conf.level = 0.95,
  interval = c("two.sided", "lower", "upper"),
  method = "wilson", n.max = 1e6
)
sample_size_proportion(
  p0, p1, alpha = 0.05, power = 0.80,
  alternative = c("greater", "less", "two.sided"),
  test = "exact", power_method = "binomial", n.max = 1e6
)

```

For Bland–Altman, the target LoA are

$$\mu \pm z_{(1+\gamma)/2}\sigma,$$

where $\gamma = \text{agree.level}$. The function uses the Lu et al. noncentral-t calculation for the probability that both LoA confidence bounds remain within $\pm\delta$ and searches for the smallest integer meeting target power. It requires $\delta > |\mu| + z_{(1+\gamma)/2}\sigma$.

For a proportion interval, the target is a half-width d . For two-sided intervals,

$$\frac{U - L}{2} \leq d,$$

and for one-sided intervals the distance from the expected proportion to the relevant limit is at most d . The interval method affects the required sample size and must be stated in the protocol.

For a one-arm test, `test` determines the rejection region and `power_method` chooses continuous normal planning or discrete binomial enumeration. The final integer sample size is checked using actual alpha and achieved power; binomial planning additionally controls actual alpha rather than relying on an approximation.

19.2 Analysis Examples

19.2.1 Bland–Altman Agreement

```

n_ba <- sample_size_bland_altman(
  power = 0.80, mu = 0.2, sd = 1, delta = 2.5
)
print(n_ba)

```

Sample size for Bland-Altman agreement assessment

```

Method           : Lu et al. (2016), noncentral-t approximation
Mean difference   : 0.2000
SD of differences : 1.0000
Clinical limit    : 2.5000
Confidence level  : 0.9500
Agreement level   : 0.9500
Sample size (n)   : 201
Target power      : 0.800000
Achieved power    : 0.800300

```

Under these teaching parameters, approximately 201 pairs are needed for 80% power. `mu` is mean difference, `sd` is difference SD, and `delta` is the clinical agreement limit. `conf.level` and

agree.level describe different layers: reliability of the LoA estimate versus coverage of the LoA itself.

19.2.2 Single-Proportion Confidence Interval

```
n_pci <- sample_size_proportion_ci(p = 0.3, half_width = 0.05)
print(n_pci)
```

Sample size for a single-proportion confidence interval

```
Method          : wilson
Interval         : two.sided
Anticipated proportion: 0.3000
Confidence level  : 0.9500
Sample size (n)   : 320
Target half-width : 0.050000
Achieved half-width : 0.049967
Target total width : 0.100000
Achieved total width : 0.099934
Planning interval  : [0.252406, 0.352339]
Actual coverage    : 0.956096
Actual alpha       : 0.043904
```

For expected proportion 0.3 and a 95% Wilson half-width of 0.05, approximately 320 observations are needed. Available methods include Wilson, Wald, continuity-corrected Wald/Wilson, Agresti–Coull, Jeffreys, and Clopper–Pearson.

19.2.3 One-Arm Target-Value Test

```
n_sp <- sample_size_proportion(
  p0 = 0.2, p1 = 0.35, alpha = 0.05, power = 0.8,
  alternative = "greater", test = "exact", power_method = "binomial"
)
print(n_sp)
```

Sample size for a single-proportion target-value test

```
Test              : Exact binomial test
Power calculation  : Binomial enumeration
H0: p <= 0.2000
H1: p > 0.2000
Expected p        : 0.3500
Sample size (n)   : 56
Target alpha      : 0.050000
Actual alpha      : 0.043209
Alpha criterion    : met
Target power      : 0.800000
Achieved power     : 0.806416 (binomial evaluation)
Critical successes : >= 17
```

alternative="greater" requires $p_1 > p_0$; use "less" or "two.sided" as appropriate. Exact, score, continuity-corrected score, Wald, and related test definitions are not interchangeable. Specify both test and power_method in the protocol.

19.2.4 Standard and Guidance Cross-Checks

The supplied examples compare the functions with PASS, SAS, and CMDE calculations. Small differences can arise from continuous normal planning versus discrete binomial enumeration, exact quantiles versus rounded handbook quantiles, interval definitions, and

integer search rules. For example, a handbook may report a rounded approximate sample size while the package returns the first integer whose actual operating characteristics meet the stated constraint. Report both the selected method and achieved alpha/power.

19.3 Interpretation Checklist

1. Define the primary endpoint, analysis unit, expected proportion or difference, allowable error, and tail direction.
2. Run sensitivity analyses for uncertain SDs, proportions, prevalence, and agreement limits.
3. Select interval and test methods before calculating the sample size.
4. Add attrition, invalid-sample, stratification, clustering, and multiple-endpoint allowances outside these functions.
5. Save parameters, function calls, outputs, and the software version in the protocol.

19.4 Complete Standard Cases

19.4.1 Bland–Altman: PASS Examples 1 and 2

PASS Example 1 fixes $\mu = 0.5$, $\delta = 7$, and $\text{conf.level} = \text{agree.level} = 0.95$, then crosses two power levels with three difference SDs. The comparison below keeps every row, including rows that do not agree, and reports both sample-size and achieved-power differences.

```
ba_manual <- data.frame(
  target_power = c(0.8, 0.8, 0.8, 0.9, 0.9, 0.9),
  sd = c(2.5, 2.6, 2.7, 2.5, 2.6, 2.7),
  pass_n = c(60L, 82L, 118L, 78L, 108L, 156L),
  pass_power = c(0.8059, 0.8019, 0.8024, 0.9014, 0.9003, 0.9002)
)
ba_results <- lapply(
  seq_len(nrow(ba_manual)),
  function(i) sample_size_bland_altman(
    power = ba_manual$target_power[i],
    mu = 0.5, sd = ba_manual$sd[i], delta = 7,
    conf.level = 0.95, agree.level = 0.95
  )
)
```

Table 19.1: All parameter combinations in PASS Bland–Altman Example 1

Standard location	Statistic	Standard	ivdtools
PASS BA Example 1 / power=0.8 / SD=2.5	sample size	60.0000	59.000000
PASS BA Example 1 / power=0.8 / SD=2.5	achieved power	0.8059	0.803126
PASS BA Example 1 / power=0.8 / SD=2.6	sample size	82.0000	82.000000
PASS BA Example 1 / power=0.8 / SD=2.6	achieved power	0.8019	0.803997
PASS BA Example 1 / power=0.8 / SD=2.7	sample size	118.0000	118.000000
PASS BA Example 1 / power=0.8 / SD=2.7	achieved power	0.8024	0.803101
PASS BA Example 1 / power=0.9 / SD=2.5	sample size	78.0000	78.000000
PASS BA Example 1 / power=0.9 / SD=2.5	achieved power	0.9014	0.901856
PASS BA Example 1 / power=0.9 / SD=2.6	sample size	108.0000	108.000000
PASS BA Example 1 / power=0.9 / SD=2.6	achieved power	0.9003	0.900492
PASS BA Example 1 / power=0.9 / SD=2.7	sample size	156.0000	156.000000
PASS BA Example 1 / power=0.9 / SD=2.7	achieved power	0.9002	0.900235

For power = 0.8 and SD = 2.5, PASS reports 60, while the Lu noncentral-t formula as listed in

the manual gives 59 as the first searched integer meeting the target. PASS does not publish its internal values at $n = 59$, so this comparison does not attribute the difference to rounding or force the package result upward. The power column is retained as reported: several small- n rows differ by approximately 0.00004–0.0028, with the difference shrinking as n increases.

PASS Example 2 uses $\mu = 0.2$, $SD = 1$, and $\delta = 2.5$ to validate the result in the Lu et al. paper. The paper table gives 203, while PASS and ivdtools both obtain 201 when searching for the first integer that reaches the target; this difference is between the paper's approximate result and an integer search.

19.4.2 Proportion Confidence Intervals: PASS Examples 1–4

PASS two-sided cases specify total width, whereas the package takes half_width, so the comparison uses $\text{half_width} = \text{target_width} / 2$. Example 1 gives three widths and five expected proportions for 15 Clopper–Pearson combinations.

```
ci_manual <- expand.grid(
  p = seq(0.1, 0.5, by = 0.1),
  target_width = c(0.04, 0.06, 0.10)
)
ci_manual$pass_n <- c(
  914L, 1585L, 2065L, 2353L, 2449L,
  417L, 715L, 928L, 1056L, 1098L,
  158L, 264L, 341L, 387L, 402L
)
ci_results <- lapply(
  seq_len(nrow(ci_manual)),
  function(i) sample_size_proportion_ci(
    p = ci_manual$p[i],
    half_width = ci_manual$target_width[i] / 2,
    conf.level = 0.95,
    interval = "two.sided",
    method = "clopper-pearson"
  )
)
```

Table 19.2: The 15 combinations in PASS exact proportion-interval Example 1

Standard location	Statistic	Standard	ivdtools
PASS CI Example 1 / $p=0.1$ / width=0.04	sample size	914	914
PASS CI Example 1 / $p=0.2$ / width=0.04	sample size	1585	1585
PASS CI Example 1 / $p=0.3$ / width=0.04	sample size	2065	2065
PASS CI Example 1 / $p=0.4$ / width=0.04	sample size	2353	2353
PASS CI Example 1 / $p=0.5$ / width=0.04	sample size	2449	2449
PASS CI Example 1 / $p=0.1$ / width=0.06	sample size	417	417
PASS CI Example 1 / $p=0.2$ / width=0.06	sample size	715	715
PASS CI Example 1 / $p=0.3$ / width=0.06	sample size	928	928
PASS CI Example 1 / $p=0.4$ / width=0.06	sample size	1056	1056
PASS CI Example 1 / $p=0.5$ / width=0.06	sample size	1098	1098
PASS CI Example 1 / $p=0.1$ / width=0.10	sample size	158	158
PASS CI Example 1 / $p=0.2$ / width=0.10	sample size	264	264
PASS CI Example 1 / $p=0.3$ / width=0.10	sample size	341	341
PASS CI Example 1 / $p=0.4$ / width=0.10	sample size	387	387
PASS CI Example 1 / $p=0.5$ / width=0.10	sample size	402	402

The package also enumerates coverage at the final n ; this is a diagnostic separate from the width condition used to select n in PASS.

```

ci_special <- data.frame(
  case = c("PASS Example 2", "PASS Example 3", "PASS Example 4"),
  design = c(
    "one-sided lower limit, Clopper--Pearson",
    "two-sided, Wilson continuity correction",
    "one-sided upper limit, Clopper--Pearson, p=0"
  ),
  pass_n = c(25L, 29L, 299L)
)
ci_special_results <- list(
  sample_size_proportion_ci(
    p = 0.92, half_width = 0.15104,
    interval = "lower", method = "clopper-pearson"
  ),
  sample_size_proportion_ci(
    p = 0.034483, half_width = 0.1945 / 2,
    interval = "two.sided", method = "wilson-cc"
  ),
  sample_size_proportion_ci(
    p = 0, half_width = 0.01,
    interval = "upper", method = "clopper-pearson"
  )
)

```

Table 19.3: One-sided, continuity-corrected, and boundary cases from PASS

Standard location	Statistic	Standard	ivdtools
PASS Example 2	sample size	25	25
PASS Example 3	sample size	29	29
PASS Example 4	sample size	299	299

Example 3 shows that Wilson continuity correction must use the corrected score-inversion formula; mechanically adding or subtracting $1/(2n)$ from ordinary Wilson endpoints does not reproduce PASS. Example 4 validates a one-sided exact interval with $p = 0$, so the package allows expected proportions of 0 and 1 for interval planning.

19.4.3 One-Sample Proportion Target Tests: SAS and PASS Comparison

The software options map to package parameters as follows.

SAS/PASS option	ivdtools parameter	Meaning
SAS TEST=Z VAREST=NULL; PASS Z-test using S(P0)	test = "score"	p0 variance in the alpha term
SAS TEST=ADJZ; PASS S(P0) with C.C.	test = "score-cc"	S(P0) Z test with continuity correction
PASS Z-test using S(Phat)	test = "wald"	Wald approximation using p1 variance in the alpha term
METHOD=NORMAL; Normal Approximation	power_method = "normal"	Continuous normal approximation
Exact; Binomial Enumeration	power_method = "binomial"	Discrete binomial rejection region and power

The comparison covers the uncorrected and corrected SAS Example 90.2 results, PASS Examples 9 and 11, and six effect sizes from PASS Example 2.

```

target_manual <- data.frame(
  case = c(
    "SAS 90.2 Z", "SAS 90.2 ADJZ", "PASS 9", "PASS 11",
    paste0("PASS 2, p1=", sprintf("%.2f", seq(0.55, 0.80, 0.05)))
  ),
  alternative = c(
    "less", "less", "greater", "two.sided", rep("two.sided", 6)
  ),
  test = c(
    "score", "score-cc", "score", "wald", rep("score", 6)
  ),
  p0 = c(0.30, 0.30, 0.50, 0.30, rep(0.50, 6)),
  p1 = c(0.20, 0.20, 0.60, 0.50, seq(0.55, 0.80, 0.05)),
  target_power = c(0.80, 0.80, 0.80, 0.80, rep(0.90, 6)),
  manual_n = c(119L, 129L, 153L, 50L, 1047L, 259L, 113L, 62L, 38L, 25L),
  manual_power = c(
    0.80000, 0.80100, 0.80125, 0.80743,
    0.90011, 0.90055, 0.90120, 0.90286, 0.90249, 0.90321
  )
)
target_results <- lapply(
  seq_len(nrow(target_manual)),
  function(i) sample_size_proportion(
    p0 = target_manual$p0[i], p1 = target_manual$p1[i],
    alpha = 0.05, power = target_manual$target_power[i],
    alternative = target_manual$alternative[i],
    test = target_manual$test[i], power_method = "normal"
  )
)

```

Table 19.4: SAS and PASS one-sample proportion target-test comparison

Standard location	Statistic	Standard	ivdtools
SAS 90.2 Z	sample size	119.00000	119.00000
SAS 90.2 Z	normal planning power	0.80000	0.80032
SAS 90.2 ADJZ	sample size	129.00000	129.00000
SAS 90.2 ADJZ	normal planning power	0.80100	0.80094
PASS 9	sample size	153.00000	153.00000
PASS 9	normal planning power	0.80125	0.80125
PASS 11	sample size	50.00000	50.00000
PASS 11	normal planning power	0.80743	0.80743
PASS 2, p1=0.55	sample size	1047.00000	1047.00000
PASS 2, p1=0.55	normal planning power	0.90011	0.90011
PASS 2, p1=0.60	sample size	259.00000	259.00000
PASS 2, p1=0.60	normal planning power	0.90055	0.90055
PASS 2, p1=0.65	sample size	113.00000	113.00000
PASS 2, p1=0.65	normal planning power	0.90120	0.90120
PASS 2, p1=0.70	sample size	62.00000	62.00000
PASS 2, p1=0.70	normal planning power	0.90286	0.90286
PASS 2, p1=0.75	sample size	38.00000	38.00000
PASS 2, p1=0.75	normal planning power	0.90249	0.90249
PASS 2, p1=0.80	sample size	25.00000	25.00000
PASS 2, p1=0.80	normal planning power	0.90321	0.90321

Check the discrete operating characteristics at the final integer sample size:

Table 19.5: Discrete binomial check of continuous-normal planning

Standard location	Statistic	Standard	ivdtools
SAS 90.2 Z	binomial achieved power	NA	0.80363
SAS 90.2 Z	actual alpha	0.05	0.04776
SAS 90.2 ADJZ	binomial achieved power	NA	0.79446
SAS 90.2 ADJZ	actual alpha	0.05	0.03595
PASS 9	binomial achieved power	NA	0.80944
PASS 9	actual alpha	0.05	0.05280
PASS 11	binomial achieved power	NA	0.83888
PASS 11	actual alpha	0.05	0.06532
PASS 2, p1=0.55	binomial achieved power	NA	0.89681
PASS 2, p1=0.55	actual alpha	0.05	0.04789
PASS 2, p1=0.60	binomial achieved power	NA	0.89499
PASS 2, p1=0.60	actual alpha	0.05	0.04656
PASS 2, p1=0.65	binomial achieved power	NA	0.91362
PASS 2, p1=0.65	actual alpha	0.05	0.05943
PASS 2, p1=0.70	binomial achieved power	NA	0.91073
PASS 2, p1=0.70	actual alpha	0.05	0.05590
PASS 2, p1=0.75	binomial achieved power	NA	0.86830
PASS 2, p1=0.75	actual alpha	0.05	0.03355
PASS 2, p1=0.80	binomial achieved power	NA	0.89088
PASS 2, p1=0.80	actual alpha	0.05	0.04329

binomial_power and actual_alpha are additional diagnostics at the final integer n ; they do not change the normal-approximation sample size. Wald and some score designs can have actual alpha above 0.05; for example PASS Example 11 is approximately 0.06532. If the protocol requires actual alpha control, use power_method = “binomial”; the resulting n may differ from the manual’s continuous approximation because the design method is different.

test	power_method = “normal”	power_method = “binomial”	Standard cross-check
exact	Supported; infinite-population planning equals score	Supported	PASS Examples 5 and 12 for discrete rejection; Example 13 for normal power
score	Supported	Supported	SAS 90.2; PASS 2/5/9
score-cc	Supported	Supported	SAS 90.2 ADJZ; PASS 5
wald	Supported	Supported	PASS 5/11
wald-cc	Supported	Supported	PASS 5; no separate manual sample-size case
agresti-coull, jeffreys	Not supported	Supported	No matching PASS one-sample-test procedure found

The testing layer also reproduces PASS Example 5: six fixed n values across five tests, for 30 discrete power values and 30 actual-alpha values. These validate rejection-region implementation and do not change the public sample-size interface.

19.4.4 CMDE Appendix Case 1: Target-Value Method with Clinical Acceptance Criteria

The appendix uses the normal-approximation formula

$$n = \frac{\left[Z_{1-\alpha/2} \sqrt{P_0(1-P_0)} + Z_{1-\beta} \sqrt{P_T(1-P_T)} \right]^2}{(P_T - P_0)^2}.$$

The case uses two-sided $\alpha = 0.05$ and power = $1 - \beta = 0.80$. The acceptable-standard/expected-value pairs are 85%/90% for positive agreement and 90%/94% for negative agreement. Using $Z_{0.975} = 1.96$ and $Z_{0.80} = 0.84$, the appendix reports 362 positive cases and 388 negative cases, for 750 total; after a 10% dropout allowance, at least 834 participants are enrolled.

`sample_size_proportion(test = "score", power_method = "normal")` performs the same continuous normal approximation. The package uses unrounded normal quantiles and rounds the theoretical result upward; actual alpha and power are then recalculated separately from the discrete binomial rejection region.

```
cmde_target_value <- data.frame(
  group = c("Positive group", "Negative group"),
  p0 = c(0.85, 0.90),
  p1 = c(0.90, 0.94),
  cmde_n = c(362L, 388L)
)

cmde_target_results <- lapply(
  seq_len(nrow(cmde_target_value)),
  function(i) {
    sample_size_proportion(
      p0 = cmde_target_value$p0[i],
      p1 = cmde_target_value$p1[i],
      alpha = 0.05,
      power = 0.80,
      alternative = "two.sided",
      test = "score",
      power_method = "normal"
    )
  }
)
```

Table 19.6: CMDE target-value method: published and ivdtools normal-approximation results

Standard location	Statistic	Standard	ivdtools
CMDE target-value / Positive group	sample size	362	363.00000
CMDE target-value / Positive group	actual alpha	NA	0.05552
CMDE target-value / Negative group	sample size	388	388.00000
CMDE target-value / Negative group	actual alpha	NA	0.05094
CMDE target-value / Total	sample size	750	751.00000
CMDE target-value / Total	actual alpha	NA	NA

The positive-group formula value is 362.7756, so the smallest integer meeting the target power is 363; the appendix writes 362 because it uses approximate quantiles and ordinary rounding. `actual_alpha` is a diagnostic obtained by placing the continuous-approximation sample size back into the discrete rejection region; it is not part of the CMDE formula's sample-size search.

19.4.5 CMDE Appendix Case 2: Allowable Error Method Requiring CI Precision

The appendix formula is

$$n = \frac{Z_{1-\alpha/2}^2 P(1 - P)}{\Delta^2}.$$

The expected sensitivity is 85%, the expected specificity is 90%, and the allowable error is $\Delta = 0.05$. This corresponds to `sample_size_proportion_ci(method = "wald")`. The package uses the exact `qnorm(0.975)` and rounds upward to the smallest integer satisfying the half-width requirement.

```
cmde_ci_precision <- data.frame(
  group = c("Positive group (sensitivity)", "Negative group (specificity)"),
  p = c(0.85, 0.90),
  cmde_n = c(196L, 138L)
)
cmde_ci_precision$ivdtools_n <- vapply(
  cmde_ci_precision$p,
  function(p) sample_size_proportion_ci(
    p = p, half_width = 0.05, conf.level = 0.95, method = "wald"
  )["n"],
  integer(1)
)
```

Table 19.7: CMDE allowable-error method and 10% addition

Standard location	Statistic	Standard	ivdtools
CMDE allowable-error / Positive group (sensitivity)	sample size	196	196
CMDE allowable-error / Negative group (specificity)	sample size	138	139
CMDE allowable-error / Total	sample size	334	335
CMDE allowable-error / 10% addition	enrolment	371	373

The package gives 139 for the negative group while the appendix reports 138. The difference has two causes: the appendix substitutes the approximate value 1.96 and reports 138.30 as 138, whereas the package uses the exact quantile and rounds upward. The appendix then reports 371 after adding 10% to 334 cases; if “at least enroll” is handled conservatively, the package’s 335 analyzable cases require $\text{ceil}(335 / 0.90) = 373$. This is a quantile-precision and integer-rounding difference, not a difference in the CI method.

When the expected proportion is close to 100%, the guidance also warns that the normal approximation may be unsuitable. Use `sample_size_proportion_ci(method = "clopper-pearson")` or another method for sensitivity analysis, but label the result as a supplementary analysis under a different interval method rather than as a recalculation of the appendix’s Wald formula.

Chapter 20

Outlier and Normality Tests

Outlier and distribution tests help identify problems in data quality, the experimental process, or model assumptions. They are diagnostic tools before analysis, not automatic tools for deleting data or declaring a product acceptable.

This chapter introduces the independent-data functions `outliers_test()` and `normal_test()`, together with `outlier()` and `normal()` methods for `mcr` and `precision` objects. Always trace conclusions back to the original records and retain sensitivity analyses with and without candidate outliers.

! Scope of Use

An outlier test produces candidate flags but does not delete observations. The null hypothesis of a normality test concerns a particular distribution, and failure to reject it does not prove normality. Determine the method, significance level, maximum number of outliers, and tail direction before viewing results. Object-level methods may test paired differences or separate sample sequences; the statistical unit must match the study design.

20.1 Main Functions

Function	Main purpose
<code>outliers_test()</code>	Apply Grubbs, ESD, Dixon, or IQR to an ordinary data column
<code>normal_test()</code>	Apply Shapiro, AD, Lilliefors, or CvM to an ordinary data column
<code>outlier()</code>	Mark candidate outliers in an <code>mcr</code> or <code>precision</code> object
<code>normal()</code>	Perform normality diagnostics for a <code>precision</code> object
<code>plot()</code>	Plot diagnostic figures for test or analysis objects

20.1.1 Parameters and Formulas for Ordinary Data

```
outliers_test(  
  data, col, method = c("grubbs", "esd", "dixon", "iqr"), ...  
)  
normal_test(  
  data, col, method = c("shapiro", "ad", "lilliefors", "cvm"), ...  
)
```

```
data, col,
method = c("auto", "shapiro", "sw", "ad", "lillie", "cvm"),
level = 0.95
)
```

Method	Core definition	Main boundary
Grubbs	$G = \max x_i - \bar{x} /s$	At most one point per test; α controls the test
Generalized ESD	Sequentially remove the largest standardized deviation and compare each R_i with its critical value	r is the maximum candidate count, default $\min(5, n-2)$
Dixon	Endpoint spacing divided by the range, Q	Only $3 \leq n \leq 30$; α only 0.01 or 0.05
IQR	Below $Q_1 - kIQR$ or above $Q_3 + kIQR$	Descriptive rule; default coef is 1.5

Missing and nonfinite values are ignored, but `indices` reports one-based positions in the original vector. Grubbs, ESD, and Dixon rely on assumptions such as independence and approximate normality; IQR makes no normality assumption but is not a probability-significance test.

The deterministic mapping used by `normal_test(method="auto")` is:

Finite sample size	Actual method
$n < 3$	No test
$3 \leq n \leq 50$	Shapiro–Wilk
$51 \leq n \leq 5000$	Anderson–Darling
$n > 5000$	Lilliefors

The significance level is $\alpha = 1 - \text{level}$. Q-Q plots and histograms from `plot()` help identify skewness, heavy tails, and mixtures. P values vary strongly with sample size and must be interpreted together with the plots and data-generating process.

`outliers_test()` returns the method, statistic, P value, candidate outliers, and indices. `normal_test()` returns the actual method, statistic, P value, and sample size.

When an ordinary data frame contains multiple analytes, levels, or lots, `outliers_test()` and `normal_test()` do not group automatically. Use `split_by_factors(data, variable="analyte")` first, then apply the same prespecified test with `lapply()` to each subset. Report each group separately; do not combine all groups and test them as one sample.

20.1.2 Statistical Unit for Object-Level Methods

`outlier(mcr, type="difference")` tests paired differences, whereas `type="candidate"` or `"reference"` tests one method's results; percent differences also depend on the denominator. `outlier(precision)` and `normal(precision)` process results according to groups defined by `by` and do not automatically reconstruct independent residuals by day or run. Object-level diagnostics are saved in object slots, so reassign the object after each call.

Object-level `outlier()` and `normal()` return updated objects with diagnostic slots; `plot()` returns a `ggplot` object.

20.2 Analysis Examples

20.2.1 Test an Ordinary Data Column

```
library(ivdttools)
```

Construct teaching data containing extreme observations. In a formal project, save the raw data and data dictionary before any exclusion or transformation.

```
diagnostic_data <- data.frame(
  sample_id = seq_len(22),
  result = c(99.8, 100.4, 99.7, 100.1, 100.0, 99.9,
             100.3, 100.2, 99.6, 100.1, 100.0, 99.8,
             100.2, 99.9, 100.1, 100.0, 99.7, 100.3,
             100.1, 99.9, 108.0, 92.0)
)

grubbs_result <- outliers_test(diagnostic_data, "result", method = "grubbs")
iqr_result <- outliers_test(diagnostic_data, "result", method = "iqr")
normality_result <- normal_test(diagnostic_data, "result", method = "shapiro")

grubbs_result
```

Grubbs Outliers Test

Data: diagnostic_data
 Column: result (n = 22, missing = 0)
 Parameters: alpha = 0.05
 Outliers: 1

Index	Value	G stat	Critical
22	92.0000	3.2308	2.7577

```
iqr_result
```

IQR Outliers Test

Data: diagnostic_data
 Column: result (n = 22, missing = 0)
 Parameters: coef = 1.5
 Outliers: 2

Index	Value
21	108.0000
22	92.0000

fences: [99.3000, 100.7000]

```
normality_result
```

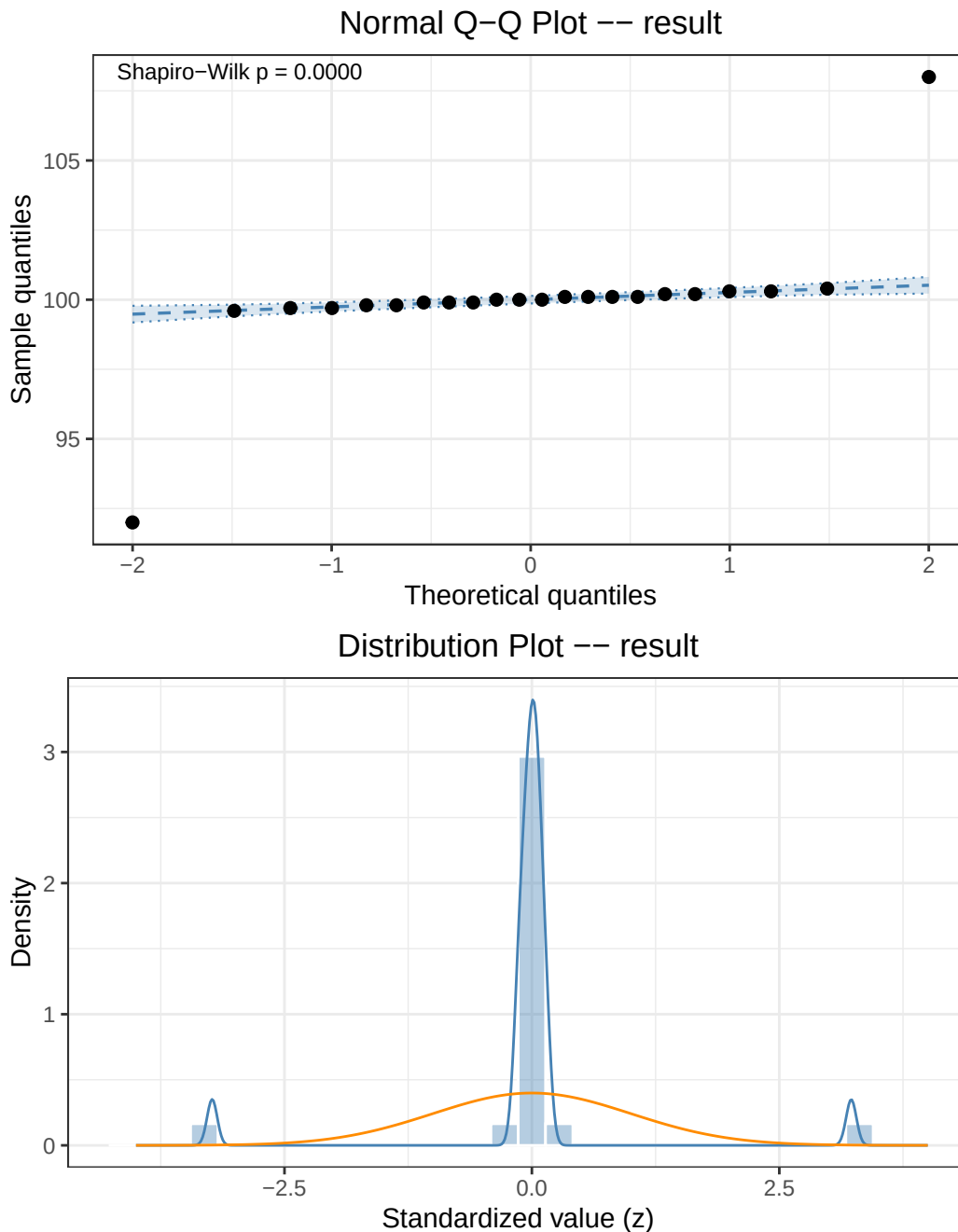
Normality Test

Data: diagnostic_data
 Column: result (n = 22, missing = 0)
 Method: Shapiro-Wilk

W = 0.5059, p = 0.0000

`outliers_test()` returns candidate outlier row numbers and values; `normal_test()` returns the method used, statistic, and P value. A P value alone cannot decide whether to delete an observation, especially for very large or very small samples; use plots, experimental records, and the study objective.

```
plot(normality_result, type = c("qq", "his"))
```



20.2.2 Outliers in a Method-Comparison Object

Make clear whether the abnormal quantity is the candidate method, the reference method, or the paired difference. First create an `mcr` object and then specify the detection type:

```
mcr_data <- data.frame(
  id = 1:12,
```

```

candidate = c(10.1, 11.2, 12.0, 13.1, 14.0, 15.2,
              16.1, 17.0, 18.2, 19.0, 20.1, 28.0),
reference = c(10.0, 11.0, 12.1, 13.0, 14.1, 15.0,
              16.0, 17.1, 18.0, 19.1, 20.0, 21.0)
)

mcr_result <- mcr(
  mcr_data, id = "id", candidate = "candidate", reference = "reference"
)
mcr_result <- outlier(mcr_result, method = "grubbs", type = "difference")

```

Outlier

Method: grubbs
 Data: Difference (candidate - reference)
 Alpha: 0.05

1 outlier(s) found:

Row	Value	Statistic	Critical
12	7.0000	3.1694	2.4116

After detecting a candidate outlier, review sample identity, run order, reagent lot, instrument alarms, and the original trace. Exclusion requires a traceable technical reason, and key results with and without the observation should both be reported.

20.2.3 Outliers and Normality in a Precision Object

Diagnostics for a precision design should preserve the experimental hierarchy; replicate results from one run must not be treated as independent patient samples.

```

precision_data <- data.frame(
  sample = rep("level_1", 20),
  day = factor(rep(1:5, each = 4)),
  run = factor(rep(rep(1:2, each = 2), 5)),
  result = c(99.8, 100.1, 100.0, 99.9, 100.2, 100.0, 100.1, 99.8,
              99.7, 100.0, 99.9, 100.1, 100.3, 100.1, 100.0, 100.2,
              99.8, 100.0, 100.1, 100.0)
)

precision_result <- precision(precision_data, result ~ day/run, by = "sample")
precision_result <- outlier(precision_result, method = "grubbs")

```

Outlier detection -- Grubbs (alpha = 0.05)
 No outliers detected.

```
precision_result <- normal(precision_result, method = "shapiro")
```

Normality test -- Shapiro-Wilk

Sample	n	Method	Statistic	p-value
level_1	20	Shapiro-Wilk	W=0.9550	0.4497

20.2.4 Standard Case: YY/T 1789.4-2022 Appendix A.2

The standard uses 18.85, 20.38, 21.97, and 22.32 ng/mL to demonstrate Grubbs and two-sided Dixon tests, with no outlier in either test. These two entry points correspond directly to

outliers_test().

```
yyt1789_4_outlier <- data.frame(
  sample = 1:4,
  result = c(18.85, 20.38, 21.97, 22.32)
)

yyt1789_4_grubbs <- ivdtools::outliers_test(
  yyt1789_4_outlier, "result", method = "grubbs", alpha = 0.05
)
yyt1789_4_grubbs
```

Grubbs Outliers Test

Data: yyt1789_4_outlier
Column: result (n = 4, missing = 0)
Parameters: alpha = 0.05
Outliers: 0

No outliers detected.

```
yyt1789_4_dixon <- ivdtools::outliers_test(
  yyt1789_4_outlier, "result", method = "dixon", alpha = 0.05,
  type = "both"
)
yyt1789_4_dixon
```

Dixon Q Outliers Test

Data: yyt1789_4_outlier
Column: result (n = 4, missing = 0)
Parameters: alpha = 0.05, type = both
Outliers: 0

No outliers detected.

Both package functions report zero outliers, consistent with the standard. The standard displays Grubbs statistics of 1.2727 at the low end and 0.9028 at the high end, and Dixon statistics of 0.441 at the low end and 0.101 at the high end. The current print method reports only outlier positions and conclusions, not these intermediate statistics; “conclusions agree” must not be written as “all intermediate values were reproduced.”

Chapter 21

Introduction to Algorithm Principles

This chapter explains how to choose statistical units, connect the main `ivdtools` workflows, and understand what key arguments change mathematically. It is intended as a bridge between function documentation and audit of the implementation.

21.1 Replicates, Multiple Samples, and Statistical Units

21.1.1 First Decide What One Row Represents

Before choosing a function, identify whether one row is a patient/sample result, a replicate measurement, a run, a bottle, a time point, or a paired method result. Replicates are not automatically independent subjects. If a function expects sample-level means, aggregate according to the protocol and preserve replicate count and SD.

Functions that can receive replicate results directly include `precision()`, `linearity()`, `interference_*`(), `dilution_recovery()`, `spike_recovery()`, `sensitivity_*`(), and `c5_c95()` when their grouping arguments are supplied. Functions that require paired or sample-level results include `mcr()`, `roc()`, `raw_to_table()`, `reference_interval()`, and many sample-size calculations. Check the help page and define the analysis unit before reshaping data.

21.1.2 Joint or Separate Analysis for Multiple Samples

Use `by` when the function is designed to estimate every sample or concentration separately, such as `precision(..., by="sample")`. Use a common model only when the scientific question and variance structure require pooling. `split_by_factors()` and `split_by_cutpoints()` create named data lists for independent, prespecified analyses; they must not be used repeatedly to search for a favorable result.

21.2 Outliers and Normality

21.2.1 Grubbs Test

For a finite sample, the Grubbs statistic is

$$G = \frac{\max_i |x_i - \bar{x}|}{s}.$$

The test generally identifies at most one point per call and assumes independent, approximately normal observations. A flagged observation is a candidate for investigation, not an instruction to delete it.

21.2.2 Generalized ESD

Generalized ESD removes the most extreme standardized observation sequentially and compares each R_i with a critical value. r is the maximum number of candidate points; it does not mean that r points will be removed.

21.2.3 Dixon and IQR

Dixon uses an endpoint spacing divided by the range and is limited to small samples, currently $3 \leq n \leq 30$ with alpha 0.01 or 0.05. IQR flags values below $Q_1 - kIQR$ or above $Q_3 + kIQR$ and is descriptive rather than a probability test.

21.2.4 Automatic Normality Branch

`normal_test(method="auto")` uses a deterministic sample-size rule:

Sample size	Test
$n < 3$	No test
$3 \leq n \leq 50$	Shapiro–Wilk
$51 \leq n \leq 5000$	Anderson–Darling
$n > 5000$	Lilliefors

P values depend strongly on sample size. Interpret them with Q-Q plots, histograms, experimental records, and the assumptions of the downstream model.

21.3 Method Comparison and Reference-Material Bias

21.3.1 Direction, Units, and Preprocessing

In `mcr()`, the candidate is Y and the reference is X; candidate-minus-reference differences and percent denominators must be recorded. A log transformation, unit conversion, or prespecified cutpoint changes the estimand and should be treated as a separate analysis.

21.3.2 Correlation Options

Pearson measures linear co-variation, Spearman measures monotone rank association, and Kendall measures concordance of ranks. None is an agreement measure. Use regression and Bland–Altman methods for bias and agreement.

21.3.3 Regression Methods

Method	X error	Y variance	Point-estimate principle	Typical use
ols	Negligible	Constant or optional weights	Minimize vertical squared residuals	Reference X has negligible error
wls	Negligible	Changes with concentration	Weighted vertical residuals	Difference SD/CV changes with concentration

Method	X error	Y variance	Point-estimate principle	Typical use
deming	Included	Constant error ratio λ	Errors-in-variables regression	Both procedures have measurement error
wdeming	Included	Common weights, constant CV, or precision profiles	Weighted/iterative errors-in-variables fit	Error changes with concentration
pb	Included	No normal-error assumption	Rank estimate of pairwise slopes	Linear relation with outliers or mixed variation

OLS minimizes $\sum_i (y_i - a - bx_i)^2$. Weighted regression minimizes a weighted version. Deming uses the relative error variance λ and should estimate it from independent precision evidence. Passing–Bablok is robust to some outliers but is not a guarantee against every data problem.

Common WLS weights include $1/y$, $1/y^2$, $1/x$, $1/x^2$, `constant_cv`, `residual`, and a user-supplied nonnegative finite column. $1/x^2$ represents constant CV on the reference scale; `residual` updates a concentration-dependent residual SD and is iterative.

21.3.4 Confidence Intervals and Bland–Altman

`ci_method="auto"` uses analytic covariance for OLS/WLS, jackknife refitting for Deming variants, and rank intervals for Passing–Bablok. Bootstrap methods require enough replicates and a fixed seed for reproducibility. Bland–Altman center, SD, and LoA describe the distribution of paired differences; `agree.level` controls the proportion covered by LoA, whereas `conf.level` controls uncertainty in the estimated LoA.

21.3.5 Reference-Material Bias

Reference-material bias is not the same as clinical-sample method-comparison bias. `reference_bias()` combines the measurement-mean uncertainty and assigned-value uncertainty when supplied. State the assigned-value source, its uncertainty, the bias direction, and whether the material is commutable.

21.4 Reference-Material Commutability

EP14/EP30 methods compare materials with the clinical-sample procedure relationship, usually using Deming prediction intervals. IFCC Part 2 compares reference-material bias with clinical-sample bias and combines uncertainty. IFCC Part 3 evaluates before/after calibration. Different paths can produce different conclusions because they target different estimands.

21.5 Qualitative Performance, C5/C95, and ROC

For a four-fold table $a = TP$, $b = FP$, $c = FN$, $d = TN$,

$$Se = \frac{a}{a + c}, \quad Sp = \frac{d}{b + d}, \quad PPV = \frac{a}{a + b}, \quad NPV = \frac{d}{c + d}.$$

Kappa adjusts observed agreement for marginal agreement; PABAK uses $2P_o - 1$ and is not the same estimand. McNemar uses discordant pairs. C5/C95 solve for mean responses whose modeled positive probability is 0.05 or 0.95, using a precision profile. ROC AUC is the probability of correctly ranking a random positive above a random negative under the selected direction; it is not accuracy at a cutoff.

21.6 Precision, Bottle Variation, and Reference Intervals

For nested precision designs, total variance is the sum of variance components. `variance()` estimates components, `ci()` estimates intervals, and `profile()` models concentration-related SD/CV. Bottle ANOVA tests group means; it does not quantify whether a difference is practically acceptable. `uncertainty_homogeneity()` produces a standard uncertainty component.

Reference intervals have two independent settings: central coverage interval and endpoint confidence `ci`. Nonparametric, biweight, type-6, parametric, and Huber methods make different distributional and robust-estimation assumptions. A normality or outlier test is not an automatic method-selection or deletion rule.

21.7 Response-Equation Fitting and Linearity

`fit_equation()` fits registered response equations such as linear, quadratic, exponential, power, 4PLC, 5PLC, Gaussian, Michaelis–Menten, sinusoidal, and cubic forms. Equation fitting describes calibration response and should not be confused with EP06 method linearity. `linearity()` specifically evaluates level means, residuals, weights, and ADL. A high calibration R^2 does not establish method linearity.

21.8 Interference, Dilution, Spike Recovery, and Hook Effect

Paired interference estimates a test-minus-control effect; dose response estimates how that effect changes with interferent concentration; patient-sample comparison assesses method differences in the presence of a factor. Dilution recovery corrects results by the dilution factor, spike recovery compares an observed increment with a theoretical mass-balance increment, and hook-effect analysis uses raw response at known high concentrations.

21.9 Analytical Sensitivity

LoB describes the blank upper tail, LoD describes distinguishability from blank, and LoQ adds a precision or total-error criterion. `sensitivity_lob()`, `sensitivity_lod()`, `sensitivity_loq()`, and `sensitivity_verify()` keep establishment and verification separate. A reported LoQ is constrained not to be below LoD when LoD is supplied.

21.10 Quality Control and Stability

For QC, $z_i = (x_i - \mu_0)/\sigma_0$. The current rules use strict > 2 , > 3 , and > 1 thresholds for 1-2s, 1-3s, and 4-1s, while R-4s uses $|z| \geq 2$ and a range $|\Delta z| \geq 4$. `qc_chart()` scans one sequence in row order; it does not implement cross-level multirule decisions. Stability functions estimate node bias, time trends, time to a limit, Arrhenius acceleration, and MKT; they do not turn a model extrapolation into observed evidence.

21.11 Measurement Uncertainty and Traceability

Type A components come from statistical repeatability or replication; Type B components come from certificates, limits, distributions, or prior information. `uncertainty_combine()` combines standard uncertainties, `uncertainty_propagate()` applies a measurement model, and `uncertainty_traceability()` accumulates components across a traceability chain. Record the input quantity, standard uncertainty, sensitivity coefficient, degrees of freedom, covariance assumptions, and coverage factor.

21.12 Parameter Choices Change the Estimand

Parameter	Mathematical object changed	Not changed	Selection principle
<code>method</code>	Loss function or rank estimator	X/Y direction	Match error and difference structure
<code>lambda</code>	Relative Deming penalty	Raw observations	Estimate independently, never tune to slope
<code>weights</code>	Contribution or concentration-dependent variance	Difference direction	Support with precision evidence
<code>ci_method</code>	Parameter uncertainty estimator	Point estimates	Prespecify the default or sensitivity path
<code>conf.level</code>	Coverage of parameter/center/LoA interval	<code>agree.level</code>	Usually 0.95
<code>agree.level</code>	Fraction of individual differences within LoA	CI confidence	Define agreement clinically
<code>percent_denominator</code>	Percent-difference denominator	Absolute difference	Choose reference or symmetric mean deliberately
<code>bootstrap_replicates</code>	Monte Carlo error	Target distribution	Use enough replicates and a fixed seed
<code>cutpoints</code>	Data ranges analyzed	Downstream estimator	Define clinical intervals before results

The safest audit trail records the data snapshot, analysis unit, transformations, function arguments, package version, random seed, complete output, and any deviations from the standard algorithm.

21.13 Detailed Algorithm Specification

21.13.1 OLS and WLS Point Estimates

Ordinary least squares minimizes

$$Q(a, b) = \sum_{i=1}^n (y_i - a - bx_i)^2.$$

Weighted least squares minimizes

$$Q_w(a, b) = \sum_{i=1}^n w_i (y_i - a - bx_i)^2.$$

Let

$$W = \sum_i w_i, \quad \bar{x}_w = \frac{\sum_i w_i x_i}{W}, \quad \bar{y}_w = \frac{\sum_i w_i y_i}{W},$$

$$S_{xx,w} = \sum_i w_i (x_i - \bar{x}_w)^2, \quad S_{xy,w} = \sum_i w_i (x_i - \bar{x}_w)(y_i - \bar{y}_w).$$

Then

$$\hat{b} = \frac{S_{xy,w}}{S_{xx,w}}, \quad \hat{a} = \bar{y}_w - \hat{b}\bar{x}_w.$$

OLS is the special case $w_i = 1$. The weight choices change the assumed concentration dependence of the error variance:

weights	Actual weight	Implied variance shape	Main distinction
NULL	unweighted	Var(Y	X) is constant under OLS
equal	1	Same as OLS	Explicit entry to the WLS interface
1/y	1/abs(y_i)	Var(Y_i) proportional to abs(y_i)	Depends on candidate observations
1/y^2	1/y_i^2	SD(Y_i) proportional to abs(y_i)	Empirical constant-CV candidate weighting
1/x	1/abs(x_i)	Var(Y_i) proportional to abs(x_i)	Depends on reference observations
1/x^2	1/x_i^2	SD(Y_i) proportional to abs(x_i)	Empirical constant-CV reference weighting
constant_cv	1/x_i^2	Same as above	Requires $x_i > 0$ in WLS
residual	1 / estimated SD_i^2	SD described by a linear function of concentration	WLS only; iteratively updated
user column	user-supplied w_i	Defined by the user	Must be finite and nonnegative; zero excludes the point from the objective

Generic built-in weights such as 1/x protect division by zero with a machine-precision floor; this prevents a computational failure but does not make a relative-error model scientifically

meaningful at zero concentration. `constant_cv` therefore rejects nonpositive concentrations directly.

Residual weighting first fits OLS. With residual $e_i^{(k)}$ at iteration k , it fits

$$|e_i^{(k)}| = \gamma_0^{(k)} + \gamma_1^{(k)} x_i + \varepsilon_i, \quad w_i^{(k+1)} = \frac{1}{(\hat{\gamma}_0^{(k)} + \hat{\gamma}_1^{(k)} x_i)^2}.$$

It then refits WLS. `weight_iterations` is the number of updates and defaults to four. Changing it can change the final weights and regression coefficients, but should not be used to match a published interval. A nonpositive or nonfinite predicted SD stops the procedure.

The analytic OLS/WLS covariance is

$$\hat{\sigma}_w^2 = \frac{\sum_i w_i e_i^2}{n-2}, \quad \widehat{\text{Cov}}(\hat{a}, \hat{b}) = \hat{\sigma}_w^2 (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1},$$

and coefficient intervals are

$$\hat{\theta} \pm t_{1-\alpha/2, n-2} SE(\hat{\theta}).$$

This is equivalent to the EP09c Appendix H calculation when the same final weights are used. YY/T 1789.2 Equation (8) does not multiply the medical-decision-level bias interval by the weighted residual scale; the two expressions need no residual-scale adjustment only when the weights are known absolute inverse variances.

21.13.2 Deming Regression and lambda

Deming regression defines

$$S_{xx} = \sum_i (x_i - \bar{x})^2, \quad S_{yy} = \sum_i (y_i - \bar{y})^2, \quad S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}).$$

In the source, `lambda` is the ratio of Y-error variance to X-error variance. The slope and intercept are

$$U = S_{yy} - \lambda S_{xx}, \quad b = \frac{U + \sqrt{U^2 + 4\lambda S_{xy}^2}}{2S_{xy}}, \quad a = \bar{y} - b\bar{x}.$$

Thus

$$\lambda = \frac{\sigma_Y^2}{\sigma_X^2}.$$

`lambda = 1` means equal error variances; `lambda > 1` means the candidate Y has greater error; `lambda < 1` means the reference X has greater error. Reversing the ratio changes the Deming slope, so always enter candidate variance divided by reference variance. The reciprocal required by the lower-level algorithm is handled inside the package and does not change the user interface.

`lambda = "replicates"` estimates the variance ratio from summarized replicate SDs. If sample i has `n_Xi` and `n_Yi` replicates, first calculate degrees-of-freedom-pooled single-measurement variances:

$$s_{p,X}^2 = \frac{\sum_i (n_{Xi} - 1) s_{Xi}^2}{\sum_i (n_{Xi} - 1)}, \quad s_{p,Y}^2 = \frac{\sum_i (n_{Yi} - 1) s_{Yi}^2}{\sum_i (n_{Yi} - 1)}.$$

Because the regression uses sample means, the input variance ratio is

$$\hat{\lambda}_{\bar{X}, \bar{Y}} = \frac{s_{p,Y}^2 / n_Y}{s_{p,X}^2 / n_X} = \frac{s_{p,Y}^2}{s_{p,X}^2} \frac{n_X}{n_Y}.$$

The two methods may have different replicate counts, but each method's count must be constant across samples. Otherwise one scalar lambda cannot describe different sample-mean variances and a precision profile should be used. Without replicate counts, the source assumes equal counts and n_X/n_Y cancels. lambda is estimated once; jackknife and bootstrap refits treat it as fixed external input rather than re-estimating it in each subsample.

21.13.3 Three Weighted-Deming Variants

For fixed observation weights, method = “wdeming” first sets $w_{\text{tilde}_i} = w_i / \sum_j w_j$ and replaces the ordinary central moments with

$$\bar{x}_w = \sum_i \tilde{w}_i x_i, \quad \bar{y}_w = \sum_i \tilde{w}_i y_i,$$

$$S_{xx,w} = \sum_i \tilde{w}_i (x_i - \bar{x}_w)^2, \quad S_{yy,w} = \sum_i \tilde{w}_i (y_i - \bar{y}_w)^2, \quad S_{xy,w} = \sum_i \tilde{w}_i (x_i - \bar{x}_w)(y_i - \bar{y}_w),$$

then applies the closed-form Deming solution. This gives one common weight to each pair and cannot represent separate concentration-dependent variance functions for X and Y.

constant_cv in WLS is simply $1/x_i^2$; in weighted Deming it calls Linnet's iterative constant-CV algorithm. At iteration k, the weight is formed from the mean of the two latent readings:

$$w_i^{(k)} = \left(\frac{x_i^{*(k)} + y_i^{*(k)}}{2} \right)^{-2}.$$

The Deming line is updated with this weight. If $d_i = y_i - a - b x_i$ and $\lambda = \sigma_Y^2 / \sigma_X^2$, the latent updates are

$$x_i^* = x_i + \frac{b d_i}{\lambda + b^2}, \quad y_i^* = y_i - \frac{\lambda d_i}{\lambda + b^2}.$$

Weights, line, and latent values are updated until relative change is below epsilon. Both concentrations must be strictly positive. After deleting a sample or drawing a bootstrap sample, the complete iteration starts again rather than reusing full-data weights.

precision_profile permits separate variance functions

$$g(\mu) = \text{Var}(X \mid \mu), \quad h(\eta) = \text{Var}(Y \mid \eta), \quad \eta = a + b\mu.$$

For each latent true value μ_i , the algorithm minimizes

$$L(a, b, \mu) = \sum_i \left[\frac{(x_i - \mu_i)^2}{g(\mu_i)} + \frac{(y_i - a - b\mu_i)^2}{h(a + b\mu_i)} + \log\{g(\mu_i)h(a + b\mu_i)\} \right].$$

Given a and b and current variances, the latent value update is

$$\mu_i = \frac{h_i x_i + g_i b (y_i - a)}{h_i + g_i b^2}.$$

variance_models can contain variance-returning functions or precision objects that have already been processed by variance() and profile(). profile_components selects the variance component on each side; the default is “total” for both sides, and the predicted SD from the VFP is squared to obtain variance. Choosing “error”, “day”, or another component changes g or h directly; the available names depend on the precision() formula and must match the intended error layer. Values outside the precision-study range generate an extrapolation warning, while nonpositive or nonfinite predicted variances stop the fit. External precision profiles remain fixed during resampling.

These three variants are not interchangeable: fixed weights change each point’s empirical influence, constant-CV Deming models relative error on both axes, and precision-profile Deming allows different nonconstant variance functions on the two axes.

21.13.4 Regression-Parameter Confidence Intervals

ci_method determines the uncertainty algorithm for the intercept and slope. The auto mapping is:

Regression method	auto method	Main basis
OLS, WLS	analytic	Analytic covariance from the regression design EP09c Appendix K1
Deming, constant-CV Deming, precision-profile Deming	jackknife	
Passing-Bablok	rank	EP09c Appendix I

Jackknife deletes observation i and completely refits, producing $\theta_{(-i)} = (a_{(-i)}, b_{(-i)})^T$. Its covariance is

$$\widehat{\text{Cov}}_{\text{jack}}(\hat{\theta}) = \frac{n-1}{n} \sum_{i=1}^n (\hat{\theta}_{(-i)} - \bar{\theta}_{(-)}) (\hat{\theta}_{(-i)} - \bar{\theta}_{(-)})^T.$$

The interval remains centered on the full-data estimate:

$$\hat{\theta} \pm t_{1-\alpha/2, n-2} SE_{\text{jack}}(\hat{\theta}).$$

Bootstrap always samples complete (x_i, y_i) pairs with replacement. Because mcr() accepts one row per ID, there is no splitting of replicate rows from the same sample. Every bootstrap sample completely refits the regression. For bootstrap parameter estimates $\theta_{\text{star}}^*(r)$,

$$CI_{\text{percentile}} = [q_{\alpha/2}(\hat{\theta}^*), q_{1-\alpha/2}(\hat{\theta}^*)],$$

$$SE_{\text{boot}}(\hat{\theta}) = SD(\hat{\theta}^*), \quad CI_{\text{standard}} = \hat{\theta} \pm t_{1-\alpha/2, n-2} SE_{\text{boot}}(\hat{\theta}).$$

Both methods also use the joint bootstrap intercept–slope draws to calculate bias at any medical decision level. The source requires at least 5,000 replicates; seed controls reproducibility and is not a tuning parameter. Failed fits are removed and counted; the process stops if fewer than half or fewer than 100 fits are valid.

analytic is allowed only for OLS/WLS, and rank only for Passing–Bablok. Although other methods can explicitly request jackknife or bootstrap, that changes the interval question and should not be selected merely because it is closer to a published value.

21.13.5 Passing–Bablok

The source forms all pairwise slopes with x_j not equal to x_i :

$$b_{ij} = \frac{y_j - y_i}{x_j - x_i}.$$

Exactly identical pairs contribute no slope; vertical pairs are retained as signed infinities. Algorithm I first counts slopes below -1 as K , removes slopes exactly equal to -1 , and cyclically shifts the ranks of the point estimate and interval by K . The shifted median is b_{hat} , and

$$\hat{a} = \text{median}_i(y_i - \hat{b}x_i).$$

If m valid slopes remain, define

$$C_\gamma = z_{1-\alpha/2} \sqrt{\frac{n(n-1)(2n+5)}{18}}, \quad M_1 = \max \left\{ 1, \text{round} \left(\frac{m - C_\gamma}{2} \right) \right\},$$

and $M_2 = m - M_1 + 1$. The slope limits are the shifted M_1 -th and M_2 -th ordered slopes; intercept limits are

$$a_L = \text{median}_i(y_i - b_U x_i), \quad a_U = \text{median}_i(y_i - b_L x_i).$$

This is `ci_method = “rank”` for regression parameters. A rank interval does not directly provide a joint interval for bias at a medical decision level. Therefore `bias()` and `predict()` use paired percentile bootstrap for Passing–Bablok under `ci_method = “auto”`, rather than the jackknife excluded by EP09c Appendix K1. The object may retain a jackknife parameter covariance for explicitly requested extended prediction calculations, but it is not the default Passing–Bablok bias interval.

21.13.6 Differences, Percentage Differences, and Outlier Analysis

Absolute differences always use candidate minus reference:

$$d_i = y_i - x_i.$$

The three `percent_denominator` choices target different estimands:

$$p_{i,\text{reference}} = 100 \frac{y_i - x_i}{x_i}, \quad p_{i,\text{mean}} = 100 \frac{y_i - x_i}{(x_i + y_i)/2}, \quad p_{i,\text{candidate}} = 100 \frac{y_i - x_i}{y_i}.$$

reference answers how far the candidate is above or below the reference and is the default. mean changes only sign when the two methods are exchanged and is useful without a natural reference. candidate uses the candidate as the denominator. The denominator changes

the value, distribution, and outlier status, not just the plot label. `outlier(type = "percent" )` and `bland_altman(type = "percent" )` use the same options so the detected quantity and difference plot remain aligned.

`outlier(type = "difference" )` tests d_i ; `type = "percent"` tests the selected p_i . method calls the Grubbs, generalized ESD, Dixon, or IQR rule described earlier. The first three use `alpha`; IQR uses `coef` for the fence width. Outlier tests flag observations and do not automatically remove them from later regression or Bland–Altman analyses.

21.13.7 Bland–Altman Axis, Center, and Limits of Agreement

`x_axis` changes only the horizontal coordinate:

$$m_i = \frac{x_i + y_i}{2}, \quad x_i, \quad y_i$$

for mean, reference, and candidate, respectively; it does not change the vertical difference. `type` selects

$$\text{difference} : y_i - x_i, \quad \text{ratio} : \frac{y_i}{x_i}, \quad \text{percent} : p_i.$$

The parametric method centers on the mean:

$$\hat{\delta} = \bar{d}, \quad CI_{\delta} = \bar{d} \pm t_{1-\alpha/2, n-1} \frac{s_d}{\sqrt{n}}.$$

For agreement level p , let $z_p = z_{(1+p)/2}$. The limits of agreement are

$$L_- = \bar{d} - z_p s_d, \quad L_+ = \bar{d} + z_p s_d.$$

The standard error and interval for each limit are

$$SE_L = s_d \sqrt{\frac{1}{n} + \frac{z_p^2}{2(n-1)}}, \quad CI(L_{\pm}) = L_{\pm} \pm t_{1-\alpha/2, n-1} SE_L.$$

`agree.level` controls the proportion of individual differences intended to lie within the LoA; `conf.level` controls statistical confidence in the center and LoA estimates. They have different meanings: changing `agree.level` from 0.95 to 0.99 widens the LoA, while changing `conf.level` from 0.95 to 0.99 primarily widens the confidence bands around the LoA.

The nonparametric method uses the ordinary sample median $d_{(\text{ceil}(n/2))}$, averaging the two middle values for even n . `median_ci_method = "order_statistic"` finds the largest k whose binomial coverage

$$C(k) = 1 - 2P\{\text{Bin}(n, 0.5) \leq k - 1\}$$

is at least `conf.level`, and reports

$$[d_{(k)}, d_{(n-k+1)}].$$

This reaches at least the requested coverage and is usually conservative. `interpolated` uses the EP09c Appendix F continuous-rank normal approximation

$$r_L = \frac{n+1 - z_{1-\alpha/2}\sqrt{n}}{2}, \quad r_U = \frac{n+1 + z_{1-\alpha/2}\sqrt{n}}{2}.$$

If $r = j + q$ with $j = \text{floor}(r)$ and $0 \leq q < 1$, linear interpolation gives

$$d_{(r)} = d_{(j)} + q\{d_{(j+1)} - d_{(j)}\}.$$

The interpolated interval has nominal coverage and its endpoint need not equal an observed value; the order-statistic interval is discrete and conservative.

method = “hodges_lehmann” uses the median of all Walsh averages rather than the ordinary median:

$$\hat{\delta}_{HL} = \text{median}_{i \leq j} \left(\frac{d_i + d_j}{2} \right).$$

Its center interval uses a Wilcoxon/Tukey normal approximation without continuity correction. The HL estimator is a pseudomedian of the difference distribution; it has an interpretation similar to the ordinary median only under conditions such as symmetry, and `hl_estimate` should not be labeled `median_diff`.

Nonparametric and HL LoA use empirical quantiles

$$L_- = Q_{(1-p)/2}(d), \quad L_+ = Q_{1-(1-p)/2}(d),$$

with R type-7 linear quantiles. LoA confidence intervals are obtained by resampling `d_i` with replacement, recalculating both empirical quantiles, and taking bootstrap percentiles. `bootstrap_replicates` defaults to 5,000 with a minimum of 100; `seed` is only for reproducibility. These empirical LoA and bootstrap intervals are `ivdtools` extensions and should not be confused with the center-estimation intervals in EP09c Appendix F.

21.13.8 Bias at Medical Decision Levels and Prediction Intervals

For x_0 , the fitted response is $\hat{y}_0 = \hat{a} + \hat{b}x_0$, with fitted-mean variance

$$V_{\text{fit}}(x_0) = SE_a^2 + x_0^2 SE_b^2 + 2x_0 \text{Cov}(a, b).$$

The mean interval uses `sqrt(V_fit)`, while a single-result prediction interval uses `sqrt(V_fit + s_e^2)`. Candidate-minus-reference bias is

$$\widehat{\text{bias}}(x_0) = (\hat{a} + \hat{b}x_0) - x_0 = \hat{a} + (\hat{b} - 1)x_0.$$

Subtracting fixed x_0 adds no variance, so under analytic or jackknife parameter covariance

$$SE\{\widehat{\text{bias}}(x_0)\} = \sqrt{SE_a^2 + x_0^2 SE_b^2 + 2x_0 \text{Cov}(a, b)}.$$

Percentile bootstrap calculates $B_{\text{star}}^{\wedge}(r)(x_0) = a_{\text{star}}^{\wedge}(r) + [b_{\text{star}}^{\wedge}(r) - 1]x_0$ for each parameter draw and takes its quantiles. Standard bootstrap uses the SD of these bias draws and `t_(n-2)`. With `ci_method = “auto”`, `bias()` and `predict()` use percentile bootstrap for Passing-Bablok and reuse the regression object’s selected analytic, jackknife, or bootstrap uncertainty for other methods. Existing compatible bootstrap samples are reused.

interval = “confidence” describes uncertainty in the mean regression relation or mean bias at x_0 ; interval = “prediction” adds `s_e^2` and describes the scatter of one future candidate result; “both” returns both. Prediction intervals are an `ivdtools` extension and are not EP09c regression-parameter or bias-confidence intervals.

21.13.9 Reference-Material Bias

At each level, bias is the measurement mean minus the assigned reference value:

$$b = \bar{x} - x_{\text{ref}}, \quad u_{\bar{x}} = \frac{s}{\sqrt{n}}, \quad U_{\bar{x}} = k u_{\bar{x}}.$$

The source combines the supplied reference-value uncertainty directly with $U_{\bar{x}}$:

$$h = \sqrt{U_{\bar{x}}^2 + U_{\text{ref}}^2}, \quad [b - h, b + h].$$

It does not divide reference uncertainty by an additional coverage factor. Relative bias is $100 b / x_{\text{ref}}$; a zero reference value returns missing.

21.14 Reference-Material Commutability

21.14.1 EP14-Style Prediction Interval

For each clinical sample, first calculate replicate means and SDs in both procedures. The pooled repeatability variance for a procedure is

$$s_p^2 = \frac{\sum_j (n_j - 1) s_j^2}{\sum_j (n_j - 1)}.$$

The source sets $\lambda = s_{p,Y}^2 / s_{p,X}^2$ and fits a Deming line to clinical-sample means, using population central moments divided by n . The slope variance is

$$\text{Var}(\hat{b}) = \frac{\hat{b}^2 \max(0, S_{xx} S_{yy} - S_{xy}^2)}{n S_{xy}^2}.$$

At reference material x_R , the prediction variance is

$$V_R = (x_R - \bar{x})^2 \text{Var}(\hat{b}) + \left(1 + \frac{1}{n}\right) \left(\frac{\hat{b}^2 s_{p,X}^2}{n_{R,X}} + \frac{s_{p,Y}^2}{n_{R,Y}} \right).$$

Degrees of freedom are the smaller integer among the two procedures' pooled repeatability degrees of freedom. The material is labeled commutable when its Y mean lies within $\hat{b} x_R \pm t \sqrt{V_R}$.

21.14.2 IFCC-Style Difference Evaluation

Clinical-sample bias is $b_i = y_i - x_i$. When a reference measurement procedure is specified, the sorting coordinate is x_i ; otherwise it is $(x_i + y_i) / 2$. After sorting, calculate

$$s_{\text{MSSD}} = \sqrt{\frac{\sum_{i=1}^{n-1} (b_{i+1} - b_i)^2}{2(n-1)}}.$$

The difference between reference-material bias and the selected clinical-sample mean bias is

$$d = (y_R - x_R) - \bar{b}_C.$$

The clinical subset may be all samples, the q closest samples, or half on either side of the target. Its standard uncertainty is

$$u_d = \sqrt{\frac{s_{X,R}^2}{p_X} + \frac{s_{Y,R}^2}{p_Y} + \frac{s_C^2}{q}}.$$

The expanded interval is $d \pm k u_d$, with $k = 1.9$ by default. If the complete interval is within $[-\Delta, \Delta]$, the material is commutable; if it is entirely on one side outside the allowable band, it is noncommutable; otherwise the result is indeterminate.

21.15 Qualitative Performance, C5/C95, and ROC

21.15.1 Four-Fold Proportions and Intervals

The source treats the first row and first column as positive in the four-fold table and calculates

$$Se = \frac{TP}{TP + FN}, \quad Sp = \frac{TN}{TN + FP}, \quad PPV = \frac{TP}{TP + FP}, \quad NPV = \frac{TN}{TN + FN},$$

$$Acc = \frac{TP + TN}{n}, \quad Prev = \frac{TP + FN}{n}.$$

For x successes, the Wald interval is

$$\hat{p} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}};$$

the continuity-corrected version adds $1/(2n)$ to the half-width. The Wilson center and half-width are

$$c = \frac{\hat{p} + z^2/(2n)}{1 + z^2/n}, \quad h = \frac{z}{1 + z^2/n} \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}.$$

Wilson continuity correction subtracts and adds $1/(2n)$ outside $c \pm h$. Agresti–Coull uses

$$\tilde{n} = n + z^2, \quad \tilde{p} = \frac{x + z^2/2}{\tilde{n}}, \quad \tilde{p} \pm z \sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{n}}}.$$

Jeffreys uses $\text{Beta}(x + 1/2, n - x + 1/2)$ quantiles. Clopper–Pearson uses $\text{Beta}(x, n - x + 1)$ and $\text{Beta}(x + 1, n - x)$ quantiles, with separate handling at $x = 0$ and $x = n$. All proportion intervals are finally clipped to $[0, 1]$.

Given prevalence π , predictive values are recalculated as

$$PPV_\pi = \frac{Se \pi}{Se \pi + (1 - Sp)(1 - \pi)}, \quad NPV_\pi = \frac{Sp(1 - \pi)}{(1 - Se)\pi + Sp(1 - \pi)}.$$

Intervals simply substitute sensitivity and specificity endpoints and do not propagate their covariance.

21.15.2 Likelihood Ratios, Agreement, and McNemar

$$LR^+ = \frac{Se}{1 - Sp}, \quad LR^- = \frac{1 - Se}{Sp}.$$

Intervals are calculated on the log scale, for example

$$SE\{\log(LR^+)\} = \sqrt{\frac{1}{TP} - \frac{1}{TP + FN} + \frac{1}{FP} - \frac{1}{FP + TN}}.$$

The odds ratio is $OR = TP * TN / (FP * FN)$, with log-scale SE equal to the square root of the sum of the four cell reciprocals. A zero cell returns a missing interval; no continuity correction is added automatically.

Observed and chance agreement are

$$p_o = \frac{TP + TN}{n}, \quad p_e = \frac{(TP + FP)(TP + FN) + (FN + TN)(FP + TN)}{n^2},$$

$$\kappa = \frac{p_o - p_e}{1 - p_e}.$$

Kappa intervals use an explicit asymptotic variance of the four-cell joint proportions and are clipped to [-1, 1]. The PABAK branch uses

$$PABAK = 2p_o - 1, \quad \text{Var}(PABAK) = \frac{4p_o(1 - p_o)}{n}.$$

This branch is triggered by supplying prevalence, but the prevalence value does not enter the formula.

When $b + c < 25$, McNemar uses a two-sided exact test with $B \sim \text{Binomial}(b + c, 0.5)$; otherwise it calculates

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}$$

with one degree of freedom. When $b + c = 0$, $p = 1$.

21.15.3 C5/C95

Raw replicates are first summarized by sample-level mean μ_j , SD s_j , and n_j . At least three levels with $n_j \geq 2$ and $s_j > 0$ are required. A concentration-dependent SD $s(\mu)$ is then fitted.

If a positive event is defined as $\text{result} \geq \text{cutoff } c$,

$$P(Y \geq c \mid \mu) = 1 - \Phi\left(\frac{c - \mu}{s(\mu)}\right);$$

the reverse event uses the corresponding Phi expression. The source searches for sign changes between the target probability and the model on a 1,001-point grid, then refines the root by one-dimensional solving; if no root is found, it expands the search range up to 16 times. If multiple roots are found, it selects the root closest to the cutoff. This is inversion of a normal response distribution with a precision profile, not binomial probit regression.

21.15.4 Empirical ROC, AUC, and Multivariable Models

Candidate thresholds are -infinity, all unique observations, and +infinity. After sorting ROC points by increasing false-positive rate, AUC is trapezoidal area:

$$A = \sum_i \frac{TPR_i + TPR_{i+1}}{2} (FPR_{i+1} - FPR_i).$$

For $0 < A < 1$, the source uses the Hanley–McNeil approximation

$$Q_1 = \frac{A}{2 - A}, \quad Q_2 = \frac{2A^2}{1 + A},$$

$$SE(A) = \sqrt{\frac{A(1 - A) + (n_+ - 1)(Q_1 - A^2) + (n_- - 1)(Q_2 - A^2)}{n_+ n_-}}.$$

The AUC interval is $A \pm 1.96 SE(A)$, clipped to $[0, 1]$; no SE is computed when $A = 0$ or 1 . The Youden threshold maximizes $Se + Sp - 1$; the upper-left-corner rule minimizes $\sqrt{(1 - Se)^2 + (1 - Sp)^2}$. With ties, the first point in the current order is returned.

Two AUCs from the same subjects cannot be compared as independent samples. `auc_compare(method = "ep24")` uses the EP24-A2 Hanley–McNeil paired approximation: calculate Pearson correlations between the two scores within positive and negative subjects (Kendall can be selected for ordinal scores), average them, linearly interpolate the area correlation r from EP24 Table 5 using the mean AUC, and calculate

$$SE(A_1 - A_2) = \sqrt{SE_1^2 + SE_2^2 - 2rSE_1SE_2}.$$

`auc_compare(method = "delong")` builds a covariance matrix Σ from rank contributions of both scores for each positive and negative subject and sets $c = (1, -1)^T$:

$$SE(A_1 - A_2) = \sqrt{c^T \widehat{\Sigma} c}.$$

Both methods use a normal-approximation test of $H_0: A_1 - A_2 = 0$. When scores are fitted probabilities from a training-set logistic model, they are treated as fixed scores; model-fit uncertainty and apparent-performance optimism are not corrected.

A multivariable model fits

$$\text{logit}(p_i) = \beta_0 + \sum_{j=1}^q \beta_j x_{ij}$$

by binomial logistic regression and uses training predictions to construct the same empirical ROC. The linear-predictor interval is $\eta_{\text{hat}} \pm 1.96 SE(\eta_{\text{hat}})$, followed by the logistic inverse. Probability SE is $SE(p_{\text{hat}}) = SE(\eta_{\text{hat}}) p_{\text{hat}} (1 - p_{\text{hat}})$, and the classification threshold is fixed at 0.5.

21.16 Precision, Bottle Variation, and Reference Intervals

21.16.1 Variance Components and Precision Profiles

The source first checks balance: balanced designs use ANOVA variance components and unbalanced designs use REML. Each random level returns $SD_j = \sqrt{VC_j}$ and $CV_j = 100 SD_j / \bar{y}$; total SD is

$$SD_T = \sqrt{\sum_j \widehat{VC}_j}.$$

Component intervals use the two-sided SD/CV intervals from the VCA inference object; the source does not implement a separate Satterthwaite–chi-square formula.

Concentration-related variance profiles use VFP. The ten registered models are:

Model	Variance or CV form
1	$\sigma^2 = \text{beta_1}$
2	$\sigma^2 = \text{beta_1} \mu^2$
3	$\sigma^2 = \text{beta_1} + \text{beta_2} \mu^2$
4	$\sigma^2 = (\text{beta_1} + \text{beta_2} \mu)^2$
5	$\sigma^2 = \text{beta_1} + \text{beta_2} \mu^K$
6	$\sigma^2 = \text{beta_1} + \text{beta_2} \mu + \text{beta_3} \mu^J$
7	$\sigma^2 = \text{beta_1} + \text{beta_2} \mu^J$
8	$\sigma^2 = (\text{beta_1} + \text{beta_2} \mu)^J$
9	$\sigma^2 = \text{beta_1} \mu^J$
10	$\text{CV} = \text{beta_1} \mu^J$

When more than one candidate fits, the model with minimum AIC is selected. For CV input, SD is first reconstructed from the mean and then squared as VC for fitting.

21.16.2 Bottle Variation

Bottle, lot, or other grouping variables enter a fixed-effect analysis of variance. Within-group descriptive intervals are

$$\bar{x}_j \pm t_{1-\alpha/2, n_j-1} \frac{s_j}{\sqrt{n_j}}.$$

Residuals receive a Shapiro–Wilk test, between-group variances receive Bartlett’s test, and pairwise comparisons use Tukey HSD. Bottle mean squares are not automatically converted into a homogeneity uncertainty; that calculation exists in the random-effects ANOVA branch of the uncertainty module.

21.16.3 Reference Intervals

For the default nonparametric method, $\alpha = (1 - p) / 2$ and EP28 continuous ranks are

$$h_q = q(n + 1),$$

with linear interpolation between the order statistics at $\text{floor}(h_q)$ and $\text{ceil}(h_q)$. Reference-limit confidence intervals use binomial rank inversion. With $\delta = (1 - \gamma)/2$ and $B \sim \text{Bin}(n, q)$, endpoint ranks are

$$a = Q_B(\delta), \quad b = Q_B(1 - \delta) + 1,$$

then clipped to $[1, n]$. Actual coverage is $P(a \leq B \leq b - 1)$. If the unclipped rank lies outside the sample, the function marks that the requested coverage cannot be fully attained. `type = “type6”` retains the R type-6 point estimate but its CI still uses the existing binomial-rank normal approximation; it should not be called the exact-rank CI of the default nonparametric method.

The parametric method first tests the raw values. If $p < 0.05$ and all values are positive, it tests log values and uses the log scale only when the log-test p is at least 0.05. On the selected scale, limits are

$$\bar{x} \pm z_{1-\alpha} s.$$

Endpoint SE is

$$SE_L = s \sqrt{\frac{1}{n} + \frac{z_{1-\alpha}^2}{2n}},$$

with $t_{(1 - \gamma/2, \max(n - 2, 1))}$ used for the endpoint interval; log-scale limits are exponentiated at the end.

biweight implements EP28 Appendix B. Initialize T_{bi} with the median and $s = \text{MAD} / 0.6745$. At each iteration use

$$u_i = \frac{x_i - T_{bi}}{3.7s}, \quad w_i = (1 - u_i^2)^2 I(|u_i| < 1), \quad T_{bi} \leftarrow \frac{\sum_i w_i x_i}{\sum_i w_i}.$$

Stop when relative change is below 0.001%. For tuning constant c ,

$$A = \sum_{|u_i| < 1} (1 - u_i^2)(1 - 5u_i^2), \quad B = \sum_{|u_i| < 1} u_i^2(1 - u_i^2)^4,$$

$$s_{bi}[c] = cs \sqrt{\frac{nB}{A \max(1, -1 + A)}}.$$

Calculate $s_{bi}[205.6]$ and $s_{bi}[3.7]$ separately. $S_T[3.7]$ uses $u_i = (x_i - T_{bi}) / (3.7 s_{bi}[3.7])$ and removes n from the numerator under the square root. The final limits are

$$T_{bi} \pm t_{n-1, 1-\alpha} \sqrt{s_{bi}^2[205.6] + S_T^2[3.7]}.$$

Endpoint intervals use percentile bootstrap from the same biweight estimator.

huber retains the package extension. Initialize location T with the median and scale S with $\text{MAD} / 0.6745$, then iterate

$$u_i = \frac{x_i - T}{S}, \quad w_i = \begin{cases} 1, & |u_i| \leq 1.5, \\ 1.5/|u_i|, & |u_i| > 1.5, \end{cases}$$

$$T \leftarrow \frac{\sum_i w_i x_i}{\sum_i w_i}, \quad S \leftarrow \sqrt{\frac{\sum_i w_i (x_i - T)^2}{n - 1}}.$$

The tolerance is $1e-8$ with at most 100 iterations; limits are $T \pm z_{(1 - \alpha)} S$. biweight and Huber endpoint intervals use 1,999 bootstrap draws by default; seed fixes the calculation without changing the caller's existing random-number state.

21.17 Response-Equation Fitting and Linearity

21.17.1 Twelve Registered Response Equations

ID	Equation in the source
E01	$a + b x$
E02	$a + b x + c x^2$
E03	$a \exp(b x)$
E04	$a \exp(-b x)$
E05	$a x^b$
E06	$a + b \log x$
E07	$A + (B - A) / [1 + (x/C)^D]$
E08	$A + (B - A) / [1 + (x/C)^D]^E$
E09	$a \exp[-(x - b)^2 / (2 c^2)]$
E10	$V_{\max} x / (K_m + x)$
E11	$a \sin(bx + c) + d$
E12	$a + b x + c x^2 + d x^3$

E01, E02, E06, and E12 use a linear-model matrix; the remaining equations use Levenberg-Marquardt nonlinear least squares. The objective is

$$RSS_w = \sum_i w_i [y_i - f(x_i, \theta)]^2.$$

When only parameter bounds are supplied, the source first attempts bounded nonlinear least squares. General inequalities use COBYLA and equalities use an augmented Lagrangian. A failed constrained optimization receives the penalty objective

$$RSS_w + 10^8 \sum_r \{\text{constraint violation}_r\}^2.$$

For constrained fits, covariance uses the central-difference residual Jacobian J:

$$\widehat{V}(\hat{\theta}) \approx (J^T J)^{-1} \frac{RSS}{n - p}.$$

This Jacobian covariance does not multiply J by the fitting weights. Constrained-model AIC is $n \log(RSS/n) + 2p$, and $R^2 = 1 - RSS/TSS$.

Nonlinear prediction uses a central-difference gradient $g(x)$:

$$V\{\hat{f}(x)\} = g(x)^T \widehat{V}(\hat{\theta}) g(x).$$

A single-result prediction interval adds $RSS/(n - p)$ to this variance.

The source uses analytic inverses for E01–E07. E09 takes the left Gaussian branch, E11 uses the principal asin branch, and E12 takes real polynomial roots and selects the root closest to the observed X mean. E08 currently uses

$$x = C \left(\frac{B - A}{y - A} - 1 \right)^{1/(DE)},$$

which is not a strict algebraic inverse of the current E08 equation when E is not 1. Other cases search an extended grid for sign changes, solve one-dimensional roots, and select the root closest to the observed X center.

21.17.2 Linearity Design Quantities and Regression Comparison

Expected concentration in a dilution panel is $C(p) = p C_H + (1 - p) C_L$. Replicate planning first sets $p_L = (1 - \alpha_F)^{1/L}$ and $z = z_{(1 + p_L)/2}$:

$$n = \left\lceil \left(\frac{z\sigma}{ADL - d_{\text{true}}} \right)^2 \right\rceil.$$

Linearity first summarizes each level with its mean, SD, and n_j . Equal weighting uses 1; level-variance weighting uses $1/s_j^2$ for raw observations and n_j/s_j^2 for level means. pooled requires `variance_group` to define contiguous groups in `x`. For group `g`,

$$\hat{\sigma}_g^2 = \frac{\sum_{j \in g} (n_j - 1) s_j^2}{\sum_{j \in g} (n_j - 1)}, \quad w_j = \frac{n_j}{\hat{\sigma}_g^2}.$$

The function does not form groups automatically because EP06 grouping also depends on the repeatability-SD versus concentration relationship. A precision-profile weight first fits SD against mean and then uses $1 / \text{estimated } SD_j^2$.

The line fits level means; quadratic and cubic polynomials retain all observations. If any nonlinear term is significant, the model with the smallest $s_{y|x}$ among models containing significant nonlinear terms is selected; otherwise the line is selected. Nonlinear deviation is

$$DL_j = \hat{y}_{\text{poly},j} - \hat{y}_{\text{linear},j},$$

and linear residual is $y_{\text{bar}_j} - \hat{y}_{\text{linear},j}$. Percentage metrics can use the linear prediction, X , or the observed mean as denominator.

Each level's point interval uses family coverage $\gamma^{1/L}$:

$$h_j = z_{[1+\gamma^{1/L}]/2} \frac{\hat{\sigma}_j}{\sqrt{n_j}}.$$

The source centers the interval on the deviation. Allowable deviation is the maximum of direct absolute limits and absolute limits converted from relative limits; the final comparison is $|\text{metric}_j| > ADL_j$.

21.18 Interference, Dilution, Spike Recovery, and Hook Effect

21.18.1 Two-Group Interference and Difference-in-Differences

For a two-group interference study,

$$D = \bar{y}_T - \bar{y}_C, \quad SE(D) = \sqrt{\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}}.$$

The interval uses a t critical value with $n_T + n_C - 2$ degrees of freedom, even though the standard error is unpooled. Percentage effect is $100 D / \bar{y}_{\text{bar}_C}$, with the SE and interval endpoints scaled directly.

For an analyte-spike design,

$$D = (\bar{y}_{I,S} - \bar{y}_{I,U}) - (\bar{y}_{N,S} - \bar{y}_{N,U}),$$

with variance equal to the sum of s_g^2 / n_g across four cells and Welch–Satterthwaite degrees of freedom. Percentage effect is $100(I/N - 1)$, where $I = \bar{y}_{I,S} - \bar{y}_{I,U}$ and $N = \bar{y}_{N,S} - \bar{y}_{N,U}$. The source calculates the variance by differentiating this expression with respect to the four means and applying the delta method.

When at least two control results exist, blank interference still uses the two-group difference; otherwise the theoretical control mean is zero and a one-sample standard error is used. Dose response calculates effects at each concentration relative to the observed baseline and can additionally fit an OLS line of raw result versus concentration. Forward and reverse conversions use linear interpolation between points and do not extrapolate; reverse interpolation requires a monotone response.

21.18.2 Dilution Recovery

For dilution factor D , the source sets $f = 1 / D$. With original assignment C_0 and diluent background C_d ,

$$C_{\text{target}} = fC_0 + (1 - f)C_d,$$

$$C_{\text{corrected}} = \frac{\bar{y} - (1 - f)C_d}{f}, \quad \text{Recovery} = 100 \frac{\bar{y}}{C_{\text{target}}}.$$

Recovery is measured diluted result divided by target; it is not corrected concentration divided by original assignment. For multiple samples, the overall interval is the t interval of sample-level recovery means.

21.18.3 Spike Recovery and Hook Effect

$$C_{\text{added}} = C_s \frac{V_s}{V_0 + V_s}, \quad C_{\text{final}} = \frac{C_0 V_0 + C_s V_s}{V_0 + V_s},$$

$$\text{Recovery} = 100 \frac{\bar{y}_{\text{spike}} - \bar{y}_{\text{solvent}}}{C_{\text{added}}}.$$

Hook-effect analysis calculates a mean and t interval at each concentration and identifies the concentration with the maximum mean as the peak. After the peak, when the first level has concentration above ULoQ and a mean response no higher than the threshold, the source reports the preceding measured concentration as hook concentration; it does not interpolate between levels. If no decline is observed, it reports the highest tested concentration as a lower bound.

21.19 Analytical Sensitivity

21.19.1 Limits of Blank and Detection

The nonparametric LoB uses rank $r = 0.5 + B(1 - \alpha)$ and linearly interpolates between adjacent integer ranks. The parametric method sets $\nu = B - K$ and calculates

$$c_p = \frac{z_{1-\alpha}}{\sqrt{1 - 1/(4\nu)}}, \quad \text{LoB} = \bar{y}_B + c_p s_B.$$

The zero-blank branch sets $LoB = 0$ directly. With two or three reagent lots, the default calculates each lot and takes the maximum LoB ; with other lot counts it pools the lots.

Classical LoD pools low-level SDs by degrees of freedom:

$$s_p = \sqrt{\frac{\sum_j (n_j - 1) s_j^2}{\sum_j (n_j - 1)}}, \quad LoD = LoB + c_p s_p.$$

The nonparametric validation method uses the median of all low-level results as a candidate LoD and retains it only when the fraction below LoB is no more than β . The precision-profile method fits $s(x)$ and solves $x - LoB - c_p s(x) = 0$, using the smallest qualifying root found.

The probit method fits

$$\Phi^{-1}\{P(\text{detected} \mid x)\} = \beta_0 + \beta_1 g(x),$$

where $g(x)$ is $\log_{10}(x)$ or x . For target detection probability $1 - \beta$,

$$g(LoD) = \frac{z_{1-\beta} - \hat{\beta}_0}{\hat{\beta}_1}.$$

When fitting by lot or stratum, the final LoD is the maximum group-specific LoD .

21.19.2 Limit of Quantitation and Limit Verification

For a low-level sample, bias is $b = \bar{y} - x_a$. The implemented error quantities are

$$E_W = |b| + ks, \quad E_{RMS} = \sqrt{s^2 + b^2}, \quad E_P = s.$$

For a relative target, divide by $\text{abs}(x_a)$. The observed candidate LoQ is the smallest measured mean whose error reaches the target. With a profile enabled, the candidate is the smallest root of a linear or quadratic regression of error against assigned value and the target line. The maximum candidate across lots is reported, and when both are finite

$$LoQ_{\text{reported}} = \max(LoQ_{\text{candidate}}, LoD).$$

LoB verification calls $y \leq LoB$ consistent; LoD verification calls $y \geq LoB$ consistent; LoQ verification calls a result consistent when it lies within the allowed window around its assigned value. The source calculates one-sided Wilson intervals and uses an internal sample-size boundary table; a sample size not in the table takes the larger boundary from neighboring rows. It returns proportions, intervals, and differences from boundaries but does not generate a pass/fail conclusion.

21.20 Quality Control and Stability

21.20.1 Levey–Jennings and Westgard

QC results are standardized as $z_i = (x_i - \mu_0) / \sigma_0$. If target values are not supplied, the source uses the sample mean and sample SD of all finite results. The rules are:

Rule	Source decision
1-2s	$\text{abs}(z_i) > 2$

Rule	Source decision
1-3s	$\text{abs}(z_i) > 3$
2-2s	two adjacent points on the same side, both strictly beyond 2s
R-4s	adjacent points each with $\text{abs}(z) \geq 2$ and $\text{abs}(\Delta z) \geq 4$
4-1s	four consecutive points on the same side, all strictly beyond 1s
8x/10x	eight/ten consecutive points strictly on the same side of the mean

The grouping variable is used only for plot facets; target estimation and rule scanning run across all finite results in row order. 1-3s is labeled out of control and other violations are warnings. Youden uses $\text{abs}(z_{1i}) \leq 2$ and $\text{abs}(z_{2i}) \leq 2$ to test membership in the $\pm 2s$ rectangle and calculates Pearson correlation; it does not fit an error ellipse.

21.20.2 Node Bias, Time Regression, and Time to Limit

Each condition uses its earliest time-point mean as baseline:

$$b_t = y_t - y_0, \quad b_{t,\%} = 100 \frac{y_t - y_0}{y_0}.$$

Self-comparison first averages by time point. If all time-point SEs are finite and positive, it fits with $1 / \text{SE}_t^2$ weights; otherwise it falls back to OLS. Absolute drift rate is b_1 , relative drift rate is b_1 / b_0 , with variance

$$V\left(\frac{b_1}{b_0}\right) = \frac{V(b_1)}{b_0^2} + \frac{b_1^2 V(b_0)}{b_0^4} - \frac{2b_1 \text{Cov}(b_0, b_1)}{b_0^3}.$$

The source uses a one-sided t critical value $t_{\text{gamma}, \text{nu}}$. For a decline it takes the lower bound and for an increase the upper bound; if direction is uncertain, it takes the larger absolute side.

Condition comparison uses test-condition minus reference-condition bias at matched time points. Absolute-bias SE is $\sqrt{\text{SE}_T^2 + \text{SE}_R^2}$; relative-bias SE uses the ratio delta method. WLS or OLS is then selected according to SE availability.

Time to limit searches a 501-point grid for the first sign change of $\text{abs}(\hat{b}(t)) - L$ or of a one-sided confidence bound versus L , then refines it with one-dimensional root finding. The default upper search is $\max(100 t_{\text{max}}, t_{\text{max}} + 100)$; a root beyond the largest observed time is marked extrapolated.

21.20.3 Mean Kinetic Temperature and Arrhenius

Let $q_i = \exp[-E_a / (R T_i)]$. For equally spaced times,

$$\bar{q} = \frac{1}{n} \sum_i q_i, \quad T_{MKT} = -\frac{E_a}{R \log \bar{q}}.$$

With timestamps, trapezoidal integration is used:

$$\bar{q} = \frac{\sum_i \Delta t_i (q_i + q_{i+1}) / 2}{\sum_i \Delta t_i}.$$

Multiple probes combine q by effective observation count or time duration, not by directly averaging probe-specific MKT values.

The Arrhenius branch fits zero-order $y = a + b t$ or first-order $\log(y) = a + b t$ at each temperature and converts the slope to a positive rate according to the degradation direction. Automatic order selects the smaller root-mean-square normalized RMSE across temperatures; exact ties select first order. Positive rates then fit

$$\log k = \alpha + \beta/T, \quad E_a = -\hat{\beta}R.$$

Zero-order shelf life is allowable change divided by the target-temperature rate. First-order relative decreases and increases are

$$t = -\frac{\log(1-r)}{k_*}, \quad t = \frac{\log(1+r)}{k_*}.$$

Shelf-life intervals convert both endpoints of the target-rate interval and sort them.

The stability design is not a continuous power formula; it uses two lookup matrices embedded in the source. Repeatability is reduced by $\sigma / \sqrt{r}$, or combined as $\sqrt{\sum_j \sigma_j^2 / L_j}$. The resulting two ratios are rounded to two significant figures and looked up in the 80% or 90% power table.

21.21 Measurement Uncertainty and Traceability

21.21.1 Type A and Type B Standardization

The standard uncertainty of one repeat result is s ; the standard uncertainty of a mean is $s / \sqrt{n}$, with $n - 1$ degrees of freedom. Relative components are stored internally as $u / \text{abs}(\bar{x})$.

For interval-based Type B components, the estimate must be the midpoint of the limits. With half-width $a = (U - L) / 2$:

Distribution	Standard uncertainty
rectangular	$a / \sqrt{3}$
triangular	$a / \sqrt{6}$
U-shaped	$a / \sqrt{2}$
normal with central probability p	$a / z_{(1+p)/2}$
t with central probability p	$a / t_{(1+p)/2, \nu}$

A directly supplied uncertainty is treated as a standard uncertainty. Only when coverage factor k is separately supplied does the source calculate $u = U / k$. This path accepts standard or normal labels and does not divide by $\sqrt{3}$, $\sqrt{6}$, and so on according to a distribution label. Distribution divisors occur only on the limit-input path.

21.21.2 IQC, Homogeneity, Stability, and Characterization

For a one-way random-effects ANOVA,

$$MS_W = SS_W / (N - m), \quad MS_B = SS_B / (m - 1),$$

$$n_{\text{eff}} = \frac{N - \sum_j n_j^2 / N}{m - 1}, \quad s_B^2 = \max \left(0, \frac{MS_B - MS_W}{n_{\text{eff}}} \right).$$

When IQC has within-run replicates, single-result uncertainty is

$$u_{\text{single}} = \sqrt{s_W^2 + s_B^2}.$$

If $s_B > 0$, mean uncertainty is $\sqrt{\text{MS}_B / N}$; otherwise the source falls back to the SD of all results divided by $\sqrt{N}$. Without a run column or with one result per run, it uses long-term SD and its mean standard error directly.

The homogeneity branch also calculates

$$u_{\text{hidden}} = \frac{s_W}{\sqrt{\bar{n}}} \left(\frac{2}{\nu_W} \right)^{1/4}, \quad u_{bb} = \max(s_B, u_{\text{hidden}}).$$

Long-term stability uncertainty fits $y = a + b t$ for each condition and takes $u_{\text{Lts}} = \text{SE}(\hat{b}) T_s$ at shelf life T_s . Characterization first averages each laboratory, then assigns the un-weighted mean of laboratory means:

$$u_{\text{char}} = \frac{SD(\bar{x}_1, \dots, \bar{x}_L)}{\sqrt{L}}.$$

21.21.3 First-Order Combination and Effective Degrees of Freedom

For standard-uncertainty vector u and correlation matrix R ,

$$\Sigma = \text{diag}(u) R \text{diag}(u), \quad u_c^2 = c^T \Sigma c.$$

Signed marginal contribution from component i is $m_i = c_i (\Sigma c)_i$, and the contribution percentage is $100 m_i / u_c^2$; with correlation, a contribution can be negative.

Only when R is the identity does the source calculate

$$\nu_{\text{eff}} = \frac{u_c^4}{\sum_i (c_i u_i)^4 / \nu_i}.$$

Without a supplied coverage factor, finite effective degrees of freedom use a t quantile; otherwise a normal quantile is used. Expanded uncertainty is $U = k u_c$. When relative and absolute components are mixed, an output value must be supplied so relative components can be converted to absolute units.

21.21.4 General Measurement Models and Monte Carlo

For $y = f(x_1, \dots, x_q)$, sensitivity coefficients use central differences:

$$c_i \approx \frac{f(\dots, x_i + h_i, \dots) - f(\dots, x_i - h_i, \dots)}{2h_i},$$

$$h_i = \epsilon^{1/3} \max\{1, |x_i|, u(x_i)\}.$$

The delta branch propagates coefficients through the covariance matrix and returns a symmetric interval $\hat{y} \pm U$.

The Monte Carlo branch samples from stored distributions: normal and standard use $N(x_i, u_i^2)$; rectangular uses a uniform distribution; triangular uses the limits and estimate as

mode; U-shaped uses $x_i + a_i \cos(U(0, \pi))$; t uses $x_i + u_i T(\nu_i)$, so u_i is the t random-variable scale. Non-independent Monte Carlo accepts only normal or standard labels and generates correlated normal draws.

Simulated-output SD is the standard uncertainty, and the coverage interval is the central interval from R type-8 quantiles.

21.21.5 Stepwise Traceability

At stage r , all components from stages 1 through r are recombined. Sensitivity coefficients default to 1 and this interface has no between-stage correlation input, so

$$u_{c,r} = \sqrt{\sum_{i \in 1:r} u_i^2}.$$

Effective degrees of freedom, coverage factor, and expanded uncertainty are recalculated at every stage; the final stage is the budget for all accumulated components.

21.22 Sample Size and Power

21.22.1 Bland–Altman Agreement

For difference mean μ , SD σ , allowable limit Δ , and agreement quantile z_γ ,

$$SE_L = \sigma \sqrt{\frac{1}{n} + \frac{z_\gamma^2}{2(n-1)}},$$

$$\tau_1 = \frac{\Delta - \mu - z_\gamma \sigma}{SE_L}, \quad \tau_2 = \frac{\Delta + \mu - z_\gamma \sigma}{SE_L}.$$

With $t_c = t_{(1 - (1 - C)/2, n - 1)}$, power is

$$\Pr\{T_{n-1}(\tau_1) > t_c\} + \Pr\{T_{n-1}(\tau_2) < -t_c\}.$$

Sample-size search uses a binary search within the stated bounds for the smallest integer reaching the target power.

21.22.2 Proportion Intervals and Target Tests

The closed-form Wald half-width sample size is

$$n = \left\lceil \frac{z_{1-\alpha/2}^2 p(1-p)}{d^2} \right\rceil.$$

Other implemented intervals calculate the maximum actual endpoint half-width, then use exponential and binary search for the smallest n with half-width $\leq d$. Methods involving $x = \text{round}(np)$ can have nonmonotone jumps; the source scans downward to correct the result.

For a one-sample proportion target test, first find the critical success count. For an upper alternative it is the smallest integer whose selected one-sided interval lower limit $L(x, n)$ is greater than p_0 ; for a lower alternative it is the largest integer whose upper limit $U(x, n)$ is less than p_0 . Therefore

$$Power_{>} = 1 - F_{\text{Bin}(n, p_1)}(x_c - 1), \quad Power_{<} = F_{\text{Bin}(n, p_1)}(x_c).$$

A two-sided test allocates $\alpha / 2$ to each tail. For sample-size search, a normal-approximation starting point is followed by upward search and downward backtracking; for alpha search, the discrete critical count is fixed first and alpha is solved so that the interval endpoint equals p_0 . Power is therefore a step function determined by the binomial critical count.

Chapter 22

Standards Mapping

This chapter answers one question: in the current `ivdtools` version, which statistical tasks from specified standards are implemented, which are available only as component calculations, which are not implemented, and which have similar names but different algorithms. “Supported” here means auditable statistical calculation capability; it does not mean that study design, sample representativeness, acceptance criteria, quality systems, or registration conclusions have been completed automatically.

The Handbook covers all CLSI EP documents, YY/T 1789.1–1789.6, and the statistical tasks mentioned in the CMDE guidance. Other standards systems are outside this chapter.

22.1 Interpretation Categories

Four statuses are used:

- **Implemented:** a public function calculates the core statistic required by the standard, with standard data or an auditable example available for checking;
- **Partially implemented:** core components can be calculated, but a standard-specific statistic, input structure, complete workflow, or final decision is missing;
- **Not implemented:** the current package has no public entry point that answers the standard’s statistical question;
- **Different algorithm:** both the package and the standard produce a result, but the estimator, error model, interval, rejection region, or integerization rule differs.

“Different algorithm” can occur together with “partially implemented.” A difference in displayed digits that does not change the statistical definition is recorded as display rounding, not as an algorithm error. When the published standard value cannot be obtained from the public formula and same data, it is reported honestly as a published-value or source-material limitation; parameters are not adjusted merely to force agreement.

22.2 CLSI EP Series

22.2.1 Core Statistical Tasks Implemented or Directly Reusable

EP05 (precision) — core implemented. `precision()`, `variance()`, `ci()`, `profile()`, and `report()` can estimate repeatability, within-laboratory precision, reproducibility, CVs, variance components, and intervals by day, run, lot, site, or center. Single-center and multicenter EP05-A3 examples have been checked.

EP06 (linearity) — core implemented. `linearity()` is the dedicated linearity entry point and supports replicate-level aggregation, straight lines, weights, deviations, and allowable deviations; `linearity_replicates()`, `linearity_panel()`, and `linearity_endpoints()` support design and endpoint checks.

EP07 (interference) — core implemented. Replicate planning, paired interference, dose response, patient-sample comparison, and cross-reactivity are handled by `interference_replicates()`, `interference_paired()`, `interference_dose_response()`, and `interference_patient()`.

EP09 (method comparison and bias) — core implemented. `mcr()`, `describe()`, `correlation()`, `regression()`, `bland_altman()`, `outlier()`, `bias()`, `predict()`, and `reference_bias()` cover OLS, WLS, Deming, weighted Deming, Passing-Bablok, difference plots, outliers, and bias at medical decision levels. Main paths in EP09c Appendices C, D, F, H, I, J, and K2 have examples.

EP12 (qualitative performance) — binary core implemented; standard-specific portions missing. `raw_to_table()`, `counts_to_table()`, `diagnostics()`, `kappa()`, `mcnemar()`, and `c5_c95()` support 2×2 diagnostic performance, agreement, Kappa, McNemar, and C5/C95 from a continuous response. ANA/APA, semi-quantitative $R \times C$ weighted Kappa, fixed-CV C95, and special intervals in some appendices have no corresponding entry point. Coding positive counts as 0/1 and passing them to a continuous-response VFP model changes the statistical model and is not a substitute implementation.

EP14 (commutability of processed samples) — core implemented. `commutability()` supports EP14-type Deming regression, clinical-sample repeatability, prediction intervals at specified reference points, and material classification. The statistical unit of samples, procedures, replicates, and prediction intervals is auditable; sample source, processing, reference procedure, and final suitability judgment are not completed automatically.

EP15 (user precision and bias verification) — core implemented. Precision can use the `precision()` family, and bias can use `reference_bias()` or the method-comparison workflow.

EP17 (detection capability) — core implemented. `sensitivity_lob()`, `sensitivity_lod()`, `sensitivity_loq()`, `sensitivity_verify()`, and `sensitivity_design()` cover parametric/nonparametric LoB, classical/Probit LoD, precision or total-error LoQ, and claim verification.

EP24 (ROC and diagnostic accuracy) — core implemented, with result-format differences. `roc()`, `auc()`, `cutoff()`, `mlr()`, and `auc_compare()` support empirical ROC, AUC, cutoff, multivariate logistic regression, and EP24 Hanley-McNeil and DeLong paired AUC comparisons. Single-curve intervals use the package's fixed Hanley-McNeil path; cutoff notation between adjacent values, Youden tie handling, and standard hand tables may differ.

EP25 (stability) — core implemented. `stability_plan()`, `stability_bias()`, `stability_regression()`, `stability_time()`, `arrhenius()`, and `mkt()` handle node bias, trends, one-sided limits, time to exceed a limit, and accelerated stability.

EP28 (reference intervals) — core implemented. `reference_interval()` supports the EP28 continuous-rank nonparametric method, exact binomial-rank confidence intervals, Appendix B biweight, and type 6, parametric, and Huber extensions.

EP29 (measurement uncertainty) — core implemented, with path differences. Type A/B components, IQC, homogeneity, stability, characterization, covariance combination, effective degrees of freedom, general measurement models, delta, Monte Carlo, and stepwise traceability are available. The approximately 6.9%/7.1% difference between the Kragten approximation and analytic delta in EP29 Example 10 is a path difference.

EP30 (reference-material commutability) — core implemented. Prediction intervals, relative residuals, and material classification in EP30 Appendix B can be calculated separately.

EP34 (extended measurement interval) — core implemented. `dilution_recovery()`, `spike_concentration()`, `spike_recovery()`, and `hook_effect()` calculate dilution, spike recovery, mass balance, and hook effect. The standard requires precision, LoQ, recovery, and hook-effect evidence together; the package does not automatically integrate these into a final EMI decision.

22.2.2 EP Documents with Only Related Tools or Not Yet Implemented

EP10 has component tools for precision, linearity, method comparison, and outliers, but no integrated preliminary-evaluation workflow.

EP18 has no FMEA, fault-tree, FRACAS, or risk-control monitoring interface.

EP19 is a method-evaluation framework rather than one statistical calculation; package modules do not automatically select a Test Life Phases Model.

EP21 can produce bias and precision separately but has no standard-defined total analytical error combination, allowable-total-error comparison, and decision workflow.

EP23 `qc_chart()`, `list_westgard()`, and `youden_plot()` support calculation and display for specified rules only; they do not replace risk identification, QCP design, residual-risk assessment, or CAPA.

EP26 can compare general lots or groups but has no dedicated patient-sample lot-to-lot design, critical difference, or rejection limit.

EP27 can use `mcr()` to obtain method-comparison results but has no error-region definition, error-grid construction, or region-counting workflow.

EP31 can analyze paired results from two procedures but has no hierarchical sampling, multicenter design, or continuing-monitoring workflow for one healthcare system.

EP32 has uncertainty and stepwise-combination components but no dedicated management of the calibration hierarchy, reference measurement procedure, commutability, and traceability evidence chain.

EP33 has no patient-history, time-window, rate-of-change, segmented-limit, or Delta Check investigation interface.

EP35 can analyze method differences or interference but has no equivalence/suitability design and argument for alternative specimen types.

EP36 is a symbols-and-formulas harmonization document and has no “EP36 statistical analysis” that the package can automate.

EP37 is a database of interference materials, not an `ivdttools` estimator; the package does not provide its substance-selection or recommended-concentration library.

EP39 has no workflow for alternative-sample selection hierarchy, preparation, records, and suitability justification.

EP43 has no interface for emergency-use implementation, laboratory confirmation, information updates, and records management.

EP46 can accept user-specified allowable limits but does not establish allowable total error from clinical outcomes, biological variation, or technical capability.

EP47 has no risk assessment, sequence design, or carryover-quantity estimation function for reagent carryover.

EP49 has no clinical-validity evidence framework, control of sample-selection bias, ethics workflow, or post-market evidence-management process.

22.3 YY/T 1789.1–1789.6

YY/T 1789.1 (precision) — core implemented. `precision()` can perform the main calculations for repeatability, within-laboratory precision, and variance components.

YY/T 1789.2 (trueness—method comparison) — core implemented, with explicit algorithm differences. Method comparison, reference-material bias, segmented patient-sample bias, and OLS/WLS/Deming/Passing–Bablok parameters can be calculated. Whether a WLS bias interval is multiplied again by a weighted residual scale, and whether a Deming interval uses analytic covariance or delete-one jackknife, produces different endpoints. These differences affect interval width and decisions at medical decision levels; similar point estimates do not establish complete reproduction.

YY/T 1789.3 (limit of detection/quantitation) — core implemented, with finite-sample algorithm differences. LoB, classical LoD, and parametric/nonparametric paths are available, but the package uses continuous ranks, its own finite-sample coefficients, and pooled SD from raw data. Some standard examples use discrete ranks, displayed SDs, and different coefficients. Intermediate values may differ even when final rounded values are close; the algorithms are not identical.

YY/T 1789.4 (linearity) — partially implemented with model differences. `linearity()` calculates straight lines, deviations, and allowable limits, but the standard often fits higher-order models to level means and tests higher-order terms, whereas the package's exploratory polynomial mode is organized at the observation level. Degrees of freedom, residual standard errors, and model-selection evidence differ. There is no dedicated ADL-PctBnd critical-table entry point.

YY/T 1789.5 (interference) — main effects available; standard intervals differ. Paired effects, dose response, patient samples, and cross-reactivity rates can be obtained separately. The standard's B_c uses an SD from an external precision study, whereas the package's paired interval uses observed SD from the current experimental units; the intervals therefore do not have the same decision meaning. The package does not choose interferents, concentrations, or acceptance limits automatically.

YY/T 1789.6 (diagnostic performance) — binary core partly implemented. Four-fold tables, diagnostic sensitivity/specificity, Wilson intervals, and some appendix data can be checked. The standard's binomial hit-rate verification differs from the continuous internal-response/VFP model required by `c5_c95()`; they must not be treated as the same C5/C95 algorithm.

22.4 CMDE Guidance Support Level

Quantitative method comparison, bias, ROC, four-fold tables, recovery, precision, detection capability, interference, linearity, stability, reference intervals, and QC-material components in the CMDE guidance can be calculated separately by the corresponding modules. Not implemented or not automated are multicenter and clustered designs, semi-quantitative $R \times C$ weighted Kappa, cross-reactivity/inclusivity, grouped Z rules, reference-interval verification, complete measurement-interval evidence synthesis, transport and multi-analyte stability decisions, QC-target transfer, and database/XPT/audit-trail submission workflows.

CMDE support is not equivalent to “a function exists, therefore the study complies.” For example, a recovery function can report recovery for each sample and the mean recovery, but the allowable range and clinical acceptability come from the protocol; method comparison can provide regression and bias, but it does not automatically judge sample representativeness, clinical equivalence, or a registration conclusion.

Chapter 23

R Packages for IVD Data Analysis: A Review

IVD data analysis is broader than analytical-performance evaluation. A complete study may involve reagent and instrument performance, diagnostic accuracy, observer agreement, reference intervals, clinical prediction, follow-up outcomes, evidence synthesis, and high-dimensional biomarker data. This chapter provides a review of CRAN packages that support these tasks.

23.1 Scope and Inclusion Rules

This review includes packages available from CRAN by 2026-09-01 that directly target IVD/laboratory medicine or provide core methods used in IVD research. It also includes foundational or recommended packages such as `stats`, `survival`, `MASS`, and `nlme`. It does not attempt to list every general data-wrangling, plotting, database, or arbitrary statistics package, and it excludes packages available only from Bioconductor, GitHub, or other repositories. Package availability, versions, defaults, and maintenance status can change; inclusion is not a quality or regulatory endorsement.

23.2 Classification Framework

Analysis level	Typical question	Sections
Analytical performance	Is the method precise, linear, stable, unbiased, and sensitive?	Method comparison, precision, calibration/sensitivity, stability/QC
Clinical performance	Can the test distinguish states, and how should a cutoff or equivalence be evaluated?	Diagnostic accuracy, ROC, paired comparison, agreement
Clinical utility	Does the result improve risk stratification, decisions, or long-term prediction?	Regression, prediction, survival, decision curves
Evidence synthesis	How should sensitivity and specificity from several studies be combined?	Diagnostic-accuracy and general meta-analysis
Biomarker discovery	What structure and discrimination signals exist in multi-marker or high-dimensional data?	Multivariate analysis, chemometrics, machine learning

Analysis level	Typical question	Sections
Study support	How should missing data, outliers, repeats, sample size, and reproducibility be handled?	Design, data validation, reporting, audit

23.3 Analytical-Performance Evaluation

23.3.1 Integrated IVD Tools

Package	Main capabilities	Typical use
<code>ivdtools</code>	Method comparison, commutability, Bland–Altman, ROC, qualitative agreement, precision, linearity, interference, dilution, stability, QC, reference intervals, uncertainty, and sample size	A unified, auditable workflow for IVD reagent evaluation
<code>mcradds</code>	Method comparison, Bland–Altman, diagnostic tables, ROC comparison, precision, reference intervals, and sample size	IVD clinical and analytical-performance workflows; some features use <code>mcr</code> and <code>pROC</code>

23.3.2 Method Comparison, Bias, and Agreement

Package	Main methods	Typical IVD use
<code>ivdtools</code>	OLS/WLS, Deming, weighted Deming, Passing–Bablok, Bland–Altman, bias, commutability	Unified comparison, reference-material bias, and commutability workflow
<code>mcr</code>	Deming, weighted Deming, Passing–Bablok, Theil–Sen, bootstrap intervals	Regression and medical-decision-level bias
<code>MethComp</code>	Method-comparison regression, repeated-measurement comparison, Bland–Altman	Multiple methods or repeated measures
<code>deming</code>	Deming regression and repeated-measurement extensions	Error in both procedures
<code>blandr</code> , <code>BlandAltmanLeh</code>	Bland–Altman statistics, intervals, and plots	LoA, proportional bias, and visualization
<code>SimplyAgree</code>	Continuous and categorical agreement, equivalence-oriented analyses	Method, device, or observer agreement

Correlation measures co-variation, not agreement. Different packages may use different error ratios, weighting, repeated-measurement models, bootstrap schemes, and interval definitions; cross-package checks must harmonize these choices.

23.3.3 Precision, Variance Components, and Repeatability

Package	Main capabilities	Typical use
<code>ivdtools</code>	Nested/repeated-measurement precision, variance components, precision profiles, C5/C95	Repeatability, within-laboratory precision, reproducibility, and qualitative detection probability
<code>VCA</code>	Crossed/nested ANOVA, REML components, intervals	EP05-style precision
<code>VFP</code>	Imprecision profiles and LoQ	CV/SD versus concentration
<code>lme4</code> , <code>nlme</code> , <code>glmmTMB</code>	Linear, nonlinear, generalized, and zero-inflated mixed models	Unbalanced day, lot, instrument, center, or operator designs
<code>rptR</code>	Repeatability and intraclass correlation	Operator, device, or reader repeatability

More flexible mixed models do not automatically implement a particular precision guideline; preserve the variance model, degrees of freedom, negative-component handling, and interval method.

23.3.4 Calibration, Linearity, Dose Response, and Detection Capability

Package	Main capabilities	Typical use
<code>ivdtools</code>	Linearity, weighted regression, response curves, LoB/LoD/LoQ, dilution/spike recovery, hook effect	Measuring interval and analytical sensitivity studies
<code>stats</code>	<code>lm()</code> , <code>glm()</code> , <code>nls()</code> , polynomial and isotonic regression	Linear/nonlinear calibration and response probabilities
<code>drc</code> , <code>dr4pl</code>	Dose-response and four-parameter logistic models	Immunoassay, C5/C50/C95, concentration-response curves
<code>chemCal</code>	Analytical calibration, inverse prediction, detection limits	Calibration and LoD calculations
<code>minpack.lm</code> , <code>investr</code>	Robust nonlinear fitting, inverse estimation, calibration intervals	4PL/5PL and inverse concentration estimates
<code>segmented</code> , <code>isotone</code>	Breakpoints and monotone constraints	Measuring ranges, plateaus, and monotone calibration
<code>NADA</code> , <code>NADA2</code> , <code>EnvStats</code>	Censored and below-detection-limit methods	LoB/LoD data description and regression

LoB, LoD, and LoQ depend on study design and claim definition; a package name alone does not determine suitability.

23.3.5 Reference Intervals

Package	Main capabilities
<code>ivdtools</code> , <code>referenceIntervals</code> , <code>mcradds</code>	Direct parametric, nonparametric, and robust intervals
<code>refineR</code> , <code>reflimR</code>	Indirect intervals from routine laboratory data
<code>RIbench</code>	Simulation and benchmarking of indirect algorithms
<code>mclust</code>	Gaussian finite mixtures for exploratory subpopulations
<code>quantreg</code> , <code>gamlss</code>	Conditional quantiles and distributional reference curves

Direct and indirect reference-interval methods answer different questions. Indirect results depend on source population, pathological-value fraction, partitioning, truncation, and distributional assumptions.

23.3.6 Stability, Interference, Recovery, Lot Differences, and QC

`ivdtools` provides stability regression, interference/recovery, dilution, lot bias, uncertainty, and Westgard QC components. General-purpose alternatives include `stats`, `nlme`, `lme4`, `emmeans`, `multcomp`, `segmented`, `strucchange`, `changepoint`, `qcc`, `IQCC`, `qicharts2`, `SixSigma`, `spc`, `metRology`, `propagate`, `errors`, and `units`. These tools can support time trends, mixed models, marginal comparisons, control charts, and uncertainty propagation, but study-specific acceptance limits and design remain external.

23.4 Clinical Diagnostic Performance

23.4.1 Binary Metrics, ROC, and Cutoffs

Package	Main capabilities
<code>ivdtools</code>	ROC, AUC, cutoff, C5/C95, qualitative agreement, classification summaries
<code>epiR</code>	Sensitivity, specificity, predictive values, likelihood ratios, prevalence adjustment
<code>pROC</code>	ROC, AUC/pAUC, DeLong/bootstrap intervals, ROC comparison
<code>ROCR</code> , <code>ROCit</code> , <code>cutpointR</code>	Threshold metrics, performance curves, bootstrap cutoffs
<code>OptimalCutpoints</code> , <code>ThresholdROC</code> , <code>nsROC</code>	Optimal threshold and nonparametric ROC methods
<code>PRROC</code> , <code>precRec</code>	Precision–recall curves for imbalance
<code>yardstick</code> , <code>mlr3measures</code> , <code>EvalTest</code>	Consistent classification and probability metrics
<code>bdpv</code> , <code>presize</code>	Prevalence-based predictive values and precision/sample-size planning

Cutoff selection must be prespecified and externally or internally validated. A cutoff optimized and evaluated on the same data has optimism.

23.4.2 Paired Tests, Multiclass, No Gold Standard, and Agreement

`DTComPair`, `testCompareR`, and `CompareTests` compare paired binary tests. `bcROCsurface` and `ROCpsych` support multiclass ROC. `poLCA`, `randomLCA`, and `BayesLCA` support latent-class models

when no perfect reference exists. `vcd`, `DescTools`, `irr`, and `irrCAC` support Kappa and categorical agreement; `psych`, `irr`, and `SimplyAgree` support ICC and continuous agreement. `RJafroc` and `MRMCaov` support multi-reader multi-case designs.

23.5 Diagnostic Accuracy and General Meta-Analysis

23.5.1 Diagnostic-Accuracy Meta-Analysis

`mada` provides diagnostic odds ratios, bivariate Reitsma models, meta-regression, and SROC; `meta4diag` and `bandit` provide Bayesian bivariate models; `diagmeta` models multiple cutoffs within a study; `CopulaREMADA` handles more complex dependence; `DTAplots` and `dmetatools` support plots and supplementary SROC inference. Verify TP/FP/FN/TN, thresholds, spectrum, reference standards, and continuity corrections before selecting a model.

23.5.2 General Effect Sizes and Prognostic Meta-Analysis

`meta`, `metafor`, and `rmeta` provide fixed/random effects, heterogeneity, meta-regression, and plots. `metasens`, `robustmeta`, `netmeta`, `mvmeta`, `metaSEM`, `metavcov`, `robmeta`, `clubSandwich`, `bayesmeta`, `MetaStan`, `metamisc`, `metaSurvival`, `metagear`, `revtools`, `citationchaser`, and `PRISMA2020` cover sensitivity/bias analyses, network and multivariate meta-analysis, Bayesian models, prognostic synthesis, survival-curve reconstruction, and review workflows.

23.6 Survival and Event Outcomes

`survival`, `prodlim`, and `rms` support Kaplan–Meier and Cox models; `survminer` and `ggsurvfit` support presentation; `flexsurv`, `rstpm2`, and `eha` support flexible parametric models; `cmprsk`, `riskRegression`, `mstate`, and `msm` support competing risks and multistate models; `timeROC` and `survivalROC` support time-dependent ROC; `pec` and `riskRegression` support prediction error and calibration; `glmnet`, `randomForestSRC`, and `ranger` support penalized or machine-learning survival models; `JMbayes2` and `joineR` support joint longitudinal-event models; `powerSurvEpi`, `gsDesign`, and `rpact` support design and power.

Baseline prognostic value, time-varying biomarkers, treatment-effect modification, and diagnostic accuracy are different estimands. Report time origin, censoring, competing events, prediction horizon, validation, and proportional-hazards checks.

23.7 Multivariate Analysis, Chemometrics, and High-Dimensional Biomarkers

`FactoMineR`, `factoextra`, `psych`, `GPArotation`, `ade4`, `RGCCA`, `cluster`, `mclust`, `dbscan`, `fpc`, `compositions`, `robCompositions`, `rrcov`, `robustbase`, `mvoutlier`, `MVN`, and `mvnrmtest` support PCA, factor analysis, multi-table analysis, clustering, compositional data, robust covariance, and multivariate normality. `pls`, `mdatools`, `chemometrics`, `prospectr`, `ChemoSpec`, `ChemoSpec2D`, `hyperSpec`, and `ALS` support PLS/PCR, spectral preprocessing, chemometric calibration, and multivariate curve resolution.

For prediction, `glmnet`, `e1071`, `kernlab`, `rpart`, `randomForest`, `ranger`, `xgboost`, `caret`, `tidymodels`, and `mlr3` support regularization, SVM, trees, ensembles, and complete resampling workflows. `themis` and `ROSE` support imbalance handling; `iml` and `DALEX` support explanations. Preprocessing, feature selection, and threshold selection must occur inside resampling folds to avoid leakage.

23.8 Regression, Clinical Prediction, and Utility

stats and MASS provide basic regression; logistf and brglm2 address separation; mgcv and rms support smooth and spline effects; geepack, lme4, nlme, and glmmTMB support repeated measures and clustering; emmeans, multcomp, sandwich, and clubSandwich support contrasts and robust SEs; pROC, yardstick, rms, riskRegression, and pec support model performance; pmcalibration supports calibration; and dcurves supports decision-curve analysis. Adding a marker should be assessed for discrimination, calibration, clinical thresholds, net benefit, overfitting, and external validation—not AUC increment alone.

23.9 Sample Size, Missing Data, and Statistical Support

ivdtools, presize, pROC, MKmisc, pmsampsize, WebPower, pwr, longpower, powerSurvEpi, gsDesign, and rpact support different sample-size and design problems. Partition total sample size into positive and negative targets, and account separately for prevalence, attrition, invalid samples, clustering, pairing, and multiple endpoints.

naniar and VIM diagnose missingness; mice, Amelia, missForest, and missMDA support imputation; outliers, EnvStats, nortest, rrcov, mvoutlier, and robustbase support distribution and outlier diagnostics; boot and rsample support resampling; multcomp and emmeans support multiplicity adjustments. Below-detection-limit results, instrument errors, missing visits, and random missing values are different mechanisms and should not all be treated as ordinary NA.

23.10 Data Management, Reporting, and Reproducibility

readr, readxl, haven, data.table, dplyr, and tidyr support import and data preparation. checkmate, pointblank, validate, and assertr support validation rules. broom, parameters, and performance tidy and audit model results. gtsummary, tableone, arsenal, and finalfit generate statistical tables. ggplot2, patchwork, and forestplot support visualization. knitr and rmarkdown create dynamic reports, renv and sessioninfo lock environments, and targets orchestrates reproducible pipelines.

A formal IVD project should preserve a read-only raw-data copy, data-generation rules, package versions, arguments, random seeds, exclusions, output objects, and the final report. CRAN availability is not the same as project-level software validation or regulatory approval.

23.11 Complete Review

23.11.1 Scope and Inclusion Rules

This review uses the following reproducible scope:

1. Include packages available from CRAN before 2026-09-01 that directly target IVD/laboratory medicine or provide core statistical methods for IVD research. Foundational or recommended packages such as stats, survival, MASS, and nlme are included.
2. Cover analytical performance, clinical performance, meta-analysis, survival analysis, multivariate analysis, and prediction modeling; one package may appear in several categories.
3. Do not call every data-wrangling, plotting, database, or general statistics package an IVD package. Otherwise the set of all packages usable for IVD would have no finite boundary.
4. Exclude packages released only through Bioconductor, GitHub, or other repositories, and packages withdrawn from CRAN. Many omics tools are hosted by Bioconductor and should be searched separately.

This is a thematic, broad directory as of the stated date, not a software-quality ranking or regulatory endorsement. CRAN availability, versions, defaults, and maintenance may change. A CRAN Task View is navigation by a defined topic, not a certification of the packages it lists.

23.12 Analytical-Performance Evaluation

23.12.1 Integrated IVD Packages

Package	Main capabilities	Use
ivdtools	Method comparison, commutability, Bland–Altman, ROC, qualitative agreement, precision, linearity, interference, dilution, stability, QC, reference intervals, uncertainty, and sample size	A unified auditable workflow for IVD reagent evaluation
mcradds	Method comparison, Bland–Altman, diagnostic tables, ROC comparison, precision, reference intervals, and sample size	IVD clinical and analytical-performance workflows, partly built around mcr and pROC

An integrated package helps organize a coherent workflow, but offering a calculation does not mean that every element of a CLSI document is implemented. Study design, sample selection, acceptance limits, and reporting remain matters for the applicable standard or protocol.

23.12.2 Method Comparison, Bias, and Agreement

Package	Main methods	Typical IVD use
ivdtools	OLS/WLS, Deming, weighted Deming, Passing–Bablok, Bland–Altman, bias, and commutability	One workflow from method comparison to reference-material and patient-sample bias
mcr	Deming, weighted Deming, Passing–Bablok, Theil–Sen, and bootstrap intervals	Candidate versus comparison-method regression and bias at medical decision levels
MethComp	Method-comparison regression, repeated-measurement comparison, and Bland–Altman	Multiple methods or repeated measures
deming	Deming regression and repeated-measurement extensions	Two methods with measurement error
blandr	Bland–Altman statistics, confidence intervals, and ggplot2 figures	Limits of agreement and proportional bias
BlandAltmanLeh	Bland–Altman and sunflower plots	Overlapping points and basic agreement summaries

Package	Main methods	Typical IVD use
SimplyAgree	Continuous and categorical agreement and equivalence-oriented methods	Method, device, or observer agreement
mcradds	Wrappers around mcr comparison, bias, and plots	Workflows that also need other IVD modules

Correlation is only a description of linear association and cannot replace agreement analysis. Before cross-checking packages, harmonize error ratios, replicate handling, weights, bootstrap settings, outlier rules, and confidence-interval definitions.

23.12.3 Precision, Variance Components, and Repeatability

Package	Main capabilities	Typical use
ivdtools	Nested/repeated-measurement precision, variance components, precision profiles, and C5/C95	Repeatability, within-laboratory precision, reproducibility, and qualitative detection probability
VCA	Crossed/nested ANOVA, REML variance components, and intervals	Repeatability, within-laboratory precision, and reproducibility
VFP	Imprecision profiles and LoQ	SD/CV versus concentration
lme4	Linear and generalized mixed models	Day, lot, instrument, center, or operator random effects
nlme	Linear/nonlinear mixed models, correlation, and heteroscedasticity	Concentration-dependent variance and repeated-measurement correlation
glmmTMB	Flexible GLMM, dispersion, and zero-inflation	Qualitative, count, or complex precision data
rptR	Repeatability and within-group correlation	Operator, device, or reader repeatability

A more flexible model is not automatically compliant with a precision guideline; check degrees of freedom, variance-component intervals, and negative estimates.

23.12.4 Calibration, Linearity, Dose Response, and Detection Capability

Package	Main capabilities	Typical IVD use
ivdtools	Linearity, weighted regression, response equations, analytical sensitivity, dilution recovery, spike recovery, and hook effect	Measuring interval, dose response, LoB/LoD/LoQ, and extended-range studies
stats	lm, glm, nls, polynomial, and isotonic regression	Linear/nonlinear calibration and qualitative response probability
drc	Dose-response models and effective doses	Immunoassays and C5/C50/C95 curves

Package	Main capabilities	Typical IVD use
dr4pl	Four-parameter logistic curves	4PL calibration and dose response
chemCal	Analytical calibration, inverse prediction, and detection limits	Calibration and LoD
minpack.lm	Nonlinear least-squares solvers	4PL/5PL convergence support
investr	Inverse estimation and calibration intervals	Estimate unknown concentrations from responses
segmented	Segmented regression and breakpoints	Measuring interval, plateau, and hook-effect transitions
isotone	Monotone regression	Monotone calibration response
NADA/NADA2 EnvStats	Censored-data methods Distribution, censoring, outlier, and interval methods	Below-detection-limit data Supplementary LoB/LoD data analysis

LoB, LoD, and LoQ depend on the design and intended claim. VFP focuses on imprecision profiles; chemCal focuses on analytical calibration and detection limits; detection-probability curves can use glm, drc, brglm2, or logistf.

23.12.5 Reference Intervals

Package	Main capabilities	Typical use
ivdtools	Parametric, nonparametric, robust, outlier, and reference-limit intervals	Reference-interval establishment
referenceIntervals	Parametric, nonparametric, robust, outlier, and reference-limit intervals	Reference-interval establishment
mcradds	Direct reference intervals and related figures	Reference-interval establishment
refineR	Indirect intervals from routine laboratory data	Mixed healthy/pathological populations
reflimR	Indirect limits using truncated Q-Q methods	Historical laboratory data
RIbench	Simulation and benchmarking	Compare indirect reference-limit algorithms
mclust	Gaussian finite mixtures	Exploratory mixed subpopulations
quantreg/gamlss	Conditional quantiles and distributional regression	Age/sex or other covariate-dependent reference curves

Direct and indirect methods answer different questions. Indirect results depend on source population, pathological-value fraction, partitioning, truncation, and distribution assumptions.

23.12.6 Stability, Interference, Recovery, Lot Differences, and QC

CRAN has no single universal IVD package for these tasks. ivdtools provides stability regression, interference/recovery, dilution, lot bias, uncertainty, and Westgard QC compo-

nents. General-purpose alternatives include stats, nlme, lme4, emmeans, multcomp, segmented, strucchange, changepoint, qcc, IQCC, qicharts2, SixSigma, spc, metRology, propagate, errors, and units. These support regression, ANOVA, repeated measures, mixed models, marginal comparisons, change points, control charts, and uncertainty propagation.

Stability, interference, and lot differences are not only comparisons of means. Time correlation, pairing, concentration-dependent bias, allowable error, and worst-case decisions must be expressed in the model and protocol.

23.13 Clinical Diagnostic Performance

23.13.1 Binary Metrics, ROC, and Cutoffs

Package	Main capabilities	Use
ivdtools	ROC, AUC, cutoff, C5/C95, qualitative agreement, and classification summaries	Integrated IVD workflow; cutoff still requires prespecification and validation
epiR	Sensitivity, specificity, predictive values, likelihood ratios, and prevalence adjustment	Four-fold tables
pROC	ROC, AUC/pAUC, DeLong/bootstrap intervals, and ROC comparison	Continuous markers and paired ROC
ROCR/ROCit	Classification curves and threshold metrics	Model/marker visualization
cutpointr	Cutoff criteria, bootstrap, and subgroup analysis	Exploratory and validated thresholds
OptimalCutpoints/ThresholdROC	Optimal thresholds and ROC inference	Cutoff sensitivity analysis
PRROC/precrc	Precision-recall curves	Low prevalence or class imbalance
yardstick/mlr3measures	Unified classification and probability metrics	Resampling and machine learning
EvalTest	Diagnostic metrics, cutoff, ROC, and interactive exploration	Rapid exploratory diagnostic report
bdpv	Prevalence-based Bayesian predictive values	PPV/NPV and prevalence uncertainty
presize	Precision-based design for sensitivity, specificity, predictive values, and AUC	Sample-size planning

A cutoff is part of the research question, not an attachment to an ROC plot. Prespecify population, positive direction, reference standard, error costs, and internal/external validation; selecting and evaluating the cutoff on the same data creates optimism.

23.13.2 Paired Tests, Multiclass, No Gold Standard, and Observer Agreement

Problem	Packages	Use
Paired binary tests	DTComPair, testCompareR, CompareTests	Paired sensitivity, specificity, predictive value, or accuracy comparisons
Multiclass ROC	bcROCsurface, ROCpsych	ROC surface, VUS, and multiclass ROC
No perfect reference	poLCA, randomLCA, BayesLCA	Latent-state and latent-class estimation
Binary agreement	vcd, DescTools, irr, irrCAC	Kappa, weighted Kappa, and observer agreement
Continuous agreement	psych, irr, SimplyAgree	ICC and device/reader agreement
Multi-reader multi-case	RJafroc, MRMCAov	Imaging or reader studies

Latent-class models do not create a gold standard; identifiability depends on assumptions such as conditional independence. Multi-reader data must not be analyzed as independent ordinary observations.

23.14 Diagnostic Accuracy and General Meta-Analysis

23.14.1 Diagnostic-Accuracy Meta-Analysis

Pooling sensitivity and specificity separately ignores their dependence and threshold effects. Diagnostic meta-analysis often needs bivariate random-effects, HSROC, or multi-threshold models.

Package	Main capabilities	Typical use
mada	Diagnostic odds ratio, bivariate Reitsma, meta-regression, SROC	Standard diagnostic-accuracy meta-analysis
meta4diag	Bayesian bivariate diagnostic meta-analysis	Bayesian inference
bamdit	JAGS-based Bayesian bivariate random-effects models	Bayesian DTA
diagmeta	Within-study multiple cutoffs	Multi-threshold studies
CopulaREMADA	Copula bivariate mixture models	More complex dependence
DTAplots	Forest and SROC plots	Presentation
MVPBT	Small-study-effect tests	Supplementary bias analysis
dmetatools	SROC AUC intervals and related methods	Supplementary SROC inference

Before selecting a model, verify TP, FP, FN, TN, thresholds, spectrum, reference standard, zero-cell correction, multiple cutoffs, case-control sampling, and inter-study threshold differences.

23.14.2 General Effect Sizes, Prognosis, and IPD Meta-Analysis

meta, metafor, and rmeta provide fixed/random effects, heterogeneity, meta-regression, and plots. metasens and robustmeta support sensitivity and robust analyses; netmeta supports network meta-analysis; mvmeta, metaSEM, and metavcov support multivariate outcomes; robumeta and clubSandwich support dependent effects and robust variance;

bayesmeta and MetaStan support Bayesian synthesis; metamisc supports prognostic studies; metaSurvival supports survival-curve reconstruction; metagear, revtools, citation-chaser, and PRISMA2020 support review workflows.

Diagnostic performance, treatment effects, and prognostic effects are different problems even when all are called meta-analysis. IPD meta-analysis must model the study level explicitly and may combine with lme4, nlme, survival, or Bayesian hierarchical models.

23.15 Survival and Event Outcomes

IVD biomarkers are often used for prognostic stratification, companion diagnostics, or treatment monitoring, where outcomes are censored recurrence, progression, or death rather than a simple positive/negative label.

Area	Packages	Main capabilities
Basic survival	survival, prodlim, rms	Kaplan–Meier, Cox, survival probabilities, regression
Presentation	survminer, ggsurvfit	Curves and risk tables
Flexible parametric	flexsurv, rstpm2, eha	Parametric and Royston–Parmar models
Nonproportional hazards	timereg, coxphw, nph	Additive, weighted Cox, and non-PH methods
Competing risks	cmprsk, riskRegression	CIF, cause-specific and Fine–Gray models
Multistate	mstate, etm, msm	Disease progression and transitions
Time-dependent ROC	timeROC, survivalROC, riskRegression	Time-specific AUC and ROC
Prediction error	pec, riskRegression, rms	Brier score, calibration, resampling
Penalized/ML survival	glmnet, randomForestSRC, ranger	High-dimensional Cox and survival forests
Frailty/clustered events	coxme, frailtypack, frailtyEM	Center, family, or recurrent-event dependence
Interval censoring	icenReg, interval	Events known only within an interval
Cure models	smcure, flexsurvcure	Long-term event-free subgroups
Joint models	JMbayes2, joineR, joineRML, rstanarm	Longitudinal markers and event time
Design	powerSurvEpi, gsDesign, rpact	Events and sequential designs

Baseline prognostic value, time-varying biomarkers, treatment modification, and diagnostic accuracy are distinct estimands. Report time origin, censoring, competing events, prediction time, validation, and proportional-hazards assumptions.

23.16 Multivariate Analysis, Chemometrics, and High-Dimensional Biomarkers

23.16.1 Exploratory Multivariate Analysis

FactoMineR, factoextra, psych, GPARotation, ade4, and RGCCA support PCA, factor analysis, rotations, multiple-table analysis, and multi-block integration. cluster, mclust, dbSCAN, and fpc support clustering. compositions and robCompositions address compositional data.

rrcov, robustbase, and mvoutlier support robust covariance and multivariate outliers. MVN, mvnrmtest, and ICSNP support multivariate normality and inference.

PCA, clustering, and heatmaps explore structure but do not prove generalizable diagnostic performance. If variables are selected and AUC is reported on the same data, use nested resampling or independent data to correct selection bias.

23.16.2 Chemometrics, Spectra, and Multimarker Quantitation

Package	Main capabilities	IVD scenario
pls	PLS/PCR and cross-validation	Multichannel and collinear calibration
mdatools	PCA, SIMCA, PLS/PLS-DA, preprocessing, validation	Spectra, metabolites, biomarker panels
chemometrics	Multivariate calibration and variable selection	Analytical chemistry
prospectr	Smoothing, derivatives, and scatter correction	Spectral preprocessing
ChemoSpec/ChemoSpec2D	Spectral exploration, PCA, clustering	NMR/IR data
hyperSpec	Spectral and hyperspectral data objects	Multichannel instrument data
ALS	MCR-ALS	Overlapping-signal decomposition

Preprocessing, peak selection, batch correction, and tuning are part of the analysis pipeline and must occur inside cross-validation; selecting wavelengths or variables on all data before validation leaks outcome information.

23.16.3 Classification, Prediction, and Machine Learning

Family	Packages	Main capabilities
Classical discrimination	MASS, klaR, mda, nnet	LDA/QDA, regularized discrimination, flexible discrimination
Regularization	glmnet	Lasso, Ridge, Elastic Net
Kernel methods	e1071, kernlab	SVM and kernel classification
Trees/ensembles	rpart, randomForest, ranger, xgboost	Trees, random forest, boosting
Unified workflows	caret, tidymodels, mlr3	Preprocessing, tuning, resampling, metrics
Imbalance	themis, ROSE	Sampling and synthetic cases, within training folds only
Explainability	iml, DALEX	Importance and local explanations

Machine-learning packages optimize prediction and do not automatically satisfy diagnostic requirements for target population, cutoff, calibration, interpretability, and external validation. Compare algorithms under the same leakage-free resampling plan.

23.17 Regression, Clinical Prediction, and Clinical Utility

Task	Packages	Main capabilities
Basic regression	stats, MASS	Linear, logistic, Poisson, and extensions
Small sample/separation	logistf, brglm2	Firth and bias-reduced logistic regression
Nonlinear effects	mgcv, rms	Smooth terms and restricted cubic splines
Repeated/clustered data	geepack, lme4, nlme, glmmTMB	GEE and mixed models
Adjusted comparisons	emmeans, multcomp, sandwich, clubSandwich	Marginal means, simultaneous inference, robust SE
Model performance	pROC, yardstick, rms, riskRegression, pec	Discrimination, calibration, Brier, validation
Calibration	pmcalibration, rms	Calibration curves, intercept, slope
Clinical utility	dcurses	Decision curves and net benefit
Causal adjustment	MatchIt, WeightIt, survey, cobalt	Matching, weighting, balance

After adding a new marker, assess calibration, clinically relevant sensitivity/specificity, net benefit, overfitting, and external validation rather than AUC increment alone. For causal questions, distinguish predictors, confounders, and mediators.

23.18 Sample Size, Missing Data, and Statistical Support

Package	Main capabilities	Typical IVD use
ivdtools	Method-comparison, precision, linearity, ROC, reference-interval, stability, QC, and sample-size modules	Modular performance-evaluation planning
presize	Precision-based sample size	Sensitivity, specificity, predictive values, AUC, means, proportions
pROC	ROC/AUC power and sample size	One or two ROC curves
MKmisc	Diagnostic and common design power tools	Basic sensitivity/specificity planning
pmsampsize	Prediction-model sample size	Continuous, binary, and survival outcomes
WebPower, pwr	General power calculations	Initial planning and sensitivity
longpower, powerSurvEpi	Longitudinal and survival power	Repeated biomarker and event outcomes
gsDesign, rpact	Group-sequential and adaptive designs	Prospective clinical performance or utility

Total sample size should be divided into positive and negative targets, with separate additions for prevalence, attrition, invalid samples, clustering, pairing, and multiple primary endpoints.

nanianr and VIM inspect missingness; mice and Amelia support chained-equation or joint-model imputation; missForest and missMDA support nonlinear or multivariate imputation; outliers, EnvStats, and nortest support univariate distribution diagnostics; rrcov, mvoutlier, and robustbase support multivariate diagnostics; boot and rsample support resampling; multcomp and emmeans support multiplicity.

Below-detection-limit results, instrument errors, lost follow-up, and random missingness are different mechanisms. The imputation model should include the outcome, design variables, and auxiliary variables and should be compatible with the primary analysis.

23.19 Data Management, Reporting, and Reproducibility

Stage	Packages	Main use
Import and reshaping	readr, readxl, haven, data.table, dplyr, tidyr	Instrument exports, EDC data, analysis datasets
Data validation	checkmate, pointblank, validate, assertr	Type, range, uniqueness, logic, and cross-field checks
Model tidying	broom, parameters, performance	Traceable result tables and model checks
Statistical tables	gtsummary, tableone, arsenal, finalfit	Baseline and regression tables
Figures	ggplot2, patchwork, forestplot	Performance and forest plots
Dynamic reports	knitr, rmarkdown	Integrated code, results, and narrative
Environment locking	renv, sessioninfo	Dependency and runtime records
Workflow orchestration	targets	Incremental auditable pipelines

A formal IVD project should retain a read-only raw-data copy, data-generation rules, package versions, function arguments, random seeds, exclusion records, output objects, and the final report. CRAN publication status is not project-level software validation or regulatory approval.

